# A central-charge-eight CFT with Haagerup symmetry

Expanded verification draft

Construction, spectrum, observables and explicit data

Research manuscript

29 September 2026

**Abstract**

We study a unitary two-dimensional conformal field theory at $c_L = c_R = 8$ with Haagerup non-invertible symmetry. Its construction uses an explicit six-current chiral algebra $U \subset V_{E_6 \oplus A_2}$, with affine levels 1 and 39. Under the stated exact construction inputs, we establish rationality and complete unitarity, identify the twelve Evans–Gannon characters, and obtain both a diagonal $U$ theory and a chosen holomorphic $E_8$ extension with Haagerup defects. These full theories have different physics: their first scalar dimensions are respectively 2/3 and 2, while the first scalars invariant under the specified full categorical symmetry have dimension two. We compare these points with published modular-bootstrap bounds and compute spectra with spin through dimension 24, finite-volume thermodynamics with explicit truncation bounds, exact chiral three-point data and four-point functions, and defect and Cardy-boundary observables. The accompanying tables retain exact arithmetic and normalization conventions. We distinguish actual lattice correlators from character consequences and from nonvacuum bulk sewing data that remain uncomputed. The mathematical arguments, finite certificate inputs, and limited Lean formalizations are documented separately for review.

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**REVIEW — Status of the construction**

This is a research draft. The realization statements depend on the explicit finite inputs in Section M; the subsequent algebraic and analytic implications are argued in the text. Exact replays check the specified finite identities, and the available Lean lemmas cover only limited parts of the argument. Neither is a machine verification of the complete CFT construction. The physics tables state which construction results each observable uses.

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## About this expanded verification draft

This volume starts from the current 78-page manuscript of 27 September 2026 and gathers the accompanying human verification guide, formalization assessment, and additional recorded results into one document. The original manuscript and its data remain unchanged. Its construction, physics discussion, mathematical arguments, and bibliography are retained; the added material is intended to make the finite inputs and the recorded outputs inspectable while reading.

The main text and Appendices A–O retain the mathematical organization of the base revision. Supplementary appendices have numbers beginning with S. All ordinary section, equation, and page references are internal to this volume, including the references in the incorporated companions. Original artifact names remain visible so that a displayed assertion can still be traced to the precise archived input. Those names are provenance, not a request to run the scripts before reading the formulas.

| Verification task | Where the details are printed |
|---|---|
| Read the construction and its current claim boundaries | Section 1, Appendices A–K. |
| Check small lattice identities, stresses, and finite witness formats | The integrated human verification guide, beginning on p. 85; its structural inputs start in Appendix S5. |
| Inspect all character coefficients and the recurrence input | Appendix S20, including the complete characteristic matrix in S20.14. |
| Look up scalars or a state of a given dimension and spin | Appendices S21 and S22; all saved levels through total dimension 24 are represented. |
| Check thermal normalization and error bars | Appendix S23, with outward-rounded endpoints and explicit moment-tail bounds. |
| Inspect actual coordinate vectors without opening TSV files | Appendix S24, with the sparse monomials of the stresses and the two weight-three structural fields. Additional charged highest vectors appear in the finite construction supplement. |
| Inspect spectral eigenvectors and all finite top-generation recipes | Appendix S25, including the eighty-current complement and the four mixed 81-dimensional tops. |
| Check the scope of machine verification | Appendices S26 and S27; the claims of those archived assessments retain their original scope. |
| Find a source or reconstruct this document | The source concordance and assembly record in Appendix S28. |

**How to read the evidence.** An exact coefficient or coordinate table records finite arithmetic. A membership identity additionally requires a recipe identifying its columns with actual lattice modes. A rank lower bound requires a nonzero minor or an elimination witness; a rank upper bound requires spanning expressions. None of these alone proves the all-weight, categorical, or analytic conclusions. The base draft supplies the arguments connecting those statements, with the finite and external hypotheses exposed. The added appendices preserve that division of work.

**Version choices.** The completed physics tensor has 113 weight-two Virasoro primaries and 18,365 nonzero ordered couplings. The older 96-field export is an intermediate record. Similarly,

early exploratory reports that describe properness, rationality, or the annulus comparison as open are historical records: the current manuscript gives its later arguments. An archived extra identification is described with its own hypotheses and is not silently promoted to an independently checked theorem by inclusion here.

**What was done for this edition.** Existing results were transcribed, organized, and cross-referenced; compact exact descriptions replace repeated large arrays where they retain the full information. The assembly checks compare printed table rows with their saved sources and check document integrity. They are not new lattice computations, a replay of the research certificates, or a new proof of the construction. Historical timing, memory, and success records refer to their original runs. The absence of bulk nonvacuum OPE and sewing data remains explicit in the physics discussion.

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# 1 Introduction and statement of results

Non-invertible topological lines constrain the spectrum and interactions of a two-dimensional CFT, but a consistent set of line fusion rules does not by itself produce a local theory. The Haagerup categories offer a particularly concrete test: they have small fusion rings, nonintegral quantum dimensions, and no immediate realization as an ordinary finite group symmetry. For physics, the questions are both constructive and quantitative. Does a unitary CFT realize the category, where does that theory lie in bootstrap exclusion plots, and which normalized observables can be computed beyond a torus partition function?

Evans and Gannon supplied the modular data and candidate character vectors at central charge eight [9, Sections 2.4 and 5]. Albert–Honda–Kaidi–Zheng studied a proposed Haagerup action in $(E_8)_1$ and the corresponding symmetry-resolved bootstrap [11]. Here an explicit lattice construction supplies the operator products needed to go beyond modular data. It also makes an essential distinction visible: the diagonal full CFT of the smaller chiral algebra $U$ is different from the full CFT of its holomorphic $E_8$ extension. The former has many relevant scalars; the latter has none. Their symmetry-invariant scalar gaps nevertheless agree at two.

The principal spectral comparison is as follows. Degeneracies count individual Virasoro-primary components, rather than extended-algebra multiplets. In the last row the diagonal theory uses all Verlinde lines, and the extension uses the specified first chiral Haagerup factor.

| Observable | Diagonal $U$ | Selected $E_8$ full theory |
|---|---:|---:|
| Currents per chirality | 6 | 248 |
| First nonidentity Virasoro-primary dimension | 2/3 | 1 |
| First nonidentity scalar dimension | 2/3 | 2 |
| Relevant scalar components | 2941 | 0 |
| Dimension-two scalar components | 19558 | 61504 |
| Invariant scalar gap (multiplicity) | 2 (36) | 2 (1488) |

Our interaction data include a complete three-point tensor for the 113 weight-two chiral Virasoro primaries of $U$, and explicit normalized four-point functions. The spectrum export reaches total dimension 24. Defect-resolved partition functions and boundary annuli give observables that distinguish the categorical action even when the ordinary $E_8$ partition function is already known.

We start with actual vectors in a positive-definite lattice VOA. The ambient vertex operators fix all products and their associativity from the outset. The first substantive problem is then to prove that these vectors generate a proper subalgebra. Generating additional fields in weights two and three does not settle that question: singular products of higher fields could still create new currents. We address this issue by a finite strong-generation criterion. Its input consists of exact normal-product identities and the recursive provenance of the higher generators. Its output controls every conformal weight.

The construction has three distinct stages. The first constructs the subalgebra and proves properness from explicit finite data. The second uses positive conformal vectors, branching, and fusion-category arguments to analyze its representation theory. The third compares VOA modules with conformal net sectors, proves the net realization, and establishes complete VOA unitarity through bounded annulus insertions. In particular, the existence of a positive vacuum inner product is not substituted for positivity of all fusion pairings.

Our presentation uses the normal-product conventions of De Sole–Kac [8, Section 2], separates character-theoretic information as in Evans–Gannon [9], and states analytic transport hypotheses in the operator-theoretic setting of Carpi–Kawahigashi–Longo–Weiner [10]. These works also guide the level of detail: finite input is recorded as such, and each passage to an

infinite-dimensional conclusion is made by a separate lemma. The exact arithmetic certificates are accompanying data, not substitutes for these implications.

## 1.1 The candidate and its finite construction

Let

$$B = V_{E_6 \oplus A_2}, \qquad H = V_{E_8}, \qquad B \subset H$$

be the standard index-three lattice extension. The notation $A_{1,k}$ means the simple affine VOA $L_k(\mathfrak{sl}_2)$. Fix the embedding $\mathcal{K} = A_{1,1} \otimes A_{1,39} \subset B$ described in Section A. Let $u \in B_2$ be the field in (83), and define the *actual generated* vertex algebra

$$U = \langle \mathcal{K}, \omega_B, u \rangle \subset B. \tag{1}$$

All products are computed in this lattice VOA. Thus associativity is inherited from the ambient algebra, rather than assumed for an abstract presentation inferred from character coefficients. Throughout the normal-space and mode formulas, $K = \mathcal{K}_1$ denotes only the six-dimensional current space, not the full affine VOA.

**Proposition 1.1** (Computational properness conclusion)**.** *Accept the exact lattice identities and provenance recorded in Section M. The algebra $U$ is strongly generated by actual fields of weights at most four. Its weight-one space is*

$$U_1 = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2, \qquad \dim U_1 = 6.$$

*Consequently $U \subsetneq B$, since* $\dim B_1 = 86$.

The finite certificates and the all-weight implication are presented in Sections B–E. This is the most directly checkable part of the construction.

**Corollary 1.2** (Initial graded pieces)**.** *Under the same finite-certificate hypotheses, $U$ is a simple unitary VOA of CFT type with the ambient conformal vector and central charge eight. Moreover,*

$$U_2 = S_2, \qquad U_3 = S_3, \qquad \mathrm{ch}_U(\tau) = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + O(q^4)).$$

*Here $S_2, S_3$ are the concrete spaces in Certificate A.2. This assertion does not require a modular character calculation.*

The proof is given after the properness argument. In particular the coefficient 6010 in the proposed character below has a different dependency from the first four coefficients.

We write $\mathcal{H}_3$ for the unitary Haagerup fusion category with six simple objects $\mathbf{1}, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho$ and fusion rules

$$\alpha^3 = \mathbf{1}, \qquad \alpha\rho = \rho\alpha^{-1}, \qquad \rho^2 = \mathbf{1} + \rho + \alpha\rho + \alpha^2\rho.$$

Here $\mathrm{FPdim}(\rho) = (3 + \sqrt{13})/2$. The other unitary category with this fusion ring, denoted $\mathcal{H}_2$, is Morita equivalent to $\mathcal{H}_3$. Consequently their Drinfeld centers are equivalent even though the fusion categories are not. In the statement below $Z$ denotes this center, not merely its modular matrices.

**Theorem 1.3** (Conformal-net realization)**.** *Assume the exact finite construction inputs recorded in Section M and Certificate F.1. The VOA $U$ is strongly rational and strongly unitary, with central charge eight. Its completely rational smeared-field conformal net $\mathcal{N}_U$ satisfies*

$$\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{H}_3)$$

*as unitary ribbon categories. The twelve-component character vector is the $\gamma = 0$ Evans–Gannon vector, and in particular*

$$\mathrm{ch}_U(\tau) = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots). \tag{2}$$

*There is a diagonal unitary full conformal net with $c_L = c_R = 8$ and partition function $\sum_{i=1}^{12} |\chi_i(\tau)|^2$.*

The proof is assembled in Theorem F.8, Proposition I.4, Corollary J.3, and Proposition K.2. The character argument is given before the analytic comparison and does not presume it.

**Theorem 1.4** (VOA realization and holomorphic extension)**.** *Under the same construction inputs, $U$ is completely unitary and*

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3) \tag{3}$$

*as unitary ribbon categories. There is a unitary holomorphic extension $H' \cong V_{E_8}$ whose category of all algebra modules is $\mathcal{H}_3$; its algebra-bimodule category is $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$.*

The additional analytic step is the pairwise annulus comparison in Proposition I.5. Together with module exhaustion, it gives Theorem I.8; the categorical identification and extension are then proved in Section J and Section K.

Here $H'$ is obtained by transporting the canonical $\mathcal{H}_3$ Lagrangian algebra. It need not coincide with the originally embedded extension $H/U$; the argument identifies that original all-module category only up to the alternatives $\mathcal{H}_2, \mathcal{H}_3$. The full-CFT defects in Theorem 1.4 preserve $U$, and need not commute with the enlarged $E_8$ chiral algebra.

## 1.2 How the qualifications are used

Amber **REVIEW** notes identify the boundary of exact construction inputs and their replay coverage. Red **HIGH PRIORITY REVIEW** notes identify the scope of the construction and the limits of its verification. A proved lemma is not made into an additional hypothesis merely because its argument is new; its proof and external dependencies are supplied for scrutiny. Exact construction identities that are accepted as finite inputs are stated separately from these deductions.

| Part of the argument | Nature of its evidence |
|---|---|
| Explicit fields and finite closure | Exact arithmetic over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$, saved recipes and full residuals. |
| Properness in all weights | Recursive algebraic theorem plus finite certificates; no character upper-bound extrapolation. |
| Rationality and branching | Positive-stress and category arguments, embedded-field and finite-top inputs, and cited external theorems. |
| Haagerup modular data | Rank exclusions, indicator and character arguments printed below, with exact finite arithmetic checked separately. |
| Net realization and strong unitarity | Actual branching, linear reconstruction, sector counting, and unitary classification. |
| Complete VOA unitarity and extension | Pairwise annulus comparison, fusion positivity, unitary classification, and extension theorems. |

### 1.3 A reading guide and the two full theories

Sections 2–7 collect the physical observables before the construction proofs. All central charges quoted in a bootstrap plot mean the common chiral value $c = c_L = c_R$, so our point is $c = 8$, not $c_L + c_R = 16$. We use $\Delta = h + \bar{h}$ and $s = h - \bar{h}$; a scalar has $s = 0$. A "primary" without further qualification is primary for the two Virasoro algebras. An extended-algebra sector can contain many such primaries, so a sector count is not an operator degeneracy.

Write $M_0 = U, M_b, M_a, M_{c_0}, \ldots, M_{d_6}$ for the twelve simple $U$ modules, and $H'$ for the selected extension in Theorem 1.4. The two bulk Hilbert spaces are

$$\mathcal{H}_{\mathrm{diag}} = \bigoplus_i M_i \otimes \overline{M_i}, \qquad \mathcal{H}_{E_8} = H' \otimes \overline{H'}. \tag{4}$$

The canonical Verlinde lines of the first theory form $\mathrm{Rep}(U) = Z(\mathcal{H}_3)$. In the second theory we specify the chosen first chiral factor $\mathcal{H}_3 \subset \mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ when discussing the Haagerup bootstrap. Statements about invariance under both factors are labeled separately. These lines preserve $U$ and the stress tensor; they need not preserve the enlarged $E_8$ current algebra pointwise.

## 2 Characters and a spin-resolved physical spectrum

The chiral algebra fixes considerably more physical data than the twelve lowest conformal weights. In this section $c_L = c_R = 8$, so the quantity often denoted simply by $c$ in two-dimensional bootstrap plots is 8; the sum of the two central charges is 16. We distinguish two local full CFTs throughout:

$$Z_{\mathrm{diag}}(\tau, \bar{\tau}) = \sum_{i=0}^{11} \chi_i(\tau)\overline{\chi_i(\tau)}, \qquad Z_{E_8}(\tau, \bar{\tau}) = \left| \frac{E_4(\tau)}{\eta(\tau)^8} \right|^2 . \tag{5}$$

The first uses the diagonal modular invariant of $U$. The second is the full theory of the selected holomorphic extension carrying the Haagerup action constructed later in the manuscript. They have the same central charge and different spectra. In particular a spectral statement about the diagonal theory must not be assigned to the holomorphic extension.

### 2.1 An exact recurrence and its normalization

Set $q = e^{2\pi i \tau}$ and write

$$\chi_i(q) = q^{h_i - 1/3} A_i(q), \qquad A_i(q) = \sum_{n \geq 0} a_i(n) q^n .$$

The grading variable $n$ measures distance above the actual lowest weight, including for $d_3$, whose lowest weight is $15/13$ rather than its twist representative $2/13$. The parameter-zero solution of Evans–Gannon [9, Sections 1.4 and 2.4] has the following initial coefficients; the two rows $a, c_0$ represent distinct simple modules with equal unflavoured characters.

| $i$ | $h_i$ | $a_i(0)$ | $a_i(1)$ | $a_i(2)$ | $a_i(3)$ |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 6 | 120 | 956 |
| $b$ | 1 | 80 | 1250 | 10630 | 65042 |
| $a$ | 1 | 81 | 1377 | 11583 | 71037 |
| $c_0$ | 1 | 81 | 1377 | 11583 | 71037 |
| $c_1$ | 1/3 | 3 | 243 | 2916 | 21870 |
| $c_2$ | 2/3 | 27 | 594 | 5967 | 39852 |
| $d_1$ | 6/13 | 7 | 292 | 3204 | 23010 |
| $d_2$ | 11/13 | 42 | 777 | 7147 | 45367 |
| $d_3$ | 15/13 | 119 | 1623 | 12996 | 76429 |
| $d_4$ | 5/13 | 5 | 229 | 2738 | 19942 |
| $d_5$ | 7/13 | 13 | 347 | 3804 | 26390 |
| $d_6$ | 8/13 | 14 | 441 | 4445 | 30329 |

Here is an explicit recurrence that extends this table without a lattice-state calculation. Write

$$\Xi = q^{\Lambda} \sum_{n \geq -1} X_n q^n, \quad X_{-1} = I, \quad X_0 = \chi, \quad C = \chi + [\Lambda, \chi],$$

where $\Lambda, \chi$ are the rational exponent matrix and integral characteristic matrix printed in [9, Section 2.4]. They are also recorded in full in the machine-readable dataset. With $J = q^{-1} + 196884q + \cdots$, define

$$g = \frac{\eta^{24}}{E_4 E_6} = \sum_{r \geq 0} g_r q^r, \qquad f = (J - 240)g = \frac{E_4^2}{E_6} - 984g = \sum_{r \geq 0} f_r q^r.$$

Thus $f_0 = 1$, $f_1 = 0$, and $g_0 = 0, g_1 = 1, g_2 = 240$. The differential equation $q\, d\Xi/dq = \Xi(f(\Lambda - I) + gC)$ gives

$$(n + 1 + \lambda_i - \lambda_j)(X_n)_{ij} = \sum_{r=1}^{n+1} \left[ f_r(\lambda_j - 1)(X_{n-r})_{ij} + g_r(X_{n-r}C)_{ij} \right]. \tag{6}$$

All denominators are nonzero. In particular this specifies the convolution index directly, independently of typographical renderings of the $g$ index in equation (1.24) of the cited paper. The desired characters are column zero. Vacuum grade $n$ uses $(X_{n-1})_{00}$, whereas nonvacuum grade $n$ uses $(X_n)_{i0}$.

The exporter uses rational arithmetic, caches $X_n C$ once per grade, and compares the first four printed coefficient columns with the published data. An independent oscillator-product calculation checks

$$\frac{E_4}{\eta^8} = \chi_0 + \chi_b + 2\chi_a = \chi_0 + \chi_b + 2\chi_{c_0} = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + \cdots) \tag{7}$$

through every exported grade. Equal scalar characters do not choose between possible extension algebra structures.

## 2.2 Removing Virasoro descendants

We count individual primary components, including their internal multiplicities, rather than only the twelve extended-algebra sectors. Let $P(q) = \prod_{m \geq 1}(1 - q^m)$, and let $v_i(n)$ count Virasoro highest vectors of weight $h_i + n$ in the $i$th chiral module. Include the identity once by setting $v_0(0) = 1$. Then

$$\sum_{n \geq 0} v_i(n) q^n = P(q) A_i(q) \quad (i \neq 0), \qquad \sum_{n \geq 0} v_0(n) q^n = P(q) A_0(q) + q. \tag{8}$$

The correction $+q$ removes the nonexistent vacuum state $L_{-1}\mathbf{1}$ from the descendant subtraction.

For completeness, the relevant absence of further null vectors can be seen directly from the Kac determinant [7, Lemmas 2–3]. Parametrize $c = 13 - 6(t + t^{-1})$; at $c = 8$, one has $|t| = 1$, $t + t^{-1} = 5/6$, and $\operatorname{Im} t \neq 0$. Its possible singular weights are

$$h_{r,s} = \tfrac{1}{4}((r^2 - 1)t + (s^2 - 1)t^{-1} - 2rs + 2).$$

For positive integers $r, s$, reality forces $r = s$, in which case $h_{r,r} = -7(r^2 - 1)/24 \leq 0$. A positive real highest weight therefore has an irreducible Verma module. At weight zero the only primitive null is $L_{-1}\mathbf{1}$. Unitarity permits orthogonal removal of the descendant submodules at each finite grade, giving (8). For example,

$$(v_0(0), v_0(1), v_0(2), v_0(3), v_0(4)) = (1, 6, 113, 830, 4934).$$

The corresponding first nonvacuum grades are

| $i$ | $v_i(0)$ | $v_i(1)$ | $v_i(2)$ | $v_i(3)$ | $i$ | $v_i(0)$ | $v_i(1)$ | $v_i(2)$ | $v_i(3)$ |
|---|---|---|---|---|---|---|---|---|---|
| $b$ | 80 | 1170 | 9300 | 53162 | $d_1$ | 7 | 285 | 2905 | 19514 |
| $a, c_0$ | 81 | 1296 | 10125 | 58077 | $d_2$ | 42 | 735 | 6328 | 37443 |
| $c_1$ | 3 | 240 | 2670 | 18711 | $d_3$ | 119 | 1504 | 11254 | 61810 |
| $c_2$ | 27 | 567 | 5346 | 33291 | $d_4$ | 5 | 224 | 2504 | 16975 |
| | | | | | $d_5$ | 13 | 334 | 3444 | 22239 |
| | | | | | $d_6$ | 14 | 427 | 3990 | 25443 |

Every exported chiral character is reconstructed independently by convolving these multiplicities with the partition numbers and using the vacuum correction. This verifies descendant subtraction through the stated grade; it is not a separate existence proof for $U$.

### 2.3 The diagonal theory: scalars and spinning operators

An ordered pair of grades $(n, m)$ in sector $i$ has

$$\begin{gathered}(h, \bar{h}) = (h_i + n, h_i + m), \quad \Delta = 2h_i + n + m, \quad s = n - m, \\ d_{\text{states}} = a_i(n)a_i(m), \quad d_{\text{prim}} = v_i(n)v_i(m).\end{gathered} \tag{9}$$

The last quantity counts primaries under both Virasoro algebras. It includes chiral currents paired with the right vacuum, and vice versa. Summing equal $(\Delta, s)$ entries produces the spin-resolved tables in the dataset. Character grades through $N$ suffice for the complete spectrum through $\Delta = N$, because all lowest weights are nonnegative. Rectangular products at larger $\Delta$ would generally be incomplete and are not reported as complete levels.

The seven relevant scalar levels are

| $\Delta$ | 2/3 | 10/13 | 12/13 | 14/13 | 16/13 | 4/3 | 22/13 |
|---|---|---|---|---|---|---|---|
| $d_{\text{prim}}$ | 9 | 25 | 49 | 169 | 196 | 729 | 1764 |

(10)

and contain 2941 primary components in total. These are the squares of the seven relevant chiral ground-space dimensions. No scalar descendant can enter below dimension two: equality of left and right weights requires $n = m$, so a positive grade raises $\Delta$ by at least two.

The complete additional relevant primary levels with nonzero spin are

| $\Delta$ | $s$ | multiplicity for each displayed spin |
|---|---|---|
| 1 | $\pm 1$ | 6 |
| 5/3 | $\pm 1$ | 720 |
| 23/13 | $\pm 1$ | 1120 |
| 25/13 | $\pm 1$ | 1995 |

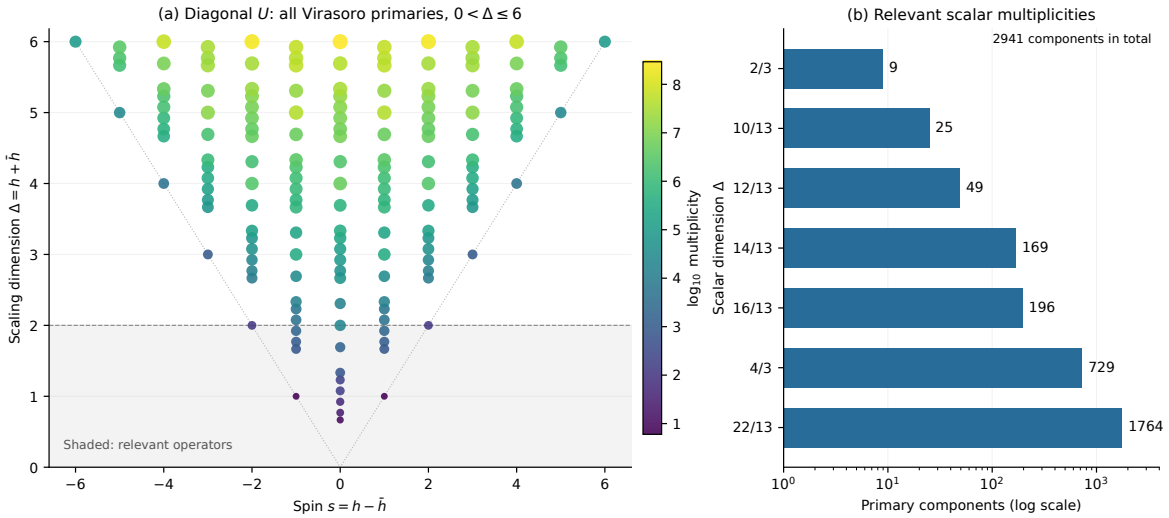



Figure 1: Exact Virasoro-primary spectrum of the diagonal $U$ theory. Left: all nonidentity $(\Delta, s)$ levels with $0 < \Delta \leq 6$, aggregated over the twelve extended sectors; color and marker area increase with the logarithm of the component multiplicity. The shaded region has $\Delta < 2$. Right: all seven relevant scalar levels, with their exact multiplicities and a logarithmic count axis. The identity and Virasoro descendants are excluded.

For example, the second line is $3(243 - 3) = 720$: the three $c_1$ ground states pair with its 240 new grade-one Virasoro primaries. Including scalars, there are 10623 nonidentity relevant Virasoro-primary components. The ordinary primary gap and the scalar primary gap are both $2/3$. There are six currents in each chirality, generating the affine subalgebra $\mathfrak{sl}_{2,1} \oplus \mathfrak{sl}_{2,39}$.

At dimension two the scalar-primary multiplicity is

$$80^2 + 81^2 + 81^2 + 6^2 = 19558. \tag{11}$$

The first three terms are the ground spaces of $b, a, c_0$; the final term comprises the current bilinears $J^A \bar{J}^B$. In addition there are 113 chiral weight-two primaries in each spin $\pm 2$ channel. Thus dimension two contains 19784 Virasoro-primary components and 19798 states. The difference consists of the two stress tensors and twelve current derivatives. We call dimension-two scalars *marginal by dimension*; their beta functions and exact marginality require OPE data not fixed by the character vector.

## 2.4 The holomorphic extension and symmetry-preserving perturbations

For the full $E_8$ theory, the same descendant subtraction gives 248 chiral weight-one and 3875 chiral weight-two Virasoro primaries. Consequently its ordinary primary gap is one, its scalar primary gap is two, and it has no relevant scalar. The $\Delta = 2$ scalar space has dimension $248^2 = 61504$; the total dimension-two state multiplicity is $2 \cdot 4124 + 248^2 = 69752$.

For the selected canonical extension, the defect construction below specifies a first chiral $\mathcal{H}_3$ action inside the full $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ defect category. Its invariant Hilbert space is $U \otimes \overline{H'}$: the first factor acts in its vacuum representation on $U$ and leaves the antiholomorphic extension unrestricted. It therefore has exactly $6 \cdot 248 = 1488$ invariant scalar components at dimension two. Requiring invariance under *both independent factors* instead gives $U \otimes \overline{U}$ and $6^2 = 36$ such components. Each convention has invariant scalar gap exactly two. These invariant counts use the specified canonical-extension defect action, not only the unflavoured character

identity. Neither count establishes an exactly marginal family: that requires the corresponding OPE coefficients and beta functions.

### 2.5 Finite-temperature observables with controlled tails

On a circle of circumference $L$ and velocity $v$, set $t = \beta v/L$, $b = 2\pi t$ and $E = \Delta - 2/3$. Then

$$H = \frac{2\pi v}{L}E, \qquad P = \frac{2\pi}{L}s, \qquad Z(b) = \sum_{\Delta,s} d_{\Delta,s} e^{-b(\Delta-2/3)}.$$

The ground energy is $-4\pi v/(3L)$. The exact finite spectral table already gives a lower bound on $Z$ at every positive temperature. The following elementary modular estimate also controls the omitted tail and makes a finite-temperature table quantitatively meaningful.

**Proposition 2.1** (A tail bound from modular invariance)**.** *Suppose $Z(b) = Z(4\pi^2/b)$ has nonnegative degeneracies, and that all states with $\Delta \le D$ are known, where $D > 2/3$. Write $Z = L_D + H_D$ for its known and omitted sums, and put $E_D = D - 2/3$. If $b > 2\pi$ and $b' = 4\pi^2/b$, then*

$$0 \le H_D(b) \le \frac{L_D(b) - L_D(b')}{e^{(b-b')E_D} - 1}. \tag{12}$$

*At $b = 2\pi$ one has $0 \le H_D(b) \le L_D'(b)/E_D$. For $b < 2\pi$ the same bounds apply to $Z(b') = Z(b)$.*

*Proof.* Every omitted energy exceeds $E_D > 0$, hence $H_D(b) \le rH_D(b')$ with $r = e^{-(b-b')E_D} < 1$. Modular invariance gives $H_D(b') = H_D(b) + L_D(b) - L_D(b')$; rearranging yields (12). At the fixed point $Z'(2\pi) = 0$, so $L_D'(2\pi) = -H_D'(2\pi) \ge E_D H_D(2\pi)$. □

The real-number rearrangement and the positive-division step at the fixed point are formalized as `modular_heavy_tail_comparison` and `selfdual_tail_first_moment_comparison` in `lean/HaagerupIzumi/CFT/PhysicsBounds.lean`. Their hypotheses explicitly include the balance, contraction, and first-moment inequalities used above. They do not formalize spectral convergence, the modular functional equation, differentiation of the trace, or the decimal interval implementation. The guarded compiler and axiom report is `c8_physics_20260927/physics_lean.log`.

To control energy and heat capacity, choose $0 < b_0 < b$ and bound $H_D(b_0)$ by the same proposition. For $k = 1, 2$,

$$\sum_{\Delta > D, s} d_{\Delta,s} E^k e^{-bE} \le H_D(b_0) \sup_{E \ge E_D} E^k e^{-(b-b_0)E}. \tag{13}$$

The supremum occurs at $E_D$ when $(b - b_0)E_D \ge k$, and otherwise at $k/(b - b_0)$. Together with the exact low-energy moments this bounds $\langle E\rangle$, $C_V = b^2 \operatorname{Var}(E)$, and $S = \log Z + b\langle E\rangle$. At $t = 1$, modular invariance fixes $\langle E\rangle = 0$ exactly.

The supplied thermal tables use $D = 24$ and $t = 1/2, 3/4, 1, 3/2, 2$, with $b_0 = 0.9b$ for moment bounds. The implementation propagates closed decimal intervals with outward rounding and a rational enclosure of $\pi$; the tables report both endpoints. The supplied 100-digit enclosure of $\pi$ is independently checked by rational alternating sums in Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, using 80 and 25 terms with explicit next-term bounds. Their certification is conditional on positivity and modular invariance of the full CFT, which are mathematical premises established elsewhere, and not consequences of finite coefficient agreement. No unbounded-temperature accuracy claim is made.

For direct comparison the following values are rounded from those intervals. Every displayed number has absolute error less than $10^{-6}$, including both rounding and the omitted-state tail; the exporter checks this containment before producing the table.

| theory | $t$ | $Z$ | $\langle\Delta - 2/3\rangle$ | $C_V$ | $S$ |
|---|---|---|---|---|---|
| $U$ diagonal | 1/2 | 4367.819471 | 2.653128 | 17.089150 | 16.717068 |
| $U$ diagonal | 3/4 | 303.785760 | 0.983857 | 16.343538 | 10.352640 |
| $U$ diagonal | 1 | 147.219258 | 0.000000 | 19.266968 | 4.991923 |
| $U$ diagonal | 3/2 | 565.179187 | -0.621644 | 3.473060 | 0.478284 |
| $U$ diagonal | 2 | 4367.819471 | -0.663282 | 0.419054 | 0.046971 |
| $E_8$ full | 1/2 | 4356.000000 | 2.659753 | 16.984459 | 16.735170 |
| $E_8$ full | 3/4 | 297.792228 | 0.991916 | 16.622062 | 10.370689 |
| $E_8$ full | 1 | 144.000000 | 0.000000 | 19.157072 | 4.969813 |
| $E_8$ full | 3/2 | 557.169604 | -0.627321 | 3.435582 | 0.410512 |
| $E_8$ full | 2 | 4356.000000 | -0.664938 | 0.272738 | 0.023449 |

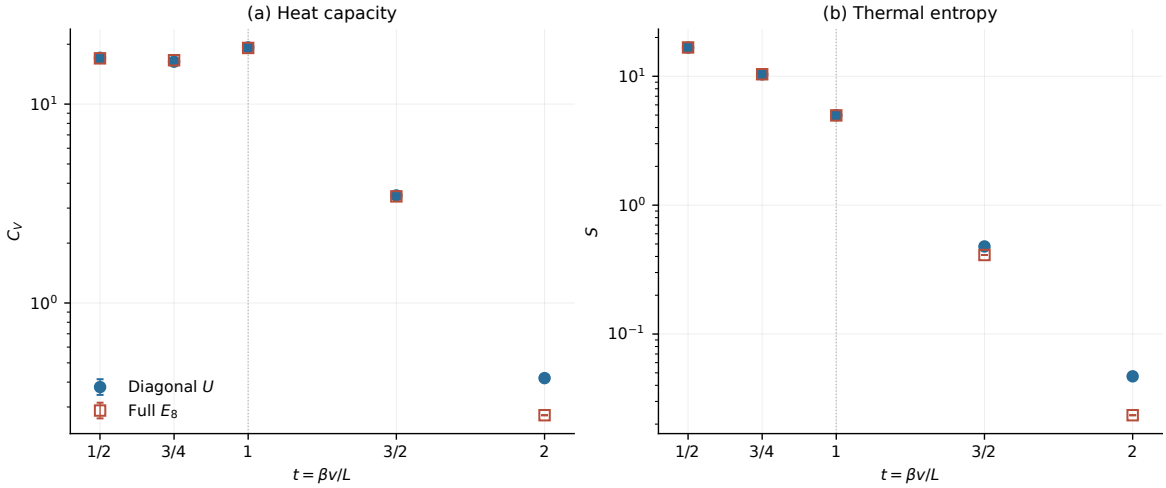



Figure 2: Heat capacity and thermal entropy of the two full theories at the five computed temperatures $t = \beta v/L$. Both vertical axes are logarithmic. No interpolation is drawn. Error bars come from the conditional certified intervals using all states through $\Delta = 24$ and the modular/moment tail bounds; they are smaller than the markers. The dotted line marks the modular fixed point $t = 1$.

In the thermodynamic limit both theories have

$$f(T) - f(0) = -\frac{4\pi}{3v}T^2, \qquad s(T) = \frac{C_V(T)}{L} = \frac{8\pi}{3v}T.$$

Their universal single-interval ground-state entanglement coefficient is $8/3$ in front of $\log(\ell/\epsilon)$, and the Rényi coefficient is $\frac{4}{3}(1+n^{-1})$. These central-charge observables do not distinguish the diagonal theory from its holomorphic extension; the low-energy spectrum and defect-resolved observables do.

## 2.6 Dataset, reproducibility, and limits

The standalone standard-library producer is `c8_physics_20260927/spectrum.py`. Its default run computes 24 grades of all twelve characters, including Virasoro-primary multiplicities, and complete spectra through $\Delta = 24$ for both full theories. The output directory contains `characters.json`, `characters.tsv`, `fundamental_matrix.json`, `diagonal_sector_spectrum.tsv`, `diagonal_spectrum.tsv`, `e8_spectrum.tsv`, `scalars.tsv`, `low_spectrum.json`, `thermal.json`, and `thermal.tsv`. The run summary records coefficient controls, nonnegative integrality, descendant reconstruction, spin reflection, the $E_8$ product check, and source/output

hashes. Paths in this paragraph are relative to `c8_physics_20260927/data/`, except the explicitly named producer.

For a replay into a fresh directory, run serially from the repository root (ordinary Python, without optimization):

```
python hi_cft_safe_run.py --limit-mib 512 -- \
  python c8_physics_20260927/spectrum.py --order 24 --delta-cut 24 \
  --outdir /tmp/c8-physics-spectrum-replay
```

Existing output files are not overwritten. The address cap in this command is lower than the permanent machine limit; the launcher also retains its host-memory reserve, process-group watchdog, and two-CPU limit.

The spectrum exporter determines unflavoured graded dimensions and the stated thermal observables. It does not compute normalized three-point coefficients, crossing matrices of conformal blocks, defect eigenvectors, or branching characters with Cartan fugacities. The separate defect and correlator datasets below provide further information of those kinds in their stated scopes. In particular an exclusion plot that resolves Haagerup charges uses the specified defect-sector decomposition; ordinary unflavoured gaps alone do not supply its inputs.

# 3 Exact correlators and channel data

The vacuum algebra gives several explicit interacting chiral correlators, in addition to current and stress-tensor Ward identities. We give two four-point functions of the weight-two generator, two spin-three four-point functions, and selected intertwiner blocks for the lightest nonvacuum module. All normalizations below are fixed. The last family consists of chiral blocks obtained from the containing lattice theory; it does not specify the diagonal $U$ theory's nonvacuum bulk four-point function.

## 3.1 Conventions and Ward identities

Write $T$ for the total stress tensor, with

$$T(z)T(0) \sim \frac{4}{z^4} + \frac{2T(0)}{z^2} + \frac{\partial T(0)}{z}. \tag{14}$$

The two commuting affine $\mathfrak{sl}_2$ algebras have levels $k_L = 1$ and $k_R = 39$. We use Cartan currents with $H_L(z)H_L(0) \sim 2z^{-2}$ and $H_R(z)H_R(0) \sim 78z^{-2}$; their mixed two-point function vanishes. More generally, for invariant form $\kappa$ normalized by $\kappa(H, H) = 2$, the currents obey

$$J^a(z)J^b(0) \sim \frac{k\kappa^{ab}}{z^2} + \frac{f^{ab}{}_c J^c(0)}{z}. \tag{15}$$

This fixes every vacuum current correlator recursively. For example, with $f^{abc} = f^{ab}{}_d\kappa^{dc}$, $\langle J^a(z_1)J^b(z_2)J^c(z_3)\rangle = kf^{abc}/(z_{12}z_{13}z_{23})$.

Let $\widehat{u} = u/\sqrt{12}$ and let $\widehat{u}^\dagger$ denote its chiral adjoint in the transported positive lattice form. The symbol † does not introduce an antiholomorphic field. This primary has weight two, Cartan charges $(q_L, q_R) = (1, 13)$, and

$$\langle \widehat{u}(z)\widehat{u}^\dagger(0)\rangle = z^{-4}, \tag{16}$$

$$\langle H_a(z_1)\widehat{u}(z_2)\widehat{u}^\dagger(z_3)\rangle = \frac{q_a}{z_{12}z_{13}z_{23}^3}, \tag{17}$$

$$\langle T(z_1)\widehat{u}(z_2)\widehat{u}^\dagger(z_3)\rangle = \frac{2}{z_{12}^2z_{13}^2z_{23}^2}. \tag{18}$$

Consequently the squared couplings to the unit Cartan currents are $1/2$ and $13/6$. A useful mixed four-point function is

$$\langle H_a(z_1)H_b(z_2)\widehat{u}(z_3)\widehat{u}^\dagger(z_4)\rangle = \frac{1}{z_{34}^4}\left[\frac{K_{ab}}{z_{12}^2} + q_a q_b\left(\frac{1}{z_{13}}-\frac{1}{z_{14}}\right)\left(\frac{1}{z_{23}}-\frac{1}{z_{24}}\right)\right]. \tag{19}$$

Here $K = \mathrm{diag}(2, 78)$. The unit stress tensor is $\widehat{T} = T/2$; both $C_{\widehat{T}\widehat{u}\widehat{u}^\dagger}$ and $C_{\widehat{T}\widehat{T}\widehat{T}}$ equal one in these conventions. Writing $y = 1 - x$, the normalized four-point functions of any unit Cartan current $\widehat{H}$ and of $\widehat{T}$ are

$$\langle \widehat{H}(\infty)\widehat{H}(1)\widehat{H}(x)\widehat{H}(0)\rangle = 1 + x^{-2} + y^{-2}, \tag{20}$$

$$\langle \widehat{T}(\infty)\widehat{T}(1)\widehat{T}(x)\widehat{T}(0)\rangle = 1 + x^{-4} + y^{-4} + x^{-2} + y^{-2} + x^{-1} + y^{-1}. \tag{21}$$

The first follows from Heisenberg Wick contractions, the second from the Virasoro Ward identity at central charge eight.

### 3.2 A three-exponential four-point function

In simple-root coordinates of $E_6 \oplus A_2$, put

$$a_1 = (0, 0, 0, 0, 0, -1, 1, 1), \tag{22}$$

$$a_2 = (0, 0, 0, 1, 1, -1, 0, 0), \tag{23}$$

$$a_3 = (1, 1, 0, 0, 0, -1, 0, 0), \tag{24}$$

$$u = e^{a_1} + \frac{\sqrt{13} + 3\sqrt{-3}}{20}e^{a_2} + \frac{\sqrt{13} - 3\sqrt{-3}}{20}e^{a_3}. \tag{25}$$

The simple-root Gram matrix has diagonal entries two and entries minus one along the edges $(0,1),(1,2),(2,3),(3,4),(2,5),(6,7)$, with indices starting at zero. Thus $(a_i, a_i) = 4$ and $(a_i, a_j) = 2$ for $i \neq j$. In the torus gauge $(5, 8, 9, 8, 5, 1, 2, 2)$, the three exponential squared norms are $4, 40, 40$. Their coefficients therefore contribute $4, 4, 4$ to $\|u\|^2 = 12$. In particular this calculation uses the transported positive form, rather than treating the three displayed coefficients as coordinates in an orthonormal basis.

The lattice Gaussian formula in a specified cocycle gauge is

$$\left\langle \prod_{j=1}^{n} e^{b_j}(z_j)\right\rangle = \delta_{\sum_j b_j, 0}\,\varepsilon(b_1, \ldots, b_n)\prod_{i<j}(z_i - z_j)^{(b_i, b_j)}. \tag{26}$$

Here the integer-lattice cocycle is

$$\varepsilon(a, b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}, \tag{27}$$

and the multivariable factor is its iterated product. These formulas also give a short independent derivation: expand each of the four fields, discard nonzero total charge, and multiply their six pairwise Gaussian factors. Of the $3^4 = 81$ terms, fifteen survive. All fifteen have coefficient $1/9$ after inserting the adjoint gauge and normalizing the four external fields.

Set the insertion points to $(\infty, 1, x, 0)$, with $\phi(\infty) = \lim_{R\to\infty} R^{2h}\phi(R)$. The two useful reflection-positive channels are

$$G_0(x) = \langle \widehat{u}^\dagger(\infty)\widehat{u}(1)\widehat{u}^\dagger(x)\widehat{u}(0)\rangle = \boxed{\frac{3 - 4x + 4x^2}{3x^4(1-x)^4}}, \tag{28}$$

$$G_+(x) = \langle \widehat{u}^\dagger(\infty)\widehat{u}^\dagger(1)\widehat{u}(x)\widehat{u}(0)\rangle = \boxed{\frac{x^2(4 - 4x + 3x^2)}{3(1-x)^4}}. \tag{29}$$

For example, the fifteen neutral contributions to the first formula sum to

$$\frac{1}{3x^4(1-x)^4}+\frac{2}{3x^2(1-x)^4}+\frac{2}{3x^4(1-x)^2}. \tag{30}$$

The three inequivalent orderings follow from $G_0(1-x)=G_0(x)$, $G_+(x)=x^{-4}G_0(1/x)$, and $G_+(1-x)$; their adjoint orderings agree. These are exact rational identities, not truncated series fits.

To display positive channel data, expand in global conformal blocks:

$$x^{2h_{\mathrm{ext}}}G(x)=A_0+\sum_{n\geq 1}A_n x^n\,{}_2F_1(n,n;2n;x). \tag{31}$$

For the two channels above, $A_n$ is the sum of squared couplings to orthonormal global quasiprimaries of weight $n$ in the appropriate charge sector. This statement uses the positive radial pairing. It does not identify $A_n$ with a single primary coupling or a Virasoro block coefficient. The first neutral coefficients are

$$(A_0,A_1,A_2,A_3,A_4)=\left(1,\frac{8}{3},\frac{14}{3},\frac{58}{9},\frac{107}{15}\right). \tag{32}$$

The value $A_1=1/2+13/6$ reproduces the two Cartan-current exchanges. At weight two the total stress tensor contributes one, leaving $11/3$ as the squared-coupling sum to ambient Virasoro primaries. In the equal-charge channel the first nonzero coefficient is $A_6=4/3$, followed by $A_8=19/13$; odd weights vanish. Indeed, the leading state $\widehat{u}_{(-3)}\widehat{u}$ is supported on the three momenta $a_i+a_j$ with $i<j$, each of norm squared twelve. It is a weight-six Virasoro primary of squared norm $4/3$. The normalized leading OPE coefficient can therefore be chosen $2/\sqrt{3}$. In particular the equal-charge OPE vanishes through orders $x^0$ and $x^1$ and starts at $x^2$.

The four orthogonal commuting stress tensors give additional resolved weight-two data. If $T_i$ has central charge $c_i$ and $u$ has $T_i$-weight $h_i$, then $C^2_{\widehat{T}_i\widehat{u}\widehat{u}^\dagger}=2h_i^2/c_i$. Here the entries are

| stress | $c_i$ | $h_i$ | squared unit-field coupling |
|---|---|---|---|
| $T_L$ | 1 | 1/4 | 1/8 |
| $T_R$ | 117/41 | 195/164 | 325/328 |
| $e$ | 2400/1271 | 180/1271 | 27/1271 |
| $f$ | 70/31 | 13/31 | 169/1085 |

Their charges and weights sum to eight and two, respectively.

### 3.3 Potts and parafermion spin-three four-point functions

The embedded Potts and parafermion algebras contain canonical spin-three generators with parameters

$$(c,\lambda)=\left(\frac{4}{5},\frac{4}{13}\right)\quad\text{and}\quad(c,\lambda)=\left(\frac{8}{5},\frac{14}{473}\right). \tag{33}$$

These are the Potts and level-thirteen parafermion cases, in the normalization of the universal $\mathcal{W}(c,\lambda)$ algebra [47, 48]. Denote either spin-three field by $a$, its intrinsic stress tensor by $T_c$, and put

$$N=\frac{c}{3},\qquad b=a_{(1)}a,\qquad A=31-16\lambda(c+2),\qquad B=NA. \tag{34}$$

The exact checked singular self-OPE is

$$a(z)a(0)\sim\frac{N}{z^6}+\frac{2T_c}{z^4}+\frac{\partial T_c}{z^3}+\frac{b}{z^2}+\frac{\frac{1}{2}\partial b-\frac{1}{12}\partial^3T_c}{z}, \tag{35}$$

$$a_{(3)}b=Aa,\qquad \|b\|^2=B. \tag{36}$$

The positive normalization is chosen so that $a$ has two-point coefficient $N$; write $\widehat{a} = a/\sqrt{N}$. The exact values of $(A, B)$ are $(1119/65, 1492/325)$ for Potts and $(69283/2365, 554264/35475)$ for the parafermion generator.

Meromorphy, the OPE and crossing determine the complete correlator:

$$\begin{aligned} N^2\langle\widehat{a}(\infty)\widehat{a}(1)\widehat{a}(x)\widehat{a}(0)\rangle = {} & N^2\big[1 + x^{-6} + (1-x)^{-6}\big] \\ & + 2c\big[x^{-4} + (1-x)^{-4} + x^{-3} + (1-x)^{-3}\big] \\ & + B\big[x^{-2} + (1-x)^{-2}\big] + (2B - 2c)\big[x^{-1} + (1-x)^{-1}\big]. \end{aligned} \tag{37}$$

For completeness, the principal part at zero follows by pairing the five terms of (35) with the outer two fields. The Ward identity gives $\langle a(\infty)a(1)T_c(0)\rangle = 3N$; its first derivative is $6N$, and its third derivative is $72N$. Likewise $\langle a(\infty)a(1)b(0)\rangle = B$, whose first derivative is $4B$. These give residues $N^2, 2c, 2c, B, 2B - 2c$ at pole orders $6, 4, 3, 2, 1$. Interchange of the equal fields fixes the principal part at one. The identity channel at infinity fixes the constant $N^2$. The difference between the displayed function and the correlator is therefore a holomorphic function on the sphere vanishing at infinity. Both crossing generators $x \mapsto 1 - x$ and $x \mapsto 1/x$ can also be verified directly in the displayed expression.

The self-cubic coefficient $C_{\widehat{aaa}}$ vanishes. The squared coupling to the unit intrinsic stress tensor is $18/c$. To isolate the new weight-four channel from Virasoro descendants, put

$$\Lambda =: T_cT_c : -\frac{3}{10}\partial^2 T_c, \qquad \|\Lambda\|^2 = \frac{c(5c+22)}{10}, \qquad W_4^{\mathrm{prim}} = b - \frac{32}{5c+22}\Lambda - \frac{3}{10}\partial^2 T_c. \tag{38}$$

The squared $\widehat{aa}\Lambda/\|\Lambda\|$ coupling is $4608/[5c(5c+22)]$. Subtracting this contribution from the global weight-four coefficient gives

$$\gamma_4^2 = \frac{48(c+2)\{8 - \lambda(5c+22)\}}{c(5c+22)} = \begin{cases} 0, & \text{Potts}, \\ 60552/2365, & \text{level-thirteen parafermion}. \end{cases} \tag{39}$$

Thus the Potts extra weight-four primary is null, whereas the parafermion channel has a strictly positive extra coupling. This separation refers to the intrinsic Virasoro algebra of central charge $c$, not to the total central-charge-eight Virasoro algebra. The corresponding total global weight-four coefficients are $576/13$ and $21192/473$. Equation (37) supplies all higher global coefficients without a new high-weight mode calculation.

### 3.4 Selected blocks for the lightest module

The two $B$-lattice sectors $(0, +1)$ and $(0, -1)$ both restrict to the simple $U$-module $c_1$ of lowest weight $1/3$ and top dimension three. In the $A_2$ simple-root basis its three fundamental weights are

$$\beta_0 = (2/3, 1/3), \quad \beta_1 = (-1/3, 1/3), \quad \beta_2 = (-1/3, -2/3), \qquad (\beta_i, \beta_j) = \delta_{ij} - \frac{1}{3}. \tag{40}$$

Their $H_R$ charges are $q = (2, 0, -2)$; their $H_L$ charges vanish. Choose paired lattice intertwiners with unit leading vacuum coefficient. On the branch $0 < x < 1$ with positive real powers, the nine selected four-point blocks are

$$\mathcal{F}_{ij}(x) = \langle V_{-\beta_i}(\infty)V_{\beta_i}(1)V_{-\beta_j}(x)V_{\beta_j}(0)\rangle = \boxed{x^{-2/3}(1-x)^{-(\beta_i,\beta_j)}}. \tag{41}$$

Thus the diagonal entries are $[x(1-x)]^{-2/3}$ and the off-diagonal entries are $x^{-2/3}(1-x)^{1/3}$. Analytic continuation requires a specified path; the fractional powers must not be replaced by a single-valued rational function.

In the $x \to 0$ channel these blocks factor through the $B$ vacuum sector, whose restriction is $U \oplus X$, with $X$ the module denoted $b$ in the modular table. The first coefficient of the reduced block is $(\beta_i, \beta_j)$. The decomposition of the intermediate weight-one Heisenberg vector gives the explicit split

$$(\beta_i, \beta_j) = \underbrace{\frac{q_i q_j}{78}}_{U \text{ current exchange}} + \underbrace{(\beta_i, \beta_j) - \frac{q_i q_j}{78}}_{X \text{ top exchange}} . \tag{42}$$

For $i = j = 0$ these two positive coefficients are $2/39$ and $8/13$. The three-by-three $X$ Gram matrix has eigenvalues $0, 1, 35/39$, providing a direct positive-channel check. Off-diagonal entries need not be positive: they are overlaps, rather than squared norms. The six ordered triples with three distinct weights have nonzero three-point functions proportional to $(z_{12}z_{13}z_{23})^{-1/3}$, with coefficient of absolute square one in this normalization. Their phases depend on the lattice-intertwiner cocycle and chosen bases. Two distinct positive weights add to the negative of the remaining weight and hence exhibit a $c_1$ channel.

Restriction of a lattice intertwiner gives a genuine $U$ intertwiner, so these are nonvacuum chiral data, beyond Ward identities. Their particular linear combinations and normalizations remember the extension $U \subset B$. They neither form a complete basis of all $U$ channels nor determine the diagonal theory's local bulk correlator $\langle \Phi\Phi\Phi\Phi \rangle$ for its lightest scalar. That quantity additionally requires channel and multiplicity bases, the full relevant intertwining operators and the left-right sewing coefficients. In particular, multiplying one selected block by its complex conjugate has not been justified as that bulk correlator.

### 3.5 Reproducible data and verification scope

The bounded standard-library script `c8_physics_20260927/correlators.py` exports `c8_physics_20260927/correlators.json`, together with the tables `c8_physics_20260927/correlators.terms.tsv`, `c8_physics_20260927/correlators.global_blocks.tsv`, and `c8_physics_20260927/correlators.samples.tsv`. The first table retains all 486 assignments across the six charge orderings, including vanishing terms, their cocycle signs and exact coefficients in $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$. The second gives the exact global-block decomposition of the two $u$ channels, both spin-three correlators and the two Ward correlators through the recorded cutoff, with default weight 32; the third gives exact values at five rational cross-ratios. Polynomial crossing identities, reconstruction of the truncated series, positivity in the specified channels and the lightest-module Gram decomposition are checked exactly.

The script checks the three-exponential input against `hi_cft_20260923_c8_exact_fields.json`. The spin-three parameters and norms are read from `hi_cft_20260923_proof_pf13_ope.json`; their singular OPE provenance is `hi_cft_20260923_proof_pf13_upper_poles.py` and its saved report. Source hashes are included. The export does not import the lattice mode engine or repeat its higher-grade computations. Its finite positivity tests provide reproducible consistency checks; all-order positivity follows here from the positive VOA realization, rather than from extrapolating those tests. The separately exported low-weight coupling tensor resolves many of the summed weight-two channels above into individual normalized primary couplings.

## 4 A complete weight-two chiral three-point dataset

The coefficient 120 of the vacuum character is small enough to resolve its first interacting primary space completely. Its 113 Virasoro primaries comprise the 96 fields in the eight charged current multiplets of the construction, together with 17 fields obtained from the currents and

stress tensors. Here "charged" refers to a nontrivial current multiplet; it does not exclude zero-weight components within that multiplet.

### 4.1 Actual fields and normalization

Label the eight original highest fields by $v_{m,0,0}$, $m = 0, \dots, 7$, in the order

$$(n_m, r_m) = (1,5), (1,9), (1,9), (1,13), (2,2), (2,6), (2,8), (2,14).$$

These are representation dimensions for the two current algebras. The two $(1,9)$ copies are distinct and have different coset-stress eigenvalues. In the last multiplet the highest field is the defining field $u$, with the normalization inherited from its saved lattice vector. Define all components by

$$v_{m,p,q} = (F_L)_0^p (F_R)_0^q v_{m,0,0}, \quad 0 \le p < n_m, \quad 0 \le q < r_m. \tag{43}$$

No factorials divide the lowering powers. The Cartan charges are $(n_m - 1 - 2p, r_m - 1 - 2q)$.

The remaining highest fields are

$$v_{8,0,0} = (E_R)_{-1} E_R, \qquad v_{9,0,0} = (E_L)_{-1} E_R,$$

of current types $(1,5)$ and $(3,3)$, and three current singlets. To specify the latter, use the four mutually commuting stress tensors

$$(T_L, T_R, T_e, T_f), \qquad (c_L^{\mathrm{aff}}, c_R^{\mathrm{aff}}, c_e, c_f) = \left(1, \frac{117}{41}, \frac{2400}{1271}, \frac{70}{31}\right).$$

In the next formula abbreviate the first two central charges by $c_l, c_r$, to distinguish them from the spacetime charges $c_L = c_R = 8$. The singlets are

$$v_{10,0,0} = c_r T_L - c_l T_R, \tag{44}$$
$$v_{11,0,0} = c_e (T_L + T_R) - (c_l + c_r) T_e, \tag{45}$$
$$v_{12,0,0} = c_f (T_L + T_R + T_e) - (c_l + c_r + c_e) T_f. \tag{46}$$

Because $\langle T_i, T_j \rangle = \delta_{ij} c_i / 2$, these three vectors are mutually orthogonal and orthogonal to the total stress tensor. A combination $\sum a_i T_i$ is a total-Virasoro primary exactly when $\sum a_i c_i = 0$, proving their primarity directly.

The right spin-two multiplet is the traceless symmetric current product. The cross-current multiplet is primary because the two current factors commute. At level one the left spin-two current product is null, so there is no additional $(5,1)$ multiplet. Together these add $5 + 9 + 3 = 17$ fields. The exact calculation checks the Virasoro-primary conditions, all possible off-diagonal Gram entries within equal Cartan weights, and positive norms for all 113 vectors. Distinct Cartan weights are orthogonal by the compact real form. Their independence and the character count $120 - 6 - 1 = 113$ prove completeness: the subtractions remove six current derivatives and the total stress tensor.

Put $N_i = \langle v_i, v_i \rangle_g$ and $\widehat{v}_i = v_i / \sqrt{N_i}$, choosing the positive square root. The exported coefficients use the antilinear first-slot Hermitian pairing and the transported lattice PCT of (84):

$$C_{ij}^{\ k} = \langle \widehat{v}_k, (\widehat{v}_i)_1 \widehat{v}_j \rangle_g = \frac{\langle v_k, (v_i)_1 v_j \rangle_g}{\sqrt{N_i N_j N_k}}. \tag{47}$$

Equivalently this is the three-point coefficient with the conjugate of $\widehat{v}_k$ at infinity. The singular OPE contains $z^{-2} C_{ij}^{\ k} \widehat{v}_k(0)$, in addition to current, stress-tensor and descendant contributions. The unnormalized numerator and norm product lie in $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$; the positive square-root normalization is left explicit. Thus the table retains the complex phase, not only $|C|^2$.

### 4.2 From directly computed seeds to the full tensor

Let $P$ be orthogonal projection onto this primary space and write $P((v_i)_1 v_j) = \sum_k p_{ij}^k v_k$. Since current zero modes commute with the total Virasoro algebra, this projection is equivariant. For either lowering operator $F$ the derivation identity gives

$$P((Fv_i)_1 v_j) = F\,P((v_i)_1 v_j) - P((v_i)_1 F v_j). \tag{48}$$

Starting with $i$ highest in each of the thirteen multiplets determines the entire tensor recursively. In the raw basis (43), $F$ simply advances one lowering exponent; its raising partner acts by $Ev_p = p(n-p)v_{p-1}$. Equation (48) therefore uses only exact additions and multiplications of known coefficients. Orthogonal projection is essential here: we do not discard descendants before justifying equivariance.

The direct seed calculation evaluates 1137 actual lattice mode products and 3512 charge-compatible projections, of which 1639 are nonzero. The completed tensor has

$$113^3 = 1442897 \quad \text{ordered entries,} \qquad 18365 \quad \text{nonzero entries.}$$

All omitted entries in the sparse table are exactly zero within this complete weight-two primary tensor. Every raising and lowering Ward identity is checked, including at the tops and bottoms of multiplets: there are 51076 pairwise checks. Primary skew-symmetry gives a further 12769 checks, and 33 additional direct lattice products test components outside the initial seed slice. These checks verify the specified finite tensor; they are not a test of arbitrary-weight associativity or full-CFT sewing.

A cross-check connects this tensor to an independent four-point calculation. The weight $(-1,-13)$ component is unique, so $\widehat{v}_{7,1,13}$ equals $\widehat{u}^\dagger$ up to a phase. Summing its nine nonzero primary projections gives exactly

$$\sum_{k=1}^{113} |C_{\widehat{u},\widehat{u}^\dagger}^{k}|^2 = \frac{11}{3}.$$

Adding the total-stress contribution one reproduces the global weight-two coefficient $14/3$ in the Gaussian four-point function (28). The three stress-difference contributions separately agree with their Ward values. The small standalone checker `check_physics_crosslinks.py` verifies these sums without importing the lattice mode engine.

Several entries are short enough to check without navigating a large file. The indices below are the explicit $(m,p,q)$ labels above.

| $i$ | $j$ | $k$ | $\lvert C_{ij}^{k}\rvert^2$ |
|---|---|---|---|
| $(0,0,0)$ | $(0,0,2)$ | $(0,0,0)$ | 4356/8645 |
| $(1,0,0)$ | $(1,0,6)$ | $(0,0,0)$ | 39605/717288 |
| $(7,0,0)$ | $(7,1,13)$ | $(10,0,0)$ | 13/316 |
| $(8,0,0)$ | $(8,0,3)$ | $(8,0,1)$ | 1225/741 |
| $(9,0,0)$ | $(9,2,0)$ | $(8,0,0)$ | 76/39 |
| $(10,0,0)$ | $(10,0,0)$ | $(10,0,0)$ | 23104/9243 |
| $(12,0,0)$ | $(12,0,0)$ | $(12,0,0)$ | 2916/3115 |

For example, for $A = c_r T_L - c_l T_R$ one has

$$\langle A, A\rangle = \frac{c_l c_r (c_l + c_r)}{2}, \qquad |C_{AAA}|^2 = \frac{8(c_r - c_l)^2}{c_l c_r (c_l + c_r)} = \frac{23104}{9243}.$$

This follows just from the two commuting stress OPEs and provides a human-sized independent normalization check on the lattice table.

The principal files are `c8_physics_20260927/full_weight_two/low_weight_basis.json`, which stores every actual lattice vector and norm; `low_weight_products.jsonl`, which stores each seed product; and `complete_primary_couplings.tsv`, which stores every nonzero normalized coefficient by its exact numerator and norm product. The accompanying JSON report documents the recurrence checks and source hashes. The earlier 96-field slice remains available as an intermediate audit, but the 113-field table is the complete release.

These are chiral vacuum-module observables of $U$, hence also observables of its selected extension. They give local spinning fields in either bulk theory. Factorized products of these fields give additional full-CFT vacuum-sector correlators. They do not supply the three-point constants of generic nonvacuum diagonal-sector primaries.

## 5 Modular data, topological defects, and conformal boundaries

We use the positive spherical structure and the Evans–Gannon ordering

$$(0, b, a, c_0, c_1, c_2, d_1, d_2, d_3, d_4, d_5, d_6).$$

The translation to the labels used in the Haagerup bootstrap literature is

$$(1, \pi_1, \pi_2, \sigma_1, \sigma_2, \sigma_3, \mu_5, \mu_2, \mu_1, \mu_3, \mu_6, \mu_4).$$

The final permutation is fixed by the distinct thirteenth-root twists in [11, arXiv version 2, Eq. (30)]; retaining the same numerical subscripts for the $d$ and $\mu$ labels would be incorrect. In particular, $a$ is a simple object of the center, whereas $\alpha$ below denotes the invertible generator of the six-object Haagerup category. These two objects have different dimensions and different roles.

Put $r = \sqrt{13}$, $d = (3 + r)/2$, $s = 2 + 3d$, and

$$A = \frac{1}{6} - \frac{1}{2r}, \qquad B = \frac{1}{6} + \frac{1}{2r}, \qquad u = \frac{1}{3}, \qquad v = \frac{1}{r}.$$

In blocks of sizes $1, 1, 4, 6$, the complete normalized matrix is

$$S = \begin{pmatrix} A & B & u\mathbf{1}_4^{\mathrm{t}} & v\mathbf{1}_6^{\mathrm{t}} \\ B & A & u\mathbf{1}_4^{\mathrm{t}} & -v\mathbf{1}_6^{\mathrm{t}} \\ u\mathbf{1}_4 & u\mathbf{1}_4 & C & 0 \\ v\mathbf{1}_6 & -v\mathbf{1}_6 & 0 & D \end{pmatrix}, \qquad C = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 & -1 \\ -1 & 2 & -1 & -1 \\ -1 & -1 & -1 & 2 \\ -1 & -1 & 2 & -1 \end{pmatrix}, \tag{49}$$

where

$$D_{ij} = -\frac{2}{\sqrt{13}} \cos \frac{2\pi ij}{13}, \qquad 1 \le i, j \le 6. \tag{50}$$

The topological twist matrix and the character matrix are distinct:

$$\Theta = \operatorname{diag}\left(1, 1, 1, 1, e^{2\pi i/3}, e^{4\pi i/3}, e^{2\pi i 6/13}, e^{2\pi i 11/13}, e^{2\pi i 2/13}, e^{2\pi i 5/13}, e^{2\pi i 7/13}, e^{2\pi i 8/13}\right), \tag{51}$$

$$T_{\mathrm{ch}} = e^{-2\pi i/3}\Theta. \tag{52}$$

Thus the $d_j$ twist is $e^{2\pi i 6 j^2/13}$. The conformal weight of $d_3$ is $15/13$, although its twist exponent is $2/13$. These are the standard positive Haagerup modular data [9]; no numerical eigenvector choice enters these formulas.

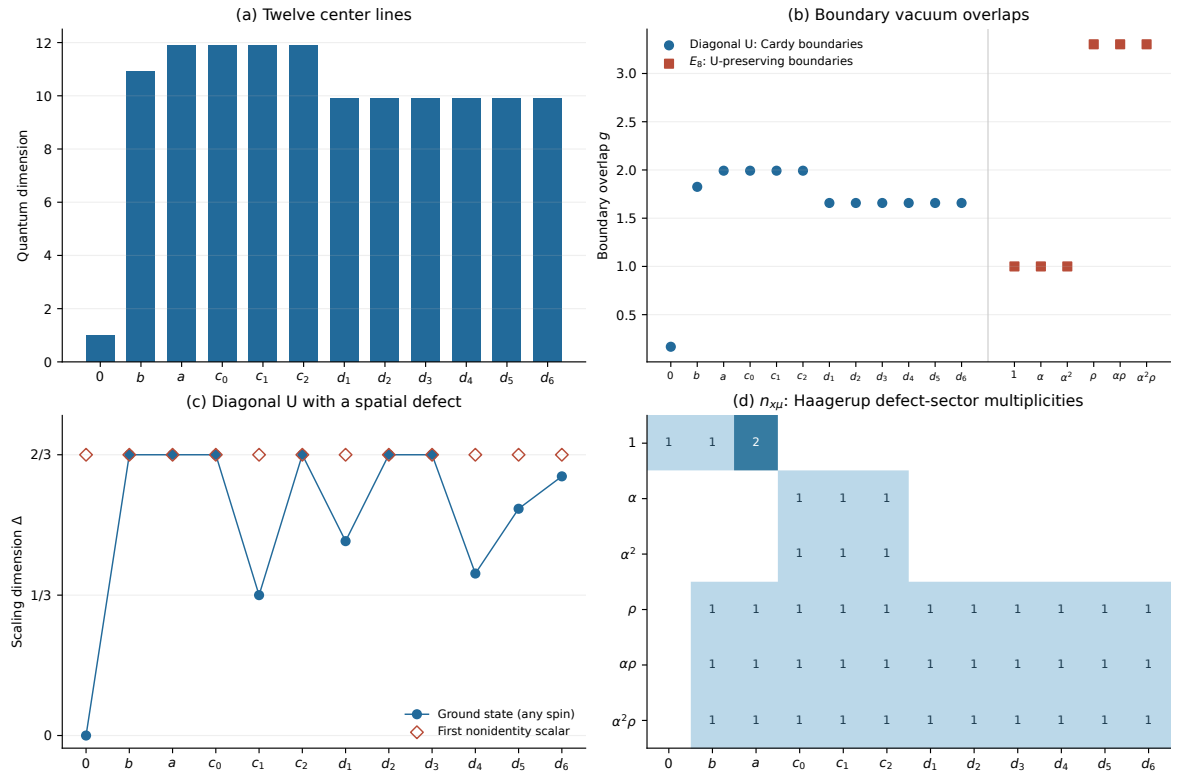



Figure 3: Exact modular and defect observables, displayed using the positive embedding of the exported algebraic data. The ground-state gap in a twisted space and its scalar gap are distinct; the latter is 2/3 for all twelve diagonal-theory lines. The final panel records the integer forgetful multiplicity matrix for the selected extension.

The categorical dimensions and total quantum dimension are

$$(\mathrm{qdim}\, M_i)_i = (1, d^2, s, s, s, s, 3d, 3d, 3d, 3d, 3d, 3d), \qquad \mathcal{D} = \frac{1}{S_{00}} = 3s. \tag{53}$$

Here $\mathcal{D}^2$, rather than $\mathcal{D}$, is the global dimension of $\mathrm{Rep}(U)$. Every simple object is self-dual because $S^2 = 1$. The six-object Haagerup category has global dimension $3 + 3d^2 = 3s = \mathcal{D}$.

## 5.1 Exact fusion and diagonal-theory defect spectra

Consider first the diagonal full CFT

$$\mathcal{H}_U = \bigoplus_i M_i \otimes \overline{M_i}, \qquad Z_U(\tau, \bar\tau) = \sum_i |\chi_i(\tau)|^2. \tag{54}$$

Its twelve chiral-algebra-preserving Verlinde lines form the fusion category underlying $\mathrm{Rep}(U) = Z(\mathcal{H}_3)$. Their multiplication and their action on bulk sectors are

$$N_{ab}{}^{c} = \sum_i \frac{S_{ai} S_{bi} S_{ci}}{S_{0i}}, \qquad \widehat{L}_a\big|_{M_i \otimes \overline{M_i}} = \lambda_a(i)\,\mathrm{id}, \qquad \lambda_a(i) = \frac{S_{ai}}{S_{0i}}. \tag{55}$$

Complex conjugation is absent from the displayed Verlinde sum because $S$ is real. In particular $\lambda_a(0) = \mathrm{qdim}(M_a)$. The line insertion and its modular transform give

$$Z_U^{[a]}(\tau,\bar\tau) = \sum_i \lambda_a(i)|\chi_i(\tau)|^2, \tag{56}$$

$$Z_{U,[a]}(\tau,\bar\tau) = \sum_{i,j} N_{ai}{}^{j}\chi_i(\tau)\overline{\chi_j(\tau)}. \tag{57}$$

The first expression is a trace with a line operator and its coefficients need not be positive. The second is the partition function of the Hilbert space with an $a$ defect intersecting the spatial circle; all its multiplicities are nonnegative integers. This is the usual distinction between a twined trace and a twisted Hilbert space [4, 1]. Indeed, applying $S$ to the first expression makes the coefficient of $\chi_i\overline{\chi_j}$ equal to the sum in (55), proving the second formula directly.

The attached data give all 1728 integers $N_{ab}{}^{c}$, all 144 line eigenvalues, and all twelve matrices in (57). They also give the lowest scaling dimension and the lowest nonidentity scalar dimension in each twisted space. For orientation, some compact fusion products are

$$b^2 = 1 + b + a + \sum_{j=0}^{2} c_j + \sum_{j=1}^{6} d_j, \tag{58}$$

$$a^2 = 1 + 2b + 2a + \sum_{j=0}^{2} c_j + \sum_{j=1}^{6} d_j, \tag{59}$$

$$d_i^2 = 1 + a + \sum_{j=0}^{2} c_j + \sum_{\substack{1\le j\le 6\\ j\neq[2i]}} d_j. \tag{60}$$

Here $[m]$ is the representative of $\{m,-m\}$ modulo 13 in $\{1,\dots,6\}$. The gap data are particularly simple:

| Line | ground $\Delta$ | multiplicity | scalar gap | multiplicity |
|---|---|---|---|---|
| 0 | 0 | 1 | 2/3 | 9 |
| $b$ | 2/3 | 18 | 2/3 | 18 |
| $a, c_0$ | 2/3 | 9 | 2/3 | 9 |
| $c_1$ | 1/3 | 6 | 2/3 | 18 |
| $c_2$ | 2/3 | 63 | 2/3 | 9 |
| $d_1$ | 6/13 | 14 | 2/3 | 9 |
| $d_2, d_3$ | 2/3 | 9 | 2/3 | 9 |
| $d_4$ | 5/13 | 10 | 2/3 | 9 |
| $d_5$ | 7/13 | 26 | 2/3 | 9 |
| $d_6$ | 8/13 | 28 | 2/3 | 9 |

The scalar condition is $h = \bar h$, not merely integral spin. For a summand $M_i \otimes \overline{M_j}$ it first requires $h_i - h_j \in \mathbb{Z}$. Among the supplied lowest weights the only unequal pair with this property is 0 versus 1; its scalar ground states use $U_1$ on the vacuum side. Consequently all the reported lowest scalar multiplicities follow from the twelve top dimensions and $\dim U_1 = 6$, without an extrapolation of character coefficients.

A useful symmetry test follows immediately from invertibility of $S$. If a bulk sector $i$ is fixed by every line, its eigenvalues must obey $\lambda_a(i) = \lambda_a(0)$ for every $a$. The $i$th column of $S$ would then be proportional to the vacuum column; hence $i = 0$. Thus the fully Verlinde-invariant space is $U \otimes \overline{U}$. Its first nonidentity scalars have weights $(1,1)$ and multiplicity $6^2 = 36$. Its scalar gap is 2, even though the unrestricted diagonal theory has scalar gap 2/3.

### 5.2 Cardy boundaries and their annuli

The diagonal theory has twelve elementary boundaries preserving $U$. With the usual normalized Ishibashi states, Cardy's solution reads

$$|a\rangle = \sum_i \frac{S_{ai}}{\sqrt{S_{0i}}}|i\rangle\!\rangle, \qquad Z_{ab}^{\text{open}}(q) = \sum_i N_{ab}{}^i \chi_i(q). \tag{61}$$

All labels are self-dual, so no conjugate label is needed in the annulus formula. Modular transformation of the closed-channel overlap gives the same Verlinde sum as above, providing a direct check of the open channel multiplicities [3].

The boundary vacuum overlaps, conventionally denoted by $g_a$, are

$$g_a = \frac{S_{a0}}{\sqrt{S_{00}}} = \frac{\text{qdim}(M_a)}{\sqrt{3s}}, \qquad \log g_a \text{ is the boundary entropy.} \tag{62}$$

They therefore take four values, in the proportions $1 : d^2 : s : 3d$. The exact squared values lie in $\mathbb{Q}(\sqrt{13})$; the dataset supplies these as exact algebraic numbers as well as decimal values of $g_a$ for plotting. The 144 annulus entries include the identity multiplicity, the lowest boundary weight, and the first positive boundary weight. The last quantity includes weight-one currents in the vacuum representation. Removing the entire vacuum representation would define a different gap. For the vacuum Cardy boundary the first positive boundary weight is 1, with six currents. Every other elementary Cardy boundary has boundary-preserving fields of weight $1/3$; there are six for boundary $c_1$ and three for each of the other ten boundaries. These are relevant boundary perturbations, without implying a particular endpoint of their renormalization-group flows.

## 6 The Haagerup symmetry of the holomorphic extension

The twelve diagonal-theory lines above must be distinguished from the six Haagerup lines obtained after extension. Let $H' = V_{E_8}$ be the selected holomorphic extension, with underlying $U$-module

$$H'|_U \simeq M_0 \oplus M_b \oplus 2M_a, \qquad \chi_{E_8} = \chi_0 + \chi_b + 2\chi_a. \tag{63}$$

Its algebra object is denoted by $A$, and $\mathcal{C} = \text{Rep}(U)_A \simeq \mathcal{H}_3$ is the category of all right $A$-modules, including the nonlocal ones. The local $A$-module category is trivial. The six simples of $\mathcal{C}$ are $1, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho$, with

$$\alpha^3 = 1, \qquad \rho\alpha = \alpha^2\rho, \qquad \rho^2 = 1 + \rho + \alpha\rho + \alpha^2\rho. \tag{64}$$

The full $E_8$ theory has the corresponding $\mathcal{C} \boxtimes \mathcal{C}^{\text{op}}$ defects. In this section we select the first factor, leaving the antiholomorphic $H'$ theory as a spectator. This convention specifies a single $\mathcal{H}_3$ action to which the twelve-sector Haagerup bootstrap applies.

### 6.1 Forgetful multiplicities and the physical sector assignment

Under the center equivalence $\text{Rep}(U) \simeq Z(\mathcal{C})$, the free-module functor is the central forgetful functor $F : Z(\mathcal{C}) \to \mathcal{C}$. Writing $X = \rho \oplus \alpha\rho \oplus \alpha^2\rho$, its values are

$$F(0) = 1, \quad F(b) = 1 \oplus X, \quad F(a) = 2 \cdot 1 \oplus X, \quad F(c_j) = \alpha \oplus \alpha^2 \oplus X, \quad F(d_j) = X. \tag{65}$$

These are the Haagerup branching rules of [2, Eq. (4.49)]. They specify the entire $6 \times 12$ matrix $n_{x\mu} = \dim \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x)$. For example its first three rows are

$$n_1 = (1,1,2,0,0,0,0,0,0,0,0,0), \tag{66}$$
$$n_\alpha = n_{\alpha^2} = (0,0,0,1,1,1,0,0,0,0,0,0), \tag{67}$$

and each of the other three rows is $(0,1,1,1,1,1,1,1,1,1,1,1)$.

Here is an intrinsic reason for the sector identification, beyond a check of modular covariance. If $R$ is restriction of an $A$-module to $U$, free–forgetful adjunction gives

$$\mathrm{Hom}_U(M_\mu, Rx) \simeq \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x), \qquad Rx \simeq \bigoplus_\mu n_{x\mu} M_\mu. \tag{68}$$

In the standard rational-CFT construction of the selected first-factor defects, their Hilbert spaces are $Rx \otimes \overline{H'}$. The tube-algebra decomposition is therefore

$$\mathcal{H}_x \simeq \bigoplus_\mu W_x^\mu \otimes \mathcal{V}_\mu, \qquad W_x^\mu = \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x), \qquad \mathcal{V}_\mu = M_\mu \otimes \overline{H'}. \tag{69}$$

This is the multiplicity-space convention of [1, Eq. (4.1)]. A full tube-projector image also contains the finite-dimensional factor $W_x^\mu$; it is not in general a single copy of $\mathcal{V}_\mu$. In particular $\dim W_1^a = 2$. The physical interpretation in (69) uses the canonical extension and its rational-CFT defect construction; modular covariance alone would not establish that interpretation for an arbitrary vector-valued function.

It follows that the twelve bootstrap partition functions and the six twisted Hilbert-space partition functions are

$$Z_\mu^{3d}(\tau,\bar\tau) = \chi_\mu(\tau)\overline{\chi_{E_8}(\tau)}, \qquad Z_x(\tau,\bar\tau) = \sum_\mu n_{x\mu} Z_\mu^{3d}(\tau,\bar\tau). \tag{70}$$

For $x = 1$, (63) recovers $Z_1 = |\chi_{E_8}|^2$. Since $\chi_{E_8}$ is $S$-invariant and has $T$ multiplier $e^{-2\pi i/3}$, these functions transform by $(S,\Theta)$: the left and right central-charge phases cancel. This also checks the spin convention used in the bootstrap.

### 6.2 Untwisted charges and scalar gaps

On the three untwisted multiplicity factors, one may choose bases in which the line actions are

| | $\widehat{\alpha}$ | $\widehat{\rho}$ |
|---|---|---|
| $M_0$ | $1$ | $d$ |
| $M_b$ | $1$ | $-d^{-1}$ |
| $\mathbb{C}^2 \otimes M_a$ | $\begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}$ | $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ |

$\omega = e^{2\pi i/3}$. (71)

The matrices obey (64); for the two-dimensional block the three noninvertible matrices sum to zero, so $\rho^2 = 1$ on that block is consistent with the categorical fusion rule. The resulting first-factor twined traces include

$$Z^{[\alpha]} = (\chi_0 + \chi_b - \chi_a)\overline{\chi_{E_8}}, \tag{72}$$
$$Z^{[\rho]} = (d\chi_0 - d^{-1}\chi_b)\overline{\chi_{E_8}}. \tag{73}$$

They act trivially on the antiholomorphic factor. Squaring their holomorphic factors in absolute value instead would correspond to particular paired left/right insertions, rather than the selected first-factor action.

Every antiholomorphic $E_8$ weight is an integer. Thus the sectors $c_1, c_2, d_1, \dots, d_6$ in (70) contain no scalar states at any grade. Their nonintegral left weights cannot equal an integer right weight. The remaining sectors $0, b, a, c_0$ have their first nonidentity scalars at $(h, \bar{h}) = (1, 1)$:

| $\mu$ | $0$ | $b$ | $a$ | $c_0$ |
|---|---|---|---|---|
| $\Delta_{\text{scalar}}$ | 2 | 2 | 2 | 2 |
| multiplicity in $\mathcal{V}_\mu$ | $6 \cdot 248$ | $80 \cdot 248$ | $81 \cdot 248$ | $81 \cdot 248$. |

(74)

The $\mathcal{H}_3$-invariant scalar gap is therefore 2, and so is the scalar gap in $\mathcal{V}_{\pi_2} \cup \mathcal{V}_{\sigma_1}$. The physical untwisted $\mathbb{Z}_3$-charged space contains two copies of $\mathcal{V}_{\pi_2}$, with scalar multiplicity $2 \cdot 81 \cdot 248$. The union used in the bootstrap is a union of tube multiplicity spaces; it must not be identified with that entire untwisted charged space.

### 6.3 Six boundaries of the extended theory

The same module category supplies six elementary boundaries of the full $E_8$ theory which preserve $U$. For $x, y \in \mathcal{C}$, their open state space is $R(x^* \otimes_A y)$, hence

$$Z_{xy}^{E_8,\text{open}}(q) = \sum_{z,\mu} N_{x^*y}{}^{z} n_{z\mu} \chi_\mu(q). \tag{75}$$

The invertible duals are $\alpha^* = \alpha^2$; the three noninvertible objects are self-dual. This dual must be retained in the annulus formula, even though it could be omitted for the twelve self-dual center labels.

The normalization of the boundary entropy can be derived without an unknown proportionality constant. Restriction multiplies categorical dimensions by $\operatorname{qdim}(A) = \mathcal{D}$, so the coefficient of the vacuum in the modular transform of (75) is

$$S_{00} \operatorname{qdim}_{\text{Rep}(U)} R(x^* \otimes_A y) = \operatorname{qdim}_{\mathcal{C}}(x) \operatorname{qdim}_{\mathcal{C}}(y).$$

Reflection positivity chooses the positive overlaps. Thus

$$(g_1, g_\alpha, g_{\alpha^2}, g_\rho, g_{\alpha\rho}, g_{\alpha^2\rho}) = (1, 1, 1, d, d, d). \tag{76}$$

In particular the ordinary $E_8$ boundary has $g_1 = 1$ and self-annulus $\chi_{E_8}$. Its first positive boundary weight is 1, with 248 currents. The same holds for the other two invertible boundaries. Each noninvertible boundary instead has nine fields of weight $1/3$ in its self-annulus. Between distinct noninvertible boundaries there are twelve ground fields of weight $1/3$; between two distinct invertible boundaries, or an invertible and a noninvertible boundary, there are three. All 36 annuli and their exact $g$ values are supplied.

### 6.4 What the attached exact data check

The exporter `c8_physics_20260927/defects.py` works in $\mathbb{Q}(\zeta_{13})$ and represents every field element by twelve rational coefficients. Its square root is the explicit Gauss sum

$$\sqrt{13} = \sum_{j=1}^{12} \left(\frac{j}{13}\right) \zeta_{13}^{j}.$$

It checks $S^2 = 1$, $(S\Theta)^3 = 1$, exact nonnegative integrality of all Verlinde coefficients, associativity, and multiplicativity of dimensions. The $6 \times 12$ forgetful table is independently

checked as a based-ring homomorphism, and its vacuum row is checked to be $S$- and $\Theta$-invariant. No coefficient is inferred by rounding. Decimal columns serve only as plot coordinates. The guarded run completed all these checks successfully; the saved JSON includes the generator hash and the precise normalization of every exported array.

The output files contain the complete fusion, line, annulus, twisted, and tube-sector recipes. These checks verify consequences of the given modular data and character tops. They neither replace the construction of $U$ nor reconstruct associators, braidings, or operator-product coefficients from modular data alone.

# 7 Perturbations, bootstrap coordinates, and use of the data

## 7.1 Dimension-two operators and controlled deformations

The absence of a relevant symmetry-invariant scalar has a precise but limited meaning: it excludes an invariant scalar perturbation with linearized RG eigenvalue $2-\Delta>0$. It does not settle the flow of the dimension-two operators. For the diagonal theory all 36 products of vacuum currents are invariant under the full Verlinde category; their nonabelian OPEs must still be taken into account.

There is nevertheless a concrete exactly marginal subfamily. Choose the two Cartan currents and normalize them by

$$j^1 = H_L/\sqrt{2}, \qquad j^2 = H_R/\sqrt{78}, \qquad j^a(z)j^b(0) \sim \frac{\delta^{ab}}{z^2}.$$

Here $L, R$ label the two affine factors within one chirality, not the two spacetime chiralities. The four local scalar fields

$$\mathcal{O}_{ab}(z,\bar{z}) = j^a(z)\overline{j^b}(\bar{z}), \qquad a,b \in \{1,2\}, \tag{77}$$

have the seed-point two-point functions

$$\langle \mathcal{O}_{ab}(z,\bar{z})\mathcal{O}_{cd}(0)\rangle = \frac{\delta^{ac}\delta^{bd}}{|z|^4}.$$

Thus the Zamolodchikov metric in this operator normalization is the identity. The abelian current-current deformation criterion supplies four simultaneous local exactly marginal directions [6, Section 2]. The associated current deformation space is locally $O(2,2)/(O(2)\times O(2))$, before duality identifications. We do not identify the complete conformal manifold or its global quotient. Although these operators are categorical singlets at the rational point, extending all topological lines and their junction data along the deformation requires a separate argument. Exact marginality alone does not prove that the entire deformed family retains the specified Haagerup symmetry. In particular the count 36 is not the dimension of a claimed Haagerup-preserving conformal manifold.

The ordinary full $E_{8,1}$ theory has eight Cartan currents in each chirality. Applying the same criterion gives the familiar 64 toroidal deformation directions, locally $O(8,8)/(O(8)\times O(8))$, at its enhanced-symmetry point. This statement concerns the ordinary CFT; it does not identify which deformations transport the specified Haagerup defect category.

## 7.2 Coordinates for published exclusion plots

For the ordinary scalar-gap bootstrap, allowing conserved currents, the diagonal theory gives $(c, \Delta_{\text{scalar}}) = (8, 2/3)$. The holomorphic extension gives $(8,2)$, the familiar $E_{8,1}$ point at the

scalar-bound kink in Collier–Lin–Yin [5, Figure 8 and Section 3.3]. Its ordinary bulk partition function is the known $E_8$ one; the additional information here is the specified Haagerup action and its relation to the explicit subalgebra. The diagonal theory lies well below that scalar upper bound. Chiral currents instead give the $E_8$ theory an all-primary gap of one, so substituting that number in a scalar-gap plot would compare different observables.

For the Haagerup bootstrap of Albert–Honda–Kaidi–Zheng, the physical sector assignment in Section 6 supplies

$$(c, \Delta^{\text{scalar}}_{\mathcal{V}_1}) = (8, 2), \qquad (c, \Delta^{\text{scalar}}_{\mathcal{V}_{\pi_2} \cup \mathcal{V}_{\sigma_1}}) = (8, 2). \tag{78}$$

These are respectively the observables in Figures 2 and 3 of [11]. The multiplicity spaces $\mathcal{V}_\mu$ exclude the finite-dimensional tube representation factors; the factor two in the untwisted $\pi_2$ block must not be counted twice when constructing the bootstrap vector. The 12-sector vector for this comparison is $\chi_\mu \overline{\chi_{E_8}}$, not the diagonal collection $|\chi_\mu|^2$. The plots are numerical exclusions at the published truncation, and our data do not rerun the associated semidefinite program. In particular, the common value two is not by itself a claim of saturation of either Haagerup bound.

Figure 4 places these two points on the original published panels. Each point lies visibly below its exclusion curve, in the shaded allowed region. We use this as a consistency comparison, without extracting a numerical curve height or claiming that the bounds single out this theory. The marker positions are calibrated from the original axis ticks; the curves themselves are reproduced without interpolation or refitting.

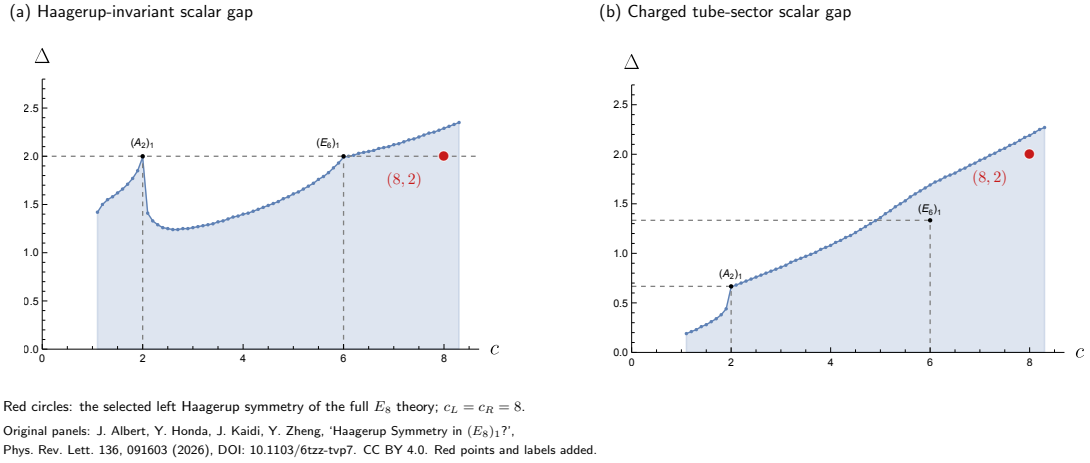



Figure 4: The $c_L = c_R = 8$ Haagerup comparison points, shown in red. Original panels from J. Albert, Y. Honda, J. Kaidi, and Y. Zheng, *Haagerup Symmetry in* $(E_8)_1$*?*, Phys. Rev. Lett. 136, 091603 (2026), doi:10.1103/6tzz-tvp7, reproduced under CC BY 4.0. Red markers and labels have been added. The left panel bounds $\mathcal{V}_1$; the right panel bounds $\mathcal{V}_{\pi_2} \cup \mathcal{V}_{\sigma_1}$.

### 7.3 Which observables are now supplied

The release directory `c8_physics_20260927/` contains plain TSV tables for inspection and JSON records retaining the algebraic numbers, basis conventions, and exact derivations. There are four distinct kinds of evidence.

1. The character recurrence supplies complete spin-resolved state and Virasoro-primary spectra through $\Delta = 24$, for both full theories. Thermodynamic intervals add a proved bound on the omitted tail, conditional on modular invariance and positivity.

2. The actual lattice fields supply chiral vacuum-module three-point coefficients and the displayed four-point functions. Unit normalization and PCT are explicit. The current Ward completion of the weight-two tensor is checked against direct additional mode calculations.

3. The positive modular data supply all Verlinde coefficients, diagonal-theory line eigenvalues, twisted partition functions and Cardy annuli. The selected extension and its free–forgetful adjunction supply the Haagerup-resolved $E_8$ sectors and boundaries.

4. Universal CFT identities then give finite-size energies, momenta, entropy coefficients and the stress-tensor and current Ward observables. They do not determine the microscopic velocity $v$.

The central unresolved interaction problem is the normalized bulk OPE of the nonvacuum diagonal sectors, beginning with the nine scalar components of dimension 2/3. Neither their character coefficients nor their fusion multiplicities determine their three-point constants. An actual lattice intertwiner block, such as the example given above, is useful input but must still be projected into channels and sewn with consistent Hermitian pairings to obtain the full diagonal-CFT four-point function. The corresponding numerical crossing tests, general defect-junction correlators, and RG endpoints are not claimed as outputs of this release. These are concrete follow-up calculations, with the spectrum and normalization data here providing their starting point.

Each exported number therefore has a mathematical location. Checking a rational recurrence is separate from proving that its coefficients are the characters of the constructed VOA; checking a lattice contraction is separate from proving full-CFT sewing. This separation is intended to make remaining gaps visible to both physicists and mathematicians.

# A The explicit lattice construction

## A.1 Lattice, cocycle, and currents

Use simple lattice vectors indexed $0, \dots, 7$. The Gram matrix has diagonal entries two and off-diagonal entries minus one on

$$(0,1), (1,2), (2,3), (3,4), (2,5), (6,7),$$

with all remaining off-diagonal entries zero. Fix the cocycle

$$\epsilon(a,b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}. \tag{79}$$

Write $L = E_6 \oplus A_2$ and $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$. The underlying vector space is

$$B = S\big(\mathfrak{h} \otimes t^{-1}\mathbb{C}[t^{-1}]\big) \otimes \mathbb{C}_\epsilon[L].$$

A monomial $h_1(-n_1)\cdots h_r(-n_r)e^\alpha$ has weight $(\alpha,\alpha)/2 + \sum_i n_i$. In particular its weight is a nonnegative integer. The exponent in (79) is bilinear modulo two, and hence defines a normalized two-cocycle. Directly,

$$\frac{\epsilon(\alpha,\beta)}{\epsilon(\beta,\alpha)} = (-1)^{(\alpha,\beta)}, \qquad \epsilon(\alpha,-\alpha) = (-1)^{(\alpha,\alpha)/2}.$$

These identities fix the signs of all root brackets. For example, $e_0^\alpha e^{-\alpha} = -\alpha(-1)\mathbf{1}$ for a root $\alpha$. The exponential fields obey

$$Y(e^\alpha, z)e^\beta = \epsilon(\alpha,\beta)z^{(\alpha,\beta)} \exp\left(\sum_{n>0} \frac{\alpha(-n)}{n} z^n\right) e^{\alpha+\beta}. \tag{80}$$

Together with the Heisenberg commutation relations, this formula specifies the exact mode engine used by the certificates.

Put $\alpha_i = (0,\dots,1,\dots,0)$ and $\alpha_L = (-1,-2,-3,-2,-1,-2,0,0)$. With

$$(c_0,\dots,c_7) = (5,8,9,8,5,0,2,2), \qquad h_R = \sum_i c_i\alpha_i,$$

the two current triples, in the order $(H,E,F)$, are

$$(H_L, E_L, F_L) = (\alpha_L(-1)\mathbf{1}, e^{\alpha_L}, -e^{-\alpha_L}), \tag{81}$$

$$(H_R, E_R, F_R) = \left(h_R(-1)\mathbf{1}, \sum_{c_i\neq 0} e^{\alpha_i}, -\sum_i c_i e^{-\alpha_i}\right). \tag{82}$$

Current letters also denote their zero-mode actions when applied to a state. Thus, for example, $F_R E_R v$ means $(F_R)_0(E_R)_0 v$.

**Lemma A.1** (The two affine levels)**.** *The triples* (81)–(82) *commute and generate* $A_{1,1}\otimes A_{1,39}$.

*Proof.* We have $(\alpha_L,\alpha_L) = 2$ and $(\alpha_L,\alpha_i) = 0$ whenever $c_i \neq 0$. Also $(h_R,\alpha_i) = 2$ for every $c_i \neq 0$, and $(h_R,h_R) = 78$. Formula (80) gives $[H_0,E_0] = 2E_0$, $[H_0,F_0] = -2F_0$ and $[E_0,F_0] = H_0$ for both triples. The double poles are

$$(H_L)_1H_L = 2\mathbf{1}, \quad (E_L)_1F_L = \mathbf{1}, \quad (H_R)_1H_R = 78\mathbf{1}, \quad (E_R)_1F_R = 39\mathbf{1}.$$

Orthogonality gives all mixed singular products equal to zero. These are the affine relations at levels one and thirty-nine. The positive-definite lattice representation is integrable; equivalently one may use the principal $\mathfrak{sl}_2$ embeddings in $A_5$ and $A_2$, whose levels 35 and 4 add in the diagonal. Thus the generated factors are the simple positive-integral-level affine VOAs. Their tensor product is simple and embeds in $B$. □

## A.2 The defining field and the positive real form

The exact coefficient convention uses unnormalized simple-root Heisenberg oscillators and a torus gauge in which the principal raising current is the sum of the corresponding simple-root vertices. The lowering current is $-\sum_i c_i e^{-\alpha_i}$, with

$$(c_0,\dots,c_7) = (5,8,9,8,5,0,2,2).$$

The torus rescales simple-root vertices by $(1/\sqrt{5}, 1/\sqrt{8}, 1/3, 1/\sqrt{8}, 1/\sqrt{5}, 1, 1/\sqrt{2}, 1/\sqrt{2})$. In this convention define

$$\begin{aligned} u = {} & e^{(0,0,0,0,0,-1,1,1)} \\ & + \frac{\sqrt{13}+3\sqrt{-3}}{20} e^{(0,0,0,1,1,-1,0,0)} \\ & + \frac{\sqrt{13}-3\sqrt{-3}}{20} e^{(1,1,0,0,0,-1,0,0)}. \end{aligned} \tag{83}$$

The compact real form and PCT operation must be transported through this gauge. Ordinary coefficient conjugation in the displayed gauge is not a replacement for the ambient PCT map.

Here is the transport explicitly. Set $\widetilde{c} = (5, 8, 9, 8, 5, 1, 2, 2)$ and let $g$ be the lattice torus automorphism which fixes the oscillators and multiplies $e^{\alpha}$ by $\prod_i \widetilde{c}_i^{-\alpha_i/2}$. If $\theta$ is the ordinary lattice PCT involution, the transported involution is $\theta_g = g\theta g^{-1}$. For real lattice oscillators,

$$\theta_g\big(h_1(-n_1)\cdots h_r(-n_r)e^{\alpha}\big) = (-1)^r\Big(\prod_i \widetilde{c}_i^{\alpha_i}\Big)h_1(-n_1)\cdots h_r(-n_r)e^{-\alpha}. \tag{84}$$

It acts antilinearly on coefficients, so fixes $\sqrt{13}$ and negates $\sqrt{-3}$. The transported Hermitian form is $\langle v, w\rangle_g = \langle g^{-1}v, g^{-1}w\rangle$ and is positive definite. Formula (84) squares to the identity. The same torus element acts on $V_{E_8}$: the positive real character on $L$ extends by its real linear logarithm to the common Cartan space. We transport the form and PCT on $H$ by that element as well, so the entire inclusion $U \subset B \subset H$ uses one unitary convention. It preserves the current algebra, including the root factor even though its two root vectors have different norms in this gauge.

All three momenta in (83) have squared norm four and Cartan weight $(1, 13)$. Substitution in the lattice formula verifies that $u$ is a highest vector of current type $(2, 14)$, where the entries now denote representation dimensions. In the original positive unitary basis, $g^{-1}u$ is, up to a common scalar,

$$e^a + \zeta e^b + \overline{\zeta}e^c, \qquad \zeta = \frac{\sqrt{13} + 3i\sqrt{3}}{\sqrt{40}}, \qquad |\zeta| = 1,$$

with $a, b, c$ the three displayed momenta. Thus the exact construction fixes a relative phase; there is no remaining continuous parameter.

The ambient and affine conformal vectors are fixed by these conventions:

$$\omega_B = \frac{1}{2}\sum_{i,j}(G^{-1})_{ij}\alpha_i(-1)\alpha_j(-1)\mathbf{1}, \tag{85}$$

$$\omega_{A_{1,k}} = \frac{1}{2(k+2)}\left(\tfrac{1}{2}H_{-1}H + E_{-1}F + F_{-1}E\right). \tag{86}$$

The conformal vector of $U$ is $\omega_B$, not the sum of the two affine conformal vectors. Their central charges sum to $1 + 117/41 < 8$.

### A.3 The constructed spaces in weights two and three

**Certificate input A.2** (Actual initial spaces)**.** The saved exact construction gives a PCT-stable 120-dimensional space $S_2 \subset U_2$, containing $u, \omega_B$, the affine vacuum descendants, two neutral primitive fields, and eight charged current multiplets. The current multiplet dimensions are

$$(1, 5), (1, 9), (1, 9), (1, 13), (2, 2), (2, 6), (2, 8), (2, 14).$$

These contribute 96 vectors, in addition to the 22 affine vacuum vectors and two neutral fields. The exact generated grade-three space $S_3 \subset U_3$ has dimension 956. Its normal subspace $N_3 = K_{-1}S_2 + TS_2$ has dimension 634, and

$$S_3 = N_3 + \operatorname{span}\{a_0 b : a, b \in S_2\}, \qquad \dim(S_3/N_3) = 322.$$

The spaces are current-stable. These are dimensions of constructed finite spaces, not initial assertions that they exhaust $U_2, U_3$.

The source records are `hi_cft_20260923_c8_exact_fields.json`, `hi_cft_20260923_c8_exact_unitary_certificate.json`, and `hi_cft_20260923_c8_minimal_generators.json`, together with the exact grade-three checkpoint bound into the final ledger.

The minimal-generator record is logically prior to the closure records. It starts with the six specified currents, $\omega_B$, and $u$, then closes their weight-two span under current zero modes and $u_1$. Every intermediate vector is therefore an actual $U$-vector. The final span has dimension 120 and contains each independently specified highest vector and neutral conformal vector. The equality with the saved space $S_2$ is checked only after this construction. This is the step that relates the large certificate presentation to the short definition (1); an ambient weight-two dimension count alone would not do so.

For reference the two neutral fields, denoted $e, f$, satisfy

$$e + f = \omega_B - \omega_{A_{1,1}} - \omega_{A_{1,39}}, \qquad c_e = \frac{2400}{1271}, \quad c_f = \frac{70}{31}. \tag{87}$$

They commute as Virasoro fields. The exact current and stress eigenvalues on the eight primitive highest vectors are as follows. The two occurrences of $(1, 9)$ are distinct vectors.

| Current dimensions $(n, m)$ | $e_1$ eigenvalue | $f_1$ eigenvalue |
|---|---|---|
| $(1, 5)$ | 798/1271 | 38/31 |
| $(1, 9)$ | 200/1271 | 42/31 |
| $(1, 9)$ | 1512/1271 | 10/31 |
| $(1, 13)$ | 420/1271 | 20/31 |
| $(2, 2)$ | 192/1271 | 49/31 |
| $(2, 6)$ | 1830/1271 | 3/31 |
| $(2, 8)$ | 752/1271 | 24/31 |
| $(2, 14)$ | 180/1271 | 13/31 |

Each row obeys

$$\frac{n^2 - 1}{12} + \frac{m^2 - 1}{164} + h_e + h_f = 2.$$

The affine vacuum space contributes 22 dimensions in weight two: 4 from $A_{1,1}$, 9 from $A_{1,39}$, and $3 \cdot 3$ mixed current products. The level-one null multiplet removes five of the nine universal weight-two states of its affine factor. Together with the two singlets and the 96 charged states, this gives 120. The exact normal-space and rank calculations certify these spaces inside the lattice algebra, including their intersections.

# B Finite closure and the all-weight criterion

Write $Y(a, z) = \sum_{n \in \mathbb{Z}} a_n z^{-n-1}$ and $T = L(-1)$. A singular product has mode index $n \geq 0$. The zero mode $a_0$ is a derivation and commutes with $T$.

**Definition B.1** (Differential normal span)**.** For a homogeneous positive-weight subspace $S \subset V$, let $\mathcal{N}(S)$ be the linear span of the vacuum and every expression obtained from vectors of $S$ by translation and arbitrary parenthesized $(-1)$ products. Let $\mathcal{N}_d(S) = \mathcal{N}(S) \cap V_d$. This definition makes translation and normal multiplication closed operations on $\mathcal{N}(S)$; closure under singular products is a separate assertion. We use the usual convention that a set strongly generates when its differential normal span is the whole algebra.

**Lemma B.2** (Filtered Wick reduction)**.** *Let $V$ be a nonnegatively integer-graded vertex algebra with $V_0 = \mathbb{C}\mathbf{1}$, and let $S$ have strictly positive weights. Fix $w$. Suppose every singular product*

*of two homogeneous generators of total weight less than $w$ belongs to $\mathcal{N}(S)$. Then the same is true for two normal expressions of total weight less than $w$. It is also true at total weight $w$ if at least one expression is a derivative of a positive-weight expression or a normal product of two positive-weight expressions.*

*Proof.* Use induction on total input weight. A vacuum factor has zero singular bracket. Sesquilinearity removes a derivative and lowers the total input weight; the resulting powers of the formal variable and translations preserve coefficientwise membership in $\mathcal{N}(S)$. Skew symmetry allows a composite input to be placed on the right. The noncommutative Wick identity is

$$[x_\lambda : yz :] =: [x_\lambda y]z : + : y[x_\lambda z] : + \int_0^\lambda [[x_\lambda y]_\mu z]\, d\mu. \tag{88}$$

The first two brackets have smaller total input weight, since an omitted factor has positive weight. Their coefficients therefore admit homogeneous expressions in $\mathcal{N}(S)$. A coefficient $x_j y$ in the integral has weight $\mathrm{wt}(x) + \mathrm{wt}(y) - j - 1$. Its pairing with $z$ has total weight at most $\mathrm{wt}(x) + \mathrm{wt}(y) + \mathrm{wt}(z) - 1$, so induction applies again. If the coefficient is scalar, its bracket is zero. Normal multiplication preserves the first two terms by definition. This proves both assertions, including the case in which both inputs are composite. Neither a preferred parenthesization nor independence of normal words was used. □

**Theorem B.3** (Recursive strong-generation criterion)**.** *Let $V$ be a nonnegatively graded vertex algebra with $V_0 = \mathbb{C}\mathbf{1}$. Choose finite homogeneous subspaces $S_d \subset V_d$, $1 \le d \le h$, where $h \ge 2$, and let $\mathcal{N}$ be their differential normally ordered span, allowing every parenthesization. Put $K = S_1$ and $A = S_2$. Assume*

1. *$k_n s, a_n s \in \mathcal{N}$ for every $k \in K$, $a \in A$, $s \in S_d$, and $n \ge 0$;*
2. *for $d \ge 3$,*

$$S_d \subset A_0 S_{d-1} + \mathcal{N}_d(S_1 + \cdots + S_{d-1}).$$

*Then $\mathcal{N}$ is a vertex subalgebra. In particular, the chosen spaces strongly generate the vertex algebra that they generate.*

*Proof.* The first assumption, the derivation property and $[T, a_0] = 0$ give $a_0\mathcal{N} \subset \mathcal{N}$ for $a \in A$. Explicitly, $a_0(x_{-1}y) = (a_0x)_{-1}y + x_{-1}(a_0y)$, so induction on the construction of a normal expression proves this stability before any singular-product closure is known.

Now induct lexicographically on $(\mathrm{wt}(s) + \mathrm{wt}(t), \mathrm{wt}(s))$ for generator pairs $s, t$. Left weight one or two is the first assumption. Otherwise the second assumption reduces the left input to $s = a_0 b$, with $\mathrm{wt}(b) = \mathrm{wt}(s) - 1$, plus normal remainders. The latter are covered by Lemma B.2: a homogeneous remainder of weight $\mathrm{wt}(s)$ in lower-weight generators has no single undifferentiated generator of that weight. Use

$$(a_0 b)_n t = a_0(b_n t) - b_n(a_0 t). \tag{89}$$

The first term has smaller total pair weight before applying $a_0$, which preserves $\mathcal{N}$. The known normal expression for $a_0 t$ has weight $\mathrm{wt}(t) + 1$. Every derivative or composite term reduces total pair weight in the second term. The only remaining possibility is a single undifferentiated generator of weight $\mathrm{wt}(t) + 1$; its pairing with $b$ has the original total weight and smaller left weight. It is therefore an earlier induction case. If $\mathrm{wt}(t) = h$, this single-generator possibility is absent.

All generator singular products close. Wick induction then gives closure for all normal expressions. Translation and normal multiplication preserve $\mathcal{N}$ by definition. For $r \geq 1$,

$$x_{-r}y = \frac{1}{(r-1)!}(T^{r-1}x)_{-1}y,$$

so all negative modes close too. □

For the Wick identities see [8, Section 2]. The full local derivation is `hi_cft_20260923_next_structure_recursive_strong_certificate.md`.

**Remark B.4** (Finite scope of the input)**.** Every membership in the first hypothesis has output weight at most $h+1$. Thus the criterion avoids checking all generator pairs through weight $2h-1$. The second hypothesis is essential: it asserts actual zero-product identities, rather than agreement of a dimension or a character coefficient. The proof takes place in the given ambient vertex algebra and makes no assertion that its normal words are free. This resolves the all-weight implication formerly marked F2.

## B.1 The finite spaces and prerequisite controls

Choose actual quotient lifts $P_3 \subset S_3$ and an actual triplet $P_4 \subset U_4$, with joint current highest weight $(0,2)$, and set

$$N_4 = K_{-1}S_3 + TS_3 + (S_2)_{-1}S_2, \qquad W_4 = N_4 + P_4, \tag{90}$$

$$N_5 = K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3. \tag{91}$$

These denote full differential normal spaces, using the lower closure identities to reorder expressions. A computational solver may use a proper subset of actual words in these spaces.

The triplet is obtained from an actual $a_0p$ by current-zero-mode projectors and lowering. It satisfies $P_4 \subset (S_2)_0S_3 + N_4$.

**Lemma B.5** (Bounds for the current projectors)**.** *On ambient weight $d$, a Cartan element represented by $h \in L \otimes \mathbb{R}$ has eigenvalues $m$ satisfying $m^2 \leq 2d(h,h)$. At weight four the two current highest weights are therefore at most 4 and 24.*

*Proof.* Cartan zero modes act on a Fock monomial of momentum $\alpha$ by $(h,\alpha)$, and commute with its oscillators. The grading gives $(\alpha,\alpha) \leq 2d$, so the estimate is Cauchy–Schwarz in the positive lattice form. Every eigenvalue is integral. Apply the estimate to $(h,h) = 2, 78$ and $d = 4$, and then to the highest vectors in the finite-dimensional semisimple current decomposition. □

Consequently finite polynomials in $FE$ give exact projectors. For an input of weight $(0,2)$, one may use

$$\prod_{\ell=1}^{2}\left(1 - \frac{F_LE_L}{\ell(\ell+1)}\right), \qquad \prod_{\ell=2}^{12}\left(1 - \frac{F_RE_R}{\ell(\ell+1)-2}\right).$$

The raising operator acts first in every $FE$ factor. Indeed, on the weight-$m$ line in an $\mathfrak{sl}_2$ representation of highest weight $j$, $FE$ acts by $(j-m)(j+m+2)/4$. The first polynomial retains left highest weight zero. At right weight two the second retains highest weight two and kills every possible larger highest weight. Thus all denominators are nonzero and the projector has a finite, justified range.

**Certificate input B.6** (Lower identities and the actual triplet)**.** The lower-closure artifact contains 3,970 completed exact identities. The actual $P_4$ vectors satisfy 18 current-zero-mode identities and six independently checked inclusions

$$L(1)P_4 \subset S_3, \qquad L(2)P_4 \subset S_2.$$

These direct Virasoro checks are available before the weight-four gate.

The two gates left for the all-weight argument are

$$(S_2)_0 S_3 \subset W_4 \qquad \text{(Gate 4)}, \tag{92}$$
$$(S_2)_0 P_4 \subset N_5 \qquad \text{(Gate 5)}. \tag{93}$$

# C Gate 4: exact seeds, raising, and singlet recovery

Put $P = S_3/N_3$, of independently known dimension 322. For a fixed joint-highest primitive field $a \in S_2$, the map

$$f_a : P \longrightarrow B_4/W_4, \qquad [p] \longmapsto [a_0 p]$$

is well defined because the lower relations give $a_0 N_3 \subset N_4$. Current covariance implies that its kernel is stable under both commuting raising operators $E_L, E_R$.

**Lemma C.1** (Raising saturation)**.** *If $C_a \subset \ker f_a$ and $P = E_L P + E_R P + C_a$, then $f_a = 0$.*

*Proof.* On $M = P/\ker f_a$ the hypothesis says $M = E_L M + E_R M$. The commuting ideal $(E_L, E_R)$ acts nilpotently because all current weights are bounded. Iterating gives $M = (E_L, E_R)^r M = 0$ for sufficiently large $r$. □

In this application nilpotence is entirely finite-dimensional. At weight three Lemma B.5 bounds the two Cartan eigenvalues by $|m_L| \le 3$ and $|m_R| \le 21$. Either raising operator increases $m_L + m_R$ by two. Every word of length 25 in the two raising operators therefore vanishes on $P$. The quotient by $\ker f_a$ inherits these operators because the kernel is stable. This gives an explicit termination bound for the argument; there is no limiting step.

The exact raising coordinates comprise 644 images. Their rank is 294. For each of nine independent highest left fields, the completed seed sets and these images have rank 322. A nonzero minor at the good split prime $p = 1{,}000{,}003$ proves a nonzero characteristic-zero minor. This proves spanning because the exact ambient quotient dimension is already 322. It does not give an upper rank for an unknown weight-four normal space.

The original exact run supplies 2,145 seed identities. The saturation calculation asks for 130 further inputs; 118 have direct exact residual certificates. The remaining twelve outputs have weight $(0,0)$. An exact replay of the known kernel and twelve additional identities at output weight $(0,2)$ shows that both raising currents kill each of these twelve remaining classes modulo $W_4$.

**Lemma C.2** (Recovery from a joint-singlet projection)**.** *Let $t \in B_4$ have weight $(0,0)$, with $E_L t, E_R t \in W_4$. Define*

$$\Pi = \prod_{j \in \{2,4\}} \left(1 - \frac{4F_L E_L}{j(j+2)}\right) \prod_{j \in \{2,4,\ldots,24\}} \left(1 - \frac{4F_R E_R}{j(j+2)}\right). \tag{94}$$

*If $\Pi t$ is a linear combination of $\Pi n_i$, with $n_i \in N_4$, then $t \in W_4$.*

*Proof.* Both $W_4$ and $N_4$ are stable under current zero modes. Every $FE$ factor kills $[t] \in B_4/W_4$, so $[\Pi t] = [t]$. Each $\Pi n_i$ remains in $N_4 \subset W_4$. Thus $[t] = 0$. Equivalently, the weight-zero highest class is a singlet in the finite semisimple current module. No dimension assertion about the selected normal columns is needed. □

The prime search selects 21 independent projected normal columns. All twelve projected identities have complete residual zero over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$. The final verifier repeats those residuals by independent integer denominator clearing, authenticates the raising premise, and repeats all nine saturation ranks. By Lemmas C.1 and C.2, all $9 \cdot 322 = 2898$ required highest-left identities hold. Current covariance gives the charged descendants. The second neutral field follows from $\omega_B - \omega_{K_L} - \omega_{K_R}$ minus the retained neutral field. This proves (92) with the certified lower controls.

## C.1 Positive modes available after Gate 4

The normal-product argument at the next gate uses positive modes on $S_3$. We establish them here, before their first use. This also explains why no additional unverified positive-mode gate is required.

**Lemma C.3** (Virasoro and current bootstrap)**.** *The lower identities, the direct $P_4$ controls, and Gate 4 imply*

$$(S_2)_1 S_3 \subset S_3, \qquad (S_2)_2 S_3 \subset S_2, \qquad (S_2)_3 S_3 \subset K, \tag{95}$$

$$K_1 W_4 \subset S_3, \qquad K_2 W_4 \subset S_2, \qquad K_3 W_4 \subset K, \tag{96}$$

$$L(1) N_5 \subset W_4, \qquad L(2) N_5 \subset S_3. \tag{97}$$

*The next mode in each of the first two lines is scalar, and all further modes vanish.*

*Proof.* The exact lower inputs give all singular products of $K$ and $S_2$ with $S_2$ in the expected lower spaces. The provenance $S_3 = N_3 + (S_2)_0 S_2$ and the current commutator give

$$K_0 S_3 \subset S_3, \qquad K_1 S_3 \subset S_2, \qquad K_2 S_3 \subset K.$$

For instance,

$$k_1(a_0 b) = (k_0 a)_1 b + (k_1 a)_0 b + a_0 (k_1 b). \tag{98}$$

The Virasoro commutators for a weight-two vector are

$$[L(1), a_n] = (2 - n) a_{n+1} + (L(1)a)_n, \tag{99}$$

$$[L(2), a_n] = (3 - n) a_{n+2} + 3(L(1)a)_{n+1} + (L(2)a)_n. \tag{100}$$

They show $L(1)S_3 \subset S_2$ and $L(2)S_3 \subset K$. Applied to the three summands of $N_4$, they give $L(1)N_4 \subset S_3$ and $L(2)N_4 \subset S_2$. For a normal input $a_{-1}b$ the latter formula includes the scalar term $(L(2)a)_{-1}b$; it must not be omitted. It lies in $S_2$. The six direct controls of Certificate B.6 extend these inclusions to $W_4$ without using Gate 5.

Each of the eight charged primitive multiplets and the two neutral singlets in $S_2$ is quasiprimary. Indeed, $L(1)$ is equivariant for current zero modes, and none of their current types is $(3,1)$ or $(1,3)$, the two types in $K$. For such a vector $a$ and $p \in S_3$, Gate 4 and the preceding controls give

$$\begin{aligned} 2a_1 p &= L(1)(a_0 p) - a_0 L(1) p \in S_3, \\ a_2 p &= L(1)(a_1 p) - a_1 L(1) p \in S_2, \\ 2a_3 p &= L(2)(a_1 p) - a_1 L(2) p \in K. \end{aligned}$$

In the last line the possible $(L(2)a)_1$ term is zero since $L(2)a$ is scalar. The affine normal and derivative part of $S_2$ is handled by the current iterate identities. This proves (95) on all of $S_2$.

The actual construction gives $P_4 \subset (S_2)_0 S_3 + N_4$. Equation (98), now with $b \in S_3$, gives $K_1 P_4 \subset S_3$; the same inclusion for $N_4$ follows on its normal generators. Since currents are primary of weight one,

$$[L(1), k_1] = -k_2, \qquad [L(2), k_1] = -k_3.$$

The controls on $L(1)W_4, L(2)W_4$ consequently give (96). The remaining modes follow from grading.

Finally apply $L(1), L(2)$ to $N_5 = K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3$. The current and translation summands follow from the established inclusions and the Virasoro relations. For the third summand, the required formulas are

$$\begin{aligned} L(1)(a_{-1}p) &= 3a_0 p + (L(1)a)_{-1}p + a_{-1}L(1)p, \\ L(2)(a_{-1}p) &= 4a_1 p + 3(L(1)a)_0 p + (L(2)a)_{-1}p + a_{-1}L(2)p. \end{aligned}$$

Every term belongs respectively to $W_4$ or $S_3$ by Gate 4, (95), and the lower identities. Reordering the last normal term uses only already established lower products. This proves (97). □

# D Gate 5 from Jacobi identities

The proof uses the full normal spaces in (90)–(91), not only the selected solver columns. To see that the displayed formula for $N_5$ includes every lower normal word of weight five, remove a leading derivative, or split a normal product into weights $1+4$ or $2+3$. Reversed splits are reordered by skew symmetry for normal products. Its correction terms are derivatives of singular products, already controlled by Lemma C.3 and the lower identities. Quasi-associativity changes the parenthesization with correction terms of the same type. Thus the resulting words lie in $K_{-1}W_4 + TW_4 + (S_2)_{-1}S_3$.

**Lemma D.1** (Normal-space controls)**.** *Before using Gate 5, the lower controls and Gate 4 give*

$$(S_2)_0 N_4 \subset N_5, \qquad (N_3)_0 S_3 \subset N_5. \tag{101}$$

*Proof.* The first follows by zero-mode derivation and skew reordering. For the second, $(Ta)_0 = 0$ and, for a current $x$,

$$(x_{-1}a)_0 p = \sum_{i \geq 0} x_{-1-i}(a_i p) + \sum_{i \geq 0} a_{-1-i}(x_i p). \tag{102}$$

All inner products belong to the required lower spaces. To make the translation reduction explicit, repeatedly use

$$x_{-r}v = \frac{T(x_{-(r-1)}v) - x_{-(r-1)}Tv}{r-1}, \qquad r \geq 2.$$

The base terms are precisely those in (91); the same identity applies to $a_{-r}v$ with $a \in S_2$. Thus no unproved $P_3$-by-$P_3$ closure enters (101). All positive inner modes in (102) are supplied by (95) and the current controls established before this section. More explicitly the first sum has inner vectors in $W_4, S_3, S_2, K, \mathbb{C}\mathbf{1}$, in that order; the second has inner vectors in $S_3, S_2, K, \mathbb{C}\mathbf{1}$. Both sums are finite by grading. □

If $P_4 \subset N_4$, Gate 5 already follows. Otherwise $Q = W_4/N_4$ is the irreducible current triplet of highest weight $(0,2)$, since the actual $P_4$ is irreducible. Let $q_-, q_0, q_+$ be its weight basis, and set

$$\beta(a,p) = [a_0 p] \in Q, \qquad F(a,q) = [a_0 \widetilde{q}] \in B_5/N_5.$$

The second map does not depend on the representative $\widetilde{q} \in W_4$, by (101).

**Lemma D.2** (Jacobi reduction)**.** *If $a, b \in S_2$ satisfy $a_0 b \in N_3$, then for $p \in S_3$,*

$$F(a, \beta(b,p)) = F(b, \beta(a,p)). \tag{103}$$

*Proof.* The difference is the class of $(a_0 b)_0 p$, which vanishes by (101) and $[a_0, b_0] = (a_0 b)_0$. □

## D.1 The charged families

Let $a_0, \dots, a_7$ denote the eight charged highest fields, with current weights

$$(0,4), (0,8), (0,8), (0,12), (1,1), (1,5), (1,7), (1,13).$$

In a product such as $(a_i)_0 p$, the parenthesized subscript denotes the field label and the next subscript denotes its mode. The saved exact relations give

$$\beta(a_3, p_3) = c_3 q_-, \qquad \beta(a_6, p_6) = c_6 q_-, \qquad c_3 c_6 \neq 0,$$

where $p_3 =$ `r3_000910` has weight $(0,-14)$, and $p_6 =$ `r3_000899` has weight $(-1,-9)$. The coefficients are authenticated in the final Jacobi certificate. Exact combinations of the lower relations establish $(a_i)_0 a_j \in N_3$ for the following pairs:

| $i$ | $j$ | Input | Weight of $(a_i)_0 p_j$ |
|---|---|---|---|
| 0 | 3 | $p_3$ | $(0,-10)$ |
| 1,2 | 3 | $p_3$ | $(0,-6)$ |
| 3 | 6 | $p_6$ | $(-1,3)$ |
| 4 | 6 | $p_6$ | $(0,-8)$ |
| 5 | 3 | $p_3$ | $(1,-9)$ |
| 6 | 3 | $p_3$ | $(1,-7)$ |
| 7 | 3 | $p_3$ | $(1,-1)$ |

Every listed output weight lies outside $Q$, so $\beta(a_i, p_j) = 0$. Equation (103) and $c_j \neq 0$ give $F(a_i, q_-) = 0$. Raise this resulting zero with $a_i$ held highest to obtain the other two right inputs. Current covariance gives all left descendants.

**Remark D.3.** The weight exclusion is used only at the displayed bottom inputs. It need not hold after raising $p_j$; using it there would be an incorrect stronger assertion. We raise the proved zero identity instead.

## D.2 The neutral family

Let $C$ be the full current span, inside $P = S_3/N_3$, of brackets of charged weight-two fields. The charged-span certificate reconstructs 28 actual charged seeds from the exact lower relations. Together with the 644 lowering images, of rank 294, these have rank 322. The independent verifier also checks both $\mathfrak{sl}_2$ commutators and all cross-axis commutators against the saved exact raising matrices. Since $C$ is a current submodule, the nilpotent-ideal argument gives

$$P = F_L P + F_R P + C \quad \Longrightarrow \quad P/C = 0.$$

Once again the finite-prime full rank is a positive spanning witness inside an independently exact 322-dimensional space.

For the retained neutral singlet $e$, write

$$e_0 a_3 = \sum_i t_i (x_i)_0 y_i + n, \qquad n \in N_3,$$

with charged $x_i, y_i$. Apply zero modes to $p_3$. By Jacobi, the charged conclusion and (101),

$$(e_0 a_3)_0 p_3 = \sum_i t_i [(x_i)_0, (y_i)_0] p_3 + n_0 p_3 \in N_5.$$

Indeed Gate 4 puts each inner product in $W_4$, and the next charged product is controlled on both $N_4$ and $P_4$. A second Jacobi identity gives

$$e_0((a_3)_0 p_3) = (a_3)_0 (e_0 p_3) + (e_0 a_3)_0 p_3 \in N_5.$$

The nonzero coefficient of $q_-$ proves the neutral bottom identity; covariance gives the triplet. The other neutral field is obtained from the total stress minus the two affine Sugawara stresses and $e$. This proves Gate 5. Its 24 independent highest channels comprise 23 charged channels and one neutral channel; they are consequences of the displayed identities, not 24 separately recomputed grade-five lattice products.

# E From the finite gates to properness

Lemma C.3 supplies the positive-current products and $L(1)N_5 \subset W_4$, $L(2)N_5 \subset S_3$ before Gate 5. For a quasiprimary primitive $a \in S_2$, Gate 5 now yields, for $p \in P_4$,

$$\begin{aligned} 2a_1 p &= L(1)(a_0 p) - a_0 L(1) p \in W_4, \\ a_2 p &= L(1)(a_1 p) - a_1 L(1) p \in S_3, \\ 2a_3 p &= L(2)(a_1 p) - a_1 L(2) p \in S_2, \\ a_4 p &= L(2)(a_2 p) - a_2 L(2) p \in K. \end{aligned}$$

The next mode is scalar. The 30 earlier prime-only $a_1 P_4$ tests are therefore derived exactly here; they are not an omitted gate.

All hypotheses of Theorem B.3 hold for $K, S_2, S_3, P_4$. Their normal span is a vertex subalgebra of $U$ containing $K, u, \omega_B$, hence equals $U$. A normal word in positive-weight generators cannot acquire weight one unless it is a weight-one generator. Thus $U_1 = K$, proving Proposition 1.1.

*Proof of Corollary 1.2.* A differential normal word of weight two uses only $K$ and $S_2$; its current descendants are already in $S_2$. A word of weight three uses only $K, S_2, S_3$, and the lower normal words lie in $N_3$. Thus the strong-generation result proves $U_2 = S_2$ and $U_3 = S_3$. The exact PCT certificate gives $\theta_g S_2 = S_2$, while $\theta_g K = K$. Since $S_2 \subset U$ contains the original generators, $\theta_g U = U$. The inherited positive form makes $U$ a unitary subalgebra of $B$, hence simple by [10, Proposition 5.29]. It contains $\omega_B$, so has central charge eight, the same nonnegative grading, and $U_0 = \mathbb{C}\mathbf{1}$. The character assertion is now the definition of the graded trace. □

**REVIEW — F1: what the finite certificate establishes**

This proof of properness has the strongest computational support in the manuscript. It still relies on correctness of the lattice mode engine, the saved actual-vector recipes, and the stated algebraic lemmas.

Hashes authenticate the records; hashes alone do not prove those lemmas or independently reconstruct every earlier mode calculation. The Lean checks in Appendix N cover selected abstract implications, not the VOA construction or the complete strong-generation argument. No second implementation of the entire construction is claimed. In particular, $\dim U_4 = 6010$ is not used as an input to this proof.

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# F Regularity from a positive conformal-vector identity

This section concerns the actual PCT-stable subalgebra $U \subset B$ defined in the construction. Its conclusions do not require properness. We use the convention $Y(a,z) = \sum_{n\in\mathbb{Z}} a_n z^{-n-1}$ and write $L^e(n) = e_{n+1}$ for the Virasoro operators of a conformal vector $e$. The subalgebras used below need not contain the ambient conformal vector. We call an inclusion $A \subset V$ *grading compatible* if the $A$-weight of each homogeneous vector of $A$ equals its $V$-weight. Thus a mode $a_n$ changes both gradings by $\mathrm{wt}(a) - n - 1$; the equality of the two energy operators is not required on all of $V$. All subalgebras in the positive identity below are grading compatible and stable under the ambient PCT involution.

**Certificate input F.1** (Embedded structural data)**.** The following are assertions about actual fields and their exact modes in the lattice construction, in addition to the finite closure ledger.

1. The Potts fields lie in $U$, are PCT stable, and generate the regular unitary algebra $K(\mathfrak{sl}_2, 3)$, with its translation and intrinsic grading inherited from the ambient VOA. The exact records are `hi_cft_20260923_proof_potts.json` and `hi_cft_20260923_proof_potts_exact.json`.

2. The actual fields $z, a$ of weights two and three lie in $U$, are PCT stable, have central charge $8/5$, and satisfy the stress, primary, and translation identities in `hi_cft_20260923_proof_pf13_exact.json`, `hi_cft_20260923_proof_pf13_membership.json`, and `hi_cft_20260923_proof_remaining_coset.json`. Their full singular first rows agree with the parafermion rows at $c = 8/5$, $\lambda = 14/473$, as polynomials in $z, a, a_1 a, a_1(a_1 a)$. The polynomial records are `hi_cft_20260923_proof_pf13_upper_poles.json`, `hi_cft_20260923_proof_pf13_nonlinear.json`, `hi_cft_20260923_proof_pf13_truncation.json`, `hi_cft_20260923_proof_pf13_simple_pole_algebra.json`, and `hi_cft_20260923_proof_pf13_dly.json`. The membership record also gives the positive identity (104).

3. The lattice–Potts core $T$ below is embedded with its stated charge realization. The six joint spectral vectors used to construct its order-thirteen extension are actual vectors of $U_2$, with the eigenvalues printed below. Their exact source is `hi_cft_20260923_next_structure_core_extension.json`; the construction of $T$ is described in `hi_cft_20260923_proof_report.md`, Section 4.

4. The zero-mode action on the complementary eighty currents is irreducible, and the four mixed lattice tops of dimension eighty-one are irreducible as $U$-tops. These are exact field-action assertions, as recorded in `hi_cft_20260923_proof_current_module_v2.json` and `hi_cft_20260923_next_81top_all.json`, respectively.

All polynomial equalities here mean exact vector equalities in the fixed lattice realization, not projected or floating-point matches.

The input from the local construction is the identity

$$\omega_U = \omega_{A_{1,1}} + \frac{3403}{4563}\omega_{A_{1,39}} + \frac{25}{6}\overline{p} + \frac{25}{26}\overline{z}. \tag{104}$$

Here $p$ and $z$ are the conformal vectors of actual embedded copies of the three-state Potts algebra $K(\mathfrak{sl}_2, 3)$ and the parafermion algebra $K(\mathfrak{sl}_2, 13)$; a bar denotes normalized averaging over the compact current group. Each subalgebra is regular and unitary, and its intrinsic grading agrees with the ambient grading on its states. The averages take place in the finite-dimensional space $U_2$. More concretely, $p, z$ lie in the already constructed current-stable space $S_2$, whose invariant subspace is spanned by the four mutually orthogonal stresses $\omega_{A_{1,1}}, \omega_{A_{1,39}}, e, f$. For either vector $v$, its Haar average is therefore

$$\overline{v} = \sum_{w \in \{\omega_{A_{1,1}}, \omega_{A_{1,39}}, e, f\}} \frac{\langle v, w \rangle_g}{\langle w, w \rangle_g} w.$$

Thus the exact coefficients in (104) can be verified inside $S_2$ without assuming $S_2 = U_2$ or properness. The compact orbit of a conformal vector has compact convex hull in the real space of PCT-fixed weight-two vectors. Its Haar average belongs to that hull. Carathéodory's theorem therefore replaces each average by a finite convex combination of orbit points. In particular,

$$\omega_U = \sum_{i=1}^{N} c_i e_i, \qquad c_i > 0. \tag{105}$$

Zero coefficients, if any, are omitted. All simple modules of each regular subalgebra $A_i$ have nonnegative lowest weights, with zero occurring only for its vacuum module.

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**REVIEW — R1: the embedded regular factors are local inputs**

Certificate F.1 states the model-specific field inputs explicitly. Proposition F.2 below proves the transport from the complete checked parafermion rows to the actual factor identification. It does not replace verification of the listed lattice identities, Potts embedding, or positive stress identity. These inputs are separate from the finite closure theorem.

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**Proposition F.2** (Finite reconstruction of the parafermion factor)**.** *Let $V, V'$ be simple vertex algebras of CFT type generated by a conformal vector $L$ and a weight-three primary $a$. Set $b = a_1 a$ and $d = a_1 b$, of weights four and five. Suppose the full singular products $a_n a, a_n b, a_n d$, $n \geq 0$, agree as homogeneous differential normal polynomials in $L, a, b, d$, with the same normal ordering. Suppose also that their self-row is*

$$a_5 a = \frac{c}{3}\mathbf{1}, \quad a_4 a = 0, \quad a_3 a = 2L, \quad a_2 a = TL, \quad a_1 a = b, \quad a_0 a = \frac{1}{2}Tb - \frac{1}{12}T^3 L,$$

*and that $a_n b = 0$ for $n \geq 4$, $a_3 b \in \mathbb{C}a$, and $a_2 b \in \mathbb{C}Ta$. Then there is an isomorphism $V \cong V'$ identifying $L, a, b, d$. In particular $a_1 d$ and both simple poles must be specified polynomials; they cannot be left as independent new generators.*

*Proof.* We first recover the remaining generator OPEs without a circular normal-product reduction. The Virasoro rows follow by commuting Virasoro modes through $b = a_1 a$ and $d = a_1 b$:

$$L(m)b = (2m+1)a_{m+1}a, \qquad L(m)d = (2m+1)a_{m+1}b + a_1 L(m)b \quad (m \geq 1).$$

These are linear in generators and derivatives under the displayed hypotheses. Thus Virasoro creation operators can be moved to the outside of any normal word using already known

commutators. Sesquilinearity removes derivatives. Since three non-Virasoro factors have weight at least nine, the possible two-factor cores through weight eight are exactly

| weight bound | cores besides individual fields |
|---|---|
| 6 | $:aa:$ |
| 7 | $:aa:,\ :ab:$ |
| 8 | $:aa:,\ :ab:,\ :bb:,\ :ad:$ . |

Reversing the order of a core uses skew symmetry and its already known pair bracket. Commuting an external $a_n$ or $b_n$ through outer Virasoro creation operators reduces their number; any resulting negative mode is itself a normal product.

Write $A_r$ for the evaluation of $a_nP$, $n \geq 0$, on every normal polynomial $P$ of weight at most $r$, and $B_r$ for the corresponding assertion with $b_n$. We use the noncommutative Wick identity [8, Section 2]

$$[x_\lambda : yz :] =: [x_\lambda y]z : + : y[x_\lambda z] : + \int_0^\lambda [[x_\lambda y]_\mu z]\, d\mu.$$

The following order is acyclic. First $A_6$ is known: the only two-factor core is $:aa:$, and $[a_\lambda a]$ is linear in the vacuum and derivatives of $L, b$, whose brackets with $a$ are given. The iterate identity

$$b_n s = a_1(a_n s) - a_0(a_{n+1}s) + a_{n+1}(a_0 s) - a_n(a_1 s) \tag{106}$$

with $s = b$ now determines $b_n b$, because each inner $a_j b$ has weight at most six. Next $A_7$ follows: its new core $:ab:$ uses only the just-computed $b, b$ row in the Wick integral. Equation (106) with $s = d$ then determines $b_n d$, since every inner $a_j d$ has weight at most seven.

At this point $B_6$ is known. For its core $:aa:$ the Wick integral brackets a coefficient of $[b_\lambda a]$, of weight at most six, with $a$, which is $A_6$ after skew symmetry. Single-field inputs use the known $b, b$ and $b, d$ rows. We obtain $A_8$ next: its core $:bb:$ uses $B_6$ on coefficients of $[a_\lambda b]$, and its core $:ad:$ uses the known $L, d$ and $b, d$ rows. Finally $B_7$ follows from $B_6$ on the coefficients of $[b_\lambda a]$ in the new core $:ab:$. With $s = d$, the second iterate identity

$$d_n s = a_1(b_n s) - a_0(b_{n+1}s) + b_{n+1}(a_0 s) - b_n(a_1 s)$$

computes $d_n d$: the first two terms use $A_8$, and the last two use $B_7$. All generator singular products have therefore been expressed by the same polynomials in the two algebras.

For the uniqueness step, take the vertex subalgebra $D \subset V \times V'$ generated by the diagonal pairs of the four named fields. Common polynomial OPE closure and filtered Wick induction show that $D$ is their differential normal span. Every generator has positive weight, so $D_0 = \mathbb{C}(\mathbf{1}, \mathbf{1})$ and $D_n = 0$ for $n < 0$. It follows that $D$ has a unique maximal proper graded ideal: every proper graded ideal has zero degree-zero part, and so does their sum. The coordinate projections are surjective because $L, a$ generate each target. They preserve the conformal vector and hence the grading. Their kernels are maximal proper graded ideals because $V, V'$ are simple. The kernels coincide, which gives the asserted isomorphism. No freeness or independence of normal monomials has been used. □

The exact parafermion records instantiate this proposition at $c = 8/5$ and $\lambda = 14/473$. The upper-pole record supplies the self-row and the remaining positive poles, the nonlinear record supplies $a_0 b$, and the truncation record supplies $a_1 d$. The algebraic simple-pole derivation supplies $a_0 d$ from the independently checked upper poles and skew symmetry. The Dong–Lam–Yamada change-of-generators record matches the seven weight-six decoupling coefficients with those of $K(\mathfrak{sl}_2, 13)$, using [48, Appendix B]. The common first rows are the recursive formulas

(5.4), (5.7), and (A.2) of Linshaw [47]. Simplicity of the actual generated subalgebra follows from its PCT stability; its translation operator agrees with the ambient one by the complementary positive energy argument. Thus the proposition transports the checked polynomial identities to an identification with the parafermion algebra. The exact membership, PCT, translation, and polynomial identities remain the explicit finite inputs to this identification.

**Lemma F.3** (Orthogonal energy bound)**.** *Let $V$ be a unitary VOA of CFT type admitting an identity of the form (105), with the properties just stated. Then $V$ is $C_2$-cofinite.*

*Proof.* For an ordinary simple $A_i$-module $M$, put

$$C_{2,A_i}(M) = \operatorname{span}\{a_{-2}m : a \in A_i,\ m \in M\}.$$

These quotients are finite-dimensional by [13, Proposition 5.2]. They are graded for the intrinsic operator $L^{e_i}(0)$. There are finitely many simple $A_i$-modules. Hence there is a finite number $D_i$ bounding the intrinsic energy in all their $C_{2,A_i}$ quotients; this includes their lowest weights. Regularity decomposes the weak $A_i$-module $V$ into ordinary simples. Consequently every intrinsic eigenvector of energy above $D_i$ lies in $C_{2,A_i}(V)$. The intrinsic energy preserves each finite-dimensional ambient grade $V_n$ and is self-adjoint for its positive form. We may therefore diagonalize it on $V_n$. Since

$$C_{2,A_i}(V) \subset C_2(V),$$

every vector in $V_n \cap C_2(V)^{\perp}$ has zero projection to the intrinsic eigenvectors of energy above $D_i$. It follows that

$$\langle L^{e_i}(0)v, v\rangle \le D_i\|v\|^2 \qquad (v \in V_n \cap C_2(V)^{\perp}).$$

For $v \in V_n \cap C_2(V)^{\perp}$, the conformal-vector identity gives

$$n\|v\|^2 = \sum_i c_i\langle L^{e_i}(0)v, v\rangle \le \Big(\sum_i c_i D_i\Big)\|v\|^2.$$

The subspace $C_2(V)$ is ambient graded. Its orthogonal complement in each finite-dimensional $V_n$ represents $(V/C_2(V))_n$, so the last inequality makes the quotient vanish for $n > \sum_i c_i D_i$. Finiteness of each $V_n$ proves the claim. No simultaneous diagonalization of the different energy operators is required. The same proof applies directly to averages, since every conjugate factor has the same bound $D_i$. □

**Lemma F.4** (Highest spaces for a compatible subalgebra)**.** *Let $A \subset V$ be grading compatible and regular, and let $M$ be a weak $V$-module. Suppose a finite-dimensional subspace $E \subset M$ is stable under every ambient degree-zero mode from $A$ and is killed by every mode from $A$ of negative ambient degree. Then $E$ is a direct sum of irreducible Zhu-algebra modules belonging to the tops of simple ordinary $A$-modules. In particular $L^A(0)|_E$ is diagonalizable, with eigenvalues among their lowest weights.*

*Proof.* Write

$$\Omega_A(M) = \{v \in M : a_n v = 0 \text{ for all homogeneous } a \in A,\ n \ge \mathrm{wt}_A(a)\}.$$

Compatibility puts $E$ inside this space. Regularity decomposes $M\downarrow_A$ as an algebraic direct sum of ordinary simples, and $\Omega_A$ commutes with that sum. The highest space of an ordinary simple is its top. Zero modes endow $\Omega_A(M)$ with the corresponding semisimple Zhu-algebra action; its invariant finite-dimensional subspace $E$ is therefore a sum of simple tops. This also explains why one must impose the lowering condition for all states of $A$, rather than only for a chosen generating set. □

**Lemma F.5** (Vacuum-like vectors)**.** *Let $M$ be a weak module over a VOA $V$. If $v \in M$ satisfies $L_M(-1)v = 0$, then $a_n v = 0$ for all $a \in V$ and $n \geq 0$, and $a \mapsto a_{-1}v$ is a $V$-module homomorphism $V \to M$ sending $\mathbf{1}$ to $v$.*

*Proof.* Translation covariance gives $L_M(-1)a_{n+1}v = -(n+1)a_n v$. Lower truncation and downward induction give $a_n v = 0$ for every $n \geq 0$. The module Jacobi identity then gives $b_m(a_{-1}v) = (b_m a)_{-1}v$; equivalently this follows by taking the constant coefficient in weak associativity, since the fields applied to $v$ have no negative powers. This is the asserted homomorphism. □

**Lemma F.6** (Lowest weights and projectivity of the vacuum)**.** *Let $V$ satisfy Lemma F.3, and suppose every component algebra has nonnegative simple lowest weights and its vacuum is the unique simple of weight zero. Every simple ordinary $V$-module has nonnegative lowest conformal weight, the only weight-zero simple is $V$, and $V$ is projective among grading-restricted generalized modules.*

*Proof.* Let $E = M_h$ be the lowest space of a simple ordinary module. For a homogeneous state $a$, $[L_V(0), a_n] = (\mathrm{wt}_V(a) - n - 1)a_n$ on every module. Thus degree-zero modes preserve $E$, and lowering modes kill $E$. Lemma F.4 implies that every $L^{e_i}(0)|_E$ is diagonalizable with nonnegative real spectrum. Taking traces, rather than assuming the operators commute, gives

$$h \dim E = \sum_i c_i \operatorname{Tr}_E L^{e_i}(0) \geq 0. \tag{107}$$

In particular $h$ is real and nonnegative. If $h = 0$, every summand vanishes. Hence every partial energy is zero on $E$, which lies in sums of vacuum tops for every $A_i$. Consequently $L^{e_i}(-1)E = 0$, and (105) gives $L_V(-1)E = 0$. By Lemma F.5, any nonzero $v \in E$ gives a nonzero module map $V \to M$, hence an isomorphism.

Now let $\pi : N \twoheadrightarrow V$ be an epimorphism in the generalized module category. $C_2$-cofiniteness gives finite length, and the preceding argument excludes negative weights in all its composition factors. The finite generalized weight-zero space $N_{[0]}$ therefore has no lower-weight target for a lowering mode. It is again in every $\Omega_{A_i}(N)$ and is invariant under degree-zero modes. On this space $L_V(0)$ is nilpotent, so

$$0 = \operatorname{Tr}_{N_{[0]}} L_V(0) = \sum_i c_i \operatorname{Tr}_{N_{[0]}} L^{e_i}(0).$$

Diagonalizability and nonnegativity of each partial energy imply it is zero. The vacuum-top argument gives $L_V(-1)N_{[0]} = 0$. Generalized weight decompositions are preserved by $\pi$, so one can choose $v \in N_{[0]}$ with $\pi(v) = \mathbf{1}$. Then

$$s : V \longrightarrow N, \qquad s(a) = a_{-1}v, \qquad \pi s = \mathrm{id}_V,$$

is a section. Every epimorphism onto the vacuum splits. □

**Lemma F.7** (A projective unit in a finite braided $r$-category)**.** *Let $\mathcal{C}$ be a finite complex linear abelian braided monoidal category with simple unit, right-exact tensor product, and Grothendieck–Verdier duality with dualizing object $\mathbf{1}$. If $\mathbf{1}$ is projective, then $\mathcal{C}$ is rigid and semisimple.*

*Proof.* Realize $\mathcal{C}$ as finite-dimensional modules for a finite-dimensional algebra, denote underlying vector-space dimension by $d_0$, and let $P$ correspond to its regular module. Every $Z$ is a quotient of $P^{\oplus d_0(Z)}$. For fixed $X$, right exactness gives

$$d_0(Z \otimes X) \leq C_X d_0(Z), \qquad \dim \mathrm{End}(X^{\otimes n}) \leq d_0(\mathbf{1})^2 C_X^{2n} < n!$$

for large $n$, where $C_X = \max\{1, d_0(P \otimes X)\}$. This uses only right exactness. For nonzero indecomposable $X$, Grothendieck–Verdier evaluation $DX \otimes X \to \mathbf{1}$ is nonzero, hence surjective onto the simple unit, and splits by projectivity. A finite abelian category is idempotent complete, and its Hom spaces are finite-dimensional. Thus $X$ is non-negligible and has moderate growth in the precise sense of [14, Theorem 1.1]; that theorem proves $X$ rigid. Finite direct sums give rigidity of all objects. Tensoring is now exact, and

$$\mathrm{Hom}(X, -) \cong \mathrm{Hom}(\mathbf{1}, - \otimes X^*)$$

is exact. Every object is projective, so $\mathcal{C}$ is semisimple. □

**Theorem F.8** (Positive stresses imply strong rationality)**.** *A simple unitary VOA of CFT type satisfying the hypotheses of Lemma F.6 is strongly rational. In particular, the actual generated algebra $U$ is strongly rational, independently of its properness in $B$.*

*Proof.* Unitarity gives self-contragredience. Lemmas F.3 and F.6 give $C_2$-cofiniteness and a projective vacuum. The category of grading-restricted generalized modules is finite abelian and braided, its tensor product is right exact, and contragredient duality is Grothendieck–Verdier duality with dualizing object $V' = V$; see [15, Sections 2.3 and 3]. Lemma F.7 proves semisimplicity, hence rationality in the $C_2$-cofinite setting. This argument does not assume that the nonvacuum modules are unitarizable. The local argument is recorded in `hi_cft_20260923_next_structure_rationality.md`. □

# G Branching and the boundary fusion ring

From this point we use the properness conclusion and $\dim U_1 = 6$ of the finite closure argument. Before that conclusion is supplied, every statement in this section is conditional on the proper branch.

## G.1 A universal top bound and actual lattice branching

The sharper bound uses an actual regular subalgebra $A_L \subset U$. In the notation of the local construction, a rational lattice–Potts core $\mathcal{T}$ of central charge $14/5$ and $K(\mathfrak{sl}_2, 13)$ of central charge $8/5$ have pointed charges $(j, k) \in (\mathbb{Z}/13)^2$ with quadratic form $-(j^2 + k^2)/13$. Put $A = \mathcal{T} \otimes K(\mathfrak{sl}_2, 13)$. The field certificate supplies nonzero weight-two vectors $v$ with

$$(\omega_U - \omega_A)_1 v = 0, \qquad z_1 v = \frac{12}{13} v, \qquad (\omega_{\mathcal{T}})_1 v = \frac{14}{13} v, \qquad p_1 v = \frac{2}{3} v,$$

at Cartan weights $(\pm 1, \pm 5)$ and $(0, \pm 8)$. These are the top characters of the simple-current charges $(5, 1)$ and its inverse, up to reversing the second charge. The label is forced by actual, unreduced lowest weights: the $\tau$-labelled $\mathcal{T}$ module of charge 5 starts at $31/65$, whereas $14/13 - 31/65$ is not integral, and the only parafermion modules starting at $12/13$ are the two inverse simple currents.

Here is why the witness gives an embedded extension. The kernel of $(\omega_U - \omega_A)_1$ in $U$ is a connected unitary extension of $A$ with conformal vector $\omega_A$. Regularity of $A$ and finite ambient homogeneous spaces make this extension finite. In a connected unitary extension, invertible constituents have multiplicity at most one and their support is a subgroup: the invariant pairing gives a nonzero product of a constituent with its inverse into the vacuum line; associativity then makes multiplication by it invertible on each supported invertible line. Integer grading makes that support isotropic. The only isotropic lines for the displayed quadratic form are

$k = 5j$ and $k = -5j$, since $5^2 = -1 \pmod{13}$. The witnessed nonzero charge generates one of these lines, and both cannot occur because together they generate nonintegral-spin charges. Consequently, after the harmless choice of parafermion charge orientation,

$$A_L = \bigoplus_{k=-5j} \mathcal{T}_{1,j} \otimes \mathrm{PF}_{0,k} \subset U, \qquad \mathrm{Rep}(A_L) \simeq \mathrm{Fib} \boxtimes \mathrm{SO}(3)_{13}.$$

The field witness and extension argument are in `hi_cft_20260923_next_structure_order13_core.md` and `hi_cft_20260923_next_structure_core_extension.json`. There are fourteen simple modules, labelled $x_a = (1, a)$ and $y_a = (\tau, a)$ for $0 \le a \le 6$. Indeed the free order-thirteen extension removes the hyperbolic pointed factor; no orbit has a stabilizer, so there is no fixed-point resolution. The two remaining labels are the Fibonacci label and the even affine label $2a$.

We give enough charge data to reconstruct their tops. For $\epsilon \in \{1, \tau\}$ put

$$p_1(m) = \begin{cases} 0, & 3 \mid m, \\ 2/3, & 3 \nmid m, \end{cases} \qquad p_\tau(m) = \begin{cases} 2/5, & 3 \mid m, \\ 1/15, & 3 \nmid m. \end{cases}$$

The actual lowest weights of the lattice–Potts modules are

$$h_{\mathcal{T}}(\epsilon, j) = \min_{m \equiv j \ (13)} \left( \frac{m^2}{156} + \frac{\mathbf{1}_{m \text{ odd}}}{4} + p_\epsilon(m) \right). \tag{108}$$

An even minimizing $m$ contributes a left affine singlet and an odd one a left doublet; their right Cartan charge is $m$. Choosing $|m| \le 6$ bounds the minimum by $14/13$, while $|m| \ge 13$ gives $m^2/156 > 14/13$. Thus $-12 \le m \le 12$ contains every minimizer. For the representative $-6 \le k \le 6$, the parafermion formula is

$$h_{\mathrm{PF}}(a, k) = \frac{a(a+1)}{15} - \frac{k^2}{13} + \max(0, |k| - a). \tag{109}$$

The last term is the least affine grade needed for charge $2k$ outside the finite $\mathfrak{sl}_2$ top. This is equivalently the actual-top formula in [16, Section 2.4]; the parafermion top is one-dimensional. Finally minimize $h_{\mathcal{T}}(\epsilon, j) + h_{\mathrm{PF}}(a, -5j)$ over $j \in \mathbb{Z}/13$, using the representative in (109).

Let $E$ be any simple $U$-module top and write $x_a, y_a$ also for the nonnegative integer multiplicities of these $A_L$ tops in $E$. Lemma F.4 justifies taking complete $A_L$ tops here: the finite space $E$ is invariant under all their Zhu zero modes, not merely contained in the sum of their highest spaces. Set $D = \dim E$. For each top, let $L$ be the trace of the left affine energy, $Q = \mathrm{Tr}\, H_R(0)^2$, and $P, Z$ the Potts and parafermion energy traces. Its contribution to (107) is

$$W = L + \frac{83}{6084} Q + \frac{25}{6} P + \frac{25}{26} Z. \tag{110}$$

This is a trace formula on a complete right $\mathfrak{sl}_2$ representation, not an assertion about a single partial-energy eigenvector. In fact, for an irreducible right $\mathfrak{sl}_2$ representation of highest weight $r$,

$$\frac{\mathrm{Tr}\, H_R(0)^2}{\dim} = \frac{r(r+2)}{3}, \qquad L^{A_{1,39}}(0) = \frac{r(r+2)}{164}.$$

Thus the Sugawara trace is $3Q/164$, and multiplying by $3403/4563$ gives $83Q/6084$. Haar

conjugation preserves traces on $E$. The exact dimensions and trace contributions are

| top | $d$ | $4680W$ | top | $d$ | $4680W$ |
|---|---|---|---|---|---|
| $x_0$ | 1 | 0 | $y_0$ | 1 | 7800 |
| $x_1$ | 1 | 600 | $y_1$ | 7 | 36960 |
| $x_2$ | 5 | 10440 | $y_2$ | 15 | 110880 |
| $x_3$ | 9 | 56640 | $y_3$ | 2 | 4080 |
| $x_4$ | 2 | 5520 | $y_4$ | 14 | 81480 |
| $x_5$ | 14 | 148560 | $y_5$ | 6 | 18240 |
| $x_6$ | 6 | 54240 | $y_6$ | 2 | 4920 |

For clarity, the lowest core weights and all minimizing absolute charges used in this table are

| | $h_{A_L}$ | $\lvert j\rvert$ | | $h_{A_L}$ | $\lvert j\rvert$ |
|---|---|---|---|---|---|
| $x_0$ | 0 | 0 | $y_0$ | 2/5 | 0 |
| $x_1$ | 2/15 | 0 | $y_1$ | 8/15 | 0, 5 |
| $x_2$ | 2/5 | 0, 3 | $y_2$ | 4/5 | 0, 2, 3, 5 |
| $x_3$ | 4/5 | 0, 2, 3, 6 | $y_3$ | 1/5 | 2 |
| $x_4$ | 1/3 | 6 | $y_4$ | 11/15 | 1, 2, 4, 6 |
| $x_5$ | 1 | 1, 4, 6 | $y_5$ | 2/5 | 1, 4 |
| $x_6$ | 4/5 | 4 | $y_6$ | 1/5 | 4 |

For each positive $j$ both signs occur. A convenient independent check on every row is

$$Z = h_{A_L} d - L - Q/156 - P, \qquad W = \frac{L}{26} + \frac{7Q}{936} + \frac{125P}{39} + \frac{25}{26} h_{A_L} d.$$

Complete right $\mathfrak{sl}_2$ characters imply $y_2 \geq x_5$ and $y_4 \geq x_5 + x_6$: these compare charges $10, 12$ and charges $7, 9$, respectively, with the common left doublet removed in the second comparison. Direct comparison of the displayed table gives

$$\begin{aligned} 4680(\tfrac{17}{10} D - W_{\text{total}}) = {} & 8460(y_2 - x_5) + 28716(y_4 - x_5 - x_6) \\ & + 7956x_0 + 7356x_1 + 29340x_2 + 14964x_3 + 10392x_4 \\ & + 22212x_6 + 156y_0 + 18732y_1 + 11832y_3 \\ & + 1188y_4 + 29496y_5 + 10992y_6 \geq 0. \end{aligned} \tag{111}$$

Since $W_{\text{total}} = hD$, this proves $0 \leq h \leq 17/10 < 2$ for *every* simple $U$-module, without assuming it is unitarizable.

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**REVIEW — R2: scope of the core and finite-top certificates**

The charge minimizations and the nonnegative identity (111) are specified above; their finite exact check is `hi_cft_20260923_next_structure_order13_top_check.json`. The embedded lattice–Potts core and the spectral field witness remain construction inputs, supplied by the exact core-extension certificate. The finite-top irreducibility below is likewise imported from exact field actions, particularly `hi_cft_20260923_next_81top_all.json`; it should not be replaced by numerical cyclicity alone.

---

The strongly rational restriction $B \downarrow_U$ is semisimple. Its simple summands have integer lowest weights, hence only weights $0, 1$ are possible. The vacuum occurs once, while the complementary current space has dimension $86 - 6 = 80$. Its certified irreducible zero-mode action therefore implies

$$B \downarrow_U = U \oplus X, \qquad h_X = 1, \qquad \dim X_{\text{top}} = 80. \tag{112}$$

The four mixed nontrivial $E_6 \oplus A_2$ lattice sectors start at weight 1, with 81-dimensional tops. Their next possible top weight would be 2, excluded by (111). Their exact top-action certificates use a one-dimensional Cartan eigenline, positive-metric adjoint identities, and a cyclic span of dimension 81. The adjoint identities make the generated operator algebra a star algebra; orthogonal complements are therefore invariant, and the unique eigenline plus cyclicity proves irreducibility.

Write the actual lattice extension as $H = B \oplus J \oplus J^2$. Its restriction is consequently

$$H \downarrow_U = \mathbf{1} \oplus X \oplus Y \oplus Y^*, \qquad h_Y = 1, \quad \dim Y_{\mathrm{top}} = 81. \tag{113}$$

At this stage $Y \cong Y^*$ has not been assumed. The three types $\mathbf{1}, X, Y$ are distinguished by their lowest spaces.

## G.2 The rank-four or rank-six alternative

Put $\mathcal{C} = \mathrm{Rep}(U)$ and $\mathcal{F} = \mathcal{C}_H$, the category of *all* right modules over the extension algebra $H$. The strict positivity of nonvacuum lowest weights gives the positive spherical structure through character asymptotics. Since $H$ is holomorphic, it is Lagrangian and the center correspondence [39, Section 4] gives $\mathcal{C} \simeq Z(\mathcal{F})$. Write $s = \mathrm{FPdim}_{\mathcal{C}} B$. Extension dimension identities give

$$d_X = s - 1, \qquad d_Y = d_{Y^*} = s, \qquad \mathrm{Dim}\,\mathcal{C} = 9s^2, \quad \mathrm{Dim}\,\mathcal{F} = 3s.$$

Pseudo-unitarity of $\mathcal{F}$ follows from that of its center and the squared global-dimension and Frobenius–Perron-dimension identities. No $C^*$ structure is needed in this section. All categorical dimensions below use the positive spherical structure and therefore equal Frobenius–Perron dimensions.

For center induction $I$ and forgetful functor $F$, one has $H = I(\mathbf{1})$ and $FI(\mathbf{1}) = \bigoplus_{a \in \mathrm{Irr}\,\mathcal{F}} a \otimes a^*$. Adjunction therefore gives

$$\mathrm{rank}\,\mathcal{F} = \dim \mathrm{End}_{\mathcal{C}}(H) = \begin{cases} 4, & Y \ncong Y^*, \\ 6, & Y \cong Y^*. \end{cases} \tag{114}$$

**Lemma G.1** (Automorphisms give invertible boundary objects)**.** *For a pseudo-unitary fusion category $\mathcal{F}$, algebra automorphisms of its canonical Lagrangian algebra $I(\mathbf{1})$ inject into* $\mathrm{Inv}(\mathcal{F})$.

*Proof.* Normalize the special Frobenius structure so that $m\Delta = \mathrm{id}$ and set $f * g = m(f \otimes g)\Delta$. If $f$ is an algebra automorphism, then $f * f = f$, and the same holds for its composition inverse. Under the two-product description [17, Lemma 3.1 and Corollary 3.4], composition is fusion-ring multiplication and the primitive convolution idempotents are $d_a[a]$. Thus both $f$ and $f^{-1}$ have expansions with coefficients 0 or $d_a > 0$. Their fusion product is $[\mathbf{1}]$. Nonnegative fusion coefficients rule out cancellation. Every pair of objects in the two supports must therefore multiply to copies of the unit; rigidity then forces each pair to consist of inverse invertible objects. The two supports are singletons and their dimensions are one. Composition preserves this assignment, proving the injection. □

The actual order-three automorphisms of $H/B$ consequently inject into $\mathrm{Inv}(\mathcal{F})$. This injection is already spherical: the proof used positivity of dimensions and fusion coefficients, without using adjoints of morphisms.

**Lemma G.2** (No sign extension)**.** *The nonvacuum summand $X$ of $B \downarrow_U = U \oplus X$ is not invertible.*

*Proof.* If $X$ were invertible, self-duality of $B$ and connectedness would give $X^2 = \mathbf{1}$. The extension would have its $\mathbb{Z}/2$ grading, and its sign automorphism $g$ would satisfy $B^g = U$. On currents this would give $(E_6 \oplus A_2)^g = U_1 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, of rank two.

Every finite-order automorphism of a complex simple Lie algebra is conjugate to a toral automorphism times a diagram automorphism; see [18, Section 5.2]. It therefore fixes a toral subalgebra whose dimension is the number of orbits of the diagram automorphism on simple roots. In type $E_6$ this number is at least four, and in type $A_2$ at least one. Since the nonisomorphic ideals $E_6, A_2$ are preserved separately, $(E_6 \oplus A_2)^g$ contains a toral subalgebra of dimension at least five. Such a subalgebra cannot lie in the rank-two Lie algebra $U_1$, a contradiction. □

If rank six had six invertibles, then $3s = 6$ and $d_X = s - 1 = 1$, contradicting Lemma G.2. Thus there are exactly three invertibles in either rank. Their associator is also determined: the local $B$-module category has metric group

$$(\mathbb{Z}/3)^2, \qquad q(a,b) = \frac{2a^2 + b^2}{3} \pmod 1.$$

The actual extension $H/B$ selects the diagonal isotropic line $L_+ = \{(a,a)\}$. Its complement $L_- = \{(a,-a)\}$ is also isotropic, and $(\mathbb{Z}/3)^2 = L_+ \oplus L_-$. The pointed subcategory on $L_-$ is symmetric with trivial twists, hence Tannakian; as a tensor category it is the untwisted $\mathrm{Vec}_{\mathbb{Z}/3}$. Free $H/B$-module induction is tensor on this subcategory. It is fully faithful since

$$\mathrm{Hom}_{H/B}(x \otimes (H/B), y \otimes (H/B)) \cong \mathrm{Hom}_{\mathrm{Rep}(B)}(x, y \otimes (H/B))$$

has dimension $\delta_{x,y}$ for $x, y \in L_-$: distinct elements of $L_-$ represent distinct cosets of $L_+$. Every free module is a direct sum of these three simple modules, and separability makes every module a summand of a free module. Thus induction identifies the entire all-module category for $H/B$ with this untwisted pointed category. Viewing these modules in $\mathrm{Rep}(U)$ forgets only the locality condition for the intermediate algebra $B$; it preserves $H$-module morphisms and relative tensor products. This gives the required full tensor embedding into $\mathcal{F}$. Merely knowing the group of invertibles would not determine this associator.

## G.3 Excluding rank four

At rank four the fusion ring is near-group:

$$a^3 = 1, \qquad a\rho = \rho a = \rho, \qquad \rho^2 = 1 + a + a^2 + m\rho.$$

There are exactly two self-dual simples, $\mathbf{1}$ and $\rho$. Consequently [19, Theorem 1.1], which assumes pseudo-unitarity, restricts $m$ to $0, 2, 3, 6$. For $m = 0$, $s = 2$ contradicts Lemma G.2. For $m = 2$, the categorical reduction [20, Theorem 7.1 and Corollary 7.4] supplies the twisted $A_4$ double. Its electric regular algebra gives a holomorphic $c = 8$ extension $H'$ with $U = (H')^{A_4}$ by the extension correspondence [40]. The nonvacuum lowest weights are positive, so $H'$ is of CFT type, and [41, Theorem 1] identifies it with $V_{E_8}$. For completeness, the nontrivial order-two fixed Lie algebras are $D_8$ and $E_7 + A_1$, with dimensions $120, 136$ and adjoint traces $-8, 24$. At order three the possibilities are $A_8$, $E_6 + A_2$, $D_7 + \mathbb{C}$, and $E_7 + \mathbb{C}$, with dimensions $80, 86, 92, 134$ and traces $-4, 5, 14, 77$. These lists follow by selecting Kac coordinates of total mark two or three in the affine $E_8$ diagram, whose marks are $1, 2, 2, 3, 3, 4, 4, 5, 6$. For order three the trace is $(3 \dim \mathfrak{g}^g - 248)/2$, since the two nontrivial eigenspaces are dual. Thus the two relevant traces are bounded below by $-8$ and $-4$. Averaging over $A_4$ gives

$$\dim(E_8^{A_4}) \geq \frac{248 + 3(-8) + 8(-4)}{12} = 16,$$

contradicting $\dim U_1 = 6$.

The remaining cases require a spherical induction argument before any comparison with affine characters. We give this reduction explicitly.

**Lemma G.3** (Near-group induction)**.** *Suppose $m = 3k$, and put $d^2 = 3 + 3kd$, $s = 2 + kd$. There are integers $p, q \geq 0$, $p + q = k$, and a primitive cube root $\zeta$ such that the known center objects have forgetful images*

| *object* | $F(\text{object})$ | *twist* |
|---|---|---|
| $X$ | $1 + k\rho$ | $1$ |
| $Y, Y'$ | $1 + a + k\rho,\ 1 + a^2 + k\rho$ | $1, 1$ |
| $C$ | $a + a^2 + k\rho$ | $\zeta$ |
| $D, D'$ | $a + p\rho,\ a^2 + p\rho$ | $\zeta^2, \zeta^2$ |
| $E, E'$ | $a + q\rho,\ a^2 + q\rho$ | $\zeta^2, \zeta^2$ |

*Every further simple $Z_j$ forgets to $n_j\rho$, $n_j > 0$. Its twist $\theta_j$ satisfies*

$$\sum_j n_j^2 = 6 + 3k^2 + 4pq, \tag{115}$$

$$S_1 := \sum_j n_j^2\theta_j = k^2 - kd - (p-q)^2\zeta^2,$$

$$\sum_j n_j^2\theta_j^2 = \overline{S_1} + \varepsilon\sqrt{9k^2 + 12}, \qquad \varepsilon = \nu_2(\rho) \in \{1, -1\}. \tag{116}$$

*Proof.* The four one-dimensional representations of the fusion algebra have formal codegrees $3s, 9s/d^2, 3, 3$. Thus induction of the unit is $1 + X + Y + Y'$, with dimensions $1, d^2/3, s, s$ and trivial twists. Since $FI(1) = 4 \cdot 1 + a + a^2 + 3k\rho$, adjunction and comparison of rational and irrational dimension coefficients give the first two rows. Also

$$FI(a) = 1 + 4a + a^2 + 3k\rho, \qquad FI(a^2) = 1 + a + 4a^2 + 3k\rho.$$

The two inductions have norm four, share one constituent, and contain $Y, Y'$ respectively. Their remaining norm-three parts are therefore multiplicity-free, with a single common constituent $C$. Write its $\rho$ multiplicity as $c$ and the other two as $p, q$. Then $c + p + q = 2k$.

The untwisted pointed subcategory gives $\nu_3(a) = 1$, so positivity in the twist-trace formula forces every constituent of $I(a)$ to have cube-root twist. Its first indicator, separated over the disjoint fields $\mathbb{Q}(d)$ and $\mathbb{Q}(\zeta)$, gives

$$2 + 2\theta_C + \theta_D + \theta_E = 0, \qquad k + c\theta_C + p\theta_D + q\theta_E = 0.$$

The first equality forces $\theta_C = \zeta$ and $\theta_D = \theta_E = \zeta^2$, up to simultaneous conjugation; the second gives $c = k$ and $p + q = k$. Ribbon duality preserves the primed twists. Subtracting the nine known squared dimensions from $9s^2$ gives (115). The first two twist traces of $I(\rho)$ give (116), using $3/d = d - 3k$ and $3s/d = \sqrt{9k^2 + 12}$. □

**Proposition G.4** (The multiplicity-three modular representation)**.** *For $k = 1$, the center has the modular data of $G_{2,3} \otimes A_{2,1}$ or their simultaneous reversal.*

*Proof.* One of $p, q$ is zero. The corresponding $D, D'$ are invertible and generate a nondegenerate pointed category $\mathcal{P}$ with twists $1, \zeta^2, \zeta^2$. Write $Z(\mathcal{F}) = \mathcal{M} \boxtimes \mathcal{P}$ using its modular centralizer. The known objects form three $\mathcal{P}$-orbits; representatives $1, X, C$ in $\mathcal{M}$ have dimensions $1, 1+d, s$ and twists $1, 1, \zeta$. Balancing verifies their trivial monodromy with $D$ from $DX = E$ and

$DC \in \{Y, Y'\}$. The remaining free orbits, together with (115), leave three simples in $\mathcal{M}$, all of dimension $d$.

Take $\zeta = e^{2\pi i/3}$, the other choice conjugating the answer. For their twists $z_1, z_2, z_3$, the two moments become

$$\sum_i z_i = \alpha := \frac{-1 - i\sqrt{7}}{2}, \qquad \sum_i z_i^2 = -\frac{1}{2} - i(\varepsilon - \tfrac{1}{2})\sqrt{7}.$$

The second sum would have absolute value four for $\varepsilon = -1$, so $\varepsilon = 1$. Since $\alpha^2 + \alpha + 2 = 0$, Newton's identity and the unit modulus of all $z_i$ give elementary symmetric functions $e_2 = \overline{\alpha}$, $e_3 = 1$. Thus the twists are the roots of $t^3 - \alpha t^2 + \overline{\alpha} t - 1$, namely $\zeta_7^3, \zeta_7^5, \zeta_7^6$.

The actual extension algebra $1 + X$ lies in $\mathcal{M}$. Its local module category has dimension three and is the pointed category with nontrivial twist $\zeta$, fixed by its multiplicative central charge. Its two nonvacuum simples must both restrict to $C$: each underlying object has dimension $s$, cannot contain the unit by induction reciprocity, and no other combination of the six known dimensions has these properties. Hence the branching columns are $1 + X, C, C$. Modular intertwining for this extension [21, Theorems 4.1 and 4.5] fixes the first three rows of $\widehat{S} = \sqrt{3} s S_{\mathcal{M}}$:

$$\begin{pmatrix} 1 & 1+d & s & d & d & d \\ 1+d & 1 & s & -d & -d & -d \\ s & s & -s & 0 & 0 & 0 \end{pmatrix}.$$

All six objects are self-dual, the last three by their distinct twists. Their fusion matrices are real symmetric, so Verlinde diagonalization and the positive vacuum row make $S_{\mathcal{M}}$ real. For the remaining block $K$, orthogonality and the modular relation yield

$$\sum_j K_{ij} = d, \qquad \sum_j K_{ij} z_j = d + i\sqrt{3} s\, z_i^{-1}.$$

These determine each real row: three distinct points on a circle are not collinear. The affine $G_{2,3}$ data [22, Table 4 and equation (3.8)] realize these rows and twists in the reversed orientation; conjugating gives the other possibility. Thus the entire representation, rather than merely its twist list, is forced. □

**Lemma G.5** (The balanced multiplicity-six pattern is impossible)**.** *In Lemma G.3 with $k = 2$, one has $\{p, q\} = \{0, 2\}$.*

*Proof.* Suppose $p = q = 1$, and put $d = 3 + 2\sqrt{3}$, $r = 1 + d$. The known dimensions are $1$, $1 + 2d$, three copies of $2r$, and four copies of $r$. The remaining dimensions are $n_j d$, with $\sum_j n_j^2 = 22$. Write $u = (2 - \sqrt{3})/12$, $v = (2 + \sqrt{3})/12$. The normalized vacuum row is

$$(u, v, 1/3, 1/3, 1/3, 1/6, 1/6, 1/6, 1/6, n_j\sqrt{3}/12).$$

Signed Galois symmetry [23, Theorem II(iii)], for an automorphism reversing $\sqrt{3}$, sends this row to the $X$ row: its first entry becomes $v > 0$, uniquely identifying $X$ and a positive sign. The forced row is

$$(v, u, 1/3, 1/3, 1/3, 1/6, 1/6, 1/6, 1/6, -n_j\sqrt{3}/12).$$

The object $X$ is self-dual because its dimension is unique. Verlinde's formula consequently gives

$$N_{XX}^X = \frac{v^3}{u} + \frac{u^3}{v} + \frac{3}{9} + \frac{4}{36} - \frac{22}{48} = \frac{97}{72} + \frac{32}{72} - \frac{33}{72} = \frac{4}{3},$$

contradicting integrality. Here $u + v = 1/3$, $uv = 1/144$ verify the first two terms directly. □

**Lemma G.6** (A universal multiplicity-six character projection)**.** *The unbalanced case has a nondegenerate pointed factor $\mathcal{P}$ with twists $1, \zeta^2, \zeta^2$. In its modular centralizer there is a character projection with matrices*

$$S_v = -\frac{1}{2}\begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & -1 \\ 1 & -2 & 1 \end{pmatrix}, \qquad T_v = \operatorname{diag}(1, -i, -1), \tag{117}$$

*before the common central-charge scalar, or the conjugate matrices.*

*Proof.* The zero multiplicity gives the pointed factor just as for $k = 1$. In the centralizer the known simples have dimensions $1, 2d+1, 2d+2$ and twists $1, 1, \zeta$; all others have dimensions $n_j d$, with $\sum n_j^2 = 6$. The Gauss sum and the second indicator give

$$\sum n_j^2 z_j = -2 - 4i, \qquad \sum n_j^2 z_j^2 = -2 + 4i(1-\varepsilon).$$

The second equality excludes $\varepsilon = -1$ by absolute values. Since $(1 + \Re z)(1 + \Im z) \geq 0$ on the unit circle, summing gives $6 - 2 - 4 + \frac{1}{2}\Im(-2) = 0$. Thus each $z_j$ is $-1$ or $-i$, with total squared multiplicity two or four, respectively.

Put

$$v_0 = \chi_1 - \chi_X, \qquad v_A = -\frac{1}{2}\sum_{z_j=-i} n_j \chi_j, \qquad v_B = -\sum_{z_j=-1} n_j \chi_j.$$

The vacuum row is $(u, v, 1/\sqrt{3}, n_j/4)$, with $u = (2\sqrt{3}-3)/12$, $v = (2\sqrt{3}+3)/12$; Galois symmetry gives the $X$ row $(v, u, 1/\sqrt{3}, -n_j/4)$. Subtracting these rows yields $Sv_0 = -v_0/2 - v_A - v_B/2$. All three combinations are fixed by duality. Applying $S^2 = 1$ and $STS = iT^{-1}ST^{-1}$ to $v_0$ then determines

$$Sv_A = -v_0/2 + v_B/2, \qquad Sv_B = -v_0/2 + v_A - v_B/2,$$

which is (117). This proof permits either one dimension-$2d$ object or four dimension-$d$ objects at twist $-i$; no unproved rank-nine classification has been used. □

**Lemma G.7** (Pole cancellation)**.** *Let $\sigma$ be a finite-image representation of $\mathrm{SL}_2(\mathbb{Z})$ with no invariant vector. Two weight-zero vector-valued modular functions, holomorphic on the upper half-plane, with representation $\sigma$ and equal principal parts at infinity are equal.*

*Proof.* Their difference has no pole at infinity. Covariance makes every component holomorphic at every cusp of the finite-index kernel of $\sigma$. It descends to a holomorphic function on the compactified modular curve and is constant. The resulting constant vector must be invariant under $\sigma$, hence zero. □

**Proposition G.8** (Current obstructions for the remaining near groups)**.** *Under the hypotheses above, $m = 3$ forces*

$$\dim U_1 = 15 + A + 10B \quad \text{or} \quad 14 + 42A, \qquad A, B \in \mathbb{Z}_{\geq 0},$$

*whereas $m = 6$ forces*

$$\dim U_1 = 8 + A \quad \text{or} \quad 5 + 12P, \qquad A, P \in \mathbb{Z}_{\geq 0}.$$

*In particular none allows $\dim U_1 = 6$.*

*Proof.* We give the theta normalizations, since a factor of two in the vacuum coordinate would change these obstructions. For $m = 3$, use the affine $G_{2,3}$ labels $1, 7, 27, 77, 14, 64$, and for each $A_{2,1}$ label put

$$v = (\chi_1 - \chi_{64}, -\chi_{77}, -\chi_{14}, -\chi_7).$$

The complement of the extension character is the even Fourier block on $\mathbb{Z}/7$, with entries

$$S_{00} = -1/\sqrt{7}, \quad S_{0r} = -2/\sqrt{7}, \quad S_{r0} = -1/\sqrt{7}, \quad S_{rs} = -2\cos(4\pi rs/7)/\sqrt{7},$$

and twist exponents $r^2/7$, $0 \le r \le 3$. Together with the even $A_2$ block, multiplication by $\eta^8$ identifies it with the Weil representation of a positive rank-four lattice with discriminant form $\mathbb{Z}_7(r^2/7) \oplus \mathbb{Z}_3(s^2/3)$. The actual extension gives $\chi_1 + \chi_{64} = \chi_B$. Thus a difference of two such character vectors has vacuum projection $\delta v_0 = 2\delta\chi_U$.

Take $L = L_7 \oplus A_2$, where $L_7$ has Gram matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 4 \end{smallmatrix}\right)$. For a basis $u, v$ of $L_7$, write $w = v - u/2$ and $x = pu + tw$. Set $a = p^2$, $b = 7t^2/4$, and $c = (y, y)/2$ on the $A_2$ factor. The quadratics $H_1 = a - b$ and $H_2 = a + b - c$ are harmonic. Poisson summation gives vector-valued theta series of weight four. Their required coefficients are

| $(r, s)$ | $q$-power | $\Theta_{H_1}$ | $\Theta_{H_2}$ |
|---|---|---|---|
| $(1, 0)$ | $1/7$ | $-1/7$ | $1/7$ |
| $(3, 0)$ | $2/7$ | $3/7$ | $4/7$ |
| $(0, 0)$ | $1$ | $2$ | $-4$ |

The first row comes from $2w/7$, the second from the two shortest vectors with $a = 1/4, b = 1/28$, and the last from two $L_7$ roots and six $A_2$ roots. The only nonvacuum polar top dimensions are $A$ at $2/7$ and $B$ at $1/7$, with zero allowed for an integer-shifted top. Subtract the affine reference, which has 22 currents and $A = 7, B = 0$. Pole matching gives

$$\eta^8 \delta v = \lambda\Theta_{H_1} + \mu\Theta_{H_2}, \qquad \lambda = 7 - A + 4B, \quad \mu = 7 - A - 3B.$$

This identity follows from Lemma G.7: after division by $\eta^8$, the remainder has no poles; its only possible $T$-fixed line is $(r, s) = (0, 1)$, which the Fourier $S$ matrix does not preserve. The vacuum coefficient therefore gives $2(\dim U_1 - 22) = 2\lambda - 4\mu$, the first formula.

For the reverse orientation, realize the negative discriminant form by the positive lattice with Gram matrix

$$G_7^- = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 6 \end{pmatrix}.$$

Its dual generator $\ell = (16, 11, 6, 1)/21$ has energy $8/21$; $(r, s) \mapsto (9r + 7s)\ell$ identifies the discriminant form. Project the theta series to functions even separately in $r, s$. Let $H = \|x_{A_3}\|^2 - 3\|x_{A_3^\perp}\|^2$ and $F = -6D_2\Theta$, where $D_2 = q\,d/dq - E_2/6$ is the modular derivative. The necessary coefficients after projection are

| $(r, s)$ | $q$-power | $F$ | $\Theta_H$ |
|---|---|---|---|
| $(0, 0)$ | $0$ | $1$ | $0$ |
| $(0, 0)$ | $1$ | $-84$ | $24$ |
| $(2, 1)$ | $2/21$ | $3/14$ | $-2/7$ |

There are twelve roots, all in $A_3$. The polar coset has shortest vector $(1, 2, 3, 4)/21$, perpendicular to $A_3$; the separately even projection halves its coefficient, since the paired coset has no vector of the same norm. If $A$ is its actual polar top dimension, matching the coefficient $-A$ gives $\eta^8 v = F + (3 + 14A)\Theta_H/4$. Here $T$ has no fixed vector, so pole cancellation again

proves equality. Since $\eta^8\chi_B = 1 + 78q + \cdots$, its vacuum coefficient is $2\dim U_1 - 94$. Thus $2\dim U_1 - 94 = -84 + 6(3 + 14A)$, giving $14 + 42A$.

For $m = 6$, tensor the projection in Lemma G.6 with its pointed factor and first take the orientation with twists $1, i, -1$. After multiplication by $\eta^8$ this is the Weil representation for $L = A_1 \oplus A_1 \oplus A_2$, restricted by interchange of the two $A_1$ factors and conjugation of the $A_2$ charge. The harmonic quadratic $H = \|x_{A_1^2}\|^2/2 - \|y_{A_2}\|^2/2$ has coefficients

$$[q^{1/4}]\Theta_{H,(10,0)} = \frac{1}{2}, \qquad [q]\Theta_{H,(00,0)} = -2.$$

The first counts the two half-roots in a nontrivial $A_1$ coset; the second counts four $A_1$ roots and six $A_2$ roots. Let $A = \sum n_j \dim(M_j)_{\mathrm{top}}$ over the quarter-twist polar modules. This is a nonnegative integer, and its projected polar coefficient is $-A/2$. Use as reference the order-three simple-current extension of $A_{2,9}$ tensored with $A_{2,1}$: it has 16 currents and $A = 8$. Its projected representation agrees with (117); its matrices are also given in [24, Lemma 3.4 and Proposition 3.5]. Thus cancellation gives

$$\eta^8\delta v = (8 - A)\Theta_H, \qquad 2(\dim U_1 - 16) = -2(8 - A).$$

As in the first orientation above, the sole $T$-fixed coordinate is not $S$-invariant. This gives $\dim U_1 = 8 + A$.

For the reverse orientation use the positive lattice with Gram matrix

$$G_6^- = \begin{pmatrix} 2 & -1 & -1 & -1 \\ -1 & 2 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ -1 & 0 & 0 & 4 \end{pmatrix}.$$

Its first three basis vectors span $A_3$; if the basis is $a, b, c, e$, then $w = e + a + (b + c)/2$ is perpendicular to $A_3$ and has norm three. The discriminant generators $(b + e)/2, (c + e)/2, 2w/3$ have quadratic values $3/4, 3/4, 2/3$ and mutually integral pairings. The polar coset is represented by $-w/3$, with energy $1/6$. This shortest vector is unique, and the projection does not halve its coefficient. With $H = \|x_{A_3}\|^2 - 3\|x_{A_3^\perp}\|^2$ and $F = -6D_2\Theta$,

| coordinate | $q$-power | $F$ | $\Theta_H$ |
|---|---|---|---|
| vacuum | 0 | 1 | 0 |
| vacuum | 1 | $-84$ | 24 |
| polar | 1/6 | 0 | $-1$ |

Let $P$ be the weighted top dimension in the half-twist coordinate with nonvacuum pointed charge; the weighting is by the integers $n_j$. Contragredience equates the two pointed-charge sums, so averaging does not change this integer. Pole cancellation gives $\eta^8 v = F + P\Theta_H$: there is no $T$-fixed coordinate in this orientation. Comparing the vacuum coefficient yields $2\dim U_1 - 94 = -84 + 24P$, hence $\dim U_1 = 5 + 12P$. □

The representations used in the last proof have finite image because they are Weil representations, with the finite-order eta multiplier removed. Thus the proof does not assume a dimension formula for a space of vector-valued modular forms. The coefficients $A, B, P$ are weighted dimensions of actual polar tops, and distinct modules need not have linearly independent characters.

**Remark G.9** (Scope of the rank-four exclusion)**.** The universal induction, balanced-pattern obstruction, weighted projection, and theta coefficient normalizations are printed above. Their

source calculations are `hi_cft_20260923_next_m3_universal_data.md`, `hi_cft_20260923_next_structure_m6_universal_projection.md`, `hi_cft_20260923_next_structure_m3_character_obstruction.md`, and `hi_cft_20260923_next_structure_m6_character_obstruction.md`. Lemma G.2 removes the formerly used pointwise stabilizer calculation $\mathrm{Aut}(B/U) = 1$ from this proof. No classification restricted to $C^*$ categories is used here.

### G.4 The rank-six ring and exclusion of higher multiplicity

Rank four being excluded, (114) gives $H = \mathbf{1} \oplus X \oplus 2Y$. The composition algebra of $\mathrm{End}_{\mathcal{C}}(H)$, identified with the complex fusion algebra by [17, Corollary 3.4], is

$$K_0(\mathcal{F}) \otimes \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}),$$

In particular it is noncommutative. Let $a$ generate the three invertibles. A permutation of order three on three noninvertibles is either transitive or trivial. If the left action were trivial, duality would make the right action trivial as well. The two nontrivial character idempotents of $\mathbb{C}[\mathbb{Z}/3]$ would then be one-dimensional direct ideals. Their four-dimensional complementary ideal would have to be $M_2(\mathbb{C})$, but the dimension character is nonzero on that ideal, a contradiction.

Choose a noninvertible $\rho$. The right action commutes with the free left cyclic action, so it is either the same cycle or its inverse. The same cycle would make the fusion ring generated by two commuting elements $a, \rho$, which is impossible. Thus $\rho a = a^2\rho$. If $\rho^* = a^k\rho$, then $(a^i\rho)^* = a^{i+k}\rho$; involutivity gives $2k = 0$ in $\mathbb{Z}/3$, hence $k = 0$. All three noninvertibles are self-dual. Frobenius reciprocity gives one copy of the unit and no other invertible in $\rho^2$. Write its remaining coefficients as $m_0, m_1, m_2$. The identity $\rho^2 a = a\rho^2$ cyclically equates these coefficients, so

$$a^3 = 1, \qquad \rho a = a^2\rho, \qquad \rho^2 = 1 + m(\rho + a\rho + a^2\rho), \qquad m \geq 1. \tag{118}$$

The following induction calculation determines the multiplicity. Put

$$d = \frac{3m + \sqrt{9m^2+4}}{2}, \quad s = 1 + d^2, \quad \Delta = 9m^2 + 4, \quad R = \rho + a\rho + a^2\rho.$$

**Lemma G.10** (Center induction and two moments)**.** *For a pseudo-unitary categorification of* (118) *whose pointed subcategory is untwisted, center induction satisfies*

$$I(1) = 1 + X + 2Y, \quad d_X = d^2, \quad d_Y = s, \qquad I(a) = I(a^2) = C_0 + C_1 + C_2,$$

*where $\theta_X = \theta_Y = 1$, $d_{C_j} = s$, and the $C_j$ twists are $1, \zeta_3, \zeta_3^2$. Every other central simple $Z_j$ satisfies $F(Z_j) = n_j R$, with $n_j \in \mathbb{Z}_{>0}$, and*

$$\sum_j n_j^2 = 4m^2 + 2,$$

$$\sum_j n_j^2 \theta_j = -\frac{m^2 + m\sqrt{\Delta}}{2}, \tag{119}$$

$$\sum_j n_j^2 \theta_j^2 = \frac{-m^2 + (2\varepsilon - m)\sqrt{\Delta}}{2}, \qquad \varepsilon = \nu_2(\rho) \in \{1, -1\}. \tag{120}$$

*Proof.* The complex fusion algebra has two characters $a \mapsto 1$, $\rho \mapsto d, -d^{-1}$, and the two-dimensional representation

$$a \longmapsto \begin{pmatrix} \zeta_3 & 0 \\ 0 & \zeta_3^2 \end{pmatrix}, \qquad \rho \longmapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Their formal codegrees are respectively $3s, 3s/d^2, 3$. The induction–formal-codegree correspondence [25, Theorems 2.5 and 2.13] gives $I(1) = 1 + X + 2Y$, with the stated dimensions and trivial twists. Conjugation by $a$ cycles the three noninvertibles, so their multiplicities in the forgetful image of a central object coincide. Since

$$FI(1) = 6 \cdot 1 + 3mR,$$

adjunction and dimensions give $F(X) = 1 + mR$ and $F(Y) = 2 \cdot 1 + mR$.

Similarly,

$$FI(a) = FI(a^2) = 3a + 3a^2 + 3mR.$$

Thus the two induced objects have self-inner-products and mutual inner-product equal to three. They are equal, multiplicity-free sums of three simples $C_j$, disjoint from $I(1)$. Adjunction gives $F(C_j) = a + a^2 + b_j R$, where $b_j \geq 0$ and $\sum_j b_j = 3m$. Untwistedness gives $\nu_3(a) = 1$. The center-twist formula [26, Theorem 4.1] therefore reads

$$\sum_j d_{C_j} \theta_{C_j}^3 = \operatorname{Dim} \mathcal{F} = \sum_j d_{C_j}.$$

Positive dimensions and equality in the triangle inequality imply $\theta_{C_j}^3 = 1$. The first indicator of $a$ is zero, so

$$\sum_j (2 + 3b_j d)\theta_{C_j} = 0.$$

The real quadratic field $\mathbb{Q}(d)$ is disjoint from $\mathbb{Q}(\zeta_3)$. Equating its two coefficients shows that the three twists are $1, \zeta_3, \zeta_3^2$ once each and $b_j = m$ for all $j$. All invertible constituents have now been exhausted by induction of $1, a, a^2$; every remaining forgetful image is $n_j R$.

The center dimension identity is

$$9s^2 = 1 + d^4 + 4s^2 + 9d^2 \sum_j n_j^2,$$

which simplifies to the first asserted equation. Finally adjunction gives

$$I(\rho) = mX + mY + m(C_0 + C_1 + C_2) + \sum_j n_j Z_j.$$

Its first and second twist traces are $\operatorname{Dim} \mathcal{F}\, \nu_1(\rho) = 0$ and $\operatorname{Dim} \mathcal{F}\, \nu_2(\rho) = 3s\varepsilon$. Using $d^{-1} = d - 3m$ gives (119) and (120). □

The roots in these equations form a multiset of exactly $4m^2 + 2$ elements, with each $\theta_j$ repeated $n_j^2$ times. No distinctness of the twists has been assumed.

Write $\Delta = t^2 D$ with $D$ squarefree. Since $(3m)^2 < \Delta < (3m + 1)^2$, we have $D > 1$. Reduction modulo three and seven gives $D \equiv 1 \pmod 3$ and $7 \nmid D$: the latter uses that $m^2 \equiv 5 \pmod 7$ has no solution. Except when $D = 10$,

$$\frac{\varphi(2D)}{\sqrt{D}} \geq \frac{12}{\sqrt{13}}. \tag{121}$$

Indeed for odd $D$ the left side is the product of $(p-1)/\sqrt{p}$ over its prime factors. A single prime must be at least 13, and two distinct primes contribute at least $40/\sqrt{55} > 12/\sqrt{13}$; all further factors exceed one. For even $D$ there is an additional factor $\sqrt{2}$. A single odd prime is 5 (the exceptional $D = 10$) or at least 11, and two odd primes again give a larger product.

Apply [19, Theorem 1.3] to two copies of the root multiset in (119), to remove any half-integral coefficients. Together with (121) this gives

$$4m^2+2 \geq \frac{mt}{2}\varphi(2D) \geq \frac{6m\sqrt{9m^2+4}}{\sqrt{13}}.$$

For $m>1$ this is impossible, since

$$36m^2(9m^2+4)-13(4m^2+2)^2 = 4(m^2-1)(29m^2+13) > 0.$$

The exceptional squareclass requires the following equality statement.

**Lemma G.11** (The conductor-forty bound)**.** *If a multiset of $N$ roots of unity has sum $-a-b\sqrt{10}$, where $a,b\in\mathbb{Z}_{\geq 0}$, then $N\geq a+8b$. If equality holds and the sum of the squared roots lies in $\mathbb{Q}(\sqrt{10})$, that second sum is $a$.*

*Proof.* Choose a cyclotomic field containing all roots and $\mathbb{Q}(\sqrt{10})$. Take the union of the multiset over its Galois group $G$ fixing $\sqrt{10}$, and put $M=|G|$. This gives a multiset of $MN$ roots, with sum $-Ma-Mb\sqrt{10}$, whose multiplicities are constant on $G$-orbits. Multiplicities remain integral throughout.

If an orbit consists of primitive $n$th roots and $40\nmid n$, its sum is the Ramanujan sum $\mu(n)\in\{0,1,-1\}$ and its size is $\varphi(n)$. Its rational contribution therefore costs at least one root per unit. A negative contribution realizes equality only for the singleton orbit $\{-1\}$.

If $40\mid n$, the primitive roots split into two equal orbits, distinguished by the primitive quadratic character of conductor 40. Their sums are

$$\frac{\mu(n)\pm\tau_n(\chi_{40})}{2}.$$

Here $\mu(n)=0$, and the induced Gauss sum vanishes unless $n=40u$ with $u$ squarefree and coprime to 40. In that case

$$\tau_n(\chi_{40}) = \mu(u)\chi_{40}(u)\sqrt{40}.$$

This formula follows by separating the factors coprime to 40 in the Chinese remainder theorem; an extra power of a prime dividing the conductor gives zero. Thus an orbit contributes zero or $\pm\sqrt{10}$, at cost at least eight roots. Equality for a nonzero contribution requires $n=40$.

The rational and irrational contributions come from disjoint types of orbit, so $MN\geq Ma+8Mb$. Equality leaves only $Ma$ copies of $\{-1\}$ and $Mb$ copies of the negative quadratic orbit of primitive fortieth roots. The latter orbit has squared sum zero: reduction of its exponents modulo 20 gives every unit once, since $\chi_{40}(x+20)=-\chi_{40}(x)$, and the resulting sum is $\mu(20)=0$. The averaged squared sum is therefore $Ma$. By the additional hypothesis, Galois averaging multiplied the original second sum by $M$. Dividing by $M$ gives the assertion. □

In our application $m,t$ are even and $a=m^2/2$, $b=mt/2$. Lemma G.11 gives

$$4m^2+2 \geq \frac{m^2}{2}+4mt$$

For $m\geq 4$ one has $t/m>\sqrt{9/10}>15/16$, making the right side strictly larger than $17m^2/4>4m^2+2$. The remaining $m=t=2$ is the equality case, so the squared sum must be 2, whereas (120) gives $-2$ or $-2-4\sqrt{10}$. Thus $m=1$. The same argument, including the indispensable second-moment hypothesis, is recorded in `hi_cft_20260923_next_structure_rank6_indicators.md` and `hi_cft_20260923_next_rank6_root_audit.md`.

Consequently

$$d=\frac{3+\sqrt{13}}{2}, \qquad s=\frac{13+3\sqrt{13}}{2}, \qquad d_X=\frac{11+3\sqrt{13}}{2}. \tag{122}$$

**Lemma G.12** (The six residual twists)**.** *For $m = 1$, there are six residual central simples, each with $n_j = 1$. Their twists are the six primitive thirteenth roots with quadratic nonresidue exponents, and $\nu_2(\rho) = 1$.*

*Proof.* Equation (119) gives a multiset of six roots with sum $(-1 - \sqrt{13})/2$. Average it over a cyclotomic Galois group fixing $\sqrt{13}$, as in Lemma G.11. An orbit with nonzero $\sqrt{13}$ coefficient has coefficient of absolute value $1/2$ and at least six members. To see the equality condition, the induced quadratic Gauss-sum formula is nonzero only at primitive orders $13u$, with $u$ squarefree and coprime to 13. The orbit size is $6\varphi(u)$, so equality permits only $u = 1, 2$.

The entire available length is already needed for the irrational coefficient. Hence no rational-only orbit or cancellation can occur. The two possible negative quadratic orbits, of orders 13 and 26, have sums $(-1 - \sqrt{13})/2$ and $(1 - \sqrt{13})/2$, respectively. The required rational coefficient forces every orbit to be the former. Consequently every original root is one of the six nonresidue thirteenth roots. These six roots are linearly independent over $\mathbb{Q}$: a rational relation would give a polynomial of degree at most twelve, with zero constant term, divisible by $\Phi_{13}(t) = 1 + t + \cdots + t^{12}$, and hence zero. Comparison with the quadratic period shows that each root occurs once. As each object was repeated $n_j^2$ times, all six objects have $n_j = 1$ and distinct twists.

Multiplication by two sends nonresidues modulo thirteen to residues, so the second sum is $(-1 + \sqrt{13})/2$. Equation (120) now gives $\varepsilon = 1$. □

The center thus has twelve simples. Its dimensions and twist exponents are

| number | 1 | 1 | 4 | 6 |
|---|---|---|---|---|
| dimension | 1 | $d^2$ | $s$ | $3d$ |

$$0, 0, 0, 0, \tfrac{1}{3}, \tfrac{2}{3}, \tfrac{2}{13}, \tfrac{5}{13}, \tfrac{6}{13}, \tfrac{7}{13}, \tfrac{8}{13}, \tfrac{11}{13}.$$

The complete $S, T$ data follow from [27, Theorem A(2), equivalently Theorem 5.1]: this statement assumes a complex spherical untwisted HI category. The auxiliary metric group has prime order 13 and its negative quadratic Gauss sum fixes its isometry class. This identifies the standard Haagerup modular data. It does not by itself identify a braided category; that stronger conclusion is treated separately below and in the analytic section.

## G.5 An optional algebraic classification route

There is a second possible route from this fusion ring to an abstract ribbon identification. We isolate its extra input rather than folding an unreplayed classification into the preceding modular-data argument. Vercleyen [43, Section 4.4 and Table 9.1, p. 171] reports an exhaustive classification of complex pivotal multiplicity-free fusion categories of rank at most seven. Its list for $\mathrm{HI}(\mathbb{Z}_3)$ has four entries. Gauge equivalence there permits complex nonzero scalars and ring automorphisms; no unitarity assumption is imposed. The solution method uses exact polynomial and algebraic-number operations, with a separate numerical redundancy check. The complete notebooks and decision logs are available on request rather than included in the thesis [43, Sections 4.1.7–4.1.8 and 4.4.1]. We have not independently replayed that exhaustion calculation.

**Proposition G.13** (Ribbon identification with an additional classification input)**.** *Assume, in addition to the hypotheses used above, that the four-entry list in [43, Table 9.1] exhausts complex pivotal fusion categories with the* $\mathrm{HI}(\mathbb{Z}_3)$ *fusion ring. Then $\mathcal{F}$ is tensor equivalent to $\mathcal{H}_2$ or $\mathcal{H}_3$, and there is an abstract ribbon equivalence*

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3).$$

*In particular this modular category is abstractly unitarizable.*

*Proof.* Evans–Gannon [44, Theorem 3(ii)] construct four inequivalent pivotal categories with this ring. Two have noninvertible dimension $(3+\sqrt{13})/2$ and are the unitary categories $\mathcal{H}_2$ and $\mathcal{H}_3$; the other two have dimension $(3-\sqrt{13})/2$. Their theorem classifies the endomorphism realizations considered there; the additional exhaustion hypothesis is what makes this a complete list for the present abstract $\mathcal{F}$.

A pivotal structure on any categorification of this ring is unique when it exists. Indeed, the ratio of two such structures is a tensor automorphism of the identity. Write its scalar on a simple $Z$ as $t_Z$. Since $\rho^2$ contains both $\mathbf{1}$ and $\rho$, $t_\rho^2 = 1 = t_\rho$. It also contains $a\rho$ and $a^2\rho$, so their scalars are one, and $a \otimes \rho = a\rho$ gives $t_a = 1$. Thus all six scalars are one. The negative-dimensional examples cannot acquire a positive spherical structure by repivotalization. The positive spherical category $\mathcal{F}$ is therefore one of the two positive examples.

The categories $\mathcal{H}_2$ and $\mathcal{H}_3$ are Morita equivalent by [36, Theorem 1.4], so their centers are braided equivalent. Combining this with $\mathrm{Rep}(U) \simeq Z(\mathcal{F})$ gives the claimed braided equivalence. Both sides have the positive spherical structures, which are unique; the induced pivotal and twist structures are consequently preserved. This makes the equivalence ribbon. Transporting the unitary structure from the Haagerup center proves abstract unitarizability. □

This proposition does not establish positivity of the canonical VOA fusion pairings or the analytic realization of the transported adjoint. Those are separate complete-unitarity assertions addressed later. Neither this classification input nor the proposition is needed for the character computation that follows.

## H The complete character vector

Let $q = e^{2\pi i\tau}$. Character exponents are $h - c/24 = h - 1/3$. Using the preceding twist list and strict positivity of nonvacuum lowest weights, the only allowed nonvacuum pole is $\gamma q^{-7/39}$ in the twist-$2/13$ coordinate. Here $\gamma \in \mathbb{Z}_{\geq 0}$ is its top dimension if that module starts at $2/13$, and is zero if its lowest weight is shifted upward. The vacuum pole is $q^{-1/3}$ with coefficient one.

**Lemma H.1** (Uniqueness from the principal part)**.** *Two character vectors for this modular representation with the same $\gamma$ coincide in every weight.*

*Proof.* The modular representation has finite image by the congruence theorem [23, Theorem I]. A constant vector must be fixed by both $S$ and the *character* $T$ matrix, which includes $e^{-2\pi i/3}$. The only $T$-fixed coordinate is the object of categorical twist $e^{2\pi i/3}$. Its $S$ image has a nonzero vacuum component, so it cannot span an invariant line. There are therefore no invariant constants, and Lemma G.7 applies. □

Evans–Gannon [9, Sections 1.4 and 2.4, equation (2.7)] provide a fundamental matrix $\Xi$ whose vacuum and $d_3$ columns have precisely the two principal parts above. Thus, in their labeling,

$$\boldsymbol{\chi} = \Xi_{:,0} + \gamma \Xi_{:,d_3}, \qquad \dim U_1 = 6 + 13\gamma.$$

The actual six-current result forces $\gamma = 0$. Their recursive construction of $\Xi$ and the uniqueness lemma give the entire vector, not just a finite coefficient match. In particular,

$$\mathrm{ch}\, U = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots). \tag{123}$$

The lowest weights and top dimensions are

| label | $h$ | $\dim M_{\mathrm{top}}$ | label | $h$ | $\dim M_{\mathrm{top}}$ |
|---|---|---|---|---|---|
| 0 | 0 | 1 | $d_1$ | 6/13 | 7 |
| $b$ | 1 | 80 | $d_2$ | 11/13 | 42 |
| $a$ | 1 | 81 | $d_3$ | 15/13 | 119 |
| $c_0$ | 1 | 81 | $d_4$ | 5/13 | 5 |
| $c_1$ | 1/3 | 3 | $d_5$ | 7/13 | 13 |
| $c_2$ | 2/3 | 27 | $d_6$ | 8/13 | 14 |

In particular the final maximum is 15/13, improving the preliminary 17/10 bound. The proof record is `hi_cft_20260923_next_structure_conditional_characters.md`. Although this vector fixes every unflavoured homogeneous dimension, it does not determine the separate branching functions for the fourteen-module core $A_L$, and it was not used to prove properness.

**Corollary H.2** (The four unmixed lattice restrictions)**.** *The two lattice modules labelled* $(0,\pm1)\in(\mathbb{Z}/3)^2$ *restrict to the simple* $U$*-module* $c_1$*, and the two labelled* $(\pm1,0)$ *restrict to* $c_2$*.*

*Proof.* The first pair has lowest weight 1/3, top dimension three, and weights in $1/3+\mathbb{Z}_{\geq0}$. The next possible summand top, 4/3, exceeds the established maximum 15/13. The character list contains a unique simple of lowest weight 1/3, with top dimension three, so semisimplicity gives precisely one copy of $c_1$. The second pair has lowest weight 2/3, top dimension 27, and next possible top 5/3; the same argument gives $c_2$. No additional finite-top irreducibility calculation is required. □

# I From vacuum unitarity to complete unitarity

The exact strong-generation certificates concern the actual subVOA $U\subset B\subset H=V_{E_8}$. In particular, they establish $\dim U_1=6$ and $U\neq B$; they do not establish positivity on all modules or on fusion products. We now separate those questions. *Vacuum unitarity* is the positive invariant form inherited by the PCT-stable subVOA $U$. *Strong unitarity* means that all simple ordinary $U$-modules admit compatible positive invariant forms. *Complete unitarity* additionally includes positivity of the canonical Hermitian fusion forms, giving the VOA representation category its unitary modular tensor structure. An abstract ribbon equivalence to a unitary category does not, by itself, prove this last positivity statement.

We use the preceding algebraic analysis as input: regularity of $U$, the actual restrictions

$$B{\downarrow_U}=U\oplus X,\qquad H{\downarrow_U}=U\oplus X\oplus Y^{\oplus2},\qquad h_X=h_Y=1, \tag{124}$$

the twelve simple algebraic $U$-modules, and the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring of the all-module category for the original extension. Every use of these inputs inherits the qualifications attached to their preceding proofs. Agreement of modular data alone is never used as a classification of tensor categories.

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**REVIEW — Analytic comparison**

The independent sector count and linear reconstruction resolve the earlier obligations A2 and A3. A separate pairwise comparison of bounded annulus insertions resolves A1 and supplies the positivity hypothesis. It uses a common localizer for each pair; a global choice of lifts for all individual annuli is unnecessary. Complete unitarity and the specified VOA extensions then follow from the same construction inputs. The net category also has a proof independent of A1. These arguments are mathematical deductions for review, not an assertion of independent expert validation.

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## I.1 The subnet and its sector count

There are two net constructions in use. Write $\mathcal{N}_V$ for the net generated by the closed smeared fields of a strongly local VOA $V$, and temporarily write $\mathcal{A}_V^{\mathrm{bl}}$ for the net generated by Tener's bounded localized insertions. For $V = U, B, H$ both exist: strong locality passes to unitary subalgebras, as do bounded localized vertex operators. The conformal vectors are the ambient ones, so the rotations and central charge agree in each construction.

First work with the bounded-insertion inclusion. Regularity of $U$ and Tener [29, Corollary 4.24] give

$$[\mathcal{A}_H^{\mathrm{bl}} : \mathcal{A}_U^{\mathrm{bl}}] < \infty.$$

The ambient net is the holomorphic lattice net, by Tener [28, Proposition 7.5]. Hence Longo [32, Theorem 24] makes $\mathcal{A}_U^{\mathrm{bl}}$ completely rational. Tener's Lemma 7.6 gives a vacuum net isomorphism $\mathcal{A}_U^{\mathrm{bl}} \cong \mathcal{N}_U$, using the ambient lattice comparison. It follows that $\mathcal{N}_U$ is completely rational as well. Only this isomorphism-invariant property is transferred here: we do not infer equality of the two concrete net constructions or an identification of their embedded extensions. In particular, this step does not assume strong unitarity of $U$.

The actual smeared-field inclusion $\mathcal{N}_U \subset \mathcal{N}_H$ will be used for the sector calculation. Its branching and index are established below, after the elementary domain lemma needed for reconstruction.

**Proposition I.1** (The actual net branching and sector count; A2)**.** *Assume the algebraic branching and rank-six ring deductions of Section G, including their certificate hypotheses. The restriction of the holomorphic $H$ net has precisely the three inequivalent irreducible $\mathcal{N}_U$ summands corresponding to (124), with multiplicities $1, 1, 2$. Its boundary fusion category $\mathcal{F}_{\mathrm{net}}$ has rank six, untwisted invertible subcategory $\mathrm{Vec}_{\mathbb{Z}/3}$, and the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring. Consequently*

$$\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{F}_{\mathrm{net}}), \qquad \#\,\mathrm{Irr}\,\mathrm{Rep}(\mathcal{N}_U) = 12.$$

*Proof.* Every field $v \in U$ satisfies polynomial energy bounds on $H$, since it is a field of the strongly local lattice VOA and the conformal Hamiltonian is unchanged. Its restriction to an algebraic unitary summand satisfies the same bound. Consequently the finite-energy spaces are cores for the smeared fields in the respective Hilbert completions. Apply Lemma I.2 below to the orthogonal summand projections. This proves that the algebraic decomposition of $H$ is a Hilbert direct-sum decomposition for the actual represented subnet. Conversely, a subnet intertwiner commutes with its local diffeomorphisms and hence with $L_0$. The equality of closed fields in reconstruction, or Fourier coefficients of their matrix elements, then makes its finite-energy restriction a $U$-module map. Thus simple summands stay irreducible and inequivalent, and the multiplicities remain $1, 1, 2$.

The inclusion is irreducible. To see the local assertion explicitly, let $x \in \mathcal{N}_H(I) \cap \mathcal{N}_U(I)'$. Locality makes $x$ commute with $\mathcal{N}_U(I')$ as well. Strong additivity of $\mathcal{N}_U$ shows that the algebras of $I$ and its complementary interval generate the global represented subnet: remove the two endpoints from any test interval and apply strong additivity twice. Thus $x$ commutes with the global subnet and, in particular, its local diffeomorphisms and common rotations. Its action on the vacuum lies in the one-dimensional zero-energy space of $H$, so the separating property of the vacuum makes it scalar. The precise finite-index input is now Kawahigashi–Longo [33, Proposition 2.3]: an irreducible local extension of a net with finitely many irreducible DHR sectors, and in which every sector is a direct sum of finite-statistical-dimension sectors, has finite index. These hypotheses follow from the complete rationality already established for $\mathcal{N}_U$. Thus the actual inclusion has finite index. Its dual canonical sector is the restriction of the $H$ vacuum computed above, and its dimension is the inclusion index. Its algebra $A_{\mathrm{net}}$

is Lagrangian because $\mathcal{N}_H$ is holomorphic. The preceding comparison also gives the algebra identity

$$\mathrm{End}(A_{\mathrm{net}}) \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{125}$$

Here the multiplication is composition, rather than a dimension count alone. Center induction identifies its dimension with $\mathrm{rank}(\mathcal{F}_{\mathrm{net}})$, and identifies the algebra with the complex fusion algebra of $\mathcal{F}_{\mathrm{net}}$. Thus the latter has rank six and is noncommutative.

We next identify the invertible objects using only the actual $\mathbb{Z}/3$ lattice-extension automorphisms of $H/B$. They act on the net inclusion and hence on its canonical algebra. The automorphism–invertible-object injection of Lemma G.1 embeds this group in $\mathrm{Inv}(\mathcal{F}_{\mathrm{net}})$. Since the category has rank six, the latter group has order three or six. If its order were six, the category would be pointed of global dimension six. The intermediate lattice inclusion has $[\mathcal{N}_H : \mathcal{N}_B] = 3$, so multiplicativity of index would give $[\mathcal{N}_B : \mathcal{N}_U] = 2$. An irreducible local index-two extension is the fixed-point extension of its sign involution: its canonical algebra is $\mathbf{1} \oplus \beta$, where $\beta^2 = \mathbf{1}$, and $\mathbf{1} \oplus (-\mathrm{id}_\beta)$ is its nontrivial algebra automorphism. Theorem 6.9 of [10] lifts this vacuum-preserving net automorphism to a unitary automorphism $g$ of the actual $B$. Their Proposition 7.6 gives $\mathcal{N}_{B^g} = \mathcal{N}_B^g = \mathcal{N}_U$, and injectivity of their unitary-subalgebra/subnet correspondence (Theorem 7.5) gives $B^g = U$. This contradicts Lemma G.2. Consequently

$$\mathrm{Inv}(\mathcal{F}_{\mathrm{net}}) \cong \mathbb{Z}/3.$$

This argument needs no assertion about the full pointwise stabilizer $\mathrm{Aut}(B/U)$.

The lattice net $\mathcal{N}_B$ has discriminant metric group $\mathbb{Z}/3 \times \mathbb{Z}/3$ with $q(a,b) = (2a^2 + b^2)/3 \pmod{\mathbb{Z}}$. The extension $H/B$ is its diagonal isotropic line $L_+ = \{(a,a)\}$, and $L_- = \{(a,-a)\}$ is a transverse isotropic line. The subcategory $\mathcal{T}_-$ supported on $L_-$ is Tannakian and is tensor equivalent to the untwisted $\mathrm{Vec}_{\mathbb{Z}/3}$. Let $Q_+$ be the extension algebra supported on $L_+$. Free-module induction $t \mapsto t \otimes Q_+$ is a tensor functor from $\mathcal{T}_-$ to the category of all $Q_+$-modules. For simple $t, t' \in \mathcal{T}_-$, its adjunction gives

$$\mathrm{Hom}_{Q_+}(t \otimes Q_+, t' \otimes Q_+) \cong \mathrm{Hom}(t, t' \otimes Q_+) = \begin{cases} \mathbb{C}, & t = t', \\ 0, & t \neq t'. \end{cases}$$

Here transversality $L_+ \cap L_- = \{0\}$ gives the last equality. The target has dimension $9/3 = 3$, so these three distinct invertible images exhaust its simple objects. Induction is therefore a tensor equivalence. Forgetting the intermediate $B$-action embeds this category fully faithfully and tensorially in $\mathcal{F}_{\mathrm{net}}$; relative tensor product over the same $H$ algebra is unchanged. Thus the entire pointed subcategory is untwisted.

We may now apply the rank-six ring and indicator argument of Section G to $\mathcal{F}_{\mathrm{net}}$ itself. Its hypotheses are the fusion-algebra decomposition above, the three invertibles, and their untwisted associator. They have just been proved for this category, and its spherical structure is positive because it is a $C^*$ fusion category. The argument gives (118) with multiplicity one. In particular, for $d = (3 + \sqrt{13})/2$ and $s = 1 + d^2$, the center calculation gives dimensions

$$1,\ d^2,\ \underbrace{s, s, s, s}_{4},\ \underbrace{3d, 3d, 3d, 3d, 3d, 3d}_{6}. \tag{126}$$

The same center-induction and indicator calculation gives the last six twists with exponents $\{2, 5, 6, 7, 8, 11\}/13$. This is the source of the net count twelve. Finally the Lagrangian-algebra theorem gives $\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{F}_{\mathrm{net}})$, and the displayed center dimensions give its rank twelve. No VOA character transformation matrix was assigned to the net category. □

For comparison, the first paragraph also follows for the bounded-insertion inclusion directly from Tener [28, Proposition 5.5 and Theorem 5.6]. The argument above has the additional

advantage that the later automorphism comparison refers throughout to the original lattice fields. The purely categorical deductions used in the proof are recorded in `hi_cft_20260923_next_structure_rank46.md`, `hi_cft_20260923_next_structure_rank6_indicators.md`, and `hi_cft_20260923_next_structure_rank6_center.md`.

## I.2 Reverse reconstruction and exhaustion

Henriques–Tener [30, Theorem 9.5 and Corollary 9.7] reconstruct unitary modules and identify their closed smeared fields. Remark 9.8 extends the construction beyond the finite-multiplicity hypothesis. This extension matters: finite-dimensional energy eigenspaces must not be silently assumed for the net representation one is trying to reconstruct. Their Theorem 8.9 identifies the reconstructed vacuum VOA of $\mathcal{N}_U$ with the original $U$, since $U$ is strongly local.

The closure step can be isolated from the VOA theory. Its formulation also explains why a formal mode identity alone is insufficient.

**Lemma I.2** (Reduction on a common core)**.** *Let $T_0 : D \to \mathcal{K}$ be a densely defined closable operator and $P$ an orthogonal projection. Suppose $PD \subset D$ and $T_0 P\xi = PT_0\xi$ for $\xi \in D$. Then $P$ reduces $T = \overline{T_0}$ and $T^*$, and it commutes with the von Neumann algebra generated by $T$. In particular, this holds simultaneously for a family of fields satisfying these hypotheses on a common core.*

*Proof.* If $\xi_j \in D$ converges to $\xi \in D(T)$ in graph norm, then $P\xi_j \to P\xi$ and $T_0 P\xi_j = PT_0\xi_j \to PT\xi$. Thus $P\xi \in D(T)$ and $TP\xi = PT\xi$. The same assertion holds for $1 - P$. It follows that $T$ is the direct sum of its closed restrictions to $P\mathcal{K}$ and $(1 - P)\mathcal{K}$; taking adjoints gives the corresponding direct sum for $T^*$. Its polar partial isometry and the spectral projections of $|T|$ are therefore block diagonal. These generate the von Neumann algebra of $T$, proving the last assertion. Notice that no separate assertion that $D$ is a core for $T^*$ is required. □

**Proposition I.3** (Reverse module comparison; A3)**.** *For this regular $U$, reconstruction sends each irreducible $\mathcal{N}_U$ sector to an irreducible ordinary unitary $U$-module. It preserves $L_0$ and induces a fully faithful linear $*$-functor on finite direct sums,*

$$\mathcal{R} : \mathrm{Rep}(\mathcal{N}_U) \longrightarrow \mathrm{Rep}^u(U).$$

*The functor preserves conformal Hamiltonians. No tensorator or compatibility with braiding is asserted.*

*Proof.* Let $\mathcal{K}$ be an irreducible net representation and put

$$M = \bigcup_{R>0} 1_{[0,R]}(L_0)\mathcal{K}$$

as in Henriques–Tener, Remark 9.8. This is a dense admissible weak $U$-module with a positive invariant Hermitian form. Theorem 9.5(ii) and its extension in that remark identify the closed reconstructed smeared field with $\pi_I(Y_U(v, f))$ for $\mathrm{supp}(f) \subset I$. By Definition 9.4, $M$ is a core for this closed operator. The proof of Corollary 9.6 applies unchanged on bounded-energy vectors, so the VOA zero mode of the conformal vector is precisely the self-adjoint rotation generator on this domain.

Regularity decomposes the weak module $M$ algebraically into ordinary simple modules. Choose a nonzero simple summand $S$. Its inherited form is positive and invariant. Since its $L_0$ is the restriction of the net Hamiltonian, its energy decomposition is

$$S = \bigoplus_{n\geq 0} S_{h+n}, \qquad \dim S_{h+n} < \infty, \quad h \geq 0.$$

The Hilbert closure $\overline{S}$ is the orthogonal Hilbert sum of these finite-dimensional spaces. In particular it reduces $L_0$, and the projection $P$ onto $\overline{S}$ satisfies

$$P1_{[0,R]}(L_0)\mathcal{K} \subset \bigoplus_{h+n\le R} S_{h+n} \subset S, \qquad PM \subset S \subset M. \tag{127}$$

This establishes the domain invariance before any commutation with fields is used. It requires finite energy multiplicities only for the chosen ordinary summand, not for $\mathcal{K}$.

For a homogeneous $v \in U$, the unitary invariance identity expresses the adjoint of a mode $v_{(n)}$, on $M$, as a finite linear combination of modes of the vectors $L_1^j\theta v$, where $\theta$ is PCT. Every such mode preserves $S$. Testing against $s \in S$ therefore gives

$$\langle Pv_{(n)}\xi, s\rangle = \langle \xi, v^\dagger_{(n)}s\rangle = \langle P\xi, v^\dagger_{(n)}s\rangle = \langle v_{(n)}P\xi, s\rangle.$$

Both outer vectors lie in $S$, by (127), whence $Pv_{(n)}\xi = v_{(n)}P\xi$ for $\xi \in M$. The smeared fields on $M$ are Hilbert-space vectors, as in the paragraph preceding Henriques–Tener Definition 9.4. Taking their energy components, or equivalently their convergent Fourier sums, now yields

$$PY^M(v,f)\xi = Y^M(v,f)P\xi, \qquad \xi \in M.$$

There is no assertion here that a smeared field maps $M$ into itself. Only its value in $\mathcal{K}$ and the boundedness of $P$ are needed. Lemma I.2 extends this identity to the closures and their adjoints.

The local algebras are generated by the closed smeared fields. Normal representations preserve this generation statement (equivalently, apply them to the polar decompositions and spectral projections). Thus $P$ is a net intertwiner. Irreducibility and $S \neq 0$ imply $P = 1$. Equation (127) then implies $M = S$. This proves ordinaryness and simplicity, including the removal of the infinite multiplicities permitted in Remark 9.8.

Finally consider morphisms. A net intertwiner commutes with the canonical diffeomorphism action, because that action is generated by the local diffeomorphism groups. It therefore preserves bounded-energy vectors. Commutation with the reconstructed fields gives commutation with their Fourier coefficients, hence a module map. Conversely, an isomorphism of simple ordinary unitary modules is a scalar multiple of a unitary. To verify this without presupposing boundedness, compare the two invariant forms on each finite-dimensional energy space. The resulting grade-preserving positive operator is an algebraic $U$-endomorphism, by invariance of the forms. An eigenvector in the finite-dimensional lowest space has an eigenspace that is a nonzero $U$-submodule. Simplicity makes that eigenspace the whole module, so the comparison operator is a single positive scalar on all grades. The isomorphism therefore has a bounded unitary rescaling. Its unitary extension intertwines the smeared fields on their common cores and hence their closures, so it intertwines the nets. Between inequivalent simples both morphism spaces vanish. Matrices of these maps handle finite direct sums and respect adjoints. This proves full faithfulness and the asserted linear $*$-functor. □

The preliminary version of this deduction is recorded in `hi_cft_20260923_next_m3_unitarity_scope.md`.

**Proposition I.4** (Strong unitarity from sector exhaustion)**.** *Under the preceding algebraic inputs, including the count of twelve simple $U$-modules, every simple $U$-module is unitarizable, and its character is the character of a net sector.*

*Proof.* Full faithfulness makes the twelve net simples pairwise inequivalent simple ordinary $U$-modules. There are exactly twelve algebraic simple modules, so this injection is a bijection. Each image carries its reconstructed positive form, and $L_0$ is preserved. Finite direct sums give the corresponding statement for ordinary modules. □

This counting step supplies essential surjectivity only as a linear comparison. Neither fusion compatibility, equality of categorical $S$ matrices, nor positivity of VOA fusion forms follows from it.

### I.3 Pairwise annulus compatibility

The next step concerns the specific geometric annulus system of Tener [28, Definition 3.18 and Corollaries 3.20–3.22]. Let $(B, A)$ be a localized pair for an interval $I$, with incoming factor $A$ and outgoing factor $B$. Subscripts distinguish their actions on the vacuum and on a reconstructed module. For an insertion point $z \in \mathrm{int}(B, A)$, put

$$X_v(B, z, A) = \overline{B_U Y_U(v, z) A_U} \in \mathcal{A}_U^{\mathrm{bl}}(I).$$

Here the bar denotes the bounded operator represented by the sesquilinear insertion form. At a point outside the incoming annulus one must not silently treat the unsmeared formal field as an everywhere defined Hilbert-space operator. The earlier abstract isomorphism $\mathcal{A}_U^{\mathrm{bl}} \cong \mathcal{N}_U$ does not identify these operators. The following proposition includes the specific vacuum identification needed to compare them, and then its module version.

**Proposition I.5** (Pairwise annulus compatibility; A1)**.** *For Tener's geometric system above, the bounded-insertion and smeared-field nets agree on the original vacuum Hilbert space: $\mathcal{A}_U^{\mathrm{bl}}(I) = \mathcal{N}_U(I)$ for every interval. The Henriques–Tener geometric point insertion for the corresponding localized annulus pair is $X_v(B, z, A)$. If $M$ is an ordinary unitary module reconstructed from a sector $\pi_M$, then*

$$\pi_M\big(X_v(B, z, A)\big) = \overline{B_M Y_M(v, z) A_M}, \qquad v \in U, \quad z \in \mathrm{int}(B, A). \tag{128}$$

*The right side exists as a bounded extension of the insertion form. The assertion includes the generators $B_U A_U$ when the insertion interior is empty. It requires no global choice of central lifts for individual incoming annuli.*

We first record the domain and continuation arguments used in the proof.

**Lemma I.6** (Dense inverse-energy domains)**.** *Let $M$ be an ordinary unitary module and let $A_M, C_M$ be the actions of geometric annuli on its Hilbert completion. Suppose*

$$A_M C_M = \lambda r^{L_0}, \qquad 0 < r < 1, \quad \lambda \neq 0. \tag{129}$$

*Then $M^{\mathrm{fin}} \subset \mathrm{Ran}\, A_M$ and $A_M^{-1} M^{\mathrm{fin}} = C_M M^{\mathrm{fin}}$ is dense. The same conclusion holds for the outgoing adjoint factor.*

*Proof.* The geometric annulus actions are injective and have dense range by Henriques–Tener Lemma 4.15. Equation (129) gives, for $v \in M^{\mathrm{fin}}$, $v = A_M C_M(\lambda^{-1} r^{-L_0} v)$, proving the range assertion. It also gives $C_M M^{\mathrm{fin}} \subset A_M^{-1} M^{\mathrm{fin}}$. Conversely, if $A_M \xi \in M^{\mathrm{fin}}$, then

$$A_M \xi = A_M C_M \lambda^{-1} r^{-L_0} A_M \xi, \qquad \xi = C_M \lambda^{-1} r^{-L_0} A_M \xi.$$

The last equality uses injectivity of $A_M$, and proves the reverse inclusion. Finally $M^{\mathrm{fin}}$ is dense and $C_M$ is bounded with dense range; hence $C_M M^{\mathrm{fin}}$ is dense. This is the argument of Henriques–Tener Lemma 7.9 with a central scalar retained. □

For the geometric annuli under consideration the factorization itself is elementary. The incoming annulus is embedded in the unit disc, its smooth inner Jordan domain contains zero, and its outer parametrization is the identity. Choose a sufficiently small round disc $r\mathbb{D}$

inside that inner domain. The intervening annulus $C$ has its outer parametrization chosen to sew to the inner boundary of $A$, and its inner parametrization is the standard round one. Then $AC$ is the round annulus. Adjusting the lift of $C$ by the central $\mathbb{C}^\times \times \mathbb{Z}$ fiber makes the product a lift of the standard round annulus, giving (129). In particular the range and density requirements in Tener's definition of incoming annulus [28, Definition 4.2] hold on each reconstructed module.

Define

$$\mathcal{D}_A = \{\xi \in \mathcal{K}_M : A_M\xi \in M^{\mathrm{fin}}\}, \qquad \mathcal{D}_{B^*} = \{\eta \in \mathcal{K}_M : B_M^*\eta \in M^{\mathrm{fin}}\}. \tag{130}$$

Both domains are dense by Lemma I.6.

**Lemma I.7** (Continuation of the insertion form)**.** *Let $O_{\mathrm{in}}$ and $O_{\mathrm{out}}$ be the incoming and outgoing insertion interiors, and put*

$$\Omega = \mathrm{int}(\overline{O_{\mathrm{in}}} \cup \overline{O_{\mathrm{out}}}).$$

*Suppose a bounded geometric insertion $G_v(z)$ is holomorphic on $\Omega$. If its matrix elements equal the algebraic insertion form on $O_{\mathrm{in}} \cup O_{\mathrm{out}}$, for vectors in the two domains (130), the equality holds throughout $\Omega$. It uniquely defines a bounded extension of that form.*

*Proof.* For $\xi \in \mathcal{D}_A$ and $\eta \in \mathcal{D}_{B^*}$, consider

$$\langle Y_M(v,z)A_M\xi, B_M^*\eta\rangle \tag{131}$$

as a formal matrix coefficient. It is a finite Laurent polynomial: if $v$ has weight $k$, and the input and output energies are $p, q$, only the mode with $q = p + k - n - 1$ can contribute. Both vectors have finite energy support. Thus the form and the geometric matrix coefficient are holomorphic on $\Omega \subset \mathbb{C}^\times$.

Every component of $\Omega$ meets $O_{\mathrm{in}} \cup O_{\mathrm{out}}$. Indeed, its complement inside $\Omega$ is contained in the union of the two boundaries; boundaries of open sets are closed with empty interior, and a finite union of such sets contains no nonempty open component. The identity theorem therefore extends the equality across every component, including any part of the sewing circle lying in $\Omega$. For fixed $z$, the form is now the restriction of the bounded operator $G_v(z)$ to two dense test domains. This proves existence and uniqueness of its bounded extension. A graph-core assertion is neither used nor needed. □

*Proof of Proposition I.5.* Write $X = (\psi, \rho, t, \gamma)$ for the incoming tuple and $\widetilde{X}$ for the tuple whose adjoint gives the outgoing factor. If $(\varphi_s)_{s\geq 0}$ is the semigroup generated by $-z\rho(z)\partial_z$, put $F_X = \psi \circ \varphi_t$. The corresponding embedded annulus is

$$A_X = \overline{\mathbb{D}} \setminus F_X(\mathbb{D}), \qquad \varphi_{\mathrm{out}} = \mathrm{id}, \qquad \varphi_{\mathrm{in}} = F_X \circ \gamma^{-1}.$$

Its inner Jordan domain contains zero by Tener's Definition 3.16.

First identify the annulus operator. For the semigroup part, the smooth framing $h(\theta, s) = \varphi_{t-s}(e^{i\theta})$, $0 \leq s \leq t$, runs from the inner to the outer boundary. The flow identity

$$\varphi_u(z)\rho(\varphi_u(z)) = z\rho(z)\varphi_u'(z)$$

gives $(-h_s/h_\theta)\partial_\theta = i\rho(e^{i\theta})\partial_\theta$. In the convention of [31, Remark 4.2], $\pi(f\partial_\theta) = T(if)$, so the time-ordered exponential [31, formula (5.2) and Theorem 5.5] is exactly $e^{-tL(\rho)}$. Since $\rho$ has only nonnegative Fourier modes, every bounded-energy subspace is invariant under $L(\rho)$; uniqueness of the evolution identifies this exponential with Tener's finite-energy definition.

Thin factors implement boundary reparametrization. Thus the annulus $A_X$ has a central lift whose vacuum action is exactly $A_U = U(\psi)e^{-tL(\rho)}U(\gamma)^*$, with the prescribed scalar. The case $t = 0$ consists only of thin factors.

We choose lifts for the pair as follows. By [28, Lemma 3.19 and Corollary 3.20], there is a common localizer $\delta$ such that $\delta F_X \gamma^{-1} = \mathrm{id}$ on $I'$, and likewise for $\widetilde{X}$. Hence $\delta A_X$ and $\delta A_{\widetilde{X}}$ are localized annuli. Their localized central extensions have $\mathbb{C}^\times$ fibers and embed canonically in $\widetilde{\mathrm{Ann}}_c$, as in [31, Section 7]. Choose their lifts $D_A, D_{\widetilde{A}}$ with vacuum operators $U(\delta)A_U, U(\delta)B_U^*$, respectively. Such normalization is possible by the preceding operator identification and is unique in the scalar fiber, since annulus operators are nonzero. Choose a common universal lift $\widetilde{\delta}$ projecting to the fixed central lift used in $U(\delta)$, and set

$$\underline{A} = \widetilde{\delta}^{-1} D_A, \qquad \underline{B} = D_{\widetilde{A}}^\dagger \widetilde{\delta}.$$

The product $\underline{BA} = D_{\widetilde{A}}^\dagger D_A$ is localized in $I$. The local transport theorem [31, Theorem 7.4], equivalently [30, Proposition 4.17], identifies their actions on a sector:

$$A_M = U^M(\widetilde{\delta})^* \pi_{M,I}(U(\delta)A_U), \qquad B_M = \pi_{M,I}(B_U U(\delta)^*) U^M(\widetilde{\delta}).$$

These are precisely the paired actions of [28, Section 5.1, formula (5.2)]. Changing $\widetilde{\delta}$ by the deck transformation $Z^m$, where $Z = e^{2\pi i L_0}$, replaces $A_M, B_M$ by $Z^{-m}A_M, B_M Z^m$. The operator $Z$ commutes with every $U$-mode, since these change energy by integers. The insertion form is therefore unchanged, even at fractional lowest conformal weights. No scalar comparison in the vacuum is being used to erase a deck transformation in a nonvacuum sector.

Next reflect $A_{\widetilde{X}}$ by $j(z) = \bar{z}^{-1}$ and sew it to $A_X$ along their standard unit circles. Since $j$ fixes the circle pointwise, the pair condition in Tener's formula (3.2) makes the resulting annulus $BA$ localized in $I$. Its insertion interior for this geometric representative is the union of the incoming and outgoing interiors with their interior seam points, namely the set $\Omega$ in Lemma I.7. For $z \in \Omega$, the geometric operator

$$G_v(z) = (\underline{BA})[v(z)]$$

is bounded, lies in $\mathcal{N}_U(I)$, and acts in any sector as $\pi_{M,I}(G_v(z))$, by [30, Lemma 6.8]. It is holomorphic in $z$ by their Corollary 6.15.

For $z$ in the incoming interior, sewing [30, Section 6.2, formula (76)] and Theorem 9.5(i) give

$$G_v^M(z)\xi = B_M Y_M(v, z)A_M\xi, \qquad \xi \in \mathcal{D}_A.$$

The theorem applies because $A_X$ is a divisor of a round annulus, as shown above; it permits every central lift. The outgoing equality of matrix coefficients follows by taking adjoints, using [30, Proposition 6.19 and formula (81)] and the invariant unitary form of $M$. The reflected state is $e^{\bar{z}L_1}(-\bar{z}^{-2})^{L_0}\Theta v$, the usual antiholomorphic change of local coordinate, a finite expression on $U$. Lemma I.7 now identifies the matrix coefficients on all of $\Omega$ and gives their unique bounded extension. In the vacuum the reconstructed VOA is the original $U$ by [30, Theorem 8.9]; hence this extension is precisely $X_v(B, z, A)$. In the sector the same calculation proves (128). Repeat the construction for every geometric pair representing the same generalized operator pair. The union of these sewn interiors is the prescribed $\mathrm{int}(B, A)$, as explained in [28, proof of Theorem 4.28]. The operators agree on overlaps by the uniqueness of the bounded extension of the insertion form, so the identity holds on that entire prescribed interior.

For a pair with no insertion points, local transport directly gives $\pi_{M,I}(B_U A_U) = B_M A_M$. This also covers empty insertion interiors and supplies all no-insertion generators. We have

thus proved $\mathcal{A}_U^{\mathrm{bl}}(I) \subset \mathcal{N}_U(I)$. Both nets act on the same vacuum Hilbert space; the bounded-insertion net has cyclic vacuum and the same conformal covariance by [28, Proposition 4.13]. Haag duality for this vacuum conformal net, together with locality of $\mathcal{N}_U$, gives

$$\mathcal{A}_U^{\mathrm{bl}}(I) \subset \mathcal{N}_U(I) \subset \mathcal{A}_U^{\mathrm{bl}}(I')' = \mathcal{A}_U^{\mathrm{bl}}(I).$$

This proves equality of the concrete nets and completes the proof. □

**Theorem I.8** (Complete unitarity)**.** *Under the preceding algebraic construction inputs, $U$ is completely unitary.*

*Proof.* Proposition I.4 unitarizes every simple module. Proposition I.5 supplies the representation required by Tener [29, Definition 2.18], for each such module; finite direct sums preserve the property. His Corollary 4.6 then supplies positivity of the canonical fusion forms. Strong unitarity must be established first: the unitarizable-fusion-product hypothesis in Theorem 4.4 cannot be used to bypass the exhaustion argument. □

The pairwise proof above does not require Gui's Condition II. His extension comparison assumes that condition, including complete unitarity and energy bounds and strong intertwining for a generating family of charged fields; his inheritance theorem concerns regular cosets [49, Definition 4.15 and Theorems 4.20, 6.11]. Those results alone do not supply the present subalgebra comparison. Henriques–Tener also construct bounded vacuum multidisc operators and prove the vacuum trace-class condition [50, Theorems 3.6 and 4.1]. Our comparison instead uses the local transport and module point-insertion theorems explicitly, with a common localizer cancelling the pair's lift ambiguity. A linear equivalence of unitary module categories alone would not give the positivity of the canonical VOA fusion forms.

# J Unitary categorical identifications

**Lemma J.1** (All modules over a commutative Q-system)**.** *Let $\mathcal{C}$ be a unitary braided fusion category and $A$ a connected commutative Q-system in $\mathcal{C}$. The category $\mathcal{C}_A$ of all right $A$-modules, with relative tensor product, is a unitarizable fusion category. This assertion does not require the modules to be local.*

*Proof.* If $r : M \otimes A \to M$ is a right action, the braiding gives a left action $l = r \circ c_{A,M}$. Commutativity of $A$ and the hexagon identity make these commuting actions, and hence an $A$-bimodule. Naturality of the braiding makes every right-module morphism a bimodule morphism for these choices. This realizes $\mathcal{C}_A$ as the full subcategory of bimodules with $l = r \circ c_{A,M}$. The relative tensor product is obtained by splitting the separability idempotent on $M \otimes N$. Its induced left and right actions again satisfy this relation, by the same hexagon identity and the balancing relation. Thus the subcategory is closed under relative tensor products, sums and summands, and its unit is $A$.

Every such bimodule is unitarizable by Gui [34, Theorem 2.28]; an isomorphism transporting the actions preserves the displayed relation. The unitary relative tensor product and its associativity and unit constraints are supplied by Gui, Theorem 3.13. The above full subcategory therefore inherits a $C^*$ tensor structure. Separability gives semisimplicity, connectedness makes its unit simple, and the usual module-category construction over a connected separable algebra gives rigidity and finiteness. It is consequently a unitary fusion category up to tensor equivalence.

For clarity, locality is the additional monodromy condition $r \circ c_{A,M} \circ c_{M,A} = r$. We have not imposed it. Only the local subcategory inherits the ambient braiding by descent. □

**Theorem J.2** (Unitary Haagerup-center identification)**.** *Under the hypotheses of Theorem I.8, and the preceding determination of the all-module fusion ring and its untwisted pointed subcategory, there is a unitary ribbon equivalence*

$$\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3).$$

*For the original embedded extension $H/U$, its all-module category is tensor equivalent to one of $\mathcal{H}_2$ and $\mathcal{H}_3$.*

*Proof.* Put $\mathcal{C} = \mathrm{Rep}(U)$ and let $A$ be the algebra of the actual unitary extension $H$. Complete unitarity allows application of Gui [34, Theorems 2.21 and 2.28]: $A$ is a standard commutative Q-system and its modules are unitarizable. Lemma J.1 makes $\mathcal{F} = \mathcal{C}_A$ a $C^*$ fusion category. Here $\mathcal{C}_A$ means *all* modules. The local modules form $\mathcal{C}_A^0 \simeq \mathrm{Vec}$ because the extension is holomorphic, so substituting local modules would lose the Haagerup category.

Its already determined fusion ring is $\mathrm{HI}(\mathbb{Z}/3)$. The actual untwisted pointed subcategory makes the obstruction $c^{(0,3)}$ vanish, while the other obstruction is in

$$H^1\Big(\mathbb{Z}/2, H^2(\mathbb{Z}/3, \mathbb{C}^\times)\Big) = 0.$$

Indeed, the cyclic-group calculation gives $H^2(\mathbb{Z}/3, \mathbb{C}^\times) = \mathbb{C}^\times/(\mathbb{C}^\times)^3 = 0$; every nonzero complex number has a cube root. Also $(\mathbb{Z}/3)/2(\mathbb{Z}/3) = 0$, so no choice of distinguished noninvertible object introduces an additional classification orbit. These are the hypotheses in Izumi [35, Definition 2.7 and Theorems 5.9 and 9.1], giving the two unitary tensor classes $\mathcal{H}_2, \mathcal{H}_3$. No algebra structure on $1+\rho$ is inserted as an extra assumption. Grossman–Snyder [36, Theorem 1.4] place these categories in the same Morita class. Their centers are therefore braided equivalent. The positive spherical structures make this a ribbon equivalence, and the Lagrangian-center theorem gives

$$\mathcal{C} \simeq Z(\mathcal{F}) \simeq Z(\mathcal{H}_3).$$

Reutter [37, Theorem 5] provides the unitary braided realization of the equivalence, preserving the associated positive ribbon structures. □

This uses the published unitary classification as an external input; its classification equations have not been replayed here. It avoids the broader pivotal census, but it does not prove that the natural reconstruction functor $\mathcal{R}$ preserves Connes fusion, the VOA tensor product, or braiding. That particular comparison is a distinct statement and is not needed for the theorem just proved.

**Corollary J.3** (Haagerup-center identification for the conformal net)**.** *Under the preceding algebraic inputs, the completely rational central-charge-eight net $\mathcal{N}_U$ satisfies*

$$\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{H}_3)$$

*as a unitary ribbon category. Its actual extension to the $E_8$ lattice net has boundary category equivalent to $\mathcal{H}_2$ or $\mathcal{H}_3$. Neither conclusion uses Proposition I.5.*

*Proof.* Proposition I.1 constructs $\mathcal{F}_{\mathrm{net}}$ as a unitary fusion category with the $\mathrm{HI}(\mathbb{Z}/3)$ fusion ring and an untwisted pointed subcategory. The Izumi classification and the vanishing obstruction groups used in the preceding proof therefore apply directly to this category. Thus $\mathcal{F}_{\mathrm{net}}$ is tensor equivalent to $\mathcal{H}_2$ or $\mathcal{H}_3$. Their Morita equivalence, the unitary Lagrangian-center theorem, and uniqueness of the unitary braided structure give $\mathrm{Rep}(\mathcal{N}_U) \simeq Z(\mathcal{F}_{\mathrm{net}}) \simeq Z(\mathcal{H}_3)$. No positivity statement about VOA intertwining-operator spaces is used in this argument. □

# K Full net theory and a chosen $E_8$ VOA extension

**Corollary K.1** (A chosen holomorphic net extension)**.** *Under the preceding algebraic inputs, there is an irreducible local conformal-net extension $\mathcal{N}_U \subset \mathcal{E}_3$ of central charge eight. It is holomorphic and has index*

$$[\mathcal{E}_3 : \mathcal{N}_U] = \mathrm{Dim}(\mathcal{H}_3) = 3(1+d^2), \qquad d = (3+\sqrt{13})/2.$$

*Its all-module boundary category is $\mathcal{H}_3$, and its algebra-bimodule category is $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$, with* 36 *simple objects. This construction does not require A1.*

*Proof.* Transport the canonical connected commutative Q-system $I_{\mathcal{H}_3}(\mathbf{1}) \in Z(\mathcal{H}_3)$ along Corollary J.3; call the result $Q_3 \in \mathrm{Rep}(\mathcal{N}_U)$. The local-net reconstruction theorem of Longo–Rehren [45, Theorem 4.9] produces the extension. Connectedness is precisely the multiplicity-one hypothesis for the identity sector, hence gives irreducibility. The Q-system has finite statistical dimension, which equals the displayed index; its commutativity is precisely the locality condition in that theorem. The conformal-net form of the correspondence is recalled in [46, Section 2.6]. The represented local diffeomorphism unitaries of the subnet give the conformal action of the extension, so the central charge remains eight.

The finite-index theorem [32, Theorem 24] gives complete rationality. Since $Q_3$ is Lagrangian, the index relation gives

$$\mu(\mathcal{E}_3) = \frac{\mu(\mathcal{N}_U)}{\dim(Q_3)^2} = 1,$$

and the extension is holomorphic. The canonical-algebra equivalences of [39, Corollary 4.1 and Proposition 4.4] identify the all-module category with $\mathcal{H}_3$. The corresponding bimodule category is $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ by the double-centralizer description [42, Sections 4 and 8]; its simple objects are the $6 \cdot 6$ ordered pairs. As below, these algebra-bimodule defects preserve the original $\mathcal{N}_U$ subnet. □

No identification of $\mathcal{E}_3$ with the $E_8$ lattice net is made here. Holomorphicity and central charge eight are the conclusions of the cited net theorems; the VOA classification used later does not by itself classify arbitrary holomorphic conformal nets. Nor does the present construction identify a reconstructed VOA extension with the selected tensor category of VOA modules.

**Proposition K.2** (Diagonal full theory)**.** *Under the preceding algebraic inputs, the completely rational net $\mathcal{N}_U$ admits its canonical unitary local diagonal full theory. With the twelve-module character determination, its partition function is*

$$Z_{\mathrm{diag}}(\tau,\bar\tau) = \sum_{i=0}^{11} \chi_i(\tau)\overline{\chi_i(\tau)}, \qquad c_L = c_R = 8. \tag{132}$$

*Proof.* Apply the diagonal Longo–Rehren construction, in the identity-coupling case of Rehren [38, Theorem 1.4, Proposition 1.5, and Corollary 1.6]. Propositions I.1 and I.3 identify all sector Hilbert spaces with the twelve ordinary modules and preserve their Hamiltonians. Hence their traces give (132). The annulus compatibility claim is not needed for this net construction or this linear character matching. □

**Theorem K.3** (Chosen $\mathcal{H}_3$ extension and defects)**.** *Assume the hypotheses of Theorem J.2 and the preceding strict positivity of all nonvacuum lowest conformal weights. There exists a*

*unitary holomorphic conformal extension $H' \supset U$ isomorphic to $V_{E_8}$ whose all-module category is $\mathcal{H}_3$. For its extension algebra $A_3 \in \mathrm{Rep}(U)$,*

$$\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3, \qquad {}_{A_3}\mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}. \tag{133}$$

*The second category describes algebra-bimodule defects preserving the $U$ chiral subalgebra in the corresponding full extension theory and contains an $\mathcal{H}_3$ fusion subcategory. It has 36 simple objects, labelled by ordered pairs of simple objects of $\mathcal{H}_3$.*

*Proof.* Let $I_{\mathcal{H}_3}$ be the right adjoint of the forgetful functor $Z(\mathcal{H}_3) \to \mathcal{H}_3$. Transport the canonical algebra $I_{\mathcal{H}_3}(\mathbf{1})$ along the unitary ribbon equivalence. By the canonical Lagrangian-algebra construction, as in Davydov–Müger–Nikshych–Ostrik [39, Corollary 4.1 and Proposition 4.4], the resulting $A_3$ is connected, commutative, separable, and twist-free, with

$$\mathrm{FPdim}(A_3)^2 = \mathrm{FPdim}(\mathrm{Rep}(U)), \qquad \mathrm{Rep}(U)^0_{A_3} \simeq \mathrm{Vec}, \qquad \mathrm{Rep}(U)_{A_3} \simeq \mathcal{H}_3.$$

Trivial twist makes its constituent conformal weights integral. Connectedness and strict positivity away from the vacuum give a one-dimensional weight-zero space. The extension correspondence of Huang–Kirillov–Lepowsky [40, Theorems 3.2–3.6], together with Gui's unitary Q-system theorem, produces a unitary conformal VOA extension $H'$. Here the transported algebra is a Q-system: the canonical algebra in the unitary center has its standard positive Frobenius structure, and the equivalence is unitary. Thus Gui's theorem is applied to a specified positive algebra structure, rather than to an arbitrary separable algebra without a chosen $*$-structure.

For completeness, the finiteness condition on the extension can be checked at the level of $C_2$ spaces. Write $H' = \bigoplus_i M_i^{\oplus m_i}$ as a finite sum of ordinary $U$-modules and set

$$C_2^U(H') = \mathrm{span}\{u_{(-2)}w : u \in U,\ w \in H'\}.$$

The module $C_2$-cofiniteness theorem of Abe–Buhl–Dong [13, Proposition 5.2] implies $\dim H'/C_2^U(H') < \infty$. Since $C_2^U(H') \subset C_2(H')$, the extension is $C_2$-cofinite. Complete reducibility for the extension follows from the separable algebra/module correspondence in Huang–Kirillov–Lepowsky, Theorems 3.5–3.6. The unitary invariant form makes the simple extension self-contragredient. Together with the already verified CFT-type grading, these are the strong-rationality hypotheses. The Lagrangian condition gives $\mathrm{Rep}(H') \simeq \mathrm{Rep}(U)^0_{A_3} \simeq \mathrm{Vec}$, so it is holomorphic. Its central charge is eight, so Dong–Mason [41, Theorem 1] identifies $H'$ with $V_{E_8}$.

For the canonical algebra, the algebra-bimodule duality gives the second equivalence in (133); see Müger [42, Sections 4 and 8]. One way to keep track of the opposite is to view $\mathcal{H}_3$ as a module category over its center. The two commuting actions on it are

$$(x \boxtimes y) \cdot m = x \otimes m \otimes y.$$

Composing two such functors reverses the order of the right factors, which accounts for "op" in the formula. The double-centralizer description identifies these endofunctors with the bimodule category of the canonical algebra. The equivalence transports with $A_3$. The first-factor embedding $x \mapsto x \boxtimes \mathbf{1}$ is fully faithful and tensor, and supplies the stated $\mathcal{H}_3$ fusion subcategory. There are $6 \cdot 6 = 36$ simple pairs, with dimensions $\dim(x)\dim(y)$. As a consistency check, their total squared dimension is $\mathrm{Dim}(\mathcal{H}_3)^2 = \mathrm{Dim}(\mathrm{Rep}(U))$, as required by Morita duality. □

**Remark K.4** (Extension choice and preserved symmetry)**.** The selection of the canonical $\mathcal{H}_3$ algebra proves existence of $H'$ with the specified all-module category. It does not identify $A_3$ with the original embedded $H/U$ algebra; the original boundary category may be $\mathcal{H}_2$ or

$\mathcal{H}_3$. Both holomorphic VOAs being isomorphic to $V_{E_8}$ does not identify their embeddings of $U$. Moreover, the defects in (133) preserve $U$ and need not preserve the enlarged $E_8$ chiral algebra. The local chiral $H'$-module category is Vec.

The finite certificates and algebraic arguments supply the stated representation-theory inputs. Comparisons A2–A3 give sector exhaustion, strong unitarity, and the unitary Haagerup-center identification for the conformal net. The pairwise annulus comparison A1 then proves complete unitarity of the VOA, its specified unitary ribbon structure, and the chosen $\mathcal{H}_3$ VOA extension. The diagonal net theory and its character interpretation also have a proof without this last comparison. The earlier synthesis sources are `hi_cft_20260923_c8two_final_theorem.md` and `hi_cft_20260923_c8two_certificate_audit.md`; the explicit domain, lift, and insertion arguments above supply the analytic intermediate steps needed for those conclusions.

## L Relation to the proposed Haagerup theories

Albert–Honda–Kaidi–Zheng explicitly assume the existence of a $c = 8$ Haagerup-center theory in Section II, footnote 4 of [11]. Their equation (12) displays the $\gamma$-dependent characters, and their extension discussion relates the center theory to the $E_8$ theory with Haagerup defects.

Under the stated construction inputs, Theorem 1.3 supplies a $\gamma = 0$ realization as a conformal net and a diagonal full net theory. Theorem 1.4 supplies the completely unitary VOA realization and a compatible $E_8$ VOA extension. The distinction between $U$-preserving and full-$E_8$-preserving defects is maintained. Neither this construction nor that finite gauging identifies the continuum limit of the $\rho$-projector chain. Section III of the cited paper proposes further asymmetric quotients towards charges two and six; those require additional conformal and defect data. No uniqueness theorem for all central-charge-eight CFTs is asserted here.

The ambient lattice is also compatible with the conditional characterization in [12, Section 4.3]: a rational chiral algebra with the Haagerup center at central charge eight must arise from a rank-two hypergroup action on $E_{6,1} \otimes A_{2,1}$. That characterization constrains a realization; it does not supply its generating fields. Conversely, the concrete inclusion constructed here does not identify the original extension's fusion category as $\mathcal{H}_3$ without the separate distinction between $\mathcal{H}_2$ and $\mathcal{H}_3$ discussed above. Keeping these two statements separate avoids making an existence argument depend on a desired choice of Morita representative.

## M Certificate ledger and reproducibility

The following frozen records supply the finite construction inputs. The revision of 27 September 2026 additionally reran the original provenance check, minimal generation, grade-two exact OPE and PCT checks, the exact grade-three construction, and the final Gate 4, charged-span, and Gate 5 verifiers. Fresh outputs were written under `c8_revision_20260927/`; the frozen inputs were not overwritten. All paths are relative to the project directory. The filenames preserve an unambiguous connection to the actual records.

| Result | Record and scope |
|---|---|
| Minimal generation | `hi_cft_20260923_c8_minimal_generators.json`: the short generating set in (1) generates all 120 vectors of $S_2$. This necessary provenance step was absent from the original 640-file bundle and is explicitly included in the supplemented ledger. |

| Result | Record and scope |
|---|---|
| Initial spaces | `hi_cft_20260923_c8_exact_unitary_certificate.json` and `hi_cft_20260923_c8_grade3_exact_v2.json`: PCT, stress and singular OPE identities, and exact dimension 956. |
| Lower identities | `hi_cft_20260923_next_structure_lower_closure.json`: 3,970 exact relations. |
| Actual triplet | `hi_cft_20260923_next_structure_p4_controls.json`: 18 current-zero and six direct Virasoro controls. |
| Raising saturation | `hi_cft_20260923_c8two_structure_saturation.json`: 644 exact images in a quotient of dimension 322. |
| Additional direct seeds | `hi_cft_20260923_c8two_targeted_h4_trimmed_v2/summary.json`: 118 completed exact identities. The unfinished final block in this historical directory is not counted as a completed identity. |
| Raised seeds | `hi_cft_20260923_c8two_structure_h4_raised_exact/summary.json`: twelve additional ordinary exact identities. |
| Singlet premise | `hi_cft_20260923_c8two_h4_singlet_premise_verified.json`: all 24 raising conditions replayed. |
| Projected residuals | `hi_cft_20260923_c8two_h4_singlet_exact.json`: twelve full exact residuals, using 21 selected columns. |
| Full Gate 4 | `hi_cft_20260923_c8two_gate4_verified.json`: all 2,898 identities, using singlet recovery and raising saturation. |
| Charged spanning | `hi_cft_20260923_c8two_charged_span_verified.json`: 28 reconstructed seeds, rank 322, and current commutators. |
| Full Gate 5 | `hi_cft_20260923_c8two_gate5_verified.json`: all 24 channels by the stated Jacobi argument. |
| Combined provenance | `hi_cft_20260923_c8two_proof_bundle.json`: 640 authenticated files, including earlier exact inputs and proof notes. It checks provenance and completion, not formal correctness of every mathematical argument. |

The final Gate 4 verifier replays the twelve projected identities using integer-cleared coefficients, separately from the rational producer. A field element is represented in the basis $1, s, t, st$, with $s^2 = 13$, $t^2 = -3$; all residuals are exact four-component rational identities. Prime calculations are used to select candidate columns and to certify positive spanning against an independently known upper dimension. A vanishing prime residual alone is never treated as a characteristic-zero containment proof.

The provenance-ledger replay can be written to a fresh output without changing the original certificate:

```
python hi_cft_safe_run.py --limit-mib 1024 -- \
  python hi_cft_20260923_c8two_proof_bundle.py \
  --output /tmp/haagerup-c8-proof-review.json
```

These lines represent a single shell command. The output path must not already exist. This rechecks the frozen source hashes and completion flags; it does not rerun all original mode calculations. The original and supplemented ledgers together bind the listed certificate producers and verifiers, including the additional dependencies described below.

### M.1 Fresh replay coverage

The supplemented ledger `c8_revision_20260927/finite_input_ledger_complete.json` retains the original provenance graph and explicitly binds the minimal-generation and exact grade-three records. This repairs an omission in the original bundle's declared roots; it does not modify the old bundle or infer a mathematical identity from a hash. The principal fresh outputs are listed below. A "replay" means the specific operation in the right column, rather than a full independent reconstruction of every earlier lattice calculation.

| Output under `c8_revision_20260927/` | Recomputed assertion |
|---|---|
| `provenance_replay.json` | All 640 original source hashes and required completion flags agree. |
| `minimal_generators_replay.json` | The specified short generating set gives the whole exact 120-dimensional $S_2$. |
| `unitary_replay.json` | The PCT, stress, primarity and grade-two singular-product tests, including all eight primary multiplets, pass in the exact coefficient field. |
| `grade3_replay.json` | The characteristic-zero generated span has dimension 956, with normal subspace of dimension 634. |
| `gate4_replay.json` | Integer-cleared projected residuals, raising premises and saturation establish all $2,898$ required Gate 4 inputs. |
| `charged_span_replay.json` | The reconstructed charged seeds and lowering images span dimension 322; the saved current matrices satisfy both axis and cross-axis commutators. |
| `gate5_replay.json` | Exact quotient relations, nonzero bottom coefficients and authenticated inputs support all 24 channels of the printed Jacobi deduction. |
| `top_bound_replay.json` | Independent rational enumeration reconstructs all fourteen core tops, trace coefficients, the two current-character inequalities and the 17/10 certificate. |
| `core_extension_replay.json` | The zero-complementary-energy space in $S_2$ has dimension 25 and contains all six required order-thirteen spectral witnesses with their exact eigenvalues. |
| `top81_replay.json` | In each of four mixed lattice sectors, exact zero-mode actions and weighted adjoints generate the entire 81-dimensional top from a unique Cartan eigenline. |
| `potts_replay.json` | The explicit Potts primary has the required stress action and lies in the actual generated $S_3$. |
| `pf13_membership_replay.json` | The parafermion candidate lies in the generated Cartan-zero $S_3$, its PCT line is stable, and the positive stress identity has the stated exact coefficients. |

The last two producers originally had fixed output paths. The adapter `c8_revision_20260927/replay_structural_input.py` redirects only that output construction to a fresh file, requires exactly one write, and compares the complete mathematical output with the frozen

record, excluding timing and memory statistics. It authenticates the unchanged producer and checks that the original record remains unchanged. The parafermion polynomial OPE and decoupling records are authenticated inputs, not claimed as newly reconstructed by this membership replay.

These calculations ran serially with at most two compute threads and CPU affinity enforced. The permanent shared-desktop guard imposed a 6 GiB address ceiling, an 8 GiB process-group RSS watchdog and a 32 GiB host-available-memory reserve; the exact lattice replays used smaller address caps of at most 2 GiB. The grade-three replay took about 493 seconds and peaked at 0.190 GiB sampled group RSS. No swap capacity was included in the allocation budget. Per-job estimates and logs are recorded in `c8_revision_20260927/resource_review.md`.

# N Limited formalization

The construction's two Lean files isolate implications that are useful independently of the VOA implementation. They use the existing Lean 4.33.1 and Mathlib 4.33.1 environment and were checked without `sorry` or new axioms. The axiom reports contain only Lean's standard propositional extensionality, quotient soundness and, for some statements, classical choice.

1. `lean/HaagerupIzumi/CFT/FiniteClosure.lean` proves raising saturation for a stable submodule, under an explicit joint nilpotence hypothesis. It also proves recovery from a projected residual and the abstract linear Jacobi-bottom-witness implication. The hypotheses identify exactly where the current-representation bounds and finite identities enter the manuscript's use of these lemmas.

2. `lean/HaagerupIzumi/CFT/TopBound.lean` proves the displayed fourteen-variable deficit identity and derives the cleared-denominator 17/10 inequality from nonnegative multiplicities and the two stated $\mathfrak{sl}_2$ inequalities. The independent rational checker separately reconstructs the table to which this identity is applied.

The physics revision adds `lean/HaagerupIzumi/CFT/PhysicsBounds.lean`, with two checked real-number implications. The theorem `modular_heavy_tail_comparison` derives $H \le r(L - L_d)/(1 - r)$ from $L + H = L_d + H_d$, $H \le rH_d$, and $0 \le r < 1$. The theorem `selfdual_tail_first_moment_comparison` derives $H \le L'/E_{\mathrm{cut}}$ from $E_{\mathrm{cut}} > 0$ and $E_{\mathrm{cut}}H \le L'$. These are the algebraic steps in the thermal tail argument; they do not prove its infinite-sum, modularity, differentiability, or interval-arithmetic premises. The successful compiler and standard-axiom report is `c8_physics_20260927/physics_lean.log`.

The safe compiler commands, issued serially from the project root, are

```
python lean/safe_lean.py \
  lean/HaagerupIzumi/CFT/FiniteClosure.lean
python lean/safe_lean.py \
  lean/HaagerupIzumi/CFT/TopBound.lean
python lean/safe_lean.py \
  lean/HaagerupIzumi/CFT/PhysicsBounds.lean
```

The driver invokes the same resource guard. These formalizations do not define a VOA, validate the lattice engine, prove filtered Wick reduction, construct the embedded parafermion factor, or formalize the analytic comparison. Their value is to check the stated algebraic implications with their premises exposed, rather than to certify the entire manuscript.

# O Compilation and review scope of the base revision

The following source names describe the preserved 27 September revision. This expanded draft has separate build instructions in Appendix S28. The main source is `hi_cft_c8_proof.tex`; its companion sources are `hi_cft_c8_manuscript_structure.tex` and `hi_cft_c8_manuscript_analytic.tex`, with references in `hi_cft_c8_bibliography.tex`. Run `pdflatex` twice on the main source for cross-references and the table of contents. The document uses standard TeX packages and requires no shell escape.

The frozen computational records remain unchanged. Fresh replays, the supplemented provenance ledger and the two Lean modules are additional checks with the scopes stated above. The original ledger does not authenticate this revised exposition. The analytic comparison proof and the exact embedded-field inputs should be examined directly when assessing the VOA conclusion; neither a successful PDF build nor a modular-character match verifies these arguments.

# Part I
# Integrated human verification guide

## S1 How to audit the inputs

The following guide is incorporated from the 27 September companion. All manuscript references now resolve within this volume. Original artifact paths identify the archived sources. This companion makes selected inputs inspectable without asking the reader to trust a hash, a rank counter, or a printed `true` value as a mathematical proof.

A useful audit proceeds in three passes.

1. Check the small equations below: the current levels, the three momenta, the stress projection matrix, the explicit parafermion rows, and the short Gate 5 relation. These are literal formulas, with all coefficient conventions fixed.

2. For larger finite equalities, inspect a witness rather than repeat the search that discovered it. The necessary format is an equation $v = \sum_j c_j w_j$, the coordinates of each vector, and a recipe identifying $w_j$ with the claimed lattice product. A rank lower bound needs a nonzero minor or an elimination transcript; a rank upper bound needs spanning expressions. These are different obligations.

3. Follow the printed implications in manuscript Appendices B–K. An exact residual of zero proves its finite identity. It does not by itself prove strong generation, regularity, module exhaustion, or complete unitarity.

| Input | Human-facing evidence supplied here |
|---|---|
| Lattice, currents, $u$ | An eight-dimensional integer calculation and explicit coefficient extraction; all 72 current singular products are independently checked. |
| Positive stress identity | Four orthogonal stress norms and eight actual overlaps; the resulting vector identity is independently checked. |
| Low-grade fields | All 337 monomials (also printed in Appendix S24) of eight relevant grade-two fields are exported in plain tab-separated form. |
| Short generation | A 120-row construction graph, exact elimination identities and expansions of all ten target fields in two bases. |
| Factor membership | A 33-word Potts expression and a 42-word parafermion expression, with full coordinates and an independent check. |
| Gates 4 and 5 | Denominator-cleared equations, named normal words, nonzero scalar witnesses and a precise residual format. |
| Embedded factors and tops | Full first-row OPE formulas, charge and spectral tables, an 80-row cyclic basis and the precise distinction between finite matrix evidence and its mode interpretation. |

### S1.1 Navigation in the expanded manuscript

The following references locate the current arguments supported by these finite inputs. Page numbers refer to the manuscript, not to this guide. The Lean companion supplies a section-by-section assessment of the remaining formalization work.

| Audit topic | Current manuscript location |
|---|---|
| Lattice and initial spaces | Appendix A, p. 35; Lemma A.1 and Certificate A.2. |
| All-weight closure | Appendix B, p. 38; Theorem B.3 and Certificate B.6. |
| Gates and properness | Appendices C–E, pp. 41–45; Lemmas C.2, C.3, and D.2. |
| Embedded factors and stresses | Appendix F, p. 46; Certificate F.1, Proposition F.2, and Theorem F.8. |
| Branching and characters | Appendices G–H, pp. 51–65. |
| Analytic and categorical conclusions | Appendices I–K, pp. 66–76. |
| Physical observables | Sections 2–7, pp. 14–33; the data guide is Section 7. |
| Evidence and formalization scope | Appendices M–O, pp. 78–82. |

## S2 The small lattice calculation

Put $s^2 = 13$, $t^2 = -3$, and use the coefficient field $K = \mathbb{Q}(s,t)$ with ordered basis $(1, s, t, st)$. Complex conjugation fixes $s$ and negates $t$. If $x = (a,b,c,d)$ and $y = (e,f,g,h)$, then

$$xy = (ae+13bf-3cg-39dh,\quad af+be-3ch-3dg,\quad ag+ce+13bh+13df,\quad ah+de+bg+cf).$$

Thus every vector equality is exactly four rational equalities per lattice monomial. No numerical recognition tolerance occurs in the checker. The historical field file also contains floating-point discovery data; only its exact rational coefficient arrays are read.

The lattice Gram matrix is

$$G = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

Let $\ell = (-1,-2,-3,-2,-1,-2,0,0)$ and $r = (5,8,9,8,5,0,2,2)$. Two matrix multiplications give

$$G\ell = (0,0,0,0,0,-1,0,0), \qquad Gr = (2,2,2,2,2,-9,2,2).$$

Consequently $\ell^T G\ell = 2$, $r^T Gr = 78$, and $\ell^T G\alpha_i = 0$ for every $i$ with $r_i \neq 0$. These are the geometric ingredients of levels 1 and 39. Explicitly, with $I = \{0,1,2,3,4,6,7\}$, the six currents are

$$\begin{aligned} H_L &= \ell(-1)\mathbf{1}, \quad E_L = e^{\ell}, \qquad F_L = -e^{-\ell}, \\ H_R &= r(-1)\mathbf{1}, \quad E_R = \sum_{i\in I} e^{\alpha_i}, \quad F_R = -\sum_{i\in I} r_i e^{-\alpha_i}. \end{aligned}$$

For signs use

$$\epsilon(\alpha,\beta) = (-1)^{\sum_i \alpha_i\beta_i + \sum_{i>j}\alpha_i\beta_j G_{ij}}.$$

For example $\epsilon(\alpha_i,-\alpha_i) = -1$, so $e_0^{\alpha_i}e^{-\alpha_i} = -\alpha_i(-1)\mathbf{1}$ and $e_1^{\alpha_i}e^{-\alpha_i} = -\mathbf{1}$. The minus signs in the lowering currents therefore give

$$(E_R)_0 F_R = \sum_i r_i\alpha_i(-1)\mathbf{1} = H_R, \qquad (E_R)_1 F_R = \sum_i r_i\mathbf{1} = 39\mathbf{1}.$$

Together with the Heisenberg brackets this checks the complete affine relations. The reader can derive the mixed vanishing from the displayed orthogonality. The independent checker evaluates all ordered pairs of the six currents at modes 0 and 1, rather than checking only these two examples. Higher singular modes vanish by weight.

### S2.1 The three-term vector without a mode engine

Write $u = e^a + (s+3t)e^b/20 + (s-3t)e^c/20$, with

$$\begin{aligned} a &= (0,0,0,0,0,-1,1,1),\\ b &= (0,0,0,1,1,-1,0,0),\\ c &= (1,1,0,0,0,-1,0,0). \end{aligned}$$

The following table can be checked directly using $G$.

| $\alpha$ | $G\alpha$ | $\alpha^T G\alpha$ | $\ell^T G\alpha$ | $r^T G\alpha$ |
|---|---|---|---|---|
| $a$ | $(0,0,1,0,0,-2,1,1)$ | 4 | 1 | 13 |
| $b$ | $(0,0,0,1,1,-2,0,0)$ | 4 | 1 | 13 |
| $c$ | $(1,1,0,0,0,-2,0,0)$ | 4 | 1 | 13 |

For each summand the inner product with a raising root is nonnegative. The coefficient of $z^{-1}$ in

$$Y(e^\alpha,z)e^\beta = \epsilon(\alpha,\beta)z^{\alpha^T G\beta}\exp\left(\sum_{n>0}\frac{\alpha(-n)}{n}z^n\right)e^{\alpha+\beta}$$

therefore vanishes for every raising current. The absolute inner product with a root occurring in either current triple is at most one, so positive current modes also vanish. The Cartan positive modes kill oscillator-free exponentials. Thus $u$ is an affine primary highest vector of current dimensions $(2,14)$; this check uses no cancellation between its three coefficients.

Set $\widetilde{r} = (5,8,9,8,5,1,2,2)$ and $q(\alpha) = \prod_i \widetilde{r}_i^{\alpha_i}$. In the transported positive form, $\|e^\alpha\|^2 = q(\alpha)$. Here $q(a) = 4$ and $q(b) = q(c) = 40$, while

$$\left|\frac{s\pm 3t}{20}\right|^2 = \frac{13+27}{400} = \frac{1}{10}.$$

The three orthogonal summands each have norm squared four, and $\|u\|^2 = 12$. This explains the equal-magnitude phase convention without confusing the transported PCT with ordinary conjugation.

## S3 A four-column check of the positive stress identity

The ordered axes are $w = (\omega_L, \omega_R, e, f)$. Their Gram matrix is diagonal with diagonal

$$N = \left(\frac{1}{2}, \frac{117}{82}, \frac{1200}{1271}, \frac{35}{31}\right).$$

The entries can be checked in two independent ways: from the stated commuting Virasoro structures, or from the sparse fields and the lattice Fock inner product. The new checker uses the latter, so it does not assume the claimed central charges to compute the pairings.

For precision, if $\alpha = \beta$, the pairing of two oscillator monomials with $r$ factors is

$$\left\langle \prod_{k=1}^{r}\alpha_{i_k}(-n_k)e^\alpha, \prod_{k=1}^{r}\alpha_{j_k}(-m_k)e^\alpha \right\rangle_g = q(\alpha)\sum_{\sigma\in S_r}\prod_{k=1}^{r}\delta_{n_k,m_{\sigma(k)}}\, n_k G_{i_k j_{\sigma(k)}}.$$

Different momenta or different numbers of oscillator factors are orthogonal. Coefficients in the first argument are conjugated. In weight two there are at most two oscillator factors, so the permutation sum has at most two terms. For example the pairing of $\alpha_i(-1)\alpha_j(-1)\mathbf{1}$ with $\alpha_k(-1)\alpha_l(-1)\mathbf{1}$ is $G_{ik}G_{jl}+G_{il}G_{jk}$, while the pairing of the two mode-two oscillators is $2G_{ij}$.

The freshly recomputed overlaps and normalized projections are

| | $\omega_L$ | $\omega_R$ | $e$ | $f$ |
|---|---|---|---|---|
| $\langle w_i,p\rangle_g$ | 0 | 13/205 | 108/1271 | 39/155 |
| $\langle w_i,z\rangle_g$ | 0 | 21/205 | 780/1271 | 13/155 |
| $\langle w_i,p\rangle_g/N_i$ | 0 | 2/45 | 9/100 | 39/175 |
| $\langle w_i,z\rangle_g/N_i$ | 0 | 14/195 | 13/20 | 13/175 |

The squared norms of the actual saved fields are $\langle p,p\rangle_g=2/5$ and $\langle z,z\rangle_g=4/5$. These norm checks alone do not verify their Virasoro OPEs.

The four columns reduce the positive identity to

$$\begin{aligned}
1&=1,\\
1&=\frac{3403}{4563}+\frac{25}{6}\frac{2}{45}+\frac{25}{26}\frac{14}{195}=\frac{3403+845+315}{4563},\\
1&=\frac{25}{6}\frac{9}{100}+\frac{25}{26}\frac{13}{20}=\frac{3}{8}+\frac{5}{8},\\
1&=\frac{25}{6}\frac{39}{175}+\frac{25}{26}\frac{13}{175}=\frac{13}{14}+\frac{1}{14}.
\end{aligned}$$

Thus the exact vector identity

$$\omega_B=\omega_L+\frac{3403}{4563}\omega_R+\frac{25}{6}\overline{p}+\frac{25}{26}\overline{z}$$

has a small inspectable arithmetic certificate. The identification of the orthogonal projections with Haar averages still uses the separately certified statement that these four axes exhaust the current invariants in the actual $S_2$. The checker verifies the projected vector equality; it does not replace that representation-theoretic input.

All 337 monomials (also printed in Appendix S24) of $u,\omega_B,\omega_L,\omega_R,e,f,p,z$ are printed as machine-independent rows in `c8_reviewability_20260927/grade2_fields.tsv`. Each row has the field name, eight momentum coordinates, the oscillator indices, and the four rational coefficients. An oscillator index $8(n-1)+i$ means $\alpha_i(-n)$. Repeated indices are repeated commuting creation operators, with no divided-power convention.

## S4 Finite closure: identities that a reader can inspect

The finite closure argument in Appendices A–E of the base manuscript has three different kinds of evidence. A sparse identity gives an equality of actual vectors. A nonzero minor proves a positive rank assertion. A file hash identifies the source of either witness. Only the first two supply mathematical information; the third prevents a change of input from going unnoticed. This section makes the distinction concrete and prints the small identities responsible for the final Jacobi step.

## S4.1 How to read a coefficient and a row

Put $F = \mathbb{Q}(s,t)$, where $s^2 = 13$ and $t^2 = -3$. The four rational strings $[a,b,c,d]$ in a record mean

$$a + bs + ct + dst.$$

The order is significant. For example, the coefficient $[0, -54087/138650, 20899/138650, 0]$ is $(-54087s + 20899t)/138650$. Multiplication can be checked using only rational arithmetic: if $x = (x_0, x_1, x_2, x_3)$ and $y = (y_0, y_1, y_2, y_3)$, the product has coordinates

$$\begin{aligned} z_0 &= x_0y_0 + 13x_1y_1 - 3x_2y_2 - 39x_3y_3, \\ z_1 &= x_0y_1 + x_1y_0 - 3(x_2y_3 + x_3y_2), \\ z_2 &= x_0y_2 + x_2y_0 + 13(x_1y_3 + x_3y_1), \\ z_3 &= x_0y_3 + x_3y_0 + x_1y_2 + x_2y_1. \end{aligned}$$

In particular, zero testing requires four rational zero tests; it does not involve a numerical tolerance or a choice of decimal roots.

A vector is a finite list of triples

$$[\,\lambda,\ \mathcal{O},\ [a,b,c,d]\,].$$

Here $\lambda \in \mathbb{Z}^8$ gives the momentum in the ordered simple-root basis, and a nonnegative integer $8(n-1)+i$ in the oscillator list $\mathcal{O}$ means $\alpha_i(-n)$, with $0 \le i < 8$. The triple denotes the corresponding oscillator monomial applied to $e^\lambda$, with coefficient $a + bs + ct + dst$. Thus the assertion $v = \sum_j c_j w_j$ means that, for every momentum and oscillator monomial, all four coordinates of $v - \sum_j c_j w_j$ vanish. Checking a pivot coordinate alone is insufficient.

The grade-three checkpoint also records how its normalized vectors were made. If a row has recipe $v$, elimination list $[(d_1, r_1), \dots, (d_m, r_m)]$, and scale $\gamma$, its meaning is

$$r = \gamma \left( v - \sum_{j=1}^{m} d_j r_j \right). \tag{134}$$

The subtraction precedes the scaling. All referenced rows are earlier rows. The leading coefficient of $r$ is one, and its leading state is different from those already inserted in the same Cartan block. These data give both a construction by actual modes and an exact linear-independence witness. The recipes are not merely names for vectors obtained by approximate diagonalization.

## S4.2 The short generators really produce the saved fields

Let $K_{\mathrm{cur}}$ denote the six-dimensional current space, and let $u$ be the displayed three-vertex field in Section S2.1. There is a small, deterministic calculation connecting the definition $U = \langle K_{\mathrm{cur}}, \omega_B, u \rangle$ to the much larger saved presentation. Start with the span of

$$\{x_{-2}\mathbf{1},\ x_{-1}y : x, y \in K_{\mathrm{cur}}\} \cup \{\omega_B, u\}.$$

Its exact dimension is 24. Keep a queue of newly accepted independent vectors. For each vector $v$ removed from the queue, insert the five vectors

$$(E_L)_0 v, \quad (F_L)_0 v, \quad (E_R)_0 v, \quad (F_R)_0 v, \quad u_1 v$$

by exact sparse elimination, adding a new queue entry only when the remainder is nonzero. The queue empties after 120 accepted vectors have been processed. Every operation preserves

membership in the already defined VOA $U$, so the resulting space lies in $U_2$. Its dimension is 120, and direct exact reduction puts every saved highest vector and the retained neutral stress in this span.

Independently, the saved $S_2$ has the decomposition by current types

$$\dim S_2 = 22 + 2 + (5 + 9 + 9 + 13 + 4 + 12 + 16 + 28) = 120.$$

The 22 affine vacuum descendants, the two neutral fields, and the eight indicated multiplets are all in the queue-generated space; dimension comparison identifies that space with $S_2$. This establishes $S_2 \subset U_2$ before finite closure is used. It does not yet assert $S_2 = U_2$; that conclusion uses the later strong-generation argument.

The complete short algorithm is `hi_cft_20260923_c8_minimal_generators.py`, and its fresh exact replay is `c8_revision_20260927/minimal_generators_replay.json`. The new witness `c8_reviewability_20260927/minimal_generation_witness.json` records all 120 accepted raw vectors, their 120 normalized remainders, every elimination and scale, and the expansions of the normalized vectors in the raw vectors. It also records the eight highest fields, the retained neutral stress, and $\omega_B$ in both bases. The adjacent `minimal_generation_witness.recipes.tsv` and `minimal_generation_witness.target_coefficients.tsv` give readable recipe and coordinate tables. There were 644 candidate insertions and 8386 retained sparse entries in the generation rows; the verifier counts 8442 after including the input-field coordinates.

The independent verification record `c8_reviewability_20260927/minimal_generation_checked.json` checks 120 unit-triangular normalization identities, 120 expansions in the raw vectors, and all 20 target-coordinate identities. It also checks the six current formulas, the ambient stress and $u$, the earlier-parent condition in the recipe graph, and the source bindings. Thus the coefficient identities and the rank-120 witness are available explicitly. The exporter recomputed the mode products in 1.81 seconds; the independent coordinate verifier took 1.02 seconds, with peak RSS 35900 KiB. The latter does not recompute a raw vector from its named mode recipe. That remaining interpretation is supplied by the exporter's actual mode calculation and the lattice-mode engine.

### S4.3 Gate 4: twelve projected residuals and a finite rank argument

Write $P = S_3/N_3$. The constructed spaces have exact dimensions

$$\dim S_3 = 956, \qquad \dim N_3 = 634, \qquad \dim P = 322.$$

The Gate 4 conclusion is $(S_2)_0 S_3 \subset W_4 = N_4 + P_4$. It is reduced to nine highest left inputs, each with 322 right inputs. There are 2145 original exact seed relations and 118 additional direct relations. The last twelve targets have weight $(0, 0)$ and are the following. The integer in the first column is the primary key in the projected exact database.

| ID | Left | Right | ID | Left | Right |
|---|---|---|---|---|---|
| 0 | $a_0$ | `r3_000950` | 6 | $a_4$ | `r3_000873` |
| 1 | $a_1$ | `r3_000916` | 7 | $a_5$ | `r3_000879` |
| 2 | $a_1$ | `r3_000920` | 8 | $a_5$ | `r3_000897` |
| 3 | $a_2$ | `r3_000916` | 9 | $a_6$ | `r3_000882` |
| 4 | $a_2$ | `r3_000920` | 10 | $a_6$ | `r3_000898` |
| 5 | $a_3$ | `r3_000922` | 11 | $a_7$ | `r3_000902` |

The fields $a_0, \dots, a_7$ are the eight charged highest vectors in their saved order. In $(a_i)_0 p$, the first subscript labels a field and the second denotes its mode.

Here is the projector used in all twelve records, with every denominator exposed:

$$\Pi = \left(1 - \frac{F_L E_L}{2}\right)\left(1 - \frac{F_L E_L}{6}\right)\prod_{d \in D}\left(1 - \frac{F_R E_R}{d}\right),$$
$$D = \{2, 6, 12, 20, 30, 42, 56, 72, 90, 110, 132, 156\}.$$

The symbols in this formula are current zero modes. On the weight-zero line of a highest-weight-$j$ $\mathfrak{sl}_2$ module, $FE$ acts by $j(j+2)/4$. The current bounds at grade four are $j_L \le 4$, $j_R \le 24$, so this polynomial kills every nontrivial joint type in the weight-$(0,0)$ subspace and is one on the joint singlets.

For each target $t$, a record supplies rational quadruples $c_0, \dots, c_{20}$, with unused coefficients omitted, and asserts

$$\Pi t - \sum_{j=0}^{20} c_j \Pi n_j = 0, \qquad n_j \in N_4. \tag{135}$$

The $n_j$ are actual normal words. Their allowed recipes are $x_{-1}p$, $Tp$, or $a_{-1}b$, followed if necessary by current lowering. For example, the database recipe `normal_descendant`, with current index $i < 6$ and input $p$, means $(x_i)_{-1}p$; current index 6 means $(\omega_B)_0 p = Tp$. A `current_lowering` recipe points to an earlier normal column and applies $(F_L)_0$ or $(F_R)_0$.

The new packet `c8_reviewability_20260927/finite_witnesses.json` prints the coefficient list of each of the twelve equations, the correspondence from projected columns to original normal columns, and the recursive normal-word recipes. In the original exact database, projected target $i$ is stored as `target:`$i$; projected normal column $j$ is stored as `grade5:`$j$. The latter is a historical storage prefix: these vectors have conformal weight *four*.

Four small entries of target 11 illustrate the complete convention. The target is $t = (a_7)_0$`r3_000902`, and its recorded coefficient $c_j$ multiplies $\Pi n_j$ as follows:

| $j$ | Actual normal word $n_j$ | Coefficient $c_j$ |
|---|---|---|
| 2 | $(H_L)_{-1}$`r3_000022` | $(169 - 3st)/2240$ |
| 3 | $(H_L)_{-1}$`r3_000023` | $(-129s + 117t)/5600$ |
| 14 | $T$`r3_000536` | $197/6720 + st/560$ |
| 15 | $T$`r3_000537` | $-s/1400 - 39t/2800$ |

These are four entries of the 21-term relation, not a replacement for its other terms. The packet retains all coefficients, including the longer fractions; the checker uses the entire relation.

The independent reader `c8_reviewability_20260927/finite_witnesses.py` uses Python's standard-library rational arithmetic to recalculate all four coordinates of every residual in (135). It imports neither the field engine nor a vertex-mode routine from the research scripts. This is a full replay of these twelve *saved-vector equalities*. The recorded run passed in 21.15 seconds with peak process RSS 39512 KiB. The construction of the saved vectors by applying $\Pi$, their normal-word membership, and the separate raising premises remain inputs from the authenticated producer records. In particular, this reader does not turn a stored projected vector into a fresh calculation of its ambient vertex modes.

The 33 projected vectors are also exported as a portable text packet, `c8_reviewability_20260927/gate4_vectors.jsonl.gz`: twelve targets followed by the 21 normal columns. Every line contains its name, exact rational coefficients, and the digest of its original vector record. A header binds the separate relation packet and records the original source hashes.

The packet contains 66610 sparse state terms and is approximately 2.45 MB compressed. The standalone verifier `c8_reviewability_20260927/check_gate4_packet.py` reads only this compressed text packet and `c8_reviewability_20260927/finite_witnesses.json`. It repeats all twelve full residual checks without opening an original SQLite database. Its output is `c8_reviewability_20260927/gate4_packet_checked.json`. All twelve residuals passed in 20.28 seconds with peak RSS 35380 KiB. This makes the residual arithmetic independently reproducible from the companion bundle while preserving the preceding boundary concerning the interpretation of the projected vectors.

The mathematical recovery step is short enough to verify directly. The earlier exact raising controls give $E_L t, E_R t \in W_4$. Since $W_4$ is stable under all current zero modes, $[\Pi t] = [t]$ in $B_4/W_4$. Since $N_4$ is current-stable, the right side of (135) lies in $N_4 \subset W_4$. Thus $t \in W_4$. Neither this inference nor the residual equality requires the 21 chosen columns to span the entire normal space.

To finish Gate 4, the saved matrices of $E_L, E_R$ on $P$ provide 644 columns, of rank 294. For each of the nine highest left fields, append its certified seed columns; the augmented rank is 322. The prime used to exhibit this full rank is

$$p = 1000003, \qquad s \longmapsto 231543, \qquad t \longmapsto 1000.$$

Both defining quadratic equations hold modulo $p$, and all denominators in a specialized column must be invertible. A reader can make the witness completely explicit by retaining the accepted column indices and nonzero pivot factors during elimination: their product, with the row-swap sign, is the determinant of the selected minor. The existing verifier recomputes the elimination; the new packet does not claim to export or recheck these $322 \times 322$ minors.

The direction of the rank argument matters. A nonzero minor after specialization proves that its characteristic-zero determinant is nonzero. More precisely, the reduction is a ring homomorphism on $\mathbb{Z}[D^{-1}, s, t] \subset F$, where $D$ contains the finitely many denominators used and $p \nmid D$; it is not a homomorphism from the whole characteristic-zero field to $\mathbb{F}_p$. Because $P$ already has exact dimension 322, the nonzero minor proves spanning of $P$. It supplies no characteristic-zero upper bound for an otherwise unknown space. If $C_a$ is the kernel of $[p] \mapsto [(a)_0 p]$, the resulting equality is $P = E_L P + E_R P + C_a$. The kernel is raising-stable. Cartan weights on $P$ satisfy $|m_L| \le 3$, $|m_R| \le 21$; every raising step increases $m_L + m_R$ by two. Thus every raising word of length 25 vanishes. Iterating the spanning equality in $P/C_a$ gives $P/C_a = 0$. This is the finite step that recovers all 2898 highest-left identities from the smaller collection of witnesses.

### S4.4 Gate 5: the entire small scalar calculation

Assume the already proved normal controls $(S_2)_0 N_4 \subset N_5$ and $(N_3)_0 S_3 \subset N_5$. If $P_4 \subset N_4$, Gate 5 follows immediately. Otherwise $Q = W_4/N_4$ is the triplet with weights $(0,-2),(0,0),(0,2)$. For its lowest vector $q_-$, define

$$\beta(a,p) = [a_0 p] \in Q, \qquad F(a,q) = [a_0 \tilde{q}] \in B_5/N_5.$$

The normal controls make $F$ well defined. The two saved bottom relations are

$$\begin{aligned}
\beta(a_3,p_3) &= c_3 q_-, & p_3 &= \texttt{r3_000910},\\
\beta(a_6,p_6) &= c_6 q_-, & p_6 &= \texttt{r3_000899},\\
c_3 &= \frac{-593619 + 1349829 s + 1399567 t - 1072149 st}{9831968},\\
c_6 &= -\frac{57967}{42018} + \frac{22295}{798342} st.
\end{aligned}$$

Their input weights are $(0,-14)$ and $(-1,-9)$, respectively. The vector $q_-$ is the image of the actual stored field `P4_minus`; this specifies its normalization. Both $c_3$ and $c_6$ are visibly nonzero, since $1, s, t, st$ are a $\mathbb{Q}$-basis and their rational coordinates are nonzero. The equations are quotient identities: the full records also give the normal-space terms that were suppressed here.

The lower-closure database reduces eight brackets. Seven are exactly zero as vectors in $S_3$:

$$(a_0)_0a_3 = (a_1)_0a_3 = (a_2)_0a_3 = (a_3)_0a_6 = (a_5)_0a_3 = (a_6)_0a_3 = (a_7)_0a_3 = 0. \tag{136}$$

There is one nonzero normal-product identity. Define

$$R_1 = \frac{1+4st}{25}(E_L)_{-1}a_1,$$
$$R_2 = \frac{81-15st}{71}\left((E_L)_{-1}a_2 - \frac{-7+3st}{20}R_1\right).$$

Then the identity, with its denominators cleared, is

$$554600(a_4)_0a_6 = (-216348s+83596t)R_1 + (34265s+103895t)R_2. \tag{137}$$

Both $R_1, R_2$ belong to $N_3 = K_{\mathrm{cur},-1}S_2 + TS_2$. Their saved row names are `r3_000141` and `r3_000150`. Their definitions follow directly from the two checkpoint recipes and the normalization rule (134); thus the displayed normal terms can be understood without treating a row identifier as an unexplained new field.

The precise lower records and weight exclusions are:

| $i$ | $j$ | Lower record | $\mathrm{wt}((a_i)_0p_j)$ | Bracket |
|---|---|---|---|---|
| 0 | 3 | 537 | $(0,-10)$ | zero |
| 1 | 3 | 777 | $(0,-6)$ | zero |
| 2 | 3 | 1017 | $(0,-6)$ | zero |
| 3 | 6 | 1286 | $(-1,3)$ | zero |
| 4 | 6 | 1526 | $(0,-8)$ | equation (137) |
| 5 | 3 | 1737 | $(1,-9)$ | zero |
| 6 | 3 | 1977 | $(1,-7)$ | zero |
| 7 | 3 | 2217 | $(1,-1)$ | zero |

For direct comparison with those database rows, the primitive right inputs have the one-term expansions

$$a_3 = \frac{6+2s-4t+st}{14}\,\texttt{r2_000047}, \qquad a_6 = \frac{307+297st}{1880}\,\texttt{r2_000076}.$$

For example, row 1526 records the mode $(a_4)_0\texttt{r2_000076}$. Multiplying its coordinates by $(307+297st)/1880$ gives exactly (137). The reader script independently performs this rational recombination for all eight rows. It also checks their stored residuals and the two bottom records, but does not recalculate these eight ambient mode products.

Every weight in the table is outside $Q$, so $\beta(a_i,p_j)=0$. The zero-mode Jacobi identity and the normal controls give

$$F(a_i,\beta(a_j,p_j)) = F(a_j,\beta(a_i,p_j)) = 0.$$

The displayed nonzero scalars imply $F(a_i,q_-)=0$ for all eight charged highest fields. Raise this *proved zero identity*, holding the left highest vector fixed, to obtain the remaining right

triplet inputs; then use current covariance for the left descendants. No assertion is made that the weight exclusion continues to hold after raising $p_j$.

The retained neutral field requires one additional input: actual charged brackets span $P$ as a current module. Its certificate uses 28 charged seeds and 644 lowering columns of rank 294; the augmented rank is 322. The same finite nilpotent argument therefore gives the full current span. Writing $e_0 a_3$, modulo $N_3$, as a sum of those charged brackets and applying Jacobi twice proves the neutral bottom identity. The other neutral field follows by subtracting the affine stresses and $e$ from $\omega_B$. This completes the 24 independent highest channels as consequences of lower identities; it does not require 24 separate new grade-five mode calculations.

### S4.5 Exact location and scope of the witnesses

The compact packet retains every coefficient in the twelve projected identities and in the eight small lower combinations. For comparison with the frozen sources, the following names are unambiguous:

- The twelve full projected vectors and relation records are in `hi_cft_20260923_c8two_h4_singlet_exact.sqlite`. Their selected-column recipes are in `hi_cft_20260923_c8two_h4_singlet_prime.sqlite`; the underlying normal-word recipes are in `hi_cft_20260923_next_n4_strong_skip_v2.sqlite`.
- The separate raising premise is `hi_cft_20260923_c8two_h4_singlet_premise_verified.json`. All seed references, quotient coordinates and saturation inputs are in `hi_cft_20260923_c8two_structure_saturation.json`. The completed mathematical reduction is recorded by `hi_cft_20260923_c8two_gate4_verified.json`.
- The eight rows in the table are primary keys of `identities` in `hi_cft_20260923_next_structure_lower_closure.sqlite`. The two bottom relations are named `A3_zero_r3_000910` and `A6_zero_r3_000899` in `hi_cft_20260923_c8two_targeted_h4_trimmed_v2/block_p0_m2.sqlite`.
- The charged-span verifier output is `hi_cft_20260923_c8two_charged_span_verified.json`. The Gate 5 coefficient summary is `hi_cft_20260923_c8two_gate5_verified.json`. When replaying its original checker, the `-charged-span` argument takes the producer `hi_cft_20260923_c8two_structure_charged_span.json`, whose schema contains the actual matrices and seeds.

The independent reader's complete invocation is

```
python hi_cft_safe_run.py --limit-mib 1024 -- \
  python c8_reviewability_20260927/finite_witnesses.py \
  --output c8_reviewability_20260927/finite_witnesses_review.json
```

For a bundle-only Gate 4 replay, the invocation is

```
python hi_cft_safe_run.py --limit-mib 1024 -- \
  python c8_reviewability_20260927/check_gate4_packet.py \
  --packet c8_reviewability_20260927/gate4_vectors.jsonl.gz \
  --relations c8_reviewability_20260927/finite_witnesses.json \
  --output /tmp/c8-gate4-packet-review.json
```

Both checkers refuse to overwrite an output; choose another fresh name if a review output already exists. The SQLite reader requires its output to remain inside the repository, whereas the portable checker accepts the displayed temporary path. They must be run by the coordinating agent when the numerical slot is free. Their allocation reviews bound individual decoded records and retain only one target residual and one source vector at a time. The permanent launcher supplies the desktop memory reserve, process-group watchdog, and two-CPU limit.

The source ledger and the small reader have complementary scopes. The ledger authenticates the unchanged construction files and fresh replay outputs. The reader exposes selected mathematical content and repeats a specified set of exact equalities. The finite-to-all-weights theorem still uses the proved filtered Wick and recursive zero-product argument, together with the full lower controls and Gates 4–5. Printing a sample scalar relation would not replace that theorem, just as hashing all its inputs would not prove a residual identity.

## S5 Structural inputs in a form suitable for direct checking

The manuscript's embedded structural-data certificate has four parts: the Potts factor, the level-thirteen parafermion factor and positive stress identity, the lattice–Potts core and its six spectral witnesses, and irreducibility on five finite top spaces. This section replaces short success flags by the equations they assert. It also distinguishes an explicit saved witness from a producer's assertion that such a witness was found. The formulas below do not certify their own membership in the algebra generated by the original fields.

### S5.1 Coordinates and the meaning of a check

Use the manuscript's simple-root basis $\alpha_0, \dots, \alpha_7$, Gram matrix $G$, lattice cocycle $\epsilon$, and transported positive form. Write

$$s^2 = 13, \qquad t^2 = -3, \qquad K = \mathbb{Q}(s,t), \qquad \overline{s} = s, \quad \overline{t} = -t.$$

A stored coefficient $(q_0, q_s, q_t, q_{st})$ means $q_0 + q_s s + q_t t + q_{st} st$. The state labelled by $(\beta; (i_1, \dots, i_r))$ is

$$h_{i_1 \bmod 8}(-1 - \lfloor i_1/8 \rfloor) \cdots h_{i_r \bmod 8}(-1 - \lfloor i_r/8 \rfloor) e^{\beta}.$$

Thus index $i + 8$ means $h_i(-2)$, not a second coordinate direction. Ordered oscillator lists represent commuting Heisenberg creation operators. The coefficient tables accompanying this section retain every rational coefficient and contain no floating-point approximations.

Write $D = L_B(-1)$, $:ab := a_{-1}b$, and $Y(a,z) = \sum_n a_n z^{-n-1}$. A polynomial identity below is an equality of actual lattice vectors in this basis. A check consists of expanding both sides, collecting equal basis states, and checking that each of the four rational components of every coefficient vanishes. No numerical tolerance enters this definition.

### S5.2 The Potts fields: a compact formula and two worked checks

Set

$$\alpha = (0,0,0,1,1,0,-1,-1), \qquad \beta = (1,1,0,-1,-1,0,0,0), \qquad \gamma = \alpha + \beta.$$

Their Gram matrix is $\left(\begin{smallmatrix} 4 & -2 \\ -2 & 4 \end{smallmatrix}\right)$. The complete Potts stress vector is

$$p = \frac{h_\alpha(-1)^2 + h_\beta(-1)^2 + h_\alpha(-1)h_\beta(-1)}{15}\mathbf{1} + \sum_{\delta = \pm\alpha, \pm\beta, \pm\gamma} p_\delta e^\delta, \tag{138}$$

where

| $\delta$ | $p_\delta$ | $p_{-\delta}$ |
|---|---|---|
| $\alpha$ | $-(s+3t)/100$ | $(-s+3t)/10$ |
| $\beta$ | $(7+3st)/100$ | $(7-3st)/100$ |
| $\gamma$ | $(-s+3t)/100$ | $-(s+3t)/10$ |

Let $A=\alpha-\beta$, $C=\alpha+2\beta$, $E=-2\alpha-\beta$, and $X_\delta=5p_\delta e^\delta$. Its weight-three generator is

$$w_P =: h_A h_C h_E : -9(: h_A(X_\gamma+X_{-\gamma}) : + : h_C(X_\alpha+X_{-\alpha}) : + : h_E(X_\beta+X_{-\beta}) :). \tag{139}$$

The triple product is the indicated nested Heisenberg product; its factors commute. These formulas contain the complete fields, not only their projections to a selected subspace. The independent readable-input checker has compared their complete 27- and 92-term expansions with the saved fields and obtained exact equality.

Here is a short check of $p_3 p = 2\mathbf{1}/5$. The Heisenberg part of $p$ is $2/5$ times the rank-two plane's conformal vector, and its contribution to $p_3 p$ is $4/25$. Each root has squared norm four, so $\epsilon(\delta,-\delta)=1$ in the specified cocycle. The table gives

$$p_\alpha p_{-\alpha} = \frac{s^2-9t^2}{1000} = \frac{1}{25}, \qquad p_\beta p_{-\beta} = \frac{49-9(st)^2}{10000} = \frac{1}{25}, \qquad p_\gamma p_{-\gamma} = \frac{1}{25}.$$

Both orders of each root pair occur. The total is therefore $4/25+6/25=2/5$, giving central charge $4/5$.

The transported PCT factor on $e^\delta$ is $g(\delta)=\prod_i(5,8,9,8,5,1,2,2)_i^{\delta_i}$. For $\alpha,\beta,\gamma$ these factors are $10,1,10$. For example $10\overline{p_\alpha}=p_{-\alpha}$; the other two pairs work in the same way. The quadratic Heisenberg part is PCT fixed. This verifies $\theta p=p$ directly.

The remaining field and membership checks have a precise finite form:

$$\begin{aligned} p_1 p &= 2p, & p_2 p &= 0, & p_0 p &= Dp, \\ p_1 w_P &= 3w_P, & p_n w_P &= 0 \quad (n\ge 2), & L_B(1) w_P &= 0, \\ p &\in S_2, & w_P &\in K_{-1}S_2 + DS_2 + \operatorname{span}(S_{2,0}S_2). \end{aligned}$$

In the last expression $S_{2,0}S_2$ means all $a_0 b$ with $a,b\in S_2$; one keeps total Cartan weight zero when checking $w_P$. The saved exact primary has 92 basis terms, and $p$ has 27. The original exact membership calculation reached rank 41 after starting with a rank-40 descendant span and adding one zero-weight pair. Its old record did not save a membership expression. The new export described in Section S5.4 supplies an explicit expression with 33 nonzero coefficients. The standard lattice Potts identification applies after this membership and the stated field identities are verified; the rank 41 alone does not establish membership.

### S5.3 Parafermion normalization and the complete polynomial rows

The exact parafermion stress $z$ and unnormalized primary $w$ are given by 64 and 342 basis terms, respectively. They are transcribed in `structural_fields.tsv`. Put $L=z$ and $a=\sigma w$, where

$$\sigma^2 = -\frac{7+3st}{9223500}, \qquad w_5 w = (-86086+36894st)\mathbf{1}.$$

No choice of square root is needed in the stored even identities; odd identities have a common factor $\sigma$. Indeed, $(st)^2=-39$ gives the worked normalization check

$$(-86086+36894st)(-7-3st) = 4919200, \qquad \sigma^2 w_5 w = \frac{8}{15}\mathbf{1}.$$

Define

$$c=\frac{8}{5},\qquad \ell=\frac{14}{473},\qquad b=a_1a,\quad d=a_1b,\quad R=a_1d.$$

All normal products in the following formulas retain their indicated order and parentheses. The full singular self-row is

| $n$ | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|
| $a_na$ | $\frac{8}{15}\mathbf{1}$ | 0 | $2L$ | $DL$ | $b$ | $\frac{1}{2}Db-\frac{1}{12}D^3L.$ |

(140)

Products with larger mode index vanish. The second row is

$$\begin{aligned}
a_nb&=0\quad(n\geq 4),\\
a_3b&=\frac{69283}{2365}a,\\
a_2b&=\frac{8}{3}\big(5-2\ell(2+c)\big)Da,\\
a_1b&=d,\\
a_0b&=\frac{2}{5}Dd+\frac{32\ell}{5}:LDa:-\frac{48\ell}{5}:DLa:+\frac{2}{15}\big(-5+2\ell(c-1)\big)D^3a.
\end{aligned}\tag{141}$$

For example the single coupling check is exactly

$$\frac{31-69283/2365}{16(2+8/5)}=\frac{4032/2365}{288/5}=\frac{14}{473}.$$

This arithmetic is necessary but is not the parafermion identification: the nonlinear rows and decoupling relation are needed.

The upper part of the third row is

$$\begin{aligned}
a_7d&=c\big(55-16\ell(2+c)\big)\mathbf{1},\\
a_6d&=0,\\
a_5d&=\frac{4}{3}\big(175-64\ell(2+c)\big)L,\\
a_4d&=\big(110-32\ell(2+c)\big)DL,\\
a_3d&=\big(95-16\ell(11+4c)\big)b+128\ell:LL:+\big(10+8\ell(c-1)\big)D^2L,\\
a_2d&=64\ell:DLL:+\big(75/2-4\ell(13+5c)\big)Db+\frac{-25+8\ell(9+5c)}{12}D^3L.
\end{aligned}$$

Here $:DLL:$ means $:(DL)L:$, not $D(:LL:)$. For $a_1d=R$ the exact polynomial is

$$\begin{aligned}
R=&-\frac{148744}{473}:Lb:+\frac{566160}{473}:aa:-\frac{4800}{121}:L(:LL:):\\
&+\frac{3860}{11}:(D^2L)L:-\frac{33320}{121}:(DL)(DL):-\frac{489817}{4730}D^2b+\frac{2193061}{52030}D^4L.
\end{aligned}\tag{142}$$

The last simple pole is

$$\begin{aligned}
a_0d=&\frac{1}{3}DR+\frac{448}{1419}:LDb:-\frac{896}{1419}:(DL)b:-\frac{224}{1419}:(D^3L)L:\\
&-\frac{34019}{28380}D^3b+\frac{9473}{170280}D^5L.
\end{aligned}\tag{143}$$

Thus every entry of all three singular rows is an explicit differential normal polynomial in $L,a,b,d$.

There is a useful small algebraic alternative to directly expanding the weight-seven vector in (143). Write $a_0 b = \beta D d + Q$, where $\beta = 2/5$ and $Q$ is the last three terms of (141). The iterate identity and self-skew symmetry give

$$(1-\beta)a_0 d = \beta DR + a_1 Q - \frac{1}{4}D(b_1 b) + \frac{1}{48}D^3(b_3 b) - \frac{1}{480}D^5(b_5 b). \tag{144}$$

The terms $b_1 b, b_3 b, b_5 b$ are obtained from the already given upper poles and $a_0 b$, without using $a_0 d$. Reordering uses, for example,

$$:aDa:-:(Da)a := \frac{1}{12}D^3 b - \frac{1}{60}D^5 L, \qquad :L(D^2 L):-:(D^2 L)L := \frac{1}{6}D^4 L.$$

Substituting $c, \ell$ in (144) gives precisely (143). A scalar verification of this implication checks an algebraic consequence of the other field identities; it does not independently verify those identities in the lattice VOA.

The seven coefficients of (142) also agree with the Dong–Lam–Yamada parafermion formulas after the recorded change of generators. The manuscript's finite reconstruction proposition then transports the full polynomial presentation to the actual simple generated algebra. The required actual-field inputs include

$$z_1 z = 2z, \quad z_2 z = 0, \quad z_3 z = \tfrac{4}{5}\mathbf{1}, \quad z_1 w = 3w, \quad z_n w = 0 \ (n \geq 2), \quad z_0 w = Dw,$$

PCT stability, and $z \in S_2$, $w \in S_3$. The membership construction reaches the 58-dimensional Cartan-zero grade-three span after 154 zero-weight products. Its new export supplies an expression for $w$ with 42 nonzero coefficients, as described next. The recorded computation of the full nonlinear product $a_0 b$ involves 25,045 nonzero basis terms; $R$ has 28,342. These are the substantial vector identities behind the short polynomials above. They can be checked by a sparse exact replay, but are not replaced by the two worked scalar checks.

### S5.4 Explicit membership expressions in generated words

The new file `membership_witnesses.json` contains 120 explicit generated $S_2$ vectors $v_j$, 58 independent raw grade-three source vectors $g_i$, and the two identities

$$w_P = \sum_{i \in I_P} \mu_i g_i, \quad |I_P| = 33, \qquad w = \sum_{i \in I_{13}} \nu_i g_i, \quad |I_{13}| = 42.$$

Every $g_i$ is labelled as $J_{-1}v_j$, $Dv_j$, or $(v_j)_0 v_k$. The complete coefficient lists are in `membership_witnesses.coefficients.tsv`; the JSON also contains every source and target coordinate. Thus the proposed linear identities are independently checkable without repeating the search for a suitable span or choosing new pivots.

For example, in the Potts expression

$$\mu_0 = \frac{1053}{70}, \qquad \mu_1 = -\frac{729}{280}, \qquad \mu_2 = \frac{1647}{280}, \qquad \mu_{40} = -\frac{1}{3985293312000},$$

where

$$\begin{aligned} g_0 &= (H_{\text{left}})_{-1} v_0, & v_0 &= (H_{\text{left}})_{-2}\mathbf{1}, \\ g_1 &= (H_{\text{left}})_{-1} v_1, & v_1 &= (H_{\text{left}})_{-1} H_{\text{left}}, \\ g_2 &= (H_{\text{left}})_{-1} v_4, & v_4 &= (H_{\text{left}})_{-1} H_{\text{right}}, \\ g_{40} &= (v_{92})_0 v_{119}. \end{aligned}$$

The remaining 29 coefficients and the exact $v_{92}, v_{119}$ appear in the same files. These are raw source words, not normalized row-reduction vectors with an unstated change of basis.

The exporter uses the original low-grade mode engine and tracks all linear combinations. It finished with 43,462 persistent sparse terms and about 83 MiB peak resident memory. An independent coordinate-sum checker, `check_membership_coordinates.py`, uses only exact rational arithmetic in $K$. Its passing record checks both literal sums and all 21,772 decoded basis rows, including source grades and the Cartan-zero condition; duplicate basis rows and duplicate JSON keys are rejected. Two boundaries remain separate: checking a sum of supplied vectors is not checking that each supplied vector equals the mode word named by its recipe, and the generation of all 120 $v_j$ by the original short generating set uses the finite construction certificate. The export makes both boundaries explicit instead of encoding membership by a truth value.

### S5.5 Where the positive stress identity is checked

Section S3 prints the full four-axis Gram and overlap table and checks the resulting rational identity. The independent calculation uses the actual saved $p, z, e$ and the affine stresses; it imports no original mode engine. Its passing record is `c8_reviewability_20260927/readable_checks.json`, and the exact fields are in `grade2_fields.tsv`. Identifying these orthogonal projections with Haar averages still uses the certified fact that the four stresses span the current-invariant part of the constructed $S_2$. This is a statement about $S_2$, and does not require $S_2 = U_2$ or properness.

### S5.6 The lattice core and six small spectral witnesses

Return to the lattice vectors $\ell, r$ from the small lattice calculation, and write $H = r$ for the right Cartan vector. The rank-two lattice core has basis $\ell$ and $v = (1, 1, 0, 1, 1, -3, 1, 1)$, with

$$G_{\mathrm{core}} = \begin{pmatrix} 2 & 3 \\ 3 & 24 \end{pmatrix}, \quad \det G_{\mathrm{core}} = 48 - 9 = 39, \quad \rho = v/3 = \ell/2 + H/6, \quad \rho^2/2 = 4/3.$$

The actual three-term field $u$ supplies the compact identities

$$p_1 u = \tfrac{2}{3} u, \qquad p_2 u = p_3 u = 0, \qquad u_{-5}(u_{-3} u) = -\tfrac{3}{5} e^v.$$

The last is a leading-coefficient lattice identity. Its small coefficient can be checked directly from the three exponential terms of $u$; it does not require constructing an ambient basis in weight twelve. Together with PCT and the left currents, these identities place the rank-two lattice and the charged simple-current sectors in the generated algebra.

The resulting core stress is

$$\omega_{\mathcal{T}} = \tfrac{1}{4} h_\ell (-1)^2 + \tfrac{1}{156} h_H (-1)^2 + p, \qquad c_{\mathcal{T}} = 14/5.$$

Put $q = \omega_B - \omega_{\mathcal{T}} - z$, of central charge $18/5$. The six order-thirteen witnesses are as follows. Every row names a nonzero actual vector in $S_2$; each corresponding Cartan block inside the constructed $S_2$ has dimension three.

| $(h_\ell(0), h_H(0))$ | $\dim\ker(q_1)$ | $q_1$ | $z_1$ | $p_1$ | $(\omega_{\mathcal{T}})_1$ |
|---|---|---|---|---|---|
| $(-1, -5)$ | 1 | 0 | 12/13 | 2/3 | 14/13 |
| $(-1, 5)$ | 1 | 0 | 12/13 | 2/3 | 14/13 |
| $(0, -8)$ | 1 | 0 | 12/13 | 2/3 | 14/13 |
| $(0, 8)$ | 1 | 0 | 12/13 | 2/3 | 14/13 |
| $(1, -5)$ | 1 | 0 | 12/13 | 2/3 | 14/13 |
| $(1, 5)$ | 1 | 0 | 12/13 | 2/3 | 14/13 |

(145)

The full six vectors, each with 21 basis terms, are printed in `core_spectral_witnesses.tsv`. For example, the first vector has nonzero coefficient $35/16 + (175/1872)st$ at momentum $(-1,0,1,1,1,1,0,0)$, and coefficient $-203s/208 - 161t/48$ at $(0,0,0,0,0,1,1,1)$. The former cannot vanish, since $(st)^2 = -39$ while its putative rational value $-234/10$ would have positive square. Each line of (145) asserts the residual equations

$$q_1 v = 0, \quad (13z_1 - 12)v = 0, \quad (3p_1 - 2)v = 0, \quad (13(\omega_{\mathcal{T}})_1 - 14)v = 0, \quad v \in S_2.$$

These can be checked separately, one 21-term vector at a time; the table of eigenvalues is not by itself the verification.

For a hand check of the module label, parafermion lowest weights are

$$h_{i,j} = \frac{i(13-i)}{390} + \frac{j(i-j)}{13}, \qquad 1 \le i \le 13, \quad 0 \le j < i.$$

Solving $h_{i,j} = 12/13$ means $i(13-i) + 30j(i-j) = 360$. Reduction modulo three forces $i \equiv 0, 1 \pmod 3$. The candidates $i = 1, 3, 4, 6, 7, 9, 10, 12$ give, respectively, $i(13-i) = 12, 30, 36, 42, 42, 36, 30, 12$. Except for $i = 3, 10$, the difference from 360 is not divisible by 30. For $i = 3, 10$ it would require $j(i-j) = 11$, impossible for the permitted integral $j$. Thus $i = 13$ and $j(13-j) = 12$, giving $j = 1, 12$. These are inverse simple currents. This finite arithmetic check explains why the specific energy $12/13$ matters; it does not produce the lattice eigenvectors.

## S5.7 Finite irreducibility: the criterion and actual row recipes

The following elementary argument turns cyclicity into the precise irreducibility needed here. Let $E$ be a finite-dimensional Hilbert space and $\mathcal{A} \subset \mathrm{End}(E)$ an adjoint-closed algebra containing two commuting selfadjoint Cartan operators. Suppose a joint eigenspace is the line $\mathbb{C}v$, and $\mathcal{A}v = E$. If $W$ is invariant, its orthogonal complement is invariant, so the orthogonal projection $P_W$ commutes with $\mathcal{A}$. It preserves $\mathbb{C}v$ and acts there as zero or one. Commuting with every $A \in \mathcal{A}$ and using cyclicity then gives $P_W = 0$ or 1. Hence $E$ is irreducible.

Let $K_{\mathrm{cur}}$ be the span of the six affine currents. For the eighty-current space, $E = K_{\mathrm{cur}}^{\perp} \cap B_1$ has dimension 80. The operators are current zero modes and the zero modes $a_1$ of the named weight-two fields and their adjoints. The distinguished Cartan line has weight $(0, 10)$. The old v2 record saved only assertions and dimensions. The new `current80_witness.json` instead gives the full 86-state ambient basis and positive Hermitian Gram matrix, an explicit 80-row basis of $E$, all 30 zero-mode and adjoint matrices, and all 80 cyclic rows with their parent, operator, elimination coefficients, normalization and pivot. The matrices have only 3,486 nonzero entries altogether. The exporter recomputes invariance, adjointness and lowering-mode annihilation with the original exact low-grade engine; it took about 1.3 seconds and 36 MiB. Thus these are finite matrices and row identities available for inspection, rather than an unexplained rank assertion.

A first cyclicity step can also be checked directly. The seed is $e^{\beta}$, where $\beta = (1,1,1,1,1,0,0,0)$. The left and right Cartan weights are 0 and 10. In the prescribed cocycle,

$$(F_{\mathrm{right}})_0 e^{\beta} = 5e^{(0,1,1,1,1,0,0,0)} - 5e^{(1,1,1,1,0,0,0,0)}.$$

Only the negative simple roots $-\alpha_0, -\alpha_4$ contribute: their pairings with $\beta$ are $-1$, and the corresponding cocycle signs are $-1, +1$. Both coefficients in $F_{\mathrm{right}}$ are $-5$. Dividing this displayed result by five gives the first nontrivial normalized cyclic row in the export.

The independent stdlib checker `check_current80_matrices.py` has passed; its result is `current80_matrix_checks.json`. It checks the lattice Gram matrix by exact positive $LDL^{\mathsf{T}}$

decomposition, the root-state metric by the positive torus factors, all fifteen adjoint pairs, orthogonality and independence of the complement, invariance under all thirty matrices, the two Cartan matrices and their unique $(0, 10)$ eigenline, and the eighty cyclic row recipes. The eight positive diagonal entries of $D$ are

$$2, \quad \frac{3}{2}, \quad \frac{4}{3}, \quad \frac{5}{4}, \quad \frac{6}{5}, \quad \frac{1}{2}, \quad 2, \quad \frac{3}{2}.$$

The check used about 25 MiB of resident memory. Its scope is the supplied finite matrices. Identifying those matrices with the actual field modes and checking lowering-mode annihilation remain separate from this independent linear-algebra verification.

The four 81-dimensional mixed tops have more complete saved data. Each has 81 root states of squared norm two, a positive diagonal metric, all 81 elimination rows, and their construction recipes. All four have the same Cartan multiplicities:

| $h_\ell(0) = \pm 1,\ h_H(0)$ | $-7$ | $-5$ | $-3$ | $-1$ | $1$ | $3$ | $5$ | $7$ |
|---|---|---|---|---|---|---|---|---|
| multiplicity | 1 | 2 | 3 | 3 | 3 | 3 | 2 | 1 |

| $h_\ell(0) = 0,\ h_H(0)$ | $-10$ | $-8$ | $-6$ | $-4$ | $-2$ | $0$ | $2$ | $4$ | $6$ | $8$ | $10$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| multiplicity | 1 | 2 | 4 | 5 | 7 | 7 | 7 | 5 | 4 | 2 | 1. |

Each outer row sums to 18 and the middle row to 45, giving $18 + 45 + 18 = 81$. In particular $(0, 10)$ is a unique line. The saved seed has index 68 in the ordered root list. Multiplying its momentum by three gives

| (glue sign, sector sign) | $3\beta_{68}$ |
|---|---|
| $(+, +)$ | $(1, 2, 3, 4, 2, 0, 2, 1)$ |
| $(+, -)$ | $(2, 4, 3, 2, 1, 0, 1, 2)$ |
| $(-, +)$ | $(1, 2, 3, 4, 2, 0, 1, 2)$ |
| $(-, -)$ | $(2, 4, 3, 2, 1, 0, 2, 1)$ |

Here is the complete first nontrivial cyclicity step in the $(+, +)$ case. Denote the root basis by $e_j$ and let $M_9$ be the adjoint of the right raising-current zero mode, in the exact metric. The saved recipe is

$$r_0 = e_{68}, \qquad r_1 = \tfrac{1}{8} M_9 r_0 = e_{62} - \tfrac{1}{4} e_{67}.$$

Thus the actual matrix identity to check is $M_9 e_{68} = 8e_{62} - 2e_{67}$, with

$$3\beta_{62} = (1, 2, 3, 1, 2, 0, 2, 1), \qquad 3\beta_{67} = (1, 2, 3, 4, 2, 0, -1, 1).$$

Both output states have Cartan weight $(0, 8)$, as predicted by the current's weight. Subsequent steps have the exact format

$$r_k = \lambda_k \left( M_{j_k} r_{p_k} - \sum_{i<k} c_{ki} r_i \right), \qquad \lambda_k \neq 0, \tag{146}$$

and each $r_k$ has a new normalized pivot. The saved recipes list $p_k, j_k, \lambda_k, c_{ki}$ explicitly; all $p_k < k$. Checking (146) and distinct pivots proves that 81 independent vectors belong to the cyclic span. Counting the saved rows without checking their recipes does not establish cyclicity.

The matrices themselves need not be reconstructed by a general VOA engine. On a pure root state $e^\beta$, the relevant zero modes of the weight-one and weight-two summands have the

elementary rules

| state | zero-mode coefficient or action |
|---|---|
| $h_i(-1)\mathbf{1}$ | $(\alpha_i,\beta)$ |
| $h_i(-2)\mathbf{1}$ | $-(\alpha_i,\beta)$ |
| $h_i(-1)h_j(-1)\mathbf{1}$ | $(\alpha_i,\beta)(\alpha_j,\beta)$ |
| $e^{\alpha}$ | $g_0(\alpha)\epsilon'(\alpha,\beta)e^{\alpha+\beta}$ |
| $h_i(-1)e^{\alpha}$ | $(\alpha_i,\beta)g_0(\alpha)\epsilon'(\alpha,\beta)e^{\alpha+\beta}$. |

In the last two rows the term contributes only when $\alpha+\beta$ is in the same norm-two root list. The cocycle $\epsilon'$ of the chosen overlattice and its quadratic gauge correction $g_0$ must be used; they are distinct from the positive torus factor in the Hermitian metric. With diagonal metric $\mathrm{diag}(d_i)$, each adjoint entry is checked by

$$(M^*)_{ji}=\overline{M_{ij}}\frac{d_i}{d_j}.$$

This is a finite exact matrix problem. Passing from these irreducible tops to entire irreducible restrictions uses the independently proved bound $h<2$ and regularity. It is not a consequence of finite cyclicity alone, and the four mixed restrictions are not asserted to be mutually isomorphic.

### S5.8 Source-to-equation map and remaining verification boundary

Paths beginning `hi_cft_` refer to the repository root. Unprefixed companion filenames throughout this section, such as `structural_fields.tsv` and `current80_witness.json`, are relative to `c8_reviewability_20260927/`; explicitly prefixed paths remain repository-relative. The accompanying TSV files contain exact transcriptions, not new checks.

- `hi_cft_20260923_proof_potts_exact.json`: keys `potts_stress`, `weight_three_primary`; the compact formulas (138)–(139). The earlier `hi_cft_20260923_proof_potts.json` contains the exact stress checks but only a numerical test of primary membership; the later exact record supersedes that numerical membership test.

- `hi_cft_20260923_proof_pf13_exact.json`: keys `stress`, `primary` and `K_invariant_stress_projection_coefficients`. Membership, PCT and translation use the separate `hi_cft_20260923_proof_pf13_membership.json` and `hi_cft_20260923_proof_remaining_coset.json`.

- The parafermion rows use `hi_cft_20260923_proof_pf13_upper_poles.json`, `hi_cft_20260923_proof_pf13_nonlinear.json`, `hi_cft_20260923_proof_pf13_truncation.json`, `hi_cft_20260923_proof_pf13_simple_pole_algebra.json`, and `hi_cft_20260923_proof_pf13_dly.json`, with the producer formulas specified by those records. The upper-pole continuation imports fifteen completed assertions from the earlier producer log; its final record is not sixteen freshly independent calculations.

- `hi_cft_20260923_next_structure_core_extension.json`: for each entry of `blocks`, use its `Cartan_weight`, then the `joint_eigenspaces` entry with `PF13_weight=12/13`, then `vectors[0].vector`. These are the six full witnesses in `core_spectral_witnesses.tsv`.

- `hi_cft_20260923_proof_current_module_v2.json`: the original eighty-current assertions. These are supplemented by `current80_witness.json`, containing explicit matrices and cyclic-basis recipes, and the separate independent matrix checker.

- `hi_cft_20260923_next_81top_all.json`: each entry of `sectors` contains `roots`, `metric`, `rows`, and `recipes`. Its `operator_matrices` entries contain hashes and sparse term counts, not the matrix entries themselves. The elementary formulas above reconstruct the latter. The files `top81_seeds_and_first_steps.tsv` and `top81_pivots.tsv` make the small identifying data readable.

The most economical further certificates would therefore be: independent checks of the exported membership words' mode-product provenance and the finite top matrices' actual field actions, and independent coefficient-by-coefficient checks of the substantial parafermion normal-product identities. The small scalar and matrix checks printed here expose their mathematical content and allow parts to be verified by hand. They do not remove those larger exact-vector obligations or formalize the ambient VOA in a proof assistant.

## S6 Reproduction and the remaining large checks

The new small checker is `c8_reviewability_20260927/check_readable_inputs.py`. It uses only Python's exact rational arithmetic and the displayed formulas; it imports none of the research mode engine. Run from the repository:

```
python hi_cft_safe_run.py --limit-mib 512 -- \
  timeout 120s python \
  c8_reviewability_20260927/check_readable_inputs.py \
  --output-dir /tmp/c8-readable-review
```

This writes a fresh report in the chosen output directory; use a new directory if it already exists. The report from this revision is `c8_reviewability_20260927/readable_checks.json`. The source hashes in that report identify the exact coordinate inputs. The mathematical content is the recomputation; the hashes identify which inputs were used. The complete current singular brackets, the highest-vector conditions, the stress Gram matrix, the overlaps, the projected identity, both complete compact Potts formulas and the eight primary energy sums are checked. The script does not check every higher OPE or prove that a saved field belongs to the algebra generated by the short defining set.

The old large calculations remain relevant. In particular, dimensions 120, 956 and 322 and the full Gate 4 closure are not established by inspecting a representative row. The preceding sections specify which actual saved witnesses can be checked and which missing witness formats would make the remaining verification easier. The integrated formalization assessment in Appendix S26 explains how these finite interfaces could become kernel-checked theorems, and why doing so would still leave substantial VOA, tensor-category and analytic foundations.

# Part II

# Additional finite construction records

## S7 Expanded finite-lattice reference: coordinates, ranks, and recipes

This section brings the finite construction's scattered ingredients into one reading order. It prints the complete nine weight-two input fields, the short-generation recipe graph, the grade-three weight census, and the additional Gate 4 test list. It also records what the historical numerical calculations found before exact certificates became available. The arithmetic tables are transcriptions of saved records, with rational coefficients rewritten in the basis $1, s, t, st$, where $s^2 = 13$ and $t^2 = -3$. Their inclusion here is a documentation step, not a fresh replay of the vertex-mode calculations.

The distinction between three kinds of assertion is used throughout. An *ambient identity* compares actual lattice vectors. A *generation recipe* establishes that such a vector is in the VOA generated by the six currents, $\omega_B$, and $u$. A *closure argument* shows that all higher products stay in the normal span. The exact fields alone supply none of the last two implications. Conversely, a successful normal-space membership test need not compute the full dimension of the ambient normal space.

### S7.1 A fully specified coordinate system

For direct checking the Gram matrix is

$$G = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

All momenta are columns in this ordered simple-root basis, even though the tuple notation suppresses transposition. Thus $(\alpha, \beta) = \alpha^{\mathsf{T}} G \beta$. Let

$$[\lambda; \mathcal{O}] = \prod_{k \in \mathcal{O}} \alpha_{k \bmod 8}(-1 - \lfloor k/8 \rfloor) e^{\lambda}, \qquad \mathrm{wt}[\lambda; \mathcal{O}] = \tfrac{1}{2}\lambda^{\mathsf{T}} G \lambda + \sum_{k \in \mathcal{O}} (1 + \lfloor k/8 \rfloor).$$

The oscillator list is sorted and repeated entries are retained. The empty list means the pure exponential $e^{\lambda}$. These are unnormalized simple-root oscillators. In particular, $[0; (i, i)]$ has no hidden factor of $\sqrt{2}$ or $1/2$.

Two practical examples fix the mode conventions:

$$[0; (0, 8)] = \alpha_0(-1)\alpha_0(-2)\mathbf{1}, \qquad [\lambda; (6)] = \alpha_6(-1)e^{\lambda}.$$

In the exact engine, a coefficient encoded by $[[q_0, q_s], [q_t, q_{st}]]$ means $q_0 + q_s s + q_t t + q_{st} st$. The two inner lists separate the *rational* $+ s$ and $t + st$ parts; they are not numerical real and imaginary coordinates. Decimal fields retained in the old JSON are historical reconstruction diagnostics and are omitted from the exact tables below.

For an independent implementation, all required arithmetic can be done with rational fractions. If $x = x_0 + x_1 s + x_2 t + x_3 st$ and $y = y_0 + y_1 s + y_2 t + y_3 st$, then

$$\begin{aligned}(xy)_0 &= x_0y_0 + 13x_1y_1 - 3x_2y_2 - 39x_3y_3,\\ (xy)_1 &= x_0y_1 + x_1y_0 - 3x_2y_3 - 3x_3y_2,\\ (xy)_2 &= x_0y_2 + x_2y_0 + 13x_1y_3 + 13x_3y_1,\\ (xy)_3 &= x_0y_3 + x_3y_0 + x_1y_2 + x_2y_1.\end{aligned}$$

Inversion can be performed by multiplying the other three Galois conjugates and dividing by their product with $x$. Zero testing is four rational zero tests. Complex conjugation is the automorphism fixing $s$ and negating $t$; it must still be combined with momentum reversal and the PCT gauge factor when applied to a state.

## S7.2 Reconstructing an arbitrary mode from the printed inputs

The root cocycle in these coordinates is

$$\epsilon(\alpha,\beta) = (-1)^{\sum_i \alpha_i\beta_i + \sum_{i>j} \alpha_i\beta_j G_{ij}}.$$

Its exponent is bilinear modulo two, so associativity of the twisted lattice multiplication follows directly. For a root $\alpha$, $\epsilon(\alpha,-\alpha) = -1$; this accounts for the minus sign in both lowering-current formulas. A checker that uses the opposite root sign and leaves all field coefficients unchanged is checking a different coordinate convention.

For a pure exponential acting on an arbitrary oscillator state, the following finite coefficient extraction is sufficient:

$$\begin{aligned}Y(e^\alpha,z)\prod_{j=1}^{r} h_j(-n_j)e^\beta = \epsilon(\alpha,\beta)z^{(\alpha,\beta)}\prod_{j=1}^{r}\left(h_j(-n_j) - (h_j,\alpha)z^{-n_j}\right)\\ \cdot\exp\left(\sum_{n>0}\frac{\alpha(-n)}{n}z^n\right)e^{\alpha+\beta}.\end{aligned} \tag{147}$$

To obtain $(e^\alpha)_m$, keep the coefficient of $z^{-m-1}$. Expanding the finite product corresponds to choosing a subset of the input oscillators to contract. If $S$ is the contracted subset, the required degree in the creation exponential is

$$d_S = -m - 1 - (\alpha,\beta) + \sum_{j\in S} n_j.$$

Terms with $d_S < 0$ are absent. At a fixed target conformal weight there are only finitely many partitions of $d_S$ and finitely many momenta. This is the mathematical meaning of the exact engine's contraction-mask loop and its creation-polynomial cache.

The first creation polynomials, useful for manually checking the weight-two rows, are

$$\begin{aligned}C_0(\alpha) &= 1,\\ C_1(\alpha) &= \alpha(-1),\\ C_2(\alpha) &= \tfrac{1}{2}\alpha(-2) + \tfrac{1}{2}\alpha(-1)^2,\\ C_3(\alpha) &= \tfrac{1}{3}\alpha(-3) + \tfrac{1}{2}\alpha(-2)\alpha(-1) + \tfrac{1}{6}\alpha(-1)^3.\end{aligned}$$

In simple-root coordinates the mixed coefficient of $\alpha_i(-1)\alpha_j(-1)$ in $C_2$ is $\alpha_i\alpha_j$ for $i \neq j$, and the diagonal coefficient is $\alpha_i^2/2$.

Heisenberg modes are even simpler. For $m > 0$, $h(m)$ contracts each occurrence of $k(-m)$ with coefficient $m(h,k)$, including its multiplicity. The zero mode is multiplication by $(h,\beta)$, and negative modes insert another oscillator. General oscillator fields are obtained from normal products and derivatives:

$$Y(h(-n)a,z) = \frac{1}{(n-1)!} : (\partial_z^{n-1} h(z)) Y(a,z) :, \qquad (Ta)_m = -m a_{m-1}.$$

The convention $a_m b \in B_{\mathrm{wt}\, a+\mathrm{wt}\, b-m-1}$ provides a check on every intermediate state. No matrices for all ambient grades are required to interpret this recipe.

### S7.3 Ambient census and the affine search space

The saved lattice enumeration finds 78 norm-two vectors, 702 norm-four vectors, and 2346 norm-six vectors. The ambient graded dimensions are

$$\dim B_0 = 1, \quad \dim B_1 = 86, \quad \dim B_2 = 1370, \quad \dim B_3 = 11586.$$

The first two nontrivial counts can be checked without a VOA product:

$$86 = 8 + 78, \qquad 1370 = (8+36) + 78 \cdot 8 + 702.$$

At weight three the rank-eight oscillator counts are $8 + 64 + 120 = 192$, so

$$11586 = 192 + 78 \cdot 44 + 702 \cdot 8 + 2346.$$

Here $44 = 8 + 36$ is the number of oscillator states of degree two; 120 is the number of three-oscillator monomials of mode number one. These are ambient counts, unrelated to the later 120-dimensional selected space except through inclusion.

The original highest-space search imposed the two raising zero modes and all six positive current modes at weight two. Its saved list of ambient affine-primary multiplicities is printed below. The entries $(n,m)$ are representation dimensions; the Cartan eigenvalues of a highest vector are $(n-1, m-1)$. The last column is the dimension of the ambient joint Cartan eigenspace before imposing the kernel equations. This table is a floating kernel search result, from `hi_cft_20260923_c8_lattice_control.json`; it is useful for locating the exact selected fields, but its multiplicities are not being promoted here to a new exact certificate.

| Type $(n,m)$ | Primary multiplicity | Cartan-block dimension |
|---|---|---|
| $(1,1)$ | 4 | 86 |
| $(1,3)$ | 6 | 78 |
| $(1,5)$ | 11 | 66 |
| $(1,7)$ | 9 | 48 |
| $(1,9)$ | 11 | 34 |
| $(1,11)$ | 7 | 19 |
| $(1,13)$ | 5 | 9 |
| $(1,15)$ | 2 | 3 |
| $(1,17)$ | 1 | 1 |
| $(2,2)$ | 4 | 47 |
| $(2,4)$ | 5 | 42 |
| $(2,6)$ | 8 | 34 |

| Type $(n,m)$ | Primary multiplicity | Cartan-block dimension |
|---|---|---|
| $(2,8)$ | 7 | 23 |
| $(2,10)$ | 5 | 14 |
| $(2,12)$ | 3 | 7 |
| $(2,14)$ | 3 | 3 |

The same exploratory record gives a largest current-algebra and adjoint residual of approximately $1.16\times10^{-14}$. This measures the numerical realization in its original orthonormal basis, whereas the subsequent exact current formulas use the unnormalized simple-root basis and the transported torus gauge.

## S7.4 Complete charged highest fields and retained neutral stress

The next nine tables give every nonzero coordinate of the eight charged highest fields $a_0,\dots,a_7$ and the retained neutral stress $e$. A row $(\lambda,\mathcal{O},c)$ means $c[\lambda;\mathcal{O}]$, and the field is the sum of all its rows. They therefore specify actual ambient lattice vectors without opening an NPZ file or a recognition script. The second neutral field is recovered exactly as

$$f=\omega_B-\omega_{A_{1,1}}-\omega_{A_{1,39}}-e.$$

Its independent coordinate table is unnecessary once the currents, ambient stress, and $e$ are given.

The source is `hi_cft_20260923_c8_exact_fields.json`. That file retains its original historical scope string, "Floating reconstruction conjectures, requiring subsequent exact verification." The later exact and exact-unitary certificates verify the corrected rational entries as lattice fields. In particular, the recorded repair changes the coefficients of the two neutral oscillator states $[0;(0,7)]$ and $[0;(4,6)]$ from

$$\frac{1799-503s}{4085}\quad\text{to}\quad-\frac{260}{3813}+\frac{5s}{279}.$$

The tables below use the corrected entries, not the saved numerical recognition approximations. The repair was constrained inside the four-dimensional commutant space; its provenance is recorded in the source's `stress_repair` object.

The remaining descendants of a highest field of type $(n,m)$ are obtained by the specified current modes

$$(F_L)_0^i(F_R)_0^j a,\qquad 0\le i<n,\quad 0\le j<m.$$

There are no divided powers in this recipe. This convention matters when comparing coefficients with checkpoint vectors, which may instead have been normalized to a pivot equal to one.

### S7.4.1 Field $a_0$: type $(1,5)$, 61 terms

The $e_1$ eigenvalue is $\frac{798}{1271}$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,0,1,1,1,0,0,0)$ | $\varnothing$ | $\frac{1159+589st}{10920}$ |
| $(0,0,0,0,0,0,1,1)$ | $(0)$ | $\frac{-57s-95t}{364}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,0,0,0,0,0,1,1)$ | $(1)$ | $\frac{-19s+19t}{182}$ |
| $(0,0,0,0,0,0,1,1)$ | $(2)$ | $\frac{57t}{182}$ |
| $(0,0,0,0,0,0,1,1)$ | $(3)$ | $\frac{19s+19t}{182}$ |
| $(0,0,0,0,0,0,1,1)$ | $(4)$ | $\frac{57s-95t}{364}$ |
| $(0,0,0,0,0,0,1,1)$ | $(6)$ | $-1$ |
| $(0,0,0,0,0,0,1,1)$ | $(7)$ | $1$ |
| $(0,0,0,0,1,0,0,1)$ | $\varnothing$ | $\frac{16+19s-57t}{182}$ |
| $(0,0,0,0,1,0,1,0)$ | $\varnothing$ | $\frac{16-19s+57t}{182}$ |
| $(0,0,0,1,0,0,0,1)$ | $\varnothing$ | $\frac{140+19s+57t}{728}$ |
| $(0,0,0,1,0,0,1,0)$ | $\varnothing$ | $\frac{140-19s-57t}{728}$ |
| $(0,0,0,1,1,0,0,0)$ | $(0)$ | $\frac{703+133st}{3640}$ |
| $(0,0,0,1,1,0,0,0)$ | $(1)$ | $\frac{57+19st}{728}$ |
| $(0,0,0,1,1,0,0,0)$ | $(2)$ | $\frac{-57-57st}{1820}$ |
| $(0,0,0,1,1,0,0,0)$ | $(3)$ | $\frac{-167-57st}{1820}$ |
| $(0,0,0,1,1,0,0,0)$ | $(4)$ | $\frac{11}{182}$ |
| $(0,0,0,1,1,0,0,0)$ | $(6)$ | $\frac{-76-19s+95t}{728}$ |
| $(0,0,0,1,1,0,0,0)$ | $(7)$ | $\frac{-76+19s-95t}{728}$ |
| $(0,0,1,0,0,0,0,1)$ | $\varnothing$ | $\frac{62+57t}{273}$ |
| $(0,0,1,0,0,0,1,0)$ | $\varnothing$ | $\frac{62-57t}{273}$ |
| $(0,0,1,0,1,0,0,0)$ | $\varnothing$ | $\frac{-89-19st}{1365}$ |
| $(0,0,1,1,0,0,0,0)$ | $(0)$ | $\frac{95-19st}{1092}$ |
| $(0,0,1,1,0,0,0,0)$ | $(1)$ | $\frac{57+19st}{728}$ |
| $(0,0,1,1,0,0,0,0)$ | $(2)$ | $\frac{13+19st}{728}$ |
| $(0,0,1,1,0,0,0,0)$ | $(3)$ | $\frac{-29-19st}{1092}$ |
| $(0,0,1,1,0,0,0,0)$ | $(4)$ | $\frac{-95-19st}{1092}$ |
| $(0,0,1,1,0,0,0,0)$ | $(6)$ | $\frac{-76-19s+95t}{2184}$ |
| $(0,0,1,1,0,0,0,0)$ | $(7)$ | $\frac{-76+19s-95t}{2184}$ |
| $(0,0,1,1,1,0,-1,0)$ | $\varnothing$ | $\frac{-76-19s+95t}{1092}$ |
| $(0,0,1,1,1,0,0,-1)$ | $\varnothing$ | $\frac{-76+19s-95t}{1092}$ |
| $(0,1,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{140-19s+57t}{728}$ |
| $(0,1,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{140+19s-57t}{728}$ |
| $(0,1,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{111+171st}{3640}$ |
| $(0,1,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{1}{28}$ |
| $(0,1,1,0,0,0,0,0)$ | $(0)$ | $\frac{95-19st}{1092}$ |
| $(0,1,1,0,0,0,0,0)$ | $(1)$ | $\frac{29-19st}{1092}$ |
| $(0,1,1,0,0,0,0,0)$ | $(2)$ | $\frac{-13+19st}{728}$ |
| $(0,1,1,0,0,0,0,0)$ | $(3)$ | $\frac{-57+19st}{728}$ |
| $(0,1,1,0,0,0,0,0)$ | $(4)$ | $\frac{-95-19st}{1092}$ |
| $(0,1,1,0,0,0,0,0)$ | $(6)$ | $\frac{76-19s-95t}{2184}$ |
| $(0,1,1,0,0,0,0,0)$ | $(7)$ | $\frac{76+19s+95t}{2184}$ |
| $(0,1,1,1,0,0,-1,0)$ | $\varnothing$ | $\frac{-76+95t}{1092}$ |
| $(0,1,1,1,0,0,0,-1)$ | $\varnothing$ | $\frac{-76-95t}{1092}$ |
| $(0,1,1,1,1,0,-1,-1)$ | $\varnothing$ | $\frac{19s}{1092}$ |
| $(1,0,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{16-19s-57t}{182}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(1,0,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{16+19s+57t}{182}$ |
| $(1,0,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{22}{65}$ |
| $(1,0,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{111-171st}{3640}$ |
| $(1,0,1,0,0,0,0,0)$ | $\varnothing$ | $\frac{-89+19st}{1365}$ |
| $(1,1,0,0,0,0,0,0)$ | $(0)$ | $\frac{-11}{182}$ |
| $(1,1,0,0,0,0,0,0)$ | $(1)$ | $\frac{167-57st}{1820}$ |
| $(1,1,0,0,0,0,0,0)$ | $(2)$ | $\frac{57-57st}{1820}$ |
| $(1,1,0,0,0,0,0,0)$ | $(3)$ | $\frac{-57+19st}{728}$ |
| $(1,1,0,0,0,0,0,0)$ | $(4)$ | $\frac{-703+133st}{3640}$ |
| $(1,1,0,0,0,0,0,0)$ | $(6)$ | $\frac{76-19s-95t}{728}$ |
| $(1,1,0,0,0,0,0,0)$ | $(7)$ | $\frac{76+19s+95t}{728}$ |
| $(1,1,1,0,-1,0,0,0)$ | $\varnothing$ | $\frac{1159-589st}{10920}$ |
| $(1,1,1,0,0,0,-1,0)$ | $\varnothing$ | $\frac{-76+19s+95t}{1092}$ |
| $(1,1,1,0,0,0,0,-1)$ | $\varnothing$ | $\frac{-76-19s-95t}{1092}$ |
| $(1,1,1,1,0,0,-1,-1)$ | $\varnothing$ | $\frac{19s}{1092}$ |

### S7.4.2 Field $a_1$: type $(1,9)$, 32 terms

The $e_1$ eigenvalue is $\frac{200}{1271}$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,0,0,1,1,0,1,1)$ | $\varnothing$ | $\frac{1-4st}{25}$ |
| $(0,0,1,1,0,0,1,1)$ | $\varnothing$ | $\frac{-3-13st}{360}$ |
| $(0,0,1,1,1,0,0,1)$ | $\varnothing$ | $\frac{-87-153s+581t+223st}{3600}$ |
| $(0,0,1,1,1,0,1,0)$ | $\varnothing$ | $\frac{87-153s+581t-223st}{3600}$ |
| $(0,0,1,2,1,0,0,0)$ | $\varnothing$ | $\frac{69s+287t}{3600}$ |
| $(0,1,1,0,0,0,1,1)$ | $\varnothing$ | $\frac{81+st}{360}$ |
| $(0,1,1,1,0,0,0,1)$ | $\varnothing$ | $\frac{42-18s+36t+7st}{360}$ |
| $(0,1,1,1,0,0,1,0)$ | $\varnothing$ | $\frac{-42-18s+36t-7st}{360}$ |
| $(0,1,1,1,1,0,0,0)$ | $(0)$ | $\frac{-91s-143t}{3600}$ |
| $(0,1,1,1,1,0,0,0)$ | $(1)$ | $\frac{19s-363t}{3600}$ |
| $(0,1,1,1,1,0,0,0)$ | $(2)$ | $\frac{33s-91t}{1200}$ |
| $(0,1,1,1,1,0,0,0)$ | $(3)$ | $\frac{29s+117t}{3600}$ |
| $(0,1,1,1,1,0,0,0)$ | $(4)$ | $\frac{-11s+22t}{360}$ |
| $(0,1,1,1,1,0,0,0)$ | $(6)$ | $\frac{169+27s+221t-51st}{3600}$ |
| $(0,1,1,1,1,0,0,0)$ | $(7)$ | $\frac{-169+27s+221t+51st}{3600}$ |
| $(0,1,2,1,0,0,0,0)$ | $\varnothing$ | $\frac{-s+2t}{24}$ |
| $(1,0,1,1,1,0,0,0)$ | $\varnothing$ | $\frac{-9s-107t}{900}$ |
| $(1,1,0,0,0,0,1,1)$ | $\varnothing$ | $1$ |
| $(1,1,0,1,1,0,0,0)$ | $\varnothing$ | $\frac{13s-26t}{200}$ |
| $(1,1,1,0,0,0,0,1)$ | $\varnothing$ | $\frac{279-57s+59t-st}{720}$ |
| $(1,1,1,0,0,0,1,0)$ | $\varnothing$ | $\frac{-279-57s+59t+st}{720}$ |
| $(1,1,1,0,1,0,0,0)$ | $\varnothing$ | $\frac{51s+23t}{900}$ |
| $(1,1,1,1,0,0,0,0)$ | $(0)$ | $\frac{11s-22t}{360}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(1,1,1,1,0,0,0,0)$ | $(1)$ | $\frac{11s+13t}{720}$ |
| $(1,1,1,1,0,0,0,0)$ | $(2)$ | $\frac{-9s+13t}{240}$ |
| $(1,1,1,1,0,0,0,0)$ | $(3)$ | $\frac{-7s+t}{144}$ |
| $(1,1,1,1,0,0,0,0)$ | $(4)$ | $\frac{-13s-39t}{720}$ |
| $(1,1,1,1,0,0,0,0)$ | $(6)$ | $\frac{65+21s+13t-5st}{720}$ |
| $(1,1,1,1,0,0,0,0)$ | $(7)$ | $\frac{-65+21s+13t+5st}{720}$ |
| $(1,1,1,1,1,0,-1,0)$ | $\varnothing$ | $\frac{-78-39s+78t-13st}{1800}$ |
| $(1,1,1,1,1,0,0,-1)$ | $\varnothing$ | $\frac{78-39s+78t+13st}{1800}$ |
| $(1,2,1,0,0,0,0,0)$ | $\varnothing$ | $\frac{-27s-31t}{720}$ |

### S7.4.3 Field $a_2$: type $(1,9)$, 32 terms

The $e_1$ eigenvalue is $\frac{1512}{1271}$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,0,0,1,1,0,1,1)$ | $\varnothing$ | $\frac{-7+3st}{20}$ |
| $(0,0,1,1,0,0,1,1)$ | $\varnothing$ | $\frac{15+5st}{48}$ |
| $(0,0,1,1,1,0,0,1)$ | $\varnothing$ | $\frac{159-43s+11t-11st}{480}$ |
| $(0,0,1,1,1,0,1,0)$ | $\varnothing$ | $\frac{-159-43s+11t+11st}{480}$ |
| $(0,0,1,2,1,0,0,0)$ | $\varnothing$ | $\frac{11s+53t}{480}$ |
| $(0,1,1,0,0,0,1,1)$ | $\varnothing$ | $\frac{6+st}{12}$ |
| $(0,1,1,1,0,0,0,1)$ | $\varnothing$ | $\frac{9-2s-6t-st}{96}$ |
| $(0,1,1,1,0,0,1,0)$ | $\varnothing$ | $\frac{-9-2s-6t+st}{96}$ |
| $(0,1,1,1,1,0,0,0)$ | $(0)$ | $\frac{s-2t}{24}$ |
| $(0,1,1,1,1,0,0,0)$ | $(1)$ | $\frac{3s-11t}{96}$ |
| $(0,1,1,1,1,0,0,0)$ | $(2)$ | $\frac{s-2t}{40}$ |
| $(0,1,1,1,1,0,0,0)$ | $(3)$ | $\frac{7s+11t}{240}$ |
| $(0,1,1,1,1,0,0,0)$ | $(4)$ | $\frac{s+3t}{96}$ |
| $(0,1,1,1,1,0,0,0)$ | $(6)$ | $\frac{-38-33s+41t+2st}{480}$ |
| $(0,1,1,1,1,0,0,0)$ | $(7)$ | $\frac{38-33s+41t-2st}{480}$ |
| $(0,1,2,1,0,0,0,0)$ | $\varnothing$ | $\frac{s+3t}{96}$ |
| $(1,0,1,1,1,0,0,0)$ | $\varnothing$ | $\frac{s-t}{24}$ |
| $(1,1,0,0,0,0,1,1)$ | $\varnothing$ | $1$ |
| $(1,1,0,1,1,0,0,0)$ | $\varnothing$ | $\frac{-s-3t}{160}$ |
| $(1,1,1,0,0,0,0,1)$ | $\varnothing$ | $\frac{6+s-4t-st}{24}$ |
| $(1,1,1,0,0,0,1,0)$ | $\varnothing$ | $\frac{-6+s-4t+st}{24}$ |
| $(1,1,1,0,1,0,0,0)$ | $\varnothing$ | $\frac{-s+2t}{30}$ |
| $(1,1,1,1,0,0,0,0)$ | $(0)$ | $\frac{-s-3t}{96}$ |
| $(1,1,1,1,0,0,0,0)$ | $(1)$ | $\frac{-s-7t}{96}$ |
| $(1,1,1,1,0,0,0,0)$ | $(2)$ | $\frac{s-t}{32}$ |
| $(1,1,1,1,0,0,0,0)$ | $(3)$ | $\frac{3s-t}{48}$ |
| $(1,1,1,1,0,0,0,0)$ | $(4)$ | $\frac{5s-5t}{96}$ |
| $(1,1,1,1,0,0,0,0)$ | $(6)$ | $\frac{5-6s+10t-st}{96}$ |
| $(1,1,1,1,0,0,0,0)$ | $(7)$ | $\frac{-5-6s+10t+st}{96}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(1,1,1,1,1,0,-1,0)$ | $\varnothing$ | $\frac{-63-3s-9t+7st}{480}$ |
| $(1,1,1,1,1,0,0,-1)$ | $\varnothing$ | $\frac{63-3s-9t-7st}{480}$ |
| $(1,2,1,0,0,0,0,0)$ | $\varnothing$ | $\frac{s+2t}{24}$ |

### S7.4.4 Field $a_3$: type $(1,13)$, 9 terms

The $e_1$ eigenvalue is $\frac{420}{1271}$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,1,1,1,1,0,1,1)$ | $\varnothing$ | $\frac{6+2s-4t+st}{14}$ |
| $(0,1,2,2,1,0,0,0)$ | $\varnothing$ | $\frac{-26+9s-13t-6st}{168}$ |
| $(1,1,1,1,0,0,1,1)$ | $\varnothing$ | $\frac{-6+2s+4t+st}{14}$ |
| $(1,1,1,1,1,0,0,1)$ | $\varnothing$ | $\frac{1+4t}{7}$ |
| $(1,1,1,1,1,0,1,0)$ | $\varnothing$ | $1$ |
| $(1,1,1,2,1,0,0,0)$ | $\varnothing$ | $\frac{-2-3s-t+2st}{56}$ |
| $(1,1,2,1,1,0,0,0)$ | $\varnothing$ | $\frac{-4-2t}{21}$ |
| $(1,2,1,1,1,0,0,0)$ | $\varnothing$ | $\frac{-2+3s-t-2st}{56}$ |
| $(1,2,2,1,0,0,0,0)$ | $\varnothing$ | $\frac{-26-9s-13t+6st}{168}$ |

### S7.4.5 Field $a_4$: type $(2,2)$, 42 terms

The $e_1$ eigenvalue is $\frac{192}{1271}$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,-2,-2,-1,0,-1,1,1)$ | $\varnothing$ | $\frac{217+32st}{295}$ |
| $(-1,-2,-1,0,0,-1,0,0)$ | $\varnothing$ | $\frac{-29s-42t}{885}$ |
| $(-1,-1,-2,-1,-1,-1,1,1)$ | $\varnothing$ | $\frac{32+2st}{59}$ |
| $(-1,-1,-2,-1,0,-1,0,1)$ | $\varnothing$ | $\frac{114-15s+80t+44st}{1180}$ |
| $(-1,-1,-2,-1,0,-1,1,0)$ | $\varnothing$ | $\frac{-114-15s+80t-44st}{1180}$ |
| $(-1,-1,-2,0,0,-1,0,0)$ | $\varnothing$ | $\frac{-29s-42t}{1180}$ |
| $(-1,-1,-1,-1,-1,-1,0,1)$ | $\varnothing$ | $\frac{192-15s+80t+12st}{708}$ |
| $(-1,-1,-1,-1,-1,-1,1,0)$ | $\varnothing$ | $\frac{-192-15s+80t-12st}{708}$ |
| $(-1,-1,-1,-1,0,-1,0,0)$ | (1) | $\frac{-29s-42t}{885}$ |
| $(-1,-1,-1,-1,0,-1,0,0)$ | (2) | $\frac{-29s-42t}{1180}$ |
| $(-1,-1,-1,-1,0,-1,0,0)$ | (3) | $\frac{22s-19t}{885}$ |
| $(-1,-1,-1,-1,0,-1,0,0)$ | (4) | $\frac{35s+10t}{708}$ |
| $(-1,-1,-1,-1,0,-1,0,0)$ | (6) | $\frac{-103-15s+80t+12st}{1770}$ |
| $(-1,-1,-1,-1,0,-1,0,0)$ | (7) | $\frac{103-15s+80t-12st}{1770}$ |
| $(-1,-1,-1,0,-1,-1,0,0)$ | $\varnothing$ | $\frac{35s+10t}{708}$ |
| $(-1,-1,-1,0,0,-1,-1,0)$ | $\varnothing$ | $\frac{-103-15s+80t+12st}{1770}$ |
| $(-1,-1,-1,0,0,-1,0,-1)$ | $\varnothing$ | $\frac{103-15s+80t-12st}{1770}$ |
| $(-1,0,-1,-1,-1,-1,0,0)$ | $\varnothing$ | $\frac{-29s-42t}{708}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,-1,-2,-2,-1,-1,1,1)$ | $\varnothing$ | $1$ |
| $(0,-1,-2,-1,-1,-1,0,1)$ | $\varnothing$ | $\frac{-54+3s-16t+4st}{236}$ |
| $(0,-1,-2,-1,-1,-1,1,0)$ | $\varnothing$ | $\frac{54+3s-16t-4st}{236}$ |
| $(0,-1,-2,-1,0,-1,0,0)$ | $(0)$ | $\frac{87s+126t}{1180}$ |
| $(0,-1,-2,-1,0,-1,0,0)$ | $(1)$ | $\frac{13s+29t}{295}$ |
| $(0,-1,-2,-1,0,-1,0,0)$ | $(2)$ | $\frac{-9s+48t}{590}$ |
| $(0,-1,-2,-1,0,-1,0,0)$ | $(3)$ | $\frac{-19s+3t}{295}$ |
| $(0,-1,-2,-1,0,-1,0,0)$ | $(4)$ | $\frac{-21s-6t}{236}$ |
| $(0,-1,-2,-1,0,-1,0,0)$ | $(6)$ | $\frac{64+15s-80t+4st}{590}$ |
| $(0,-1,-2,-1,0,-1,0,0)$ | $(7)$ | $\frac{-64+15s-80t-4st}{590}$ |
| $(0,-1,-1,-1,-1,-1,0,0)$ | $(0)$ | $\frac{29s+42t}{708}$ |
| $(0,-1,-1,-1,-1,-1,0,0)$ | $(1)$ | $\frac{2s+9t}{177}$ |
| $(0,-1,-1,-1,-1,-1,0,0)$ | $(2)$ | $\frac{-7s-2t}{236}$ |
| $(0,-1,-1,-1,-1,-1,0,0)$ | $(3)$ | $\frac{-7s-2t}{177}$ |
| $(0,-1,-1,-1,-1,-1,0,0)$ | $(6)$ | $\frac{5+3s-16t+4st}{354}$ |
| $(0,-1,-1,-1,-1,-1,0,0)$ | $(7)$ | $\frac{-5+3s-16t-4st}{354}$ |
| $(0,-1,-1,-1,0,-1,-1,0)$ | $\varnothing$ | $\frac{-64-15s+80t-4st}{1770}$ |
| $(0,-1,-1,-1,0,-1,0,-1)$ | $\varnothing$ | $\frac{64-15s+80t+4st}{1770}$ |
| $(0,-1,-1,0,0,-1,-1,-1)$ | $\varnothing$ | $\frac{-39+16st}{1770}$ |
| $(0,0,-2,-1,-1,-1,0,0)$ | $\varnothing$ | $\frac{-7s-2t}{236}$ |
| $(0,0,-1,-2,-1,-1,0,0)$ | $\varnothing$ | $\frac{7s+2t}{177}$ |
| $(0,0,-1,-1,-1,-1,-1,0)$ | $\varnothing$ | $\frac{-5-3s+16t-4st}{354}$ |
| $(0,0,-1,-1,-1,-1,0,-1)$ | $\varnothing$ | $\frac{5-3s+16t+4st}{354}$ |
| $(0,0,-1,-1,0,-1,-1,-1)$ | $\varnothing$ | $\frac{-39+16st}{1770}$ |

### S7.4.6 Field $a_5$: type $(2,6)$, 32 terms

The $e_1$ eigenvalue is $\frac{1830}{1271}$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,-1,-1,-1,0,-1,1,1)$ | $\varnothing$ | $\frac{-55+15st}{126}$ |
| $(-1,-1,-1,0,0,-1,0,1)$ | $\varnothing$ | $\frac{55-20s+40t-15st}{252}$ |
| $(-1,-1,-1,0,0,-1,1,0)$ | $\varnothing$ | $\frac{-55-20s+40t+15st}{252}$ |
| $(-1,-1,-1,0,1,-1,0,0)$ | $\varnothing$ | $\frac{-s+17t}{126}$ |
| $(-1,-1,0,0,0,-1,0,0)$ | $\varnothing$ | $\frac{5s-5t}{252}$ |
| $(-1,0,-1,0,0,-1,0,0)$ | $\varnothing$ | $\frac{5s-5t}{252}$ |
| $(0,-1,-2,-1,0,-1,1,1)$ | $\varnothing$ | $1$ |
| $(0,-1,-1,-1,-1,-1,1,1)$ | $\varnothing$ | $\frac{55+15st}{126}$ |
| $(0,-1,-1,-1,0,-1,0,1)$ | $\varnothing$ | $\frac{4+10t}{63}$ |
| $(0,-1,-1,-1,0,-1,1,0)$ | $\varnothing$ | $\frac{-4+10t}{63}$ |
| $(0,-1,-1,0,0,-1,0,0)$ | $(0)$ | $\frac{-7s+31t}{252}$ |
| $(0,-1,-1,0,0,-1,0,0)$ | $(1)$ | $\frac{-13s+13t}{252}$ |
| $(0,-1,-1,0,0,-1,0,0)$ | $(2)$ | $\frac{-s}{21}$ |
| $(0,-1,-1,0,0,-1,0,0)$ | $(3)$ | $\frac{-s-3t}{42}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,-1,-1,0,0,-1,0,0)$ | (4) | $\frac{-s-13t}{126}$ |
| $(0,-1,-1,0,0,-1,0,0)$ | (6) | $\frac{13+20s-5st}{252}$ |
| $(0,-1,-1,0,0,-1,0,0)$ | (7) | $\frac{-13+20s+5st}{252}$ |
| $(0,0,-1,-1,-1,-1,0,1)$ | $\varnothing$ | $\frac{55+20s+40t+15st}{252}$ |
| $(0,0,-1,-1,-1,-1,1,0)$ | $\varnothing$ | $\frac{-55+20s+40t-15st}{252}$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (0) | $\frac{-s+13t}{126}$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (1) | $\frac{-s+3t}{42}$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (2) | $\frac{-s}{21}$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (3) | $\frac{-13s-13t}{252}$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (4) | $\frac{-7s-31t}{252}$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (6) | $\frac{-13+20s-5st}{252}$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (7) | $\frac{13+20s+5st}{252}$ |
| $(0,0,-1,0,-1,-1,0,0)$ | $\varnothing$ | $\frac{-5s-5t}{252}$ |
| $(0,0,-1,0,0,-1,-1,0)$ | $\varnothing$ | $\frac{-13}{126}$ |
| $(0,0,-1,0,0,-1,0,-1)$ | $\varnothing$ | $\frac{13}{126}$ |
| $(0,0,0,-1,-1,-1,0,0)$ | $\varnothing$ | $\frac{5s+5t}{252}$ |
| $(0,0,0,0,0,-1,-1,-1)$ | $\varnothing$ | $\frac{-13}{126}$ |
| $(1,0,-1,-1,-1,-1,0,0)$ | $\varnothing$ | $\frac{s+17t}{126}$ |

### S7.4.7 Field $a_6$: type $(2,8)$, 22 terms

The $e_1$ eigenvalue is $\frac{752}{1271}$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,-1,-1,0,0,-1,1,1)$ | $\varnothing$ | $\frac{307+297st}{1880}$ |
| $(-1,0,0,0,0,-1,0,0)$ | $\varnothing$ | $\frac{-267s-541t}{5640}$ |
| $(0,-1,-1,-1,0,-1,1,1)$ | $\varnothing$ | $\frac{567+77st}{940}$ |
| $(0,-1,-1,0,0,-1,0,1)$ | $\varnothing$ | $\frac{-827-385s-935t+143st}{3760}$ |
| $(0,-1,-1,0,0,-1,1,0)$ | $\varnothing$ | $\frac{827-385s-935t-143st}{3760}$ |
| $(0,-1,-1,0,1,-1,0,0)$ | $\varnothing$ | $\frac{103s+89t}{940}$ |
| $(0,-1,0,0,0,-1,0,0)$ | $\varnothing$ | $\frac{-9s+5t}{282}$ |
| $(0,0,-1,-1,-1,-1,1,1)$ | $\varnothing$ | 1 |
| $(0,0,-1,-1,0,-1,0,1)$ | $\varnothing$ | $\frac{-373-253s-319t+77st}{1880}$ |
| $(0,0,-1,-1,0,-1,1,0)$ | $\varnothing$ | $\frac{373-253s-319t-77st}{1880}$ |
| $(0,0,-1,0,0,-1,0,0)$ | (0) | $\frac{267s+541t}{5640}$ |
| $(0,0,-1,0,0,-1,0,0)$ | (1) | $\frac{9s-5t}{282}$ |
| $(0,0,-1,0,0,-1,0,0)$ | (2) | $\frac{33s-81t}{940}$ |
| $(0,0,-1,0,0,-1,0,0)$ | (3) | $\frac{9s-193t}{2820}$ |
| $(0,0,-1,0,0,-1,0,0)$ | (4) | $\frac{-15s-23t}{282}$ |
| $(0,0,-1,0,0,-1,0,0)$ | (6) | $\frac{-81-363s+891t-11st}{11280}$ |
| $(0,0,-1,0,0,-1,0,0)$ | (7) | $\frac{81-363s+891t+11st}{11280}$ |
| $(0,0,0,-1,0,-1,0,0)$ | $\varnothing$ | $\frac{-9s+193t}{2820}$ |
| $(0,0,0,0,-1,-1,0,0)$ | $\varnothing$ | $\frac{15s+23t}{282}$ |
| $(0,0,0,0,0,-1,-1,0)$ | $\varnothing$ | $\frac{81+363s-891t+11st}{11280}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,0,0,0,0,-1,0,-1)$ | $\varnothing$ | $\frac{-81+363s-891t-11st}{11280}$ |
| $(1,0,-1,-1,0,-1,0,0)$ | $\varnothing$ | $\frac{59s+197t}{940}$ |

### S7.4.8 Field $a_7$: type $(2,14)$, 3 terms

The $e_1$ eigenvalue is $\frac{180}{1271}$. This is the original defining field $u$.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,0,0,0,0,-1,1,1)$ | $\varnothing$ | $1$ |
| $(0,0,0,1,1,-1,0,0)$ | $\varnothing$ | $\frac{s+3t}{20}$ |
| $(1,1,0,0,0,-1,0,0)$ | $\varnothing$ | $\frac{s-3t}{20}$ |

### S7.4.9 Field $e$: type $(1,1)$, 70 terms

The $e_1$ eigenvalue is 2.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,-1,0,0,0,0,1,1)$ | $\varnothing$ | $\frac{10s}{93}$ |
| $(-1,-1,0,1,1,0,0,0)$ | $\varnothing$ | $\frac{-10}{93}$ |
| $(-1,0,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{260-205s}{3813}$ |
| $(-1,0,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{260+205s}{3813}$ |
| $(-1,0,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{5}{41}$ |
| $(-1,0,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(-1,0,1,0,0,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(0,-1,-1,0,0,0,1,1)$ | $\varnothing$ | $\frac{6s}{31}$ |
| $(0,-1,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{416-164s}{3813}$ |
| $(0,-1,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{416+164s}{3813}$ |
| $(0,-1,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{88}{3813}$ |
| $(0,-1,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{166}{1271}$ |
| $(0,0,-1,-1,0,0,1,1)$ | $\varnothing$ | $\frac{6s}{31}$ |
| $(0,0,-1,0,0,0,0,1)$ | $\varnothing$ | $\frac{156}{1271}$ |
| $(0,0,-1,0,0,0,1,0)$ | $\varnothing$ | $\frac{156}{1271}$ |
| $(0,0,-1,0,1,0,0,0)$ | $\varnothing$ | $\frac{33}{1271}$ |
| $(0,0,0,-1,-1,0,1,1)$ | $\varnothing$ | $\frac{10s}{93}$ |
| $(0,0,0,-1,0,0,0,1)$ | $\varnothing$ | $\frac{416+164s}{3813}$ |
| $(0,0,0,-1,0,0,1,0)$ | $\varnothing$ | $\frac{416-164s}{3813}$ |
| $(0,0,0,0,-1,0,0,1)$ | $\varnothing$ | $\frac{260+205s}{3813}$ |
| $(0,0,0,0,-1,0,1,0)$ | $\varnothing$ | $\frac{260-205s}{3813}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{890}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2)$ | $\frac{20}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $\frac{-455}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $\frac{-925}{11439}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,0,0,0,0,0,0,0)$ | $(0,6)$ | $\frac{-780-205s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7)$ | $\frac{-780+205s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2)$ | $\frac{73}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $\frac{-646}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $\frac{-455}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6)$ | $\frac{-1248-164s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,7)$ | $\frac{-1248+164s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2)$ | $\frac{63}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3)$ | $\frac{73}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4)$ | $\frac{20}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,6)$ | $\frac{-156}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,7)$ | $\frac{-156}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6)$ | $\frac{-1248+164s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7)$ | $\frac{-1248-164s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{890}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6)$ | $\frac{-780+205s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7)$ | $\frac{-780-205s}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6)$ | $\frac{1820}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7)$ | $\frac{1820}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7)$ | $\frac{1820}{11439}$ |
| $(0,0,0,0,1,0,-1,0)$ | $\varnothing$ | $\frac{104-82s}{3813}$ |
| $(0,0,0,0,1,0,0,-1)$ | $\varnothing$ | $\frac{104+82s}{3813}$ |
| $(0,0,0,1,0,0,-1,0)$ | $\varnothing$ | $\frac{104-41s}{3813}$ |
| $(0,0,0,1,0,0,0,-1)$ | $\varnothing$ | $\frac{104+41s}{3813}$ |
| $(0,0,0,1,1,0,-1,-1)$ | $\varnothing$ | $\frac{s}{93}$ |
| $(0,0,1,0,-1,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(0,0,1,0,0,0,-1,0)$ | $\varnothing$ | $\frac{104}{3813}$ |
| $(0,0,1,0,0,0,0,-1)$ | $\varnothing$ | $\frac{104}{3813}$ |
| $(0,0,1,1,0,0,-1,-1)$ | $\varnothing$ | $\frac{s}{93}$ |
| $(0,1,0,-1,0,0,0,0)$ | $\varnothing$ | $\frac{166}{1271}$ |
| $(0,1,0,0,-1,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(0,1,0,0,0,0,-1,0)$ | $\varnothing$ | $\frac{104+41s}{3813}$ |
| $(0,1,0,0,0,0,0,-1)$ | $\varnothing$ | $\frac{104-41s}{3813}$ |
| $(0,1,1,0,0,0,-1,-1)$ | $\varnothing$ | $\frac{s}{93}$ |
| $(1,0,-1,0,0,0,0,0)$ | $\varnothing$ | $\frac{33}{1271}$ |
| $(1,0,0,-1,0,0,0,0)$ | $\varnothing$ | $\frac{88}{3813}$ |
| $(1,0,0,0,-1,0,0,0)$ | $\varnothing$ | $\frac{5}{41}$ |
| $(1,0,0,0,0,0,-1,0)$ | $\varnothing$ | $\frac{104+82s}{3813}$ |
| $(1,0,0,0,0,0,0,-1)$ | $\varnothing$ | $\frac{104-82s}{3813}$ |
| $(1,1,0,-1,-1,0,0,0)$ | $\varnothing$ | $\frac{-10}{93}$ |
| $(1,1,0,0,0,0,-1,-1)$ | $\varnothing$ | $\frac{s}{93}$ |

### S7.5 Exactly what the low-weight checker asserts

The implementation `hi_cft_20260923_c8_exact_check.py` reconstructs the printed fields in the exact coefficient field and checks the following equations. With $T_K = \omega_B - \omega_{A_{1,1}} - \omega_{A_{1,39}}$ and $f = T_K - e$, it tests

$$e_1 e = 2e, \quad e_2 e = 0, \quad e_3 e = \frac{1200}{1271}\mathbf{1}, \qquad e_n f = 0 \quad (n = 1, 2, 3), \quad f_3 f = \frac{35}{31}\mathbf{1}.$$

For a highest input $a_i$ of type $(n, m)$ it tests

$$(E_L)_0 a_i = (E_R)_0 a_i = 0, \qquad x_1 a_i = 0 \quad (x \in K),$$
$$e_1 a_i = h_e a_i, \quad e_2 a_i = 0, \quad (\omega_B)_2 a_i = 0, \quad (\omega_B)_1 a_i = 2a_i.$$

The lowering strings terminate at the stated dimensions. Exact sparse elimination finds 22 affine normal/translation descendants, 24 vectors after adjoining $e, f$, and 120 after the eight multiplets. The transported PCT fixes $e, f$ and sends every charged highest field into this same 120-dimensional span.

When the closure option is used, every one of the eight left highest fields is paired with all 96 charged descendants. For each pair $(a_i, v)$ the checker tests both

$$(a_i)_1 v \in S_2, \qquad (a_i)_2 v \in K.$$

Thus the eight rows labelled "96 exact pairs" mean 768 pairs and two membership tests per pair. The boolean called `exact_OPE_closure` in this record has this low-weight scope; it does not assert the later Gate 4, Gate 5, or all-weight closure. The current, neutral-stress and descendant consequences use covariance and the separately saved lower-closure inputs.

Three useful scalar checks on the selected stress eigenvalues are

$$\mathrm{Tr}_{S_2}(e_1) = \frac{67200}{1271},$$
$$\mathrm{Tr}_{S_2}(e_1 (H_L)_0^2/2) = \frac{19900}{1271},$$
$$\mathrm{Tr}_{S_2}(e_1 (H_R)_0^2/2) = \frac{792900}{1271}.$$

They appear in `hi_cft_20260923_c8_trace_certificate.json`. For a current multiplet of dimensions $(n, m)$ and stress eigenvalue $h$, its contributions to these three sums are respectively

$$nmh, \qquad \frac{mn(n^2-1)}{6}h, \qquad \frac{nm(m^2-1)}{6}h.$$

The neutral field $e$ contributes 2 only to the first sum; the affine vacuum descendants and $f$ contribute zero. These formulas make all three traces a short rational calculation from the eight eigenvalue rows. They are necessary consistency checks and do not select the fields uniquely or establish any OPE closure.

### S7.6 The complete accepted short-generation graph

To connect the explicit tables to the short definition of $U$, write $v_i$ for an accepted *raw* vector and $r_i$ for its normalized remainder. The indices in this subsection are the local indices of `c8_reviewability_20260927/minimal_generation_witness.json`. They must not be identified with the independently constructed `r2_000000` or `r3_000000` checkpoint labels.

Here is a deterministic interpretation of every row of the next table. Compute its raw vector $v_i$ using the indicated mode product on inputs or earlier $r_j$. Select all earlier normalized rows of the same joint Cartan weight. In increasing lexicographic order of their pivot states, subtract the current coefficient at that pivot times its normalized row. If the result is $b_i$, define

$$\pi_i = \min \operatorname{supp}(b_i), \qquad r_i = b_i / b_i[\pi_i].$$

The state order first compares momentum tuples, then oscillator tuples. The witness records all subtractions explicitly; the stated algorithm specifies them without a large coefficient matrix. Distinct unit pivots supply the independence proof. Since only earlier rows enter a recipe, every $v_i$ and every $r_i$ belongs to the originally generated $U$.

The six current labels below are $H_L, E_L, F_L, H_R, E_R, F_R$. The table gives the raw recipe, its Cartan weight, and its raw and normalized support sizes. A support size counts lattice-oscillator basis states, not rational components. The initial 24 independent rows are followed by 96 accepted descendants. There are 644 attempted insertions in the full queue, so rejected dependent or zero candidates are not extra generators. The complete input conformal vectors are also tabulated in Section S24.

| $i$ | Raw vector $v_i$ | Weight | $\lvert v_i\rvert$ | $\lvert r_i\rvert$ |
|---|---|---|---|---|
| 0 | $(H_L)_{-2}\mathbf{1}$ | $(0,0)$ | 6 | 6 |
| 1 | $(H_L)_{-1}H_L$ | $(0,0)$ | 21 | 21 |
| 2 | $(H_L)_{-1}E_L$ | $(2,0)$ | 6 | 6 |
| 3 | $(H_L)_{-1}F_L$ | $(-2,0)$ | 6 | 6 |
| 4 | $(H_L)_{-1}H_R$ | $(0,0)$ | 32 | 30 |
| 5 | $(H_L)_{-1}E_R$ | $(0,2)$ | 42 | 42 |
| 6 | $(H_L)_{-1}F_R$ | $(0,-2)$ | 42 | 42 |
| 7 | $(E_L)_{-1}H_R$ | $(2,0)$ | 7 | 6 |
| 8 | $(E_L)_{-1}E_R$ | $(2,2)$ | 7 | 7 |
| 9 | $(E_L)_{-1}F_R$ | $(2,-2)$ | 7 | 7 |
| 10 | $(F_L)_{-1}H_R$ | $(-2,0)$ | 7 | 6 |
| 11 | $(F_L)_{-1}E_R$ | $(-2,2)$ | 7 | 7 |
| 12 | $(F_L)_{-1}F_R$ | $(-2,-2)$ | 7 | 7 |
| 13 | $(H_R)_{-2}\mathbf{1}$ | $(0,0)$ | 7 | 6 |
| 14 | $(H_R)_{-1}H_R$ | $(0,0)$ | 28 | 21 |
| 15 | $(H_R)_{-1}E_R$ | $(0,2)$ | 49 | 42 |
| 16 | $(H_R)_{-1}F_R$ | $(0,-2)$ | 49 | 42 |
| 17 | $(E_R)_{-2}\mathbf{1}$ | $(0,2)$ | 7 | 7 |
| 18 | $(E_R)_{-1}E_R$ | $(0,4)$ | 26 | 26 |
| 19 | $(E_R)_{-1}F_R$ | $(0,0)$ | 46 | 60 |
| 20 | $(F_R)_{-2}\mathbf{1}$ | $(0,-2)$ | 7 | 42 |
| 21 | $(F_R)_{-1}F_R$ | $(0,-4)$ | 26 | 26 |
| 22 | $\omega_B$ | $(0,0)$ | 24 | 26 |
| 23 | $u$ | $(1,13)$ | 3 | 3 |
| 24 | $u_1 r_3$ | $(-1,13)$ | 3 | 3 |
| 25 | $u_1 r_6$ | $(1,11)$ | 7 | 7 |
| 26 | $u_1 r_{12}$ | $(-1,11)$ | 7 | 7 |
| 27 | $u_1 r_{21}$ | $(1,9)$ | 12 | 12 |
| 28 | $(F_R)_0 r_{26}$ | $(-1,9)$ | 12 | 12 |
| 29 | $(F_R)_0 r_{27}$ | $(1,7)$ | 22 | 22 |
| 30 | $(F_R)_0 r_{28}$ | $(-1,7)$ | 22 | 22 |

| $i$ | Raw vector $v_i$ | Weight | $\lvert v_i\rvert$ | $\lvert r_i\rvert$ |
|---|---|---|---|---|
| 31 | $(F_R)_0 r_{29}$ | $(1,5)$ | 32 | 32 |
| 32 | $(F_R)_0 r_{30}$ | $(-1,5)$ | 32 | 32 |
| 33 | $(F_R)_0 r_{31}$ | $(1,3)$ | 39 | 39 |
| 34 | $(F_R)_0 r_{32}$ | $(-1,3)$ | 39 | 39 |
| 35 | $(F_R)_0 r_{33}$ | $(1,1)$ | 44 | 44 |
| 36 | $(F_R)_0 r_{34}$ | $(-1,1)$ | 44 | 44 |
| 37 | $(F_R)_0 r_{35}$ | $(1,-1)$ | 44 | 44 |
| 38 | $(F_R)_0 r_{36}$ | $(-1,-1)$ | 44 | 44 |
| 39 | $(F_R)_0 r_{37}$ | $(1,-3)$ | 39 | 39 |
| 40 | $(F_R)_0 r_{38}$ | $(-1,-3)$ | 39 | 39 |
| 41 | $u_1 r_{38}$ | $(0,12)$ | 9 | 9 |
| 42 | $(F_R)_0 r_{39}$ | $(1,-5)$ | 32 | 32 |
| 43 | $(F_R)_0 r_{40}$ | $(-1,-5)$ | 32 | 32 |
| 44 | $u_1 r_{40}$ | $(0,10)$ | 16 | 16 |
| 45 | $(F_R)_0 r_{42}$ | $(1,-7)$ | 22 | 22 |
| 46 | $(F_R)_0 r_{43}$ | $(-1,-7)$ | 22 | 22 |
| 47 | $u_1 r_{43}$ | $(0,8)$ | 32 | 32 |
| 48 | $(F_R)_0 r_{44}$ | $(0,8)$ | 30 | 31 |
| 49 | $(F_R)_0 r_{45}$ | $(1,-9)$ | 12 | 12 |
| 50 | $(F_R)_0 r_{46}$ | $(-1,-9)$ | 12 | 12 |
| 51 | $u_1 r_{46}$ | $(0,6)$ | 42 | 42 |
| 52 | $(F_R)_0 r_{47}$ | $(0,6)$ | 45 | 44 |
| 53 | $(F_R)_0 r_{49}$ | $(1,-11)$ | 7 | 7 |
| 54 | $(F_R)_0 r_{50}$ | $(-1,-11)$ | 7 | 7 |
| 55 | $u_1 r_{50}$ | $(0,4)$ | 36 | 42 |
| 56 | $(F_R)_0 r_{51}$ | $(0,4)$ | 59 | 57 |
| 57 | $(F_R)_0 r_{52}$ | $(0,4)$ | 58 | 57 |
| 58 | $(F_R)_0 r_{53}$ | $(1,-13)$ | 3 | 3 |
| 59 | $(F_R)_0 r_{54}$ | $(-1,-13)$ | 3 | 3 |
| 60 | $u_1 r_{54}$ | $(0,2)$ | 35 | 30 |
| 61 | $(F_R)_0 r_{55}$ | $(0,2)$ | 59 | 72 |
| 62 | $(F_R)_0 r_{56}$ | $(0,2)$ | 67 | 70 |
| 63 | $u_1 r_{59}$ | $(0,0)$ | 29 | 25 |
| 64 | $(F_R)_0 r_{60}$ | $(0,0)$ | 64 | 62 |
| 65 | $(F_R)_0 r_{61}$ | $(0,0)$ | 82 | 58 |
| 66 | $(F_R)_0 r_{62}$ | $(0,0)$ | 82 | 63 |
| 67 | $(F_R)_0 r_{63}$ | $(0,-2)$ | 49 | 49 |
| 68 | $(F_R)_0 r_{64}$ | $(0,-2)$ | 72 | 68 |
| 69 | $(F_R)_0 r_{65}$ | $(0,-2)$ | 70 | 71 |
| 70 | $(F_R)_0 r_{67}$ | $(0,-4)$ | 42 | 42 |
| 71 | $(F_R)_0 r_{68}$ | $(0,-4)$ | 61 | 58 |
| 72 | $(F_R)_0 r_{69}$ | $(0,-4)$ | 61 | 57 |
| 73 | $(F_R)_0 r_{70}$ | $(0,-6)$ | 42 | 42 |
| 74 | $(F_R)_0 r_{71}$ | $(0,-6)$ | 45 | 44 |
| 75 | $(F_R)_0 r_{73}$ | $(0,-8)$ | 32 | 32 |
| 76 | $(F_R)_0 r_{74}$ | $(0,-8)$ | 32 | 31 |
| 77 | $u_1 r_{74}$ | $(1,7)$ | 22 | 21 |
| 78 | $(F_R)_0 r_{75}$ | $(0,-10)$ | 16 | 16 |
| 79 | $u_1 r_{75}$ | $(1,5)$ | 32 | 31 |

| $i$ | Raw vector $v_i$ | Weight | $\lvert v_i\rvert$ | $\lvert r_i\rvert$ |
|---|---|---|---|---|
| 80 | $u_1 r_{76}$ | $(1,5)$ | 32 | 30 |
| 81 | $(F_L)_0 r_{77}$ | $(-1,7)$ | 21 | 21 |
| 82 | $(F_R)_0 r_{78}$ | $(0,-12)$ | 9 | 9 |
| 83 | $u_1 r_{78}$ | $(1,3)$ | 30 | 38 |
| 84 | $(F_L)_0 r_{79}$ | $(-1,5)$ | 31 | 31 |
| 85 | $(F_L)_0 r_{80}$ | $(-1,5)$ | 30 | 30 |
| 86 | $(F_R)_0 r_{80}$ | $(1,3)$ | 39 | 37 |
| 87 | $u_1 r_{82}$ | $(1,1)$ | 27 | 42 |
| 88 | $(F_L)_0 r_{83}$ | $(-1,3)$ | 38 | 38 |
| 89 | $(F_R)_0 r_{83}$ | $(1,1)$ | 43 | 42 |
| 90 | $(F_R)_0 r_{85}$ | $(-1,3)$ | 39 | 37 |
| 91 | $(F_R)_0 r_{86}$ | $(1,1)$ | 43 | 41 |
| 92 | $(F_L)_0 r_{87}$ | $(-1,1)$ | 42 | 42 |
| 93 | $(F_R)_0 r_{87}$ | $(1,-1)$ | 44 | 43 |
| 94 | $(F_R)_0 r_{88}$ | $(-1,1)$ | 43 | 42 |
| 95 | $(F_R)_0 r_{89}$ | $(1,-1)$ | 44 | 42 |
| 96 | $(F_R)_0 r_{90}$ | $(-1,1)$ | 43 | 41 |
| 97 | $(F_R)_0 r_{91}$ | $(1,-1)$ | 43 | 41 |
| 98 | $(F_R)_0 r_{92}$ | $(-1,-1)$ | 44 | 43 |
| 99 | $(F_R)_0 r_{93}$ | $(1,-3)$ | 39 | 38 |
| 100 | $(F_R)_0 r_{94}$ | $(-1,-1)$ | 44 | 42 |
| 101 | $(F_R)_0 r_{95}$ | $(1,-3)$ | 39 | 37 |
| 102 | $(F_R)_0 r_{96}$ | $(-1,-1)$ | 43 | 41 |
| 103 | $(F_R)_0 r_{98}$ | $(-1,-3)$ | 39 | 38 |
| 104 | $(F_R)_0 r_{99}$ | $(1,-5)$ | 31 | 31 |
| 105 | $(F_R)_0 r_{100}$ | $(-1,-3)$ | 39 | 37 |
| 106 | $(F_R)_0 r_{101}$ | $(1,-5)$ | 31 | 30 |
| 107 | $(F_R)_0 r_{103}$ | $(-1,-5)$ | 31 | 31 |
| 108 | $(F_R)_0 r_{104}$ | $(1,-7)$ | 22 | 21 |
| 109 | $(F_R)_0 r_{105}$ | $(-1,-5)$ | 31 | 30 |
| 110 | $(F_R)_0 r_{107}$ | $(-1,-7)$ | 22 | 21 |
| 111 | $u_1 r_{107}$ | $(0,8)$ | 32 | 30 |
| 112 | $u_1 r_{110}$ | $(0,6)$ | 40 | 43 |
| 113 | $(F_R)_0 r_{112}$ | $(0,4)$ | 57 | 56 |
| 114 | $(F_R)_0 r_{113}$ | $(0,2)$ | 66 | 71 |
| 115 | $(F_R)_0 r_{114}$ | $(0,0)$ | 82 | 62 |
| 116 | $(F_R)_0 r_{115}$ | $(0,-2)$ | 71 | 70 |
| 117 | $(F_R)_0 r_{116}$ | $(0,-4)$ | 61 | 56 |
| 118 | $(F_R)_0 r_{117}$ | $(0,-6)$ | 45 | 43 |
| 119 | $(F_R)_0 r_{118}$ | $(0,-8)$ | 32 | 30 |

The recipe inventory has four accepted current derivatives, eighteen accepted current normal products, $\omega_B$, $u$, seventy-eight current-zero-mode descendants, and eighteen $u_1$ descendants. The absence of separate accepted derivatives for $E_L$ and $F_L$ is a linear-dependence outcome of the insertion order; it does not omit those vectors from the initial affine span.

### S7.6.1 Coordinates of all target fields in the generation basis

For each target the next table lists all nonzero $c_i$ in target $= \sum_i c_i r_i$. Together with the accepted recipe graph and the normalization rule, these are explicit membership expressions for every field in the preceding coordinate tables. They are often much shorter than the fields' ambient supports. In particular the three-term $u$ is exactly $r_{23}$, and the retained stress $e$ has nine normalized coordinates.

| Target | $i$ | Coefficient of $r_i$ |
|---|---|---|
| $a_0$ | 18 | $-1$ |
| | 55 | $\frac{1159+589st}{10920}$ |
| | 56 | $\frac{38s-19t}{273}$ |
| | 57 | $\frac{-703s-171t}{1820}$ |
| $a_1$ | 47 | $\frac{1-4st}{25}$ |
| | 48 | $\frac{-3579-2609st}{16920}$ |
| | 111 | $\frac{239283+48015s+218845t-1507st}{1602000}$ |
| $a_2$ | 47 | $\frac{-7+3st}{20}$ |
| | 48 | $\frac{621+511st}{2256}$ |
| | 111 | $\frac{-11487-44835s+25795t+15323st}{213600}$ |
| $a_3$ | 41 | $\frac{6+2s-4t+st}{14}$ |
| $a_4$ | 35 | $\frac{217+32st}{295}$ |
| | 87 | $\frac{447s-319t}{1770}$ |
| | 89 | $\frac{-1953+1187st}{11800}$ |
| | 91 | $\frac{18717-2496s-12058t+377st}{143960}$ |
| $a_5$ | 31 | $\frac{-55+15st}{126}$ |
| | 79 | $\frac{275-60s+120t-75st}{756}$ |
| | 80 | $\frac{-11565-30195s+13115t+6585st}{139608}$ |
| $a_6$ | 29 | $\frac{307+297st}{1880}$ |
| | 77 | $\frac{-253s-319t}{1410}$ |
| $a_7$ | 23 | $1$ |
| $e$ | 1 | $\frac{890}{11439}$ |
| | 4 | $\frac{-860}{3813}$ |
| | 14 | $\frac{860}{3813}$ |
| | 19 | $\frac{260-205s}{3813}$ |
| | 22 | $\frac{40-15s}{93}$ |
| | 63 | $\frac{10s}{93}$ |
| | 64 | $\frac{10s}{93}$ |
| | 65 | $\frac{-11-st}{62}$ |

| Target | $i$ | Coefficient of $r_i$ |
|---|---|---|
| | 66 | $\frac{23+3st}{62}$ |
| $\omega_B$ | 1 | $\frac{2}{3}$ |
| | 4 | $-1$ |
| | 14 | 1 |
| | 22 | 1 |

The independent coordinate verifier reports 120 normalization identities, 120 raw-generator expansions, and 20 target expansions (two bases for each of ten targets), together with the earlier-parent and source-binding checks. This demonstrates the precise benefit of saving a recipe graph: membership in the original generated algebra can be reviewed separately from the linear equations in the ambient basis. A coordinate verifier that merely sums saved raw vectors does not by itself establish that those raw vectors were produced by the named vertex modes; the exact exporter and mode-engine formulas supply that remaining interpretation.

## S7.7 Grade three: exact history and complete charge census

The exact record `hi_cft_20260923_c8_grade3_exact_v2.json` constructs $N_3 = K_{-1}S_2 + TS_2$ first, obtaining dimension 634. It then adds charged zero-products and closes under currents. Its order differs from the original floating and finite-field runs, which explains why the intermediate dimensions differ even though the final space agrees. The full recorded exact progression is as follows.

| Completed stage | Dimension | Cumulative seconds |
|---|---|---|
| $N_3$ | 634 | 131.23 |
| $(2, 14)$ charged family | 924 | 253.37 |
| $(1, 13)$ charged family | 953 | 302.35 |
| $(2, 8)$ charged family | 956 | 332.99 |
| $(1, 9)$ charged family | 956 | 367.70 |
| $(1, 9)$ charged family | 956 | 402.15 |
| $(2, 6)$ charged family | 956 | 436.17 |
| $(2, 2)$ charged family | 956 | 483.15 |
| $(1, 5)$ charged family | 956 | 534.52 |

The quotient $P = S_3/N_3$ has dimension 322. The next table contains the complete saved charge census in a compressed Weyl-symmetric form. Each row represents all independent sign choices of its nonzero Cartan coordinates. Thus a row $(1, 3)$ stands for four weight spaces, a row $(0, 4)$ for two, and $(0, 0)$ for one. The orbit size is printed explicitly. The $P$ column is transcribed by counting the quotient weights in the exact saturation record, and the $N_3$ column is their difference from the exact $S_3$ multiplicity.

| Weight representative | Sign orbit | $\dim S_3[w]$ | $\dim N_3[w]$ | $\dim P[w]$ |
|---|---|---|---|---|
| $(0, 0)$ | 1 | 58 | 40 | 18 |

| Weight representative | Sign orbit | $\dim S_3[w]$ | $\dim N_3[w]$ | $\dim P[w]$ |
|---|---|---|---|---|
| $(0,2)$ | 2 | 49 | 31 | 18 |
| $(0,4)$ | 2 | 37 | 22 | 15 |
| $(0,6)$ | 2 | 31 | 17 | 14 |
| $(0,8)$ | 2 | 22 | 13 | 9 |
| $(0,10)$ | 2 | 14 | 7 | 7 |
| $(0,12)$ | 2 | 8 | 4 | 4 |
| $(0,14)$ | 2 | 4 | 1 | 3 |
| $(0,16)$ | 2 | 1 | 0 | 1 |
| $(0,18)$ | 2 | 1 | 0 | 1 |
| $(1,1)$ | 4 | 29 | 19 | 10 |
| $(1,3)$ | 4 | 25 | 16 | 9 |
| $(1,5)$ | 4 | 22 | 14 | 8 |
| $(1,7)$ | 4 | 16 | 10 | 6 |
| $(1,9)$ | 4 | 10 | 6 | 4 |
| $(1,11)$ | 4 | 7 | 5 | 2 |
| $(1,13)$ | 4 | 5 | 4 | 1 |
| $(1,15)$ | 4 | 1 | 1 | 0 |
| $(2,0)$ | 2 | 12 | 12 | 0 |
| $(2,2)$ | 4 | 7 | 7 | 0 |
| $(2,4)$ | 4 | 5 | 5 | 0 |
| $(2,6)$ | 4 | 3 | 3 | 0 |
| $(2,8)$ | 4 | 3 | 3 | 0 |
| $(2,10)$ | 4 | 1 | 1 | 0 |
| $(2,12)$ | 4 | 1 | 1 | 0 |
| Total (with sign orbits) | | 956 | 634 | 322 |

The exact quotient current-character decomposition gives a second way to check the 322 count. Here $(j_L, j_R)$ denotes highest Cartan weights, so the irreducible has dimension $(j_L+1)(j_R+1)$. The sum of multiplicities is 28. Equivalently, the quotient by the two raising images has dimension 28, and those images have rank $322 - 28 = 294$.

| Highest weight | Multiplicity | Irrep dimension | Total dimension |
|---|---|---|---|
| $(1,13)$ | 1 | 28 | 28 |
| $(1,11)$ | 1 | 24 | 24 |
| $(1,9)$ | 2 | 20 | 40 |
| $(1,7)$ | 2 | 16 | 32 |
| $(1,5)$ | 2 | 12 | 24 |
| $(1,3)$ | 1 | 8 | 8 |
| $(1,1)$ | 1 | 4 | 4 |
| $(0,18)$ | 1 | 19 | 19 |
| $(0,14)$ | 2 | 15 | 30 |
| $(0,12)$ | 1 | 13 | 13 |
| $(0,10)$ | 3 | 11 | 33 |

| Highest weight | Multiplicity | Irrep dimension | Total dimension |
|---|---|---|---|
| $(0,8)$ | 2 | 9 | 18 |
| $(0,6)$ | 5 | 7 | 35 |
| $(0,4)$ | 1 | 5 | 5 |
| $(0,2)$ | 3 | 3 | 9 |
| Total | 28 | | 322 |

The ambient weight-three space has dimension 11586, so these counts are dimensions of explicitly constructed subspaces of a much larger space. Before the two closure gates, they do not rule out extra vectors appearing from higher-weight products. After strong generation, weight bookkeeping gives $U_3 = S_3$, just as it gives $U_2 = S_2$.

### S7.7.1 What the preceding numerical and modular records establish

The floating span record reached $634 \to 902 \to 956$ by using the $(1,5)$ and then the first $(1,9)$ families. Its smallest accepted relative residual was $9.7730725 \times 10^{-5}$; its largest rejected relative residual was $2.6155948 \times 10^{-9}$. These provide evidence of numerical separation for that run, not an exact upper bound.

The modular run used $p = 1000003$ with $s \mapsto 231543$ and $t \mapsto 1000$. At a prime where all required denominators remain invertible, a nonzero minor survives as a proof of a characteristic-zero lower bound. A modular rank of 956 therefore shows that at least 956 vectors are independent. It does not show that a characteristic-zero span has no further vectors. The separate number-field run is what records exact dimension 956.

The historical single-generator action test used only the three-term $u$, on all 956 numerical source vectors. Its diagnostics are:

| Mode | Target grade | Total leakage | Largest block relative leakage |
|---|---|---|---|
| 1 | 3 | $4.111 \times 10^{-10}$ | $6.343 \times 10^{-11}$ |
| 2 | 2 | $1.524 \times 10^{-10}$ | $5.408 \times 10^{-11}$ |
| 3 | 1 | $6.015 \times 10^{-11}$ | $5.237 \times 10^{-11}$ |

These small floating residuals motivated the exact closure work. They do not establish the action of every composite generator, and they do not replace the finite gates. The exact positive-mode consequences proved after Gate 4 apply to the whole $S_2$ and therefore have a strictly different logical scope from this early single-generator test.

## S7.8 The 3970 lower identities: an explicit task inventory

The number 3970 in the closure ledger is the size of a concrete ordered task list. It is useful to unpack that number because an unqualified "all lower checks passed" conceals several different target spaces. The implementation `hi_cft_20260923_next_structure_lower_closure.py` uses ten weight-two primitive fields: the eight $a_i$, followed by $e, f$. Its tasks are exactly:

| Task family | Number | Target grade |
|---|---|---|
| $L(1)a = 0$ for the ten primitive fields | 10 | 1 |
| $a_0 v \in S_3$ for ten fields and 120 basis vectors | 1200 | 3 |
| $a_1 v \in S_2$ for the same pairs | 1200 | 2 |
| $x_0 v \in S_2$ for six currents and 120 basis vectors | 720 | 2 |
| $x_1 v \in K$ for the same pairs | 720 | 1 |
| $L(1)v \in K$ for all 120 basis vectors | 120 | 1 |
| Total | 3970 | |

The first two left fields processed in the weight-two product portion are $e, f$, since their zero-product families were not seeds in the old grade-three construction. The source then processes $a_0, \dots, a_7$. This order is why database primary keys cannot be interpreted by simply multiplying the charged-field index by 120.

There is a further basis convention in this database. The six `current_basis` rows are unit-pivot echelon rows, and are

$$\begin{array}{c|l} 0 & -H_L \\ 1 & E_L \\ 2 & -F_L \\ 3 & -(H_R + 5H_L)/2 \\ 4 & E_R \\ 5 & -F_R/5 \end{array}$$

in the canonical current notation. The left current in a stored `current` task is still the original canonical current indexed $0, \dots, 5$; the transformed rows appear when recording the target's coordinates in $K$. Interchanging these two roles changes coefficients and can appear to create a sign failure even when the exact identity is correct.

The completed record has cursor 3970, all residual lists empty, 224480 sparse input terms, and a largest recorded coefficient size of 290 bits. Its exact arithmetic uses four GMP rational components. Its saved elapsed time is 77.88 seconds and peak RSS 239784 KiB. These resource figures describe the old run; they are not estimates for arbitrarily larger grades.

## S7.9 The actual triplet before any quotient-dimension assertion

The triplet record `hi_cft_20260923_next_structure_p4_controls.json` starts with the actual vector

$$t = (a_0)_0 \texttt{r3_000705} \in B_4, \qquad \mathrm{wt}_{\mathrm{Cartan}}(t) = (0, 2).$$

It constructs its highest triplet vector by

$$v = \left(1 - \frac{F_L E_L}{2}\right)\left(1 - \frac{F_L E_L}{6}\right) \prod_{d \in D_2} \left(1 - \frac{F_R E_R}{d}\right) t,$$

$$D_2 = \{4, 10, 18, 28, 40, 54, 70, 88, 108, 130, 154\}.$$

Every operator here is a zero mode, with $E$ applied before $F$. The three stored fields are

$$P4_+ = v, \qquad P4_0 = (F_R)_0 v, \qquad P4_- = (F_R)_0^2 v.$$

They have 2313, 2363, and 2313 nonzero lattice-oscillator coordinates, respectively. The left projection retains 2669 terms; the first right projection lowers that support to 2313, and

the remaining right factors retain this count. Support counts are a useful fingerprint for a reconstruction, but are not an equality test.

There are eighteen current-zero-mode identities, six for each of the three actual vectors. They establish the current type $(1,3)$ in representation-dimension notation. Six further exact checks put $L(1)P4_{\pm,0}$ in $S_3$ and $L(2)P4_{\pm,0}$ in $S_2$. These direct positive-Virasoro controls precede Gate 4 in the logical order, so subsequent positive-mode arguments do not use their own conclusion as a premise.

The separate modular replacement control finds nonzero multipliers 779530, 591896, and 665783 at $p = 1000003$ for the three weights $(0,2),(0,0),(0,-2)$. It shows that these actual fields replace the selected discovery columns. It does not prove that $N_4$ has a particular characteristic-zero dimension. The later argument therefore allows both possibilities: $P_4 \subset N_4$, or an irreducible triplet survives in $W_4/N_4$.

### S7.10 Gate 4 saturation: every operator deficit and every added input

The exact saturation record contains 644 raising-coordinate columns, two for each basis vector of $P = S_3/N_3$. Their rank is 294. For each highest left input, the old exact seed inputs are appended to these columns. The next table displays the resulting rank and the exact deficit. The index 8 denotes the retained neutral stress $e$; indices $0,\dots,7$ denote the charged highest fields. Each augmented rank becomes 322 after the listed additional requests are supplied.

| Left index | Old seed inputs | Rank with raising images | Deficit | Completed rank |
|---:|---:|---:|---:|---:|
| 0 | 197 | 307 | 15 | 322 |
| 1 | 236 | 303 | 19 | 322 |
| 2 | 234 | 302 | 20 | 322 |
| 3 | 277 | 306 | 16 | 322 |
| 4 | 226 | 313 | 9 | 322 |
| 5 | 244 | 307 | 15 | 322 |
| 6 | 259 | 305 | 17 | 322 |
| 7 | 302 | 313 | 9 | 322 |
| 8 | 170 | 312 | 10 | 322 |
| Total seed inputs | 2145 | | 130 | |

The 130 requested input pairs are printed next. “Direct” means one of the 118 completed additional ordinary exact identities. “Singlet” means one of the twelve weight-zero targets recovered by raising controls and the projected exact identity. These are descriptions of the completed later coverage, not success claims made by the original request file. The source list is `hi_cft_20260923_c8two_structure_saturation.json`; final coverage is recorded in `hi_cft_20260923_c8two_gate4_verified.json`.

| Left | Exact right input | Input weight | Output weight | Coverage |
|---:|---|---:|---:|---|
| 0 | `r3_000921` | $(0,-10)$ | $(0,-6)$ | Direct |
| 0 | `r3_000917` | $(0,-10)$ | $(0,-6)$ | Direct |
| 0 | `r3_000915` | $(0,-10)$ | $(0,-6)$ | Direct |
| 0 | `r3_000920` | $(0,-8)$ | $(0,-4)$ | Direct |
| 0 | `r3_000916` | $(0,-8)$ | $(0,-4)$ | Direct |

| Left | Exact right input | Input weight | Output weight | Coverage |
|---|---|---|---|---|
| 0 | r3_000874 | $(-1,-3)$ | $(-1,1)$ | Direct |
| 0 | r3_000952 | $(0,-6)$ | $(0,-2)$ | Direct |
| 0 | r3_000951 | $(0,-6)$ | $(0,-2)$ | Direct |
| 0 | r3_000946 | $(0,-6)$ | $(0,-2)$ | Direct |
| 0 | r3_000913 | $(0,-6)$ | $(0,-2)$ | Direct |
| 0 | r3_000912 | $(0,-6)$ | $(0,-2)$ | Direct |
| 0 | r3_000955 | $(0,-2)$ | $(0,2)$ | Direct |
| 0 | r3_000947 | $(0,-2)$ | $(0,2)$ | Direct |
| 0 | r3_000945 | $(0,-2)$ | $(0,2)$ | Direct |
| 0 | r3_000950 | $(0,-4)$ | $(0,0)$ | Singlet |
| 1 | r3_000923 | $(0,-14)$ | $(0,-6)$ | Direct |
| 1 | r3_000910 | $(0,-14)$ | $(0,-6)$ | Direct |
| 1 | r3_000955 | $(0,-2)$ | $(0,6)$ | Direct |
| 1 | r3_000947 | $(0,-2)$ | $(0,6)$ | Direct |
| 1 | r3_000945 | $(0,-2)$ | $(0,6)$ | Direct |
| 1 | r3_000922 | $(0,-12)$ | $(0,-4)$ | Direct |
| 1 | r3_000950 | $(0,-4)$ | $(0,4)$ | Direct |
| 1 | r3_000921 | $(0,-10)$ | $(0,-2)$ | Direct |
| 1 | r3_000917 | $(0,-10)$ | $(0,-2)$ | Direct |
| 1 | r3_000915 | $(0,-10)$ | $(0,-2)$ | Direct |
| 1 | r3_000898 | $(-1,-7)$ | $(-1,1)$ | Direct |
| 1 | r3_000882 | $(-1,-7)$ | $(-1,1)$ | Direct |
| 1 | r3_000952 | $(0,-6)$ | $(0,2)$ | Direct |
| 1 | r3_000951 | $(0,-6)$ | $(0,2)$ | Direct |
| 1 | r3_000946 | $(0,-6)$ | $(0,2)$ | Direct |
| 1 | r3_000913 | $(0,-6)$ | $(0,2)$ | Direct |
| 1 | r3_000912 | $(0,-6)$ | $(0,2)$ | Direct |
| 1 | r3_000920 | $(0,-8)$ | $(0,0)$ | Singlet |
| 1 | r3_000916 | $(0,-8)$ | $(0,0)$ | Singlet |
| 2 | r3_000923 | $(0,-14)$ | $(0,-6)$ | Direct |
| 2 | r3_000910 | $(0,-14)$ | $(0,-6)$ | Direct |
| 2 | r3_000955 | $(0,-2)$ | $(0,6)$ | Direct |
| 2 | r3_000947 | $(0,-2)$ | $(0,6)$ | Direct |
| 2 | r3_000945 | $(0,-2)$ | $(0,6)$ | Direct |
| 2 | r3_000922 | $(0,-12)$ | $(0,-4)$ | Direct |
| 2 | r3_000950 | $(0,-4)$ | $(0,4)$ | Direct |
| 2 | r3_000900 | $(-1,-9)$ | $(-1,-1)$ | Direct |
| 2 | r3_000921 | $(0,-10)$ | $(0,-2)$ | Direct |
| 2 | r3_000917 | $(0,-10)$ | $(0,-2)$ | Direct |
| 2 | r3_000915 | $(0,-10)$ | $(0,-2)$ | Direct |
| 2 | r3_000898 | $(-1,-7)$ | $(-1,1)$ | Direct |
| 2 | r3_000882 | $(-1,-7)$ | $(-1,1)$ | Direct |
| 2 | r3_000952 | $(0,-6)$ | $(0,2)$ | Direct |
| 2 | r3_000951 | $(0,-6)$ | $(0,2)$ | Direct |
| 2 | r3_000946 | $(0,-6)$ | $(0,2)$ | Direct |
| 2 | r3_000913 | $(0,-6)$ | $(0,2)$ | Direct |
| 2 | r3_000912 | $(0,-6)$ | $(0,2)$ | Direct |
| 2 | r3_000920 | $(0,-8)$ | $(0,0)$ | Singlet |
| 2 | r3_000916 | $(0,-8)$ | $(0,0)$ | Singlet |

| Left | Exact right input | Input weight | Output weight | Coverage |
|---|---|---|---|---|
| 3 | r3_000919 | $(0,-18)$ | $(0,-6)$ | Direct |
| 3 | r3_000952 | $(0,-6)$ | $(0,6)$ | Direct |
| 3 | r3_000951 | $(0,-6)$ | $(0,6)$ | Direct |
| 3 | r3_000946 | $(0,-6)$ | $(0,6)$ | Direct |
| 3 | r3_000913 | $(0,-6)$ | $(0,6)$ | Direct |
| 3 | r3_000912 | $(0,-6)$ | $(0,6)$ | Direct |
| 3 | r3_000920 | $(0,-8)$ | $(0,4)$ | Direct |
| 3 | r3_000916 | $(0,-8)$ | $(0,4)$ | Direct |
| 3 | r3_000902 | $(-1,-13)$ | $(-1,-1)$ | Direct |
| 3 | r3_000923 | $(0,-14)$ | $(0,-2)$ | Direct |
| 3 | r3_000910 | $(0,-14)$ | $(0,-2)$ | Direct |
| 3 | r3_000901 | $(-1,-11)$ | $(-1,1)$ | Direct |
| 3 | r3_000921 | $(0,-10)$ | $(0,2)$ | Direct |
| 3 | r3_000917 | $(0,-10)$ | $(0,2)$ | Direct |
| 3 | r3_000915 | $(0,-10)$ | $(0,2)$ | Direct |
| 3 | r3_000922 | $(0,-12)$ | $(0,0)$ | Singlet |
| 4 | r3_000898 | $(-1,-7)$ | $(0,-6)$ | Direct |
| 4 | r3_000882 | $(-1,-7)$ | $(0,-6)$ | Direct |
| 4 | r3_000897 | $(-1,-5)$ | $(0,-4)$ | Direct |
| 4 | r3_000879 | $(-1,-5)$ | $(0,-4)$ | Direct |
| 4 | r3_000874 | $(-1,-3)$ | $(0,-2)$ | Direct |
| 4 | r3_000955 | $(0,-2)$ | $(1,-1)$ | Direct |
| 4 | r3_000947 | $(0,-2)$ | $(1,-1)$ | Direct |
| 4 | r3_000945 | $(0,-2)$ | $(1,-1)$ | Direct |
| 4 | r3_000873 | $(-1,-1)$ | $(0,0)$ | Singlet |
| 5 | r3_000901 | $(-1,-11)$ | $(0,-6)$ | Direct |
| 5 | r3_000900 | $(-1,-9)$ | $(0,-4)$ | Direct |
| 5 | r3_000899 | $(-1,-9)$ | $(0,-4)$ | Direct |
| 5 | r3_000873 | $(-1,-1)$ | $(0,4)$ | Direct |
| 5 | r3_000898 | $(-1,-7)$ | $(0,-2)$ | Direct |
| 5 | r3_000882 | $(-1,-7)$ | $(0,-2)$ | Direct |
| 5 | r3_000874 | $(-1,-3)$ | $(0,2)$ | Direct |
| 5 | r3_000952 | $(0,-6)$ | $(1,-1)$ | Direct |
| 5 | r3_000951 | $(0,-6)$ | $(1,-1)$ | Direct |
| 5 | r3_000946 | $(0,-6)$ | $(1,-1)$ | Direct |
| 5 | r3_000913 | $(0,-6)$ | $(1,-1)$ | Direct |
| 5 | r3_000912 | $(0,-6)$ | $(1,-1)$ | Direct |
| 5 | r3_000950 | $(0,-4)$ | $(1,1)$ | Direct |
| 5 | r3_000897 | $(-1,-5)$ | $(0,0)$ | Singlet |
| 5 | r3_000879 | $(-1,-5)$ | $(0,0)$ | Singlet |
| 6 | r3_000902 | $(-1,-13)$ | $(0,-6)$ | Direct |
| 6 | r3_000873 | $(-1,-1)$ | $(0,6)$ | Direct |
| 6 | r3_000901 | $(-1,-11)$ | $(0,-4)$ | Direct |
| 6 | r3_000874 | $(-1,-3)$ | $(0,4)$ | Direct |
| 6 | r3_000900 | $(-1,-9)$ | $(0,-2)$ | Direct |
| 6 | r3_000899 | $(-1,-9)$ | $(0,-2)$ | Direct |
| 6 | r3_000897 | $(-1,-5)$ | $(0,2)$ | Direct |
| 6 | r3_000920 | $(0,-8)$ | $(1,-1)$ | Direct |
| 6 | r3_000879 | $(-1,-5)$ | $(0,2)$ | Direct |

| Left | Exact right input | Input weight | Output weight | Coverage |
|---|---|---|---|---|
| 6 | `r3_000916` | $(0,-8)$ | $(1,-1)$ | Direct |
| 6 | `r3_000952` | $(0,-6)$ | $(1,1)$ | Direct |
| 6 | `r3_000951` | $(0,-6)$ | $(1,1)$ | Direct |
| 6 | `r3_000946` | $(0,-6)$ | $(1,1)$ | Direct |
| 6 | `r3_000913` | $(0,-6)$ | $(1,1)$ | Direct |
| 6 | `r3_000912` | $(0,-6)$ | $(1,1)$ | Direct |
| 6 | `r3_000898` | $(-1,-7)$ | $(0,0)$ | Singlet |
| 6 | `r3_000882` | $(-1,-7)$ | $(0,0)$ | Singlet |
| 7 | `r3_000898` | $(-1,-7)$ | $(0,6)$ | Direct |
| 7 | `r3_000882` | $(-1,-7)$ | $(0,6)$ | Direct |
| 7 | `r3_000900` | $(-1,-9)$ | $(0,4)$ | Direct |
| 7 | `r3_000899` | $(-1,-9)$ | $(0,4)$ | Direct |
| 7 | `r3_000923` | $(0,-14)$ | $(1,-1)$ | Direct |
| 7 | `r3_000910` | $(0,-14)$ | $(1,-1)$ | Direct |
| 7 | `r3_000901` | $(-1,-11)$ | $(0,2)$ | Direct |
| 7 | `r3_000922` | $(0,-12)$ | $(1,1)$ | Direct |
| 7 | `r3_000902` | $(-1,-13)$ | $(0,0)$ | Singlet |
| 8 | `r3_000952` | $(0,-6)$ | $(0,-6)$ | Direct |
| 8 | `r3_000951` | $(0,-6)$ | $(0,-6)$ | Direct |
| 8 | `r3_000946` | $(0,-6)$ | $(0,-6)$ | Direct |
| 8 | `r3_000913` | $(0,-6)$ | $(0,-6)$ | Direct |
| 8 | `r3_000912` | $(0,-6)$ | $(0,-6)$ | Direct |
| 8 | `r3_000950` | $(0,-4)$ | $(0,-4)$ | Direct |
| 8 | `r3_000873` | $(-1,-1)$ | $(-1,-1)$ | Direct |
| 8 | `r3_000955` | $(0,-2)$ | $(0,-2)$ | Direct |
| 8 | `r3_000947` | $(0,-2)$ | $(0,-2)$ | Direct |
| 8 | `r3_000945` | $(0,-2)$ | $(0,-2)$ | Direct |

There is no assertion that these seed inputs are stable under both lowering operators, or that they exhaust all 322 quotient basis vectors for a fixed left field. The deduction uses stability of the *kernel* of $[p] \mapsto [a_0 p]$ under the raising operators. For a highest $a$, covariance makes that stability immediate. If the raising images and certified kernel inputs span $P$, then the quotient by this kernel equals its own image under the raising ideal. Because every raising word of length 25 vanishes on grade three, iteration forces the quotient to vanish. This explains why the table's nine completed rank-322 assertions imply $9 \cdot 322 = 2898$ highest-left identities, despite far fewer direct computations for each family.

### S7.10.1 The twelve additional raising-premise inputs

The singlet route also uses twelve ordinary identities with output Cartan weight $(0,2)$. These belong to a separate request file and should not be silently identified with the twelve projected weight-zero targets. The following table gives the complete list from `hi_cft_20260923_c8two_structure_h4_raise_request.json`. The request file itself has status "request only"; the completed raised-input summary and singlet-premise verifier are the later proof records.

| Left | Exact right input | Input weight | Output weight |
|---|---|---|---|
| 0 | `r3_000944` | $(0,-2)$ | $(0,2)$ |

| Left | Exact right input | Input weight | Output weight |
|---|---|---|---|
| 1 | `r3_000905` | $(0,-6)$ | $(0,2)$ |
| 1 | `r3_000907` | $(0,-6)$ | $(0,2)$ |
| 2 | `r3_000905` | $(0,-6)$ | $(0,2)$ |
| 2 | `r3_000907` | $(0,-6)$ | $(0,2)$ |
| 3 | `r3_000909` | $(0,-10)$ | $(0,2)$ |
| 4 | `r3_000842` | $(-1,1)$ | $(0,2)$ |
| 5 | `r3_000850` | $(-1,-3)$ | $(0,2)$ |
| 5 | `r3_000871` | $(-1,-3)$ | $(0,2)$ |
| 6 | `r3_000854` | $(-1,-5)$ | $(0,2)$ |
| 6 | `r3_000875` | $(-1,-5)$ | $(0,2)$ |
| 7 | `r3_000885` | $(-1,-11)$ | $(0,2)$ |

Together with the earlier exact seed relations, these imply the 24 raising conditions $E_L t, E_R t \in W_4$ for the twelve remaining weight-zero targets. The source-bound replay is `hi_cft_20260923_c8two_h4_singlet_premise_verified.json`. Only after this premise is available does a projected residual prove an unprojected membership statement.

### S7.11 All twenty-one normal columns in the singlet certificate

For weight-zero inputs the joint-singlet projector has the right-axis denominators

$$D_0 = \{2, 6, 12, 20, 30, 42, 56, 72, 90, 110, 132, 156\},$$

with the same left-axis denominators $2, 6$. Thus $D_0$ differs from $D_2$ above: they project different Cartan weight spaces. The exact projected equations have the form

$$\Pi t_i = \sum_{j=0}^{20} c_{ij} \Pi n_j.$$

The following table exposes every unprojected word $n_j$ and its original normal-column ID. The projected-column number is a separate index. This translation is enough to interpret the saved coefficients as actual elements of $N_4$, rather than as arbitrary vectors in a linear solver.

| Projected $j$ | Original normal ID | Actual word $n_j$ |
|---|---|---|
| 0 | 1 | $(H_L)_{-1}$`r3_000001` |
| 1 | 14 | $(H_L)_{-1}$`r3_000014` |
| 2 | 22 | $(H_L)_{-1}$`r3_000022` |
| 3 | 23 | $(H_L)_{-1}$`r3_000023` |
| 4 | 121 | $(H_L)_{-1}$`r3_000121` |
| 5 | 1811 | $(H_R)_{-1}$`r3_000221` |
| 6 | 1816 | $(H_R)_{-1}$`r3_000226` |
| 7 | 1819 | $(H_R)_{-1}$`r3_000229` |
| 8 | 1820 | $(H_R)_{-1}$`r3_000230` |
| 9 | 1823 | $(H_R)_{-1}$`r3_000233` |
| 10 | 1920 | $(H_R)_{-1}$`r3_000330` |

| Projected $j$ | Original normal ID | Actual word $n_j$ |
|---:|---:|---|
| 11 | 2256 | $(H_R)_{-1}$r3_000666 |
| 12 | 2521 | $(H_R)_{-1}$r3_000931 |
| 13 | 2544 | $(H_R)_{-1}$r3_000954 |
| 14 | 4003 | $T$r3_000536 |
| 15 | 4004 | $T$r3_000537 |
| 16 | 4429 | $(a_0)_{-1}$r2_000028 |
| 17 | 4535 | $(a_1)_{-1}$r2_000037 |
| 18 | 4544 | $(a_1)_{-1}$r2_000046 |
| 19 | 4640 | $(a_2)_{-1}$r2_000046 |
| 20 | 4735 | $(a_3)_{-1}$r2_000059 |

The exact projected coefficients can be long: some have numerators and denominators of approximately one hundred decimal digits. A compact exact description of their reconstruction is to choose 21 independent pivot coordinates of the projected normal columns, form the square matrix $M$ of their coefficients, and solve

$$c_i = M^{-1}(\Pi t_i)|_{\text{pivots}}.$$

This produces candidate coefficients only. The decisive check is the entire sparse identity $\Pi t_i - \sum_j c_{ij}\Pi n_j = 0$, on every ambient basis state and all four rational components. A pivot solve without that last check could pass while a nonzero residual remains outside the chosen coordinates.

The following target-size table is transcribed from the portable independent replay. All twelve full residuals were zero.

| Target ID | Left | Right input | Target terms | Columns used |
|---:|---:|---|---:|---:|
| 0 | 0 | r3_000950 | 2343 | 19 |
| 1 | 1 | r3_000916 | 2343 | 19 |
| 2 | 1 | r3_000920 | 2343 | 19 |
| 3 | 2 | r3_000916 | 2343 | 19 |
| 4 | 2 | r3_000920 | 2343 | 19 |
| 5 | 3 | r3_000922 | 2343 | 19 |
| 6 | 4 | r3_000873 | 2726 | 21 |
| 7 | 5 | r3_000879 | 2726 | 21 |
| 8 | 5 | r3_000897 | 2726 | 21 |
| 9 | 6 | r3_000882 | 2726 | 21 |
| 10 | 6 | r3_000898 | 2726 | 21 |
| 11 | 7 | r3_000902 | 2726 | 21 |

The portable packet contains the twelve projected targets and the 21 projected normal columns, with 66610 sparse state terms in all. It permits exact residual checking without the original SQLite stores. Its historical grade5: prefix for projected normal vectors is a storage name; the vectors in this packet have weight four. The portable residual replay does not reconstruct the projector action, the normal-word provenance, or the raising premises, all of which remain separately source-bound inputs.

The final recovery is a quotient calculation. Since $E_L t_i, E_R t_i$ lie in $W_4$ and $W_4$ is current-stable, every $FE$ factor acts as zero on $[t_i]$. Hence $[\Pi t_i] = [t_i]$ in $B_4/W_4$. Every $\Pi n_j$ stays in the current-stable space $N_4$. Therefore the full residual equality implies $[t_i] = 0$. The argument never needs the 21 selected columns to span all of $N_4$.

### S7.12 Gate 5: the charged-span witness behind the neutral step

The charged part of Gate 5 is unusually small after Gate 4 is known. The eight charged highest weights are

$$(0,4),(0,8),(0,8),(0,12),(1,1),(1,5),(1,7),(1,13).$$

If $Q = W_4/N_4$ is nonzero, it is the triplet with weights $(0,-2),(0,0),(0,2)$. Two previously computed bottom relations give nonzero coefficients of its lowest vector. Seven of the eight required charged brackets vanish exactly, and the remaining bracket is an explicit current normal product. The complete small coefficients and normal-word identities are printed in the accompanying finite-witness section. They imply $a_0 P_4 \subset N_5$ for every charged $a$.

A separate spanning witness is required for the retained neutral field. This subsection exposes the 28 accepted seeds of that witness. For a checkpoint row $r$ of $S_2$, the seed indexed by $(i,x,r)$ is

$$\big[(a_i)_0(x_0 r)\big] \in P = S_3/N_3.$$

The saved lower identities express $x_0 r$ in the exact $S_2$ basis, and express every $(a_i)_0 r'$ in the exact $S_3$ basis. Composing these two saved coordinate maps gives the displayed seed. Thus no new grade-three mode calculation is needed for this particular spanning replay. The seed lies in the current span of charged brackets: a current zero mode kills the neutral singlets, while the affine-normal right parts contribute zero modulo $N_3$ by the lower normal controls.

Before inserting these seeds, the 644 lowering images of the 322 quotient basis vectors have rank 294. Each row below adds one to that rank; the final rank is 322. The original producer tested 1129 candidates before obtaining this accepted list.

| Seed | $i$ | Current $x$ | Right input $r$ | Rank after insertion |
|---|---|---|---|---|
| 0 | 0 | $H_L$ | `r2_000060` | 295 |
| 1 | 0 | $H_L$ | `r2_000061` | 296 |
| 2 | 0 | $H_L$ | `r2_000064` | 297 |
| 3 | 0 | $H_L$ | `r2_000065` | 298 |
| 4 | 0 | $H_L$ | `r2_000066` | 299 |
| 5 | 0 | $H_L$ | `r2_000068` | 300 |
| 6 | 0 | $H_L$ | `r2_000076` | 301 |
| 7 | 0 | $H_L$ | `r2_000077` | 302 |
| 8 | 0 | $H_L$ | `r2_000078` | 303 |
| 9 | 0 | $H_L$ | `r2_000094` | 304 |
| 10 | 0 | $H_R$ | `r2_000025` | 305 |
| 11 | 0 | $H_R$ | `r2_000027` | 306 |
| 12 | 0 | $H_R$ | `r2_000029` | 307 |
| 13 | 0 | $H_R$ | `r2_000030` | 308 |
| 14 | 0 | $H_R$ | `r2_000031` | 309 |
| 15 | 0 | $H_R$ | `r2_000032` | 310 |
| 16 | 0 | $H_R$ | `r2_000039` | 311 |

| Seed | $i$ | Current $x$ | Right input $r$ | Rank after insertion |
|---|---|---|---|---|
| 17 | 0 | $H_R$ | `r2_000040` | 312 |
| 18 | 0 | $H_R$ | `r2_000041` | 313 |
| 19 | 0 | $H_R$ | `r2_000048` | 314 |
| 20 | 0 | $H_R$ | `r2_000050` | 315 |
| 21 | 0 | $E_R$ | `r2_000034` | 316 |
| 22 | 1 | $H_R$ | `r2_000030` | 317 |
| 23 | 1 | $H_R$ | `r2_000034` | 318 |
| 24 | 1 | $H_R$ | `r2_000036` | 319 |
| 25 | 1 | $H_R$ | `r2_000043` | 320 |
| 26 | 1 | $H_R$ | `r2_000045` | 321 |
| 27 | 1 | $H_R$ | `r2_000048` | 322 |

These columns are specialized at the same good split prime $p = 1000003$, with $s \mapsto 231543$, $t \mapsto 1000$. The independent verifier reconstructs the seeds from the exact lower relations, checks the two $\mathfrak{sl}_2$ commutators and the cross-axis commutators against the saved raising maps, and recomputes the positive full-rank witness. In particular, the lowering maps are not merely arbitrary matrices with the desired rank.

Let $C$ be the full current submodule generated by charged brackets. The rank witness gives

$$P = F_L P + F_R P + C.$$

On $P/C$ the right side has no $C$ term. The lowering ideal is nilpotent for the same bounded-weight reason as the raising ideal, so iteration gives $P/C = 0$. The proof needs full current stability of $C$; the list of 28 seeds itself need not be a current-stable basis. The independent result is `hi_cft_20260923_c8two_charged_span_verified.json`.

### S7.12.1 How the neutral identity uses this particular spanning claim

Because $P$ is generated by charged brackets, one may write

$$e_0 a_3 = \sum_j d_j (x_j)_0 y_j + n, \qquad n \in N_3,$$

where $x_j, y_j$ are charged fields, including their current descendants. For the bottom input $p_3 =$ `r3_000910`, the zero-mode commutator gives

$$(e_0 a_3)_0 p_3 = \sum_j d_j \big((x_j)_0 (y_j)_0 - (y_j)_0 (x_j)_0\big) p_3 + n_0 p_3 \in N_5.$$

Here the inner products are in $W_4$ by Gate 4. The outer charged products preserve membership modulo $N_5$ on its normal and triplet parts by the charged conclusion and the normal-space identities. A second application of Jacobi to $e, a_3, p_3$ then forces the neutral bottom identity, since the coefficient of $P4_-$ in $(a_3)_0 p_3$ is nonzero. Current covariance propagates it along the triplet, and $f$ follows from the total ambient stress minus $e$ and the affine Sugawara stresses.

This makes the dependencies of the saved Gate 5 flag explicit. The record `hi_cft_20260923_c8two_jacobi_checked.json` already has the exact small scalar data, but marks Gate 4 authentication, charged-family derivation, neutral-family derivation and all-channel completion false. Its status is conditional prerequisites. In contrast, `hi_cft_20260923_c8two_gate5_ve`

`rified.json` authenticates Gate 4 and the charged-span certificate and records all those flags true. The older conditional record is not evidence of a contradiction: its prerequisites had not yet been supplied when it was written.

### S7.13 Why the unsuccessful direct routes remain in the archive

Several earlier files describe direct weight-five normal-product searches, including the `n5_direct`, `n5_primitive`, `n5_axisstop`, and `n5_priority` families. Their mathematical role was to search for positive containment identities using a selected subset of actual normal words. They were useful exploratory and engineering work. The completed Gate 5 conclusion uses the Jacobi reduction above; it does not silently assume that all of those direct searches completed.

The distinctions are especially relevant at weight four. The direct pipeline retained 6007 selected raw normal words, then adjoined the actual triplet. Their characteristic-zero span had not been separately shown to exhaust the full formal $N_4$. A prime rank over these columns can establish a positive lower bound or a successful exact relation. It cannot establish failure of containment in the full $N_4$, nor prove a full characteristic-zero normal-space dimension. Consequently "rank 6007 plus three" is not, by itself, a proof that $\dim U_4 = 6010$.

The final strong-generation argument identifies $U_4 = W_4$ by weight bookkeeping, but the coefficient 6010 in the full vacuum character has a later character-theoretic dependency. This distinction is retained even though the various numerical and modular counts are consistent with 6010.

The current-projector optimization called "axis stop" has a similarly precise meaning. If a raising action is exactly zero on the current vector, every remaining same-axis $FE$ factor acts as the identity. The other commuting axis must still be processed. This is an algebraic identity optimization and does not change projector ranges or permit small numerical residuals to be treated as zero.

Finally, an apparently attractive shortcut through the single weak generator $u$ was rejected in the source audits. Trying to show that a Griess product preserves a not-yet-certified normal space introduces

$$(a_0x)_0p = a_0(x_0p) - x_0(a_0p).$$

Even after Gate 4, $a_0p$ can contain a triplet component, so the second term asks for exactly an unknown $x_0P_4$ identity. The eventual Jacobi argument avoids this circle by choosing two concrete bottom witnesses, using weight exclusion there, and only then propagating proved zero identities by covariance. Weight exclusion is not asserted anew after raising the bottom input.

### S7.14 Finite membership of the structural weight-three fields

The Potts and parafermion primaries used later in the structural argument also have finite generation obligations. It is not enough to recognize their ambient OPEs: they must belong to the actual $U$. The portable membership witness uses the same 120 weight-two source fields and 58 accepted Cartan-zero weight-three words. Its source-word inventory is

34 current normal products, 6 translations, 18 zero-products.

The two target expansions have the following sizes:

| Target | Ambient terms | Selected words | Rank at discovery | Pairs tried |
|---|---|---|---|---|
| Potts primary | 92 | 33 | 41 | 1 |
| Parafermion primary | 342 | 42 | 58 | 154 |

Here "pairs tried" is the recorded cumulative number of Cartan-zero weight-two pairs at discovery. The membership exporter has 43462 persistent sparse terms in all of its source and target material. The independent coordinate verifier decodes 21772 basis rows, checks every stored grade, confirms Cartan weight zero for all weight-three sources and targets, and checks both exact coordinate sums. These differing counts reflect different retained and decoded objects; they are not competing claims for a vector-space dimension.

The complete expressions are in `c8_reviewability_20260927/membership_witnesses.json` and are printed in full, with their source-word conventions, in Section S7.16. Its separate check record is `c8_reviewability_20260927/membership_coordinate_check.json`. That checker establishes saved coordinate sums and labels. It does not recompute the source modes or prove the weight-two generation step; those premises are supplied by the preceding finite construction. The former rank-41 Potts membership assertion lacked a saved expression; the 33-word export is the additional concrete witness that repairs that reviewability limitation.

## S7.15 A compact provenance map for checking claims in this section

The next list orders the source records by their mathematical role, not by filename or execution time. Every entry is a local artifact; no claim of a new research replay is made by this transcription.

| Claim or object | Principal saved source |
|---|---|
| Ambient root enumeration and numerical primary kernels | `hi_cft_20260923_c8_lattice_control.json` |
| Historical Virasoro-axis and primary-eigenvalue search | `hi_cft_20260923_c8_ope_control.json` |
| All nine exact field-coordinate tables | `hi_cft_20260923_c8_exact_fields.json` |
| Exact low-mode closure and transported PCT | `hi_cft_20260923_c8_exact_unitary_certificate.json` |
| Short-generator membership graph | `c8_reviewability_20260927/minimal_generation_witness.json` |
| Independent graph/coordinate verification | `c8_reviewability_20260927/minimal_generation_checked.json` |
| Exact grade-three dimension and charge census | `hi_cft_20260923_c8_grade3_exact_v2.json` |
| All lower-mode identities | `hi_cft_20260923_next_structure_lower_closure.json` |
| Actual triplet, current action and Virasoro controls | `hi_cft_20260923_next_structure_p4_controls.json` |
| Exact quotient raising maps and 130 requests | `hi_cft_20260923_c8two_structure_saturation.json` |
| Completed 118 direct additional relations | `hi_cft_20260923_c8two_targeted_h4_trimmed_v2/summary.json` |

| Claim or object | Principal saved source |
|---|---|
| Completed twelve raised-input relations | `hi_cft_20260923_c8two_structure_h4_raised_exact/summary.json` |
| Twenty-four raising premises for the singlet targets | `hi_cft_20260923_c8two_h4_singlet_premise_verified.json` |
| Full projected exact residuals | `hi_cft_20260923_c8two_h4_singlet_exact.json` |
| Portable residual-only independent replay | `c8_reviewability_20260927/gate4_packet_checked.json` |
| Final Gate 4 recovery and saturation | `hi_cft_20260923_c8two_gate4_verified.json` |
| Charged-span seeds and independent replay | `hi_cft_20260923_c8two_structure_charged_span.json`; `hi_cft_20260923_c8two_charged_span_verified.json` |
| Final Gate 5 Jacobi prerequisites and conclusion | `hi_cft_20260923_c8two_gate5_verified.json` |
| Potts/parafermion generated-field membership | `c8_reviewability_20260927/membership_witnesses.json` |
| Repaired finite-input authentication ledger | `c8_revision_20260927/finite_input_ledger.json` |

The original proof-bundle ledger authenticated 640 files but omitted the independent short-generator record and its producer. That omission mattered because the grade-four initializer imports saved highest fields directly, whereas the short-generator record proves their membership in $\langle K, \omega_B, u\rangle$. The independent exact grade-three dimension record was also absent from that old ledger. The 27 September supplementary authentication bound these records, the relevant fresh replay outputs, and their sources, reaching 650 authenticated source/output files. Source authentication prevents unnoticed changes of input; the equality and rank witnesses supply the mathematical content.

For an efficient manual review, first check one full charged field from its coordinate table and the elementary mode formula; then check the short-generation coordinate expressions for that field; then inspect its Gate 4 row and the common saturation argument. The remaining Gate 5 work is the small bottom-witness/Jacobi calculation together with the independent charged-span argument. This order keeps ambient identities, actual generation, finite closure, and all-weight deductions separate while making their connection explicit.

### S7.16 Complete membership expressions for both structural primaries

Here are all 33 Potts and all 42 parafermion membership coefficients, so neither expansion requires consulting the TSV file. The source is `c8_reviewability_20260927/membership_witnesses.coefficients.tsv`, whose coefficients are also stored in the portable membership JSON. The parafermion target is the *unnormalized* saved primary $w$; the normalized field $a = \sigma w$ used in the nonlinear OPE presentation is obtained by multiplying the complete expression by $\sigma$. The Potts target is the saved $w_P$.

The $S_2$ basis used here differs from the short-generation normalized basis $r_i$. Denote it by $b_0, \dots, b_{119}$. Its first 23 vectors are the *raw* vectors $v_0, \dots, v_{22}$ of Section S7.6; the saved coordinates agree entry by entry. Next, $b_{23} = e$. The remaining 96 are the charged multiplets,

in the following completely specified order:

$$b_{q_i+m_i\ell+j} = (F_L)_0^{\ell}(F_R)_0^{j}a_i, \qquad 0 \le \ell < n_i, \quad 0 \le j < m_i.$$

There are no pivot normalizations or divided powers in this formula.

| $i$ | $q_i$ | $n_i$ | $m_i$ | Range of $b$ indices |
|---|---|---|---|---|
| 0 | 24 | 1 | 5 | 24–28 |
| 1 | 29 | 1 | 9 | 29–37 |
| 2 | 38 | 1 | 9 | 38–46 |
| 3 | 47 | 1 | 13 | 47–59 |
| 4 | 60 | 2 | 2 | 60–63 |
| 5 | 64 | 2 | 6 | 64–75 |
| 6 | 76 | 2 | 8 | 76–91 |
| 7 | 92 | 2 | 14 | 92–119 |

Thus $b_{23} = e$, whereas $r_{23} = v_{23} = u$ in the short-generation graph. This single example makes the index mismatch visible. The checkpoint labels beginning `r2_` provide yet a third basis convention and are not used in the tables below.

Write $z_j$ for the indicated unreduced weight-three source word. Each table asserts the actual lattice-vector equality target $= \sum_j c_j z_j$, with precisely the listed nonzero coefficients. A current factor such as $(H_L)_{-1}b_0$ is a normal product; $Tb_i$ is an ambient translation; $(b_i)_0 b_k$ is a zero-product. The source ID $j$ is retained solely to allow a direct comparison with the exported certificate.

### S7.16.1 Potts primary

| Source $j$ | Actual source word $z_j$ | Coefficient $c_j$ |
|---|---|---|
| 0 | $(H_L)_{-1}b_0$ | $\frac{1053}{70}$ |
| 1 | $(H_L)_{-1}b_1$ | $\frac{-729}{280}$ |
| 2 | $(H_L)_{-1}b_4$ | $\frac{1647}{280}$ |
| 3 | $(H_L)_{-1}b_{13}$ | $\frac{-190971}{27265}$ |
| 4 | $(H_L)_{-1}b_{14}$ | $\frac{-204741}{218120}$ |
| 5 | $(H_L)_{-1}b_{19}$ | $\frac{27729}{54530}$ |
| 6 | $(H_L)_{-1}b_{22}$ | $\frac{-1053}{35}$ |
| 7 | $(H_L)_{-1}b_{23}$ | $\frac{2511}{140}$ |
| 8 | $(H_L)_{-1}b_{26}$ | $\frac{117}{152}$ |
| 9 | $(H_L)_{-1}b_{33}$ | $\frac{-27s-54t}{35200}$ |
| 10 | $(H_L)_{-1}b_{42}$ | $\frac{9s-27t}{44800}$ |
| 11 | $(H_L)_{-1}b_{53}$ | $\frac{-2+t}{985600}$ |
| 12 | $(E_L)_{-1}b_3$ | $\frac{81}{2}$ |
| 13 | $(E_L)_{-1}b_{10}$ | $\frac{-27}{2}$ |

| Source $j$ | Actual source word $z_j$ | Coefficient $c_j$ |
|---|---|---|
| 14 | $(H_R)_{-1}b_{13}$ | $\frac{60903}{21812}$ |
| 15 | $(H_R)_{-1}b_{14}$ | $\frac{-13717}{218120}$ |
| 16 | $(H_R)_{-1}b_{19}$ | $\frac{-118071}{27265}$ |
| 17 | $(H_R)_{-1}b_{22}$ | $\frac{-351}{35}$ |
| 18 | $(H_R)_{-1}b_{23}$ | $\frac{837}{140}$ |
| 19 | $(H_R)_{-1}b_{26}$ | $\frac{39}{152}$ |
| 20 | $(H_R)_{-1}b_{33}$ | $\frac{-9s-18t}{35200}$ |
| 21 | $(H_R)_{-1}b_{42}$ | $\frac{3s-9t}{44800}$ |
| 22 | $(H_R)_{-1}b_{53}$ | $\frac{-2+t}{2956800}$ |
| 23 | $(E_R)_{-1}b_{16}$ | $\frac{-27729}{109060}$ |
| 24 | $(E_R)_{-1}b_{20}$ | $-9$ |
| 29 | $(F_R)_{-1}b_{15}$ | $\frac{518499}{109060}$ |
| 34 | $Tb_{22}$ | $\frac{-1053}{35}$ |
| 35 | $Tb_{23}$ | $\frac{2511}{140}$ |
| 36 | $Tb_{26}$ | $\frac{117}{152}$ |
| 37 | $Tb_{33}$ | $\frac{-27s-54t}{35200}$ |
| 38 | $Tb_{42}$ | $\frac{9s-27t}{44800}$ |
| 39 | $Tb_{53}$ | $\frac{-2+t}{985600}$ |
| 40 | $(b_{92})_0b_{119}$ | $\frac{-1}{3985293312000}$ |

### S7.16.2 Unnormalized parafermion primary

| Source $j$ | Actual source word $z_j$ | Coefficient $c_j$ |
|---|---|---|
| 0 | $(H_L)_{-1}b_0$ | $\frac{-4901s-14703t}{280}$ |
| 1 | $(H_L)_{-1}b_1$ | $\frac{1521s+4563t}{1120}$ |
| 2 | $(H_L)_{-1}b_4$ | $\frac{-13637s-40911t}{3360}$ |
| 3 | $(H_L)_{-1}b_{13}$ | $\frac{1195337s+3586011t}{327180}$ |
| 4 | $(H_L)_{-1}b_{14}$ | $\frac{1281527s+3844581t}{2617440}$ |
| 5 | $(H_L)_{-1}b_{19}$ | $\frac{-173563s-520689t}{654360}$ |
| 6 | $(H_L)_{-1}b_{22}$ | $\frac{2197s+6591t}{140}$ |
| 7 | $(H_L)_{-1}b_{23}$ | $\frac{-5239s-15717t}{560}$ |
| 8 | $(H_L)_{-1}b_{26}$ | $\frac{-2197s-6591t}{5472}$ |
| 9 | $(H_L)_{-1}b_{33}$ | $\frac{-169+169st}{84480}$ |

| Source $j$ | Actual source word $z_j$ | Coefficient $c_j$ |
|---|---|---|
| 10 | $(H_L)_{-1}b_{42}$ | $\frac{-169}{40320}$ |
| 11 | $(H_L)_{-1}b_{53}$ | $\frac{1521+338s+1014t-169st}{319334400}$ |
| 12 | $(E_L)_{-1}b_3$ | $\frac{-169s-507t}{8}$ |
| 13 | $(E_L)_{-1}b_{10}$ | $\frac{169s+507t}{24}$ |
| 14 | $(H_R)_{-1}b_{13}$ | $\frac{-729383s-2188149t}{261744}$ |
| 15 | $(H_R)_{-1}b_{14}$ | $\frac{788891s+2366673t}{34026720}$ |
| 16 | $(H_R)_{-1}b_{19}$ | $\frac{398907s+1196721t}{109060}$ |
| 17 | $(H_R)_{-1}b_{22}$ | $\frac{1287s+3861t}{140}$ |
| 18 | $(H_R)_{-1}b_{23}$ | $\frac{403s+1209t}{560}$ |
| 19 | $(H_R)_{-1}b_{26}$ | $\frac{-1079s-3237t}{5472}$ |
| 20 | $(H_R)_{-1}b_{33}$ | $\frac{-3+3st}{2560}$ |
| 21 | $(H_R)_{-1}b_{42}$ | $\frac{29}{40320}$ |
| 22 | $(H_R)_{-1}b_{53}$ | $\frac{9+2s+6t-st}{3225600}$ |
| 23 | $(E_R)_{-1}b_{16}$ | $\frac{318929s+956787t}{1308720}$ |
| 24 | $(E_R)_{-1}b_{20}$ | $\frac{91s+273t}{12}$ |
| 29 | $(F_R)_{-1}b_{15}$ | $\frac{-5281159s-15843477t}{1308720}$ |
| 34 | $Tb_{22}$ | $\frac{4901s+14703t}{140}$ |
| 35 | $Tb_{23}$ | $\frac{5239s+15717t}{560}$ |
| 36 | $Tb_{26}$ | $\frac{-2866747s-8600241t}{5296896}$ |
| 37 | $Tb_{33}$ | $\frac{-111163+111163st}{40888320}$ |
| 38 | $Tb_{42}$ | $\frac{-377}{19514880}$ |
| 39 | $Tb_{53}$ | $\frac{1453959+323102s+969306t-161551st}{309115699200}$ |
| 40 | $(b_{92})_0b_{119}$ | $\frac{13s+39t}{99325771776000}$ |
| 49 | $(b_{49})_0b_{37}$ | $\frac{-26+39s+13t+26st}{37468569600}$ |
| 50 | $(b_{49})_0b_{46}$ | $\frac{26-13t}{2554675200}$ |
| 51 | $(b_{49})_0b_{57}$ | $\frac{-468-13s-39t+52st}{303495413760000}$ |
| 52 | $(b_{57})_0b_{29}$ | $\frac{26-39s-13t-26st}{1686085632000}$ |
| 53 | $(b_{57})_0b_{38}$ | $\frac{-26+13t}{114960384000}$ |
| 54 | $(b_{29})_0b_{37}$ | $\frac{91s-143t}{1040793600}$ |
| 55 | $(b_{29})_0b_{46}$ | $\frac{-13s-26t}{70963200}$ |
| 56 | $(b_{37})_0b_{38}$ | $\frac{13s+26t}{70963200}$ |
| 57 | $(b_{38})_0b_{46}$ | $\frac{-13s+39t}{38707200}$ |

Each word on the right is in $U_3$ because every $b_i$ is in $U_2$ and the currents, translation and

vertex modes preserve $U$. Thus, once the mode-word provenance and exact coordinate sum are accepted, these two tables prove membership of the actual saved primary fields. The OPE identifications of those fields remain separate structural calculations; a membership expression alone does not identify a Potts or parafermion algebra.

### S7.17 Historical discovery spectra, retained as numerical diagnostics

For completeness, the following two tables retain the exploratory spectral information in `hi_cft_20260923_c8_ope_control.json`. They are rounded decimal transcriptions of that floating run and carry no exact status. Their usefulness is diagnostic: a reviewer can see the candidate eigenvalues available before the exact field selection, and can distinguish a repeated numerical eigenvalue from a selected vector.

The commutant Griess calculation reported the following sorted list of axis central charges. The saved record calls this list `all_axis_charges`; this is the output of that search, rather than a new assertion here that every possible Virasoro axis has been classified. The selected stress has exact charge 2400/1271 and its commuting complement charge 70/31 in the later exact certificate.

| Sorted index | Numerical axis charge |
|---:|---:|
| 0 | 0 |
| 1 | 0.799999999999999 |
| 2 | 1.36216216216216 |
| 3 | 1.80343063282447 |
| 4 | 1.80343063282447 |
| 5 | 1.88827694728558 |
| 6 | 1.8882769472856 |
| 7 | 1.98417930125248 |
| 8 | 2.16216216216196 |
| 9 | 2.25806451612902 |
| 10 | 2.25806451612904 |
| 11 | 2.34291083059015 |
| 12 | 2.34291083059016 |
| 13 | 2.78417930125247 |
| 14 | 3.34634146341464 |
| 15 | 4.14634146341463 |

The next table gives all eigenvalues of the selected numerical $e_1$ on each relevant ambient affine-highest space. The final column is $\|e_2 v\|$ for the corresponding normalized numerical eigenvector. It was computed in the original orthonormal oscillator convention, so it should not be compared directly with an untransported norm of the exact-gauge coordinate table. An eigenvector with a visibly nonzero final entry is not primary for this Virasoro stress, even though it belongs to the affine-highest space. Tiny entries are floating residuals and do not themselves prove exact vanishing.

For repeated numerical eigenvalues the particular eigenbasis returned by a diagonalizer need not equal the later exact chosen basis. In particular, multiplicity and phase information cannot be recovered from this list alone. The exact charged fields, their $e_1$ eigenvalues, and their positive-mode checks supply the subsequent selection.

| Current type | Eigenvector index | Numerical $e_1$ eigenvalue | $\|e_2 v\|$ |
|---|---|---|---|
| $(1,5)$ | 0 | 0.0524850686944 | $9.08421 \times 10^{-16}$ |
| $(1,5)$ | 1 | 0.122084992219 | $3.92597 \times 10^{-15}$ |
| $(1,5)$ | 2 | 0.26983331241 | $6.95716 \times 10^{-16}$ |
| $(1,5)$ | 3 | 0.627852084972 | $1.44018 \times 10^{-14}$ |
| $(1,5)$ | 4 | 0.627852084972 | $8.67494 \times 10^{-16}$ |
| $(1,5)$ | 5 | 0.714264339016 | $1.60186 \times 10^{-14}$ |
| $(1,5)$ | 6 | 0.966153927523 | $2.24913 \times 10^{-15}$ |
| $(1,5)$ | 7 | 1.12208499222 | 0.494135593171 |
| $(1,5)$ | 8 | 1.46124613193 | $2.33481 \times 10^{-16}$ |
| $(1,5)$ | 9 | 1.71426433902 | 1.19521072537 |
| $(1,5)$ | 10 | 1.7766387585 | $9.22827 \times 10^{-15}$ |
| $(1,9)$ | 0 | 0.0881552109736 | $2.23261 \times 10^{-16}$ |
| $(1,9)$ | 1 | 0.157356412274 | $6.10418 \times 10^{-16}$ |
| $(1,9)$ | 2 | 0.157356412274 | $3.21692 \times 10^{-15}$ |
| $(1,9)$ | 3 | 0.318646734854 | $4.48097 \times 10^{-15}$ |
| $(1,9)$ | 4 | 0.580108555496 | $2.60578 \times 10^{-16}$ |
| $(1,9)$ | 5 | 0.738001573564 | $6.55504 \times 10^{-15}$ |
| $(1,9)$ | 6 | 0.775315891617 | $6.97350 \times 10^{-17}$ |
| $(1,9)$ | 7 | 1.18961447679 | $2.41474 \times 10^{-15}$ |
| $(1,9)$ | 8 | 1.18961447679 | $4.12746 \times 10^{-15}$ |
| $(1,9)$ | 9 | 1.31864673485 | 0.798306626372 |
| $(1,9)$ | 10 | 1.44391050714 | $3.10381 \times 10^{-15}$ |
| $(1,13)$ | 0 | 0.0930995402395 | 0 |
| $(1,13)$ | 1 | 0.330448465775 | 0 |
| $(1,13)$ | 2 | 0.330448465775 | 0 |
| $(1,13)$ | 3 | 0.556254917388 | 0 |
| $(1,13)$ | 4 | 0.82586190744 | 0 |
| $(2,2)$ | 0 | 0.151062155783 | $1.97606 \times 10^{-16}$ |
| $(2,2)$ | 1 | 0.151062155783 | $3.36752 \times 10^{-16}$ |
| $(2,2)$ | 2 | 1.05428796223 | $2.53151 \times 10^{-16}$ |
| $(2,2)$ | 3 | 1.53815892998 | $1.13555 \times 10^{-16}$ |
| $(2,6)$ | 0 | 0.0701906421897 | $2.02179 \times 10^{-15}$ |
| $(2,6)$ | 1 | 0.166719040056 | $6.82318 \times 10^{-16}$ |
| $(2,6)$ | 2 | 0.310778914241 | $1.67610 \times 10^{-15}$ |
| $(2,6)$ | 3 | 0.64814137984 | $9.57819 \times 10^{-15}$ |
| $(2,6)$ | 4 | 0.874193627135 | $6.99816 \times 10^{-16}$ |
| $(2,6)$ | 5 | 1.31077891424 | 0.788389388869 |
| $(2,6)$ | 6 | 1.43981117231 | $7.96599 \times 10^{-16}$ |
| $(2,6)$ | 7 | 1.43981117231 | $6.90936 \times 10^{-15}$ |
| $(2,8)$ | 0 | 0.052056041325 | $2.08167 \times 10^{-17}$ |
| $(2,8)$ | 1 | 0.164636264896 | $6.59195 \times 10^{-17}$ |
| $(2,8)$ | 2 | 0.591660110149 | $1.73472 \times 10^{-16}$ |
| $(2,8)$ | 3 | 0.591660110149 | $6.93889 \times 10^{-17}$ |
| $(2,8)$ | 4 | 0.591660110149 | $2.94903 \times 10^{-16}$ |
| $(2,8)$ | 5 | 0.986425890887 | $2.77556 \times 10^{-17}$ |
| $(2,8)$ | 6 | 1.22803837252 | $6.24500 \times 10^{-17}$ |
| $(2,14)$ | 0 | 0.141620771046 | 0 |
| $(2,14)$ | 1 | 0.141620771046 | 0 |
| $(2,14)$ | 2 | 0.496459480724 | 0 |

The largest leakage of $e_1$ out of these affine-highest spaces is on the order of $10^{-15}$ in this saved run. The two conspicuously nonzero $e_2$ norms in the $(1,5)$ space, and the nonzero norms in the $(1,9)$ and $(2,6)$ spaces, explain why diagonalizing $e_1$ alone would have been insufficient. The exact primary checks test both the eigenvalue equation and the required positive-mode annihilation.

# Part III
# Additional structure and analytic records

## S8 Additional structure: an atlas of the commuting algebras

Several different commutants occur in the construction. Their central charges and their ambient algebras must be specified together. This section collects the scattered structural calculations, including useful routes that the main proof does not need. The exact inputs have the same status as the finite certificate inputs in the main text. The additional all-weight arguments below are reproduced from the named research notes for verification; they are not new numerical replays or independent external validation. In particular, identifying an ambient commutant does not identify its intersection with the actual generated algebra $U$.

We use $\mathcal{T}$ for the lattice–Potts core of charge 14/5, $P = K(\mathfrak{sl}_2, 3)$, $Q = K(\mathfrak{sl}_2, 13)$, and $R = A_L$ for the order-thirteen extension of $\mathcal{T} \otimes Q$. Write $s_R = \omega_B - \omega_R$, so $c(s_R) = 18/5$. This notation avoids confusing the core $\mathcal{T}$ with the translation operator $D$. The three ambient algebras remain $U \subset B = V_{E_6 \oplus A_2} \subset H = V_{E_8}$.

| Algebra or stress | Charge | Information available |
|---|---|---|
| $P$ | 4/5 | Actual Potts embedding, including its weight-three generator. |
| $\mathcal{T}$ | 14/5 | Actual order-three extension of a rank-two lattice and $P$; four currents. |
| $Q$ | 8/5 | Actual parafermion embedding obtained from complete finite polynomial OPE rows. |
| $R = A_L$ | 22/5 | Actual order-thirteen simple-current extension; fourteen simple modules and four currents. |
| $\mathrm{Com}(\mathcal{T}, B)$ | 26/5 | An ambient rational frame $V_M \otimes L(1/2, 0) \otimes L(7/10, 0)$. |
| $C_B = \mathrm{Com}(R, B)$ | 18/5 | Five weight-two states; supplementary all-weight identification with the $(4, 8, 1; 13)$ diagonal coset. |
| $C_H = \mathrm{Com}(R, H)$ | 18/5 | Thirteen weight-two states; supplementary all-weight identification with an $F_4$ diagonal coset. |
| $C_U = \mathrm{Com}(R, U)$ | 18/5 | Two weight-two states and two weight-three states under the properness and finite-grade conclusions. No complete character is supplied here. |
| $C_6 = \mathrm{Com}(V_L, U)$ | 6 | Lattice double commutant; supplementary regularity and category factorization; seven weight-two states. |

The charge-26/5 and charge-18/5 ambient cosets are different objects. The two charge-18/5 ambient cosets are also different from $C_U$. Finally, the charge-6 lattice coset removes only the Cartan plane; it retains the Potts and parafermion factors. These distinctions explain several otherwise incompatible-looking dimensions in the old records.

## S8.1 The determinant-thirty-nine lattice and its actual extension

In the simple-root coordinates of the construction put

$$\begin{aligned} a_0 &= (-1,-2,-3,-2,-1,-2,0,0),\\ H_R &= (5,8,9,8,5,0,2,2),\\ t_0 &= (1,1,0,1,1,-3,1,1) = \tfrac{1}{2}(3a_0 + H_R). \end{aligned}$$

Their pairings are $a_0^2 = 2$, $H_R^2 = 78$, $(a_0, H_R) = 0$, $t_0^2 = 24$ and $(a_0, t_0) = 3$. Thus

$$L = \mathbb{Z}a_0 + \mathbb{Z}t_0, \qquad G_L = \begin{pmatrix} 2 & 3 \\ 3 & 24 \end{pmatrix}, \qquad \det G_L = 39.$$

The vector $r_0 = t_0/3 = a_0/2 + H_R/6$ has norm $8/3$. The exact generating field $u$ has Potts weight $2/3$ and plane-Heisenberg weight $4/3$. The identity

$$u_{-5}(u_{-3}u) = -\frac{3}{5}e^{t_0} \tag{148}$$

places the lattice field $e^{t_0}$ in $U$. PCT gives $e^{-t_0}$, and the left affine factor already supplies $e^{\pm a_0}$ and the required Heisenberg directions. This is the actual field-level reason for $V_L \subset U$; it is not inferred from a character. The inner product formula fixes the leading power in a product of lattice exponentials. For the triple product in (148), only the leading terms at the specified powers contribute. A basis of the full weight-twelve ambient space is therefore unnecessary.

Let $\psi$ be the Potts simple current of weight $2/3$. The extension is

$$\mathcal{T} = (V_L \otimes P) \oplus (V_{L+r_0} \otimes \psi) \oplus (V_{L+2r_0} \otimes \psi^2). \tag{149}$$

The order-three support is closed under fusion. In the existing unitary ambient algebra, nonzero multiplication with the conjugate sector has a nonzero vacuum pairing, so multiplication by a supported invertible sector is invertible. Connectedness then gives multiplicity one. The commutant-energy argument removes a possible hidden positive-energy complementary factor. These observations promote the three witnessed charge sectors to the actual embedded extension.

The core stress and its complement are

$$\omega_{\mathcal{T}} = \frac{1}{4}h_{a_0}(-1)^2\mathbf{1} + \frac{1}{156}h_{H_R}(-1)^2\mathbf{1} + p, \qquad c_{\mathcal{T}} = 2 + \frac{4}{5} = \frac{14}{5}.$$

Here the two Heisenberg terms have central charge one each. The source is `hi_cft_20260923_proof_rational_core.json`, whose exact record includes the three terms of $u_{-3}u$, the triple-product coefficient, the lattice Gram matrix and the stress commutators. Its historical caveat that this core alone does not prove rationality of $U$ remains correct; the positive-stress proof uses additional factors.

## S8.2 Discriminant conventions and why two quadratic forms agree

The discriminant generator $g = H_R/39$ has $g^2/2 = 1/39$. Its pairings with the lattice basis are $0, 1$, so it lies in $L^*$, and its order is 39. The class of $r_0$ is $26g$: indeed $r_0 - 26g = a_0/2 - H_R/2 = 2a_0 - t_0 \in L$. Condensing the order-three diagonal current in (149) leaves the order-thirteen pointed generator $3g$, with quadratic form

$$q(j_{\mathrm{old}}) = \frac{3j_{\mathrm{old}}^2}{13} \pmod 1.$$

The top-bound and core-extension files instead use the actual right Cartan charge $j = 6j_{\mathrm{old}}$ (mod 13). Since $6^{-1} = 11$ and $3 \cdot 11^2 \equiv -1 \pmod{13}$, the same form becomes

$$q(j) = -\frac{j^2}{13} \pmod 1.$$

Thus the two formulas in the archive are a change of labels, not a disagreement of discriminant forms. The remaining category is

$$\mathrm{Rep}(\mathcal{T}) \simeq \mathrm{Fib} \boxtimes \mathcal{C}(\mathbb{Z}/13, -j^2/13), \qquad \mathrm{Dim}\,\mathrm{Rep}(\mathcal{T}) = 13\frac{5+\sqrt{5}}{2}.$$

The equality of this core's central charge with that of $G_{2,1}$ does not identify the VOAs: $\mathcal{T}_1$ has dimension four, while the affine $G_2$ vacuum algebra has fourteen currents.

### S8.3 An explicit rational frame in the larger ambient commutant

Let $\alpha = b - a$, $\beta = c - b$, and $\gamma = \alpha + \beta$, using the three momenta in the original $u$. The Gram matrix is $\left(\begin{smallmatrix} 4 & -2 \\ -2 & 4 \end{smallmatrix}\right)$. If $p_\delta$ denotes the exponential coefficient of the actual Potts stress, set

$$e_I = \frac{1}{16}h_\alpha(-1)^2\mathbf{1} - \frac{5}{4}(p_\alpha e^\alpha + p_{-\alpha}e^{-\alpha}), \qquad e_{TI} = \omega_{\sqrt{2}A_2} - p - e_I.$$

The saved exact calculations give mutually commuting conformal vectors with charges $1/2$ and $7/10$. In the unitary realization they generate the simple Ising and tricritical Ising Virasoro VOAs. The rank-four perpendicular lattice has basis

$$\begin{aligned} m_1 &= (-1,1,0,0,0,0,0,0), \quad m_2 = (0,0,0,-1,1,0,0,0),\\ m_3 &= (0,0,-1,0,0,0,1,0), \quad m_4 = (0,0,-1,0,0,0,0,1) \end{aligned}$$

and Gram matrix

$$G_M = \begin{pmatrix} 6 & 0 & 1 & 1 \\ 0 & 6 & -1 & -1 \\ 1 & -1 & 4 & 1 \\ 1 & -1 & 1 & 4 \end{pmatrix}.$$

The lattice $L \oplus \sqrt{2}A_2 \oplus M$ has index 156 in $E_6 \oplus A_2$. Consequently the actual ambient commutant contains

$$C_0 = V_M \otimes L(1/2,0) \otimes L(7/10,0), \qquad \omega_{C_0} = \omega_B - \omega_{\mathcal{T}}.$$

The charge check is $4 + 1/2 + 7/10 = 26/5$. Because the conformal vector is shared, restrictions to this regular frame have no independently shifted copies of its simple modules: every copy starts at that module's prescribed lowest weight. Finite-dimensional ambient graded spaces then make the extension finite. The archived argument applies the unitary extension theorem to obtain strong rationality of $\mathrm{Com}(\mathcal{T}, B)$.

The exact record `hi_cft_20260923_proof_ambient_coset.json` also explicitly reports $e_I \notin S_2$ and $e_{TI} \notin S_2$. It therefore does not give this frame inside $\mathrm{Com}(\mathcal{T}, U)$. The older numerical record `hi_cft_20260923_proof_core_coset.json` found dimensions 4 and 15 in the intersections with the selected $S_2, S_3$, with a Potts-kernel residual approximately $7.665 \times 10^{-11}$. Those two numerical dimensions are retained only as historical finite span observations. Neither is used here as a certified full coset character.

### S8.4 Order-thirteen extension: the ten additional weight-two states

The zero-$s_R$-energy part of the exact 120-dimensional $S_2$ has dimension 25, while $(\mathcal{T}\otimes Q)_2$ has dimension 15. The ten additional states separate into the following simultaneous energy types:

| number | $(H_L, H_R)$ | $h_{\mathcal{T}}$ | $h_Q$ | $h_P$ |
|---|---|---|---|---|
| 6 | $(\pm1,\pm5),(0,\pm8)$ | 14/13 | 12/13 | 2/3 |
| 4 | $(\pm1,\pm3)$ | 4/13 | 22/13 | not needed for the witness |

In the first row, the four signs in $(\pm1,\pm5)$ are independent. The saved vectors are actual joint eigenvectors in $S_2$. For charge $j=5$, the Fibonacci-$\tau$ core module begins at 31/65; $14/13-31/65=3/5$ is not integral. Thus the witnessed core factor has Fibonacci label 1. The actual parafermion lowest-weight formula singles out the two inverse simple currents at 12/13. By contrast, 22/13 alone does not distinguish a simple current from the $i=10$ parafermion sector. The second row is a consistency check, not the step forcing the extension.

The pointed form of $\mathrm{Rep}(\mathcal{T}\otimes Q)$ is $-(j^2+k^2)/13$. Its two maximal isotropic lines are $k=\pm5j$. The witnessed nonzero pair generates one full line; both lines cannot coexist in an integer-graded local extension since they generate the entire pointed group, including nonintegral twists. After reversing the parafermion charge convention if necessary,

$$R=\bigoplus_{j\in\mathbb{Z}/13}\mathcal{T}_{1,j}\otimes Q_{0,-5j}.$$

There is no fixed-point resolution: the order-thirteen action on the charge pairs is free. The remaining simple labels are $x_a=(1,a)$ and $y_a=(\tau,a)$, $0\le a\le 6$, with the second entry corresponding to even affine label $2a$. The category is $\mathrm{Fib}\boxtimes \mathrm{SO}(3)_{13}$.

The thirteen vacuum-extension sectors have lowest weights

| $\lvert j\rvert$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $h$ | 0 | 4 | 3 | 2 | 4 | 2 | 3 |

where positive $|j|$ denotes two conjugate sectors. Only the charge $\pm3$ and $\pm5$ sectors add weight-two states. This supplies an independent explanation of $\dim R_1=4$ and $\dim R_2=25$. The source is `hi_cft_20260923_next_structure_order13_core.md` and its exact spectral record `hi_cft_20260923_next_structure_core_extension.json`.

### S8.5 Why the core is already its double commutant

The actual lowest-weight table has only two integer-weight labels: $x_0$ of weight 0 and $x_5$ of weight 1. Hence a connected integer-graded conformal extension $D$ of $R$, with finite graded spaces, has underlying module

$$D=R\oplus mX,\qquad X=x_5,\qquad m\in\mathbb{Z}_{\ge0}.$$

Regularity of $R$ supplies the decomposition; finite multiplicity and the absence of shifted copies use the common conformal vector. The truncated level-thirteen fusion rule gives

$$X\boxtimes X=x_0\oplus x_1\oplus x_2\oplus x_3.$$

In particular the odd summand cannot multiply into itself. If $m>0$, the transformation $+1$ on $R$ and $-1$ on $mX$ preserves every vertex product and has fixed algebra $R$. The compact-orbifold tensor theorem applied to this order-two automorphism forces the odd part

to be an invertible simple $R$-module. This contradicts the four displayed fusion channels (or $\mathrm{FPdim}\, X = \sin(11\pi/15)/\sin(\pi/15) > 1$). Therefore $m = 0$.

For $V = B$ or $H$, positivity identifies

$$\ker_V(s_R)_1 = \mathrm{Com}(\mathrm{Vir}(s_R), V).$$

This kernel has conformal vector $\omega_R$, ambient integer grading, and one-dimensional grade zero, so the extension obstruction gives

$$\mathrm{Com}(\mathrm{Vir}(s_R), V) = R, \qquad \mathrm{Com}(\mathrm{Com}(R, V), V) = R.$$

Restriction gives the same result inside $U$. This argument does not assume rationality of the opposite commutant. It isolates the double-commutant input needed later for coset and mirror theorems. The archived proof is `hi_cft_20260923_next_structure_core_maximality.md`; its external input is McRae's compact-orbifold tensor equivalence, Corollary 4.8 in the source cited in that note.

## S9 Residual conformal vectors and their complete finite algebra

The residual stress splits as $s_R = q + r$ into commuting PCT-fixed Virasoro vectors with

$$c_q = \frac{32}{19}, \qquad c_r = \frac{182}{95}, \qquad \frac{32}{19} + \frac{182}{95} = \frac{18}{5}.$$

These central charges agree with those of $K(\mathfrak{sl}_2, 17)$ and the diagonal affine coset at levels $(13, 4; 17)$, respectively. Only the Virasoro fields, their membership and their commutation are certified by `hi_cft_20260923_proof_remaining_axes.json`; the larger VOA identifications do not follow from these numbers.

The right raising and lowering currents have the exact simultaneous weights

$$(h_{\mathcal{T}}, h_Q, h_{s_R}) = \left(\frac{6}{65}, \frac{7}{65}, \frac{4}{5}\right).$$

They sum to one. In particular their $R$ energy is $1/5$. This small branching datum later identifies a tensor-generating core label. The translation and PCT checks that justify using the actual parafermion stress in this decomposition are stored in `hi_cft_20260923_proof_remaining_coset.json`.

### S9.1 Five ambient directions, but only two permitted in the proper algebra

The full ambient grade-two kernel of $(\omega_R)_1$ in $B$ has dimension five. Its producer enumerated a complete momentum ball using rational LDL bounds and worked in the entire 1370-dimensional ambient grade, whose Cartan-zero block has dimension 86. Thus this is a full ambient result, rather than a kernel in $S_2$. For $H$, the three lattice-glue sectors have grade-two dimensions $1370, 1377, 1377$ and Cartan-zero dimensions $86, 83, 83$. Their residual kernels have dimensions $5, 4, 4$, totaling 13.

| Ambient space | Grade two | Cartan zero | Residual kernel | Scope |
|---|---|---|---|---|
| $B$ | 1370 | 86 | 5 | Full grade |
| $H/B$ glue sector 1 | 1377 | 83 | 4 | Full grade |
| $H/B$ glue sector 2 | 1377 | 83 | 4 | Full grade |
| $H$ | 4124 | 252 | 13 | Full grade |
| $S_2 \subset U$ | 120 | selected block | 2 | Generated span |

Sources: `hi_cft_20260923_next_residual_ambient2.json` and `hi_cft_20260923_next_residual_e8_ambient2.json`. The complete momentum enumeration is part of the finite certification boundary; neither count by itself identifies a VOA at every weight.

There is also a direct properness-sensitive test. In the full five-dimensional residual space consider

$$\Phi(v) = (v_1 E_+ \bmod K_1,\ v_1 E_- \bmod K_1).$$

Its exact rank is three, and its kernel is exactly $\operatorname{span}\{q, r\}$. The kernel is separately checked to preserve all six known currents. If $v \in C_U$ and $U_1 = K_1$, then $v_1 E_\pm \in U_1$, so $\Phi(v) = 0$. Since $q, r \in C_U$, this proves

$$(C_U)_2 = \operatorname{span}\{q, r\}$$

once properness is supplied. This is a stronger statement than the absence of three directions from the selected $S_2$: it excludes them from the actual algebra using the current-preservation condition. See `hi_cft_20260923_next_structure_residual_current_kernel.json`.

The exact grade-three computation in the authenticated $S_3$ has 58 Cartan-zero dimensions, a residual kernel of dimension two, and no residual Virasoro primary of weight three. Its kernel is precisely $\operatorname{span}\{Dq, Dr\}$. Using the separate conclusion $U_3 = S_3$ therefore gives

$$\operatorname{ch} C_U = q_{\mathrm{ser}}^{-3/20}(1 + 2q_{\mathrm{ser}}^2 + 2q_{\mathrm{ser}}^3 + O(q_{\mathrm{ser}}^4)). \tag{150}$$

Here $q_{\mathrm{ser}}$ is the character variable, distinct from the stress $q$. The source `hi_cft_20260923_next_structure_residual_grade3.json` checks the positive core modes as well as the zero-energy kernel. The formula does not prove generation by $q, r$ or identify higher character coefficients; any new residual strong generator must start at weight at least four.

## S9.2 The finite Griess algebra in a reproducible basis

Use the exact basis $(q, r, v_1, v_2, v_3)$ of `hi_cft_20260923_next_structure_residual_tensor.json`. The last three basis vectors are obtained from the saved complete ambient kernel by subtracting their invariant-form projections to $q$ and $r$, in the producer's pivot order. This fixes their gauge; the sparse lattice embedding of the five vectors remains in that source. The following tables specify the entire finite algebra in this basis without importing its mode engine.

Write $s = \sqrt{13}$, $t = \sqrt{-3}$ and abbreviate

$$[a, b, c, d] = a + bs + ct + dst.$$

All four entries are rational. The invariant bilinear form is $\langle x, y\rangle \mathbf{1} = x_3 y$; it is not the positive Hermitian Gram matrix in this non-real basis. In particular a complex or negative-looking diagonal entry below is not a failure of vacuum unitarity. We have

$$\langle q, q\rangle = 16/19, \qquad \langle r, r\rangle = 91/95, \qquad \langle q, r\rangle = \langle q, v_i\rangle = \langle r, v_i\rangle = 0.$$

The residual weight-one space vanishes, so $x_2 y = 0$ for these weight-two vectors and the Griess product $x \cdot y = x_1 y$ is commutative. Its unit is $(q + r)/2$. The producer checks every product in the full ambient space and verifies invariance $\langle x \cdot y, z\rangle = \langle x, y \cdot z\rangle$.

**Invariant form on the three complementary directions.**

| Entry | Exact coefficient |
|---|---|
| $\langle v_1, v_1\rangle$ | $[-\frac{1350}{91}, 0, -\frac{1350}{91}, 0]$ |
| $\langle v_1, v_2\rangle$ | $[\frac{470}{91}, \frac{30}{91}, \frac{30}{7}, \frac{470}{91}]$ |
| $\langle v_1, v_3\rangle$ | $[-\frac{820}{91}, \frac{720}{91}, -\frac{40}{91}, -\frac{220}{91}]$ |
| $\langle v_2, v_2\rangle$ | $[\frac{22940}{273}, -\frac{8095}{546}, -\frac{14145}{182}, -\frac{170}{91}]$ |
| $\langle v_2, v_3\rangle$ | $[-\frac{34960}{273}, 0, 0, \frac{1840}{273}]$ |
| $\langle v_3, v_3\rangle$ | $[\frac{22940}{273}, \frac{8095}{546}, \frac{14145}{182}, -\frac{170}{91}]$ |

**Action of the first stress.** Write $q\cdot v_j = \sum_{i=1}^{3} Q_{ij} v_i$. The row index is the output coordinate and the column index is the input vector. Then $r \cdot v_j = 2v_j - q \cdot v_j$, while $q \cdot q = 2q$, $r \cdot r = 2r$ and $q \cdot r = 0$.

**The nine entries of $Q$.**

| Entry | Exact coefficient |
|---|---|
| $Q_{11}$ | $\frac{4}{57}$ |
| $Q_{12}$ | $[\frac{55}{1368}, \frac{107}{1368}, -\frac{29}{1368}, \frac{35}{1368}]$ |
| $Q_{13}$ | $[-\frac{71}{1368}, \frac{1}{1368}, \frac{13}{1368}, -\frac{71}{1368}]$ |
| $Q_{21}$ | $[-\frac{11}{190}, -\frac{2}{95}, \frac{1}{95}, \frac{1}{190}]$ |
| $Q_{22}$ | $[\frac{64}{57}, -\frac{10}{57}, 0, 0]$ |
| $Q_{23}$ | $[-\frac{9}{76}, \frac{5}{76}, -\frac{9}{76}, \frac{5}{76}]$ |
| $Q_{31}$ | $[\frac{1}{76}, -\frac{1}{380}, \frac{13}{380}, -\frac{1}{76}]$ |
| $Q_{32}$ | $[-\frac{9}{76}, -\frac{5}{76}, \frac{9}{76}, \frac{5}{76}]$ |
| $Q_{33}$ | $[\frac{64}{57}, \frac{10}{57}, 0, 0]$ |

Its exact characteristic polynomial is

$$\det(xI - Q) = x^3 - \frac{44}{19}x^2 + \frac{449}{361}x - \frac{910}{6859}.$$

The remaining six products are recorded coordinate by coordinate below. Together with commutativity and the preceding stress actions, this is the complete multiplication table. Define $v_i \cdot v_j = c_{ij}^{q} q + c_{ij}^{r} r + \sum_k c_{ij}^{k} v_k$.

$v_1 \cdot v_1$.

| Entry | Exact coefficient |
|---|---|
| $c_{11}^{q}$ | $[-5, 0, -5, 0]$ |
| $c_{11}^{r}$ | $[-\frac{220100}{8281}, 0, -\frac{220100}{8281}, 0]$ |
| $c_{11}^{1}$ | $[-\frac{409}{182}, 0, \frac{409}{182}, 0]$ |
| $c_{11}^{2}$ | $[\frac{9}{91}, \frac{9}{455}, -\frac{9}{35}, -\frac{9}{91}]$ |

| Entry | Exact coefficient |
|---|---|
| $c_{11}^{3}$ | $[-\frac{198}{455}, \frac{72}{455}, -\frac{36}{455}, \frac{18}{455}]$ |

$v_1 \cdot v_2$.

| Entry | Exact coefficient |
|---|---|
| $c_{12}^{q}$ | $[\frac{475}{96}, \frac{5}{96}, \frac{65}{96}, \frac{475}{96}]$ |
| $c_{12}^{r}$ | $[\frac{319675}{49686}, \frac{31925}{49686}, \frac{31925}{3822}, \frac{319675}{49686}]$ |
| $c_{12}^{1}$ | $[\frac{2813}{1092}, \frac{2117}{1092}, -\frac{1919}{1092}, -\frac{827}{1092}]$ |
| $c_{12}^{2}$ | $[\frac{3}{14}, \frac{38}{91}, -\frac{3}{14}, -\frac{38}{91}]$ |
| $c_{12}^{3}$ | $[-\frac{1381}{728}, -\frac{13}{56}, -\frac{1381}{728}, -\frac{13}{56}]$ |

$v_1 \cdot v_3$.

| Entry | Exact coefficient |
|---|---|
| $c_{13}^{q}$ | $[-\frac{335}{96}, \frac{715}{96}, -\frac{205}{96}, -\frac{235}{96}]$ |
| $c_{13}^{r}$ | $[-\frac{782375}{49686}, \frac{495475}{49686}, \frac{47675}{49686}, -\frac{143875}{49686}]$ |
| $c_{13}^{1}$ | $[\frac{2813}{1092}, -\frac{2117}{1092}, \frac{1919}{1092}, -\frac{827}{1092}]$ |
| $c_{13}^{2}$ | $[\frac{1381}{364}, -\frac{13}{28}, 0, 0]$ |
| $c_{13}^{3}$ | $[\frac{3}{14}, -\frac{38}{91}, -\frac{3}{14}, \frac{38}{91}]$ |

$v_2 \cdot v_2$.

| Entry | Exact coefficient |
|---|---|
| $c_{22}^{q}$ | $[\frac{10405}{72}, -\frac{4265}{144}, -\frac{5615}{48}, \frac{85}{24}]$ |
| $c_{22}^{r}$ | $[\frac{3607250}{74529}, -\frac{366500}{74529}, -\frac{1476500}{24843}, -\frac{174250}{24843}]$ |
| $c_{22}^{1}$ | $[-\frac{286577}{13104}, -\frac{817}{1008}, \frac{559}{1008}, \frac{15083}{13104}]$ |
| $c_{22}^{2}$ | $[-\frac{6779}{1092}, -\frac{601}{1092}, \frac{869}{364}, \frac{21}{52}]$ |
| $c_{22}^{3}$ | $[\frac{547}{364}, \frac{41}{182}, \frac{41}{14}, \frac{547}{364}]$ |

$v_2 \cdot v_3$.

| Entry | Exact coefficient |
|---|---|
| $c_{23}^{q}$ | $[-\frac{20425}{96}, 0, 0, \frac{1075}{96}]$ |
| $c_{23}^{r}$ | $[-\frac{1330475}{16562}, 0, 0, \frac{70025}{16562}]$ |
| $c_{23}^{1}$ | $[\frac{288059}{26208}, \frac{15161}{8736}, \frac{288059}{26208}, -\frac{15161}{26208}]$ |
| $c_{23}^{2}$ | $[\frac{4673}{1092}, -\frac{11}{14}, \frac{499}{273}, -\frac{457}{1092}]$ |

| Entry | Exact coefficient |
|---|---|
| $c_{23}^{3}$ | $[\frac{4673}{1092}, \frac{11}{14}, -\frac{499}{273}, -\frac{457}{1092}]$ |

$v_3 \cdot v_3$.

| Entry | Exact coefficient |
|---|---|
| $c_{33}^{q}$ | $[\frac{10405}{72}, \frac{4265}{144}, \frac{5615}{48}, \frac{85}{24}]$ |
| $c_{33}^{r}$ | $[\frac{3607250}{74529}, \frac{366500}{74529}, \frac{1476500}{24843}, -\frac{174250}{24843}]$ |
| $c_{33}^{1}$ | $[\frac{22027}{1872}, -\frac{27935}{13104}, -\frac{139655}{13104}, -\frac{2231}{13104}]$ |
| $c_{33}^{2}$ | $[\frac{547}{364}, -\frac{41}{182}, -\frac{41}{14}, \frac{547}{364}]$ |
| $c_{33}^{3}$ | $[-\frac{6779}{1092}, \frac{601}{1092}, -\frac{869}{364}, \frac{21}{52}]$ |

### S9.3 The residual charge spectrum and a unique charge-four-fifths axis

The exact idempotent equations for $e = aq + br + t_1v_1 + t_2v_2 + t_3v_3$ are $e \cdot e = 2e$. If its central charge is $c_e$, invariance against the total stress makes

$$\frac{c_e}{2} = \frac{16}{19}a + \frac{91}{95}b, \qquad b = \frac{95c_e}{182} - \frac{80a}{91}.$$

This linear elimination is useful for independently reconstructing the polynomial system from the printed multiplication table. The all-charge calculation found a zero-dimensional quotient of length 32. Multiplication by $c_e$ on that quotient has the factored characteristic polynomial encoded by the following table. A pair in the first two columns contributes both linear factors with the listed multiplicity.

| $c_e$ | $18/5 - c_e$ | Multiplicity of each factor |
|---|---|---|
| 0 | 18/5 | 1 |
| 4/5 | 14/5 | 1 |
| 6/7 | 96/35 | 1 |
| 52/55 | 146/55 | 1 |
| 34/35 | 92/35 | 1 |
| 6/5 | 12/5 | 2 |
| 10/7 | 76/35 | 2 |
| 58/35 | 68/35 | 1 |
| 32/19 | 182/95 | 2 |
| 96/55 | 102/55 | 1 |
| 62/35 | 64/35 | 1 |
| $c_e^2 - (18/5)c_e + 32168/10225$ | | 2 |

The sum of the factor degrees with multiplicity is 32. Complementation $e \mapsto q + r - e$ explains the symmetry about 9/5. These are algebraic central charges of the complex idempotent scheme. A real charge is not a certificate of a PCT-fixed stress, and a Virasoro stress alone is not a rational conformal frame. Scheme multiplicity must not be read as a count of distinct unitary axes. The record is `hi_cft_20260923_next_structure_residual_all_charges_v2.json`.

At $c_e = 4/5$, the reduced exact equations give a unique point:

$$e = \frac{1}{24}q + \frac{8}{21}r + \frac{7}{120}(1+t)v_1 + \frac{19+s+7t+st}{400}v_2 + \frac{19-s-7t+st}{400}v_3.$$

The original ideal computation, in `hi_cft_20260923_next_structure_residual_potts.json`, only established this algebraic point. The separate verifier `hi_cft_20260923_next_structure_residual_potts_verify.json` then checked its full ambient Virasoro products, PCT reality, commutation with the core, and commutation with its complement. Its $e_1$ spectrum on the five-dimensional residual algebra is

$$0^{\oplus 3}, \qquad (2/5)^{\oplus 1}, \qquad 2^{\oplus 1}.$$

The complement has central charge $14/5$. The verifier's conclusion is the unitary minimal Virasoro subalgebra $L(4/5, 0)$ in $C_B$; it does not assert a weight-three Potts extension inside $C_B$.

The field $e$ has nonzero complementary coordinates and fails membership in $S_2$. More decisively, the current-kernel argument shows that $e \notin C_U$ in the proper case. In fact every idempotent in $\mathrm{span}\{q, r\}$ is one of $0, q, r, q+r$, because the two stresses commute and each satisfies $x \cdot x = 2x$. Thus $C_U$ has no charge-4/5 conformal vector under the established properness hypotheses. This does not conflict with the original Potts algebra $P \subset U$: $P$ is part of the core, not a subalgebra of this residual commutant.

The tensor producer also performed a complete test within the rank-one lattice Ising ansatz supported by the residual momenta. It checked 21 momentum pairs with arbitrary coefficients $u, v$ satisfying $uv = 1/16$, found no such axis, and left no underdetermined cases. This excludes that ansatz only. It is not an exclusion of every Ising conformal vector by an unrestricted idempotent classification.

# S10 Supplementary all-weight identification of the ambient residual cosets

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**REVIEW — Scope of the supplementary coset route**

The following route is additional verification material from `hi_cft_20260923_next_neighbor_independent_audit.md`, `hi_cft_20260923_next_structure_affine_pair.md`, `hi_cft_20260923_next_structure_residual_identification.md`, and `hi_cft_20260923_next_structure_residual_category.md`. It is not needed to replace the main manuscript's positive-stress regularity or finite closure arguments. Its conclusions use the actual parafermion embeddings, actual simple-current sectors, lattice gluing, and the external affine-embedding, coset-regularity and mirror theorems specified below. The archived internal audits are evidence of review, not independent formal certification of these additional theorems or their application. We reproduce the steps so they can be verified in this draft.

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## S10.1 The two-neighbor lattice preserves exactly the relevant perpendicular part

Let $\Lambda = E_8$ be the actual ambient lattice and let $\Pi = \mathrm{span}_{\mathbb{R}}\{a_0, H_R\}$. Because $\det L = 39$ is squarefree, $L = \Lambda \cap \Pi$ is saturated: the square of a nontrivial saturation index would divide

39. Put

$$\Lambda_0 = \{x \in \Lambda : (x, t_0) \in 2\mathbb{Z}\}, \qquad \Lambda' = \Lambda_0 \cup (t_0/2 + \Lambda_0).$$

The parity map is nontrivial since $(a_0, t_0) = 3$. Thus $[\Lambda : \Lambda_0] = 2$. The vector $t_0/2$ has even norm six, pairs integrally with $\Lambda_0$, and has double in $\Lambda_0$. Hence $\Lambda'$ is even, integral and unimodular of rank eight, and is again an $E_8$ lattice.

Let $N = \Lambda \cap \Pi^\perp$. Every vector of $N$ already lies in $\Lambda_0$. Conversely, a new-coset vector $t_0/2 + x$ cannot be perpendicular to $a_0$: that would require $(x, a_0) = -3/2$, contrary to integrality of $\Lambda$. Therefore

$$\Lambda \cap \Pi^\perp = \Lambda' \cap \Pi^\perp = N. \tag{151}$$

This equality needs the displayed integrality argument. It is not a general property of neighboring lattices.

The plane lattice changes to

$$L' = \mathbb{Z}(2a_0) + \mathbb{Z}(t_0/2).$$

Set $\alpha' = t_0/2$, $\gamma' = 4a_0 - t_0/2$, and $\beta' = (\alpha' + \gamma')/2 = 2a_0$. Then

$$(\alpha')^2 = 6, \qquad (\gamma')^2 = 26, \qquad (\alpha', \gamma') = 0, \qquad G_{(\alpha', \beta')} = \begin{pmatrix} 6 & 3 \\ 3 & 8 \end{pmatrix}.$$

Both $L$ and $L'$ have determinant 39.

## S10.2 The sign of the discriminant transport is substantive

The natural old discriminant generator is $\mu = H_R/39$. Its natural new partner is

$$\lambda' = \mu + a_0 = \alpha'/3 + 3\gamma'/13, \qquad q_{L'}(\lambda') = 40/39 \equiv 1/39 \pmod 1.$$

If $\mu + \eta$, $\eta \in N^*$, is an old glue lift, then $\lambda' + \eta = (\mu + \eta) + a_0$ has even $t_0$ pairing $1 + 3 = 4$. Therefore the old class $n\mu$ and the new class $n\lambda'$ have exactly the same $N$ discriminant component. Unimodularity and the order 39 account for every glue sector. The expression $-\alpha'/3 + 3\gamma'/13$ is also a discriminant generator, but corresponds to $14\lambda'$ modulo $L'$. It is not the natural transport of $\mu$. The archived audit explicitly corrects an earlier sign proposal at this point.

The actual 39-sector description of the core is

$$R = \bigoplus_{n \bmod 39} V_{L+n\mu} \otimes P_{-n} \otimes Q_{3n}.$$

The symbols $P_{-n} \otimes Q_{3n}$ here denote the actual multiplicity subspaces in the corresponding $N$-lattice sectors, not formal modules with matching characters. The gluing preserves these subspaces, so it transports the sum to

$$R' = \bigoplus_{n \bmod 39} V_{L'+n\lambda'} \otimes P_{-n} \otimes Q_{3n} \subset V_{\Lambda'}.$$

The multiplicity-space intertwining operators are unchanged; only the torus factors are replaced by factors with the same charge fusion. The old product closure excludes extra output multiplicity spaces. Compatible lattice cocycles may be chosen to agree on $N$; their remaining discrepancy is a coboundary and rescales the nonzero intertwining operators. Positive lattice forms can likewise be chosen to extend the original form on $N$. These are the field-level ingredients of the transport.

### S10.3 Actual affine currents in the neighbor

The two identities

$$13\lambda' - \alpha' - 6\beta' = \alpha'/3, \qquad 9\lambda' - \alpha' - 4\beta' = \gamma'/13$$

exhibit the charged sectors supplying the affine currents. Their parafermion charges are $(-1, 0)$ and $(0, 1)$, respectively. The lattice and parafermion energies are

| factor | $h_{\text{lattice}}$ | $h_{\text{parafermion}}$ | sum |
|---|---|---|---|
| $\alpha'$ | $1/3$ | $2/3$ | $1$ |
| $\gamma'$ | $1/13$ | $12/13$ | $1$ |

with one-dimensional parafermion tops. They give actual commuting $A_{1,3}$ and $A_{1,13}$ current algebras. Their stresses are $\omega_{\alpha'} + p$ and $\omega_{\gamma'} + z$. The primitive plane field $e^{\beta'}$ supplies the order-two simple-current extension, so the standard lattice–parafermion reconstruction gives

$$R' = (A_{1,3} \otimes A_{1,13}) \oplus (J_3 \otimes J_{13}), \qquad h_{J_3 \otimes J_{13}} = 3/4 + 13/4 = 4.$$

The reconstruction uses the actual PF3 and PF13 fields already identified earlier. Matching their central charges would not supply these affine raising and lowering vectors.

Both old and new cores have plane stress plus $p + z$. A vector commuting with either core has zero plane-Heisenberg energy and therefore lies in the common $V_N$ of (151). Positivity then gives the all-weight equality

$$\mathrm{Com}(R, V_\Lambda) = \mathrm{Com}(R', V_{\Lambda'}) = \{v \in V_N : p_1 v = z_1 v = 0\}.$$

The order-two affine extension shares the affine-pair stress, so it has the same commutant. This proof does not infer a full commutant from the finite weight-two kernel.

### S10.4 Why the affine pair has a unique relevant conjugacy class

An index-three $\mathfrak{sl}_2$ in $E_8$ has Cartan characteristic of norm six. In Euclidean $E_8$ coordinates the norm-six vectors have three types, up to signs and permutations:

$$(2, 1, 1, 0^5), \qquad (1^6, 0^2), \qquad ((3/2)^2, (1/2)^6),$$

with counts 1344, 1792, 3584. The $D_8$ Weyl group is transitive on each type; reflection in $(1/2, \ldots, 1/2)$ connects the types. Thus the characteristic has one $E_8$ Weyl orbit. For fixed $h$, the centralizer group acts with an open orbit on every possible raising operator: $\mathrm{ad}(e) : \mathfrak{g}_0 \to \mathfrak{g}_2$ is surjective by $\mathfrak{sl}_2$ representation theory. Two open orbits in the vector space $\mathfrak{g}_2$ coincide. This upgrades the Cartan-orbit assertion to conjugacy of the first $\mathfrak{sl}_2$ embedding.

For its standard representative, use the conformal inclusion $G_{2,1} \otimes F_{4,1} \subset E_{8,1}$. The finite branching

$$248 = (14, 1) \oplus (1, 52) \oplus (7, 26)$$

and

$$14 \downarrow A_{1,3} \times A_{1,1} = (3, 1) \oplus (1, 3) \oplus (4, 2), \qquad 7 \downarrow A_{1,3} \times A_{1,1} = (3, 1) \oplus (2, 2)$$

show that the current centralizer is exactly $F_4 \oplus A_1$, both at level one. The finite Lie centralizer alone is not the required VOA assertion. To verify the latter, list the simple lowest weights:

| | |
|---|---|
| $A_{1,3}$ | $0, 3/20, 2/5, 3/4$ |
| $A_{1,1}$ | $0, 1/4$ |
| $F_{4,1}$ | $0, 3/5$ |

Only the vacuum triple and $(3,1,0)$, $(2,0,26)$, $(1,1,26)$ have integral sums, the last three at weight one. Only the vacuum triple has zero $A_{1,3}$ energy. Regularity and the shared conformal vector therefore imply

$$\mathrm{Com}(A_{1,3}, V_{E_8}) = F_{4,1} \otimes A_{1,1}$$

at every weight, with no additional affine-commutant extension.

The second $A_{1,13}$ projects to the external $A_1$ with index zero or one. The allowed $F_4$ indices quoted from the Penkov–Zuckerman table in the archived affine-pair note are

$$1, 2, 3, 4, 6, 8, 9, 10, 11, 12, 28, 35, 36, 60, 156,$$

and that source states uniqueness by index. There is no index thirteen and there is a unique index twelve. Hence the second factor must use indices $12 + 1$. The actual residual commutant is therefore identified, under these external embedding inputs, as

$$C_H \cong \mathrm{Com}(A_{1,13}, F_{4,1} \otimes A_{1,1}). \tag{152}$$

## S10.5 Transporting the order-three action distinguishes the $B$ coset

Let $\chi : \Lambda \to \mathbb{Z}/3$ have kernel the $E_6 \oplus A_2$ lattice. The vectors $a_0, t_0$ have charge zero. The extension $\chi'$ to $\Lambda'$ is fixed by $\chi'|_{\Lambda_0} = \chi$ and $\chi'(t_0/2) = 0$; the relation $2(t_0/2) = t_0$ is compatible. On $N$ the action is unchanged, and on the reconstructed core it is trivial. Thus its action on $C_H$ fixes precisely $C_B$.

The index-twelve $\mathfrak{sl}_2$ in $F_4$ has adjoint branching

$$52 = 6V_2 \oplus 4V_4 \oplus 2V_6.$$

Its full triple centralizer has zero-dimensional Lie algebra; the nilpotent centralizer has dimension $6 + 4 + 2 = 12$. The weighted diagram is 0200 (orbit $F_4(a_3)$), with characteristic $h = (4, 2, 2, 0)$ of norm 24. The external orbit and component-group tables cited in the archived residual identification note give the finite triple centralizer $S_4$. The actual order-three element is nontrivial because its fixed weight-two space has dimension five inside a space of dimension thirteen. All order-three elements of $S_4$ are conjugate.

A representative is the grading automorphism of

$$F_{4,1} \supset A_{2,1} \otimes A_{2,2}.$$

The two nontrivial simple-current sectors have current tops $(3, \overline{6})$ and $(\overline{3}, 6)$, of dimension 18 each. The principal $A_1$ factors in the two $A_2$ algebras have levels four and eight. Consequently the same actual fixed residual is

$$C_B \cong \mathrm{Com}(A_{1,13}, A_{2,1} \otimes A_{2,2} \otimes A_{1,1}), \tag{153}$$

with diagonal levels $4 + 8 + 1$. It is the order-three fixed algebra of (152). The conjugacy of the grading action, rather than a comparison of central charges, is the decisive step.

The finite count gives a convenient check on the identification. The nineteen currents of the numerator in (153) restrict as $3V_2 \oplus 2V_4$. Horizontal singlets at grade two come from the separate factors with multiplicities $1, 2, 1$ and from the three cross terms with multiplicities $2, 1, 1$, giving eight. The diagonal affine vacuum removes one; the two complementary $V_2$ current tops remove two descendant singlets. Thus the commutant has $8 - 1 - 2 = 5$ states at weight two. The charge-4/5 stress from the $A_{2,2}/A_{1,8}$ coset has spectrum $0^3, (2/5)^1, 2^1$, agreeing with the exact Griess result.

### S10.6 Regularity and the full category of the ambient residual

The displayed standard cosets contain a common conformal frame

$$L(4/5,0)\otimes \mathrm{Com}(A_{1,12}, A_{1,4}\otimes A_{1,8})\otimes L(34/35,0).$$

The charges are 4/5, 64/35, 34/35, totaling 18/5. The last factor is the unitary minimal model $M(14,15)$. The middle factor is the level-four diagonal $A_1$ coset; its regularity is the specific theorem of Lin (Theorem 5.7 at $k=8$) invoked in the archived note. The unitary finite extension of this regular frame yields $C_2$-cofiniteness of the ambient residual. Together with core maximality, the strong rationality of the ambient lattice VOA, and quasiprimary compatibility of the core stress, McRae's coset theorem supplies strong rationality. This order of argument avoids assuming rationality of arbitrary cosets or an unstated categorical-dimension condition on an arbitrary extension.

There is more information than regularity. The right currents have $R$ energy 1/5 and charges $\pm 2$. Among the fourteen core simples, the two lowest-weight-1/5 tops are $y_3$ with charges $\pm 2$ and $y_6$ with charges $\pm 4$. Hence the currents force $y_3$ to occur in $B\downarrow_R$. Its fusion square is

$$y_3 \boxtimes y_3 = (1\oplus\tau)\boxtimes(0\oplus 2\oplus 4\oplus 6\oplus 8\oplus 10\oplus 12),$$

so it tensor-generates every core label. Mutual commutants and the mirror theorem then give fourteen distinct simple multiplicity spaces $N_i$ and the actual diagonal branching

$$B=\bigoplus_{i\in \mathrm{Irr}(R)} M_i\otimes N_i.$$

Their braided subcategory is $\mathrm{Rep}(R)^{\mathrm{rev}}$, with inverse twists because $B$ is integer graded. This is a statement about actual branching modules, not merely fourteen proposed characters.

Splitting off this nondegenerate subcategory by the modular centralizer theorem, and using the local-module category of the existing extension algebra $B$, yields

$$\mathrm{Rep}(C_B)\simeq \mathrm{Fib}^{\mathrm{rev}}\boxtimes \mathrm{SO}(3)_{13}^{\mathrm{rev}}\boxtimes \mathcal{P}(E_6\oplus A_2). \tag{154}$$

The last factor has group $(\mathbb{Z}/3)^2$ and form $(2a^2+b^2)/3$, so there are $14\cdot 9=126$ simple residual modules. The extension algebra is diagonal in the first two mutually reversed factors; its dimension is the sum of squared core dimensions, which is positive. Thus the local-module argument does not omit a zero-dimension exception.

If $U$ contained all of $C_B$, its right currents would still force the $y_3$ summand. Fusion closure of mirror labels would force all fourteen summands, each of which has multiplicity one in $B$. Hence $U=B$. Conversely $U=B$ certainly contains $C_B$. This explains structurally why a proper $U$ must have a proper residual $C_U\subsetneq C_B$. It supplies no further regularity or character claim for $C_U$ itself.

## S11 Complete finite top data behind the universal energy bound

The main proof states the sharp bound $h\le 17/10$. This section prints the charge characters and all partial-energy traces used there, so a reviewer need not recover them from the checking script. The coefficients are the saved exact data in `hi_cft_20260923_next_structure_order13_top_check.json`, with the preceding lattice–Potts minimization explained in `hi_cft_20260923_next_structure_top_bound_sharp.md`.

### S11.1 Every lattice–Potts top, including both minimizing charges

For $\epsilon = 1, \tau$ the core energy at right Cartan charge $m$ is

$$\frac{m^2}{156} + \frac{\mathbf{1}_{m \text{ odd}}}{4} + p_\epsilon(m), \qquad m \equiv j \pmod{13},$$

where $p_1(m)$ is 0 or $2/3$ according as $3 \mid m$ or not, and $p_\tau(m)$ is $2/5$ or $1/15$ in the same two cases. An odd $m$ carries a complete left affine doublet, and an even $m$ a singlet. The minimum is attained within $-12 \le m \le 12$. For a positive $j$, the table combines the conjugate labels $j, -j$; its dimension is the dimension of that pair of tops. The absolute charges listed can belong to different signs of $j$, but both signs of every listed charge occur in the combined pair.

| Label | $\lvert m\rvert$ | Left multiplicity | Potts weight | Pair dimension |
|---|---|---|---|---|
| $(1,0)$ | $0$ | $1$ | $0$ | $1$ |
| $(\tau,0)$ | $0$ | $1$ | $2/5$ | $1$ |
| $(1,1)$ | $1,12$ | $2,1$ | $2/3, 0$ | $6$ |
| $(\tau,1)$ | $1$ | $2$ | $1/15$ | $4$ |
| $(1,2)$ | $2$ | $1$ | $2/3$ | $2$ |
| $(\tau,2)$ | $2$ | $1$ | $1/15$ | $2$ |
| $(1,3)$ | $3$ | $2$ | $0$ | $4$ |
| $(\tau,3)$ | $3,10$ | $2,1$ | $2/5, 1/15$ | $6$ |
| $(1,4)$ | $4,9$ | $1,2$ | $2/3, 0$ | $6$ |
| $(\tau,4)$ | $4$ | $1$ | $1/15$ | $2$ |
| $(1,5)$ | $5,8$ | $2,1$ | $2/3, 2/3$ | $6$ |
| $(\tau,5)$ | $5,8$ | $2,1$ | $1/15, 1/15$ | $6$ |
| $(1,6)$ | $6$ | $1$ | $0$ | $2$ |
| $(\tau,6)$ | $6,7$ | $1,2$ | $2/5, 1/15$ | $6$ |

As a worked minimization, for $(1,1)$ the charges $m = 1, -12$ both have energy $12/13$:

$$\frac{1}{156} + \frac{1}{4} + \frac{2}{3} = \frac{12}{13}, \qquad \frac{144}{156} = \frac{12}{13}.$$

Their left multiplicities are two and one. For $(\tau,3)$, $m = 3, -10$ give respectively

$$\frac{9}{156} + \frac{1}{4} + \frac{2}{5} = \frac{46}{65}, \qquad \frac{100}{156} + \frac{1}{15} = \frac{46}{65}.$$

Omitting these second minimizers would remove charge ten and invalidate the later right-$\mathfrak{sl}_2$ inequalities. The analogous second minimizer at $(\tau,6)$ restores charge seven.

### S11.2 The fourteen core tops as full Cartan characters

Let $\ell$ record the left Cartan and $z$ the right Cartan. Write $\sigma = \ell + \ell^{-1}$ and $[m] = z^m + z^{-m}$ for $m > 0$. These are Laurent characters, unrelated to the square-root normalization of the parafermion primary. The complete top characters are:

| Module | Top Cartan character | Dimension |
|---|---|---|
| $x_0$ | $1$ | 1 |
| $x_1$ | $1$ | 1 |
| $x_2$ | $1+\sigma[3]$ | 5 |
| $x_3$ | $1+[2]+\sigma[3]+[6]$ | 9 |
| $x_4$ | $[6]$ | 2 |
| $x_5$ | $\sigma[1]+[4]+[6]+\sigma[9]+[12]$ | 14 |
| $x_6$ | $[4]+\sigma[9]$ | 6 |
| $y_0$ | $1$ | 1 |
| $y_1$ | $1+\sigma[5]+[8]$ | 7 |
| $y_2$ | $1+[2]+\sigma[3]+\sigma[5]+[8]+[10]$ | 15 |
| $y_3$ | $[2]$ | 2 |
| $y_4$ | $\sigma[1]+[2]+[4]+[6]+\sigma[7]$ | 14 |
| $y_5$ | $\sigma[1]+[4]$ | 6 |
| $y_6$ | $[4]$ | 2 |

These are complete $R$-tops. They need not individually be complete right-$\mathfrak{sl}_2$ representations, because the right raising and lowering currents are in $U$ but are not currents of $R$. For instance $x_5$ has charge twelve but no charge ten. A genuine $U$-module top is a nonnegative integral sum of such core tops whose combined character must be a finite-dimensional representation of both current Lie algebras.

This gives a direct way to see the two inequalities used in the bound. The multiplicities of right charge twelve and ten are $x_5$ and $y_2$, so the usual nonincreasing multiplicity string within an $\mathfrak{sl}_2$ representation forces $y_2 \geq x_5$. In the left-doublet sector the right charge-nine multiplicity is $x_5 + x_6$, while charge seven occurs $y_4$ times. Dividing out the common doublet gives $y_4 \geq x_5 + x_6$.

### S11.3 All partial-energy traces, with the parafermion contribution retained

For one core top write $d$ for dimension, $L = \operatorname{Tr} \omega_{A_{1,1},1}$, $Q_H = \operatorname{Tr} H_R(0)^2$, $P_0 = \operatorname{Tr} p_1$, and $Z_0 = \operatorname{Tr} z_1$. The traces are taken over that entire core top. The relation

$$Z_0 = h_R d - L - Q_H/156 - P_0$$

checks each row from the common core stress. On a complete right $\mathfrak{sl}_2$ representation the affine Sugawara trace is $3Q_H/164$. Multiplying by its positive-stress coefficient gives $83Q_H/6084$. Hence the contribution to the total trace is

$$W = L + \frac{83}{6084} Q_H + \frac{25}{6} P_0 + \frac{25}{26} Z_0.$$

The following table is a transcription, not a fresh evaluation of the research checker.

| Label | $h_R$ | $d$ | $L$ | $Q_H$ | $P_0$ | $Z_0$ | $4680W$ |
|---|---|---|---|---|---|---|---|
| $x_0$ | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| $x_1$ | $\frac{2}{15}$ | 1 | 0 | 0 | 0 | $\frac{2}{15}$ | 600 |
| $x_2$ | $\frac{2}{5}$ | 5 | 1 | 36 | 0 | $\frac{10}{13}$ | 10440 |

| Label | $h_R$ | $d$ | $L$ | $Q_H$ | $P_0$ | $Z_0$ | $4680W$ |
|---|---|---|---|---|---|---|---|
| $x_3$ | $\frac{4}{5}$ | 9 | 1 | 116 | $\frac{4}{3}$ | $\frac{268}{65}$ | 56640 |
| $x_4$ | $\frac{1}{3}$ | 2 | 0 | 72 | 0 | $\frac{8}{39}$ | 5520 |
| $x_5$ | 1 | 14 | 2 | 720 | 4 | $\frac{44}{13}$ | 148560 |
| $x_6$ | $\frac{4}{5}$ | 6 | 1 | 356 | $\frac{4}{3}$ | $\frac{12}{65}$ | 54240 |
| $y_0$ | $\frac{2}{5}$ | 1 | 0 | 0 | $\frac{2}{5}$ | 0 | 7800 |
| $y_1$ | $\frac{8}{15}$ | 7 | 1 | 228 | $\frac{4}{5}$ | $\frac{92}{195}$ | 36960 |
| $y_2$ | $\frac{4}{5}$ | 15 | 2 | 472 | $\frac{8}{3}$ | $\frac{56}{13}$ | 110880 |
| $y_3$ | $\frac{1}{5}$ | 2 | 0 | 8 | $\frac{2}{15}$ | $\frac{14}{65}$ | 4080 |
| $y_4$ | $\frac{11}{15}$ | 14 | 2 | 312 | $\frac{8}{5}$ | $\frac{14}{3}$ | 81480 |
| $y_5$ | $\frac{2}{5}$ | 6 | 1 | 36 | $\frac{2}{5}$ | $\frac{10}{13}$ | 18240 |
| $y_6$ | $\frac{1}{5}$ | 2 | 0 | 32 | $\frac{2}{15}$ | $\frac{4}{65}$ | 4920 |

For example the $x_5$ character gives $Q_H = 2(2 \cdot 1^2 + 4^2 + 6^2 + 2 \cdot 9^2 + 12^2) = 720$. Its four odd-charge doublets contribute $L = 2$, while the Potts trace is 4. Therefore $Z_0 = 14 - 2 - 720/156 - 4 = 44/13$, agreeing with the table. This row alone has $W/d > 17/10$; it is the compatibility inequalities between complete core tops that lower the universal bound. Bounding every individual partial energy separately would lose that information.

## S11.4 What makes the top bound an all-module statement

The top of an arbitrary simple $U$-module is not assumed to have a positive Hermitian form at this point. Instead regularity of $R$ as a weak-module theory decomposes the restriction into ordinary $R$-modules. Intrinsic and ambient weights agree for vectors of $R$, so every $R$ mode lowering its intrinsic degree also lowers ambient degree and annihilates the lowest $U$ space. That space is stable under all $R$ Zhu zero modes and is therefore a direct sum of complete simple $R$-tops. Finite-dimensional current Lie algebra representations are semisimple algebraically. These are the precise inputs behind the character inequalities.

Writing $D_E$ for the total dimension and $W_E$ for the sum of the listed $W$ contributions, the printed dual identity in the main proof has only the nonnegative differences $y_2 - x_5$, $y_4 - x_5 - x_6$ and nonnegative core multiplicities. It gives $W_E \leq (17/10)D_E$. The stress identity gives $W_E = hD_E$, so every simple $U$-module satisfies $h < 2$. No simultaneous diagonalization of all partial stresses, and no unitarity of the unknown module, enters this trace argument.

The resulting absence of hidden branching summands is particularly concrete. Every simple summand of an integer-graded lattice module has integral lowest weight. Only zero and one are below two. The vacuum multiplicity is one; the remaining 80 currents form one exact irreducible top. Thus $B = U \oplus X$ after properness. For a mixed lattice module starting at weight one with an irreducible 81-dimensional top, an additional summand could only start at an integer weight at least two, which the same bound excludes. The finite irreducibility checks and the universal top bound have different jobs: one identifies the lowest action, the other rules out all later summands.

# S12 The supplementary charge-six lattice coset and its limits

The charge-six construction uses the same saturated plane lattice $L$ but removes only its Heisenberg/lattice sector. Put

$$C_6 = \mathrm{Com}(V_L, U), \qquad N = E_8 \cap \Pi^{\perp}.$$

Lattice positivity gives $C_6 = U \cap V_N = \mathrm{Com}(\mathcal{H}_\Pi, U)$, where $\mathcal{H}_\Pi$ is the plane Heisenberg algebra. Its conformal vector is the perpendicular stress. The zero-perpendicular-energy space in $V_{E_8}$ is exactly $V_L$: a vector of zero energy has neither perpendicular momentum nor perpendicular oscillators. This proves $\mathrm{Com}(C_6, U) = V_L$ at every weight.

The source route is `hi_cft_20260923_next_c2_asymmetric_coset.md`, with the independent local audit `hi_cft_20260923_next_structure_c6_coset_audit.md`. Its use of the lattice-coset regularity theorem is specific: it checks the positive-definite lattice double commutant required by Section 4.3 of the Creutzig–Kanade–Linshaw–Ridout source before applying its Lemma 4.6, Theorem 4.12 and Corollary 4.13. It does not claim rationality for arbitrary cosets. Under the strong rationality of $U$ already obtained in the main proof, this route yields a strongly rational $C_6$. As elsewhere in this supplementary route, these theorem applications are written out for review and are not a new external validation.

## S12.1 A fully specified rank-six lattice, not only a determinant

The lattice $N$ has determinant 39 and root system $A_2 \oplus A_2 \oplus A_1$. Its full isometry class is the even index-six extension

$$N = (A_2 \oplus A_2 \oplus A_1 \oplus \langle 78 \rangle) + \mathbb{Z}(\omega, \omega, \alpha/2, u/6),$$

where $\omega^2 = 2/3$, $\alpha^2 = 2$ and $u^2 = 78$. The glue vector has norm $2/3 + 2/3 + 1/2 + 78/36 = 4$. In a basis $(a_1, a_2, b_1, b_2, c, v)$ consisting of simple roots and this glue vector, the Gram matrix is

$$G_N = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 1 \\ -1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ 1 & 0 & 1 & 0 & 1 & 4 \end{pmatrix}, \tag{155}$$

$$\nu = 6v - 4a_1 - 2a_2 - 4b_1 - 2b_2 - 3c.$$

with $\nu^2 = 78$ and $\nu$ orthogonal to the five roots. The determinant check is the Schur complement

$$18\left(4 - \frac{2}{3} - \frac{2}{3} - \frac{1}{2}\right) = 39.$$

The letter $\nu$ here is the perpendicular vector denoted $u$ in the abstract lattice description, not the original weight-two generating field.

There is also a basis in the original ambient coordinates. Let $e_0, \ldots, e_7$ be the simple-root basis, with diagonal Gram entries two and edges $(0,1), (1,2), (2,3), (3,4), (2,5), (6,7)$ of pairing $-1$. The index-three glue vector is

$$\lambda = (4/3, 5/3, 2, 4/3, 2/3, 1, 2/3, 1/3).$$

The vectors $e_0, \dots, e_6, \lambda$ form an integral basis of $E_8$. For $x = \sum_{i=0}^{6} y_i e_i + k\lambda$, orthogonality to the plane says

$$y_5 = -k, \qquad y_6 = -\sum_{i=0}^{4} y_i - 8k.$$

Thus $n_i = e_i - e_6$, $0 \le i \le 4$, and $n_5 = \lambda - e_5 - 8e_6$ form a complete integral basis of $N$. After replacing $n_5$ by

$$w = n_5 - 2n_0 - 2n_1 - 2n_2 - n_3 - n_4 = (-2/3, -1/3, 0, 1/3, -1/3, 0, 2/3, 1/3),$$

the Gram matrix is

$$\begin{pmatrix} 4 & 1 & 2 & 2 & 2 & -2 \\ 1 & 4 & 1 & 2 & 2 & -1 \\ 2 & 1 & 4 & 1 & 2 & -1 \\ 2 & 2 & 1 & 4 & 1 & 0 \\ 2 & 2 & 2 & 1 & 4 & -2 \\ -2 & -1 & -1 & 0 & -2 & 2 \end{pmatrix}.$$

The archived proof identifies the isometry class by the root and glue argument below; it does not claim to have saved a change-of-basis matrix from this ambient basis to (155). All of this data is in `hi_cft_20260923_next_c6_lattice.md`.

## S12.2 The root-and-glue proof, including the excluded alternative

The authenticated neighbor adjoint table has 22 joint Cartan-zero vectors. Removing the eight Cartan directions leaves 14 roots in $N$. No such root lies in the $E_6 \oplus A_2$ root lattice: orthogonality to $a_0$ removes the $e_5$ coordinate of an $E_6$ root, leaving the parabolic $A_5$; $H_R$ has positive pairing two with every simple root of that $A_5$, and also with both roots of the separate $A_2$. Thus all perpendicular roots carry nonzero charge in $E_8/(E_6 \oplus A_2) = \mathbb{Z}/3$.

An $A_3$ subsystem would have consecutive simple roots $r, s, t$. Nonzero charges of $r + s$ and $s + t$ force the three simple charges to agree; then $r + s + t$ has zero charge, a contradiction. Hence every irreducible root component is $A_1$ or $A_2$. Fourteen roots and rank at most six leave $A_2 \oplus A_1^4$ or $A_2^2 \oplus A_1$. The first has full rank and determinant 48, whose index in $N$ would have square $48/39$, impossible. Therefore the rank-five root lattice is $R_0 = A_2^2 \oplus A_1$.

Its discriminant form

$$D_{R_0} = (\mathbb{Z}/3)^2 \oplus \mathbb{Z}/2, \qquad q(a, b, c) = \frac{a^2 + b^2}{3} + \frac{c^2}{4}$$

has no nonzero isotropic element, so $R_0$ is primitive in its span. Let its rank-one perpendicular have norm $2m$, and let $r$ be the glue index. Injectivity of the glue projections into the two discriminant groups gives $r \mid 6$. Determinants give

$$39 = \frac{18(2m)}{r^2}, \qquad m = \frac{13r^2}{12}.$$

Integrality forces $r = 6$ and $2m = 78$. The order-six class on the rank-one side has quadratic value $13/12 \equiv 1/12$; its partner must have value $11/12$, forcing nonzero components in both $A_2$ discriminants and in the $A_1$ discriminant. Diagram conjugation reverses either $A_2$ component, so the glue is unique up to isometry. This recovers the complete lattice above.

Its VOA has weight-one dimension $6 + 14 = 20$, with Lie algebra $A_2 \oplus A_2 \oplus A_1$ and one abelian current. Its chiral category is the pointed category on $\mathbb{Z}/39$ with form $-n^2/39$. In contrast $(C_6)_1 = 0$ in the proper branch: the non-Cartan $U$ currents have nonzero plane charges and the two Cartan currents fail the positive Heisenberg-mode condition. The known $U_2 = S_2$ conclusion gives $\dim(C_6)_2 = 7$.

### S12.3 Category factorization and the transported extension algebra

Every one of the 39 lattice labels occurs in the actual core $R$, hence in $U$. The mirror theorem for the regular mutual pair $(V_L, C_6)$ identifies the branching modules with the full reverse pointed category $\mathcal{P}_{39}^{\mathrm{rev}}$, where $q_{\mathcal{P}_{39}}(n) = n^2/39$. Modular centralizer decomposition and the existing diagonal extension give

$$\mathrm{Rep}(C_6) \simeq \mathcal{P}_{39}^{\mathrm{rev}} \boxtimes \mathrm{Rep}(U). \tag{156}$$

The actual $U$ algebra over $V_L \otimes C_6$ is the diagonal algebra in $\mathcal{P}_{39} \boxtimes \mathcal{P}_{39}^{\mathrm{rev}}$, tensored with the unit of $\mathrm{Rep}(U)$. Its local-module category is the second factor, explaining (156). Under the main Haagerup-center identification the rank is $39 \cdot 12 = 468$. Its multiplicative central charge is consistent: the reverse pointed factor has charge six modulo eight and a Drinfeld center has trivial anomaly.

Let $A_U$ be the actual extension algebra of the original $U \subset H$. Transitivity of extension algebras identifies the ambient algebra over $V_L \otimes C_6$ as $\Delta_{\mathcal{P}_{39}} \boxtimes A_U$, including its multiplication. Taking the $V_L$ vacuum multiplicity yields the actual extension $C_6 \subset V_N$ with algebra $\mathbf{1}_{\mathcal{P}_{39}^{\mathrm{rev}}} \boxtimes A_U$. Thus the $C_6$-preserving algebra-bimodule category is

$$\mathcal{P}_{39}^{\mathrm{rev}} \boxtimes_{A_U} \mathrm{Rep}(U)_{A_U} \simeq \mathcal{P}_{39}^{\mathrm{rev}} \boxtimes \mathcal{F} \boxtimes \mathcal{F}^{\mathrm{op}}, \qquad \mathcal{F} = \mathrm{Rep}(U)_{A_U}.$$

This transports the actual associativity data of the original extension, not just its fusion matrices. The categorical defect statement retains the full-CFT realization requirements: this supplementary coset argument does not independently prove complete unitarity of $C_6$ or the needed comparison of every bimodule with a physical defect operator. Nor does it select $\mathcal{F} = \mathcal{H}_3$ when the original extension is only known to have $\mathcal{F} \simeq \mathcal{H}_2$ or $\mathcal{H}_3$.

### S12.4 Precise obstructions to obtaining charge two by the immediate routes

Several negative results constrain this additional construction. They do not identify the physical chain or rule out an unrelated charge-two theory.

First, the simple dimensions of $\mathrm{Rep}(U)$ are $1, d^2, 1+d^2$ (four times), and $3d$ (six times), where $d = (3+\sqrt{13})/2$. Its only invertible is the unit and $(3d)^2 > 1 + d^2$. If two nontrivial fusion factors existed, each would have a noninvertible simple of dimension at least $3d$; their external product would exceed the largest allowed simple dimension. Thus this category has no nontrivial fusion-category tensor factorization. In particular a tensor factorization of $U$ into unitary strongly rational charges two and six would force one factor to be holomorphic, whose charge must be zero modulo eight, a contradiction. This does not exclude a finite conformal extension of a tensor product or a mutual-commutant pair.

Second, the transparent current algebra inside $U$ is contained in its two $\mathfrak{sl}_2$ factors. Their full Sugawara charge is

$$1 + \frac{3 \cdot 39}{39 + 2} = \frac{158}{41}.$$

A compact reductive current subalgebra has no larger Sugawara charge, by positivity of the complementary affine coset. An ordinary coset by such currents therefore has charge at least $8 - 158/41 = 170/41 > 2$. The precise premise is that the gauged currents commute with the specified symmetry action; generalized quotients or modified actions are outside this statement.

Third, the exact charge-one and charge-two idempotent exclusions in the complete seven-dimensional $(C_6)_2$ are recorded in `hi_cft_20260923_next_c2_reduced_exclusion_c1_checked.json` and `hi_cft_20260923_next_c2_reduced_exclusion_c2_checked.json`. Their original finite scope becomes exhaustive only after using $U_2 = S_2$ and the complete

Cartan-commutant dimension. If $e$ is any complex Virasoro vector in this weight-two space and $t_6$ is the total stress, the absence of weight-one states and invariance give

$$(t_6, e) = (e, e) = c_e/2, \qquad t_6 - e \text{ is a commuting Virasoro vector of charge } 6 - c_e.$$

Thus the exclusions also rule out charges five and four. In particular an ordinary conformal charge-four coset inside this $C_6$ cannot leave charge two. The argument does not require $e$ to be PCT real. The polynomial system and its finite certificate interpretation are described in `hi_cft_20260923_next_c2_slice_theorem.md`.

These facts distinguish an actual charge-six lattice commutant from an asserted charge-two physical theory. Finite topological gauging itself preserves the stress and central charge. Deleting a commuting conformal sector is additional structure and does not automatically transport the desired defect action.

# S13 Categorical exclusions: diagnostic calculations and their scope

The main branching argument eliminates the rank-four and higher multiplicity rank-six possibilities before invoking unitary classification. Several early notes only compared with familiar affine models; the later notes give universal replacements. This section records the small diagnostic calculations that are particularly useful when checking that upgrade.

## S13.1 The rank is an endomorphism-algebra calculation

From the actual extension restriction $H = \mathbf{1} \oplus X \oplus Y \oplus Y^*$, the alternatives are

| case | $\mathrm{End}_U(H)$ | boundary rank |
|---|---|---|
| $Y \ncong Y^*$ | $\mathbb{C}^4$ | 4 |
| $Y \cong Y^*$ | $\mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C})$ | 6 |

The dimension is the sum of squared multiplicities, not the number of distinct summands. In the second case the composition algebra is noncommutative, an essential input to the rank-six fusion-ring analysis. Center induction identifies this algebra with the complexified fusion algebra of the all-module boundary category. The local modules are a different category: for a holomorphic extension they form Vec and cannot give this rank calculation.

The actual order-three lattice grading automorphisms inject into the invertible objects of the boundary category. Untwistedness requires additional information. The metric group of $B$ has quadratic form $(2a^2 + b^2)/3$; its two transverse isotropic lines are $(a, a)$ and $(a, -a)$. Induction from the second line to modules of the first-line extension is fully faithful by adjunction: only the trivial intersection can contribute a morphism. The target dimension is $9/3 = 3$, so the three images exhaust it. This gives the tensor equivalence with untwisted $\mathrm{Vec}_{\mathbb{Z}/3}$. The existence of three invertible objects alone would not determine their associator.

## S13.2 Why the balanced multiplicity-six near-group case is impossible

For a rank-four $\mathbb{Z}/3 + 6$ near-group category, put $d = 3 + 2\sqrt{3}$ and $r = 1 + d$. In the balanced induction pattern, the known central dimensions are

$$1, \quad 1 + 2d, \quad 2r \text{ (3 times)}, \quad r \text{ (4 times)}, \quad n_j d, \qquad \sum_j n_j^2 = 22.$$

The normalized modular vacuum row has entries

$$b,\ a,\ \underbrace{1/3,1/3,1/3}_{3},\underbrace{1/6,1/6,1/6,1/6}_{4},\frac{n_j\sqrt{3}}{12},\qquad a=\frac{2+\sqrt{3}}{12},\quad b=\frac{2-\sqrt{3}}{12}.$$

Signed Galois symmetry under $\sqrt{3}\mapsto-\sqrt{3}$ forces the row of the unique dimension-$1+2d$ object to be

$$a,\ b,\ 1/3,1/3,1/3,\ 1/6,1/6,1/6,1/6,-n_j\sqrt{3}/12.$$

That object is self-dual because its dimension is unique. Its self-fusion coefficient is therefore

$$N_{XX}^{X}=\frac{a^3}{b}+\frac{b^3}{a}+\frac{3}{9}+\frac{4}{36}-\frac{22}{48}=\frac{97}{72}+\frac{32}{72}-\frac{33}{72}=\frac{4}{3}.$$

For a hand check, $a+b=1/3$, $ab=1/144$, so $a^2+b^2=7/72$ and $(a^3/b)+(b^3/a)=97/72$. Nonintegrality excludes the balanced case without specifying any unknown twist or unprinted modular row. The source is `hi_cft_20260923_next_structure_m6_balanced_exclusion.md`.

For the remaining unbalanced case, the later universal projection uses only a forced three-dimensional character subspace. With $d=3+2\sqrt{3}$, its remaining twists $z_j$ satisfy

$$\sum_j n_j^2=6,\qquad \sum_j n_j^2 z_j=-2-4i,\qquad \sum_j n_j^2 z_j^2=-2+4i(1-\epsilon),\quad \epsilon\in\{1,-1\}.$$

The second-moment modulus excludes $\epsilon=-1$. Next

$$\sum_j n_j^2(1+\Re z_j)(1+\Im z_j)=6-2-4+\tfrac{1}{2}\Im(-2)=0.$$

Each summand is nonnegative, so every twist is $-1$ or $-i$; the total squared multiplicities are two and four respectively. This argument allows a dimension-$2d$ object at the latter twist; it does not assume exhaustion by a familiar rank-nine model. In the categorical twist convention, before the common character phase $e^{-2\pi i/3}$ is included, the stable weighted projection obeys

$$S_v=-\frac{1}{2}\begin{pmatrix}1&2&1\\1&0&-1\\1&-2&1\end{pmatrix},\qquad T_v=\operatorname{diag}(1,-i,-1).$$

The harmonic-theta coefficient calculation in the two orientations then gives either $\dim U_1=8+A$, $A\in\mathbb{Z}_{\geq0}$, or $\dim U_1=5+12P$, $P\in\mathbb{Z}_{\geq0}$. Neither can be six. The universal scope is supplied by `hi_cft_20260923_next_structure_m6_universal_projection.md`; the earlier `hi_cft_20260923_next_structure_m6_character_obstruction.md` by itself only excluded the two explicit ribbon models compared there.

## S13.3 The exceptional rank-six multiplicity-two check

For the noncommutative rank-six family, $d^2=1+3md$ and the residual central objects have dimensions $3n_jd$. The first two indicator moments are

$$\sum_j n_j^2=4m^2+2,$$

$$\sum_j n_j^2\theta_j=-\frac{m^2+m\sqrt{9m^2+4}}{2},$$

$$\sum_j n_j^2\theta_j^2=\frac{-m^2+(2\epsilon-m)\sqrt{9m^2+4}}{2}.$$

The root-count estimate eliminates $m > 1$ except possibly the squareclass ten. In that squareclass, a multiset summing to $-a - b\sqrt{10}$ has at least $a + 8b$ roots. Its equality statement requires that the sum of squared roots belongs to $\mathbb{Q}(\sqrt{10})$; this is an actual hypothesis, provided here by the displayed second indicator moment. Without it, Galois averaging would not recover the original squared sum.

The last candidate is $m = 2$, $\sqrt{9m^2 + 4} = 2\sqrt{10}$. It has 18 roots, $a = 2$, $b = 2$, and attains the lower bound $2 + 8 \cdot 2 = 18$. Equality forces squared sum 2. The indicator formula instead gives $-2$ when $\epsilon = 1$ and $-2 - 4\sqrt{10}$ when $\epsilon = -1$. This explicit contradiction removes the exceptional case.

The corrected proof and its prior audit are `hi_cft_20260923_next_structure_rank6_indicators.md`. The same note records a useful failed diagnostic: the HLPW primary-$n$ positivity tests are nonnegative for the entire family, so those tests cannot prove $m = 1$. A criterion that every candidate passes is not evidence that it selects the desired multiplicity.

At $m = 1$, the remaining six-root sum is $(-1 - \sqrt{13})/2$. The minimal quadratic Gauss orbit forces six distinct thirteenth-root twists with exponents $\{2, 5, 6, 7, 8, 11\}/13$, each with $n_j = 1$; the squared sum is $(-1 + \sqrt{13})/2$ and forces second indicator $+1$. This gives the twelve central dimensions and twists. Determining these modular data and proving a braided category equivalence remain separate steps. The main manuscript's unitary route uses the separately stated unitary classification; the older optional pivotal route invokes a broader external exhaustive census whose equation-solving files were not independently replayed here.

## S14 Analytic verification map: domains, sectors and extensions

The analytic part of the current manuscript proves three bridges: A2 is the independent net sector count; A3 reconstructs and exhausts ordinary unitary modules; A1 compares the particular bounded annulus insertions. Their roles differ. The following table records both the conclusion and the premise that would be lost by a shortcut. The references point to the proofs already present in this draft.

| Step | Actual conclusion | Required distinction |
|---|---|---|
| Vacuum form | The PCT-stable $U$ has a positive invariant form. | This is a form on the vacuum VOA, not yet on every simple module. |
| Finite index | The bounded-insertion subnet is completely rational; its abstract vacuum isomorphism transfers that property to $\mathcal{N}_U$. | An isomorphism-invariant property does not identify individual concrete insertion operators. |
| A2 | The actual $H$ restriction has multiplicities $1, 1, 2$; the boundary ring gives twelve net sectors. | The count is derived for the net category, rather than copied from the VOA modular matrix. |
| A3 | Each irreducible net sector reconstructs an irreducible ordinary unitary $U$-module; the functor is fully faithful and linear $*$. | No tensorator or braiding compatibility is claimed for this functor. |
| Exhaustion | Twelve distinct net images exhaust twelve algebraic simple modules. | Equality of counts gives linear surjectivity, not positivity of canonical fusion pairings. |

| Step | Actual conclusion | Required distinction |
|---|---|---|
| A1 | The two concrete nets coincide and each localized annulus insertion transports to its bounded module form. | Central path lifts, range domains and full insertion interiors are part of the claim. |
| Fusion positivity | Tener's theorem gives positivity of canonical VOA fusion forms after strong unitarity and A1. | Unitarizability of the fusion product is supplied before applying that theorem. |
| Classification | The now unitary all-module category has the specified Haagerup classes and the center is $Z(\mathcal{H}_3)$. | Matching modular data alone is not this tensor classification. |

### S14.1 Closed-field reduction: the domain fact needed before irreducibility

For a closable smeared field $T_0$ on a common finite-energy core $D$ and an orthogonal projection $P$, the two statements $PD \subset D$ and $T_0P = PT_0$ on $D$ imply that $P$ reduces the closure and its adjoint. A graph-norm approximation establishes $TP = PT$ on the closed domain; applying the same argument to $1 - P$ decomposes the closed operator as a direct sum. Its polar partial isometry and the spectral projections of its absolute value are then block diagonal. This is the reason the formal module projection becomes an intertwiner of the generated local von Neumann algebras. Formal mode commutation without domain invariance would not give that conclusion.

In reverse reconstruction, choose one ordinary simple algebraic summand $S$ of the reconstructed weak module $M$ and let $P$ project onto its Hilbert closure. The needed domain inclusion is

$$P1_{[0,R]}(L_0)\mathcal{K} \subset \bigoplus_{h+n\le R} S_{h+n} \subset S.$$

It uses finite-dimensional energy spaces of $S$, not a prior finite-multiplicity assertion for the entire sector Hilbert space. The invariant form then gives mode commutation. A smeared field need not map $M$ into itself; its Hilbert-space value and boundedness of $P$ suffice to pass to the closed field. Irreducibility forces $P = 1$ and hence $M = S$. These points are the substantive content of Proposition I.3.

Full faithfulness also needs boundedness of an algebraic module isomorphism. Compare its two positive invariant forms grade by grade. The resulting positive grade-preserving comparison operator is a $U$-endomorphism. An eigenvector in the finite lowest space generates an eigenspace submodule, so simplicity makes the operator one scalar on the whole module. The isomorphism therefore has a bounded unitary rescaling. This is stronger than simply observing that algebraic Schur's lemma gives one-dimensional morphism spaces.

### S14.2 A fractional-energy test for the common-lift argument

Vacuum normalization cannot distinguish every universal rotation lift: $Z = e^{2\pi i L_0}$ is the identity on the integer-graded vacuum but acts by $e^{2\pi i h}$ on a simple module of lowest weight $h$. For example, on a module with $h = 2/13$ it acts by $e^{4\pi i/13}$. Identifying two annulus lifts only from their vacuum scalar would miss this phase.

The pairwise construction instead first chooses the lifts of the two localized annuli in the canonical localized extension, whose remaining ambiguity is scalar. It normalizes their nonzero

vacuum operators there, and then undoes the same universal lift $\widetilde{\delta}$ of a common localizer. On a module, changing that common lift by $Z^m$ changes the pair as

$$A_M \longmapsto Z^{-m}A_M, \qquad B_M \longmapsto B_M Z^m.$$

Every $U$ mode changes conformal energy by an integer, so it commutes with $Z$. Hence

$$B_M Z^m Y_M(v,z) Z^{-m} A_M = B_M Y_M(v,z) A_M.$$

This is a pairwise cancellation at fractional as well as integer lowest weights. It neither assumes a global choice of lifts for all individual annuli nor discards a deck transformation by vacuum normalization.

For the semigroup part of an incoming annulus, the framing $h(\theta,s) = \varphi_{t-s}(e^{i\theta})$ has logarithmic velocity $i\rho(e^{i\theta})\partial_\theta$. With the stated stress convention its integrated action is exactly $e^{-tL(\rho)}$. Thin annuli supply the boundary reparametrizations, giving the specific operator $U(\psi)e^{-tL(\rho)}U(\gamma)^*$, including its scalar. This concrete identification is what an abstract vacuum-net isomorphism alone does not provide.

## S14.3 Range, density and holomorphic continuation are three separate checks

For a geometrically complementary annulus $C_M$, round factorization gives

$$A_M C_M = \lambda r^{L_0}, \qquad 0 < r < 1, \quad \lambda \neq 0.$$

Thus every finite-energy vector lies in $\operatorname{Ran} A_M$:

$$v = A_M C_M(\lambda^{-1} r^{-L_0} v).$$

Injectivity of $A_M$ gives the equality of test domains

$$A_M^{-1} M^{\mathrm{fin}} = C_M M^{\mathrm{fin}}.$$

Boundedness and dense range of $C_M$ then give density of this domain. The outgoing adjoint has the same properties. These are range and density statements; neither follows merely from saying that an annulus operator is bounded.

On those two test domains, both the input $A_M\xi$ and output $B_M^*\eta$ have finite energy support. For a weight-$k$ state $v$, only modes with $q = p + k - n - 1$ contribute between input energy $p$ and output energy $q$. Hence

$$\langle Y_M(v,z) A_M\xi, B_M^*\eta\rangle$$

is a finite Laurent polynomial in $z$. Sewing first identifies it with the bounded geometric insertion on the incoming interior; adjoints do so on the outgoing interior. The holomorphic identity theorem extends it to the interior of the union of the two closed regions. Every component meets one of the original open regions, since the remaining set lies in their boundaries and has no open component. Uniqueness on dense test domains identifies the bounded operator, without treating the unsmeared formal field as everywhere defined.

The prescribed insertion interior is the union over all compatible geometric representatives of the same generalized operator pair. The proof repeats over that union and uses uniqueness on overlaps. Pairs with empty insertion interior still contribute the generator $B_U A_U$, handled directly by local annulus transport. Finally, locality and Haag duality on the same vacuum Hilbert space give

$$\mathcal{A}_U^{\mathrm{bl}}(I) \subset \mathcal{N}_U(I) \subset \mathcal{A}_U^{\mathrm{bl}}(I')' = \mathcal{A}_U^{\mathrm{bl}}(I).$$

That is concrete equality of the two net constructions, the conclusion needed for the module insertion representation.

### S14.4 Primary theorem chain retained by the current A1 proof

The following is the exact source map recorded by the final 27 September analytic revision. It is useful for reviewing the pairwise argument without reconstructing its citation chain.

| Source | Specific uses |
|---|---|
| Tener, fields [28] | Definition 3.16 for incoming semigroup data; Lemma 3.19 and Corollaries 3.20–3.22 for a common localizer; Definition 4.2 for module range and density; Proposition 4.13 for vacuum covariance; proof of Theorem 4.28 for the union of representative interiors; Section 5.1, formula (5.2), for paired module annulus actions. |
| Henriques–Tener, annuli [31] | Remark 4.2, formula (5.2) and Theorem 5.5 for the integrated framing operator; Section 7 and Theorem 7.4 for the localized extension and local sector transport. |
| Henriques–Tener, nets [30] | Lemmas 4.15 and 7.9 for injectivity/dense range and round factorization; Proposition 4.17 for transport; Lemma 6.8 for local point insertions; formula (76) for sewing; Corollary 6.15 for holomorphicity; Proposition 6.19 and formula (81) for adjoints; Theorems 8.9 and 9.5(i) for vacuum reconstruction and module insertions. |
| Tener, fusion [29] | Definition 2.18 is the required module representation property; Theorem 4.4 and Corollary 4.6 supply canonical fusion positivity after the separately proved strong unitarity. |

The frozen earlier part of `hi_cft_c8_revision_20260927_analytic.md` records an unsuccessful search for a ready-made theorem proving A1. Its final section explicitly supersedes that assessment with the pairwise construction. Both records remain useful: the failed shortcut explains why complete unitarity is not deduced from a linear category equivalence, while the final proof supplies the additional operator comparison. The current manuscript treats A1 as proved under its ordinary construction inputs, not as an additional unresolved assumption. External expert verification of the written argument is still distinct from those internal review records.

### S14.5 The three extension objects that must remain distinct

| Object | Established construction | Identification boundary |
|---|---|---|
| Original $H = V_{E_8}$ | The given lattice inclusion over $U$, with restriction $\mathbf{1} \oplus X \oplus Y^{\oplus 2}$. | Its all-module category is $\mathcal{H}_2$ or $\mathcal{H}_3$; the proof does not choose which. |
| Chosen net $\mathcal{E}_3$ | Transport the canonical $\mathcal{H}_3$ Q-system to $\mathrm{Rep}(\mathcal{N}_U)$ and apply local net reconstruction. | It is holomorphic of charge eight; no classification of arbitrary holomorphic charge-eight nets as the $E_8$ lattice net is asserted. |

| Object | Established construction | Identification boundary |
|---|---|---|
| Chosen VOA $H'$ | After complete unitarity, transport the canonical $\mathcal{H}_3$ algebra to $\mathrm{Rep}(U)$ and use unitary VOA extension theory. | Holomorphic charge-eight VOA classification identifies $H'$ with $V_{E_8}$ as a VOA, but does not identify its embedding over $U$ with the original lattice embedding. |

For the chosen $\mathcal{H}_3$ extension, all right algebra modules form $\mathcal{H}_3$, local modules form Vec, and algebra bimodules form $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$ with 36 simples. Only the local subcategory inherits the ambient braiding by descent. The unitarization argument for all modules uses the braided left action and the relative tensor product; it does not impose the extra local-module monodromy condition. The diagonal full net theory, by contrast, has partition function $\sum_i |\chi_i|^2$ and uses the twelve-sector linear character comparison. These constructions have different roles even though they start from the same smaller chiral algebra.

# S15 Structural provenance, replay results and historical status changes

A saved exact identity, an exact replay of its producer, and an all-weight implication are different pieces of evidence. This section records their relationship. No research producer was run for the present expansion: the coefficients and results here were read from the existing artifacts, with lightweight text and JSON transcription only.

## S15.1 The bounded structural replays of 27 September

| Replay | Exact assertions recovered | Recorded resources |
|---|---|---|
| Potts primary | Stress charge 4/5; actual weight-three primary; membership after a rank-40 descendant space and one additional zero-mode pair, reaching rank 41; $w_{P,2}$ eigenvalues $0, 2, -2$ on $H_R, E_+, E_-$. | 8.234 s; producer peak 97820 KiB. |
| PF13 membership | The saved primary lies in the generated rank-58 Cartan-zero space after 154 zero-weight pairs; PCT-stable primary line; positive-stress coefficients $1, 3403/4563, 25/6, 25/26$. | 180.622 s; adapter peak 120436 KiB. |
| Order-thirteen core | $\dim S_2 = 120$; known tensor-core grade two 15; zero-residual-energy dimension 25; ten extras and all six one-dimensional witness charge spaces. | 6.455 s; producer peak 40216 KiB. |
| Four mixed tops | All four sign choices complete; seed index 68, one-dimensional seed Cartan eigenspace, exact weighted adjoints, cyclic dimension 81, irreducible top. | 3.031 s; producer peak 26496 KiB. |

| Replay | Exact assertions recovered | Recorded resources |
|---|---|---|
| Top-bound arithmetic | All fourteen minima, dimensions, stress traces and charge characters; both $\mathfrak{sl}_2$ inequalities and the nonnegative dual identity. | Tiny exact Fraction calculation; no field action is inferred. |

The files are, respectively, `c8_revision_20260927/potts_replay.json`, `c8_revision_20260927/pf13_membership_replay.json`, `c8_revision_20260927/core_extension_replay.json`, `c8_revision_20260927/top81_replay.json`, and `c8_revision_20260927/top_bound_replay.json`. The first two corresponding logs end with the adapter status `unchanged_producer_exact_replay_passed`. The adapter compares complete mathematical outputs with the frozen certificate, ignoring only run time and memory accounting. It authenticates source and original certificate before and after, and redirects exactly one requested output path and one write. This status is stronger than an intermediate producer success message.

The core and top producers can record a bounded partial run, so process exit zero alone is insufficient. The saved mixed-top record has `status=complete`, four sectors, and in every sector `cyclic_dimension=81`, `seed_eigenspace_dimension=1`, `exact_weighted_adjoint_checks=true`, and `irreducible=true`. Even these full finite checks do not supply the $h < 2$ all-weight branching step by themselves.

The archived resource review `c8_revision_20260927/structural_replay_review.md` estimated below 0.5 GiB for core and mixed-top jobs and below 0.75 GiB for the two membership jobs, including sparse pivots, Fraction temporaries, copies and bounded mode caches. It used serial root-controlled launches with a 2 GiB address cap, 8 GiB process-group RSS watchdog, two CPUs/threads and the permanent 32 GiB host reserve. These measured historical jobs do not authorize a larger future replay or renewal of an expired resource exception.

## S15.2 PF13 checkpoints and the boundary of the replay claim

The full PF13 identification combines more than the membership record. Its direct OPE stage used a continuation of a partially completed run: fifteen completed exact assertions were retained in an authenticated log, and the remaining direct coefficient was computed by a separate continuation. The final simple pole is obtained by a checked algebraic Jacobi/skew-symmetry derivation, not by treating an interrupted weight-seven expansion as zero. The weight-six decoupling identity had already completed before that direct high-weight cross-check was stopped.

The reconstruction therefore depends on the recursive field checkpoints `hi_cft_20260923_proof_pf13_W5.json` and `hi_cft_20260923_proof_pf13_W6.json`, as well as the exact-field, DLY-conversion and weight-seven producers. A source provenance audit found these five dependencies missing from an earlier combined graph. Their historical hashes were retained in `hi_cft_20260923_proof_artifact_audit.json`; the revised `c8_revision_20260927/authenticate_inputs.py` explicitly binds them. The $W^6$ file is about 8.8 MiB and is stream-hashed rather than parsed by the revision's 8 MiB JSON cap.

Hash authentication proves identity of bytes with the recorded checkpoint, not correctness of the modes that produced it. The membership replays above do not rerun the expensive $W^5, W^6$ construction. The readable companion prints the complete nonlinear rows and the seven-term decoupling relation so their algebraic role can be checked separately. The finite-reconstruction theorem then recovers the remaining generator brackets by an acyclic sequence

of Wick and iterate reductions. It is this combination, not agreement of $(c, \lambda)$ alone, that identifies the PF13 factor.

### S15.3 Reading old scope flags without reinstating obsolete gaps

| Archived record | Original finite scope | Current use in this draft |
|---|---|---|
| `hi_cft_20260923_proof_potts.json` | Potts stress exact; weight-three membership numerical. | The separate exact primary and membership record supplies that missing finite input. |
| `hi_cft_20260923_proof_pf13_membership.json` | Membership and positive stress exact; PF identification explicitly false/conditional in that one file. | The complete OPE, decoupling and reconstruction chain supplies the identification separately. The old flag is not edited. |
| `hi_cft_20260923_proof_core_coset.json` | Numerical dimensions in selected $S_2, S_3$ only. | Retained as exploratory observations; not promoted to a full coset character. |
| `hi_cft_20260923_next_structure_order13_core.md` | Actual core established; no properness or Rep($U$) identification. | Core input to the top bound; properness uses the independent finite strong-generation argument. |
| `hi_cft_20260923_next_structure_residual_low_character.md` | Residual grades two and three conditional on properness and $U_3 = S_3$. | The present manuscript supplies those separate premises; no claim about residual grades four and higher is added. |
| `hi_cft_20260923_next_structure_m6_character_obstruction.md` | Two explicit ribbon models excluded; no general classification claimed. | The later universal weighted projection supplies the missing generality. |
| `hi_cft_20260923_next_structure_final_dependency_audit.md` | Conditional identification coherent; properness and analytic bridges not yet supplied at that date. | Useful audit of the algebraic chain; current finite closure and analytic proofs are additional work. |
| `hi_cft_c8_revision_20260927_analytic.md` | Early entries leave A1 open; final entry explicitly supersedes them. | Current pairwise A1 proof is retained with its domain and lift obligations; this appendix does not revert it to an assumption. |

A historical record can have a correct limited statement even when later work proves more. Conversely, a later global argument does not turn a numerical residual or a source hash into an independent exact field verification. The source annotations above preserve both directions of this distinction.

### S15.4 Additional primary dependencies of the supplementary coset route

The following source locations are retained from the archived notes; the expansion does not claim a fresh external source audit. They identify which theorem applications beyond the main proof a reviewer should inspect when using the supplementary results.

| Application | Primary source location recorded in the archive |
|---|---|
| Lattice–parafermion reconstruction | Arakawa–Lam–Yamada [16], Sections 2.4 and 6, equations (2.13), (2.16), (2.19) and simple-current extension Theorem 2.4. |
| No proper extension of $R$ | McRae, compact-orbifold tensor theory, Corollary 4.8 (and Theorem 4.14 for the optional regularity of a hypothetical extension). |
| Index-twelve $F_4$ embedding | Penkov–Zuckerman, generalized Harish-Chandra modules, Section 3 and Table B for the index list and uniqueness by index. |
| Finite centralizer $S_4$ | Westaway, Section 4.9, and Xue, Section 5.1 and Table 13, for the $F_4(a_3)$ diagram, nilpotent centralizer and component group. |
| Regular diagonal frame | Lin, rationality of diagonal coset VOAs, Theorem 5.7 at $k = 8$, and Theorem 3.10 for the stated finite extension $C_2$ step. |
| Ambient residual regularity | McRae, Theorem 6.11, after the actual double-commutant and $C_2$ hypotheses have been checked. |
| Mirror branching | McRae, mirror equivalence, Theorem 1.1 and the twist discussion after Theorem 4.6. |
| Charge-six regularity | Creutzig–Kanade–Linshaw–Ridout, Section 4.3, including its standing lattice hypothesis, Lemma 4.6, Theorem 4.12 and Corollary 4.13. |
| Category splitting and actual extension transport | Müger modular centralizer decomposition, and the Lagrangian/local-module and extension-transitivity results used in [39, 42, 40]. The archived notes specify the diagonal algebra and nonzero dimension rather than appealing only to a matching global dimension. |

# Part IV

# Interaction, defect and boundary data

# S16 Weight-two interactions: a complete printed reconstruction atlas

The short physics chapter identifies the fields and quotes selected couplings. This section supplies the information needed to inspect the entire weight-two primary tensor without opening its 18,365-row sparse file. It gives the actual component index dictionary, every highest-vector norm, every nonzero ordered map between current multiplets, a phase-sensitive representative of each map, and an explicit elementary formula reconstructing the other components. The compression uses current-algebra equivariance, not a numerical approximation or a choice to omit small coefficients.

The source distinction matters. The files `c8_physics_20260927/complete_primary_couplings.json` and its neighboring TSV concern the earlier 96-field slice. All interaction tables in this section use the files of the same names in `c8_physics_20260927/full_weight_two/`. That later release contains all 113 Virasoro primaries in $U_2$. Its exact sparse tensor has $113^3 = 1,442,897$ ordered entries, of which 18,365 are nonzero and 1,424,532 vanish. The additional 17 fields are retained throughout this section, including in the normalization and the $u$-pair sum rule.

## S16.1 Actual component basis, including the lowering gauge

Write $V_m$ for the current multiplet numbered $m$, with dimensions $(n_m, r_m)$ under the left and right $\mathfrak{sl}_2$ factors. The field denoted by $(m, p, q)$ is the actual saved vector

$$v_{m,p,q} = (F_L)_0^p (F_R)_0^q v_{m,0,0}, \qquad 0 \le p < n_m, \quad 0 \le q < r_m. \tag{157}$$

The integer indices $0, \dots, 112$ used in the tensor are printed in Table 51. There is no factorial division in this convention. The Cartan charges are $(n_m - 1 - 2p, r_m - 1 - 2q)$, and the algebraic raising rules are

$$(E_L)_0 v_{m,p,q} = p(n_m - p) v_{m,p-1,q}, \tag{158}$$

$$(E_R)_0 v_{m,p,q} = q(r_m - q) v_{m,p,q-1}. \tag{159}$$

These rules fix an algebraic $\mathfrak{sl}_2$ basis. They do not by themselves say that the displayed $E$ and $F$ are Hermitian adjoints.

In fact the left lowering normalization is substantial and must not be lost in a comparison of component tables. The saved positive norms give

$$\boxed{N_{m,p,q} = N_{m,0,0}\, 8640^{2p} \frac{p!\,(n_m - 1)!}{(n_m - 1 - p)!} \frac{q!\,(r_m - 1)!}{(r_m - 1 - q)!}.} \tag{160}$$

Thus on these multiplets the adjoints of the zero-mode lowering operators are $(F_L)_0^* = 8640^2 (E_L)_0$ and $(F_R)_0^* = (E_R)_0$. Equivalently one can pass to compactly normalized generators $E_L' = 8640 E_L$, $F_L' = F_L/8640$. The tabulated vectors continue to use the original saved generators, as in (157). For example, the highest defining field has $N_{7,0,0} = 12$, whereas $N_{7,1,0} = 895795200$. Replacing the latter by 12 would change normalized coefficients incorrectly. The right lowering alone has the familiar factor $q!(r_m - 1)!/(r_m - 1 - q)!$.

The factor can also be derived directly from the torus gauge, without consulting all component norms. The left root is $\alpha_L = (-1, -2, -3, -2, -1, -2, 0, 0)$ and the gauge entries are $(5, 8, 9, 8, 5, 1, 2, 2)$. Hence

$$\prod_i \tilde{c}_i^{(\alpha_L)_i} = \frac{1}{5^2\, 8^4\, 9^3} = \frac{1}{8640^2}.$$

With $E_L = e^{\alpha_L}$ and $F_L = -e^{-\alpha_L}$, transported PCT gives $\theta_g(E_L) = -F_L/8640^2$. The adjoint identity for a weight-one zero mode is $a_0^* = -(\theta_g a)_0$, yielding the adjoints just stated. The right-current coefficients similarly pair each root with its inverse gauge factor, giving the unit compact normalization on the right. Iterating these adjoints and the raising rules proves (160).

Equation (160) is also directly visible in the saved data: the transcription checks its ratio against every one of the 113 recorded norms. This is a check of an encoding convention, not a new lattice-mode calculation. The highest vectors and their norms are as follows.

Table 50: All thirteen multiplet highest-vector norms in the saved lattice gauge. The last column counts stored nonzero Fock-basis terms; it is not a representation dimension.

| $m$ | first index | $(n_m, r_m)$ | $N_{m,0,0}$ | Fock terms |
|---|---|---|---|---|
| 0 | 0 | $(1,5)$ | $\frac{20520}{91}$ | 61 |
| 1 | 5 | $(1,9)$ | 3432 | 32 |
| 2 | 14 | $(1,9)$ | 3900 | 32 |
| 3 | 23 | $(1,13)$ | $\frac{1123200}{7}$ | 9 |
| 4 | 36 | $(2,2)$ | $\frac{65}{135936}$ | 42 |
| 5 | 40 | $(2,6)$ | $\frac{65}{2268}$ | 32 |
| 6 | 52 | $(2,8)$ | $\frac{143}{752}$ | 22 |
| 7 | 68 | $(2,14)$ | 12 | 3 |
| 8 | 96 | $(1,5)$ | 2964 | 26 |
| 9 | 101 | $(3,3)$ | $\frac{13}{24883200}$ | 7 |
| 10 | 110 | $(1,1)$ | $\frac{9243}{1681}$ | 66 |
| 11 | 111 | $(1,1)$ | $\frac{33748800}{1615441}$ | 76 |
| 12 | 112 | $(1,1)$ | $\frac{49840}{961}$ | 74 |

The first eight highest vectors are the original charged multiplets, of types $(1,5)$, $(1,9)$, $(1,9)$, $(1,13)$, $(2,2)$, $(2,6)$, $(2,8)$ and $(2,14)$. The two copies of $(1,9)$ have different coset weights; their labels must remain distinct. Multiplets 8 and 9 have highest vectors $(E_R)_{-1}E_R$ and $(E_L)_{-1}E_R$ and types $(1,5),(3,3)$. The last three are the following total-Virasoro primaries:

$$v_{10} = c_r T_L - c_l T_R, \tag{161}$$

$$v_{11} = c_e(T_L + T_R) - (c_l + c_r)T_e, \tag{162}$$

$$v_{12} = c_f(T_L + T_R + T_e) - (c_l + c_r + c_e)T_f, \tag{163}$$

where

$$(c_l, c_r, c_e, c_f) = \left(1, \frac{117}{41}, \frac{2400}{1271}, \frac{70}{31}\right).$$

The stress labels here are internal commuting chiral factors, not the left and right spacetime central charges. Their sum is eight.

For a linear combination $A = \sum_i a_i T_i$, the fourth-order pole of the total stress with $A$ is $\frac{1}{2}\sum_i a_i c_i$. Hence $A$ is primary of weight two exactly when $\sum_i a_i c_i = 0$. Since the four stress tensors are orthogonal and have norms $c_i/2$, the three combinations above are mutually orthogonal and orthogonal to the total stress. This verifies both the primary condition and the normalization mechanism without referring to a mode engine.

There are $96 + 5 + 9 + 3 = 113$ displayed fields. The vacuum character gives $\dim U_2 = 120$ and $\dim U_1 = 6$. The six current derivatives and the total stress account for the remaining seven states. The positive orthogonal primary vectors therefore exhaust the primary space:

there is no omitted primary multiplet at this weight. The first column of the next table makes all subsequent tensor indices unambiguous.

Table 51: Complete index dictionary for all 113 weight-two Virasoro primaries. Multiplying the last column by Table 50 gives the actual squared norm.

| index $i$ | $(m,p,q)$ | $(H_L,H_R)$ | $N_i/N_{m,0,0}$ |
|---|---|---|---|
| 0 | $(0,0,0)$ | $(0,4)$ | 1 |
| 1 | $(0,0,1)$ | $(0,2)$ | 4 |
| 2 | $(0,0,2)$ | $(0,0)$ | 24 |
| 3 | $(0,0,3)$ | $(0,-2)$ | 144 |
| 4 | $(0,0,4)$ | $(0,-4)$ | 576 |
| 5 | $(1,0,0)$ | $(0,8)$ | 1 |
| 6 | $(1,0,1)$ | $(0,6)$ | 8 |
| 7 | $(1,0,2)$ | $(0,4)$ | 112 |
| 8 | $(1,0,3)$ | $(0,2)$ | 2016 |
| 9 | $(1,0,4)$ | $(0,0)$ | 40320 |
| 10 | $(1,0,5)$ | $(0,-2)$ | 806400 |
| 11 | $(1,0,6)$ | $(0,-4)$ | 14515200 |
| 12 | $(1,0,7)$ | $(0,-6)$ | 203212800 |
| 13 | $(1,0,8)$ | $(0,-8)$ | 1625702400 |
| 14 | $(2,0,0)$ | $(0,8)$ | 1 |
| 15 | $(2,0,1)$ | $(0,6)$ | 8 |
| 16 | $(2,0,2)$ | $(0,4)$ | 112 |
| 17 | $(2,0,3)$ | $(0,2)$ | 2016 |
| 18 | $(2,0,4)$ | $(0,0)$ | 40320 |
| 19 | $(2,0,5)$ | $(0,-2)$ | 806400 |
| 20 | $(2,0,6)$ | $(0,-4)$ | 14515200 |
| 21 | $(2,0,7)$ | $(0,-6)$ | 203212800 |
| 22 | $(2,0,8)$ | $(0,-8)$ | 1625702400 |
| 23 | $(3,0,0)$ | $(0,12)$ | 1 |
| 24 | $(3,0,1)$ | $(0,10)$ | 12 |
| 25 | $(3,0,2)$ | $(0,8)$ | 264 |
| 26 | $(3,0,3)$ | $(0,6)$ | 7920 |
| 27 | $(3,0,4)$ | $(0,4)$ | 285120 |
| 28 | $(3,0,5)$ | $(0,2)$ | 11404800 |
| 29 | $(3,0,6)$ | $(0,0)$ | 479001600 |
| 30 | $(3,0,7)$ | $(0,-2)$ | 20118067200 |
| 31 | $(3,0,8)$ | $(0,-4)$ | 804722688000 |
| 32 | $(3,0,9)$ | $(0,-6)$ | 28970016768000 |
| 33 | $(3,0,10)$ | $(0,-8)$ | 869100503040000 |
| 34 | $(3,0,11)$ | $(0,-10)$ | 19120211066880000 |
| 35 | $(3,0,12)$ | $(0,-12)$ | 229442532802560000 |
| 36 | $(4,0,0)$ | $(1,1)$ | 1 |
| 37 | $(4,0,1)$ | $(1,-1)$ | 1 |
| 38 | $(4,1,0)$ | $(-1,1)$ | 74649600 |
| 39 | $(4,1,1)$ | $(-1,-1)$ | 74649600 |
| 40 | $(5,0,0)$ | $(1,5)$ | 1 |
| 41 | $(5,0,1)$ | $(1,3)$ | 5 |

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| index $i$ | $(m,p,q)$ | $(H_L,H_R)$ | $N_i/N_{m,0,0}$ |
|---|---|---|---|
| 42 | $(5,0,2)$ | $(1,1)$ | 40 |
| 43 | $(5,0,3)$ | $(1,-1)$ | 360 |
| 44 | $(5,0,4)$ | $(1,-3)$ | 2880 |
| 45 | $(5,0,5)$ | $(1,-5)$ | 14400 |
| 46 | $(5,1,0)$ | $(-1,5)$ | 74649600 |
| 47 | $(5,1,1)$ | $(-1,3)$ | 373248000 |
| 48 | $(5,1,2)$ | $(-1,1)$ | 2985984000 |
| 49 | $(5,1,3)$ | $(-1,-1)$ | 26873856000 |
| 50 | $(5,1,4)$ | $(-1,-3)$ | 214990848000 |
| 51 | $(5,1,5)$ | $(-1,-5)$ | 1074954240000 |
| 52 | $(6,0,0)$ | $(1,7)$ | 1 |
| 53 | $(6,0,1)$ | $(1,5)$ | 7 |
| 54 | $(6,0,2)$ | $(1,3)$ | 84 |
| 55 | $(6,0,3)$ | $(1,1)$ | 1260 |
| 56 | $(6,0,4)$ | $(1,-1)$ | 20160 |
| 57 | $(6,0,5)$ | $(1,-3)$ | 302400 |
| 58 | $(6,0,6)$ | $(1,-5)$ | 3628800 |
| 59 | $(6,0,7)$ | $(1,-7)$ | 25401600 |
| 60 | $(6,1,0)$ | $(-1,7)$ | 74649600 |
| 61 | $(6,1,1)$ | $(-1,5)$ | 522547200 |
| 62 | $(6,1,2)$ | $(-1,3)$ | 6270566400 |
| 63 | $(6,1,3)$ | $(-1,1)$ | 94058496000 |
| 64 | $(6,1,4)$ | $(-1,-1)$ | 1504935936000 |
| 65 | $(6,1,5)$ | $(-1,-3)$ | 22574039040000 |
| 66 | $(6,1,6)$ | $(-1,-5)$ | 270888468480000 |
| 67 | $(6,1,7)$ | $(-1,-7)$ | 1896219279360000 |
| 68 | $(7,0,0)$ | $(1,13)$ | 1 |
| 69 | $(7,0,1)$ | $(1,11)$ | 13 |
| 70 | $(7,0,2)$ | $(1,9)$ | 312 |
| 71 | $(7,0,3)$ | $(1,7)$ | 10296 |
| 72 | $(7,0,4)$ | $(1,5)$ | 411840 |
| 73 | $(7,0,5)$ | $(1,3)$ | 18532800 |
| 74 | $(7,0,6)$ | $(1,1)$ | 889574400 |
| 75 | $(7,0,7)$ | $(1,-1)$ | 43589145600 |
| 76 | $(7,0,8)$ | $(1,-3)$ | 2092278988800 |
| 77 | $(7,0,9)$ | $(1,-5)$ | 94152554496000 |
| 78 | $(7,0,10)$ | $(1,-7)$ | 3766102179840000 |
| 79 | $(7,0,11)$ | $(1,-9)$ | 124281371934720000 |
| 80 | $(7,0,12)$ | $(1,-11)$ | 2982752926433280000 |
| 81 | $(7,0,13)$ | $(1,-13)$ | 38775788043632640000 |
| 82 | $(7,1,0)$ | $(-1,13)$ | 74649600 |
| 83 | $(7,1,1)$ | $(-1,11)$ | 970444800 |
| 84 | $(7,1,2)$ | $(-1,9)$ | 23290675200 |
| 85 | $(7,1,3)$ | $(-1,7)$ | 768592281600 |
| 86 | $(7,1,4)$ | $(-1,5)$ | 30743691264000 |
| 87 | $(7,1,5)$ | $(-1,3)$ | 1383466106880000 |
| 88 | $(7,1,6)$ | $(-1,1)$ | 66406373130240000 |

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| index $i$ | $(m,p,q)$ | $(H_L,H_R)$ | $N_i/N_{m,0,0}$ |
|---|---|---|---|
| 89 | $(7,1,7)$ | $(-1,-1)$ | 3253912283381760000 |
| 90 | $(7,1,8)$ | $(-1,-3)$ | 156187789602324480000 |
| 91 | $(7,1,9)$ | $(-1,-5)$ | 7028450532104601600000 |
| 92 | $(7,1,10)$ | $(-1,-7)$ | 2811380212841840640000000 |
| 93 | $(7,1,11)$ | $(-1,-9)$ | 927755470237807411200000 0 |
| 94 | $(7,1,12)$ | $(-1,-11)$ | 22266131285707377868800000 0 |
| 95 | $(7,1,13)$ | $(-1,-13)$ | 28945970671419591229440000 00 |
| 96 | $(8,0,0)$ | $(0,4)$ | 1 |
| 97 | $(8,0,1)$ | $(0,2)$ | 4 |
| 98 | $(8,0,2)$ | $(0,0)$ | 24 |
| 99 | $(8,0,3)$ | $(0,-2)$ | 144 |
| 100 | $(8,0,4)$ | $(0,-4)$ | 576 |
| 101 | $(9,0,0)$ | $(2,2)$ | 1 |
| 102 | $(9,0,1)$ | $(2,0)$ | 2 |
| 103 | $(9,0,2)$ | $(2,-2)$ | 4 |
| 104 | $(9,1,0)$ | $(0,2)$ | 149299200 |
| 105 | $(9,1,1)$ | $(0,0)$ | 298598400 |
| 106 | $(9,1,2)$ | $(0,-2)$ | 597196800 |
| 107 | $(9,2,0)$ | $(-2,2)$ | 22290251120640000 |
| 108 | $(9,2,1)$ | $(-2,0)$ | 44580502241280000 |
| 109 | $(9,2,2)$ | $(-2,-2)$ | 89161004482560000 |
| 110 | $(10,0,0)$ | $(0,0)$ | 1 |
| 111 | $(11,0,0)$ | $(0,0)$ | 1 |
| 112 | $(12,0,0)$ | $(0,0)$ | 1 |

## S16.2 What a phase-sensitive representative means

Let $N_i = \langle v_i, v_i\rangle_g > 0$ and $\widehat{v}_i = v_i/\sqrt{N_i}$, using the positive square root. All pairings are antilinear in their first slot. The tensor is

$$C_{ij}^{\ k} = \frac{\langle v_k, (v_i)_1 v_j\rangle_g}{\sqrt{N_i N_j N_k}}. \tag{164}$$

Thus its third entry refers to the adjoint field at infinity. In a local OPE it multiplies $z^{-2}\widehat{v}_k$. Projection to the primary space is understood: the identity, currents, their derivatives, and the total stress have their own OPE terms.

To print complex phases compactly, use only in this section

$$\sigma = \sqrt{13} > 0, \qquad \vartheta = i\sqrt{3}, \qquad \overline{\sigma} = \sigma, \quad \overline{\vartheta} = -\vartheta.$$

The numerator in (164) is a positive rational multiple of a primitive integer combination $\kappa = a + b\sigma + c\vartheta + d\sigma\vartheta$. "Primitive" means that the four integer coefficients have no common positive factor. Its sign is retained; $\kappa$ and $-\kappa$ are different phases. Each representative table row gives $\kappa$ and the saved rational number $Q = |C_{ij}^{\ k}|^2$. The exact coefficient is

$$\boxed{C_{ij}^{\ k} = \kappa\sqrt{\frac{Q}{\overline{\kappa}\kappa}}.} \tag{165}$$

Every square root in this formula is positive real. This is a complete exact complex value, including its phase; it is not merely an absolute-value table. If desired its phase denominator is obtained with rational arithmetic from

$$\overline{\kappa}\kappa = a^2 + 13b^2 + 3c^2 + 39d^2 + (2ab + 6cd)\sigma.$$

For $\kappa = \pm 1$ the coefficient is simply $\pm\sqrt{Q}$. The use of $\sigma$ here avoids confusing $\sqrt{13}$ with the dimension symbol $s = 3d + 2$ used in the modular-data discussion.

## S16.3 An explicit reconstruction formula for every component

For each fixed ordered triple $(m, n, \ell)$ the projection

$$P_\ell((\,\cdot\,)_1\,\cdot\,) : V_m \otimes V_n \longrightarrow V_\ell$$

is an $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ intertwiner. This follows because current zero modes are derivations and commute with the total Virasoro modes. Orthogonal projection onto the full primary space preserves this property. Each factor in a finite $\mathfrak{sl}_2$ tensor product occurs with multiplicity at most one. Consequently a specified triple of the labeled current multiplets has at most one intertwiner, even when two different multiplets elsewhere in the list have the same dimensions. One nonzero component fixes this map completely.

Here is a concrete formula that removes the need to choose a Clebsch–Gordan phase convention. For one current factor let $a, b, c$ be the highest Cartan weights, i.e. the representation dimensions minus one, and set $L = (a + b - c)/2$. A map is allowed only if

$$L \in \mathbb{Z}, \qquad 0 \le L \le \min(a, b).$$

Use the polynomial realization with $F = Y\partial_X$ and basis

$$w_{a,p} = F^p X^a = a^{\underline{p}} X^{a-p} Y^p, \qquad a^{\underline{p}} = \frac{a!}{(a-p)!}.$$

The $L$th transvectant is

$$\mathcal{T}_L(f, g) = \sum_{t=0}^{L} (-1)^t \binom{L}{t} (\partial_X^{L-t} \partial_Y^t f)(\partial_X^t \partial_Y^{L-t} g). \tag{166}$$

It is equivariant and lands in degree $c = a + b - 2L$. Put $r = p + q - L$. The coefficient of $w_{c,r}$ in $\mathcal{T}_L(w_{a,p}, w_{b,q})$ is

$$K_{a,b;c}(p, q; r) = \frac{a^{\underline{p}} b^{\underline{q}}}{c^{\underline{r}}} \sum_{t=0}^{L} (-1)^t \binom{L}{t} (a-p)^{\underline{L-t}} p^{\underline{t}} \, (b-q)^{\underline{t}} q^{\underline{L-t}}. \tag{167}$$

Set this coefficient to zero if $r \neq p + q - L$ or $r$ is outside $0, \ldots, c$. A falling factorial with more differentiations than its nonnegative integer degree is zero. For degree zero, $0^{\underline{0}} = 1$. These conventions cover trivial representations, tops and bottoms, without a limiting prescription.

For a full component triple $(i, j, k)$, multiply the $K$ factors for the left and right current algebras and denote the result by $K_{ijk}$. Use the lowering exponents from Table 51. If $(i_*, j_*, k_*)$ is the printed nonzero representative for the same ordered multiplet triple, the exact normalized tensor is

$$\boxed{C_{ij}{}^{k} = C_{i_* j_*}{}^{k_*} \frac{K_{ijk}}{K_{i_* j_* k_*}} \sqrt{\frac{N_k N_{i_*} N_{j_*}}{N_i N_j N_{k_*}}}.} \tag{168}$$

The denominator $K_{i_*j_*k_*}$ is nonzero because that component of the unique intertwiner is nonzero. All norm factors are explicit in (160) and the printed highest-norm table. The formula uses the raw algebraic lowering basis, so the factor $8640^{2p}$ must be included in its norm conversion.

To see the normalization directly, write $P_\ell((v_i)_1 v_j) = B^{\ell}_{mn} K_{ijk} v_k$ in a charge-compatible component. Pairing with $v_k$ gives a numerator $B^{\ell}_{mn} K_{ijk} N_k$, hence $C_{ij}^{\ k} = B^{\ell}_{mn} K_{ijk} \sqrt{N_k/(N_i N_j)}$. Taking the ratio with the representative gives (168); the unknown reduced scalar $B^{\ell}_{mn}$ cancels. This proof also explains why the phase of a representative is sufficient.

For example the first $V_0 \otimes V_0 \to V_1$ representative is $(i, j, k) = (0, 0, 5)$, with

$$C_{00}^{\ 5} = \frac{\sigma + 2\vartheta}{5} \sqrt{\frac{3971}{2808}}.$$

On the right factor its degrees are $(4, 4, 8)$, so $L = 0$. Moving to $(i, j, k) = (0, 1, 6)$ changes $K$ from 1 to $4/8 = 1/2$, while $N_6/N_5 = 8$ and $N_1/N_0 = 4$. The normalized coefficient is therefore divided by $\sqrt{2}$, and $|C_{01}^{\ 6}|^2 = 3971/5616$, agreeing with the component export. This example tests both the no-factorial convention and the distinction between raw coefficients and unit-field coefficients.

The saved computation used a second, independent way to obtain the components: the Ward recurrence

$$P((Fv_i)_1 v_j) = F\, P((v_i)_1 v_j) - P((v_i)_1 F v_j).$$

Its directly computed highest-first slice contained 1137 actual lattice mode products and 3512 charge-compatible projections, with 1639 nonzero projections. After completion, the saved report records 51,076 raising/lowering pair checks, 12,769 primary skew-symmetry pair checks, and 33 additional direct lattice product probes. The printed reconstruction formula is the representation theory underlying those Ward identities; the document transcription does not rerun the lattice calculation or extend its weight cutoff.

## S16.4 Exhaustive support and all reduced coefficients

Among the $13^3 = 2197$ ordered multiplet triples, exactly 318 have a nonzero projected map. The other 1879 maps vanish in the saved complete tensor. This includes triples forbidden by the two $\mathfrak{sl}_2$ triangle rules and any allowed triples whose reduced coefficient is zero. A zero map is stronger than a vanishing selected component. To make the distinction explicit, the support table lists every ordered pair $(m, n)$, including pairs with no target.

Table 52: Exhaustive support of the primary tensor by ordered pair of multiplets. A missing target means the entire equivariant map vanishes; the final column counts all nonzero ordered component triples for that pair.

| first $m$ | second $n$ | nonzero target multiplets | nonzero components |
|---|---|---|---|
| 0 | 0 | 0, 1, 2, 8, 10, 11, 12 | 103 |
| 0 | 1 | 0, 1, 2, 3 | 148 |
| 0 | 2 | 0, 1, 2, 3 | 148 |
| 0 | 3 | 1, 2, 3 | 149 |
| 0 | 4 | 5 | 20 |

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| first $m$ | second $n$ | nonzero target multiplets | nonzero components |
|---|---|---|---|
| 0 | 5 | $4, 5, 6$ | 120 |
| 0 | 6 | $5, 6$ | 120 |
| 0 | 7 | $7$ | 124 |
| 0 | 8 | $0$ | 19 |
| 0 | 9 | $\varnothing$ | 0 |
| 0 | 10 | $0$ | 5 |
| 0 | 11 | $0$ | 5 |
| 0 | 12 | $0$ | 5 |
| 1 | 0 | $0, 1, 2, 3$ | 148 |
| 1 | 1 | $0, 1, 2, 3, 8, 10, 11, 12$ | 298 |
| 1 | 2 | $0, 1, 2, 3$ | 232 |
| 1 | 3 | $0, 1, 2, 3$ | 284 |
| 1 | 4 | $6$ | 32 |
| 1 | 5 | $5, 6, 7$ | 252 |
| 1 | 6 | $4, 5, 6, 7$ | 352 |
| 1 | 7 | $5, 6, 7$ | 452 |
| 1 | 8 | $1$ | 39 |
| 1 | 9 | $\varnothing$ | 0 |
| 1 | 10 | $1$ | 9 |
| 1 | 11 | $1$ | 9 |
| 1 | 12 | $1$ | 9 |
| 2 | 0 | $0, 1, 2, 3$ | 148 |
| 2 | 1 | $0, 1, 2, 3$ | 232 |
| 2 | 2 | $0, 1, 2, 3, 8, 10, 11, 12$ | 298 |
| 2 | 3 | $0, 1, 2, 3$ | 284 |
| 2 | 4 | $6$ | 32 |
| 2 | 5 | $5, 6, 7$ | 252 |
| 2 | 6 | $4, 5, 6, 7$ | 352 |
| 2 | 7 | $5, 6, 7$ | 452 |
| 2 | 8 | $2$ | 39 |
| 2 | 9 | $\varnothing$ | 0 |
| 2 | 10 | $2$ | 9 |
| 2 | 11 | $2$ | 9 |
| 2 | 12 | $2$ | 9 |
| 3 | 0 | $1, 2, 3$ | 149 |
| 3 | 1 | $0, 1, 2, 3$ | 284 |
| 3 | 2 | $0, 1, 2, 3$ | 284 |
| 3 | 3 | $0, 1, 2, 3, 8, 10, 11, 12$ | 478 |
| 3 | 4 | $7$ | 52 |
| 3 | 5 | $6, 7$ | 240 |
| 3 | 6 | $5, 6, 7$ | 392 |
| 3 | 7 | $4, 5, 6, 7$ | 648 |
| 3 | 8 | $3$ | 59 |
| 3 | 9 | $\varnothing$ | 0 |
| 3 | 10 | $3$ | 13 |
| 3 | 11 | $3$ | 13 |
| 3 | 12 | $3$ | 13 |

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| first $m$ | second $n$ | nonzero target multiplets | nonzero components |
|---|---|---|---|
| 4 | 0 | 5 | 20 |
| 4 | 1 | 6 | 32 |
| 4 | 2 | 6 | 32 |
| 4 | 3 | 7 | 52 |
| 4 | 4 | 9, 10, 11, 12 | 28 |
| 4 | 5 | 0 | 20 |
| 4 | 6 | 1, 2 | 64 |
| 4 | 7 | 3 | 52 |
| 4 | 8 | ∅ | 0 |
| 4 | 9 | 4 | 16 |
| 4 | 10 | 4 | 4 |
| 4 | 11 | 4 | 4 |
| 4 | 12 | 4 | 4 |
| 5 | 0 | 4, 5, 6 | 120 |
| 5 | 1 | 5, 6, 7 | 252 |
| 5 | 2 | 5, 6, 7 | 252 |
| 5 | 3 | 6, 7 | 240 |
| 5 | 4 | 0 | 20 |
| 5 | 5 | 0, 1, 2, 8, 9, 10, 11, 12 | 308 |
| 5 | 6 | 0, 1, 2, 3 | 320 |
| 5 | 7 | 1, 2, 3 | 360 |
| 5 | 8 | 5 | 44 |
| 5 | 9 | 5 | 64 |
| 5 | 10 | 5 | 12 |
| 5 | 11 | 5 | 12 |
| 5 | 12 | 5 | 12 |
| 6 | 0 | 5, 6 | 120 |
| 6 | 1 | 4, 5, 6, 7 | 352 |
| 6 | 2 | 4, 5, 6, 7 | 352 |
| 6 | 3 | 5, 6, 7 | 392 |
| 6 | 4 | 1, 2 | 64 |
| 6 | 5 | 0, 1, 2, 3 | 320 |
| 6 | 6 | 0, 1, 2, 3, 8, 9, 10, 11, 12 | 568 |
| 6 | 7 | 1, 2, 3 | 464 |
| 6 | 8 | 6 | 64 |
| 6 | 9 | 6 | 88 |
| 6 | 10 | 6 | 16 |
| 6 | 11 | 6 | 16 |
| 6 | 12 | 6 | 16 |
| 7 | 0 | 7 | 124 |
| 7 | 1 | 5, 6, 7 | 452 |
| 7 | 2 | 5, 6, 7 | 452 |
| 7 | 3 | 4, 5, 6, 7 | 648 |
| 7 | 4 | 3 | 52 |
| 7 | 5 | 1, 2, 3 | 360 |
| 7 | 6 | 1, 2, 3 | 464 |
| 7 | 7 | 0, 1, 2, 3, 8, 9, 10, 11, 12 | 1168 |

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| first $m$ | second $n$ | nonzero target multiplets | nonzero components |
|---|---|---|---|
| 7 | 8 | 7 | 124 |
| 7 | 9 | 7 | 160 |
| 7 | 10 | 7 | 28 |
| 7 | 11 | 7 | 28 |
| 7 | 12 | 7 | 28 |
| 8 | 0 | 0 | 19 |
| 8 | 1 | 1 | 39 |
| 8 | 2 | 2 | 39 |
| 8 | 3 | 3 | 59 |
| 8 | 4 | $\varnothing$ | 0 |
| 8 | 5 | 5 | 44 |
| 8 | 6 | 6 | 64 |
| 8 | 7 | 7 | 124 |
| 8 | 8 | 8, 10, 11, 12 | 34 |
| 8 | 9 | 9 | 27 |
| 8 | 10 | 8 | 5 |
| 8 | 11 | 8 | 5 |
| 8 | 12 | 8 | 5 |
| 9 | 0 | $\varnothing$ | 0 |
| 9 | 1 | $\varnothing$ | 0 |
| 9 | 2 | $\varnothing$ | 0 |
| 9 | 3 | $\varnothing$ | 0 |
| 9 | 4 | 4 | 16 |
| 9 | 5 | 5 | 64 |
| 9 | 6 | 6 | 88 |
| 9 | 7 | 7 | 160 |
| 9 | 8 | 9 | 27 |
| 9 | 9 | 8, 9, 10, 11, 12 | 90 |
| 9 | 10 | 9 | 9 |
| 9 | 11 | 9 | 9 |
| 9 | 12 | 9 | 9 |
| 10 | 0 | 0 | 5 |
| 10 | 1 | 1 | 9 |
| 10 | 2 | 2 | 9 |
| 10 | 3 | 3 | 13 |
| 10 | 4 | 4 | 4 |
| 10 | 5 | 5 | 12 |
| 10 | 6 | 6 | 16 |
| 10 | 7 | 7 | 28 |
| 10 | 8 | 8 | 5 |
| 10 | 9 | 9 | 9 |
| 10 | 10 | 10, 11, 12 | 3 |
| 10 | 11 | 10 | 1 |
| 10 | 12 | 10 | 1 |
| 11 | 0 | 0 | 5 |
| 11 | 1 | 1 | 9 |
| 11 | 2 | 2 | 9 |

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| first $m$ | second $n$ | nonzero target multiplets | nonzero components |
|---|---|---|---|
| 11 | 3 | 3 | 13 |
| 11 | 4 | 4 | 4 |
| 11 | 5 | 5 | 12 |
| 11 | 6 | 6 | 16 |
| 11 | 7 | 7 | 28 |
| 11 | 8 | 8 | 5 |
| 11 | 9 | 9 | 9 |
| 11 | 10 | 10 | 1 |
| 11 | 11 | 11, 12 | 2 |
| 11 | 12 | 11 | 1 |
| 12 | 0 | 0 | 5 |
| 12 | 1 | 1 | 9 |
| 12 | 2 | 2 | 9 |
| 12 | 3 | 3 | 13 |
| 12 | 4 | 4 | 4 |
| 12 | 5 | 5 | 12 |
| 12 | 6 | 6 | 16 |
| 12 | 7 | 7 | 28 |
| 12 | 8 | 8 | 5 |
| 12 | 9 | 9 | 9 |
| 12 | 10 | 10 | 1 |
| 12 | 11 | 11 | 1 |
| 12 | 12 | 12 | 1 |

The next table contains exactly one row for each of the 318 nonzero maps. The choice of representative favors a short saved squared coefficient and need not put the first component at the top of its multiplet. Its first column identifies the ordered map; its second column gives the actual component indices. These are all the rows required by (168). A triple absent from this table is a zero map, as specified by the support table.

Table 53: Complete phase-sensitive representative atlas: one nonzero component for every nonzero ordered multiplet map. The coefficient is $C_{ij}^{\,k} = \kappa\sqrt{Q/(\overline{\kappa}\kappa)}$, with positive square roots. Use the index dictionary and the transvectant reconstruction in the text.

| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q = \lvert C_{ij}^{\,k}\rvert^2$ |
|---|---|---|---|
| $(0,0,0)$ | $(0,2,0)$ | $1$ | $\frac{4356}{8645}$ |
| $(0,0,1)$ | $(0,0,5)$ | $\sigma + 2\vartheta$ | $\frac{3971}{2808}$ |
| $(0,0,2)$ | $(0,0,14)$ | $-\sigma + 3\vartheta$ | $\frac{361}{1560}$ |
| $(0,0,8)$ | $(0,2,96)$ | $1$ | $\frac{2}{247}$ |
| $(0,0,10)$ | $(0,4,110)$ | $-1$ | $\frac{4}{1027}$ |
| $(0,0,11)$ | $(0,4,111)$ | $-1$ | $\frac{154587}{703100}$ |
| $(0,0,12)$ | $(0,4,112)$ | $-1$ | $\frac{1681}{3115}$ |
| $(0,1,0)$ | $(0,12,3)$ | $\sigma - 2\vartheta$ | $\frac{3971}{5616}$ |
| $(0,1,1)$ | $(0,7,5)$ | $1$ | $\frac{39605}{717288}$ |
| $(0,1,2)$ | $(0,7,14)$ | $1 + \sigma\vartheta$ | $\frac{57}{1144}$ |
| $(0,1,3)$ | $(0,5,23)$ | $6 - 2\sigma + 4\vartheta + \sigma\vartheta$ | $\frac{665}{858}$ |

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| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q=\lvert C_{ij}^{\ k}\rvert^2$ |
|---|---|---|---|
| $(0,2,0)$ | $(0,20,2)$ | $-\sigma-3\vartheta$ | $\frac{361}{7280}$ |
| $(0,2,1)$ | $(0,16,5)$ | $1-\sigma\vartheta$ | $\frac{57}{1144}$ |
| $(0,2,2)$ | $(0,16,14)$ | $-1$ | $\frac{39}{760}$ |
| $(0,2,3)$ | $(0,15,24)$ | $9+2\sigma+6\vartheta-\sigma\vartheta$ | $\frac{19}{91}$ |
| $(0,3,1)$ | $(0,35,13)$ | $6-2\sigma-4\vartheta-\sigma\vartheta$ | $\frac{665}{858}$ |
| $(0,3,2)$ | $(0,34,21)$ | $9+2\sigma-6\vartheta+\sigma\vartheta$ | $\frac{19}{91}$ |
| $(0,3,3)$ | $(2,24,24)$ | $1$ | $\frac{80}{1729}$ |
| $(0,4,5)$ | $(0,36,40)$ | $-16-\sigma\vartheta$ | $\frac{19}{78}$ |
| $(0,5,4)$ | $(0,45,37)$ | $-16+\sigma\vartheta$ | $\frac{19}{78}$ |
| $(0,5,5)$ | $(1,42,41)$ | $1$ | $\frac{1815}{84721}$ |
| $(0,5,6)$ | $(1,43,55)$ | $81-11\sigma\vartheta$ | $\frac{836}{4095}$ |
| $(0,6,5)$ | $(1,56,42)$ | $81+11\sigma\vartheta$ | $\frac{836}{4095}$ |
| $(0,6,6)$ | $(2,52,52)$ | $1$ | $\frac{208}{5985}$ |
| $(0,7,7)$ | $(0,70,68)$ | $-1$ | $\frac{5}{133}$ |
| $(0,8,0)$ | $(0,98,0)$ | $1$ | $\frac{2}{247}$ |
| $(0,10,0)$ | $(0,110,0)$ | $-1$ | $\frac{4}{1027}$ |
| $(0,11,0)$ | $(0,111,0)$ | $-1$ | $\frac{154587}{703100}$ |
| $(0,12,0)$ | $(0,112,0)$ | $-1$ | $\frac{1681}{3115}$ |
| $(1,0,0)$ | $(5,4,0)$ | $\sigma-2\vartheta$ | $\frac{3971}{2808}$ |
| $(1,0,1)$ | $(5,4,7)$ | $1$ | $\frac{39605}{717288}$ |
| $(1,0,2)$ | $(5,2,14)$ | $1+\sigma\vartheta$ | $\frac{133}{572}$ |
| $(1,0,3)$ | $(5,0,23)$ | $6-2\sigma+4\vartheta+\sigma\vartheta$ | $\frac{665}{858}$ |
| $(1,1,0)$ | $(5,11,0)$ | $1-4\sigma\vartheta$ | $\frac{39605}{717288}$ |
| $(1,1,1)$ | $(7,9,7)$ | $-\sigma+2\vartheta$ | $\frac{5041}{960960}$ |
| $(1,1,2)$ | $(7,9,16)$ | $-7\sigma-11\vartheta$ | $\frac{25}{17472}$ |
| $(1,1,3)$ | $(5,7,23)$ | $-2+12\sigma+\vartheta+8\sigma\vartheta$ | $\frac{11767}{37752}$ |
| $(1,1,8)$ | $(8,9,97)$ | $-1+4\sigma\vartheta$ | $\frac{5}{741}$ |
| $(1,1,10)$ | $(5,13,110)$ | $-1+4\sigma\vartheta$ | $\frac{400}{9243}$ |
| $(1,1,11)$ | $(5,13,111)$ | $1-4\sigma\vartheta$ | $\frac{100}{21093}$ |
| $(1,1,12)$ | $(5,13,112)$ | $-1+4\sigma\vartheta$ | $\frac{343}{445}$ |
| $(1,2,0)$ | $(5,20,0)$ | $-31-\sigma\vartheta$ | $\frac{57}{1144}$ |
| $(1,2,1)$ | $(7,18,7)$ | $\sigma+3\vartheta$ | $\frac{25}{17472}$ |
| $(1,2,2)$ | $(6,19,16)$ | $-\sigma+2\vartheta$ | $\frac{75}{4576}$ |
| $(1,2,3)$ | $(7,20,29)$ | $-62-3\sigma+31\vartheta-2\sigma\vartheta$ | $\frac{35}{416}$ |
| $(1,3,0)$ | $(5,35,4)$ | $-6-2\sigma+4\vartheta-\sigma\vartheta$ | $\frac{665}{858}$ |
| $(1,3,1)$ | $(5,34,10)$ | $-2-\vartheta$ | $\frac{11767}{25168}$ |
| $(1,3,2)$ | $(7,29,16)$ | $2-3\sigma+\vartheta+2\sigma\vartheta$ | $\frac{35}{416}$ |
| $(1,3,3)$ | $(7,26,24)$ | $-\sigma+2\vartheta$ | $\frac{70}{1573}$ |
| $(1,4,6)$ | $(6,36,52)$ | $231\sigma-937\vartheta$ | $\frac{49}{585}$ |
| $(1,5,5)$ | $(5,44,40)$ | $-\sigma+2\vartheta$ | $\frac{11}{1560}$ |
| $(1,5,6)$ | $(6,43,53)$ | $3\sigma-61\vartheta$ | $\frac{5}{3146}$ |
| $(1,5,7)$ | $(5,40,68)$ | $-\sigma+2\vartheta$ | $\frac{9}{770}$ |
| $(1,6,4)$ | $(6,59,37)$ | $459\sigma-443\vartheta$ | $\frac{49}{585}$ |
| $(1,6,5)$ | $(6,58,42)$ | $147\sigma-19\vartheta$ | $\frac{5}{3146}$ |
| $(1,6,6)$ | $(5,57,53)$ | $-\sigma+2\vartheta$ | $\frac{416}{1331}$ |
| $(1,6,7)$ | $(6,52,68)$ | $-147\sigma+19\vartheta$ | $\frac{125}{363}$ |
| $(1,7,5)$ | $(5,81,45)$ | $-\sigma+2\vartheta$ | $\frac{9}{770}$ |

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| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q=\lvert C_{ij}^{\ k}\rvert^2$ |
|---|---|---|---|
| $(1,7,6)$ | $(6,81,59)$ | $-3\sigma+61\vartheta$ | $\frac{125}{363}$ |
| $(1,7,7)$ | $(8,69,68)$ | $-\sigma+2\vartheta$ | $\frac{21}{44}$ |
| $(1,8,1)$ | $(8,99,9)$ | $1$ | $\frac{5}{741}$ |
| $(1,10,1)$ | $(5,110,5)$ | $-1$ | $\frac{400}{9243}$ |
| $(1,11,1)$ | $(5,111,5)$ | $1$ | $\frac{100}{21093}$ |
| $(1,12,1)$ | $(5,112,5)$ | $-1$ | $\frac{343}{445}$ |
| $(2,0,0)$ | $(14,4,0)$ | $-\sigma-3\vartheta$ | $\frac{361}{1560}$ |
| $(2,0,1)$ | $(14,2,5)$ | $1-\sigma\vartheta$ | $\frac{133}{572}$ |
| $(2,0,2)$ | $(14,2,14)$ | $-1$ | $\frac{91}{380}$ |
| $(2,0,3)$ | $(15,0,24)$ | $9+2\sigma+6\vartheta-\sigma\vartheta$ | $\frac{19}{91}$ |
| $(2,1,0)$ | $(14,11,0)$ | $-31-\sigma\vartheta$ | $\frac{57}{1144}$ |
| $(2,1,1)$ | $(16,9,7)$ | $\sigma+3\vartheta$ | $\frac{25}{17472}$ |
| $(2,1,2)$ | $(15,10,16)$ | $-\sigma+2\vartheta$ | $\frac{75}{4576}$ |
| $(2,1,3)$ | $(16,11,29)$ | $-62-3\sigma+31\vartheta-2\sigma\vartheta$ | $\frac{35}{416}$ |
| $(2,2,0)$ | $(14,20,0)$ | $7-3\sigma\vartheta$ | $\frac{39}{760}$ |
| $(2,2,1)$ | $(15,19,7)$ | $\sigma-\vartheta$ | $\frac{75}{4576}$ |
| $(2,2,2)$ | $(16,19,17)$ | $-\sigma-3\vartheta$ | $\frac{21}{520}$ |
| $(2,2,3)$ | $(14,16,23)$ | $14-9\sigma-7\vartheta-6\sigma\vartheta$ | $\frac{3}{728}$ |
| $(2,2,8)$ | $(17,18,97)$ | $7-3\sigma\vartheta$ | $\frac{5}{741}$ |
| $(2,2,10)$ | $(14,22,110)$ | $7-3\sigma\vartheta$ | $\frac{400}{9243}$ |
| $(2,2,11)$ | $(14,22,111)$ | $7-3\sigma\vartheta$ | $\frac{112908}{175775}$ |
| $(2,2,12)$ | $(14,22,112)$ | $-7+3\sigma\vartheta$ | $\frac{45}{623}$ |
| $(2,3,0)$ | $(15,34,4)$ | $9-2\sigma-6\vartheta-\sigma\vartheta$ | $\frac{19}{91}$ |
| $(2,3,1)$ | $(16,29,7)$ | $2+3\sigma+\vartheta-2\sigma\vartheta$ | $\frac{35}{416}$ |
| $(2,3,2)$ | $(14,35,20)$ | $-2-\vartheta$ | $\frac{3}{728}$ |
| $(2,3,3)$ | $(16,29,27)$ | $\sigma+3\vartheta$ | $\frac{9}{104}$ |
| $(2,4,6)$ | $(17,37,55)$ | $129\sigma+197\vartheta$ | $\frac{11}{65}$ |
| $(2,5,5)$ | $(14,44,40)$ | $-\sigma-3\vartheta$ | $\frac{1573}{1176}$ |
| $(2,5,6)$ | $(15,44,54)$ | $-9\sigma-5\vartheta$ | $\frac{27}{143}$ |
| $(2,5,7)$ | $(14,40,68)$ | $\sigma+3\vartheta$ | $\frac{9}{14}$ |
| $(2,6,4)$ | $(17,56,36)$ | $87\sigma+641\vartheta$ | $\frac{11}{65}$ |
| $(2,6,5)$ | $(15,57,41)$ | $9\sigma-193\vartheta$ | $\frac{27}{143}$ |
| $(2,6,6)$ | $(18,52,52)$ | $-\sigma-3\vartheta$ | $\frac{832}{63525}$ |
| $(2,6,7)$ | $(15,52,68)$ | $9\sigma-193\vartheta$ | $\frac{3}{11}$ |
| $(2,7,5)$ | $(14,81,45)$ | $\sigma+3\vartheta$ | $\frac{9}{14}$ |
| $(2,7,6)$ | $(15,76,54)$ | $9\sigma+5\vartheta$ | $\frac{9}{91}$ |
| $(2,7,7)$ | $(17,69,68)$ | $\sigma+3\vartheta$ | $\frac{5}{84}$ |
| $(2,8,2)$ | $(17,99,18)$ | $1$ | $\frac{5}{741}$ |
| $(2,10,2)$ | $(14,110,14)$ | $-1$ | $\frac{400}{9243}$ |
| $(2,11,2)$ | $(14,111,14)$ | $-1$ | $\frac{112908}{175775}$ |
| $(2,12,2)$ | $(14,112,14)$ | $1$ | $\frac{45}{623}$ |
| $(3,0,1)$ | $(23,4,5)$ | $6-2\sigma-4\vartheta-\sigma\vartheta$ | $\frac{665}{858}$ |
| $(3,0,2)$ | $(24,4,15)$ | $9+2\sigma-6\vartheta+\sigma\vartheta$ | $\frac{19}{91}$ |
| $(3,0,3)$ | $(24,2,24)$ | $1$ | $\frac{80}{1729}$ |
| $(3,1,0)$ | $(23,13,0)$ | $-6-2\sigma+4\vartheta-\sigma\vartheta$ | $\frac{665}{858}$ |
| $(3,1,1)$ | $(23,11,5)$ | $-2-\vartheta$ | $\frac{11767}{37752}$ |
| $(3,1,2)$ | $(29,7,16)$ | $2-3\sigma+\vartheta+2\sigma\vartheta$ | $\frac{35}{416}$ |

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| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q=\lvert C_{ij}^{\ k}\rvert^2$ |
|---|---|---|---|
| $(3,1,3)$ | $(24,9,24)$ | $\sigma-2\vartheta$ | $\frac{35}{1716}$ |
| $(3,2,0)$ | $(24,21,0)$ | $9-2\sigma-6\vartheta-\sigma\vartheta$ | $\frac{19}{91}$ |
| $(3,2,1)$ | $(29,16,7)$ | $2+3\sigma+\vartheta-2\sigma\vartheta$ | $\frac{35}{416}$ |
| $(3,2,2)$ | $(23,20,14)$ | $-2-\vartheta$ | $\frac{3}{728}$ |
| $(3,2,3)$ | $(27,20,29)$ | $\sigma+3\vartheta$ | $\frac{9}{104}$ |
| $(3,3,0)$ | $(24,34,2)$ | $-1-4\vartheta$ | $\frac{80}{1729}$ |
| $(3,3,1)$ | $(24,32,7)$ | $-24+\sigma+2\vartheta+4\sigma\vartheta$ | $\frac{70}{1573}$ |
| $(3,3,2)$ | $(27,29,16)$ | $36+\sigma-3\vartheta+4\sigma\vartheta$ | $\frac{9}{104}$ |
| $(3,3,3)$ | $(23,30,24)$ | $2+\vartheta$ | $\frac{9}{143}$ |
| $(3,3,8)$ | $(25,33,98)$ | $1+4\vartheta$ | $\frac{2}{247}$ |
| $(3,3,10)$ | $(23,35,110)$ | $-1-4\vartheta$ | $\frac{196}{1027}$ |
| $(3,3,11)$ | $(23,35,111)$ | $1+4\vartheta$ | $\frac{147}{7031}$ |
| $(3,3,12)$ | $(23,35,112)$ | $-1-4\vartheta$ | $\frac{5}{623}$ |
| $(3,4,7)$ | $(23,36,68)$ | $-32+3\sigma-16\vartheta-2\sigma\vartheta$ | $\frac{1}{2}$ |
| $(3,5,6)$ | $(24,45,53)$ | $-162-33\sigma-81\vartheta+22\sigma\vartheta$ | $\frac{128}{429}$ |
| $(3,5,7)$ | $(23,42,68)$ | $-2-\vartheta$ | $\frac{1}{7}$ |
| $(3,6,5)$ | $(24,58,40)$ | $-162+33\sigma-81\vartheta-22\sigma\vartheta$ | $\frac{128}{429}$ |
| $(3,6,6)$ | $(25,57,53)$ | $2+\vartheta$ | $\frac{5776}{51909}$ |
| $(3,6,7)$ | $(23,55,68)$ | $-162+33\sigma-81\vartheta-22\sigma\vartheta$ | $\frac{24}{77}$ |
| $(3,7,4)$ | $(23,81,37)$ | $-32-3\sigma-16\vartheta+2\sigma\vartheta$ | $\frac{1}{2}$ |
| $(3,7,5)$ | $(23,81,43)$ | $-2-\vartheta$ | $\frac{1}{7}$ |
| $(3,7,6)$ | $(23,81,56)$ | $162+33\sigma+81\vartheta-22\sigma\vartheta$ | $\frac{24}{77}$ |
| $(3,7,7)$ | $(23,74,68)$ | $2+\vartheta$ | $\frac{1}{22}$ |
| $(3,8,3)$ | $(25,98,25)$ | $1$ | $\frac{2}{247}$ |
| $(3,10,3)$ | $(23,110,23)$ | $-1$ | $\frac{196}{1027}$ |
| $(3,11,3)$ | $(23,111,23)$ | $1$ | $\frac{147}{7031}$ |
| $(3,12,3)$ | $(23,112,23)$ | $-1$ | $\frac{5}{623}$ |
| $(4,0,5)$ | $(36,0,40)$ | $-16-\sigma\vartheta$ | $\frac{19}{78}$ |
| $(4,1,6)$ | $(36,6,52)$ | $231\sigma-937\vartheta$ | $\frac{49}{585}$ |
| $(4,2,6)$ | $(36,19,56)$ | $-129\sigma-197\vartheta$ | $\frac{11}{65}$ |
| $(4,3,7)$ | $(36,23,68)$ | $-32+3\sigma-16\vartheta-2\sigma\vartheta$ | $\frac{1}{2}$ |
| $(4,4,9)$ | $(36,36,101)$ | $-217-32\sigma\vartheta$ | $\frac{1}{39}$ |
| $(4,4,10)$ | $(36,39,110)$ | $-217-32\sigma\vartheta$ | $\frac{361}{4108}$ |
| $(4,4,11)$ | $(36,39,111)$ | $217+32\sigma\vartheta$ | $\frac{48}{175775}$ |
| $(4,4,12)$ | $(36,39,112)$ | $217+32\sigma\vartheta$ | $\frac{567}{445}$ |
| $(4,5,0)$ | $(36,51,4)$ | $16+\sigma\vartheta$ | $\frac{19}{78}$ |
| $(4,6,1)$ | $(36,62,7)$ | $-27\sigma+203\vartheta$ | $\frac{98}{195}$ |
| $(4,6,2)$ | $(36,63,17)$ | $-159\sigma-37\vartheta$ | $\frac{11}{65}$ |
| $(4,7,3)$ | $(36,95,35)$ | $32+3\sigma-16\vartheta+2\sigma\vartheta$ | $\frac{1}{2}$ |
| $(4,9,4)$ | $(36,106,37)$ | $1$ | $\frac{1}{78}$ |
| $(4,10,4)$ | $(36,110,36)$ | $1$ | $\frac{361}{4108}$ |
| $(4,11,4)$ | $(36,111,36)$ | $-1$ | $\frac{48}{175775}$ |
| $(4,12,4)$ | $(36,112,36)$ | $-1$ | $\frac{567}{445}$ |
| $(5,0,4)$ | $(40,4,36)$ | $-16+\sigma\vartheta$ | $\frac{19}{78}$ |
| $(5,0,5)$ | $(41,2,41)$ | $1$ | $\frac{605}{338884}$ |
| $(5,0,6)$ | $(41,2,54)$ | $-81+11\sigma\vartheta$ | $\frac{836}{4459}$ |
| $(5,1,5)$ | $(40,10,41)$ | $-\sigma+2\vartheta$ | $\frac{11}{5460}$ |

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| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q=\lvert C_{ij}^{\ k}\rvert^2$ |
|---|---|---|---|
| $(5,1,6)$ | $(41,8,53)$ | $-3\sigma+61\vartheta$ | $\frac{65}{847}$ |
| $(5,1,7)$ | $(40,5,68)$ | $-\sigma+2\vartheta$ | $\frac{9}{770}$ |
| $(5,2,5)$ | $(40,19,41)$ | $-\sigma-3\vartheta$ | $\frac{1573}{4116}$ |
| $(5,2,6)$ | $(41,21,57)$ | $9\sigma+5\vartheta$ | $\frac{27}{143}$ |
| $(5,2,7)$ | $(40,14,68)$ | $\sigma+3\vartheta$ | $\frac{9}{14}$ |
| $(5,3,6)$ | $(40,30,54)$ | $162+33\sigma+81\vartheta-22\sigma\vartheta$ | $\frac{64}{4719}$ |
| $(5,3,7)$ | $(42,23,68)$ | $-2-\vartheta$ | $\frac{1}{7}$ |
| $(5,4,0)$ | $(40,39,0)$ | $16+\sigma\vartheta$ | $\frac{19}{78}$ |
| $(5,5,0)$ | $(41,49,1)$ | $-1$ | $\frac{1815}{84721}$ |
| $(5,5,1)$ | $(40,47,5)$ | $\sigma+2\vartheta$ | $\frac{11}{1560}$ |
| $(5,5,2)$ | $(40,47,14)$ | $\sigma-3\vartheta$ | $\frac{1573}{1176}$ |
| $(5,5,8)$ | $(41,48,96)$ | $1$ | $\frac{6}{247}$ |
| $(5,5,9)$ | $(40,44,101)$ | $-1$ | $\frac{5}{39}$ |
| $(5,5,10)$ | $(40,51,110)$ | $-1$ | $\frac{1681}{36972}$ |
| $(5,5,11)$ | $(40,51,111)$ | $1$ | $\frac{29403}{28124}$ |
| $(5,5,12)$ | $(40,51,112)$ | $-1$ | $\frac{841}{3115}$ |
| $(5,6,0)$ | $(41,65,2)$ | $81+11\sigma\vartheta$ | $\frac{836}{4459}$ |
| $(5,6,1)$ | $(41,66,10)$ | $3\sigma+61\vartheta$ | $\frac{65}{847}$ |
| $(5,6,2)$ | $(41,62,15)$ | $-9\sigma+5\vartheta$ | $\frac{27}{143}$ |
| $(5,6,3)$ | $(40,61,24)$ | $162+33\sigma-81\vartheta+22\sigma\vartheta$ | $\frac{128}{429}$ |
| $(5,7,1)$ | $(40,93,11)$ | $\sigma+2\vartheta$ | $\frac{3}{715}$ |
| $(5,7,2)$ | $(40,93,20)$ | $-\sigma+3\vartheta$ | $\frac{3}{13}$ |
| $(5,7,3)$ | $(42,95,35)$ | $2-\vartheta$ | $\frac{1}{7}$ |
| $(5,8,5)$ | $(41,99,42)$ | $1$ | $\frac{8}{741}$ |
| $(5,9,5)$ | $(40,106,41)$ | $1$ | $\frac{5}{78}$ |
| $(5,10,5)$ | $(40,110,40)$ | $1$ | $\frac{1681}{36972}$ |
| $(5,11,5)$ | $(40,111,40)$ | $-1$ | $\frac{29403}{28124}$ |
| $(5,12,5)$ | $(40,112,40)$ | $1$ | $\frac{841}{3115}$ |
| $(6,0,5)$ | $(54,2,41)$ | $-81-11\sigma\vartheta$ | $\frac{836}{4459}$ |
| $(6,0,6)$ | $(52,2,52)$ | $1$ | $\frac{208}{5985}$ |
| $(6,1,4)$ | $(52,12,36)$ | $-459\sigma+443\vartheta$ | $\frac{49}{585}$ |
| $(6,1,5)$ | $(53,10,41)$ | $147\sigma-19\vartheta$ | $\frac{65}{847}$ |
| $(6,1,6)$ | $(52,10,53)$ | $-\sigma+2\vartheta$ | $\frac{832}{3993}$ |
| $(6,1,7)$ | $(52,6,68)$ | $-147\sigma+19\vartheta$ | $\frac{125}{363}$ |
| $(6,2,4)$ | $(55,19,37)$ | $-87\sigma-641\vartheta$ | $\frac{11}{65}$ |
| $(6,2,5)$ | $(54,21,44)$ | $-9\sigma+193\vartheta$ | $\frac{27}{143}$ |
| $(6,2,6)$ | $(52,18,52)$ | $-\sigma-3\vartheta$ | $\frac{832}{63525}$ |
| $(6,2,7)$ | $(52,15,68)$ | $9\sigma-193\vartheta$ | $\frac{3}{11}$ |
| $(6,3,5)$ | $(53,34,45)$ | $162-33\sigma+81\vartheta+22\sigma\vartheta$ | $\frac{128}{429}$ |
| $(6,3,6)$ | $(53,33,57)$ | $2+\vartheta$ | $\frac{5776}{51909}$ |
| $(6,3,7)$ | $(52,26,68)$ | $162-33\sigma+81\vartheta+22\sigma\vartheta$ | $\frac{6}{121}$ |
| $(6,4,1)$ | $(52,39,6)$ | $27\sigma-203\vartheta$ | $\frac{49}{585}$ |
| $(6,4,2)$ | $(55,38,17)$ | $159\sigma+37\vartheta$ | $\frac{11}{65}$ |
| $(6,5,0)$ | $(54,50,2)$ | $-81-11\sigma\vartheta$ | $\frac{836}{4459}$ |
| $(6,5,1)$ | $(53,48,6)$ | $3\sigma+61\vartheta$ | $\frac{5}{3146}$ |
| $(6,5,2)$ | $(54,47,15)$ | $9\sigma-5\vartheta$ | $\frac{27}{143}$ |
| $(6,5,3)$ | $(53,46,24)$ | $-162-33\sigma+81\vartheta-22\sigma\vartheta$ | $\frac{128}{429}$ |

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| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q=\lvert C_{ij}^{\ k}\rvert^2$ |
|---|---|---|---|
| $(6,6,0)$ | $(52,67,2)$ | $-307-297\sigma\vartheta$ | $\frac{208}{5985}$ |
| $(6,6,1)$ | $(52,65,7)$ | $-59\sigma+179\vartheta$ | $\frac{416}{1331}$ |
| $(6,6,2)$ | $(52,67,18)$ | $149\sigma+147\vartheta$ | $\frac{832}{63525}$ |
| $(6,6,3)$ | $(53,62,25)$ | $614+891\sigma-307\vartheta+594\sigma\vartheta$ | $\frac{5776}{51909}$ |
| $(6,6,8)$ | $(52,65,96)$ | $-307-297\sigma\vartheta$ | $\frac{7}{247}$ |
| $(6,6,9)$ | $(52,58,101)$ | $-307-297\sigma\vartheta$ | $\frac{7}{39}$ |
| $(6,6,10)$ | $(52,67,110)$ | $-307-297\sigma\vartheta$ | $\frac{81}{4108}$ |
| $(6,6,11)$ | $(52,67,111)$ | $307+297\sigma\vartheta$ | $\frac{29584}{527325}$ |
| $(6,6,12)$ | $(52,67,112)$ | $307+297\sigma\vartheta$ | $\frac{169}{3115}$ |
| $(6,7,1)$ | $(52,95,12)$ | $-3\sigma-61\vartheta$ | $\frac{125}{363}$ |
| $(6,7,2)$ | $(52,95,21)$ | $-9\sigma+5\vartheta$ | $\frac{3}{11}$ |
| $(6,7,3)$ | $(52,95,32)$ | $162+33\sigma-81\vartheta+22\sigma\vartheta$ | $\frac{6}{121}$ |
| $(6,8,6)$ | $(52,100,54)$ | $1$ | $\frac{7}{247}$ |
| $(6,9,6)$ | $(52,106,53)$ | $1$ | $\frac{7}{78}$ |
| $(6,10,6)$ | $(52,110,52)$ | $1$ | $\frac{81}{4108}$ |
| $(6,11,6)$ | $(52,111,52)$ | $-1$ | $\frac{29584}{527325}$ |
| $(6,12,6)$ | $(52,112,52)$ | $-1$ | $\frac{169}{3115}$ |
| $(7,0,7)$ | $(68,4,70)$ | $-1$ | $\frac{5}{133}$ |
| $(7,1,5)$ | $(68,13,40)$ | $-\sigma+2\vartheta$ | $\frac{9}{770}$ |
| $(7,1,6)$ | $(68,12,52)$ | $3\sigma-61\vartheta$ | $\frac{125}{363}$ |
| $(7,1,7)$ | $(68,10,69)$ | $\sigma-2\vartheta$ | $\frac{21}{44}$ |
| $(7,2,5)$ | $(68,22,40)$ | $\sigma+3\vartheta$ | $\frac{9}{14}$ |
| $(7,2,6)$ | $(68,21,52)$ | $9\sigma+5\vartheta$ | $\frac{3}{11}$ |
| $(7,2,7)$ | $(68,19,69)$ | $-\sigma-3\vartheta$ | $\frac{5}{84}$ |
| $(7,3,4)$ | $(68,35,36)$ | $-32-3\sigma-16\vartheta+2\sigma\vartheta$ | $\frac{1}{2}$ |
| $(7,3,5)$ | $(68,35,42)$ | $-2-\vartheta$ | $\frac{1}{7}$ |
| $(7,3,6)$ | $(68,32,52)$ | $-162-33\sigma-81\vartheta+22\sigma\vartheta$ | $\frac{6}{121}$ |
| $(7,3,7)$ | $(68,30,69)$ | $2+\vartheta$ | $\frac{3}{11}$ |
| $(7,4,3)$ | $(68,39,23)$ | $32+3\sigma-16\vartheta+2\sigma\vartheta$ | $\frac{1}{2}$ |
| $(7,5,1)$ | $(68,51,5)$ | $\sigma+2\vartheta$ | $\frac{9}{770}$ |
| $(7,5,2)$ | $(68,51,14)$ | $-\sigma+3\vartheta$ | $\frac{9}{14}$ |
| $(7,5,3)$ | $(68,49,23)$ | $2-\vartheta$ | $\frac{1}{7}$ |
| $(7,6,1)$ | $(68,67,6)$ | $3\sigma+61\vartheta$ | $\frac{125}{363}$ |
| $(7,6,2)$ | $(68,67,15)$ | $9\sigma-5\vartheta$ | $\frac{3}{11}$ |
| $(7,6,3)$ | $(68,64,23)$ | $-162-33\sigma+81\vartheta-22\sigma\vartheta$ | $\frac{24}{77}$ |
| $(7,7,0)$ | $(68,93,0)$ | $1$ | $\frac{5}{133}$ |
| $(7,7,1)$ | $(68,94,8)$ | $-\sigma-2\vartheta$ | $\frac{21}{44}$ |
| $(7,7,2)$ | $(68,94,17)$ | $\sigma-3\vartheta$ | $\frac{5}{84}$ |
| $(7,7,3)$ | $(68,89,23)$ | $-2+\vartheta$ | $\frac{1}{22}$ |
| $(7,7,8)$ | $(68,93,96)$ | $-1$ | $\frac{2}{19}$ |
| $(7,7,9)$ | $(68,80,101)$ | $-1$ | $\frac{1}{3}$ |
| $(7,7,10)$ | $(68,95,110)$ | $1$ | $\frac{13}{316}$ |
| $(7,7,11)$ | $(68,95,111)$ | $-1$ | $\frac{1587}{7031}$ |
| $(7,7,12)$ | $(68,95,112)$ | $-1$ | $\frac{81}{3115}$ |
| $(7,8,7)$ | $(68,100,70)$ | $1$ | $\frac{2}{19}$ |
| $(7,9,7)$ | $(68,106,69)$ | $1$ | $\frac{1}{6}$ |
| $(7,10,7)$ | $(68,110,68)$ | $-1$ | $\frac{13}{316}$ |

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| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q=\lvert C_{ij}^{\ k}\rvert^2$ |
|---|---|---|---|
| $(7,11,7)$ | $(68,111,68)$ | $1$ | $\frac{1587}{7031}$ |
| $(7,12,7)$ | $(68,112,68)$ | $1$ | $\frac{81}{3115}$ |
| $(8,0,0)$ | $(96,2,0)$ | $1$ | $\frac{2}{247}$ |
| $(8,1,1)$ | $(97,9,8)$ | $-1$ | $\frac{5}{741}$ |
| $(8,2,2)$ | $(97,18,17)$ | $-1$ | $\frac{5}{741}$ |
| $(8,3,3)$ | $(97,29,28)$ | $-1$ | $\frac{7}{494}$ |
| $(8,5,5)$ | $(96,43,41)$ | $1$ | $\frac{6}{247}$ |
| $(8,6,6)$ | $(96,54,52)$ | $1$ | $\frac{7}{247}$ |
| $(8,7,7)$ | $(96,70,68)$ | $1$ | $\frac{2}{19}$ |
| $(8,8,8)$ | $(96,99,97)$ | $-1$ | $\frac{1225}{741}$ |
| $(8,8,10)$ | $(96,100,110)$ | $-1$ | $\frac{6724}{9243}$ |
| $(8,8,11)$ | $(96,100,111)$ | $1$ | $\frac{4800}{7031}$ |
| $(8,8,12)$ | $(96,100,112)$ | $1$ | $\frac{35}{89}$ |
| $(8,9,9)$ | $(96,103,101)$ | $-1$ | $\frac{76}{39}$ |
| $(8,10,8)$ | $(96,110,96)$ | $-1$ | $\frac{6724}{9243}$ |
| $(8,11,8)$ | $(96,111,96)$ | $1$ | $\frac{4800}{7031}$ |
| $(8,12,8)$ | $(96,112,96)$ | $1$ | $\frac{35}{89}$ |
| $(9,4,4)$ | $(101,39,36)$ | $1$ | $\frac{1}{39}$ |
| $(9,5,5)$ | $(101,47,40)$ | $1$ | $\frac{5}{39}$ |
| $(9,6,6)$ | $(101,61,52)$ | $1$ | $\frac{7}{39}$ |
| $(9,7,7)$ | $(101,83,68)$ | $1$ | $\frac{1}{3}$ |
| $(9,8,9)$ | $(101,99,102)$ | $-1$ | $\frac{38}{39}$ |
| $(9,9,8)$ | $(101,107,96)$ | $-1$ | $\frac{76}{39}$ |
| $(9,9,9)$ | $(101,105,101)$ | $1$ | $\frac{4}{39}$ |
| $(9,9,10)$ | $(101,109,110)$ | $1$ | $\frac{5776}{9243}$ |
| $(9,9,11)$ | $(101,109,111)$ | $1$ | $\frac{4800}{7031}$ |
| $(9,9,12)$ | $(101,109,112)$ | $1$ | $\frac{35}{89}$ |
| $(9,10,9)$ | $(101,110,101)$ | $1$ | $\frac{5776}{9243}$ |
| $(9,11,9)$ | $(101,111,101)$ | $1$ | $\frac{4800}{7031}$ |
| $(9,12,9)$ | $(101,112,101)$ | $1$ | $\frac{35}{89}$ |
| $(10,0,0)$ | $(110,0,0)$ | $-1$ | $\frac{4}{1027}$ |
| $(10,1,1)$ | $(110,5,5)$ | $-1$ | $\frac{400}{9243}$ |
| $(10,2,2)$ | $(110,14,14)$ | $-1$ | $\frac{400}{9243}$ |
| $(10,3,3)$ | $(110,23,23)$ | $-1$ | $\frac{196}{1027}$ |
| $(10,4,4)$ | $(110,36,36)$ | $1$ | $\frac{361}{4108}$ |
| $(10,5,5)$ | $(110,40,40)$ | $1$ | $\frac{1681}{36972}$ |
| $(10,6,6)$ | $(110,52,52)$ | $1$ | $\frac{81}{4108}$ |
| $(10,7,7)$ | $(110,68,68)$ | $-1$ | $\frac{13}{316}$ |
| $(10,8,8)$ | $(110,96,96)$ | $-1$ | $\frac{6724}{9243}$ |
| $(10,9,9)$ | $(110,101,101)$ | $1$ | $\frac{5776}{9243}$ |
| $(10,10,10)$ | $(110,110,110)$ | $1$ | $\frac{23104}{9243}$ |
| $(10,10,11)$ | $(110,110,111)$ | $1$ | $\frac{4800}{7031}$ |
| $(10,10,12)$ | $(110,110,112)$ | $1$ | $\frac{35}{89}$ |
| $(10,11,10)$ | $(110,111,110)$ | $1$ | $\frac{4800}{7031}$ |
| $(10,12,10)$ | $(110,112,110)$ | $1$ | $\frac{35}{89}$ |
| $(11,0,0)$ | $(111,0,0)$ | $-1$ | $\frac{154587}{703100}$ |
| $(11,1,1)$ | $(111,5,5)$ | $1$ | $\frac{100}{21093}$ |

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| $(m,n,\ell)$ | $(i,j,k)$ | $\kappa$ | $Q=\lvert C_{ij}^{\ k}\rvert^2$ |
|---|---|---|---|
| $(11,2,2)$ | $(111,14,14)$ | $-1$ | $\frac{112908}{175775}$ |
| $(11,3,3)$ | $(111,23,23)$ | $1$ | $\frac{147}{7031}$ |
| $(11,4,4)$ | $(111,36,36)$ | $-1$ | $\frac{48}{175775}$ |
| $(11,5,5)$ | $(111,40,40)$ | $-1$ | $\frac{29403}{28124}$ |
| $(11,6,6)$ | $(111,52,52)$ | $-1$ | $\frac{29584}{527325}$ |
| $(11,7,7)$ | $(111,68,68)$ | $1$ | $\frac{1587}{7031}$ |
| $(11,8,8)$ | $(111,96,96)$ | $1$ | $\frac{4800}{7031}$ |
| $(11,9,9)$ | $(111,101,101)$ | $1$ | $\frac{4800}{7031}$ |
| $(11,10,10)$ | $(111,110,110)$ | $1$ | $\frac{4800}{7031}$ |
| $(11,11,11)$ | $(111,111,111)$ | $-1$ | $\frac{1560001}{2109300}$ |
| $(11,11,12)$ | $(111,111,112)$ | $1$ | $\frac{35}{89}$ |
| $(11,12,11)$ | $(111,112,111)$ | $1$ | $\frac{35}{89}$ |
| $(12,0,0)$ | $(112,0,0)$ | $-1$ | $\frac{1681}{3115}$ |
| $(12,1,1)$ | $(112,5,5)$ | $-1$ | $\frac{343}{445}$ |
| $(12,2,2)$ | $(112,14,14)$ | $1$ | $\frac{45}{623}$ |
| $(12,3,3)$ | $(112,23,23)$ | $-1$ | $\frac{5}{623}$ |
| $(12,4,4)$ | $(112,36,36)$ | $-1$ | $\frac{567}{445}$ |
| $(12,5,5)$ | $(112,40,40)$ | $1$ | $\frac{841}{3115}$ |
| $(12,6,6)$ | $(112,52,52)$ | $-1$ | $\frac{169}{3115}$ |
| $(12,7,7)$ | $(112,68,68)$ | $1$ | $\frac{81}{3115}$ |
| $(12,8,8)$ | $(112,96,96)$ | $1$ | $\frac{35}{89}$ |
| $(12,9,9)$ | $(112,101,101)$ | $1$ | $\frac{35}{89}$ |
| $(12,10,10)$ | $(112,110,110)$ | $1$ | $\frac{35}{89}$ |
| $(12,11,11)$ | $(112,111,111)$ | $1$ | $\frac{35}{89}$ |
| $(12,12,12)$ | $(112,112,112)$ | $-1$ | $\frac{2916}{3115}$ |

## S16.5 Stress projections and an independent four-point cross-check

For a unit primary of internal stress weight $h_i$, the squared coupling to the unit stress $T_i/\sqrt{c_i/2}$ is $2h_i^2/c_i$. The complete saved list for the eight original current multiplets is printed below. It resolves the two distinct $(1,9)$ multiplets without relying on their names. Each multiplet's four $h_i$ values sum to two.

Table 54: Resolved stress couplings for the eight original highest multiplets. These are intrinsic-stress projections, before changing to the total stress and the three orthogonal stress-difference primaries.

| $m$ | stress | $c_i$ | $h_i$ | $2h_i^2/c_i$ |
|---|---|---|---|---|
| 0 | $T_L$ | 1 | 0 | 0 |
| 0 | $T_R$ | $\frac{117}{41}$ | $\frac{6}{41}$ | $\frac{8}{533}$ |
| 0 | $T_e$ | $\frac{2400}{1271}$ | $\frac{798}{1271}$ | $\frac{53067}{127100}$ |
| 0 | $T_f$ | $\frac{70}{31}$ | $\frac{38}{31}$ | $\frac{1444}{1085}$ |
| 1 | $T_L$ | 1 | 0 | 0 |
| 1 | $T_R$ | $\frac{117}{41}$ | $\frac{20}{41}$ | $\frac{800}{4797}$ |
| 1 | $T_e$ | $\frac{2400}{1271}$ | $\frac{200}{1271}$ | $\frac{100}{3813}$ |
| 1 | $T_f$ | $\frac{70}{31}$ | $\frac{42}{31}$ | $\frac{252}{155}$ |

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| $m$ | stress | $c_i$ | $h_i$ | $2h_i^2/c_i$ |
|---|---|---|---|---|
| 2 | $T_L$ | 1 | 0 | 0 |
| 2 | $T_R$ | $\frac{117}{41}$ | $\frac{20}{41}$ | $\frac{800}{4797}$ |
| 2 | $T_e$ | $\frac{2400}{1271}$ | $\frac{1512}{1271}$ | $\frac{47628}{31775}$ |
| 2 | $T_f$ | $\frac{70}{31}$ | $\frac{10}{31}$ | $\frac{20}{217}$ |
| 3 | $T_L$ | 1 | 0 | 0 |
| 3 | $T_R$ | $\frac{117}{41}$ | $\frac{42}{41}$ | $\frac{392}{533}$ |
| 3 | $T_e$ | $\frac{2400}{1271}$ | $\frac{420}{1271}$ | $\frac{147}{1271}$ |
| 3 | $T_f$ | $\frac{70}{31}$ | $\frac{20}{31}$ | $\frac{80}{217}$ |
| 4 | $T_L$ | 1 | $\frac{1}{4}$ | $\frac{1}{8}$ |
| 4 | $T_R$ | $\frac{117}{41}$ | $\frac{3}{164}$ | $\frac{1}{4264}$ |
| 4 | $T_e$ | $\frac{2400}{1271}$ | $\frac{192}{1271}$ | $\frac{768}{31775}$ |
| 4 | $T_f$ | $\frac{70}{31}$ | $\frac{49}{31}$ | $\frac{343}{155}$ |
| 5 | $T_L$ | 1 | $\frac{1}{4}$ | $\frac{1}{8}$ |
| 5 | $T_R$ | $\frac{117}{41}$ | $\frac{35}{164}$ | $\frac{1225}{38376}$ |
| 5 | $T_e$ | $\frac{2400}{1271}$ | $\frac{1830}{1271}$ | $\frac{11163}{5084}$ |
| 5 | $T_f$ | $\frac{70}{31}$ | $\frac{3}{31}$ | $\frac{9}{1085}$ |
| 6 | $T_L$ | 1 | $\frac{1}{4}$ | $\frac{1}{8}$ |
| 6 | $T_R$ | $\frac{117}{41}$ | $\frac{63}{164}$ | $\frac{441}{4264}$ |
| 6 | $T_e$ | $\frac{2400}{1271}$ | $\frac{752}{1271}$ | $\frac{35344}{95325}$ |
| 6 | $T_f$ | $\frac{70}{31}$ | $\frac{24}{31}$ | $\frac{576}{1085}$ |
| 7 | $T_L$ | 1 | $\frac{1}{4}$ | $\frac{1}{8}$ |
| 7 | $T_R$ | $\frac{117}{41}$ | $\frac{195}{164}$ | $\frac{325}{328}$ |
| 7 | $T_e$ | $\frac{2400}{1271}$ | $\frac{180}{1271}$ | $\frac{27}{1271}$ |
| 7 | $T_f$ | $\frac{70}{31}$ | $\frac{13}{31}$ | $\frac{169}{1085}$ |

The extremal pair $(68, 95)$ is $(v_{7,0,0}, v_{7,1,13})$. The normalized second component is the chiral adjoint of $\widehat{u}$ up to a phase, because that extremal Cartan weight occurs only once. There are exactly nine nonzero primary targets. Printing all nine makes the aggregate four-point check a short rational addition.

Table 55: All nine nonzero primary projections of the extremal $u$ pair. Their sum is 11/3; targets 110, 111, 112 sum to 41/140.

| target $k$ | $(m, p, q)$ | $\lvert C^{\,k}_{68,95}\rvert^2$ |
|---|---|---|
| 2 | $(0, 0, 2)$ | $\frac{65}{133}$ |
| 9 | $(1, 0, 4)$ | $\frac{273}{880}$ |
| 18 | $(2, 0, 4)$ | $\frac{13}{336}$ |
| 29 | $(3, 0, 6)$ | $\frac{13}{154}$ |
| 98 | $(8, 0, 2)$ | $\frac{26}{19}$ |
| 105 | $(9, 1, 1)$ | $\frac{13}{12}$ |
| 110 | $(10, 0, 0)$ | $\frac{13}{316}$ |
| 111 | $(11, 0, 0)$ | $\frac{1587}{7031}$ |
| 112 | $(12, 0, 0)$ | $\frac{81}{3115}$ |

Their sum is 11/3. The three stress-difference targets sum to 41/140; adding the total stress contribution 1 gives 181/140 for the four commuting stresses. The six remaining primary

targets sum to 1417/420. Thus the total global weight-two coefficient is

$$\frac{14}{3} = 1 + \frac{11}{3} = \frac{181}{140} + \frac{1417}{420}.$$

The two decompositions resolve different orthogonal subspaces. Neither interpretation counts the derivative of a weight-one current as a new global quasiprimary.

# S17 Correlator verification tables and exact channel expansions

This section prints the channel data of `c8_physics_20260927/correlators.json`, including all coefficients through the saved cutoff 32. Rational-function identities are exact at every cross-ratio where defined. A finite global-block table records coefficients only through its stated cutoff. Those two statements should be kept distinct when checking crossing and positivity.

## S17.1 A Gaussian calculation with every surviving term visible

Let the three lattice momenta of the defining weight-two field be

$$a_0 = (0,0,0,0,0,-1,1,1),$$
$$a_1 = (0,0,0,1,1,-1,0,0),$$
$$a_2 = (1,1,0,0,0,-1,0,0).$$

They have Gram matrix with diagonal 4 and off-diagonal 2, and

$$u = e^{a_0} + \frac{\sqrt{13}+3\sqrt{-3}}{20}e^{a_1} + \frac{\sqrt{13}-3\sqrt{-3}}{20}e^{a_2}.$$

The three squared exponential norms in the transported form are $(4,40,40)$. The coefficient absolute squares are $(1,1/10,1/10)$, so each contributes four to the norm $\|u\|^2 = 12$. The normalized field is $\widehat{u} = u/\sqrt{12}$. An adjoint at a chiral insertion is still a holomorphic field; it is not an antiholomorphic insertion.

Number the four insertion points $(\infty, 1, x, 0)$. A letter $p$ means $\widehat{u}$ and a letter $m$ its chiral adjoint. Inserting the three exponential terms at each position gives 81 choices per ordering. Momentum conservation is imposed before assigning a nonzero correlator. Exactly fifteen choices survive in each of the six two-$p$, two-$m$ orderings. Although the cocycle signs vary, the complete normalized coefficient of each surviving monomial is $1/9$. The cocycle sign in the next table is one factor in this calculation, not the sign of the final correlator contribution.

Table 56: Every surviving Gaussian term in the two independent reflection-positive $u$ orderings. Choices are numbered 0,1,2 in the momentum order of the text; each complete normalized coefficient is 1/9, including its adjoint and cocycle factors.

| ordering | momentum choices | cocycle sign | $x$ exponent | $(1-x)$ exponent |
|---|---|---|---|---|
| mpmp | $(0,0,0,0)$ | 1 | -4 | -4 |
| mpmp | $(0,0,1,1)$ | 1 | -4 | -2 |
| mpmp | $(0,0,2,2)$ | 1 | -4 | -2 |
| mpmp | $(0,1,1,0)$ | 1 | -2 | -4 |
| mpmp | $(0,2,2,0)$ | 1 | -2 | -4 |
| mpmp | $(1,0,0,1)$ | 1 | -2 | -4 |
| mpmp | $(1,1,0,0)$ | 1 | -4 | -2 |
| mpmp | $(1,1,1,1)$ | 1 | -4 | -4 |
| mpmp | $(1,1,2,2)$ | 1 | -4 | -2 |
| mpmp | $(1,2,2,1)$ | 1 | -2 | -4 |
| mpmp | $(2,0,0,2)$ | 1 | -2 | -4 |
| mpmp | $(2,1,1,2)$ | 1 | -2 | -4 |
| mpmp | $(2,2,0,0)$ | 1 | -4 | -2 |

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| ordering | momentum choices | cocycle sign | $x$ exponent | $(1-x)$ exponent |
|---|---|---|---|---|
| mpmp | $(2,2,1,1)$ | 1 | -4 | -2 |
| mpmp | $(2,2,2,2)$ | 1 | -4 | -4 |
| mmpp | $(0,0,0,0)$ | 1 | 4 | -4 |
| mmpp | $(0,1,0,1)$ | 1 | 2 | -2 |
| mmpp | $(0,1,1,0)$ | 1 | 2 | -4 |
| mmpp | $(0,2,0,2)$ | 1 | 2 | -2 |
| mmpp | $(0,2,2,0)$ | 1 | 2 | -4 |
| mmpp | $(1,0,0,1)$ | 1 | 2 | -4 |
| mmpp | $(1,0,1,0)$ | 1 | 2 | -2 |
| mmpp | $(1,1,1,1)$ | 1 | 4 | -4 |
| mmpp | $(1,2,1,2)$ | 1 | 2 | -2 |
| mmpp | $(1,2,2,1)$ | 1 | 2 | -4 |
| mmpp | $(2,0,0,2)$ | 1 | 2 | -4 |
| mmpp | $(2,0,2,0)$ | 1 | 2 | -2 |
| mmpp | $(2,1,1,2)$ | 1 | 2 | -4 |
| mmpp | $(2,1,2,1)$ | 1 | 2 | -2 |
| mmpp | $(2,2,2,2)$ | 1 | 4 | -4 |

Summing the monomials gives

$$G_0(x) = G_{mpmp}(x) = G_{pmpm}(x) = \frac{3-4x+4x^2}{3x^4(1-x)^4}, \tag{169}$$

$$G_+(x) = G_{mmpp}(x) = G_{ppmm}(x) = \frac{x^2(4-4x+3x^2)}{3(1-x)^4}, \tag{170}$$

$$G_{pmmp}(x) = G_{mppm}(x) = G_+(1-x). \tag{171}$$

The neutral sum before collecting a numerator is particularly simple:

$$G_0(x) = \frac{1}{3x^4(1-x)^4} + \frac{2}{3x^2(1-x)^4} + \frac{2}{3x^4(1-x)^2}.$$

The three terms correspond to the three exponent pairs in the surviving-term table. Crossing is a rational identity: $G_0(1-x) = G_0(x)$ and $G_+(x) = x^{-4}G_0(1/x)$.

The equal-charge OPE starts at power $x^2$. Its leading state is the weight-six primary $\widehat{u}_{(-3)}\widehat{u}$, with the following three momentum components:

$$\begin{aligned}\widehat{u}_{(-3)}\widehat{u} = &-\frac{\sqrt{13}+3\sqrt{-3}}{120}e^{(0,0,0,1,1,-2,1,1)} \\ &-\frac{\sqrt{13}-3\sqrt{-3}}{120}e^{(1,1,0,0,0,-2,1,1)} - \frac{1}{60}e^{(1,1,0,1,1,-2,0,0)}.\end{aligned} \tag{172}$$

Each momentum has squared length twelve. The saved exact norm of this state is $4/3$, agreeing with the first equal-charge global coefficient $A_6 = 4/3$. This supplies an actual state behind the leading channel, not merely a power-series coefficient.

## S17.2 How to verify the global-block columns by hand

For an external unit field of weight $h$, the convention is

$$R(x) = x^{2h}G(x) = A_0 + \sum_{n\geq 1} A_n x^n \, {}_2F_1(n,n;2n;x). \tag{173}$$

Write $R(x) = \sum_{m\geq 0} r_m x^m$. Since the hypergeometric series has leading coefficient one, this is triangular:

$$A_0 = r_0, \tag{174}$$

$$A_m = r_m - \sum_{n=1}^{m-1} A_n \frac{(n)_{m-n}^2}{(2n)_{m-n}(m-n)!}, \qquad m \geq 1. \tag{175}$$

Here $(n)_j$ is the rising Pochhammer symbol. This recurrence is enough to verify every entry of the next six tables with rational arithmetic. The $A_0$ term is constant: it has no descendant contribution to subtract.

For example the neutral generator has $r_1 = 8/3$, $r_2 = 6$ and $r_3 = 12$. The first current block contributes $A_1/2$ at order two, so $A_2 = 6 - 4/3 = 14/3$. At order three it contributes $A_1/3$, while the weight-two block contributes $A_2$, giving $A_3 = 12 - 8/9 - 14/3 = 58/9$. A bare Taylor coefficient such as 6 therefore cannot be interpreted directly as a primary coupling sum.

In a reflection-positive channel, $A_n$ is a sum of squared couplings to orthonormal global quasiprimaries at weight $n$. It includes global quasiprimaries which are Virasoro descendants of lower-weight states. It is neither the squared coupling to one distinguished field nor the coefficient of a single Virasoro conformal block. Odd coefficients vanish for the identical-boson equal-charge, spin-three, current and stress channels displayed below. The neutral charged-adjoint ordering does not have that odd-weight selection rule.

### S17.2.1 Weight-two generator: neutral channel

The first coefficient $A_1 = 8/3 = 1/2 + 13/6$ resolves the two Cartan currents, whose charges on $u$ are $(1, 13)$ and whose squared norms are $(2, 78)$. At weight two the decomposition was printed in Table 55.

Table 57: Global-block and reduced power-series coefficients through weight 32 for the neutral weight-two generator. All entries are copied as exact fractions.

| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | $\frac{8}{3}$ | $\frac{8}{3}$ |
| 2 | $\frac{14}{3}$ | 6 |
| 3 | $\frac{58}{9}$ | 12 |
| 4 | $\frac{107}{15}$ | $\frac{65}{3}$ |
| 5 | $\frac{674}{105}$ | 36 |
| 6 | $\frac{131}{27}$ | 56 |
| 7 | $\frac{2207}{693}$ | $\frac{248}{3}$ |
| 8 | $\frac{2402}{1287}$ | 117 |
| 9 | $\frac{19264}{19305}$ | 160 |
| 10 | $\frac{18047}{36465}$ | $\frac{638}{3}$ |
| 11 | $\frac{31957}{138567}$ | 276 |
| 12 | $\frac{53957}{529074}$ | 351 |
| 13 | $\frac{12497}{289731}$ | $\frac{1316}{3}$ |
| 14 | $\frac{136957}{7800450}$ | 540 |
| 15 | $\frac{208007}{30087450}$ | 656 |
| 16 | $\frac{153811}{58169070}$ | $\frac{2363}{3}$ |

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| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 17 | $\frac{888764}{901620585}$ | 936 |
| 18 | $\frac{628679}{1750204665}$ | 1102 |
| 19 | $\frac{872957}{6806351475}$ | $\frac{3860}{3}$ |
| 20 | $\frac{170281}{3786992550}$ | 1491 |
| 21 | $\frac{1603037}{103384896615}$ | 1716 |
| 22 | $\frac{2126357}{403693405830}$ | $\frac{5888}{3}$ |
| 23 | $\frac{2785279}{1578074222790}$ | 2232 |
| 24 | $\frac{901658}{1543768261425}$ | 2525 |
| 25 | $\frac{4621052}{24185702762325}$ | $\frac{8528}{3}$ |
| 26 | $\frac{5863327}{94807954828314}$ | 3186 |
| 27 | $\frac{36319}{1832211328626}$ | 3556 |
| 28 | $\frac{4549}{722515868004}$ | $\frac{11861}{3}$ |
| 29 | $\frac{11375107}{5736517950570330}$ | 4380 |
| 30 | $\frac{1996091}{3221492839236540}$ | 4836 |
| 31 | $\frac{17047367}{88698436173646068}$ | $\frac{15968}{3}$ |
| 32 | $\frac{10333541}{174535632470722908}$ | 5841 |

### S17.2.2 Weight-two generator: equal-charge channel

There is no identity or current exchange in this charge sector, and no exchange below weight six. The leading weight-six field is the explicit vector (172).

Table 58: Global-block and reduced power-series coefficients through weight 32 for the equal-charge weight-two generator. All entries are copied as exact fractions.

| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 0 |
| 2 | 0 | 0 |
| 3 | 0 | 0 |
| 4 | 0 | 0 |
| 5 | 0 | 0 |
| 6 | $\frac{4}{3}$ | $\frac{4}{3}$ |
| 7 | 0 | 4 |
| 8 | $\frac{19}{13}$ | 9 |
| 9 | 0 | $\frac{52}{3}$ |
| 10 | $\frac{48}{85}$ | 30 |
| 11 | 0 | 48 |
| 12 | $\frac{50}{357}$ | $\frac{217}{3}$ |
| 13 | 0 | 104 |
| 14 | $\frac{82}{3059}$ | 144 |
| 15 | 0 | $\frac{580}{3}$ |
| 16 | $\frac{259}{60030}$ | 253 |
| 17 | 0 | 324 |
| 18 | $\frac{6272}{10235115}$ | $\frac{1222}{3}$ |
| 19 | 0 | 504 |

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| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 20 | $\frac{724}{9147325}$ | 615 |
| 21 | 0 | $\frac{2224}{3}$ |
| 22 | $\frac{50}{5270837}$ | 884 |
| 23 | 0 | 1044 |
| 24 | $\frac{4895}{4574128182}$ | $\frac{3667}{3}$ |
| 25 | 0 | 1420 |
| 26 | $\frac{6952}{60541478179}$ | 1638 |
| 27 | 0 | $\frac{5632}{3}$ |
| 28 | $\frac{39}{3301329773}$ | 2139 |
| 29 | 0 | 2424 |
| 30 | $\frac{923}{787266089745}$ | $\frac{8200}{3}$ |
| 31 | 0 | 3068 |
| 32 | $\frac{85225}{755565508531268}$ | 3429 |

### S17.3 Spin-three self-OPEs with the residues fully specialized

For either intrinsic spin-three algebra put $N = c/3$, $b = a_{(1)}a$ and $\widehat{a} = a/\sqrt{N}$. The self-OPE is

$$a(z)a(0) \sim \frac{N}{z^6} + \frac{2T_c}{z^4} + \frac{\partial T_c}{z^3} + \frac{b}{z^2} + \frac{\frac{1}{2}\partial b - \frac{1}{12}\partial^3 T_c}{z}.$$

Here $a$ is the canonically rescaled generator. The raw spin-three coordinate fields printed in Appendix S24 must first be multiplied by the conversion factors stated there. Their raw self-OPE coefficient is not, in general, $c/3$ and is not automatically their positive Hermitian norm. The final unit field $\widehat{a}$ includes the further positive factor $1/\sqrt{c/3}$. The two cases have

| algebra | $c$ | $\lambda$ | $a_{(3)}b/a$ | $\|b\|^2$ |
|---|---|---|---|---|
| Potts | 4/5 | 4/13 | 1119/65 | 1492/325 |
| parafermion 13 | 8/5 | 14/473 | 69283/2365 | 554264/35475 |

These are intrinsic central charges. Replacing either with the ambient value eight would give different stress residues.

Let $S_j(x) = x^{-j} + (1-x)^{-j}$. Specializing every coefficient, the complete unit-normalized functions are

$$G_{\mathrm{Potts}}(x) = 1 + S_6(x) + \frac{45}{2}\{S_4(x) + S_3(x)\} + \frac{3357}{52}S_2(x) + \frac{1386}{13}S_1(x), \tag{176}$$

$$G_{\mathrm{PF13}}(x) = 1 + S_6(x) + \frac{45}{4}\{S_4(x) + S_3(x)\} + \frac{207849}{3784}S_2(x) + \frac{46641}{473}S_1(x). \tag{177}$$

Their denominators can be taken to be $x^6(1-x)^6$. For a direct polynomial comparison, write their numerators as $P(x) = \sum_{j=0}^{12} p_j x^j$. They are palindromic, $p_{12-j} = p_j$, and their first seven coefficients are

$$(p_0, \ldots, p_6)_{\mathrm{Potts}} = \left(1, -6, \frac{75}{2}, -\frac{265}{2}, \frac{14667}{52}, -\frac{5190}{13}, \frac{11319}{26}\right),$$

$$(p_0, \ldots, p_6)_{\mathrm{PF13}} = \left(1, -6, \frac{105}{4}, -\frac{305}{4}, \frac{647739}{3784}, -\frac{138690}{473}, \frac{671793}{1892}\right).$$

The principal parts at zero follow from the displayed OPE. Equal-field crossing fixes those at one, and the identity channel fixes the constant at infinity. There is no remaining holomorphic function on the sphere vanishing at infinity. This is why the rational functions are determined beyond the finite expansion cutoff.

The intrinsic Virasoro quasiprimary $\Lambda =: T_cT_c : -\frac{3}{10}\partial^2 T_c$ has norm $c(5c+22)/10$. Its squared unit-normalized coupling in the $\widehat{a}\widehat{a}$ channel is $4608/[5c(5c+22)]$. Subtracting this from the global weight-four exchange leaves

$$\gamma_4^2 = \frac{48(c+2)\{8-\lambda(5c+22)\}}{c(5c+22)} = 0 \quad \text{or} \quad \frac{60552}{2365},$$

for Potts and the parafermion case, respectively. The new weight-four primary is null in Potts. Its nonzero counterpart in the parafermion case should not be confused with the unavoidable weight-four global quasiprimary in the Virasoro vacuum module.

### S17.3.1 Potts spin-three global blocks

Table 59: Global-block and reduced power-series coefficients through weight 32 for the Potts spin-three generator. All entries are copied as exact fractions.

| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 2 | $\frac{45}{2}$ | $\frac{45}{2}$ |
| 3 | 0 | $\frac{45}{2}$ |
| 4 | $\frac{576}{13}$ | $\frac{3357}{52}$ |
| 5 | 0 | $\frac{1386}{13}$ |
| 6 | $\frac{28765}{364}$ | $\frac{11345}{52}$ |
| 7 | 0 | $\frac{5190}{13}$ |
| 8 | $\frac{211533}{3718}$ | $\frac{35427}{52}$ |
| 9 | 0 | $\frac{14246}{13}$ |
| 10 | $\frac{3271869}{126412}$ | $\frac{87381}{52}$ |
| 11 | 0 | $\frac{32220}{13}$ |
| 12 | $\frac{273438}{29393}$ | $\frac{184107}{52}$ |
| 13 | 0 | $\frac{64026}{13}$ |
| 14 | $\frac{53701083}{19315400}$ | $\frac{348381}{52}$ |
| 15 | 0 | $\frac{116242}{13}$ |
| 16 | $\frac{7887591}{11202932}$ | $\frac{46959}{4}$ |
| 17 | 0 | $\frac{197526}{13}$ |
| 18 | $\frac{16764239}{108734340}$ | $\frac{1009781}{52}$ |
| 19 | 0 | $\frac{319032}{13}$ |
| 20 | $\frac{114189576}{3829070245}$ | $\frac{1596555}{52}$ |
| 21 | 0 | $\frac{494826}{13}$ |
| 22 | $\frac{220284561}{42408196168}$ | $\frac{2433501}{52}$ |
| 23 | 0 | $\frac{742302}{13}$ |
| 24 | $\frac{492404611}{594636663660}$ | $\frac{3597475}{52}$ |
| 25 | 0 | $\frac{1082598}{13}$ |
| 26 | $\frac{22355809337}{182593098187864}$ | $\frac{5181141}{52}$ |

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| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 27 | 0 | $\frac{1541012}{13}$ |
| 28 | $\frac{295020099}{17383387729001}$ | $\frac{7294635}{52}$ |
| 29 | 0 | 165186 |
| 30 | $\frac{289881678351}{130291488164677840}$ | $\frac{10067229}{52}$ |
| 31 | 0 | $\frac{2936682}{13}$ |
| 32 | $\frac{420256422603}{1512642148079598536}$ | $\frac{13648995}{52}$ |

### S17.3.2 Parafermion spin-three global blocks

Table 60: Global-block and reduced power-series coefficients through weight 32 for the level-thirteen parafermion spin-three generator. All entries are copied as exact fractions.

| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 2 | $\frac{45}{4}$ | $\frac{45}{4}$ |
| 3 | 0 | $\frac{45}{4}$ |
| 4 | $\frac{21192}{473}$ | $\frac{207849}{3784}$ |
| 5 | 0 | $\frac{46641}{473}$ |
| 6 | $\frac{3619235}{79464}$ | $\frac{673685}{3784}$ |
| 7 | 0 | $\frac{138690}{473}$ |
| 8 | $\frac{9200541}{270556}$ | $\frac{1757259}{3784}$ |
| 9 | 0 | $\frac{336691}{473}$ |
| 10 | $\frac{15229635}{836264}$ | $\frac{4017657}{3784}$ |
| 11 | 0 | $\frac{731460}{473}$ |
| 12 | $\frac{5502941}{731731}$ | $\frac{8344119}{3784}$ |
| 13 | 0 | $\frac{1459221}{473}$ |
| 14 | $\frac{3445217631}{1405566800}$ | $\frac{1459407}{344}$ |
| 15 | 0 | $\frac{2716742}{473}$ |
| 16 | $\frac{1595203417}{2445686232}$ | $\frac{29007459}{3784}$ |
| 17 | 0 | $\frac{4776471}{473}$ |
| 18 | $\frac{108115736279}{735863828040}$ | $\frac{49733777}{3784}$ |
| 19 | 0 | $\frac{8001672}{473}$ |
| 20 | $\frac{805110996}{27863849629}$ | $\frac{81547215}{3784}$ |
| 21 | 0 | $\frac{12861561}{473}$ |
| 22 | $\frac{172584315463}{33946129948016}$ | $\frac{128670717}{3784}$ |
| 23 | 0 | $\frac{19946442}{473}$ |
| 24 | $\frac{1925254375}{2360250141912}$ | $\frac{196356475}{3784}$ |
| 25 | 0 | $\frac{2725713}{43}$ |
| 26 | $\frac{1609242967069}{13287159298901488}$ | $\frac{291007017}{3784}$ |
| 27 | 0 | $\frac{43848652}{473}$ |
| 28 | $\frac{39551299737}{2349240684519278}$ | $\frac{420296295}{3784}$ |
| 29 | 0 | $\frac{62588253}{473}$ |
| 30 | $\frac{20967212289587}{9481211369521941280}$ | $\frac{593290773}{3784}$ |
| 31 | 0 | $\frac{87427662}{473}$ |

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| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 32 | $\frac{2767264286301}{10006709594988113392}$ | $\frac{820570515}{3784}$ |

## S17.4 Current and stress Ward correlators as normalization controls

For a unit Cartan current and the unit total stress $\widehat{T} = T/2$,

$$G_{\widehat{H}}(x) = 1 + x^{-2} + (1-x)^{-2}, \tag{178}$$

$$G_{\widehat{T}}(x) = 1 + x^{-4} + (1-x)^{-4} + x^{-2} + (1-x)^{-2} + x^{-1} + (1-x)^{-1}. \tag{179}$$

These controls use the same global-block extraction and normalization as the interacting examples. The stress exchange in the first has $A_2 = 2$, whereas in the second it has $A_2 = 1$; their external weights differ. The coefficient $C_{\widehat{T}\widehat{T}\widehat{T}}$ is one at $c = 8$.

### S17.4.1 Unit Cartan current

Table 61: Global-block and reduced power-series coefficients through weight 32 for the unit Cartan current. All entries are copied as exact fractions.

| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 2 | 2 | 2 |
| 3 | 0 | 2 |
| 4 | $\frac{6}{5}$ | 3 |
| 5 | 0 | 4 |
| 6 | $\frac{5}{21}$ | 5 |
| 7 | 0 | 6 |
| 8 | $\frac{14}{429}$ | 7 |
| 9 | 0 | 8 |
| 10 | $\frac{9}{2431}$ | 9 |
| 11 | 0 | 10 |
| 12 | $\frac{11}{29393}$ | 11 |
| 13 | 0 | 12 |
| 14 | $\frac{13}{371450}$ | 13 |
| 15 | 0 | 14 |
| 16 | $\frac{2}{646323}$ | 15 |
| 17 | 0 | 16 |
| 18 | $\frac{17}{64822395}$ | 17 |
| 19 | 0 | 18 |
| 20 | $\frac{19}{883631595}$ | 19 |
| 21 | 0 | 20 |
| 22 | $\frac{7}{4077711170}$ | 21 |
| 23 | 0 | 22 |
| 24 | $\frac{23}{171529806825}$ | 23 |
| 25 | 0 | 24 |

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| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 26 | $\frac{25}{2430973200726}$ | 25 |
| 27 | 0 | 26 |
| 28 | $\frac{27}{34766775458002}$ | 27 |
| 29 | 0 | 28 |
| 30 | $\frac{29}{501121108325684}$ | 29 |
| 31 | 0 | 30 |
| 32 | $\frac{62}{14544636039226909}$ | 31 |

### S17.4.2 Unit total stress

Table 62: Global-block and reduced power-series coefficients through weight 32 for the unit total stress tensor. All entries are copied as exact fractions.

| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 2 | 1 | 1 |
| 3 | 0 | 1 |
| 4 | $\frac{31}{10}$ | 4 |
| 5 | 0 | 7 |
| 6 | $\frac{589}{126}$ | 14 |
| 7 | 0 | 25 |
| 8 | $\frac{4255}{1716}$ | 41 |
| 9 | 0 | 63 |
| 10 | $\frac{18569}{24310}$ | 92 |
| 11 | 0 | 129 |
| 12 | $\frac{60191}{352716}$ | 175 |
| 13 | 0 | 231 |
| 14 | $\frac{8439}{273700}$ | 298 |
| 15 | 0 | 377 |
| 16 | $\frac{557}{116280}$ | 469 |
| 17 | 0 | 575 |
| 18 | $\frac{155101}{233360622}$ | 696 |
| 19 | 0 | 833 |
| 20 | $\frac{1492639}{17672631900}$ | 987 |
| 21 | 0 | 1159 |
| 22 | $\frac{86851}{8681578620}$ | 1350 |
| 23 | 0 | 1561 |
| 24 | $\frac{4605151}{4116715363800}$ | 1793 |
| 25 | 0 | 2047 |
| 26 | $\frac{5653}{47415831372}$ | 2324 |
| 27 | 0 | 2625 |
| 28 | $\frac{383105}{31402248800776}$ | 2951 |
| 29 | 0 | 3303 |
| 30 | $\frac{489857}{406314412155960}$ | 3682 |

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| $n$ | $A_n$ | $[x^n]x^{2h}G(x)$ |
|---|---|---|
| 31 | 0 | 4089 |
| 32 | $\frac{26899071}{232714176627630544}$ | 4525 |

## S17.5 Nine lightest-module lattice blocks and their precise scope

The two $A_2$ lattice cosets with charges $+1$ and $-1$ restrict to the same lightest $U$-module $M_{c_1}$, of weight $1/3$ and top dimension three. In simple-root coordinates take

$$\beta_0 = (2/3, 1/3), \quad \beta_1 = (-1/3, 1/3), \quad \beta_2 = (-1/3, -2/3).$$

Their mutual Gram matrix is

$$((\beta_i, \beta_j)) = \begin{pmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{pmatrix},$$

and their right-Cartan charges are $(2, 0, -2)$. With paired lattice intertwiners normalized to unit leading vacuum coefficient,

$$\mathcal{B}_{ij}(x) = \langle V_{-\beta_i}(\infty) V_{\beta_i}(1) V_{-\beta_j}(x) V_{\beta_j}(0) \rangle = x^{-2/3}(1-x)^{-(\beta_i, \beta_j)}. \tag{180}$$

Positive real powers on $0 < x < 1$ specify the initial branch. Analytic continuation requires a path, since these blocks have fractional exponents.

Table 63: All nine selected lightest-module lattice blocks and their relative-weight-one decomposition. Each block also has the common factor $x^{-2/3}$.

| $i$ | $j$ | $(1-x)$ exponent | total weight 1 | current part | $M_b$ part |
|---|---|---|---|---|---|
| 0 | 0 | $\frac{-2}{3}$ | $\frac{2}{3}$ | $\frac{2}{39}$ | $\frac{8}{13}$ |
| 0 | 1 | $\frac{1}{3}$ | $\frac{-1}{3}$ | 0 | $\frac{-1}{3}$ |
| 0 | 2 | $\frac{1}{3}$ | $\frac{-1}{3}$ | $\frac{-2}{39}$ | $\frac{-11}{39}$ |
| 1 | 0 | $\frac{1}{3}$ | $\frac{-1}{3}$ | 0 | $\frac{-1}{3}$ |
| 1 | 1 | $\frac{-2}{3}$ | $\frac{2}{3}$ | 0 | $\frac{2}{3}$ |
| 1 | 2 | $\frac{1}{3}$ | $\frac{-1}{3}$ | 0 | $\frac{-1}{3}$ |
| 2 | 0 | $\frac{1}{3}$ | $\frac{-1}{3}$ | $\frac{-2}{39}$ | $\frac{-11}{39}$ |
| 2 | 1 | $\frac{1}{3}$ | $\frac{-1}{3}$ | 0 | $\frac{-1}{3}$ |
| 2 | 2 | $\frac{-2}{3}$ | $\frac{2}{3}$ | $\frac{2}{39}$ | $\frac{8}{13}$ |

The relative-weight-one coefficient is the Gram entry $(\beta_i, \beta_j)$. Its projection to the $U$ current is $q_i q_j/78$. The complement in the $M_b$ top is therefore

$$G_b = \begin{pmatrix} 8/13 & -1/3 & -11/39 \\ -1/3 & 2/3 & -1/3 \\ -11/39 & -1/3 & 8/13 \end{pmatrix}, \qquad \operatorname{spec}(G_b) = \left\{0, 1, \frac{35}{39}\right\}.$$

The zero eigenvector is $(1, 1, 1)$, reflecting $\beta_0 + \beta_1 + \beta_2 = 0$. The other eigenvectors can be chosen as $(1, -2, 1)$ and $(1, 0, -1)$, with eigenvalues 1 and $35/39$. Negative off-diagonal entries are therefore consistent with a positive semidefinite Gram matrix.

All six permutations of $(0,1,2)$ give charge-neutral chiral three-point functions of the three corresponding lattice intertwiners, with absolute coefficient squared one and coordinate factor $(z_{12}z_{13}z_{23})^{-1/3}$. Their phases depend on the intertwiner bases and cocycle convention. The data do not provide a complete $U$-channel basis or the sewing constants of a diagonal-$U$ bulk four-point function. In particular, taking $|\mathcal{B}_{ij}|^2$ by itself is not the claimed construction of that nonvacuum bulk correlator.

## S17.6 Exact cross-ratio samples and saved check boundaries

These rational-point evaluations are useful when comparing an independent implementation, plotting convention, or crossing transformation. They are exact values, not floating-point output.

Table 64: All fifty saved rational-point evaluations. Ordering names use $p = \widehat{u}$, $m = \widehat{u}^\dagger$ at $(\infty, 1, x, 0)$; spin-three and Ward names refer to the unit fields.

| function | $x$ | exact value |
|---|---|---|
| ppmm | $\frac{1}{10}$ | $\frac{121}{6561}$ |
| ppmm | $\frac{1}{4}$ | $\frac{17}{81}$ |
| ppmm | $\frac{1}{2}$ | $\frac{11}{3}$ |
| ppmm | $\frac{3}{4}$ | $129$ |
| ppmm | $\frac{9}{10}$ | $7641$ |
| pmpm | $\frac{1}{10}$ | $\frac{88000000}{6561}$ |
| pmpm | $\frac{1}{4}$ | $\frac{16384}{27}$ |
| pmpm | $\frac{1}{2}$ | $\frac{512}{3}$ |
| pmpm | $\frac{3}{4}$ | $\frac{16384}{27}$ |
| pmpm | $\frac{9}{10}$ | $\frac{88000000}{6561}$ |
| pmmp | $\frac{1}{10}$ | $7641$ |
| pmmp | $\frac{1}{4}$ | $129$ |
| pmmp | $\frac{1}{2}$ | $\frac{11}{3}$ |
| pmmp | $\frac{3}{4}$ | $\frac{17}{81}$ |
| pmmp | $\frac{9}{10}$ | $\frac{121}{6561}$ |
| mppm | $\frac{1}{10}$ | $7641$ |
| mppm | $\frac{1}{4}$ | $129$ |
| mppm | $\frac{1}{2}$ | $\frac{11}{3}$ |
| mppm | $\frac{3}{4}$ | $\frac{17}{81}$ |
| mppm | $\frac{9}{10}$ | $\frac{121}{6561}$ |
| mpmp | $\frac{1}{10}$ | $\frac{88000000}{6561}$ |
| mpmp | $\frac{1}{4}$ | $\frac{16384}{27}$ |
| mpmp | $\frac{1}{2}$ | $\frac{512}{3}$ |
| mpmp | $\frac{3}{4}$ | $\frac{16384}{27}$ |
| mpmp | $\frac{9}{10}$ | $\frac{88000000}{6561}$ |
| mmpp | $\frac{1}{10}$ | $\frac{121}{6561}$ |
| mmpp | $\frac{1}{4}$ | $\frac{17}{81}$ |
| mmpp | $\frac{1}{2}$ | $\frac{11}{3}$ |
| mmpp | $\frac{3}{4}$ | $129$ |
| mmpp | $\frac{9}{10}$ | $7641$ |
| Potts | $\frac{1}{10}$ | $\frac{8672450492983}{6908733}$ |
| Potts | $\frac{1}{4}$ | $\frac{124559725}{9477}$ |

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| function | $x$ | exact value |
|---|---|---|
| `Potts` | $\frac{1}{2}$ | $\frac{27975}{13}$ |
| `Potts` | $\frac{3}{4}$ | $\frac{124559725}{9477}$ |
| `Potts` | $\frac{9}{10}$ | $\frac{8672450492983}{6908733}$ |
| `PF13` | $\frac{1}{10}$ | $\frac{284160941571218}{251371593}$ |
| `PF13` | $\frac{1}{4}$ | $\frac{3195504725}{344817}$ |
| `PF13` | $\frac{1}{2}$ | $\frac{710850}{473}$ |
| `PF13` | $\frac{3}{4}$ | $\frac{3195504725}{344817}$ |
| `PF13` | $\frac{9}{10}$ | $\frac{284160941571218}{251371593}$ |
| `unit_Cartan` | $\frac{1}{10}$ | $\frac{8281}{81}$ |
| `unit_Cartan` | $\frac{1}{4}$ | $\frac{169}{9}$ |
| `unit_Cartan` | $\frac{1}{2}$ | $9$ |
| `unit_Cartan` | $\frac{3}{4}$ | $\frac{169}{9}$ |
| `unit_Cartan` | $\frac{9}{10}$ | $\frac{8281}{81}$ |
| `unit_total_stress` | $\frac{1}{10}$ | $\frac{66363661}{6561}$ |
| `unit_total_stress` | $\frac{1}{4}$ | $\frac{22945}{81}$ |
| `unit_total_stress` | $\frac{1}{2}$ | $45$ |
| `unit_total_stress` | $\frac{3}{4}$ | $\frac{22945}{81}$ |
| `unit_total_stress` | $\frac{9}{10}$ | $\frac{66363661}{6561}$ |

The saved report records 486 charge assignments, six $u$ orderings with fifteen surviving terms each, exact polynomial crossing identities for the $u$ and spin-three functions, and reconstruction of the global expansions through weight 32. Reflection-positive coefficients were checked through the same cutoff. Positivity at all weights follows from the positive VOA realization when its construction inputs hold; it is not established by extending the finite numerical-looking table by inspection. None of these checks is a replay of all lattice self-OPE certificates or a verification of arbitrary full-CFT sewing identities.

# S18 Complete modular, fusion and diagonal-defect reference tables

All rows in this section use the fixed sector order

$$(0, b, a, c_0, c_1, c_2, d_1, d_2, d_3, d_4, d_5, d_6). \tag{181}$$

The identity label 0 in a coefficient vector means the vacuum object. A zero coefficient means its absence. Confusing these two uses of zero is avoided by always printing a full twelve-entry vector. These tables concern the diagonal $U$ theory; the six-line Haagerup action after holomorphic extension is treated separately.

## S18.1 Every exact modular and line eigenvalue in compact form

For direct inspection of the cyclotomic data, put

$$r = \sqrt{13}, \quad d = \frac{3+r}{2}, \quad s = 3d+2 = d^2+1, \quad A = \frac{1}{6} - \frac{1}{2r}, \quad B = \frac{1}{6} + \frac{1}{2r}, \quad t_j = -2\cos\frac{2\pi j}{13}.$$

Indices on $t_j$ are taken modulo 13 with $t_j = t_{13-j}$. The exact $S$ matrix is completely specified by

$$S_{00} = S_{bb} = A, \qquad S_{0b} = S_{b0} = B, \tag{182}$$

$$S_{0x} = S_{bx} = \frac{1}{3} \qquad (x = a, c_0, c_1, c_2), \tag{183}$$

$$S_{0d_j} = r^{-1}, \qquad S_{bd_j} = -r^{-1}, \tag{184}$$

$$S_{xd_j} = 0 \qquad (x = a, c_0, c_1, c_2), \tag{185}$$

$$(S_{xy})_{x,y=a,c_0,c_1,c_2} = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 & -1 \\ -1 & 2 & -1 & -1 \\ -1 & -1 & -1 & 2 \\ -1 & -1 & 2 & -1 \end{pmatrix}, \tag{186}$$

$$S_{d_i d_j} = r^{-1} t_{ij}. \tag{187}$$

Symmetry supplies the transposed entries. Thus the twelve-term cyclotomic encodings in the JSON may be checked using a small real trigonometric matrix. The positive embedding is essential: $S_{00} = 1/(3s) > 0$.

The quantum dimensions are

$$(1, d^2, s, s, s, s, 3d, 3d, 3d, 3d, 3d, 3d), \qquad \mathcal{D} = \frac{1}{S_{00}} = 3s.$$

Here $\mathcal{D}$ is the total quantum dimension and the global dimension is $\mathcal{D}^2$. For a Verlinde line $x$ the eigenvalue on the diagonal sector $i$ is $\lambda_x(i) = S_{xi}/S_{0i}$. For the first six sector columns, the entire eigenvalue table is

| line $x$ | 0 | $b$ | $a$ | $c_0$ | $c_1$ | $c_2$ |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| $b$ | $d^2$ | $d^{-2}$ | 1 | 1 | 1 | 1 |
| $a$ | $s$ | $s/d^2$ | 2 | $-1$ | $-1$ | $-1$ |
| $c_0$ | $s$ | $s/d^2$ | $-1$ | 2 | $-1$ | $-1$ |
| $c_1$ | $s$ | $s/d^2$ | $-1$ | $-1$ | $-1$ | 2 |
| $c_2$ | $s$ | $s/d^2$ | $-1$ | $-1$ | 2 | $-1$ |
| $d_i$ $(1 \le i \le 6)$ | $3d$ | $-3/d$ | 0 | 0 | 0 | 0 |

For the final six sector columns, the line 0 has six entries 1, the line $b$ six entries $-1$, and each of $a, c_0, c_1, c_2$ six entries 0. The remaining $6 \times 6$ block is

$$(\lambda_{d_i}(d_j))_{i,j=1}^{6} = \begin{pmatrix} t_1 & t_2 & t_3 & t_4 & t_5 & t_6 \\ t_2 & t_4 & t_6 & t_5 & t_3 & t_1 \\ t_3 & t_6 & t_4 & t_1 & t_2 & t_5 \\ t_4 & t_5 & t_1 & t_3 & t_6 & t_2 \\ t_5 & t_3 & t_2 & t_6 & t_1 & t_4 \\ t_6 & t_1 & t_5 & t_2 & t_4 & t_3 \end{pmatrix}. \tag{188}$$

Equations and tables above specify all 144 exact line eigenvalues without decimal rounding. A line eigenvalue is not a Hilbert-space multiplicity, and may be negative or irrational.

## S18.2 All center fusion products

The full coefficient tensor is

$$N_{xy}{}^{z} = \sum_i \frac{S_{xi} S_{yi} S_{zi}}{S_{0i}}.$$

All labels are self-dual and $S$ is real in the stated basis. The table below prints every unordered product; commutativity gives the remaining ordered products. For example its $b, b$ vector is $(1,1,1,1,1,1,1,1,1,1,1,1)$, and its $a, a$ vector is $(1,2,2,1,1,1,1,1,1,1,1,1)$. All entries are exact integers copied from the saved Verlinde output.

Table 65: All 78 unordered center fusion products. Vectors are in the fixed order $(0, b, a, c_0, c_1, c_2, d_1, d_2, d_3, d_4, d_5, d_6)$; commutativity supplies the other 66 ordered products. Every entry of the 1728-entry tensor is therefore printed.

| $a$ | $b$ | coefficient vector of $a \otimes b$ |
|---|---|---|
| 0 | 0 | $(1,0,0,0,0,0,0,0,0,0,0,0)$ |
| 0 | $b$ | $(0,1,0,0,0,0,0,0,0,0,0,0)$ |
| 0 | $a$ | $(0,0,1,0,0,0,0,0,0,0,0,0)$ |
| 0 | $c_0$ | $(0,0,0,1,0,0,0,0,0,0,0,0)$ |
| 0 | $c_1$ | $(0,0,0,0,1,0,0,0,0,0,0,0)$ |
| 0 | $c_2$ | $(0,0,0,0,0,1,0,0,0,0,0,0)$ |
| 0 | $d_1$ | $(0,0,0,0,0,0,1,0,0,0,0,0)$ |
| 0 | $d_2$ | $(0,0,0,0,0,0,0,1,0,0,0,0)$ |
| 0 | $d_3$ | $(0,0,0,0,0,0,0,0,1,0,0,0)$ |
| 0 | $d_4$ | $(0,0,0,0,0,0,0,0,0,1,0,0)$ |
| 0 | $d_5$ | $(0,0,0,0,0,0,0,0,0,0,1,0)$ |
| 0 | $d_6$ | $(0,0,0,0,0,0,0,0,0,0,0,1)$ |
| $b$ | $b$ | $(1,1,1,1,1,1,1,1,1,1,1,1)$ |
| $b$ | $a$ | $(0,1,2,1,1,1,1,1,1,1,1,1)$ |
| $b$ | $c_0$ | $(0,1,1,2,1,1,1,1,1,1,1,1)$ |
| $b$ | $c_1$ | $(0,1,1,1,2,1,1,1,1,1,1,1)$ |
| $b$ | $c_2$ | $(0,1,1,1,1,2,1,1,1,1,1,1)$ |
| $b$ | $d_1$ | $(0,1,1,1,1,1,0,1,1,1,1,1)$ |
| $b$ | $d_2$ | $(0,1,1,1,1,1,1,0,1,1,1,1)$ |
| $b$ | $d_3$ | $(0,1,1,1,1,1,1,1,0,1,1,1)$ |

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| $a$ | $b$ | coefficient vector of $a \otimes b$ |
|---|---|---|
| $b$ | $d_4$ | $(0,1,1,1,1,1,1,1,1,0,1,1)$ |
| $b$ | $d_5$ | $(0,1,1,1,1,1,1,1,1,1,0,1)$ |
| $b$ | $d_6$ | $(0,1,1,1,1,1,1,1,1,1,1,0)$ |
| $a$ | $a$ | $(1,2,2,1,1,1,1,1,1,1,1,1)$ |
| $a$ | $c_0$ | $(0,1,1,1,2,2,1,1,1,1,1,1)$ |
| $a$ | $c_1$ | $(0,1,1,2,1,2,1,1,1,1,1,1)$ |
| $a$ | $c_2$ | $(0,1,1,2,2,1,1,1,1,1,1,1)$ |
| $a$ | $d_1$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $a$ | $d_2$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $a$ | $d_3$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $a$ | $d_4$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $a$ | $d_5$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $a$ | $d_6$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_0$ | $c_0$ | $(1,2,1,2,1,1,1,1,1,1,1,1)$ |
| $c_0$ | $c_1$ | $(0,1,2,1,1,2,1,1,1,1,1,1)$ |
| $c_0$ | $c_2$ | $(0,1,2,1,2,1,1,1,1,1,1,1)$ |
| $c_0$ | $d_1$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_0$ | $d_2$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_0$ | $d_3$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_0$ | $d_4$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_0$ | $d_5$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_0$ | $d_6$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_1$ | $c_1$ | $(1,2,1,1,2,1,1,1,1,1,1,1)$ |
| $c_1$ | $c_2$ | $(0,1,2,2,1,1,1,1,1,1,1,1)$ |
| $c_1$ | $d_1$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_1$ | $d_2$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_1$ | $d_3$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_1$ | $d_4$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_1$ | $d_5$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_1$ | $d_6$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_2$ | $c_2$ | $(1,2,1,1,1,2,1,1,1,1,1,1)$ |
| $c_2$ | $d_1$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_2$ | $d_2$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_2$ | $d_3$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_2$ | $d_4$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_2$ | $d_5$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $c_2$ | $d_6$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ |
| $d_1$ | $d_1$ | $(1,0,1,1,1,1,1,0,1,1,1,1)$ |
| $d_1$ | $d_2$ | $(0,1,1,1,1,1,0,1,0,1,1,1)$ |
| $d_1$ | $d_3$ | $(0,1,1,1,1,1,1,0,1,0,1,1)$ |
| $d_1$ | $d_4$ | $(0,1,1,1,1,1,1,1,0,1,0,1)$ |
| $d_1$ | $d_5$ | $(0,1,1,1,1,1,1,1,1,0,1,0)$ |
| $d_1$ | $d_6$ | $(0,1,1,1,1,1,1,1,1,1,0,0)$ |
| $d_2$ | $d_2$ | $(1,0,1,1,1,1,1,1,1,0,1,1)$ |
| $d_2$ | $d_3$ | $(0,1,1,1,1,1,0,1,1,1,0,1)$ |
| $d_2$ | $d_4$ | $(0,1,1,1,1,1,1,0,1,1,1,0)$ |
| $d_2$ | $d_5$ | $(0,1,1,1,1,1,1,1,0,1,1,0)$ |

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| $a$ | $b$ | coefficient vector of $a \otimes b$ |
|---|---|---|
| $d_2$ | $d_6$ | $(0,1,1,1,1,1,1,1,1,0,0,1)$ |
| $d_3$ | $d_3$ | $(1,0,1,1,1,1,1,1,1,1,1,0)$ |
| $d_3$ | $d_4$ | $(0,1,1,1,1,1,0,1,1,1,1,0)$ |
| $d_3$ | $d_5$ | $(0,1,1,1,1,1,1,0,1,1,0,1)$ |
| $d_3$ | $d_6$ | $(0,1,1,1,1,1,1,1,0,0,1,1)$ |
| $d_4$ | $d_4$ | $(1,0,1,1,1,1,1,1,1,1,0,1)$ |
| $d_4$ | $d_5$ | $(0,1,1,1,1,1,0,1,1,0,1,1)$ |
| $d_4$ | $d_6$ | $(0,1,1,1,1,1,1,0,0,1,1,1)$ |
| $d_5$ | $d_5$ | $(1,0,1,1,1,1,1,1,0,1,1,1)$ |
| $d_5$ | $d_6$ | $(0,1,1,1,1,1,0,0,1,1,1,1)$ |
| $d_6$ | $d_6$ | $(1,0,1,1,1,1,0,1,1,1,1,1)$ |

One way to verify an individual row independently is to check the dimension identity

$$\operatorname{qdim}(x)\operatorname{qdim}(y) = \sum_z N_{xy}{}^z \operatorname{qdim}(z).$$

It is a necessary consistency check, not a replacement for the Verlinde calculation: distinct fusion vectors can have the same weighted dimension. The saved exporter also checks exact associativity and all Verlinde coefficients, rather than relying on this single scalar test.

### S18.3 All twelve spatially twisted Hilbert spaces

For a line $x$, a temporal insertion has partition function $\sum_i \lambda_x(i)|\chi_i|^2$. Its modular transform is the spatially twisted Hilbert-space trace

$$Z_{[x]} = \sum_{i,j} (N_x)_{ij} \chi_i \overline{\chi_j}, \qquad (N_x)_{ij} = N_{xi}{}^j. \tag{189}$$

The matrices below have both row and column order (181). They print all twelve complete bilinear forms, so a reader need not transpose indices in the fusion table while verifying a twisted spectrum. Symmetry of each matrix follows from the real self-dual modular data.

If $h_i$ and $t_i$ denote a module's lowest weight and top dimension, the ground dimension is the minimum of $h_i + h_j$ over positive matrix entries. Its multiplicity is the sum of $(N_x)_{ij} t_i t_j$ over pairs attaining that minimum. Scalars require exact equality of left and right weights, not merely integer spin. A pair with $h_i - h_j \notin \mathbb{Z}$ contributes no scalar at any grade. The vacuum module has its first positive weight at one with six states, which must be used when excluding the identity. These elementary rules explain why the scalar gap can differ from the ground-state gap of a defect space.

**Line** 0

Ground dimension 0 with multiplicity 1; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

**Line** $b$

Ground dimension $\frac{2}{3}$ with multiplicity 18; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 18.

$$N_b = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

**Line** $a$

Ground dimension $\frac{2}{3}$ with multiplicity 9; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_a = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

**Line $c_0$**

Ground dimension $\frac{2}{3}$ with multiplicity 9; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{c_0} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

**Line $c_1$**

Ground dimension $\frac{1}{3}$ with multiplicity 6; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 18.

$$N_{c_1} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

**Line $c_2$**

Ground dimension $\frac{2}{3}$ with multiplicity 63; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{c_2} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

**Line $d_1$**

Ground dimension $\frac{6}{13}$ with multiplicity 14; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{d_1} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

**Line $d_2$**

Ground dimension $\frac{2}{3}$ with multiplicity 9; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{d_2} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

**Line $d_3$**

Ground dimension $\frac{2}{3}$ with multiplicity 9; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{d_3} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

**Line $d_4$**

Ground dimension $\frac{5}{13}$ with multiplicity 10; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{d_4} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

**Line $d_5$**

Ground dimension $\frac{7}{13}$ with multiplicity 26; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{d_5} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

**Line $d_6$**

Ground dimension $\frac{8}{13}$ with multiplicity 28; first nonidentity scalar dimension $\frac{2}{3}$ with multiplicity 9.

$$N_{d_6} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

## S18.4 All diagonal Cardy annuli and their positive gaps

The elementary boundary labeled $x$ has closed-channel state

$$|x\rangle = \sum_i \frac{S_{xi}}{\sqrt{S_{0i}}} |i\rangle\!\rangle.$$

The open annulus between $x$ and $y$ is $Z_{xy}^{\mathrm{open}} = \sum_i N_{xy}{}^i \chi_i$. Accordingly Table 65 already prints every full annulus character vector. The next table adds all ground-state and first-positive-weight data. A self-annulus has one vacuum, whereas distinct elementary boundaries have none. The first positive weight in a vacuum character is one, not an absent state obtained by deleting the entire vacuum representation.

Table 66: Ground states and first positive boundary weights for all 78 distinct Cardy annuli. The complete character coefficient vector is the corresponding row of Table 65. Reversing the boundaries gives the same row.

| $a$ | $b$ | $h_{\min}$ | vacua | $h_{>0}$ | multiplicity |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 6 |
| 0 | $b$ | 1 | 0 | 1 | 80 |
| 0 | $a$ | 1 | 0 | 1 | 81 |
| 0 | $c_0$ | 1 | 0 | 1 | 81 |
| 0 | $c_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| 0 | $c_2$ | $\frac{2}{3}$ | 0 | $\frac{2}{3}$ | 27 |
| 0 | $d_1$ | $\frac{6}{13}$ | 0 | $\frac{6}{13}$ | 7 |
| 0 | $d_2$ | $\frac{11}{13}$ | 0 | $\frac{11}{13}$ | 42 |
| 0 | $d_3$ | $\frac{15}{13}$ | 0 | $\frac{15}{13}$ | 119 |
| 0 | $d_4$ | $\frac{5}{13}$ | 0 | $\frac{5}{13}$ | 5 |
| 0 | $d_5$ | $\frac{7}{13}$ | 0 | $\frac{7}{13}$ | 13 |
| 0 | $d_6$ | $\frac{8}{13}$ | 0 | $\frac{8}{13}$ | 14 |
| $b$ | $b$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $b$ | $a$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $c_0$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $c_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 6 |
| $b$ | $c_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $d_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $d_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $d_3$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $b$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $a$ | $a$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $a$ | $c_0$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 6 |
| $a$ | $c_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $a$ | $c_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 6 |
| $a$ | $d_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $a$ | $d_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $a$ | $d_3$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $a$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |

*Continued on next page.*

*Continued from preceding page.*

| $a$ | $b$ | $h_{\min}$ | vacua | $h_{>0}$ | multiplicity |
|---|---|---|---|---|---|
| $a$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $a$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_0$ | $c_0$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $c_0$ | $c_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_0$ | $c_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 6 |
| $c_0$ | $d_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_0$ | $d_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_0$ | $d_3$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_0$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_0$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_0$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_1$ | $c_1$ | 0 | 1 | $\frac{1}{3}$ | 6 |
| $c_1$ | $c_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_1$ | $d_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_1$ | $d_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_1$ | $d_3$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_1$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_1$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_1$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_2$ | $c_2$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $c_2$ | $d_1$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_2$ | $d_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_2$ | $d_3$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_2$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_2$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $c_2$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_1$ | $d_1$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $d_1$ | $d_2$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_1$ | $d_3$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_1$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_1$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_1$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_2$ | $d_2$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $d_2$ | $d_3$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_2$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_2$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_2$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_3$ | $d_3$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $d_3$ | $d_4$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_3$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_3$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_4$ | $d_4$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $d_4$ | $d_5$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_4$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_5$ | $d_5$ | 0 | 1 | $\frac{1}{3}$ | 3 |
| $d_5$ | $d_6$ | $\frac{1}{3}$ | 0 | $\frac{1}{3}$ | 3 |
| $d_6$ | $d_6$ | 0 | 1 | $\frac{1}{3}$ | 3 |

Boundary overlaps in this normalization are

$$g_x = \frac{\mathrm{qdim}(x)}{\sqrt{3s}}, \qquad g_0^2 = \frac{1}{3s}, \quad g_b^2 = \frac{d^4}{3s}, \quad g_a^2 = g_{c_j}^2 = \frac{s}{3}, \quad g_{d_j}^2 = \frac{3d^2}{s}. \tag{190}$$

Their positive values, shown with the saved plotting precision, are

| boundary type | $g_x$ |
|---|---|
| 0 | 0.1673069562521514 |
| $b$ | 1.825038973650249 |
| $a, c_j$ | 1.992345929902401 |
| $d_j$ | 1.657732017398097 |

These decimals are the saved plotting values; the displayed algebraic expressions define the observables. The boundary entropy is $\log g_x$, not $g_x$ itself.

The vacuum Cardy boundary has six positive-weight-one currents. Every other self-annulus contains weight-1/3 boundary fields: six for $c_1$ and three for each other nonvacuum boundary. This identifies relevant boundary operators. Neither their degeneracy nor their conformal weight specifies a boundary renormalization group flow or its endpoint.

## S19 The six Haagerup lines and all extension boundary sectors

The selected holomorphic extension is $H' = V_{E_8}$ with $H'|_U = M_0 \oplus M_b \oplus 2M_a$. Its six right-module labels are ordered as

$$(1, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho).$$

They have dimensions $(1, 1, 1, d, d, d)$ and fusion rules $\alpha^3 = 1$, $\rho\alpha = \alpha^2\rho$, $\rho^2 = 1 + \rho + \alpha\rho + \alpha^2\rho$. Unlike the center fusion table, the ordered product table need not be symmetric.

Table 67: All 36 ordered Haagerup products, in basis $(1, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho)$. Order matters.

| $x$ | $y$ | coefficient vector of $x \otimes y$ |
|---|---|---|
| $1$ | $1$ | $(1,0,0,0,0,0)$ |
| $1$ | $\alpha$ | $(0,1,0,0,0,0)$ |
| $1$ | $\alpha^2$ | $(0,0,1,0,0,0)$ |
| $1$ | $\rho$ | $(0,0,0,1,0,0)$ |
| $1$ | $\alpha\rho$ | $(0,0,0,0,1,0)$ |
| $1$ | $\alpha^2\rho$ | $(0,0,0,0,0,1)$ |
| $\alpha$ | $1$ | $(0,1,0,0,0,0)$ |
| $\alpha$ | $\alpha$ | $(0,0,1,0,0,0)$ |
| $\alpha$ | $\alpha^2$ | $(1,0,0,0,0,0)$ |
| $\alpha$ | $\rho$ | $(0,0,0,0,1,0)$ |
| $\alpha$ | $\alpha\rho$ | $(0,0,0,0,0,1)$ |
| $\alpha$ | $\alpha^2\rho$ | $(0,0,0,1,0,0)$ |
| $\alpha^2$ | $1$ | $(0,0,1,0,0,0)$ |
| $\alpha^2$ | $\alpha$ | $(1,0,0,0,0,0)$ |
| $\alpha^2$ | $\alpha^2$ | $(0,1,0,0,0,0)$ |
| $\alpha^2$ | $\rho$ | $(0,0,0,0,0,1)$ |
| $\alpha^2$ | $\alpha\rho$ | $(0,0,0,1,0,0)$ |
| $\alpha^2$ | $\alpha^2\rho$ | $(0,0,0,0,1,0)$ |
| $\rho$ | $1$ | $(0,0,0,1,0,0)$ |
| $\rho$ | $\alpha$ | $(0,0,0,0,0,1)$ |
| $\rho$ | $\alpha^2$ | $(0,0,0,0,1,0)$ |
| $\rho$ | $\rho$ | $(1,0,0,1,1,1)$ |
| $\rho$ | $\alpha\rho$ | $(0,0,1,1,1,1)$ |
| $\rho$ | $\alpha^2\rho$ | $(0,1,0,1,1,1)$ |
| $\alpha\rho$ | $1$ | $(0,0,0,0,1,0)$ |
| $\alpha\rho$ | $\alpha$ | $(0,0,0,1,0,0)$ |
| $\alpha\rho$ | $\alpha^2$ | $(0,0,0,0,0,1)$ |
| $\alpha\rho$ | $\rho$ | $(0,1,0,1,1,1)$ |
| $\alpha\rho$ | $\alpha\rho$ | $(1,0,0,1,1,1)$ |
| $\alpha\rho$ | $\alpha^2\rho$ | $(0,0,1,1,1,1)$ |
| $\alpha^2\rho$ | $1$ | $(0,0,0,0,0,1)$ |
| $\alpha^2\rho$ | $\alpha$ | $(0,0,0,0,1,0)$ |
| $\alpha^2\rho$ | $\alpha^2$ | $(0,0,0,1,0,0)$ |
| $\alpha^2\rho$ | $\rho$ | $(0,0,1,1,1,1)$ |
| $\alpha^2\rho$ | $\alpha\rho$ | $(0,1,0,1,1,1)$ |
| $\alpha^2\rho$ | $\alpha^2\rho$ | $(1,0,0,1,1,1)$ |

### S19.1 Forgetful matrix, tube factors and first-factor traces

The entire forgetful multiplicity matrix, with the six rows above and twelve columns in (181), is

$$n = \begin{pmatrix} 1 & 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}. \tag{191}$$

Restriction of a right module $x$ is $Rx = \bigoplus_\mu n_{x\mu} M_\mu$. The selected action is on the first chiral factor, with the antiholomorphic $H'$ theory a spectator. Its Hilbert space and tube decomposition are

$$\mathcal{H}_x = Rx \otimes \overline{H'} = \bigoplus_\mu W_x^\mu \otimes \mathcal{V}_\mu, \tag{192}$$

$$\dim W_x^\mu = n_{x\mu}, \qquad \mathcal{V}_\mu = M_\mu \otimes \overline{H'}, \tag{193}$$

$$Z_x = \sum_\mu n_{x\mu} \chi_\mu \overline{\chi_{E_8}}. \tag{194}$$

The twelve functions $\chi_\mu \overline{\chi_{E_8}}$ count the $\mathcal{V}_\mu$ factors; they exclude $W_x^\mu$. For example $\dim W_1^a = 2$, and omitting that factor would incorrectly halve the corresponding physical untwisted charged multiplicity. Conversely inserting it into the bootstrap multiplicity-space function would count that factor twice.

Table 68: All twelve tube multiplicity spaces $\mathcal{V}_\mu = M_\mu \otimes \overline{V_{E_8}}$. These rows exclude the finite-dimensional $W_x^\mu$ factor. "None means there is no scalar at any grade.

| $\mu$ | $\Delta_{\min}$ | mult. | $\Delta_{\text{scalar}}$ | mult. |
|---|---|---|---|---|
| 0 | 0 | 1 | 2 | 1488 |
| $b$ | 1 | 80 | 2 | 19840 |
| $a$ | 1 | 81 | 2 | 20088 |
| $c_0$ | 1 | 81 | 2 | 20088 |
| $c_1$ | $\frac{1}{3}$ | 3 | none | 0 |
| $c_2$ | $\frac{2}{3}$ | 27 | none | 0 |
| $d_1$ | $\frac{6}{13}$ | 7 | none | 0 |
| $d_2$ | $\frac{11}{13}$ | 42 | none | 0 |
| $d_3$ | $\frac{15}{13}$ | 119 | none | 0 |
| $d_4$ | $\frac{5}{13}$ | 5 | none | 0 |
| $d_5$ | $\frac{7}{13}$ | 13 | none | 0 |
| $d_6$ | $\frac{8}{13}$ | 14 | none | 0 |

All antiholomorphic weights of $H'$ are integers. A sector with noninteger $h_\mu$ therefore has no scalar at any grade, because adding integral grades never makes its left weight equal an integer right weight. In the four integer-weight sectors, the first nonidentity scalars occur at $(1, 1)$ and have multiplicities $6 \cdot 248$, $80 \cdot 248$, $81 \cdot 248$, $81 \cdot 248$. The resulting physical six twisted spaces, including the forgetful multiplicities, are the following.

Table 69: All six first-factor Haagerup twisted Hilbert spaces, with the full antiholomorphic $E_8$ theory as spectator. Multiplicities here include the forgetful matrix.

| $x$ | $\Delta_{\min}$ | mult. | $\Delta_{\text{scalar}}$ | mult. |
|---|---|---|---|---|
| $1$ | $0$ | $1$ | $2$ | $61504$ |
| $\alpha$ | $\frac{1}{3}$ | $3$ | $2$ | $20088$ |
| $\alpha^2$ | $\frac{1}{3}$ | $3$ | $2$ | $20088$ |
| $\rho$ | $\frac{1}{3}$ | $3$ | $2$ | $60016$ |
| $\alpha\rho$ | $\frac{1}{3}$ | $3$ | $2$ | $60016$ |
| $\alpha^2\rho$ | $\frac{1}{3}$ | $3$ | $2$ | $60016$ |

For orientation, the untwisted invariant scalar multiplicity is 1488, while the entire untwisted scalar space at dimension two has multiplicity $248^2 = 61504$. The untwisted $\mathbb{Z}_3$-charged space contains two copies of $\mathcal{V}_a$, so its scalar multiplicity is $2 \cdot 81 \cdot 248 = 40176$. In contrast, the bootstrap union $\mathcal{V}_a \cup \mathcal{V}_{c_0}$ concerns two tube multiplicity-space labels and is not that entire charged subspace. Their identical scalar gap two does not identify the underlying spaces.

On the three untwisted multiplicity factors, the line matrices may be chosen as

| | $\alpha$ | $\rho$ |
|---|---|---|
| $M_0$ | $1$ | $d$ |
| $M_b$ | $1$ | $-d^{-1}$ |
| $\mathbb{C}^2 \otimes M_a$ | $\begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}$ | $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ |

, $\omega = e^{2\pi i/3}$.

Taking the traces on these multiplicity factors gives

$$Z^{[\alpha]} = (\chi_0 + \chi_b - \chi_a)\overline{\chi_{E_8}}, \qquad Z^{[\rho]} = (d\chi_0 - d^{-1}\chi_b)\overline{\chi_{E_8}}.$$

The antiholomorphic character is unchanged in a first-factor insertion. Replacing either expression with the absolute square of its holomorphic factor would describe a different paired left/right insertion.

## S19.2 All $U$-preserving annuli of the full extension

The open Hilbert space between boundary modules $x, y$ is $R(x^* \otimes_A y)$. Thus

$$Z_{xy}^{E_8,\text{open}} = \sum_{z,\mu} N_{x^*y}^{\ z} n_{z\mu} \chi_\mu. \tag{195}$$

The duals are $1^* = 1$, $\alpha^* = \alpha^2$, $(\alpha^2)^* = \alpha$ and the three noninvertible objects are self-dual. The dual on $x$ cannot be dropped just because the center labels in the earlier Cardy table were self-dual. For instance the $\alpha, \alpha$ annulus has a vacuum; using $\alpha \otimes \alpha$ instead of $\alpha^* \otimes \alpha$ would incorrectly remove it.

Table 70: All 21 distinct $U$-preserving annuli of the selected full $E_8$ theory. The vector order is the twelve-sector center order. Equal boundaries have one identity; distinct boundaries have none. Reversal gives the other 15 ordered annuli.

| $x$ | $y$ | $U$-character coefficient vector | $h_{>0}$ | mult. |
|---|---|---|---|---|
| 1 | 1 | $(1,1,2,0,0,0,0,0,0,0,0,0)$ | 1 | 248 |
| 1 | $\alpha$ | $(0,0,0,1,1,1,0,0,0,0,0,0)$ | $\frac{1}{3}$ | 3 |
| 1 | $\alpha^2$ | $(0,0,0,1,1,1,0,0,0,0,0,0)$ | $\frac{1}{3}$ | 3 |
| 1 | $\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| 1 | $\alpha\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| 1 | $\alpha^2\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| $\alpha$ | $\alpha$ | $(1,1,2,0,0,0,0,0,0,0,0,0)$ | 1 | 248 |
| $\alpha$ | $\alpha^2$ | $(0,0,0,1,1,1,0,0,0,0,0,0)$ | $\frac{1}{3}$ | 3 |
| $\alpha$ | $\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| $\alpha$ | $\alpha\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| $\alpha$ | $\alpha^2\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| $\alpha^2$ | $\alpha^2$ | $(1,1,2,0,0,0,0,0,0,0,0,0)$ | 1 | 248 |
| $\alpha^2$ | $\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| $\alpha^2$ | $\alpha\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| $\alpha^2$ | $\alpha^2\rho$ | $(0,1,1,1,1,1,1,1,1,1,1,1)$ | $\frac{1}{3}$ | 3 |
| $\rho$ | $\rho$ | $(1,4,5,3,3,3,3,3,3,3,3,3)$ | $\frac{1}{3}$ | 9 |
| $\rho$ | $\alpha\rho$ | $(0,3,3,4,4,4,3,3,3,3,3,3)$ | $\frac{1}{3}$ | 12 |
| $\rho$ | $\alpha^2\rho$ | $(0,3,3,4,4,4,3,3,3,3,3,3)$ | $\frac{1}{3}$ | 12 |
| $\alpha\rho$ | $\alpha\rho$ | $(1,4,5,3,3,3,3,3,3,3,3,3)$ | $\frac{1}{3}$ | 9 |
| $\alpha\rho$ | $\alpha^2\rho$ | $(0,3,3,4,4,4,3,3,3,3,3,3)$ | $\frac{1}{3}$ | 12 |
| $\alpha^2\rho$ | $\alpha^2\rho$ | $(1,4,5,3,3,3,3,3,3,3,3,3)$ | $\frac{1}{3}$ | 9 |

The boundary overlaps are exactly

$$(g_1, g_\alpha, g_{\alpha^2}, g_\rho, g_{\alpha\rho}, g_{\alpha^2\rho}) = (1,1,1,d,d,d).$$

This normalization is fixed by the vacuum coefficient in the closed channel, rather than by an arbitrary proportionality constant. Restriction multiplies categorical dimensions by $\mathrm{qdim}(A) = \mathcal{D}$, so multiplying by $S_{00} = 1/\mathcal{D}$ gives $\mathrm{qdim}(x)\,\mathrm{qdim}(y)$ for that coefficient. Positive boundary overlaps then give $g_x = \mathrm{qdim}(x)$.

The three invertible self-annuli equal $\chi_{E_8}$ and have 248 weight-one currents as their first positive boundary states. Each noninvertible self-annulus instead has nine weight-1/3 states. Distinct noninvertible boundaries have twelve weight-1/3 ground fields between them. All other distinct pairs have three such ground fields. These are consequences of the printed vectors and the $M_{c_1}$ top dimension three.

## S19.3 Provenance and what has actually been incorporated

The modular, fusion, defect, and boundary entries above are saved exact consequences of the positive Haagerup modular data and the supplied $\gamma = 0$ character tops. The source exporter represents numbers in $\mathbb{Q}(\zeta_{13})$ by twelve rational coefficients and uses the positive Gauss sum

$$\sqrt{13} = \sum_{j=1}^{12} \left(\frac{j}{13}\right) \zeta_{13}^{j}.$$

It records checks of $S^2 = 1$, $(S\Theta)^3 = 1$, nonnegative integrality of the complete Verlinde tensor, associativity, dimension multiplicativity, self-duality, and the forgetful matrix's compatibility

with the two fusion rings. These checks do not reconstruct associators, braidings, or generic nonvacuum OPE coefficients from modular data.

For reproducibility, the new transcription utility is `physics_tables/build_physics_tables.py`; its input hashes and coverage counts are recorded in `physics_tables/transcription_manifest.json` and `physics_sources.md`. It reads saved finite outputs and formats exact entries. It does not import the lattice engine or rerun a physics exporter. Its coverage checks include all 113 component labels, the 318 distinct nonzero multiplet triples, all 18,365 sparse component entries when identifying that support, 198 global-block rows, all center fusion products, and every boundary pair up to the stated reversal symmetry. Original artifacts remain the provenance for the finite exact calculations; the printed formulas and tables make their contents available for inspection within this expanded draft.

# Part V

# Complete spectral and coordinate tables

## S20 Character and recurrence tables through grade twenty-four

These tables transcribe the completed 27 September release. They add no new character-identification assumption. Every coefficient is an exact integer, and every grade is measured from the actual lowest weight $h_i$. The column $v_i(n)$ includes the vacuum identity once. The formulas in Section 2 reconstruct every displayed full-spectrum entry from these chiral tables. Equal $a$ and $c_0$ characters still label two different modules.

The source is `c8_physics_20260927/data/characters.json`. All thirteen rows of modules below (twelve $U$ modules and the $E_8$ vacuum module) retain all twenty-five supplied grades.

### S20.1 Module 0

$h = 0$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 6 | 6 |
| 2 | 120 | 113 |
| 3 | 956 | 830 |
| 4 | 6010 | 4934 |
| 5 | 29512 | 22547 |
| 6 | 126979 | 91463 |
| 7 | 484920 | 328550 |
| 8 | 1698317 | 1087380 |
| 9 | 5519921 | 3342814 |
| 10 | 16883159 | 9695389 |
| 11 | 48970422 | 26700331 |
| 12 | 135729020 | 70389870 |
| 13 | 361328212 | 178454060 |
| 14 | 928246226 | 437193715 |
| 15 | 2309453500 | 1038459581 |
| 16 | 5582384430 | 2399169031 |
| 17 | 13143600484 | 5404345101 |
| 18 | 30212253455 | 11896439240 |
| 19 | 67929988152 | 25637618529 |
| 20 | 149654742594 | 54181554870 |
| 21 | 323527951168 | 112448204178 |
| 22 | 687223604695 | 229476596839 |
| 23 | 1435999900866 | 460992314155 |
| 24 | 2954844830174 | 912553664428 |

### S20.2 Module b

$h = 1$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 80 | 80 |
| 1 | 1250 | 1170 |
| 2 | 10630 | 9300 |
| 3 | 65042 | 53162 |
| 4 | 322968 | 247296 |
| 5 | 1383275 | 995345 |
| 6 | 5292868 | 3587875 |
| 7 | 18520078 | 11854645 |
| 8 | 60223427 | 36476773 |
| 9 | 184149027 | 105739120 |
| 10 | 534214988 | 291290851 |
| 11 | 1480529982 | 767781803 |
| 12 | 3941558372 | 1946716675 |
| 13 | 10125500876 | 4768927567 |
| 14 | 25192449100 | 11328048327 |
| 15 | 60894204778 | 26170628095 |
| 16 | 143375095718 | 58952796631 |
| 17 | 329564542011 | 129769620970 |
| 18 | 741003385528 | 279664420747 |
| 19 | 1632481593304 | 591028830191 |
| 20 | 3529150491568 | 1226623611688 |
| 21 | 7496457100149 | 2503203118200 |
| 22 | 15664360124354 | 5028658544440 |
| 23 | 32232408099627 | 9954428604211 |
| 24 | 65373671940778 | 19434824155343 |

### S20.3 Module a

$h = 1$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 81 | 81 |
| 1 | 1377 | 1296 |
| 2 | 11583 | 10125 |
| 3 | 71037 | 58077 |
| 4 | 352512 | 269892 |
| 5 | 1510245 | 1086777 |
| 6 | 5777730 | 3916350 |
| 7 | 20218491 | 12942180 |
| 8 | 65743326 | 39819519 |
| 9 | 201032037 | 115434315 |
| 10 | 583185663 | 317991582 |
| 11 | 1616258934 | 838171476 |
| 12 | 4302886212 | 2125170270 |

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 13 | 11053747701 | 5206122234 |
| 14 | 27501902460 | 12366507492 |
| 15 | 66476588382 | 28569796092 |
| 16 | 156518697519 | 64357143759 |
| 17 | 359776795149 | 141666059295 |
| 18 | 808933371912 | 305302037112 |
| 19 | 1782136338633 | 645210389241 |
| 20 | 3852678442104 | 1339071814071 |
| 21 | 8183680701318 | 2732679710793 |
| 22 | 17100360030654 | 5489650866690 |
| 23 | 35187252928542 | 10866982265118 |
| 24 | 71366677559406 | 21216473827326 |

### S20.4 Module c0

$h = 1$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 81 | 81 |
| 1 | 1377 | 1296 |
| 2 | 11583 | 10125 |
| 3 | 71037 | 58077 |
| 4 | 352512 | 269892 |
| 5 | 1510245 | 1086777 |
| 6 | 5777730 | 3916350 |
| 7 | 20218491 | 12942180 |
| 8 | 65743326 | 39819519 |
| 9 | 201032037 | 115434315 |
| 10 | 583185663 | 317991582 |
| 11 | 1616258934 | 838171476 |
| 12 | 4302886212 | 2125170270 |
| 13 | 11053747701 | 5206122234 |
| 14 | 27501902460 | 12366507492 |
| 15 | 66476588382 | 28569796092 |
| 16 | 156518697519 | 64357143759 |
| 17 | 359776795149 | 141666059295 |
| 18 | 808933371912 | 305302037112 |
| 19 | 1782136338633 | 645210389241 |
| 20 | 3852678442104 | 1339071814071 |
| 21 | 8183680701318 | 2732679710793 |
| 22 | 17100360030654 | 5489650866690 |
| 23 | 35187252928542 | 10866982265118 |
| 24 | 71366677559406 | 21216473827326 |

### S20.5 Module c1

$h = 1/3$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 3 | 3 |
| 1 | 243 | 240 |
| 2 | 2916 | 2670 |
| 3 | 21870 | 18711 |
| 4 | 123444 | 98658 |
| 5 | 581013 | 435702 |
| 6 | 2388204 | 1683990 |
| 7 | 8852004 | 5885706 |
| 8 | 30177684 | 18959589 |
| 9 | 96033357 | 57130029 |
| 10 | 288281592 | 162673434 |
| 11 | 823102236 | 441298935 |
| 12 | 2249539452 | 1147588638 |
| 13 | 5915357991 | 2875281948 |
| 14 | 15029087328 | 6969072330 |
| 15 | 37022515776 | 16396507860 |
| 16 | 88685084628 | 37552493430 |
| 17 | 207095279364 | 83924916075 |
| 18 | 472447033668 | 183402719829 |
| 19 | 1054885895460 | 392613233760 |
| 20 | 2309034290904 | 824608476846 |
| 21 | 4961888228889 | 1701583792920 |
| 22 | 10481005218960 | 3453903360714 |
| 23 | 21786435278208 | 6903820668978 |
| 24 | 44610098518620 | 13602293626383 |

## S20.6 Module c2

$h = 2/3$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 27 | 27 |
| 1 | 594 | 567 |
| 2 | 5967 | 5346 |
| 3 | 39852 | 33291 |
| 4 | 210654 | 164835 |
| 5 | 942786 | 692307 |
| 6 | 3734964 | 2582118 |
| 7 | 13434984 | 8763228 |
| 8 | 44704413 | 27574911 |
| 9 | 139368816 | 81446040 |
| 10 | 411119766 | 228029175 |
| 11 | 1156080276 | 609537312 |
| 12 | 3117611322 | 1564789023 |
| 13 | 8101080540 | 3875827725 |
| 14 | 20364234354 | 9298340325 |
| 15 | 49684390416 | 21674859822 |

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 16 | 117979019019 | 49225632093 |
| 17 | 273305608812 | 109169981712 |
| 18 | 618924153411 | 236892911580 |
| 19 | 1372587876540 | 503826314355 |
| 20 | 2985593707386 | 1051821501192 |
| 21 | 6378266472066 | 2158285037733 |
| 22 | 13399324817472 | 4358030082525 |
| 23 | 27710297015328 | 8668408114125 |
| 24 | 56467520205534 | 17000534883915 |

## S20.7 Module d1

$h = 6/13$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 7 | 7 |
| 1 | 292 | 285 |
| 2 | 3204 | 2905 |
| 3 | 23010 | 19514 |
| 4 | 126412 | 100198 |
| 5 | 583401 | 433986 |
| 6 | 2362991 | 1653470 |
| 7 | 8656740 | 5713559 |
| 8 | 29231455 | 18235026 |
| 9 | 92280130 | 54521551 |
| 10 | 275126813 | 154221639 |
| 11 | 780894920 | 415977380 |
| 12 | 2123094648 | 1076313049 |
| 13 | 5557092027 | 2684696613 |
| 14 | 14060468760 | 6481215751 |
| 15 | 34507145308 | 15193919772 |
| 16 | 82379301472 | 34684735750 |
| 17 | 191773721536 | 77284909612 |
| 18 | 436247645366 | 168430223304 |
| 19 | 971498203390 | 359651616744 |
| 20 | 2121325001394 | 753613575117 |
| 21 | 4548184405648 | 1551706327975 |
| 22 | 9586806849289 | 3143294525545 |
| 23 | 19888216227660 | 6271041793478 |
| 24 | 40647525486405 | 12333558962808 |

## S20.8 Module d2

$h = 11/13$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 42 | 42 |
| 1 | 777 | 735 |
| 2 | 7147 | 6328 |
| 3 | 45367 | 37443 |
| 4 | 231832 | 179318 |
| 5 | 1012498 | 735341 |
| 6 | 3935828 | 2692275 |
| 7 | 13944635 | 9003498 |
| 8 | 45822436 | 27988117 |
| 9 | 141352107 | 81824015 |
| 10 | 413198624 | 227081946 |
| 11 | 1152763121 | 602380050 |
| 12 | 3087032234 | 1536027580 |
| 13 | 7971888860 | 3781850992 |
| 14 | 19927836136 | 9024204637 |
| 15 | 48374540289 | 20933790944 |
| 16 | 114341192576 | 47332698770 |
| 17 | 263764343996 | 104547822344 |
| 18 | 595004863391 | 226019997605 |
| 19 | 1314819817795 | 479051302311 |
| 20 | 2850439526659 | 996914439688 |
| 21 | 6070689103437 | 2039553499760 |
| 22 | 12716298286464 | 4106881397436 |
| 23 | 26226503718584 | 8147663799870 |
| 24 | 53307749485344 | 15940303264893 |

## S20.9 Module d3

$h = 15/13$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 119 | 119 |
| 1 | 1623 | 1504 |
| 2 | 12996 | 11254 |
| 3 | 76429 | 61810 |
| 4 | 370216 | 280791 |
| 5 | 1554659 | 1108133 |
| 6 | 5860920 | 3937668 |
| 7 | 20257614 | 12855150 |
| 8 | 65211172 | 39170690 |
| 9 | 197684485 | 112598911 |
| 10 | 569238280 | 307973711 |
| 11 | 1567342047 | 806650418 |
| 12 | 4148722812 | 2033954639 |
| 13 | 10602953025 | 4957958635 |
| 14 | 26258444747 | 11724698013 |
| 15 | 63204697021 | 26977672153 |

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 16 | 148245904818 | 60547417743 |
| 17 | 339564573378 | 132830364976 |
| 18 | 761018127008 | 285372006535 |
| 19 | 1671554559985 | 601358399328 |
| 20 | 3603564686592 | 1244732883814 |
| 21 | 7634662170707 | 2533843728290 |
| 22 | 15914429034768 | 5078381952093 |
| 23 | 32672407194325 | 10030947469080 |
| 24 | 66124154006027 | 19544090515996 |

### S20.10 Module d4

$h = 5/13$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 5 | 5 |
| 1 | 229 | 224 |
| 2 | 2738 | 2504 |
| 3 | 19942 | 16975 |
| 4 | 111748 | 89068 |
| 5 | 521123 | 389438 |
| 6 | 2130500 | 1497858 |
| 7 | 7857543 | 5208663 |
| 8 | 26688282 | 16720410 |
| 9 | 84649505 | 50218166 |
| 10 | 253420427 | 142623705 |
| 11 | 721826269 | 385998585 |
| 12 | 1968649660 | 1001781625 |
| 13 | 5167082331 | 2505424955 |
| 14 | 13106183982 | 6062956301 |
| 15 | 32237120040 | 14243942489 |
| 16 | 77116856674 | 32579916449 |
| 17 | 179856935180 | 72724504692 |
| 18 | 409838999796 | 158751966100 |
| 19 | 914127309737 | 339498239112 |
| 20 | 1998971629824 | 712382313257 |
| 21 | 4291684306837 | 1468721627927 |
| 22 | 9057634990856 | 2978811831450 |
| 23 | 18812763926905 | 5949651971360 |
| 24 | 38492434575816 | 11713966606545 |

### S20.11 Module d5

$h = 7/13$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 13 | 13 |
| 1 | 347 | 334 |
| 2 | 3804 | 3444 |
| 3 | 26390 | 22239 |
| 4 | 143074 | 112880 |
| 5 | 652171 | 482720 |
| 6 | 2620211 | 1825313 |
| 7 | 9530967 | 6262402 |
| 8 | 32007218 | 19882777 |
| 9 | 100559827 | 59168520 |
| 10 | 298613588 | 166725104 |
| 11 | 844570064 | 448159934 |
| 12 | 2289163136 | 1156162609 |
| 13 | 5975309777 | 2876203659 |
| 14 | 15081569898 | 6927183975 |
| 15 | 36930910285 | 16204625013 |
| 16 | 87987869892 | 36920376179 |
| 17 | 204452003314 | 82120343886 |
| 18 | 464299854676 | 178677214710 |
| 19 | 1032349257472 | 380958458475 |
| 20 | 2250924763754 | 797149212279 |
| 21 | 4819528623477 | 1639220862003 |
| 22 | 10145940225070 | 3316561606896 |
| 23 | 21023290194574 | 6609232493660 |
| 24 | 42919713322672 | 12984894444655 |

## S20.12 Module d6

$h = 8/13$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 14 | 14 |
| 1 | 441 | 427 |
| 2 | 4445 | 3990 |
| 3 | 30329 | 25443 |
| 4 | 161537 | 126763 |
| 5 | 729016 | 537164 |
| 6 | 2902745 | 2012633 |
| 7 | 10490312 | 6863010 |
| 8 | 35029813 | 21667526 |
| 9 | 109550894 | 64196751 |
| 10 | 324004097 | 180182735 |
| 11 | 913221244 | 482730535 |
| 12 | 2467650847 | 1241644820 |
| 13 | 6423760304 | 3080820330 |
| 14 | 16173881411 | 7402507021 |
| 15 | 39518591878 | 17279953730 |

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 16 | 93964270270 | 39294407141 |
| 17 | 217939078630 | 87247137965 |
| 18 | 494093912780 | 189524612354 |
| 19 | 1096885333844 | 403483222843 |
| 20 | 2388185132606 | 843112479335 |
| 21 | 5106550516940 | 1731505748532 |
| 22 | 10736677143348 | 3499064669915 |
| 23 | 22221194977604 | 6965077249750 |
| 24 | 45315169047136 | 13669544141362 |

### S20.13 Module E8

$h = 0$; $\chi = q^{h-1/3} \sum_n a(n) q^n$.

| Grade $n$ | States $a(n)$ | Virasoro primaries $v(n)$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 248 | 248 |
| 2 | 4124 | 3875 |
| 3 | 34752 | 30380 |
| 4 | 213126 | 174250 |
| 5 | 1057504 | 809627 |
| 6 | 4530744 | 3260362 |
| 7 | 17333248 | 11749125 |
| 8 | 60655377 | 38826385 |
| 9 | 197230000 | 119458625 |
| 10 | 603096260 | 346303139 |
| 11 | 1749556736 | 953974346 |
| 12 | 4848776870 | 2514514625 |
| 13 | 12908659008 | 6375511275 |
| 14 | 33161242504 | 15618365750 |
| 15 | 82505707520 | 37099522892 |
| 16 | 199429765972 | 85709389310 |
| 17 | 469556091240 | 193071429250 |
| 18 | 1079330385764 | 424998178800 |
| 19 | 2426800117504 | 915906113500 |
| 20 | 5346409013164 | 1935631163543 |
| 21 | 11558035326944 | 4017215444008 |
| 22 | 24551042107480 | 8198039136625 |
| 23 | 51301080086528 | 16468952591975 |
| 24 | 105561758786885 | 32600946798875 |

### S20.14 All entries of the characteristic matrix

Source: `c8_physics_20260927/data/fundamental_matrix.json`. The row and column order is $(0, b, a, c_0, c_1, c_2, d_1, d_2, d_3, d_4, d_5, d_6)$. The exponent diagonal is

$$\Lambda = \mathrm{diag}(2/3, 2/3, 2/3, 2/3, 0, 1/3, 5/39, 20/39, 32/39, 2/39, 8/39, 11/39).$$

To keep each integer legible, the twelve-by-twelve characteristic matrix $\chi = X_0$ is printed in two six-column panels. The panels have the same row ordering and must be concatenated horizontally, not vertically.

| $i$ | $j=0$ | $j=1$ | $j=2$ | $j=3$ | $j=4$ | $j=5$ |
|---|---|---|---|---|---|---|
| 0 | 6 | 80 | 81 | 81 | 8748 | 1215 |
| 1 | 80 | 6 | 81 | 81 | 8748 | 1215 |
| 2 | 81 | 81 | 167 | -81 | -8748 | -1215 |
| 3 | 81 | 81 | -81 | 167 | -8748 | -1215 |
| 4 | 3 | 3 | -3 | -3 | -12 | 18 |
| 5 | 27 | 27 | -27 | -27 | 1458 | -152 |
| 6 | 7 | -7 | 0 | 0 | 0 | 0 |
| 7 | 42 | -42 | 0 | 0 | 0 | 0 |
| 8 | 119 | -119 | 0 | 0 | 0 | 0 |
| 9 | 5 | -5 | 0 | 0 | 0 | 0 |
| 10 | 13 | -13 | 0 | 0 | 0 | 0 |
| 11 | 14 | -14 | 0 | 0 | 0 | 0 |

| $i$ | $j=6$ | $j=7$ | $j=8$ | $j=9$ | $j=10$ | $j=11$ |
|---|---|---|---|---|---|---|
| 0 | 3549 | 273 | 13 | 5538 | 2275 | 1378 |
| 1 | -3549 | -273 | -13 | -5538 | -2275 | -1378 |
| 2 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | -88 | -14 | -1 | 50 | 63 | 64 |
| 7 | -1484 | 92 | 16 | 2940 | -192 | -1041 |
| 8 | -2142 | 987 | 11 | -24990 | -6035 | 4641 |
| 9 | 17 | 13 | -3 | -2 | 35 | -14 |
| 10 | 174 | -1 | -5 | 294 | -147 | 51 |
| 11 | 448 | -77 | 7 | -343 | 125 | -24 |

## S20.15 Scalar modular-form recurrence coefficients

These are the complete saved $f_r, g_r$ used in the matrix recurrence. The values at $r = 25$ are needed by the nonvacuum grade-twenty-four calculation. The recurrence in equation 6, $X_{-1} = I$, and the matrix above are a compact exact specification of all saved $12 \times 12$ coefficient matrices. Printing their 3,744 individual entries would duplicate this finite specification without changing its scope. The two scalar sequences are printed separately to preserve legibility of the largest integers.

| $r$ | $f_r$ |
|---|---|
| 0 | 1 |
| 1 | 0 |
| 2 | 338328 |
| 3 | 111221760 |
| 4 | 75650746392 |
| 5 | 36798073344000 |
| 6 | 20561692815386208 |
| 7 | 10812935351968856064 |

| $r$ | $f_r$ |
|---|---|
| 8 | 5835854082905153568792 |
| 9 | 3114524277111188324278272 |
| 10 | 1670230994092345751087741328 |
| 11 | 893834179235372381032339264512 |
| 12 | 478770106623952080200901398569056 |
| 13 | 256347544792251392656666808780427264 |
| 14 | 137278859985672540698960546143585422528 |
| 15 | 735100943030702296029538137262580392857 60 |
| 16 | 393644089424163872402153680406630819467 10040 |
| 17 | 210792278581967625016673649708309955803966 25920 |
| 18 | 112877701583761276096557321991510447520208134 54520 |
| 19 | 604450222121180614848502111572666339863773926811 5456 |
| 20 | 323678154156506747501486251800363627320005180422046 9392 |
| 21 | 173326926619906772424247624841197095259982436908434435 2768 |
| 22 | 928151284225774669399020861969403881841019703425912195363616 |
| 23 | 4970172549834517798609009339196079630018564815909020956235 3152 |
| 24 | 266148595645567545142154085992967657479997759992229905559690018912 |
| 25 | 1425203514170564303684844432149813682156857270847244828433111101440 00 |

| $r$ | $g_r$ |
|---|---|
| 0 | 0 |
| 1 | 1 |
| 2 | 240 |
| 3 | 199044 |
| 4 | 90246400 |
| 5 | 52098501126 |
| 6 | 27027759582144 |
| 7 | 14674033425574664 |
| 8 | 7811463279291691008 |
| 9 | 4193671459446763916301 |
| 10 | 2243207340058422507726240 |
| 11 | 1201788508249633900010245260 |
| 12 | 643416252201910574480397284352 |
| 13 | 344574371704661798059584392403470 |
| 14 | 184509699906794083352364505749591936 |
| 15 | 98805020211444275075440786620713099544 |
| 16 | 52908891035920178752441109674084596514816 |
| 17 | 28332355684442128880537857679864207736007698 |
| 18 | 15171720197322904176563567096102497150342941744 |
| 19 | 8124334147005129382796302533337586075069638295444 |
| 20 | 435051208519538206246736072628312292278299580906 2400 |
| 21 | 232966316285088768747318624879153986210192772237 1151904 |
| 22 | 124751512759064877458175256258028184627102433318029622 5600 |
| 23 | 66803395395762002408529736197930451847283754200626747867 9832 |
| 24 | 35772660495344639915166260396368055636568367175712159252622 5408 |
| 25 | 19155961260362383231446465051845701256145179315760901291956636 6751 |

## S21 Full scalar spectrum through total dimension twenty-four

Source: `c8_physics_20260927/data/scalars.tsv`. These are all recorded scalar levels, not only relevant or marginal ones. “States” includes descendants; “primaries” removes both left and right Virasoro descendants. The identity occurs at zero. The counts describe undeformed theories and do not assert exact marginality for dimension-two perturbations.

### S21.1 diagonal_U

| Δ | States | Primary components |
|---|---|---|
| 0 | 1 | 1 |
| 2/3 | 9 | 9 |
| 10/13 | 25 | 25 |
| 12/13 | 49 | 49 |
| 14/13 | 169 | 169 |
| 16/13 | 196 | 196 |
| 4/3 | 729 | 729 |
| 22/13 | 1764 | 1764 |
| 2 | 19558 | 19558 |
| 30/13 | 14161 | 14161 |
| 8/3 | 59049 | 57600 |
| 36/13 | 52441 | 50176 |
| 38/13 | 85264 | 81225 |
| 40/13 | 120409 | 111556 |
| 42/13 | 194481 | 182329 |
| 10/3 | 352836 | 321489 |
| 48/13 | 603729 | 540225 |
| 4 | 5369158 | 4740901 |
| 56/13 | 2634129 | 2262016 |
| 14/3 | 8503056 | 7128900 |
| 62/13 | 7496644 | 6270016 |
| 64/13 | 10265616 | 8439025 |
| 66/13 | 14470416 | 11861136 |
| 68/13 | 19758025 | 15920100 |
| 16/3 | 35605089 | 28579716 |
| 74/13 | 51079609 | 40043584 |
| 6 | 382242614 | 292210150 |
| 82/13 | 168896016 | 126652516 |
| 20/3 | 478296900 | 350101521 |
| 88/13 | 397683364 | 288150625 |
| 90/13 | 529460100 | 380796196 |
| 92/13 | 696432100 | 494573121 |
| 94/13 | 919848241 | 647346249 |
| 22/3 | 1588181904 | 1108290681 |
| 100/13 | 2058164689 | 1401978249 |
| 8 | 14359092602 | 9596418458 |
| 108/13 | 5841392041 | 3820476100 |
| 26/3 | 15238421136 | 9733400964 |
| 114/13 | 12487615504 | 7933108624 |
| 116/13 | 15979993744 | 10039639204 |
| 118/13 | 20470169476 | 12741894400 |
| 120/13 | 26094202369 | 16068858169 |
| 28/3 | 44375107716 | 27170577225 |
| 126/13 | 53746076224 | 32154945124 |
| 10 | 353708707456 | 207347062153 |
| 134/13 | 137059886656 | 78843585681 |
| 32/3 | 337576106169 | 189836232804 |

| Δ | States | Primary components |
|---|---|---|
| 140/13 | 271569181129 | 151661955844 |
| 142/13 | 340356726801 | 188343848196 |
| 144/13 | 425327013241 | 233018598400 |
| 146/13 | 531464328256 | 288545162896 |
| 34/3 | 888845441796 | 479288982249 |
| 152/13 | 1025152200004 | 540726386281 |
| 12 | 6491253312116 | 3361245644852 |
| 160/13 | 2416964606281 | 1227958745689 |
| 38/3 | 5703518345616 | 2835822320100 |
| 166/13 | 4539030250000 | 2243578588164 |
| 168/13 | 5583726466081 | 2733963040900 |
| 170/13 | 6865505684521 | 3331767547969 |
| 172/13 | 8425928535025 | 4050691592689 |
| 40/3 | 13949956081296 | 6667333365924 |
| 178/13 | 15490742045584 | 7248344675625 |
| 14 | 95013926977624 | 43656386763125 |
| 186/13 | 34350383246400 | 15505229278224 |
| 44/3 | 78357974816016 | 34641535118436 |
| 192/13 | 61740981996849 | 27130170247569 |
| 194/13 | 74939147427600 | 32644756446481 |
| 196/13 | 90839331955089 | 39217678809604 |
| 198/13 | 110046645857344 | 47100906260100 |
| 46/3 | 180498795080256 | 76794164979984 |
| 204/13 | 194452845283225 | 81062976236004 |
| 16 | 1163452326392735 | 476715049645225 |
| 212/13 | 410370924972996 | 165254881522500 |
| 50/3 | 910692611603856 | 359466015048921 |
| 218/13 | 712264396111524 | 279572110568100 |
| 220/13 | 854477961417025 | 332516173220676 |
| 222/13 | 1024462004099524 | 395324821231729 |
| 224/13 | 1227087798814969 | 469481682960676 |
| 52/3 | 1998484541674569 | 760375716657921 |
| 230/13 | 2099695640974096 | 783334693205689 |
| 18 | 12301700514555122 | 4512917560714847 |
| 238/13 | 4252496953613584 | 1534342955076100 |
| 56/3 | 9222405656689449 | 3263840213540841 |
| 244/13 | 7165538696745025 | 2521864196403556 |
| 246/13 | 8515622392816900 | 2972599523445601 |
| 248/13 | 10112278806269929 | 3500913758990400 |
| 250/13 | 12001398376199236 | 4121222838956001 |
| 58/3 | 19423666873241856 | 6633457431681600 |
| 256/13 | 19980418153339449 | 6695169430720225 |
| 20 | 115023665003604748 | 37924924225274171 |
| 264/13 | 39079155609715225 | 12678514758385921 |
| 62/3 | 83106276286054464 | 26462646129352356 |
| 270/13 | 64221912820862329 | 20341521227927025 |
| 272/13 | 75694763231536969 | 23784313935846321 |
| 274/13 | 89170074938233744 | 27797260303810816 |
| 276/13 | 104978654872785409 | 32465817992080225 |

| Δ | States | Primary components |
|---|---|---|
| 64/3 | 169019461995894756 | 51997304651180625 |
| 282/13 | 170733102875493376 | 51566210199146916 |
| 22 | 967994790692197366 | 287800559997539210 |
| 290/13 | 324032219417358400 | 94847806667111521 |
| 68/3 | 677497290908199696 | 194744750032134225 |
| 296/13 | 521033162618460361 | 148994907622002225 |
| 298/13 | 609796876081806400 | 173037180671664400 |
| 300/13 | 713298593004964096 | 200847326442884356 |
| 302/13 | 833973040492907536 | 233028769421386225 |
| 70/3 | 1336521604556236176 | 371535734720185344 |
| 308/13 | 1328862813137660641 | 362861724638002500 |
| 24 | 7434977277940713436 | 1999506477176584861 |

### S21.2 E8_full

| Δ | States | Primary components |
|---|---|---|
| 0 | 1 | 1 |
| 2 | 61504 | 61504 |
| 4 | 17007376 | 15015625 |
| 6 | 1207701504 | 922944400 |
| 8 | 45422691876 | 30363062500 |
| 10 | 1118314710016 | 655495879129 |
| 12 | 20527641193536 | 10629960371044 |
| 14 | 300441486229504 | 138041938265625 |
| 16 | 3679074759012129 | 1507488172168225 |
| 18 | 38899672900000000 | 14270363086890625 |
| 20 | 363725098825987600 | 119925864081253321 |
| 22 | 3060948772482973696 | 910067052826127716 |
| 24 | 23510637135046996900 | 6322783799338890625 |

## S22 Complete spin-resolved spectrum through dimension twenty-four

The next tables preserve every saved $(\Delta, |s|)$ value. For $s > 0$, each printed multiplicity is the multiplicity at spin $+s$ separately and also at spin $-s$ separately: it is not their sum. The two opposite spin rows in each original TSV agree entry by entry. A spin-zero row appears once. Thus no spectral information is lost by this folding. The state and primary columns have the same meaning as in Appendix S21. Rows with zero primary multiplicity are retained when they contain descendant states.

### S22.1 Diagonal U

Source: `c8_physics_20260927/data/diagonal_spectrum.tsv`. 2582 signed-spin rows are represented by 1345 nonnegative-spin rows.

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 0 | 0 | 1 | 1 |
| 2/3 | 0 | 9 | 9 |
| 10/13 | 0 | 25 | 25 |
| 12/13 | 0 | 49 | 49 |
| 1 | 1 | 6 | 6 |
| 14/13 | 0 | 169 | 169 |
| 16/13 | 0 | 196 | 196 |
| 4/3 | 0 | 729 | 729 |
| 5/3 | 1 | 729 | 720 |
| 22/13 | 0 | 1764 | 1764 |
| 23/13 | 1 | 1145 | 1120 |
| 25/13 | 1 | 2044 | 1995 |
| 2 | 0 | 19558 | 19558 |
| 2 | 2 | 120 | 113 |
| 27/13 | 1 | 4511 | 4342 |
| 29/13 | 1 | 6174 | 5978 |
| 30/13 | 0 | 14161 | 14161 |
| 7/3 | 1 | 16038 | 15309 |
| 8/3 | 0 | 59049 | 57600 |
| 8/3 | 2 | 8748 | 8010 |
| 35/13 | 1 | 32634 | 30870 |
| 36/13 | 0 | 52441 | 50176 |
| 36/13 | 2 | 13690 | 12520 |
| 38/13 | 0 | 85264 | 81225 |
| 38/13 | 2 | 22428 | 20335 |
| 3 | 1 | 323794 | 304230 |
| 3 | 3 | 956 | 830 |
| 40/13 | 0 | 120409 | 111556 |
| 40/13 | 2 | 49452 | 44772 |
| 42/13 | 0 | 194481 | 182329 |
| 42/13 | 2 | 62230 | 55860 |
| 43/13 | 1 | 193137 | 178976 |
| 10/3 | 0 | 352836 | 321489 |
| 10/3 | 2 | 161109 | 144342 |
| 11/3 | 1 | 708588 | 640800 |
| 11/3 | 3 | 65610 | 56133 |
| 48/13 | 0 | 603729 | 540225 |
| 48/13 | 2 | 300174 | 265776 |
| 49/13 | 1 | 627002 | 560896 |
| 49/13 | 3 | 99710 | 84875 |
| 51/13 | 1 | 935568 | 827925 |
| 51/13 | 3 | 161070 | 136598 |
| 4 | 0 | 5369158 | 4740901 |
| 4 | 2 | 2732582 | 2389230 |
| 4 | 4 | 6010 | 4934 |
| 53/13 | 1 | 1319988 | 1150296 |
| 53/13 | 3 | 343070 | 289107 |
| 55/13 | 1 | 1960245 | 1703730 |
| 55/13 | 3 | 424606 | 356202 |
| 56/13 | 0 | 2634129 | 2262016 |
| 56/13 | 2 | 1546524 | 1339226 |
| 13/3 | 1 | 3544398 | 3031182 |
| 13/3 | 3 | 1076004 | 898857 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 14/3 | 0 | 8503056 | 7128900 |
| 14/3 | 2 | 5314410 | 4490640 |
| 14/3 | 4 | 370332 | 295974 |
| 61/13 | 1 | 5553219 | 4651080 |
| 61/13 | 3 | 1905414 | 1572606 |
| 62/13 | 0 | 7496644 | 6270016 |
| 62/13 | 2 | 4566718 | 3802400 |
| 62/13 | 4 | 558740 | 445340 |
| 64/13 | 0 | 10265616 | 8439025 |
| 64/13 | 2 | 6718920 | 5561490 |
| 64/13 | 4 | 884884 | 701386 |
| 5 | 1 | 45301802 | 37218790 |
| 5 | 3 | 16747414 | 13691038 |
| 5 | 5 | 29512 | 22547 |
| 66/13 | 0 | 14470416 | 11861136 |
| 66/13 | 2 | 9157330 | 7427826 |
| 66/13 | 4 | 1859962 | 1467440 |
| 68/13 | 0 | 19758025 | 15920100 |
| 68/13 | 2 | 13375089 | 10864161 |
| 68/13 | 4 | 2261518 | 1774682 |
| 69/13 | 1 | 21092508 | 16926016 |
| 69/13 | 3 | 9095051 | 7355390 |
| 16/3 | 0 | 35605089 | 28579716 |
| 16/3 | 2 | 23672088 | 18875997 |
| 16/3 | 4 | 5687658 | 4450545 |
| 17/3 | 1 | 63772920 | 49958370 |
| 17/3 | 3 | 29996892 | 23677920 |
| 17/3 | 5 | 1743039 | 1307106 |
| 74/13 | 0 | 51079609 | 40043584 |
| 74/13 | 2 | 35250159 | 27520605 |
| 74/13 | 4 | 9736944 | 7531356 |
| 75/13 | 1 | 54601196 | 42505400 |
| 75/13 | 3 | 25590292 | 19951232 |
| 75/13 | 5 | 2605615 | 1947190 |
| 77/13 | 1 | 73724040 | 56688170 |
| 77/13 | 3 | 36912304 | 28556430 |
| 77/13 | 5 | 4083807 | 3037902 |
| 6 | 0 | 382242614 | 292210150 |
| 6 | 2 | 277659598 | 213292666 |
| 6 | 4 | 83121456 | 63641466 |
| 6 | 6 | 126979 | 91463 |
| 79/13 | 1 | 100387560 | 76591116 |
| 79/13 | 3 | 49646678 | 37701920 |
| 79/13 | 5 | 8478223 | 6275360 |
| 81/13 | 1 | 134812405 | 101517570 |
| 81/13 | 3 | 71237817 | 54127801 |
| 81/13 | 5 | 10206224 | 7520296 |
| 82/13 | 0 | 168896016 | 126652516 |
| 82/13 | 2 | 124044267 | 92962240 |
| 82/13 | 4 | 44055704 | 33414129 |
| 19/3 | 1 | 237796884 | 177973686 |
| 19/3 | 3 | 125128476 | 93461445 |
| 19/3 | 5 | 25455222 | 18692289 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 20/3 | 0 | 478296900 | 350101521 |
| 20/3 | 2 | 359962704 | 263416860 |
| 20/3 | 4 | 141186159 | 104568480 |
| 20/3 | 6 | 7164612 | 5051970 |
| 87/13 | 1 | 324237949 | 236939304 |
| 87/13 | 3 | 180133464 | 131798730 |
| 87/13 | 5 | 42524916 | 30884322 |
| 88/13 | 0 | 397683364 | 288150625 |
| 88/13 | 2 | 305966024 | 223026272 |
| 88/13 | 4 | 119337167 | 87234112 |
| 88/13 | 6 | 10652500 | 7489290 |
| 90/13 | 0 | 529460100 | 380796196 |
| 90/13 | 2 | 405024048 | 291075190 |
| 90/13 | 4 | 170353092 | 123686010 |
| 90/13 | 6 | 16540937 | 11574290 |
| 7 | 1 | 2342785162 | 1674561070 |
| 7 | 3 | 1378069488 | 991444195 |
| 7 | 5 | 356083564 | 256234252 |
| 7 | 7 | 484920 | 328550 |
| 92/13 | 0 | 696432100 | 494573121 |
| 92/13 | 2 | 544253496 | 388758720 |
| 92/13 | 4 | 226303337 | 161228480 |
| 92/13 | 6 | 34062743 | 23729069 |
| 94/13 | 0 | 919848241 | 647346249 |
| 94/13 | 2 | 718031965 | 505784370 |
| 94/13 | 4 | 321496056 | 229369028 |
| 94/13 | 6 | 40638430 | 28176862 |
| 95/13 | 1 | 993271284 | 695609740 |
| 95/13 | 3 | 600860568 | 422309664 |
| 95/13 | 5 | 185004421 | 131867827 |
| 22/3 | 0 | 1588181904 | 1108290681 |
| 22/3 | 2 | 1256972418 | 881207910 |
| 22/3 | 4 | 560014884 | 392538069 |
| 22/3 | 6 | 100844028 | 69717186 |
| 23/3 | 1 | 2699720280 | 1845989838 |
| 23/3 | 3 | 1694233908 | 1163324340 |
| 23/3 | 5 | 580333572 | 404157600 |
| 23/3 | 7 | 26556012 | 17657118 |
| 100/13 | 0 | 2058164689 | 1401978249 |
| 100/13 | 2 | 1656903304 | 1134724304 |
| 100/13 | 4 | 786710946 | 540475635 |
| 100/13 | 6 | 165304776 | 113075550 |
| 101/13 | 1 | 2228478616 | 1511929300 |
| 101/13 | 3 | 1426834774 | 975152752 |
| 101/13 | 5 | 487884500 | 335520192 |
| 101/13 | 7 | 39287715 | 26043315 |
| 103/13 | 1 | 2908740120 | 1955263772 |
| 103/13 | 3 | 1869216804 | 1260729330 |
| 103/13 | 5 | 689993372 | 471238950 |
| 103/13 | 7 | 60597180 | 39994913 |
| 8 | 0 | 14359092602 | 9596418458 |
| 8 | 2 | 11627656304 | 7783879810 |
| 8 | 4 | 5903545960 | 3991814953 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 8 | 6 | 1362331220 | 923450000 |
| 8 | 8 | 1698317 | 1087380 |
| 105/13 | 1 | 3775722860 | 2510338320 |
| 105/13 | 3 | 2480858484 | 1662487680 |
| 105/13 | 5 | 909213217 | 609654542 |
| 105/13 | 7 | 123902571 | 81411226 |
| 107/13 | 1 | 4899255673 | 3225231009 |
| 107/13 | 3 | 3240476120 | 2143284360 |
| 107/13 | 5 | 1280110545 | 859394291 |
| 107/13 | 7 | 146864368 | 96082140 |
| 108/13 | 0 | 5841392041 | 3820476100 |
| 108/13 | 2 | 4811327136 | 3160021914 |
| 108/13 | 4 | 2523211557 | 1666632032 |
| 108/13 | 6 | 697449480 | 468582492 |
| 25/3 | 1 | 8394983208 | 5487521985 |
| 25/3 | 3 | 5625604062 | 3701073222 |
| 25/3 | 5 | 2218568616 | 1464060906 |
| 25/3 | 7 | 362744568 | 236607156 |
| 26/3 | 0 | 15238421136 | 9733400964 |
| 26/3 | 2 | 12706754310 | 8152420122 |
| 26/3 | 4 | 6964002864 | 4496253300 |
| 26/3 | 6 | 2151036972 | 1412569440 |
| 26/3 | 8 | 90533052 | 56878767 |
| 113/13 | 1 | 10517522344 | 6714203874 |
| 113/13 | 3 | 7236323206 | 4653237848 |
| 113/13 | 5 | 3058138356 | 1978822125 |
| 113/13 | 7 | 585674670 | 378146916 |
| 114/13 | 0 | 12487615504 | 7933108624 |
| 114/13 | 2 | 10392234866 | 6610710050 |
| 114/13 | 4 | 5833309000 | 3750636432 |
| 114/13 | 6 | 1799377347 | 1166740512 |
| 114/13 | 8 | 133441410 | 83602050 |
| 116/13 | 0 | 15979993744 | 10039639204 |
| 116/13 | 2 | 13424057010 | 8468802804 |
| 116/13 | 4 | 7571023164 | 4803330350 |
| 116/13 | 6 | 2527768080 | 1628364315 |
| 116/13 | 8 | 204620185 | 127645182 |
| 9 | 1 | 71266641664 | 44607032218 |
| 9 | 3 | 49811940844 | 31339857040 |
| 9 | 5 | 22586143820 | 14386119100 |
| 9 | 7 | 4767191684 | 3051529040 |
| 9 | 9 | 5519921 | 3342814 |
| 118/13 | 0 | 20470169476 | 12741894400 |
| 118/13 | 2 | 17210792690 | 10735210080 |
| 118/13 | 4 | 9967282644 | 6286377972 |
| 118/13 | 6 | 3307245549 | 2091642268 |
| 118/13 | 8 | 416093834 | 258476101 |
| 120/13 | 0 | 26094202369 | 16068858169 |
| 120/13 | 2 | 22110326264 | 13667063652 |
| 120/13 | 4 | 12902701525 | 8030405670 |
| 120/13 | 6 | 4626227592 | 2930505270 |
| 120/13 | 8 | 490417382 | 303345364 |
| 121/13 | 1 | 28295238664 | 17355691710 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 121/13 | 3 | 20204348364 | 12470928782 |
| 121/13 | 5 | 9512273160 | 5922252672 |
| 121/13 | 7 | 2410656066 | 1529762850 |
| 28/3 | 0 | 44375107716 | 27170577225 |
| 28/3 | 2 | 37571907672 | 23047592337 |
| 28/3 | 4 | 22286530188 | 13804002828 |
| 28/3 | 6 | 7980380496 | 4968750276 |
| 28/3 | 8 | 1207019151 | 744522597 |
| 29/3 | 1 | 71722568772 | 42985487916 |
| 29/3 | 3 | 52230021480 | 31509136890 |
| 29/3 | 5 | 25812443664 | 15714835020 |
| 29/3 | 7 | 7333177212 | 4550301360 |
| 29/3 | 9 | 288100071 | 171390087 |
| 126/13 | 0 | 53746076224 | 32154945124 |
| 126/13 | 2 | 45933996766 | 27533373063 |
| 126/13 | 4 | 28129362716 | 17036716200 |
| 126/13 | 6 | 10834981395 | 6617571030 |
| 126/13 | 8 | 1924542312 | 1175500914 |
| 127/13 | 1 | 58234453004 | 34686463784 |
| 127/13 | 3 | 42486431000 | 25426139550 |
| 127/13 | 5 | 21513952734 | 13042492152 |
| 127/13 | 7 | 6111616578 | 3745371840 |
| 127/13 | 9 | 423247525 | 251090830 |
| 129/13 | 1 | 73748887212 | 43484529228 |
| 129/13 | 3 | 54372422910 | 32265813580 |
| 129/13 | 5 | 27736194960 | 16597888895 |
| 129/13 | 7 | 8535584860 | 5196982410 |
| 129/13 | 9 | 645960910 | 381650857 |
| 10 | 0 | 353708707456 | 207347062153 |
| 10 | 2 | 305300664470 | 179599304990 |
| 10 | 4 | 190573663540 | 112946021500 |
| 10 | 6 | 79035619754 | 47538939150 |
| 10 | 8 | 15501412498 | 9388960802 |
| 10 | 10 | 16883159 | 9695389 |
| 131/13 | 1 | 93308713654 | 54489433600 |
| 131/13 | 3 | 69147368290 | 40593135807 |
| 131/13 | 5 | 36255798468 | 21567712488 |
| 131/13 | 7 | 11106504646 | 6640847518 |
| 131/13 | 9 | 1307277751 | 769190760 |
| 133/13 | 1 | 117763057592 | 68092520132 |
| 133/13 | 3 | 88037353105 | 51207421419 |
| 133/13 | 5 | 46629436840 | 27383409900 |
| 133/13 | 7 | 15448147533 | 9252033602 |
| 133/13 | 9 | 1533712516 | 898754514 |
| 134/13 | 0 | 137059886656 | 78843585681 |
| 134/13 | 2 | 118821032711 | 68493700730 |
| 134/13 | 4 | 76168516320 | 44314515672 |
| 134/13 | 6 | 32878107522 | 19334145600 |
| 134/13 | 8 | 7760129468 | 4661312110 |
| 31/3 | 1 | 198601642044 | 114116424345 |
| 31/3 | 3 | 148845785328 | 85961290338 |
| 31/3 | 5 | 80166549528 | 46848216888 |
| 31/3 | 7 | 26554421322 | 15634974537 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 31/3 | 9 | 3762958032 | 2199043080 |
| 32/3 | 0 | 337576106169 | 189836232804 |
| 32/3 | 2 | 294809454576 | 166139085420 |
| 32/3 | 4 | 193593327480 | 110127444966 |
| 32/3 | 6 | 87998126544 | 50622102630 |
| 32/3 | 8 | 23336105751 | 13711206960 |
| 32/3 | 10 | 864844776 | 488020302 |
| 139/13 | 1 | 234729436336 | 131859877438 |
| 139/13 | 3 | 178556708876 | 100806852825 |
| 139/13 | 5 | 99662306345 | 56974135344 |
| 139/13 | 7 | 35604032772 | 20571265995 |
| 139/13 | 9 | 5936788494 | 3436608630 |
| 140/13 | 0 | 271569181129 | 151661955844 |
| 140/13 | 2 | 238079114000 | 133411216344 |
| 140/13 | 4 | 156695122506 | 88417054425 |
| 140/13 | 6 | 73072516116 | 41867906640 |
| 140/13 | 8 | 19384736645 | 11248869184 |
| 140/13 | 10 | 1267102135 | 713118525 |
| 142/13 | 0 | 340356726801 | 188343848196 |
| 142/13 | 2 | 298710418292 | 165674387060 |
| 142/13 | 4 | 199191587400 | 111494390326 |
| 142/13 | 6 | 93657581820 | 52972750530 |
| 142/13 | 8 | 26945797960 | 15538642035 |
| 142/13 | 10 | 1925887691 | 1079551473 |
| 11 | 1 | 1515259935328 | 834831889549 |
| 11 | 3 | 1168038301676 | 647259394350 |
| 11 | 5 | 666873582698 | 373229868900 |
| 11 | 7 | 256998794074 | 146267755640 |
| 11 | 9 | 47400411108 | 27217660964 |
| 11 | 11 | 48970422 | 26700331 |
| 144/13 | 0 | 425327013241 | 233018598400 |
| 144/13 | 2 | 374884068614 | 206041331440 |
| 144/13 | 4 | 251522219130 | 139269558078 |
| 144/13 | 6 | 121755457272 | 68476283988 |
| 144/13 | 8 | 34894259969 | 19762285680 |
| 144/13 | 10 | 3881976644 | 2167426352 |
| 146/13 | 0 | 531464328256 | 288545162896 |
| 146/13 | 2 | 468900719065 | 255127396979 |
| 146/13 | 4 | 318160672648 | 174615563430 |
| 146/13 | 6 | 155707518785 | 86453428740 |
| 146/13 | 8 | 48311944254 | 27412012677 |
| 146/13 | 10 | 4536057358 | 2522558290 |
| 147/13 | 1 | 575559636344 | 311153773203 |
| 147/13 | 3 | 447944254680 | 243387259080 |
| 147/13 | 5 | 263267951544 | 144671858100 |
| 147/13 | 7 | 105837732156 | 58912717760 |
| 147/13 | 9 | 23524453715 | 13399270409 |
| 34/3 | 0 | 888845441796 | 479288982249 |
| 34/3 | 2 | 786785106456 | 425623420530 |
| 34/3 | 4 | 535410982368 | 291736623348 |
| 34/3 | 6 | 266751232371 | 147415474206 |
| 34/3 | 8 | 82785076704 | 46179904680 |
| 34/3 | 10 | 11100233682 | 6156787725 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 35/3 | 1 | 1387577570652 | 733717810980 |
| 35/3 | 3 | 1092726781776 | 580671982548 |
| 35/3 | 5 | 659985949080 | 354752869779 |
| 35/3 | 7 | 280033269012 | 152537177430 |
| 35/3 | 9 | 70052426856 | 39041624160 |
| 35/3 | 11 | 2469306708 | 1323896805 |
| 152/13 | 0 | 1025152200004 | 540726386281 |
| 152/13 | 2 | 912450876896 | 482773368450 |
| 152/13 | 4 | 632626256045 | 337117975614 |
| 152/13 | 6 | 327492950092 | 177108804376 |
| 152/13 | 8 | 109830587139 | 60140651025 |
| 152/13 | 10 | 17354342208 | 9537441732 |
| 153/13 | 1 | 1110252551500 | 583322823804 |
| 153/13 | 3 | 878064715164 | 463925196084 |
| 153/13 | 5 | 532217719644 | 283828959750 |
| 153/13 | 7 | 231770344690 | 125746287664 |
| 153/13 | 9 | 58033277783 | 31947709920 |
| 153/13 | 11 | 3609131345 | 1929992925 |
| 155/13 | 1 | 1378571312391 | 717582831420 |
| 155/13 | 3 | 1094315816880 | 572487184682 |
| 155/13 | 5 | 672615779550 | 355838297364 |
| 155/13 | 7 | 295665536520 | 158385105655 |
| 155/13 | 9 | 80337029396 | 43953167115 |
| 155/13 | 11 | 5466264440 | 2911841660 |
| 12 | 0 | 6491253312116 | 3361245644852 |
| 12 | 2 | 5797176266784 | 3008658021250 |
| 12 | 4 | 4087311688780 | 2138867746130 |
| 12 | 6 | 2168461963602 | 1148353784270 |
| 12 | 8 | 785854492728 | 424016093837 |
| 12 | 10 | 137507098978 | 74978106350 |
| 12 | 12 | 135729020 | 70389870 |
| 157/13 | 1 | 1708825628081 | 881115091360 |
| 157/13 | 3 | 1363633572558 | 706899937760 |
| 157/13 | 5 | 844670483020 | 442173077703 |
| 157/13 | 7 | 382529581908 | 203776382880 |
| 157/13 | 9 | 103618915036 | 55686184736 |
| 157/13 | 11 | 10979410832 | 5826079142 |
| 159/13 | 1 | 2116147548920 | 1081113992812 |
| 159/13 | 3 | 1694573529544 | 869975736630 |
| 159/13 | 5 | 1062419198477 | 551286864018 |
| 159/13 | 7 | 486953723830 | 256145036490 |
| 159/13 | 9 | 142885806777 | 76938027845 |
| 159/13 | 11 | 12785097416 | 6758227490 |
| 160/13 | 0 | 2416964606281 | 1227958745689 |
| 160/13 | 2 | 2169806358720 | 1105661735388 |
| 160/13 | 4 | 1548269180406 | 794576821500 |
| 160/13 | 6 | 847484391312 | 440826945260 |
| 160/13 | 8 | 320841919155 | 169348762144 |
| 160/13 | 10 | 67739355320 | 36648871609 |
| 37/3 | 1 | 3521271769704 | 1787618366226 |
| 37/3 | 3 | 2830133119536 | 1444486687380 |
| 37/3 | 5 | 1781560266876 | 917996362101 |
| 37/3 | 7 | 831613725072 | 435410529840 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 37/3 | 9 | 244205141004 | 129292542225 |
| 37/3 | 11 | 31214167452 | 16457507424 |
| 38/3 | 0 | 5703518345616 | 2835822320100 |
| 38/3 | 2 | 5143129400052 | 2564413875612 |
| 38/3 | 4 | 3725254023696 | 1870515131562 |
| 38/3 | 6 | 2100249517590 | 1068959972619 |
| 38/3 | 8 | 840629122272 | 434338068780 |
| 38/3 | 10 | 200013843348 | 105911744400 |
| 38/3 | 12 | 6748618356 | 3442765914 |
| 165/13 | 1 | 3985017978344 | 1979740190775 |
| 165/13 | 3 | 3232812621320 | 1614489254364 |
| 165/13 | 5 | 2078826454012 | 1047959064831 |
| 165/13 | 7 | 1010243508729 | 517782366920 |
| 165/13 | 9 | 321055330848 | 166905230310 |
| 165/13 | 11 | 48416051082 | 25299962100 |
| 166/13 | 0 | 4539030250000 | 2243578588164 |
| 166/13 | 2 | 4094746380789 | 2028451301394 |
| 166/13 | 4 | 2982362136936 | 1489253477880 |
| 166/13 | 6 | 1688080428710 | 852453367850 |
| 166/13 | 8 | 693865129126 | 357129757320 |
| 166/13 | 10 | 165298215601 | 86463683040 |
| 166/13 | 12 | 9843248300 | 5008908125 |
| 168/13 | 0 | 5583726466081 | 2733963040900 |
| 168/13 | 2 | 5050350772740 | 2479604616174 |
| 168/13 | 4 | 3695206689460 | 1827113135148 |
| 168/13 | 6 | 2123365791300 | 1063933546214 |
| 168/13 | 8 | 881506308852 | 448013861295 |
| 168/13 | 10 | 228021316640 | 118553553300 |
| 168/13 | 12 | 14861662536 | 7534191343 |
| 13 | 1 | 24834642327080 | 12113621818425 |
| 13 | 3 | 20286034681592 | 9942105135900 |
| 13 | 5 | 13290644162268 | 6580868060428 |
| 13 | 7 | 6630752626156 | 3328965867620 |
| 13 | 9 | 2279738501542 | 1168061613617 |
| 13 | 11 | 381090719988 | 197628662572 |
| 13 | 13 | 361328212 | 178454060 |
| 170/13 | 0 | 6865505684521 | 3331767547969 |
| 170/13 | 2 | 6215820279357 | 3022986693440 |
| 170/13 | 4 | 4579400708132 | 2244367867760 |
| 170/13 | 6 | 2653773834530 | 1315848716280 |
| 170/13 | 8 | 1135926088752 | 574201258176 |
| 170/13 | 10 | 293065812208 | 149685417956 |
| 170/13 | 12 | 29759120768 | 15030113917 |
| 172/13 | 0 | 8425928535025 | 4050691592689 |
| 172/13 | 2 | 7647605292992 | 3686561903640 |
| 172/13 | 4 | 5658610902581 | 2746640598338 |
| 172/13 | 6 | 3322569064126 | 1633357935693 |
| 172/13 | 8 | 1440198211165 | 718929112650 |
| 172/13 | 10 | 402730568604 | 206125938445 |
| 172/13 | 12 | 34547111858 | 17383027480 |
| 173/13 | 1 | 9111732026280 | 4363459853844 |
| 173/13 | 3 | 7499692824624 | 3609610423650 |
| 173/13 | 5 | 4984024664788 | 2421140348900 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 173/13 | 7 | 2569107567060 | 1267188144394 |
| 173/13 | 9 | 923873728440 | 463192461344 |
| 173/13 | 11 | 186513703593 | 95991399742 |
| 40/3 | 0 | 13949956081296 | 6667333365924 |
| 40/3 | 2 | 12666314825424 | 6066844086996 |
| 40/3 | 4 | 9417163416102 | 4545310454685 |
| 40/3 | 6 | 5554126055232 | 2711420117640 |
| 40/3 | 8 | 2453151643722 | 1219043969550 |
| 40/3 | 10 | 686711683944 | 345607655904 |
| 40/3 | 12 | 84175505694 | 42249303621 |
| 41/3 | 1 | 21140391360816 | 9911470046940 |
| 41/3 | 3 | 17533626713892 | 8260730846478 |
| 41/3 | 5 | 11854741721508 | 5636334401082 |
| 41/3 | 7 | 6304718417040 | 3043782623574 |
| 41/3 | 9 | 2400166120176 | 1178268156450 |
| 41/3 | 11 | 546638086836 | 275421273120 |
| 41/3 | 13 | 17746073973 | 8625845844 |
| 178/13 | 0 | 15490742045584 | 7248344675625 |
| 178/13 | 2 | 14118915048230 | 6620641222818 |
| 178/13 | 4 | 10623106982752 | 5018773164206 |
| 178/13 | 6 | 6412721038269 | 3063736593645 |
| 178/13 | 8 | 2953130565728 | 1436974554288 |
| 178/13 | 10 | 895696945017 | 442749336750 |
| 178/13 | 12 | 129655353828 | 64513158360 |
| 179/13 | 1 | 16740495361500 | 7801837543854 |
| 179/13 | 3 | 13907877580686 | 6511563029580 |
| 179/13 | 5 | 9459412884740 | 4472831609288 |
| 179/13 | 7 | 5053710155234 | 2421037392375 |
| 179/13 | 9 | 1976360324522 | 966540456840 |
| 179/13 | 11 | 450820772140 | 224399084000 |
| 179/13 | 13 | 25835411655 | 12527124775 |
| 181/13 | 1 | 20455798709340 | 9447198399730 |
| 181/13 | 3 | 17053660078455 | 7913745993636 |
| 181/13 | 5 | 11665315793560 | 5462950367098 |
| 181/13 | 7 | 6330667967130 | 3009481063446 |
| 181/13 | 9 | 2501987323680 | 1208414288900 |
| 181/13 | 11 | 619943637216 | 306749218965 |
| 181/13 | 13 | 38899644189 | 18792876291 |
| 14 | 0 | 95013926977624 | 43656386763125 |
| 14 | 2 | 86903761370383 | 40029443772185 |
| 14 | 4 | 65963766349712 | 30589869726962 |
| 14 | 6 | 40640314424462 | 19077297571276 |
| 14 | 8 | 19235600114930 | 9170495724530 |
| 14 | 10 | 6318127064136 | 3078799230612 |
| 14 | 12 | 1014560205376 | 501085642100 |
| 14 | 14 | 928246226 | 437193715 |
| 183/13 | 1 | 24973144574037 | 11430843781826 |
| 183/13 | 3 | 20874179370278 | 9597814113440 |
| 183/13 | 5 | 14387496688198 | 6678942537600 |
| 183/13 | 7 | 7880412587320 | 3707799587856 |
| 183/13 | 9 | 3212744523456 | 1543462812696 |
| 183/13 | 11 | 794339608192 | 386158311406 |
| 183/13 | 13 | 77679027101 | 37390647567 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 185/13 | 1 | 30450700706440 | 13812720405330 |
| 185/13 | 3 | 25537294154008 | 11639014936264 |
| 185/13 | 5 | 17696522764078 | 8137772747013 |
| 185/13 | 7 | 9826720257913 | 4584389326605 |
| 185/13 | 9 | 4059268429580 | 1926094834650 |
| 185/13 | 11 | 1088234023527 | 530182338140 |
| 185/13 | 13 | 89932644256 | 43131484620 |
| 186/13 | 0 | 34350383246400 | 15505229278224 |
| 186/13 | 2 | 31493681923626 | 14245215934950 |
| 186/13 | 4 | 24142219253152 | 10998777215790 |
| 186/13 | 6 | 15108827504065 | 6959738688910 |
| 186/13 | 8 | 7397820686880 | 3465936143594 |
| 186/13 | 10 | 2543796142281 | 1213202228672 |
| 186/13 | 12 | 493698014628 | 242040602041 |
| 43/3 | 1 | 50179181580576 | 22627688756904 |
| 43/3 | 3 | 42146694714618 | 19090303909677 |
| 43/3 | 5 | 29358598565664 | 13425158003400 |
| 43/3 | 7 | 16383944914632 | 7591319264925 |
| 43/3 | 9 | 6898331006892 | 3258586469952 |
| 43/3 | 11 | 1851861125268 | 887235376041 |
| 43/3 | 13 | 218729174580 | 104647348575 |
| 44/3 | 0 | 78357974816016 | 34641535118436 |
| 44/3 | 2 | 72070465639536 | 31927758280110 |
| 44/3 | 4 | 55796628850641 | 24891667895358 |
| 44/3 | 6 | 35586632842848 | 16049035651572 |
| 44/3 | 8 | 18001245901320 | 8257144372785 |
| 44/3 | 10 | 6559657042032 | 3064061663460 |
| 44/3 | 12 | 1437431991813 | 690067667520 |
| 44/3 | 14 | 45087261984 | 20907216990 |
| 191/13 | 1 | 54883684882780 | 24239892577950 |
| 191/13 | 3 | 46395124805128 | 20580809942897 |
| 191/13 | 5 | 32769941670024 | 14672518721770 |
| 191/13 | 7 | 18745581975008 | 8502629304078 |
| 191/13 | 9 | 8238798025787 | 3811860956400 |
| 191/13 | 11 | 2398624045818 | 1128980271300 |
| 191/13 | 13 | 334819332120 | 158837741664 |
| 192/13 | 0 | 61740981996849 | 27130170247569 |
| 192/13 | 2 | 56859384801000 | 25044799881780 |
| 192/13 | 4 | 44112803994115 | 19556862130708 |
| 192/13 | 6 | 28319225876396 | 12703208156940 |
| 192/13 | 8 | 14394659456398 | 6552325980375 |
| 192/13 | 10 | 5390162769080 | 2508461189000 |
| 192/13 | 12 | 1183261853799 | 561215189920 |
| 192/13 | 14 | 65530919910 | 30314781505 |
| 194/13 | 0 | 74939147427600 | 32644756446481 |
| 194/13 | 2 | 69073665081905 | 30151068440220 |
| 194/13 | 4 | 53836320122130 | 23661589832286 |
| 194/13 | 6 | 34779330684956 | 15452699784522 |
| 194/13 | 8 | 17968392109200 | 8117382593320 |
| 194/13 | 10 | 6802395252192 | 3126689407345 |
| 194/13 | 12 | 1622670871884 | 765138534705 |
| 194/13 | 14 | 98423281320 | 45368510257 |
| 15 | 1 | 332481840094204 | 144262456414375 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 15 | 3 | 282583533781824 | 123162592219093 |
| 15 | 5 | 201704909594432 | 88677059643263 |
| 15 | 7 | 117896243370778 | 52553337869644 |
| 15 | 9 | 53310045116824 | 24171766749000 |
| 15 | 11 | 16820455978288 | 7806265158370 |
| 15 | 13 | 2606316674998 | 1227529169558 |
| 15 | 15 | 2309453500 | 1038459581 |
| 196/13 | 0 | 90839331955089 | 39217678809604 |
| 196/13 | 2 | 83865664682998 | 36292291334201 |
| 196/13 | 4 | 65582202934417 | 28561827974400 |
| 196/13 | 6 | 42723840489512 | 18819929739520 |
| 196/13 | 8 | 22288203988960 | 9966628772226 |
| 196/13 | 10 | 8707976569344 | 3981824025396 |
| 196/13 | 12 | 2073432492619 | 960652022106 |
| 196/13 | 14 | 196060408674 | 90053391675 |
| 198/13 | 0 | 110046645857344 | 47100906260100 |
| 198/13 | 2 | 101682614536685 | 43608777855958 |
| 198/13 | 4 | 79864354540304 | 34484183554164 |
| 198/13 | 6 | 52338649817089 | 22840504036805 |
| 198/13 | 8 | 27697087109276 | 12282113002005 |
| 198/13 | 10 | 10968708014915 | 4954162831800 |
| 198/13 | 12 | 2832878294064 | 1315510280910 |
| 198/13 | 14 | 226434339754 | 103635098294 |
| 199/13 | 1 | 118728255044880 | 50619312790200 |
| 199/13 | 3 | 101381135450348 | 43406334221770 |
| 199/13 | 5 | 73185959298760 | 31616760818601 |
| 199/13 | 7 | 43506312502120 | 19035855076910 |
| 199/13 | 9 | 20369177242812 | 9078043804172 |
| 199/13 | 11 | 6733377123876 | 3059067777056 |
| 199/13 | 13 | 1261751409975 | 589997077565 |
| 46/3 | 0 | 180498795080256 | 76794164979984 |
| 46/3 | 2 | 166969373196132 | 71201674041498 |
| 46/3 | 4 | 131394968561376 | 56385663614280 |
| 46/3 | 6 | 86604023186964 | 37587189061125 |
| 46/3 | 8 | 46072111159152 | 20292106653792 |
| 46/3 | 10 | 18602786758374 | 8365362116958 |
| 46/3 | 12 | 4812041840760 | 2197594320075 |
| 46/3 | 14 | 549834327558 | 251055188775 |
| 47/3 | 1 | 267132979478736 | 111590566734834 |
| 47/3 | 3 | 229347247320828 | 96206397535710 |
| 47/3 | 5 | 167495352612696 | 70877140540668 |
| 47/3 | 7 | 101607032420784 | 43537670329230 |
| 47/3 | 9 | 49197427815240 | 21472531005618 |
| 47/3 | 11 | 17249183901756 | 7677002801160 |
| 47/3 | 13 | 3652068220704 | 1672577359200 |
| 47/3 | 15 | 111067547328 | 49189523580 |
| 204/13 | 0 | 194452845283225 | 81062976236004 |
| 204/13 | 2 | 180349226637008 | 75351707696175 |
| 204/13 | 4 | 143118725633286 | 60168553014115 |
| 204/13 | 6 | 95792663399168 | 40719880392828 |
| 204/13 | 8 | 52297404510407 | 22554916212150 |
| 204/13 | 10 | 22063019376398 | 9719982526240 |
| 204/13 | 12 | 6194157644220 | 2779660479120 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 204/13 | 14 | 836969117712 | 379016594754 |
| 205/13 | 1 | 209704323411126 | 87090980911830 |
| 205/13 | 3 | 180345770402500 | 75219681688428 |
| 205/13 | 5 | 132063213179521 | 55543090427790 |
| 205/13 | 7 | 80662641908212 | 34380121968780 |
| 205/13 | 9 | 39258811519720 | 17005243084375 |
| 205/13 | 11 | 14147471422278 | 6273584087320 |
| 205/13 | 13 | 3001316131878 | 1358102211424 |
| 205/13 | 15 | 161185600200 | 71219712445 |
| 207/13 | 1 | 253049105756700 | 104186896917534 |
| 207/13 | 3 | 218057116668830 | 90149748931970 |
| 207/13 | 5 | 160509257831013 | 66930032223054 |
| 207/13 | 7 | 98714488627040 | 41680101521240 |
| 207/13 | 9 | 48852407850480 | 21003172838186 |
| 207/13 | 11 | 17804922854508 | 7799043660765 |
| 207/13 | 13 | 4105656877920 | 1847146489035 |
| 207/13 | 15 | 241550017156 | 106357438404 |
| 16 | 0 | 1163452326392735 | 476715049645225 |
| 16 | 2 | 1081125743569916 | 443866729938375 |
| 16 | 4 | 864087809408216 | 357036390866017 |
| 16 | 6 | 585139450209360 | 244283842750241 |
| 16 | 8 | 326740734288560 | 138521089452970 |
| 16 | 10 | 141924857252224 | 61287279914800 |
| 16 | 12 | 43210286810674 | 19123316973713 |
| 16 | 14 | 6484560847520 | 2915848837350 |
| 16 | 16 | 5582384430 | 2399169031 |
| 209/13 | 1 | 305059738519806 | 124513942450354 |
| 209/13 | 3 | 263487964863497 | 108001068746760 |
| 209/13 | 5 | 194747122299548 | 80481542202880 |
| 209/13 | 7 | 120836017336736 | 50588293349920 |
| 209/13 | 9 | 60411015159040 | 25711900261551 |
| 209/13 | 11 | 22730078391708 | 9905645401596 |
| 209/13 | 13 | 5233304754606 | 2313679447650 |
| 209/13 | 15 | 480101833705 | 210660125169 |
| 211/13 | 1 | 367473667671656 | 148704447613260 |
| 211/13 | 3 | 317998309804030 | 129204499555383 |
| 211/13 | 5 | 236204170778552 | 96787678663540 |
| 211/13 | 7 | 147519020092028 | 61192370808205 |
| 211/13 | 9 | 74841382538663 | 31591169155260 |
| 211/13 | 11 | 28553614551280 | 12292473116700 |
| 211/13 | 13 | 7132681702251 | 3160870497967 |
| 211/13 | 15 | 553260286292 | 241919352220 |
| 212/13 | 0 | 410370924972996 | 165254881522500 |
| 212/13 | 2 | 382197462198240 | 154241172550920 |
| 212/13 | 4 | 307331963765615 | 124774569043163 |
| 212/13 | 6 | 210741119068480 | 86476246285401 |
| 212/13 | 8 | 119790385310163 | 49859062336580 |
| 212/13 | 10 | 53916801664752 | 22890125507306 |
| 212/13 | 12 | 17208592759575 | 7456769787040 |
| 212/13 | 14 | 3124754924893 | 1395239063547 |
| 49/3 | 1 | 600603073384392 | 241645232172708 |
| 49/3 | 3 | 520537510482624 | 210303285912720 |
| 49/3 | 5 | 387597959708076 | 157866194056725 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 49/3 | 7 | 243532934460504 | 100473082823520 |
| 49/3 | 9 | 124243046404344 | 52093391364693 |
| 49/3 | 11 | 48339147582180 | 20720175017850 |
| 49/3 | 13 | 12096355206276 | 5272158964275 |
| 49/3 | 15 | 1341478541232 | 585221215194 |
| 50/3 | 0 | 910692611603856 | 359466015048921 |
| 50/3 | 2 | 850087660297428 | 336250554465474 |
| 50/3 | 4 | 688475251140768 | 273940436121660 |
| 50/3 | 6 | 478233099445068 | 192274828577370 |
| 50/3 | 8 | 277692148112688 | 113218799847804 |
| 50/3 | 10 | 129368879263170 | 53799400529028 |
| 50/3 | 12 | 43824818648448 | 18607423121100 |
| 50/3 | 14 | 8996471333568 | 3935161886400 |
| 50/3 | 16 | 266055253884 | 112657480290 |
| 217/13 | 1 | 638977144830860 | 251990955433266 |
| 217/13 | 3 | 556337580589596 | 220292749984125 |
| 217/13 | 5 | 418362780402752 | 166982665253586 |
| 217/13 | 7 | 267247379867672 | 108017585805900 |
| 217/13 | 9 | 140049391359878 | 57513480677940 |
| 217/13 | 11 | 56975089682420 | 23931553077376 |
| 217/13 | 13 | 15483928677672 | 6632790408195 |
| 217/13 | 15 | 2031730692138 | 879219219648 |
| 218/13 | 0 | 712264396111524 | 279572110568100 |
| 218/13 | 2 | 665137125466215 | 261569503172058 |
| 218/13 | 4 | 539912219723500 | 213630057523890 |
| 218/13 | 6 | 376160270780087 | 150322516945230 |
| 218/13 | 8 | 219992662205680 | 89226685775500 |
| 218/13 | 10 | 103041955844802 | 42529588611125 |
| 218/13 | 12 | 35884731742716 | 15181642577704 |
| 218/13 | 14 | 7382300489160 | 3190643117536 |
| 218/13 | 16 | 385584283370 | 162899582245 |
| 220/13 | 0 | 854477961417025 | 332516173220676 |
| 220/13 | 2 | 798845092576200 | 311512098410009 |
| 220/13 | 4 | 650122182977683 | 255000853437330 |
| 220/13 | 6 | 455574877222920 | 180528359236680 |
| 220/13 | 8 | 268384640642976 | 107844414883702 |
| 220/13 | 10 | 127868687541270 | 52389169706082 |
| 220/13 | 12 | 45049741907040 | 18827931756655 |
| 220/13 | 14 | 10076086429936 | 4330267135020 |
| 220/13 | 16 | 576655110304 | 242793150250 |
| 17 | 1 | 3783178831222395 | 1466756868570745 |
| 17 | 3 | 3305881135973736 | 1286726524326450 |
| 17 | 5 | 2506690915975708 | 983548859485276 |
| 17 | 7 | 1621670780749232 | 643888001154762 |
| 17 | 9 | 869866677869432 | 351218871749970 |
| 17 | 11 | 364592596945302 | 150137872395050 |
| 17 | 13 | 107507935169840 | 45425149894507 |
| 17 | 15 | 15674238006704 | 6736352228690 |
| 17 | 17 | 13143600484 | 5404345101 |
| 222/13 | 0 | 1024462004099524 | 395324821231729 |
| 222/13 | 2 | 958432392662709 | 370537057985040 |
| 222/13 | 4 | 782430608027068 | 304325499757552 |
| 222/13 | 6 | 550804103208944 | 216335763340480 |

| Δ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 222/13 | 8 | 327519726520064 | 130507635303920 |
| 222/13 | 10 | 157688425015030 | 63963893172501 |
| 222/13 | 12 | 57370291891992 | 23857221609900 |
| 222/13 | 14 | 12815025868895 | 5412344754342 |
| 222/13 | 16 | 1143842308596 | 479964890327 |
| 224/13 | 0 | 1227087798814969 | 469481682960676 |
| 224/13 | 2 | 1149223057938928 | 440582944080510 |
| 224/13 | 4 | 940501272546265 | 362641718491255 |
| 224/13 | 6 | 665752898415904 | 259305465102740 |
| 224/13 | 8 | 398616914871839 | 157394622317660 |
| 224/13 | 10 | 194826226260016 | 78385311656190 |
| 224/13 | 12 | 71892902871895 | 29536003013790 |
| 224/13 | 14 | 17427699018198 | 7378540242710 |
| 224/13 | 16 | 1315499783780 | 550121699974 |
| 225/13 | 1 | 1321022750863608 | 503545095553500 |
| 225/13 | 3 | 1158612951826200 | 443377128679548 |
| 225/13 | 5 | 884971415146520 | 341275832291563 |
| 225/13 | 7 | 580255103272152 | 226500177520638 |
| 225/13 | 9 | 317082735798348 | 125718736236590 |
| 225/13 | 11 | 137795977512900 | 55796866478290 |
| 225/13 | 13 | 42617455824381 | 17633945811552 |
| 225/13 | 15 | 7521358945499 | 3210342986207 |
| 52/3 | 0 | 1998484541674569 | 760375716657921 |
| 52/3 | 2 | 1872417813058944 | 713730218217120 |
| 52/3 | 4 | 1535517525698424 | 588798237292650 |
| 52/3 | 6 | 1089936299088936 | 421986947858784 |
| 52/3 | 8 | 656737295424588 | 257931998606205 |
| 52/3 | 10 | 322844261680080 | 129030180792975 |
| 52/3 | 12 | 121513386390318 | 49708927377450 |
| 52/3 | 14 | 29512527907104 | 12289645519074 |
| 52/3 | 16 | 3185433513513 | 1329092066511 |
| 53/3 | 1 | 2898064301005188 | 1083161869398081 |
| 53/3 | 3 | 2551869805510368 | 957448006534404 |
| 53/3 | 5 | 1965736052424144 | 743142993550650 |
| 53/3 | 7 | 1307011665624876 | 500006664753876 |
| 53/3 | 9 | 730215451841004 | 283669566425784 |
| 53/3 | 11 | 328686139863360 | 130398312366630 |
| 53/3 | 13 | 107957656002816 | 43778675986200 |
| 53/3 | 15 | 21550475564604 | 9012598423200 |
| 53/3 | 17 | 621285838092 | 251774748225 |
| 230/13 | 0 | 2099695640974096 | 783334693205689 |
| 230/13 | 2 | 1971103538595945 | 736702355404470 |
| 230/13 | 4 | 1626278713900672 | 611367046167150 |
| 230/13 | 6 | 1167170354486258 | 442954748347050 |
| 230/13 | 8 | 715672856872688 | 275437393590440 |
| 230/13 | 10 | 361660681911620 | 141603846693456 |
| 230/13 | 12 | 142424244863992 | 57105166942936 |
| 230/13 | 14 | 37587017804553 | 15386336343840 |
| 230/13 | 16 | 4802330088192 | 1987973348340 |
| 231/13 | 1 | 2259149860600410 | 839668324968060 |
| 231/13 | 3 | 1991261902230861 | 742878815156415 |
| 231/13 | 5 | 1537850866104500 | 578171068530930 |
| 231/13 | 7 | 1025908616768180 | 390131832476750 |

| Δ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 231/13 | 9 | 577411116324588 | 223153189891940 |
| 231/13 | 11 | 261363520969044 | 102918683209475 |
| 231/13 | 13 | 88265234669520 | 35666831992456 |
| 231/13 | 15 | 17659760178346 | 7297901284576 |
| 231/13 | 17 | 899284675900 | 363622523460 |
| 233/13 | 1 | 2697482467489150 | 994201900045326 |
| 233/13 | 3 | 2381701287169620 | 881154433503201 |
| 233/13 | 5 | 1845247667905720 | 687806118508600 |
| 233/13 | 7 | 1238615540737848 | 467104794883314 |
| 233/13 | 9 | 702483117317124 | 269001231229374 |
| 233/13 | 11 | 323531386167600 | 126474444165014 |
| 233/13 | 13 | 110560893566832 | 44138336937660 |
| 233/13 | 15 | 24054756029824 | 9885149688750 |
| 233/13 | 17 | 1342416050752 | 540994367284 |
| 18 | 0 | 12301700514555122 | 4512917560714847 |
| 18 | 2 | 11568256161199843 | 4251985668116620 |
| 18 | 4 | 9590254753511804 | 3544620220113075 |
| 18 | 6 | 6947108793639290 | 2592456811732549 |
| 18 | 8 | 4317298791209728 | 1632571759573300 |
| 18 | 10 | 2234611738666926 | 860394763074700 |
| 18 | 12 | 907112643867360 | 356634547606330 |
| 18 | 14 | 259864166508128 | 104943652442117 |
| 18 | 16 | 36904898258422 | 15174507090044 |
| 18 | 18 | 30212253455 | 11896439240 |
| 235/13 | 1 | 3218640304831286 | 1176434488580040 |
| 235/13 | 3 | 2846076252979596 | 1044099624739808 |
| 235/13 | 5 | 2212951771963504 | 818032153609342 |
| 235/13 | 7 | 1492925811568256 | 558102814616480 |
| 235/13 | 9 | 854911471034498 | 324665869027920 |
| 235/13 | 11 | 398002629608220 | 154053644420025 |
| 235/13 | 13 | 140485182724140 | 55808728544772 |
| 235/13 | 15 | 30531790852524 | 12331405643786 |
| 235/13 | 17 | 2657876043082 | 1067564470518 |
| 237/13 | 1 | 3837547330802822 | 1390984771408026 |
| 237/13 | 3 | 3398904066808264 | 1236595912132350 |
| 237/13 | 5 | 2650848399914780 | 971559404848655 |
| 237/13 | 7 | 1798956949876552 | 666966898090480 |
| 237/13 | 9 | 1037674968227248 | 390534027491790 |
| 237/13 | 11 | 490537649314219 | 188341986135303 |
| 237/13 | 13 | 175660140897710 | 68947015382700 |
| 237/13 | 15 | 41438243189070 | 16778711849207 |
| 237/13 | 17 | 3051147100820 | 1221459931510 |
| 238/13 | 0 | 4252496953613584 | 1534342955076100 |
| 238/13 | 2 | 4004615990918790 | 1447475890741650 |
| 238/13 | 4 | 3336260020017600 | 1212698226645948 |
| 238/13 | 6 | 2436682419446973 | 893875947649594 |
| 238/13 | 8 | 1535923564567392 | 571116157039449 |
| 238/13 | 10 | 810373096747725 | 306451423229350 |
| 238/13 | 12 | 341254747932012 | 131949751438302 |
| 238/13 | 14 | 102581223265083 | 40574418918112 |
| 238/13 | 16 | 17641262673342 | 7205142711417 |
| 55/3 | 1 | 6230401109785008 | 2245867304302440 |
| 55/3 | 3 | 5523387478293744 | 1998271651176900 |

| Δ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 55/3 | 5 | 4317918211970064 | 1573897264986816 |
| 55/3 | 7 | 2939240307823092 | 1083314394146061 |
| 55/3 | 9 | 1706525020073160 | 638872063050375 |
| 55/3 | 11 | 811555467475608 | 309551047759575 |
| 55/3 | 13 | 296466757612272 | 115873800608412 |
| 55/3 | 15 | 70079537297286 | 27910933396731 |
| 55/3 | 17 | 7379251437924 | 2947589506224 |
| 56/3 | 0 | 9222405656689449 | 3263840213540841 |
| 56/3 | 2 | 8699670786392928 | 3084221449858626 |
| 56/3 | 4 | 7286104285480944 | 2597355789523110 |
| 56/3 | 6 | 5372359117424208 | 1932527790505620 |
| 56/3 | 8 | 3436899892424883 | 1252766095307496 |
| 56/3 | 10 | 1855250656117632 | 687554737933140 |
| 56/3 | 12 | 809682420021120 | 306795058568460 |
| 56/3 | 14 | 258605706775248 | 100265157458100 |
| 56/3 | 16 | 50324152885452 | 20141979858000 |
| 56/3 | 18 | 1417341101004 | 550208159487 |
| 243/13 | 1 | 6477097876472652 | 2290100105229755 |
| 243/13 | 3 | 5761903994182240 | 2044531846647108 |
| 243/13 | 5 | 4537077368999188 | 1621772749113750 |
| 243/13 | 7 | 3125613962860532 | 1129504056704780 |
| 243/13 | 9 | 1848138938191520 | 678153956183456 |
| 243/13 | 11 | 904066141981912 | 337893294223191 |
| 243/13 | 13 | 345732839445483 | 132469029093632 |
| 243/13 | 15 | 88843106631552 | 34789533595950 |
| 243/13 | 17 | 11078102447832 | 4391008538448 |
| 244/13 | 0 | 7165538696745025 | 2521864196403556 |
| 244/13 | 2 | 6763355820336414 | 2384726823319050 |
| 244/13 | 4 | 5671780947197067 | 2010536547741855 |
| 244/13 | 6 | 4194208100630000 | 1500526621259250 |
| 244/13 | 8 | 2692685445577713 | 975707683625290 |
| 244/13 | 10 | 1464589847620536 | 540015391817468 |
| 244/13 | 12 | 642872647837680 | 241790923750775 |
| 244/13 | 14 | 211145953573412 | 81580110788296 |
| 244/13 | 16 | 41187238156220 | 16290289051008 |
| 244/13 | 18 | 2049194998980 | 793759830500 |
| 246/13 | 0 | 8515622392816900 | 2972599523445601 |
| 246/13 | 2 | 8042357053502915 | 2812235596927614 |
| 246/13 | 4 | 6760004289760800 | 2376711303295420 |
| 246/13 | 6 | 5016853545372168 | 1779651337130030 |
| 246/13 | 8 | 3242013045643827 | 1165120744289418 |
| 246/13 | 10 | 1777411976889120 | 649404855818698 |
| 246/13 | 12 | 794009413537080 | 296494150430808 |
| 246/13 | 14 | 263943281916288 | 100759157353750 |
| 246/13 | 16 | 55997926688512 | 22026199239420 |
| 246/13 | 18 | 3053733517562 | 1179011563128 |
| 19 | 1 | 37616308678436092 | 13082509558333376 |
| 19 | 3 | 33559138704097904 | 11713191621733195 |
| 19 | 5 | 26578682942626416 | 9342977448292325 |
| 19 | 7 | 18494965026543728 | 6573148993607235 |
| 19 | 9 | 11090764588691504 | 3999375602258393 |
| 19 | 11 | 5559752379999240 | 2043764775958396 |
| 19 | 13 | 2192638802762632 | 823916522442230 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 19 | 15 | 611848594672906 | 236399179678011 |
| 19 | 17 | 84830277695748 | 33402849918830 |
| 19 | 19 | 67929988152 | 25637618529 |
| 248/13 | 0 | 10112278806269929 | 3500913758990400 |
| 248/13 | 2 | 9557790208878184 | 3314958063133808 |
| 248/13 | 4 | 8049569409171888 | 2806557667001468 |
| 248/13 | 6 | 5998090429741696 | 2110358640321617 |
| 248/13 | 8 | 3896923752575867 | 1388401030272480 |
| 248/13 | 10 | 2157780531586452 | 781940527098000 |
| 248/13 | 12 | 974606722421150 | 360374655664107 |
| 248/13 | 14 | 334705857069168 | 127153775560476 |
| 248/13 | 16 | 70944845149958 | 27428194857924 |
| 248/13 | 18 | 6035898110788 | 2322803791230 |
| 250/13 | 0 | 12001398376199236 | 4121222838956001 |
| 250/13 | 2 | 11349802929143861 | 3904114095363610 |
| 250/13 | 4 | 9579975774588128 | 3312984489010350 |
| 250/13 | 6 | 7162961157875015 | 2498975339011060 |
| 250/13 | 8 | 4683024041780864 | 1654905771744120 |
| 250/13 | 10 | 2612680281488707 | 938363997503023 |
| 250/13 | 12 | 1198559373067862 | 439653862752390 |
| 250/13 | 14 | 417671181350150 | 156784684492590 |
| 250/13 | 16 | 96111133675830 | 37254527911055 |
| 250/13 | 18 | 6917314778920 | 2653344572956 |
| 251/13 | 1 | 12891236953066420 | 4410577037118590 |
| 251/13 | 3 | 11531409350263920 | 3959048250961650 |
| 251/13 | 5 | 9186066350103240 | 3176321538145224 |
| 251/13 | 7 | 6449849258181108 | 2253892255978987 |
| 251/13 | 9 | 3925382857103400 | 1392150163080285 |
| 251/13 | 11 | 2006906673568463 | 724703584183530 |
| 251/13 | 13 | 821408242484916 | 303606722409862 |
| 251/13 | 15 | 240603103519614 | 91063316285472 |
| 251/13 | 17 | 40408184231982 | 15806813432144 |
| 58/3 | 0 | 19423666873241856 | 6633457431681600 |
| 58/3 | 2 | 18378867811727358 | 6287884206028425 |
| 58/3 | 4 | 15531920010775584 | 5341514439563136 |
| 58/3 | 6 | 11644166053662408 | 4040469902490714 |
| 58/3 | 8 | 7637585317984440 | 2683262664811575 |
| 58/3 | 10 | 4289807423607516 | 1532691927471375 |
| 58/3 | 12 | 1980022326858432 | 721577758334202 |
| 58/3 | 14 | 703980806486373 | 263160229169178 |
| 58/3 | 16 | 162343531634328 | 61899379630704 |
| 58/3 | 18 | 16710952142097 | 6396108612660 |
| 59/3 | 1 | 27684649041064344 | 9293538001949586 |
| 59/3 | 3 | 24839319177701424 | 8366846433737715 |
| 59/3 | 5 | 19912932227261808 | 6754369332208428 |
| 59/3 | 7 | 14127081615538164 | 4841946047612520 |
| 59/3 | 9 | 8732095115703264 | 3036438752325660 |
| 59/3 | 11 | 4570207437452544 | 1617646672451880 |
| 59/3 | 13 | 1939542800814360 | 702644704568730 |
| 59/3 | 15 | 603889834625424 | 224079525920250 |
| 59/3 | 17 | 114804629181324 | 44016652758960 |
| 59/3 | 19 | 3164657686380 | 1177839701280 |
| 256/13 | 0 | 19980418153339449 | 6695169430720225 |

| Δ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 256/13 | 2 | 18933767503528064 | 6355596073235682 |
| 256/13 | 4 | 16074860963805835 | 5423527575414900 |
| 256/13 | 6 | 12150027903479752 | 4135408652944500 |
| 256/13 | 8 | 8071521526972280 | 2780950090308272 |
| 256/13 | 10 | 4619910107081152 | 1618202327097566 |
| 256/13 | 12 | 2194607769291063 | 783823934316192 |
| 256/13 | 14 | 817196503340672 | 299521317816560 |
| 256/13 | 16 | 204944895284892 | 76842649422840 |
| 256/13 | 18 | 24990204262422 | 9492839899410 |
| 257/13 | 1 | 21451913702438635 | 7162300893225030 |
| 257/13 | 3 | 19264303022079858 | 6454054600619850 |
| 257/13 | 5 | 15468749355385380 | 5217942884217375 |
| 257/13 | 7 | 11008468906195500 | 3752770812246390 |
| 257/13 | 9 | 6829933915251786 | 2361145575948838 |
| 257/13 | 11 | 3602433690229920 | 1268679469610252 |
| 257/13 | 13 | 1537864355792908 | 553044081721775 |
| 257/13 | 15 | 492448288522840 | 182102159748768 |
| 257/13 | 17 | 93853130953284 | 35560440406400 |
| 257/13 | 19 | 4570636548685 | 1697491195560 |
| 259/13 | 1 | 25388738070125690 | 8408402956042089 |
| 259/13 | 3 | 22826694713708600 | 7585358339711880 |
| 259/13 | 5 | 18379078363127520 | 6149578107931391 |
| 259/13 | 7 | 13131358445972757 | 4439065308697110 |
| 259/13 | 9 | 8202891535052760 | 2812756898913486 |
| 259/13 | 11 | 4362117252674896 | 1522400373314856 |
| 259/13 | 13 | 1895547726870720 | 676837933425500 |
| 259/13 | 15 | 614443003801344 | 224512662422860 |
| 259/13 | 17 | 127384312446872 | 48002613641640 |
| 259/13 | 19 | 6800487423730 | 2517561317208 |
| 20 | 0 | 115023665003604748 | 37924924225274171 |
| 20 | 2 | 109123700515170814 | 36039148171753373 |
| 20 | 4 | 93006675071535124 | 30873853475380895 |
| 20 | 6 | 70759192961171456 | 23689028345557625 |
| 20 | 8 | 47511954395173644 | 16102503064130296 |
| 20 | 10 | 27594012772579840 | 9500038067308327 |
| 20 | 12 | 13438825815379244 | 4721616526055512 |
| 20 | 14 | 5162554696270198 | 1855978776221880 |
| 20 | 16 | 1406406441768696 | 520373179861660 |
| 20 | 18 | 190735057020896 | 71985909383078 |
| 20 | 20 | 149654742594 | 54181554870 |
| 261/13 | 1 | 30028530749129276 | 9864877650526080 |
| 261/13 | 3 | 27032338154721952 | 8910664028056718 |
| 261/13 | 5 | 21817938306832512 | 7240355034926818 |
| 261/13 | 7 | 15656572406102947 | 5249971929420267 |
| 261/13 | 9 | 9835762521948558 | 3343890248412000 |
| 261/13 | 11 | 5283853058116090 | 1829178071467440 |
| 261/13 | 13 | 2321999886449880 | 821072245844781 |
| 261/13 | 15 | 777735420606456 | 282822464343384 |
| 261/13 | 17 | 161112049572572 | 59678189713140 |
| 261/13 | 19 | 13420540347136 | 4952459960175 |
| 263/13 | 1 | 35494938486012718 | 11567146173293985 |
| 263/13 | 3 | 31989969404947372 | 10459576418106410 |
| 263/13 | 5 | 25886427292094264 | 8521420816108200 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 263/13 | 7 | 18646538103634480 | 6200560663228890 |
| 263/13 | 9 | 11791018330721576 | 3976360281428444 |
| 263/13 | 11 | 6383714776196486 | 2190458774675990 |
| 263/13 | 13 | 2849842353018830 | 999767600888463 |
| 263/13 | 15 | 968739204510350 | 348116080480350 |
| 263/13 | 17 | 217895415535980 | 80927009475158 |
| 263/13 | 19 | 15356394673816 | 5648765119802 |
| 264/13 | 0 | 39079155609715225 | 12678514758385921 |
| 264/13 | 2 | 37120695386064160 | 12063542761730590 |
| 264/13 | 4 | 31750610194095858 | 10369612120952700 |
| 264/13 | 6 | 24315332503307040 | 8009038095441852 |
| 264/13 | 8 | 16483976346893475 | 5494077576078455 |
| 264/13 | 10 | 9721296380455352 | 3292189679768283 |
| 264/13 | 12 | 4830671788618009 | 1667489915776930 |
| 264/13 | 14 | 1926603779014728 | 681400639279722 |
| 264/13 | 16 | 551113302592494 | 199776868923904 |
| 264/13 | 18 | 90561157113952 | 33959268777665 |
| 61/3 | 1 | 57297275021617056 | 18572073308217000 |
| 61/3 | 3 | 51681890119457988 | 16807937129579232 |
| 61/3 | 5 | 41885058229288848 | 13712602980446244 |
| 61/3 | 7 | 30257244178000560 | 10007844533621550 |
| 61/3 | 9 | 19199115049670244 | 6437306095379775 |
| 61/3 | 11 | 10466215578692064 | 3572775518759370 |
| 61/3 | 13 | 4701699865945188 | 1638770518008063 |
| 61/3 | 15 | 1630814567781204 | 583622722232352 |
| 61/3 | 17 | 367640947126134 | 134318280865860 |
| 61/3 | 19 | 37059872666580 | 13603310487585 |
| 62/3 | 0 | 83106276286054464 | 26462646129352356 |
| 62/3 | 2 | 79045270877286252 | 25211420954219115 |
| 62/3 | 4 | 67885890727989168 | 21757808917549782 |
| 62/3 | 6 | 52362772597763964 | 16923064213035288 |
| 62/3 | 8 | 35892526473078912 | 11735848112996700 |
| 62/3 | 10 | 21510562958561088 | 7143991267617720 |
| 62/3 | 12 | 10947641586818832 | 3704853896816940 |
| 62/3 | 14 | 4529173759690680 | 1570319104679325 |
| 62/3 | 16 | 1377655550175888 | 489685261943430 |
| 62/3 | 18 | 256337272596780 | 94227176102400 |
| 62/3 | 20 | 6927102872712 | 2473825430538 |
| 269/13 | 1 | 58406496111900768 | 18580756555733190 |
| 269/13 | 3 | 52822414335182756 | 16859483317865850 |
| 269/13 | 5 | 43047537736364590 | 13829621244474840 |
| 269/13 | 7 | 31375983388076080 | 10181782879486800 |
| 269/13 | 9 | 20176894232027728 | 6635867661976217 |
| 269/13 | 11 | 11214766424279448 | 3753805524496192 |
| 269/13 | 13 | 5187316883595392 | 1772278240045110 |
| 269/13 | 15 | 1885123766539412 | 661578619792832 |
| 269/13 | 17 | 462318778854807 | 166124698239675 |
| 269/13 | 19 | 55222432347390 | 20120154697062 |
| 270/13 | 0 | 64221912820862329 | 20341521227927025 |
| 270/13 | 2 | 61102236366846845 | 19384141017295110 |
| 270/13 | 4 | 52539877285284120 | 16750199500466250 |
| 270/13 | 6 | 40600571600372733 | 13049914262385165 |
| 270/13 | 8 | 27922724973651000 | 9081447599103258 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 270/13 | 10 | 16799504706604920 | 5547132475031182 |
| 270/13 | 12 | 8617654499606152 | 2901827998279532 |
| 270/13 | 14 | 3586707001359560 | 1234498467146700 |
| 270/13 | 16 | 1122139181441448 | 397514923114400 |
| 270/13 | 18 | 209335153929773 | 76047605561088 |
| 270/13 | 20 | 9994858149120 | 3561911566285 |
| 272/13 | 0 | 75694763231536969 | 23784313935846321 |
| 272/13 | 2 | 72061084733939600 | 22679731938516380 |
| 272/13 | 4 | 62061145663752840 | 19626596432654274 |
| 272/13 | 6 | 48106300833811980 | 15339172495475667 |
| 272/13 | 8 | 33224761135661160 | 10716495807805970 |
| 272/13 | 10 | 20131503079832508 | 6593948466171192 |
| 272/13 | 12 | 10413732257678464 | 3475341152678500 |
| 272/13 | 14 | 4412713332543360 | 1508137726168568 |
| 272/13 | 16 | 1397737455752664 | 489289798698120 |
| 272/13 | 18 | 283677475389880 | 102500710772040 |
| 272/13 | 20 | 14849275009758 | 5275295025819 |
| 21 | 1 | 333679949234110836 | 104473989232137539 |
| 21 | 3 | 302428278756844702 | 94992672812135987 |
| 21 | 5 | 247607350638704404 | 78280354803445375 |
| 21 | 7 | 181774204139461748 | 58031949611765175 |
| 21 | 9 | 118210558486184400 | 38249568715961386 |
| 21 | 11 | 66699217105470432 | 21947504523247205 |
| 21 | 13 | 31641633445913402 | 10636053289408442 |
| 21 | 15 | 11866743232301644 | 4085469220313950 |
| 21 | 17 | 3162208336733888 | 1121447303362071 |
| 21 | 19 | 420202542778430 | 152131478801542 |
| 21 | 21 | 323527951168 | 112448204178 |
| 274/13 | 0 | 89170074938233744 | 27797260303810816 |
| 274/13 | 2 | 84929819525218928 | 26516960018077680 |
| 274/13 | 4 | 73269743531515648 | 22987723330485193 |
| 274/13 | 6 | 56950480299364359 | 18011943546528918 |
| 274/13 | 8 | 39516895344008478 | 12644278962959175 |
| 274/13 | 10 | 24085268691478735 | 7822296586275360 |
| 274/13 | 12 | 12588776496928008 | 4167572063085520 |
| 274/13 | 14 | 5395488367456460 | 1826274327680754 |
| 274/13 | 16 | 1766196647187504 | 615364327461240 |
| 274/13 | 18 | 358225192342784 | 127240125130650 |
| 274/13 | 20 | 29262021928802 | 10362939759627 |
| 276/13 | 0 | 104978654872785409 | 32465817992080225 |
| 276/13 | 2 | 100044203699992136 | 30989731955491785 |
| 276/13 | 4 | 86441347719701611 | 26903371420115320 |
| 276/13 | 6 | 67387249802174848 | 21143700732993300 |
| 276/13 | 8 | 46948653396373195 | 14898529913196293 |
| 276/13 | 10 | 28809685776532048 | 9282169065421720 |
| 276/13 | 12 | 15178706326604990 | 4981076932414583 |
| 276/13 | 14 | 6609874315769270 | 2219828931243495 |
| 276/13 | 16 | 2196247442307100 | 756203203292460 |
| 276/13 | 18 | 483726432225204 | 172287336153961 |
| 276/13 | 20 | 33434591856484 | 11803574710690 |
| 277/13 | 1 | 112529576224085800 | 34677504475228721 |
| 277/13 | 3 | 102208211809749084 | 31597053461848420 |
| 277/13 | 5 | 84043225318490568 | 26146791977540850 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 277/13 | 7 | 62143059443283000 | 19522795062363180 |
| 277/13 | 9 | 40822927451926273 | 12992524783239729 |
| 277/13 | 11 | 23399390112326536 | 7575087541513023 |
| 277/13 | 13 | 11330286259334922 | 3742435890694830 |
| 277/13 | 15 | 4412981195620488 | 1494872927439904 |
| 277/13 | 17 | 1235132420133984 | 429199497828640 |
| 277/13 | 19 | 198914992638215 | 71561649520032 |
| 64/3 | 0 | 169019461995894756 | 51997304651180625 |
| 64/3 | 2 | 161121539267073216 | 49644400294644480 |
| 64/3 | 4 | 139370984112163986 | 43148918043001953 |
| 64/3 | 6 | 108837887437611360 | 33964762042896300 |
| 64/3 | 8 | 76059682199753256 | 24009411923308350 |
| 64/3 | 10 | 46841747702738976 | 15005657178789354 |
| 64/3 | 12 | 24852752272428426 | 8114107066049655 |
| 64/3 | 14 | 10891775122375824 | 3634377861174192 |
| 64/3 | 16 | 3693120423403437 | 1266429505306680 |
| 64/3 | 18 | 815317198664760 | 285669520239285 |
| 64/3 | 20 | 80611030099422 | 28399180532184 |
| 65/3 | 1 | 237285222972839712 | 71787613176992790 |
| 65/3 | 3 | 216030825279500364 | 65561772169010502 |
| 65/3 | 5 | 178511804199272844 | 54514163993199372 |
| 65/3 | 7 | 133037541143805312 | 41017910827114980 |
| 65/3 | 9 | 88417320266306304 | 27611555271161400 |
| 65/3 | 11 | 51527187074968164 | 16361696492437860 |
| 65/3 | 13 | 25564669665809616 | 8279864370127350 |
| 65/3 | 15 | 10332416626319160 | 3431648290720419 |
| 65/3 | 17 | 3076047271161360 | 1048277334139200 |
| 65/3 | 19 | 561095332689672 | 197906034443040 |
| 65/3 | 21 | 14885664686667 | 5104751378760 |
| 282/13 | 0 | 170733102875493376 | 51566210199146916 |
| 282/13 | 2 | 162945496025245947 | 49289154246900750 |
| 282/13 | 4 | 141455336972402024 | 42990519624266860 |
| 282/13 | 6 | 111165080413266100 | 34049887842770016 |
| 282/13 | 8 | 78432535443480608 | 24295640539079175 |
| 282/13 | 10 | 48979125293531922 | 15393474766551904 |
| 282/13 | 12 | 26507947357279232 | 8487604878038860 |
| 282/13 | 14 | 11966196994066532 | 3914584112026392 |
| 282/13 | 16 | 4252499758655477 | 1430254544844440 |
| 282/13 | 18 | 1021614998426715 | 352102707198585 |
| 282/13 | 20 | 119718460119678 | 41870406466896 |
| 283/13 | 1 | 182925521309796863 | 55052548317457425 |
| 283/13 | 3 | 166645219237418300 | 50307635939999750 |
| 283/13 | 5 | 137900550366945342 | 41891732471831550 |
| 283/13 | 7 | 102982404204476226 | 31579896155635563 |
| 283/13 | 9 | 68681184245220000 | 21335403208688562 |
| 283/13 | 11 | 40187367700524902 | 12687857502065662 |
| 283/13 | 13 | 20098652792494640 | 6477426183907056 |
| 283/13 | 15 | 8173009333931832 | 2694814624547500 |
| 283/13 | 17 | 2502880574059906 | 850103590736448 |
| 283/13 | 19 | 457764503229696 | 159573638169568 |
| 283/13 | 21 | 21458421534185 | 7343608139635 |
| 285/13 | 1 | 214845130627489960 | 64152713330525820 |
| 285/13 | 3 | 195919450119744240 | 58682256793018999 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 285/13 | 5 | 162441885518109285 | 48955512540166938 |
| 285/13 | 7 | 121717822333442400 | 37030808585067809 |
| 285/13 | 9 | 81540073798496228 | 25122690525408840 |
| 285/13 | 11 | 48060166858066272 | 15052689729199500 |
| 285/13 | 13 | 24242499686808832 | 7743793373303176 |
| 285/13 | 15 | 10038058319871660 | 3286747377554256 |
| 285/13 | 17 | 3112680243661560 | 1044787946641320 |
| 285/13 | 19 | 619426900407048 | 214779868908345 |
| 285/13 | 21 | 31837290839536 | 10861944295825 |
| 22 | 0 | 967994790692197366 | 287800559997539210 |
| 22 | 2 | 924769341898138810 | 275374529741740670 |
| 22 | 4 | 805140758020534320 | 240852996868414875 |
| 22 | 6 | 636081719176268352 | 191766481148175655 |
| 22 | 8 | 452257341674670400 | 137847950519161075 |
| 22 | 10 | 285734147148772100 | 88366233593251096 |
| 22 | 12 | 157042900050970288 | 49439599051460079 |
| 22 | 14 | 72732040978743038 | 23412589072568730 |
| 22 | 16 | 26681557550549344 | 8804524587844170 |
| 22 | 18 | 6966564037336562 | 2370011575936452 |
| 22 | 20 | 908407114653296 | 315734212039610 |
| 22 | 22 | 687223604695 | 229476596839 |
| 287/13 | 1 | 252200097128429632 | 74719511604783136 |
| 287/13 | 3 | 230197848930937472 | 68408430453868680 |
| 287/13 | 5 | 191253042649970386 | 57186915958481043 |
| 287/13 | 7 | 143741945006031366 | 43380810779407950 |
| 287/13 | 9 | 96766777368770135 | 29578512696354069 |
| 287/13 | 11 | 57383137095335532 | 17822203989126880 |
| 287/13 | 13 | 29251765922147236 | 9269744417851680 |
| 287/13 | 15 | 12252873164899640 | 3973602577935690 |
| 287/13 | 17 | 3927056575423488 | 1312020930987900 |
| 287/13 | 19 | 781070893022638 | 266247836901186 |
| 287/13 | 21 | 62653872105201 | 21309871206039 |
| 289/13 | 1 | 295887424523436668 | 86979708064313225 |
| 289/13 | 3 | 270333356368707218 | 79709563339979820 |
| 289/13 | 5 | 225023122205943152 | 66753754601603580 |
| 289/13 | 7 | 169669062252390232 | 50803479710193210 |
| 289/13 | 9 | 114712394980905110 | 34778205115471090 |
| 289/13 | 11 | 68501456455154320 | 21107540917488124 |
| 289/13 | 13 | 35205224944654310 | 11059708949857295 |
| 289/13 | 15 | 14985374280704620 | 4822074712122822 |
| 289/13 | 17 | 4875655308936580 | 1609898059143570 |
| 289/13 | 19 | 1053189643479246 | 360009028676045 |
| 289/13 | 21 | 71491707237160 | 24241080479448 |
| 290/13 | 0 | 324032219417358400 | 94847806667111521 |
| 290/13 | 2 | 309839205380040795 | 90827958624494798 |
| 290/13 | 4 | 270543076873655664 | 79671406638330910 |
| 290/13 | 6 | 214790529640582350 | 63735301946720250 |
| 290/13 | 8 | 153898643986587240 | 46167968175453684 |
| 290/13 | 10 | 98261751065970839 | 29894848775920349 |
| 290/13 | 12 | 54883005898100688 | 17001169975474713 |
| 290/13 | 14 | 25952580778707162 | 8210244859166560 |
| 290/13 | 16 | 9890191578595968 | 3211576561544890 |
| 290/13 | 18 | 2712933050855655 | 904443032589312 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 290/13 | 20 | 428824197704448 | 148123213173866 |
| 67/3 | 1 | 475287452546335416 | 138992290387077600 |
| 67/3 | 3 | 434497798695334752 | 127445869358818920 |
| 67/3 | 5 | 362154050206423020 | 106875604568207475 |
| 67/3 | 7 | 273593162718240336 | 81483476289569100 |
| 67/3 | 9 | 185569409565705024 | 55967045693862996 |
| 67/3 | 11 | 111228967424846934 | 34079249677408551 |
| 67/3 | 13 | 57572919718683048 | 17995033935497520 |
| 67/3 | 15 | 24665365361735172 | 7886401919409780 |
| 67/3 | 17 | 8190231859314180 | 2693455476541830 |
| 67/3 | 19 | 1773442662187284 | 596382791175864 |
| 67/3 | 21 | 172213194745782 | 58273696018791 |
| 68/3 | 0 | 677497290908199696 | 194744750032134225 |
| 68/3 | 2 | 648500814489367584 | 186682184562842892 |
| 68/3 | 4 | 568071685732505787 | 164264941072416492 |
| 68/3 | 6 | 453543048192788352 | 132130747088072370 |
| 68/3 | 8 | 327723457739215104 | 96505024690649160 |
| 68/3 | 10 | 211798073848928112 | 63238023411185700 |
| 68/3 | 12 | 120325049549115732 | 36566253783709650 |
| 68/3 | 14 | 58320751624112592 | 18094145532889482 |
| 68/3 | 16 | 23070354533710200 | 7346186216883360 |
| 68/3 | 18 | 6733143992276064 | 2201704633178820 |
| 68/3 | 20 | 1205738839620027 | 408380110300800 |
| 68/3 | 22 | 31443015656880 | 10361710082142 |
| 295/13 | 1 | 476320135395145504 | 136789633985577300 |
| 295/13 | 3 | 436358510652817038 | 125683943746333700 |
| 295/13 | 5 | 365291367086462960 | 105846888040662064 |
| 295/13 | 7 | 277886401256330360 | 81249408400820226 |
| 295/13 | 9 | 190393870156574292 | 56359522013757600 |
| 295/13 | 11 | 115770228800814848 | 34805674046230570 |
| 295/13 | 13 | 61149015397280672 | 18747306407081392 |
| 295/13 | 15 | 26993585637459497 | 8462866770324015 |
| 295/13 | 17 | 9397017237780865 | 3031436641024008 |
| 295/13 | 19 | 2214791512214043 | 732732113170680 |
| 295/13 | 21 | 254968942344354 | 85661246989920 |
| 296/13 | 0 | 521033162618460361 | 148994907622002225 |
| 296/13 | 2 | 498896037450604820 | 142877806958420625 |
| 296/13 | 4 | 437390961613396155 | 125817846290732530 |
| 296/13 | 6 | 349781534055498924 | 101375115164803410 |
| 296/13 | 8 | 253304556910461720 | 74191896216582207 |
| 296/13 | 10 | 164297463143957000 | 48800088492466242 |
| 296/13 | 12 | 93727585631807140 | 28321685658243096 |
| 296/13 | 14 | 45798688549203408 | 14139720116594800 |
| 296/13 | 16 | 18229526810775254 | 5762982608926200 |
| 296/13 | 18 | 5473184322458112 | 1783805312395528 |
| 296/13 | 20 | 982795706265673 | 328993644655648 |
| 296/13 | 22 | 45288174954280 | 14894059157250 |
| 298/13 | 0 | 609796876081806400 | 173037180671664400 |
| 298/13 | 2 | 584120264201596824 | 165990762493867311 |
| 298/13 | 4 | 512809174673523510 | 146373823305206763 |
| 298/13 | 6 | 411007959836845800 | 118185137731094526 |
| 298/13 | 8 | 298719385073575920 | 86811357058588548 |
| 298/13 | 10 | 194661547964622752 | 57350170020552500 |

| Δ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 298/13 | 12 | 111880980917823936 | 33540568782873432 |
| 298/13 | 14 | 55146937346006792 | 16876371514614192 |
| 298/13 | 16 | 22354173660003900 | 7018241649142416 |
| 298/13 | 18 | 6796725304466376 | 2189247435714885 |
| 298/13 | 20 | 1328069846449216 | 442236303472875 |
| 298/13 | 22 | 67107647945023 | 22003061678815 |
| 23 | 1 | 2682726089963743140 | 758590194887261387 |
| 23 | 3 | 2461970461077033440 | 698209438385785440 |
| 23 | 5 | 2068336485937477250 | 590027112012286193 |
| 23 | 7 | 1582582241223565020 | 455518323504241815 |
| 23 | 9 | 1093179555697384624 | 318463831111291575 |
| 23 | 11 | 672759307444338596 | 199056396224604944 |
| 23 | 13 | 360981700251301184 | 108828809353433680 |
| 23 | 15 | 163533001311331184 | 50456068764281348 |
| 23 | 17 | 58781319691513462 | 18607049193448650 |
| 23 | 19 | 15060537898154576 | 4918730414819106 |
| 23 | 21 | 1929596183253606 | 644327222185500 |
| 23 | 23 | 1435999900866 | 460992314155 |
| 300/13 | 0 | 713298593004964096 | 200847326442884356 |
| 300/13 | 2 | 683575217558291968 | 192761331226436336 |
| 300/13 | 4 | 600876117446528579 | 170180713721614680 |
| 300/13 | 6 | 482719095507523764 | 137731654212898575 |
| 300/13 | 8 | 351987287206295595 | 101479876090661226 |
| 300/13 | 10 | 230546784557587212 | 67391242604419027 |
| 300/13 | 12 | 133337667453294694 | 39641132400649920 |
| 300/13 | 14 | 66429237407914024 | 20169083996464800 |
| 300/13 | 16 | 27243696904686080 | 8472135158025525 |
| 300/13 | 18 | 8562517801320216 | 2745381887088876 |
| 300/13 | 20 | 1672376432346519 | 547499767909002 |
| 300/13 | 22 | 131897222925910 | 43115300889648 |
| 302/13 | 0 | 833973040492907536 | 233028769421386225 |
| 302/13 | 2 | 799528984393520159 | 223722959566182700 |
| 302/13 | 4 | 703728684144911776 | 197778655600747830 |
| 302/13 | 6 | 566568041311506143 | 160394013342700046 |
| 302/13 | 8 | 414562358600885936 | 118592495248527300 |
| 302/13 | 10 | 272754315704891150 | 79085220527412253 |
| 302/13 | 12 | 158881075346528080 | 46866021617831260 |
| 302/13 | 14 | 79814448388742860 | 24024708435830102 |
| 302/13 | 16 | 33267435290154676 | 10265823638794449 |
| 302/13 | 18 | 10615482914433670 | 3364018792546650 |
| 302/13 | 20 | 2251988777970540 | 739352954623164 |
| 302/13 | 22 | 150313480006872 | 48986905378810 |
| 303/13 | 1 | 892191091005959160 | 248427122711161198 |
| 303/13 | 3 | 820138132497971820 | 229021077374798129 |
| 303/13 | 5 | 691430993421195300 | 194206660724408150 |
| 303/13 | 7 | 531933437925053658 | 150722751661816950 |
| 303/13 | 9 | 370437672864319320 | 106229116351359204 |
| 303/13 | 11 | 230471830138447062 | 67094591665803819 |
| 303/13 | 13 | 125712238097709648 | 37297571011976016 |
| 303/13 | 15 | 58163854429094432 | 17638843723928350 |
| 303/13 | 17 | 21723523061565060 | 6767687426037312 |
| 303/13 | 19 | 5848585486338816 | 1872078257256256 |
| 303/13 | 21 | 908524798314133 | 301527403666510 |

| Δ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 70/3 | 0 | 1336521604556236176 | 371535734720185344 |
| 70/3 | 2 | 1281711637179590652 | 356817549963746025 |
| 70/3 | 4 | 1129038003180440640 | 315670819923459000 |
| 70/3 | 6 | 910371142990004202 | 256400906909586075 |
| 70/3 | 8 | 667508990288713344 | 189941738488225416 |
| 70/3 | 10 | 440647388791280316 | 127106390688712974 |
| 70/3 | 12 | 257668701709430232 | 75579142529089584 |
| 70/3 | 14 | 130378848612640794 | 39048243080289300 |
| 70/3 | 16 | 54700372055872080 | 16772881831192305 |
| 70/3 | 18 | 17815037651972262 | 5623037745372432 |
| 70/3 | 20 | 3788690284407204 | 1223747616394611 |
| 70/3 | 22 | 361781770071744 | 117666812228175 |
| 71/3 | 1 | 1851600952911414672 | 506429643767500530 |
| 71/3 | 3 | 1705288818895401672 | 467731988199369432 |
| 71/3 | 5 | 1443293708754000096 | 398143304315997570 |
| 71/3 | 7 | 1117253781973142784 | 310871050060869540 |
| 71/3 | 9 | 785040723867394512 | 221022935895911580 |
| 71/3 | 11 | 494585774558222256 | 141328719421139250 |
| 71/3 | 13 | 274497868372545684 | 79908931834934958 |
| 71/3 | 15 | 130219334479164240 | 38734436416294080 |
| 71/3 | 17 | 50498579942070480 | 15429249210265506 |
| 71/3 | 19 | 14468866075440324 | 4543228727096400 |
| 71/3 | 21 | 2546884268207280 | 828936806571360 |
| 71/3 | 23 | 65359305834624 | 20711462006934 |
| 308/13 | 0 | 1328862813137660641 | 362861724638002500 |
| 308/13 | 2 | 1275557471332446016 | 348804131976070680 |
| 308/13 | 4 | 1126843287130828020 | 309446232297172880 |
| 308/13 | 6 | 913141995960347296 | 252570495212298529 |
| 308/13 | 8 | 674565307622899515 | 188477344896722112 |
| 308/13 | 10 | 450027267294012928 | 127432641581001750 |
| 308/13 | 12 | 267060870767262008 | 76878300230259304 |
| 308/13 | 14 | 137941167489662312 | 40529453930533390 |
| 308/13 | 16 | 59649430673905765 | 17937117912430773 |
| 308/13 | 18 | 20372091297031873 | 6308474574345664 |
| 308/13 | 20 | 4716925433370549 | 1499071822323600 |
| 308/13 | 22 | 534084528031488 | 172489018692312 |
| 309/13 | 1 | 1421023039045918540 | 386686289729000625 |
| 309/13 | 3 | 1309444210666175337 | 357332989681558275 |
| 309/13 | 5 | 1109431986515228910 | 304470545974363966 |
| 309/13 | 7 | 860353350495371280 | 238164558432500490 |
| 309/13 | 9 | 605949017340791982 | 169697805350997687 |
| 309/13 | 11 | 383185200400990000 | 108930981148949736 |
| 309/13 | 13 | 213576529090690908 | 61824048174051800 |
| 309/13 | 15 | 102151898608490276 | 30238429161227616 |
| 309/13 | 17 | 39863492241950208 | 12092689767537575 |
| 309/13 | 19 | 11750631632119706 | 3677678956329208 |
| 309/13 | 21 | 2074198412906024 | 667253850244800 |
| 309/13 | 23 | 94063819634525 | 29748259856800 |
| 311/13 | 1 | 1657913825302388160 | 447721882182831620 |
| 311/13 | 3 | 1528905018936219951 | 414038311874608707 |
| 311/13 | 5 | 1297501885033738800 | 353365935110149801 |
| 311/13 | 7 | 1008694065249263140 | 277061522084334072 |
| 311/13 | 9 | 713136194224721280 | 198173284107034250 |

| Δ | $\|s\|$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 311/13 | 11 | 453159578026074176 | 127788279496153640 |
| 311/13 | 13 | 254507312554169766 | 73096358890809744 |
| 311/13 | 15 | 122809030886936680 | 36036372694515312 |
| 311/13 | 17 | 48811688282075940 | 14706015304833138 |
| 311/13 | 19 | 14572382835696192 | 4507706882767375 |
| 311/13 | 21 | 2799347599992388 | 895838939780325 |
| 311/13 | 23 | 139217513593620 | 43897292554346 |
| 24 | 0 | 7434977277940713436 | 1999506477176584861 |
| 24 | 2 | 7142097050138982112 | 1923398051740268565 |
| 24 | 4 | 6324587696459949460 | 1710431275189720595 |
| 24 | 6 | 5146056698294203120 | 1401538784665482401 |
| 24 | 8 | 3825358688438880118 | 1052363201086101175 |
| 24 | 10 | 2573884599302344244 | 717381062716501975 |
| 24 | 12 | 1546416925561331480 | 438172457251477200 |
| 24 | 14 | 811642572826434816 | 234535098778487283 |
| 24 | 16 | 360274530969613550 | 106631374014241636 |
| 24 | 18 | 127075311236465712 | 38617135833103890 |
| 24 | 20 | 31990894859179422 | 10037784314112263 |
| 24 | 22 | 4032023134319464 | 1294382077843910 |
| 24 | 24 | 2954844830174 | 912553664428 |

## S22.2 Holomorphic E8 full theory

Source: `c8_physics_20260927/data/e8_spectrum.tsv`. 325 signed-spin rows are represented by 169 nonnegative-spin rows.

| Δ | $\|s\|$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 0 | 0 | 1 | 1 |
| 1 | 1 | 248 | 248 |
| 2 | 0 | 61504 | 61504 |
| 2 | 2 | 4124 | 3875 |
| 3 | 1 | 1022752 | 961000 |
| 3 | 3 | 34752 | 30380 |
| 4 | 0 | 17007376 | 15015625 |
| 4 | 2 | 8618496 | 7534240 |
| 4 | 4 | 213126 | 174250 |
| 5 | 1 | 143317248 | 117722500 |
| 5 | 3 | 52855248 | 43214000 |
| 5 | 5 | 1057504 | 809627 |
| 6 | 0 | 1207701504 | 922944400 |
| 6 | 2 | 878931624 | 675218750 |
| 6 | 4 | 262260992 | 200787496 |
| 6 | 6 | 4530744 | 3260362 |
| 7 | 1 | 7406554752 | 5293715000 |
| 7 | 3 | 4361146496 | 3137304625 |
| 7 | 5 | 1123624512 | 808569776 |
| 7 | 7 | 17333248 | 11749125 |
| 8 | 0 | 45422691876 | 30363062500 |
| 8 | 2 | 36750379008 | 24596468260 |
| 8 | 4 | 18684788256 | 12633902750 |
| 8 | 6 | 4298645504 | 2913783000 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 8 | 8 | 60655377 | 38826385 |
| 9 | 1 | 225381597504 | 141077504750 |
| 9 | 3 | 157452415488 | 99049797560 |
| 9 | 5 | 71482314752 | 45527859375 |
| 9 | 7 | 15042533496 | 9628943480 |
| 9 | 9 | 197230000 | 119458625 |
| 10 | 0 | 1118314710016 | 655495879129 |
| 10 | 2 | 965619345744 | 568118078500 |
| 10 | 4 | 602365034496 | 356938417500 |
| 10 | 6 | 250142774748 | 150452241875 |
| 10 | 8 | 48913040000 | 29625739000 |
| 10 | 10 | 603096260 | 346303139 |
| 11 | 1 | 4791279902976 | 2639677104974 |
| 11 | 3 | 3694165813248 | 2047285031250 |
| 11 | 5 | 2107895661504 | 1179545576300 |
| 11 | 7 | 813376520000 | 462902171875 |
| 11 | 9 | 149567872480 | 85883178472 |
| 11 | 11 | 1749556736 | 953974346 |
| 12 | 0 | 20527641193536 | 10629960371044 |
| 12 | 2 | 18329979092992 | 9512408826375 |
| 12 | 4 | 12927237878502 | 6765497586250 |
| 12 | 6 | 6854136960000 | 3629153027500 |
| 12 | 8 | 2487168976240 | 1341924663625 |
| 12 | 10 | 433890070528 | 236585637808 |
| 12 | 12 | 4848776870 | 2514514625 |
| 13 | 1 | 78532509376512 | 38306400683250 |
| 13 | 3 | 64143303799008 | 31434889608395 |
| 13 | 5 | 42034840980000 | 20815665406250 |
| 13 | 7 | 20958801227520 | 10520689362820 |
| 13 | 9 | 7215171979264 | 3696650590750 |
| 13 | 11 | 1202496663760 | 623599627000 |
| 13 | 13 | 12908659008 | 6375511275 |
| 14 | 0 | 300441486229504 | 138041938265625 |
| 14 | 2 | 274813985410488 | 126588070251370 |
| 14 | 4 | 208571513920000 | 96716928182875 |
| 14 | 6 | 128535493508760 | 60343321970750 |
| 14 | 8 | 60800595689472 | 28981740631480 |
| 14 | 10 | 19996355811880 | 9743744171875 |
| 14 | 12 | 3201347433984 | 1581126796200 |
| 14 | 14 | 33161242504 | 15618365750 |
| 15 | 1 | 1051354692074496 | 456176050663125 |
| 15 | 3 | 893598639120000 | 389478361522250 |
| 15 | 5 | 637776707335040 | 280376371519153 |
| 15 | 7 | 372876028916736 | 166230029790500 |
| 15 | 9 | 168504693786240 | 76390954307500 |
| 15 | 11 | 53235309748992 | 24705106190625 |
| 15 | 13 | 8223988140992 | 3873354706000 |
| 15 | 15 | 82505707520 | 37099522892 |
| 16 | 0 | 3679074759012129 | 1507488172168225 |
| 16 | 2 | 3418636503040000 | 1403534317453125 |
| 16 | 4 | 2732474761417440 | 1129073594876318 |
| 16 | 6 | 1850163246546944 | 772363387828942 |
| 16 | 8 | 1033400419195620 | 438154173406250 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 16 | 10 | 448601717846016 | 193688032534500 |
| 16 | 12 | 136756964086496 | 60521167281250 |
| 16 | 14 | 20461415464960 | 9200681677216 |
| 16 | 16 | 199429765972 | 85709389310 |
| 17 | 1 | 11963060005710000 | 4638146565820625 |
| 17 | 3 | 10453617042452480 | 4068758868003375 |
| 17 | 5 | 7926793684291584 | 3110301706673252 |
| 17 | 7 | 5127600935132480 | 2035818932294875 |
| 17 | 9 | 2751170859739008 | 1110932839668750 |
| 17 | 11 | 1152419499499008 | 474485951485000 |
| 17 | 13 | 340253537812480 | 143760651206500 |
| 17 | 15 | 49458581961056 | 21255928548880 |
| 17 | 17 | 469556091240 | 193071429250 |
| 18 | 0 | 38899672900000000 | 14270363086890625 |
| 18 | 2 | 36581031017590020 | 13445699001522515 |
| 18 | 4 | 30325500795158528 | 11208363837947250 |
| 18 | 6 | 21968566711091280 | 8198227931794250 |
| 18 | 8 | 13650958535596032 | 5161786067044425 |
| 18 | 10 | 7067522969907504 | 2721500231937500 |
| 18 | 12 | 2867238347735040 | 1127083505458960 |
| 18 | 14 | 822448354868528 | 332123883576250 |
| 18 | 16 | 116449910627520 | 47881714454000 |
| 18 | 18 | 1079330385764 | 424998178800 |
| 19 | 1 | 118948675359800000 | 41368896818123875 |
| 19 | 3 | 106120023404969472 | 37039375237919210 |
| 19 | 5 | 84045051984373760 | 29543346643453125 |
| 19 | 7 | 58485829348541952 | 20786474691581550 |
| 19 | 9 | 35068146592950016 | 12645050607075250 |
| 19 | 11 | 17584111420907520 | 6464591863931000 |
| 19 | 13 | 6930583227058944 | 2603851247237800 |
| 19 | 15 | 1936449320273760 | 748151788343750 |
| 19 | 17 | 267673935669472 | 105399548342400 |
| 19 | 19 | 2426800117504 | 915906113500 |
| 20 | 0 | 363725098825987600 | 119925864081253321 |
| 20 | 2 | 345065075041280000 | 113960463658434250 |
| 20 | 4 | 294104389038729990 | 97629512918380625 |
| 20 | 6 | 223748987933097984 | 74906678908884375 |
| 20 | 8 | 150245100507542976 | 50921526193401500 |
| 20 | 10 | 87250115725230080 | 30036775420481284 |
| 20 | 12 | 42503668302548472 | 14934861087267500 |
| 20 | 14 | 16318013282772480 | 5865510020615000 |
| 20 | 16 | 4451158510890736 | 1646867942850000 |
| 20 | 18 | 601846429140992 | 227144716148000 |
| 20 | 20 | 5346409013164 | 1935631163543 |
| 21 | 1 | 1055151124139407360 | 330364310545272094 |
| 21 | 3 | 956324262070100000 | 300380459644890625 |
| 21 | 5 | 782979578694686016 | 247538055334990875 |
| 21 | 7 | 574792040309972992 | 183502131492468750 |
| 21 | 9 | 373812239311994880 | 120957874655206904 |
| 21 | 11 | 210897775234453888 | 69392635738887370 |
| 21 | 13 | 100074611501616240 | 33642696546812500 |
| 21 | 15 | 37508889566070528 | 12911444671944000 |
| 21 | 17 | 10008123684586496 | 3549136189812500 |

| $\Delta$ | $\lvert s\rvert$ | States at each spin | Primaries at each spin |
|---|---|---|---|
| 21 | 19 | 1325909435264672 | 480036528558664 |
| 21 | 21 | 11558035326944 | 4017215444008 |
| 22 | 0 | 3060948772482973696 | 910067052826127716 |
| 22 | 2 | 2924279195871506200 | 870784307698907875 |
| 22 | 4 | 2545974816147840000 | 761609810583496875 |
| 22 | 6 | 2011407665868544008 | 606404681680313750 |
| 22 | 8 | 1430091889859624960 | 435886931898469500 |
| 22 | 10 | 903565215599043168 | 279443635949530220 |
| 22 | 12 | 496557444710664960 | 156315842049389750 |
| 22 | 14 | 230033367796338264 | 74055932655900000 |
| 22 | 16 | 84336157683499008 | 27825227728130000 |
| 22 | 18 | 22048590770288336 | 7500570758729125 |
| 22 | 20 | 2866392761082112 | 996269430113984 |
| 22 | 22 | 24551042107480 | 8198039136625 |
| 23 | 1 | 8483210234269496320 | 2398782444891810250 |
| 23 | 3 | 7785163969340110080 | 2207859567262392225 |
| 23 | 5 | 6540391859063920000 | 1865748497242093750 |
| 23 | 7 | 5004414794277335040 | 1440440359121105420 |
| 23 | 9 | 3456765592174637056 | 1007010328676853750 |
| 23 | 11 | 2127438443049082560 | 629482751212388500 |
| 23 | 13 | 1141396200266973056 | 344090000507307600 |
| 23 | 15 | 517214201843157504 | 159596640277375000 |
| 23 | 17 | 185798406025475328 | 58804474748436340 |
| 23 | 19 | 47665337688317056 | 15566709845531000 |
| 23 | 21 | 6088658442655040 | 2033113705883000 |
| 23 | 23 | 51301080086528 | 16468952591975 |
| 24 | 0 | 23510637135046996900 | 6322783799338890625 |
| 24 | 2 | 22584431320173477888 | 6082074198983751150 |
| 24 | 4 | 19999421331115435040 | 5408689085275089250 |
| 24 | 6 | 16272600694169600000 | 4431857992834343500 |
| 24 | 8 | 12096487640053431444 | 3327785747464944350 |
| 24 | 10 | 8138932179373547520 | 2268420356186906250 |
| 24 | 12 | 4890169669317928416 | 1385647912228725600 |
| 24 | 14 | 2566350831460950016 | 741542318954664500 |
| 24 | 16 | 1139458767339590664 | 337283730247367750 |
| 24 | 18 | 401664843681957888 | 122043005188963040 |
| 24 | 20 | 101248497651247520 | 31767401654421875 |
| 24 | 22 | 12722667861458944 | 4084300242809800 |
| 24 | 24 | 105561758786885 | 32600946798875 |

## S23 Certified thermal intervals and explicit tail certificates

The source `c8_physics_20260927/data/thermal.json` stores eighty-digit directed-rounding intervals. Here every endpoint is rounded *outward* to twelve digits after the decimal point, so the printed interval still encloses the stored interval. These wider intervals are conditional on positivity and full modular invariance, as explained in Section 2. The table is a transcription and outward rounding of saved output, not a fresh thermal calculation.

Set $t = \beta v/L$, $b = 2\pi t$, and $E = \Delta - 2/3$. The six columns printed as individual rows are $Z$, $\log Z$, $F/(2\pi v/L) = -\log Z/b$, $\langle E\rangle$, $C_V = b^2\,\mathrm{Var}(E)$, and $S = \log Z + b\langle E\rangle$. The energy zero at $t = 1$ is exact under the modular premise.

### S23.1 diagonal_U

| $t$ | Observable | Enclosing interval |
|---|---|---|
| 1/2 | $Z$ | [4367.819470935009, 4367.819470935010] |
| 1/2 | $\log Z$ | [8.382019186656, 8.382019186657] |
| 1/2 | $F/(2\pi v/L)$ | [−2.668079573295, −2.668079573294] |
| 1/2 | $\langle E\rangle$ | [2.653128317037, 2.653128317038] |
| 1/2 | $C_V$ | [17.089150396217, 17.089150396220] |
| 1/2 | $S$ | [16.717067616493, 16.717067616494] |
| 3/4 | $Z$ | [303.785760211651, 303.785760211652] |
| 3/4 | $\log Z$ | [5.716322716817, 5.716322716818] |
| 3/4 | $F/(2\pi v/L)$ | [−1.213041355587, −1.213041355586] |
| 3/4 | $\langle E\rangle$ | [0.983857183417, 0.983857183418] |
| 3/4 | $C_V$ | [16.343538404752, 16.343538404753] |
| 3/4 | $S$ | [10.352640466228, 10.352640466229] |
| 1 | $Z$ | [147.219258296160, 147.219258296161] |
| 1 | $\log Z$ | [4.991923028553, 4.991923028554] |
| 1 | $F/(2\pi v/L)$ | [−0.794489225529, −0.794489225528] |
| 1 | $\langle E\rangle$ | [0.000000000000, 0.000000000000] |
| 1 | $C_V$ | [19.266967842265, 19.266967842266] |
| 1 | $S$ | [4.991923028553, 4.991923028554] |
| 3/2 | $Z$ | [565.179187258049, 565.179187258050] |
| 3/2 | $\log Z$ | [6.337142826455, 6.337142826456] |
| 3/2 | $F/(2\pi v/L)$ | [−0.672391737274, −0.672391737273] |
| 3/2 | $\langle E\rangle$ | [−0.621644247673, −0.621644247672] |
| 3/2 | $C_V$ | [3.473060477082, 3.473060477083] |
| 3/2 | $S$ | [0.478283821550, 0.478283821551] |
| 2 | $Z$ | [4367.819470935009, 4367.819470935010] |
| 2 | $\log Z$ | [8.382019186656, 8.382019186657] |
| 2 | $F/(2\pi v/L)$ | [−0.667019893324, −0.667019893323] |
| 2 | $\langle E\rangle$ | [−0.663282079260, −0.663282079259] |
| 2 | $C_V$ | [0.419053536545, 0.419053536546] |
| 2 | $S$ | [0.046970756819, 0.046970756820] |

The following positive bounds are rounded upward to eight significant digits. $H$ bounds the partition-function tail at the modular evaluation temperature. $M_1, M_2$ bound omitted energy moments at the requested temperature; these use the separate moment-tail certificate and its reference temperature $t_0$. The exact cutoff is $D = 24$, hence $E_D = 70/3$.

| $t$ | $t_{\mathrm{eval}}$ | $H$ upper bound | $t_0$ | $M_1$ upper bound | $M_2$ upper bound |
|---|---|---|---|---|---|
| 1/2 | 2 | $3.2646838 \times 10^{-108}$ | 9/20 | $2.9584393 \times 10^{-11}$ | $6.9030250 \times 10^{-10}$ |
| 3/4 | 4/3 | $4.6643730 \times 10^{-66}$ | 27/40 | $1.7808481 \times 10^{-27}$ | $4.1553123 \times 10^{-26}$ |
| 1 | 1 | $5.0964961 \times 10^{-45}$ | 9/10 | $1.3778210 \times 10^{-43}$ | $3.2149157 \times 10^{-42}$ |
| 3/2 | 3/2 | $1.3745711 \times 10^{-76}$ | 27/20 | $2.7126666 \times 10^{-75}$ | $6.3295552 \times 10^{-74}$ |
| 2 | 2 | $3.2646838 \times 10^{-108}$ | 9/5 | $9.9405283 \times 10^{-86}$ | $2.3194566 \times 10^{-84}$ |

### S23.2 E8_full

| $t$ | Observable | Enclosing interval |
|---|---|---|
| 1/2 | $Z$ | [4355.999999999999, 4356.000000000001] |
| 1/2 | $\log Z$ | [8.379309484052, 8.379309484053] |
| 1/2 | $F/(2\pi v/L)$ | [−2.667217048168, −2.667217048167] |
| 1/2 | $\langle E\rangle$ | [2.659752956672, 2.659752956673] |
| 1/2 | $C_V$ | [16.984458803503, 16.984458803505] |
| 1/2 | $S$ | [16.735169833099, 16.735169833100] |
| 3/4 | $Z$ | [297.792227617174, 297.792227617175] |
| 3/4 | $\log Z$ | [5.696396020571, 5.696396020572] |
| 3/4 | $F/(2\pi v/L)$ | [−1.208812779311, −1.208812779310] |
| 3/4 | $\langle E\rangle$ | [0.991915807903, 0.991915807904] |
| 3/4 | $C_V$ | [16.622062155726, 16.622062155727] |
| 3/4 | $S$ | [10.370689143206, 10.370689143207] |
| 1 | $Z$ | [143.999999999999, 144.000000000001] |
| 1 | $\log Z$ | [4.969813299576, 4.969813299577] |
| 1 | $F/(2\pi v/L)$ | [−0.790970352872, −0.790970352871] |
| 1 | $\langle E\rangle$ | [0.000000000000, 0.000000000000] |
| 1 | $C_V$ | [19.157071797970, 19.157071797971] |
| 1 | $S$ | [4.969813299576, 4.969813299577] |
| 3/2 | $Z$ | [557.169604095784, 557.169604095785] |
| 3/2 | $\log Z$ | [6.322869689261, 6.322869689262] |
| 3/2 | $F/(2\pi v/L)$ | [−0.670877310382, −0.670877310381] |
| 3/2 | $\langle E\rangle$ | [−0.627320675215, −0.627320675214] |
| 3/2 | $C_V$ | [3.435582054611, 3.435582054612] |
| 3/2 | $S$ | [0.410511615165, 0.410511615166] |
| 2 | $Z$ | [4355.999999999999, 4356.000000000001] |
| 2 | $\log Z$ | [8.379309484052, 8.379309484053] |
| 2 | $F/(2\pi v/L)$ | [−0.666804262042, −0.666804262041] |
| 2 | $\langle E\rangle$ | [−0.664938239169, −0.664938239168] |
| 2 | $C_V$ | [0.272738105411, 0.272738105412] |
| 2 | $S$ | [0.023449135006, 0.023449135007] |

The following positive bounds are rounded upward to eight significant digits. $H$ bounds the partition-function tail at the modular evaluation temperature. $M_1, M_2$ bound omitted energy moments at the requested temperature; these use the separate moment-tail certificate and its reference temperature $t_0$. The exact cutoff is $D = 24$, hence $E_D = 70/3$.

| $t$ | $t_{\mathrm{eval}}$ | $H$ upper bound | $t_0$ | $M_1$ upper bound | $M_2$ upper bound |
|---|---|---|---|---|---|
| 1/2 | 2 | $1.2347657 \times 10^{-108}$ | 9/20 | $1.3306332 \times 10^{-11}$ | $3.1048108 \times 10^{-10}$ |
| 3/4 | 4/3 | $6.5056378 \times 10^{-67}$ | 27/40 | $3.4187009 \times 10^{-28}$ | $7.9769687 \times 10^{-27}$ |
| 1 | 1 | $2.3149019 \times 10^{-46}$ | 9/10 | $9.7501039 \times 10^{-45}$ | $2.2750243 \times 10^{-43}$ |
| 3/2 | 3/2 | $2.7314984 \times 10^{-77}$ | 27/20 | $3.9388323 \times 10^{-76}$ | $9.1906085 \times 10^{-75}$ |
| 2 | 2 | $1.2347657 \times 10^{-108}$ | 9/5 | $1.8934340 \times 10^{-86}$ | $4.4180126 \times 10^{-85}$ |

## S24 Complete coordinate atlas for the small verification fields

All 337 rows of `c8_reviewability_20260927/grade2_fields.tsv` are printed below, followed by the additional 434 rows for the two weight-three structural fields in `c8_reviewability_20260927/structural_fields.tsv`. The latter also repeats the 91 rows of $p, z$; those repetitions agree exactly with the first table and are printed only once. No coefficient is rounded.

Use $s=\sqrt{13}>0$, $t=i\sqrt{3}$, in the ordered coefficient basis $(1,s,t,st)$. In a row $(\lambda;\mathcal{O};c)$, the contribution is $c\prod_{j\in\mathcal{O}}\alpha_{j \bmod 8}(-1-\lfloor j/8\rfloor)e^{\lambda}$. The momentum is always an eight-tuple in the ordered simple-root basis. An empty oscillator list is denoted $\varnothing$. Oscillators commute; the lists have no factorial or divided-power normalization. Each field is the sum of every row in its subsection. Unlisted states have zero coefficient. These are actual saved vectors, not new membership proofs. The exact products and membership equations belong to the integrated verification guide and the finite construction appendices.

## S24.1 Field u

$u$: 3 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,0,-1,1,1)$ | $\varnothing$ | $1$ |
| $(0,0,0,1,1,-1,0,0)$ | $\varnothing$ | $\frac{1}{20}s+\frac{3}{20}t$ |
| $(1,1,0,0,0,-1,0,0)$ | $\varnothing$ | $\frac{1}{20}s-\frac{3}{20}t$ |

## S24.2 Field omega

$\omega_B$: 24 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{2}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $\frac{5}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $\frac{4}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $\frac{2}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,5)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $\frac{5}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2)$ | $4$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $\frac{8}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $\frac{4}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,5)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2)$ | $3$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3)$ | $4$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,5)$ | $3$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $\frac{5}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $\frac{5}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,5)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{2}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,5)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(5,5)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6)$ | $\frac{1}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7)$ | $\frac{1}{3}$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7)$ | $\frac{1}{3}$ |

## S24.3 Field omegaL

$\omega_L$: 21 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{1}{4}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2)$ | $\frac{3}{2}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $\frac{1}{2}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,5)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2)$ | $3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,5)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2)$ | $\frac{9}{4}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3)$ | $3$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4)$ | $\frac{3}{2}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,5)$ | $3$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,5)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{1}{4}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,5)$ | $1$ |
| $(0,0,0,0,0,0,0,0)$ | $(5,5)$ | $1$ |

## S24.4 Field omegaR

$\omega_R$: 60 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(-1,0,0,0,0,0,0,1)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(-1,0,0,0,0,0,1,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(-1,0,0,0,1,0,0,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(-1,0,0,1,0,0,0,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(-1,0,1,0,0,0,0,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(0,-1,0,0,0,0,0,1)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(0,-1,0,0,0,0,1,0)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(0,-1,0,0,1,0,0,0)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(0,-1,0,1,0,0,0,0)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(0,0,-1,0,0,0,0,1)$ | $\varnothing$ | $-\frac{9}{41}$ |
| $(0,0,-1,0,0,0,1,0)$ | $\varnothing$ | $-\frac{9}{41}$ |
| $(0,0,-1,0,1,0,0,0)$ | $\varnothing$ | $-\frac{9}{41}$ |
| $(0,0,0,-1,0,0,0,1)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(0,0,0,-1,0,0,1,0)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(0,0,0,0,-1,0,0,1)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(0,0,0,0,-1,0,1,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{35}{164}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $\frac{20}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2)$ | $\frac{45}{82}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $\frac{20}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $\frac{25}{82}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6)$ | $\frac{5}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7)$ | $\frac{5}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $\frac{20}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2)$ | $\frac{36}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $\frac{32}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $\frac{20}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6)$ | $\frac{8}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,7)$ | $\frac{8}{41}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,0,0,0,0)$ | $(2,2)$ | $\frac{99}{164}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3)$ | $\frac{36}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4)$ | $\frac{45}{82}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,6)$ | $\frac{9}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,7)$ | $\frac{9}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $\frac{20}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $\frac{20}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6)$ | $\frac{8}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7)$ | $\frac{8}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{35}{164}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6)$ | $\frac{5}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7)$ | $\frac{5}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6)$ | $\frac{2}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7)$ | $\frac{2}{41}$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7)$ | $\frac{2}{41}$ |
| $(0,0,0,0,1,0,-1,0)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(0,0,0,0,1,0,0,-1)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(0,0,0,1,0,0,-1,0)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(0,0,0,1,0,0,0,-1)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(0,0,1,0,-1,0,0,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(0,0,1,0,0,0,-1,0)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(0,0,1,0,0,0,0,-1)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(0,1,0,-1,0,0,0,0)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(0,1,0,0,-1,0,0,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(0,1,0,0,0,0,-1,0)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(0,1,0,0,0,0,0,-1)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(1,0,-1,0,0,0,0,0)$ | $\varnothing$ | $-\frac{9}{41}$ |
| $(1,0,0,-1,0,0,0,0)$ | $\varnothing$ | $-\frac{8}{41}$ |
| $(1,0,0,0,-1,0,0,0)$ | $\varnothing$ | $-\frac{5}{41}$ |
| $(1,0,0,0,0,0,-1,0)$ | $\varnothing$ | $-\frac{2}{41}$ |
| $(1,0,0,0,0,0,0,-1)$ | $\varnothing$ | $-\frac{2}{41}$ |

## S24.5 Field e

$e$: 70 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(-1,-1,0,0,0,0,1,1)$ | $\varnothing$ | $\frac{10}{93}s$ |
| $(-1,-1,0,1,1,0,0,0)$ | $\varnothing$ | $-\frac{10}{93}$ |
| $(-1,0,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{260}{3813}-\frac{5}{93}s$ |
| $(-1,0,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{260}{3813}+\frac{5}{93}s$ |
| $(-1,0,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{5}{41}$ |
| $(-1,0,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(-1,0,1,0,0,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(0,-1,-1,0,0,0,1,1)$ | $\varnothing$ | $\frac{6}{31}s$ |
| $(0,-1,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{416}{3813}-\frac{4}{93}s$ |
| $(0,-1,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{416}{3813}+\frac{4}{93}s$ |
| $(0,-1,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{88}{3813}$ |
| $(0,-1,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{166}{1271}$ |
| $(0,0,-1,-1,0,0,1,1)$ | $\varnothing$ | $\frac{6}{31}s$ |
| $(0,0,-1,0,0,0,0,1)$ | $\varnothing$ | $\frac{156}{1271}$ |
| $(0,0,-1,0,0,0,1,0)$ | $\varnothing$ | $\frac{156}{1271}$ |
| $(0,0,-1,0,1,0,0,0)$ | $\varnothing$ | $\frac{33}{1271}$ |
| $(0,0,0,-1,-1,0,1,1)$ | $\varnothing$ | $\frac{10}{93}s$ |
| $(0,0,0,-1,0,0,0,1)$ | $\varnothing$ | $\frac{416}{3813}+\frac{4}{93}s$ |
| $(0,0,0,-1,0,0,1,0)$ | $\varnothing$ | $\frac{416}{3813}-\frac{4}{93}s$ |
| $(0,0,0,0,-1,0,0,1)$ | $\varnothing$ | $\frac{260}{3813}+\frac{5}{93}s$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,-1,0,1,0)$ | $\varnothing$ | $\frac{260}{3813} - \frac{5}{93}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{890}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2)$ | $\frac{20}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $-\frac{455}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $-\frac{925}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6)$ | $-\frac{260}{3813} - \frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7)$ | $-\frac{260}{3813} + \frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2)$ | $\frac{73}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $-\frac{646}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $-\frac{455}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6)$ | $-\frac{416}{3813} - \frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,7)$ | $-\frac{416}{3813} + \frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2)$ | $\frac{63}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3)$ | $\frac{73}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4)$ | $\frac{20}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,6)$ | $-\frac{156}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,7)$ | $-\frac{156}{1271}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $\frac{980}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6)$ | $-\frac{416}{3813} + \frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7)$ | $-\frac{416}{3813} - \frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{890}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6)$ | $-\frac{260}{3813} + \frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7)$ | $-\frac{260}{3813} - \frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6)$ | $\frac{1820}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7)$ | $\frac{1820}{11439}$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7)$ | $\frac{1820}{11439}$ |
| $(0,0,0,0,1,0,-1,0)$ | $\varnothing$ | $\frac{104}{3813} - \frac{2}{93}s$ |
| $(0,0,0,0,1,0,0,-1)$ | $\varnothing$ | $\frac{104}{3813} + \frac{2}{93}s$ |
| $(0,0,0,1,0,0,-1,0)$ | $\varnothing$ | $\frac{104}{3813} - \frac{1}{93}s$ |
| $(0,0,0,1,0,0,0,-1)$ | $\varnothing$ | $\frac{104}{3813} + \frac{1}{93}s$ |
| $(0,0,0,1,1,0,-1,-1)$ | $\varnothing$ | $\frac{1}{93}s$ |
| $(0,0,1,0,-1,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(0,0,1,0,0,0,-1,0)$ | $\varnothing$ | $\frac{104}{3813}$ |
| $(0,0,1,0,0,0,0,-1)$ | $\varnothing$ | $\frac{104}{3813}$ |
| $(0,0,1,1,0,0,-1,-1)$ | $\varnothing$ | $\frac{1}{93}s$ |
| $(0,1,0,-1,0,0,0,0)$ | $\varnothing$ | $\frac{166}{1271}$ |
| $(0,1,0,0,-1,0,0,0)$ | $\varnothing$ | $\frac{55}{3813}$ |
| $(0,1,0,0,0,0,-1,0)$ | $\varnothing$ | $\frac{104}{3813} + \frac{1}{93}s$ |
| $(0,1,0,0,0,0,0,-1)$ | $\varnothing$ | $\frac{104}{3813} - \frac{1}{93}s$ |
| $(0,1,1,0,0,0,-1,-1)$ | $\varnothing$ | $\frac{1}{93}s$ |
| $(1,0,-1,0,0,0,0,0)$ | $\varnothing$ | $\frac{33}{1271}$ |
| $(1,0,0,-1,0,0,0,0)$ | $\varnothing$ | $\frac{88}{3813}$ |
| $(1,0,0,0,-1,0,0,0)$ | $\varnothing$ | $\frac{5}{41}$ |
| $(1,0,0,0,0,0,-1,0)$ | $\varnothing$ | $\frac{104}{3813} + \frac{2}{93}s$ |
| $(1,0,0,0,0,0,0,-1)$ | $\varnothing$ | $\frac{104}{3813} - \frac{2}{93}s$ |
| $(1,1,0,-1,-1,0,0,0)$ | $\varnothing$ | $-\frac{10}{93}$ |
| $(1,1,0,0,0,0,-1,-1)$ | $\varnothing$ | $\frac{1}{93}s$ |

## S24.6 Field f

$f$: 68 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(-1,-1,0,0,0,0,1,1)$ | $\varnothing$ | $-\frac{10}{93}s$ |
| $(-1,-1,0,1,1,0,0,0)$ | $\varnothing$ | $\frac{10}{93}$ |
| $(-1,0,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{5}{93}+\frac{5}{93}s$ |
| $(-1,0,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{5}{93}-\frac{5}{93}s$ |
| $(-1,0,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{10}{93}$ |
| $(-1,0,1,0,0,0,0,0)$ | $\varnothing$ | $\frac{10}{93}$ |
| $(0,-1,-1,0,0,0,1,1)$ | $\varnothing$ | $-\frac{6}{31}s$ |
| $(0,-1,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{8}{93}+\frac{4}{93}s$ |
| $(0,-1,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{8}{93}-\frac{4}{93}s$ |
| $(0,-1,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{16}{93}$ |
| $(0,-1,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{2}{31}$ |
| $(0,0,-1,-1,0,0,1,1)$ | $\varnothing$ | $-\frac{6}{31}s$ |
| $(0,0,-1,0,0,0,0,1)$ | $\varnothing$ | $\frac{3}{31}$ |
| $(0,0,-1,0,0,0,1,0)$ | $\varnothing$ | $\frac{3}{31}$ |
| $(0,0,-1,0,1,0,0,0)$ | $\varnothing$ | $\frac{6}{31}$ |
| $(0,0,0,-1,-1,0,1,1)$ | $\varnothing$ | $-\frac{10}{93}s$ |
| $(0,0,0,-1,0,0,0,1)$ | $\varnothing$ | $\frac{8}{93}-\frac{4}{93}s$ |
| $(0,0,0,-1,0,0,1,0)$ | $\varnothing$ | $\frac{8}{93}+\frac{4}{93}s$ |
| $(0,0,0,0,-1,0,0,1)$ | $\varnothing$ | $\frac{5}{93}-\frac{5}{93}s$ |
| $(0,0,0,0,-1,0,1,0)$ | $\varnothing$ | $\frac{5}{93}+\frac{5}{93}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{35}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $\frac{26}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2)$ | $-\frac{2}{31}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $-\frac{32}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $-\frac{16}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6)$ | $-\frac{5}{93}+\frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7)$ | $-\frac{5}{93}-\frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $\frac{26}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2)$ | $\frac{2}{31}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $-\frac{16}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $-\frac{32}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6)$ | $-\frac{8}{93}+\frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,7)$ | $-\frac{8}{93}-\frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2)$ | $\frac{3}{31}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3)$ | $\frac{2}{31}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4)$ | $-\frac{2}{31}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,6)$ | $-\frac{3}{31}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,7)$ | $-\frac{3}{31}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $\frac{26}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $\frac{26}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6)$ | $-\frac{8}{93}-\frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7)$ | $-\frac{8}{93}+\frac{4}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{35}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6)$ | $-\frac{5}{93}-\frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7)$ | $-\frac{5}{93}+\frac{5}{279}s$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6)$ | $\frac{35}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7)$ | $\frac{35}{279}$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7)$ | $\frac{35}{279}$ |
| $(0,0,0,0,1,0,-1,0)$ | $\varnothing$ | $\frac{2}{93}+\frac{2}{93}s$ |
| $(0,0,0,0,1,0,0,-1)$ | $\varnothing$ | $\frac{2}{93}-\frac{2}{93}s$ |
| $(0,0,0,1,0,0,-1,0)$ | $\varnothing$ | $\frac{2}{93}+\frac{1}{93}s$ |
| $(0,0,0,1,0,0,0,-1)$ | $\varnothing$ | $\frac{2}{93}-\frac{1}{93}s$ |
| $(0,0,0,1,1,0,-1,-1)$ | $\varnothing$ | $-\frac{1}{93}s$ |
| $(0,0,1,0,-1,0,0,0)$ | $\varnothing$ | $\frac{10}{93}$ |
| $(0,0,1,0,0,0,-1,0)$ | $\varnothing$ | $\frac{2}{93}$ |
| $(0,0,1,0,0,0,0,-1)$ | $\varnothing$ | $\frac{2}{93}$ |
| $(0,0,1,1,0,0,-1,-1)$ | $\varnothing$ | $-\frac{1}{93}s$ |
| $(0,1,0,-1,0,0,0,0)$ | $\varnothing$ | $\frac{2}{31}$ |
| $(0,1,0,0,-1,0,0,0)$ | $\varnothing$ | $\frac{10}{93}$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,1,0,0,0,0,-1,0)$ | $\varnothing$ | $\frac{2}{93}-\frac{1}{93}s$ |
| $(0,1,0,0,0,0,0,-1)$ | $\varnothing$ | $\frac{2}{93}+\frac{1}{93}s$ |
| $(0,1,1,0,0,0,-1,-1)$ | $\varnothing$ | $-\frac{1}{93}s$ |
| $(1,0,-1,0,0,0,0,0)$ | $\varnothing$ | $\frac{6}{31}$ |
| $(1,0,0,-1,0,0,0,0)$ | $\varnothing$ | $\frac{16}{93}$ |
| $(1,0,0,0,0,0,-1,0)$ | $\varnothing$ | $\frac{2}{93}-\frac{2}{93}s$ |
| $(1,0,0,0,0,0,0,-1)$ | $\varnothing$ | $\frac{2}{93}+\frac{2}{93}s$ |
| $(1,1,0,-1,-1,0,0,0)$ | $\varnothing$ | $\frac{10}{93}$ |
| $(1,1,0,0,0,0,-1,-1)$ | $\varnothing$ | $-\frac{1}{93}s$ |

## S24.7 Field p

$p$: 27 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(-1,-1,0,0,0,0,1,1)$ | $\varnothing$ | $-\frac{1}{10}s-\frac{3}{10}t$ |
| $(-1,-1,0,1,1,0,0,0)$ | $\varnothing$ | $\frac{7}{100}-\frac{3}{100}st$ |
| $(0,0,0,-1,-1,0,1,1)$ | $\varnothing$ | $-\frac{1}{10}s+\frac{3}{10}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $\frac{2}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,7)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $\frac{2}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7)$ | $-\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6)$ | $\frac{1}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7)$ | $\frac{2}{15}$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7)$ | $\frac{1}{15}$ |
| $(0,0,0,1,1,0,-1,-1)$ | $\varnothing$ | $-\frac{1}{100}s-\frac{3}{100}t$ |
| $(1,1,0,-1,-1,0,0,0)$ | $\varnothing$ | $\frac{7}{100}+\frac{3}{100}st$ |
| $(1,1,0,0,0,0,-1,-1)$ | $\varnothing$ | $-\frac{1}{100}s+\frac{3}{100}t$ |

## S24.8 Field z

$z$: 64 nonzero monomials.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(-1,-1,0,0,0,0,1,1)$ | $\varnothing$ | $\frac{1}{15}s+\frac{1}{5}t$ |
| $(-1,0,0,0,0,0,0,1)$ | $\varnothing$ | $-\frac{1}{60}-\frac{1}{20}s+\frac{1}{60}t-\frac{1}{60}st$ |
| $(-1,0,0,0,0,0,1,0)$ | $\varnothing$ | $-\frac{1}{60}+\frac{1}{20}s-\frac{1}{60}t-\frac{1}{60}st$ |
| $(-1,0,0,0,1,0,0,0)$ | $\varnothing$ | $\frac{19}{150}-\frac{1}{150}st$ |
| $(-1,0,1,0,0,0,0,0)$ | $\varnothing$ | $\frac{1}{90}+\frac{1}{90}st$ |
| $(0,-1,-1,0,0,0,1,1)$ | $\varnothing$ | $\frac{1}{10}s-\frac{1}{10}t$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,-1,0,0,0,0,0,1)$ | $\varnothing$ | $\frac{1}{10}-\frac{1}{30}s+\frac{1}{30}t+\frac{1}{30}st$ |
| $(0,-1,0,0,0,0,1,0)$ | $\varnothing$ | $\frac{1}{10}+\frac{1}{30}s-\frac{1}{30}t+\frac{1}{30}st$ |
| $(0,-1,0,1,0,0,0,0)$ | $\varnothing$ | $\frac{1}{12}-\frac{1}{60}st$ |
| $(0,0,-1,-1,0,0,1,1)$ | $\varnothing$ | $\frac{1}{10}s+\frac{1}{10}t$ |
| $(0,0,-1,0,0,0,0,1)$ | $\varnothing$ | $\frac{1}{10}-\frac{1}{10}t$ |
| $(0,0,-1,0,0,0,1,0)$ | $\varnothing$ | $\frac{1}{10}+\frac{1}{10}t$ |
| $(0,0,-1,0,1,0,0,0)$ | $\varnothing$ | $\frac{1}{50}+\frac{1}{50}st$ |
| $(0,0,0,-1,-1,0,1,1)$ | $\varnothing$ | $\frac{1}{15}s-\frac{1}{5}t$ |
| $(0,0,0,-1,0,0,0,1)$ | $\varnothing$ | $\frac{1}{10}+\frac{1}{30}s+\frac{1}{30}t-\frac{1}{30}st$ |
| $(0,0,0,-1,0,0,1,0)$ | $\varnothing$ | $\frac{1}{10}-\frac{1}{30}s-\frac{1}{30}t-\frac{1}{30}st$ |
| $(0,0,0,0,-1,0,0,1)$ | $\varnothing$ | $-\frac{1}{60}+\frac{1}{20}s+\frac{1}{60}t+\frac{1}{60}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $\varnothing$ | $-\frac{1}{60}-\frac{1}{20}s-\frac{1}{60}t+\frac{1}{60}st$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0)$ | $\frac{11}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1)$ | $\frac{4}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2)$ | $-\frac{2}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3)$ | $\frac{4}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4)$ | $-\frac{4}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6)$ | $-\frac{4}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7)$ | $-\frac{4}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1)$ | $\frac{11}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2)$ | $\frac{2}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3)$ | $-\frac{4}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4)$ | $\frac{4}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6)$ | $-\frac{14}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,7)$ | $-\frac{14}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2)$ | $\frac{2}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3)$ | $\frac{2}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4)$ | $-\frac{2}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,6)$ | $-\frac{2}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,7)$ | $-\frac{2}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3)$ | $\frac{11}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4)$ | $\frac{4}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6)$ | $-\frac{14}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7)$ | $-\frac{14}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4)$ | $\frac{11}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6)$ | $-\frac{4}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7)$ | $-\frac{4}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6)$ | $\frac{7}{65}$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7)$ | $\frac{16}{195}$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7)$ | $\frac{7}{65}$ |
| $(0,0,0,0,1,0,-1,0)$ | $\varnothing$ | $-\frac{1}{150}-\frac{1}{50}s+\frac{1}{150}t-\frac{1}{150}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $\varnothing$ | $-\frac{1}{150}+\frac{1}{50}s-\frac{1}{150}t-\frac{1}{150}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $\varnothing$ | $\frac{1}{40}-\frac{1}{120}s+\frac{1}{120}t+\frac{1}{120}st$ |
| $(0,0,0,1,0,0,0,-1)$ | $\varnothing$ | $\frac{1}{40}+\frac{1}{120}s-\frac{1}{120}t+\frac{1}{120}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | $\varnothing$ | $\frac{1}{150}s+\frac{1}{50}t$ |
| $(0,0,1,0,-1,0,0,0)$ | $\varnothing$ | $\frac{1}{90}-\frac{1}{90}st$ |
| $(0,0,1,0,0,0,-1,0)$ | $\varnothing$ | $\frac{1}{45}-\frac{1}{45}t$ |
| $(0,0,1,0,0,0,0,-1)$ | $\varnothing$ | $\frac{1}{45}+\frac{1}{45}t$ |
| $(0,0,1,1,0,0,-1,-1)$ | $\varnothing$ | $\frac{1}{180}s-\frac{1}{180}t$ |
| $(0,1,0,-1,0,0,0,0)$ | $\varnothing$ | $\frac{1}{12}+\frac{1}{60}st$ |
| $(0,1,0,0,0,0,-1,0)$ | $\varnothing$ | $\frac{1}{40}+\frac{1}{120}s+\frac{1}{120}t-\frac{1}{120}st$ |
| $(0,1,0,0,0,0,0,-1)$ | $\varnothing$ | $\frac{1}{40}-\frac{1}{120}s-\frac{1}{120}t-\frac{1}{120}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | $\varnothing$ | $\frac{1}{180}s+\frac{1}{180}t$ |
| $(1,0,-1,0,0,0,0,0)$ | $\varnothing$ | $\frac{1}{50}-\frac{1}{50}st$ |
| $(1,0,0,0,-1,0,0,0)$ | $\varnothing$ | $\frac{19}{150}+\frac{1}{150}st$ |
| $(1,0,0,0,0,0,-1,0)$ | $\varnothing$ | $-\frac{1}{150}+\frac{1}{50}s+\frac{1}{150}t+\frac{1}{150}st$ |
| $(1,0,0,0,0,0,0,-1)$ | $\varnothing$ | $-\frac{1}{150}-\frac{1}{50}s-\frac{1}{150}t+\frac{1}{150}st$ |
| $(1,1,0,0,0,0,-1,-1)$ | $\varnothing$ | $\frac{1}{150}s-\frac{1}{50}t$ |

### S24.9 Field w_P

$w_{\mathrm{Potts}}$: 92 nonzero monomials. This is the raw Potts weight-three field. Its bilinear self-OPE coefficient is $(w_P)_5 w_P = 8424\mathbf{1}$. The canonical field used in the spin-three OPE formulas and its unit version are

$$a_{\mathrm{Potts}} = \frac{w_P}{\sqrt{31590}}, \qquad (a_{\mathrm{Potts}})_5 a_{\mathrm{Potts}} = \frac{4}{15}\mathbf{1}, \qquad \widehat{a}_{\mathrm{Potts}} = \frac{w_P}{\sqrt{8424}},$$

with the common overall sign chosen consistently. The rows below retain the raw coefficients and have not been divided by either square root.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(-1,-1,0,0,0,0,1,1)$ | $(0)$ | $-\frac{9}{2}s-\frac{27}{2}t$ |
| $(-1,-1,0,0,0,0,1,1)$ | $(1)$ | $-\frac{9}{2}s-\frac{27}{2}t$ |
| $(-1,-1,0,0,0,0,1,1)$ | $(3)$ | $9s+27t$ |
| $(-1,-1,0,0,0,0,1,1)$ | $(4)$ | $9s+27t$ |
| $(-1,-1,0,0,0,0,1,1)$ | $(6)$ | $-\frac{9}{2}s-\frac{27}{2}t$ |
| $(-1,-1,0,0,0,0,1,1)$ | $(7)$ | $-\frac{9}{2}s-\frac{27}{2}t$ |
| $(-1,-1,0,1,1,0,0,0)$ | $(0)$ | $\frac{63}{20}-\frac{27}{20}st$ |
| $(-1,-1,0,1,1,0,0,0)$ | $(1)$ | $\frac{63}{20}-\frac{27}{20}st$ |
| $(-1,-1,0,1,1,0,0,0)$ | $(3)$ | $\frac{63}{20}-\frac{27}{20}st$ |
| $(-1,-1,0,1,1,0,0,0)$ | $(4)$ | $\frac{63}{20}-\frac{27}{20}st$ |
| $(-1,-1,0,1,1,0,0,0)$ | $(6)$ | $-\frac{63}{10}+\frac{27}{10}st$ |
| $(-1,-1,0,1,1,0,0,0)$ | $(7)$ | $-\frac{63}{10}+\frac{27}{10}st$ |
| $(0,0,0,-1,-1,0,1,1)$ | $(0)$ | $9s-27t$ |
| $(0,0,0,-1,-1,0,1,1)$ | $(1)$ | $9s-27t$ |
| $(0,0,0,-1,-1,0,1,1)$ | $(3)$ | $-\frac{9}{2}s+\frac{27}{2}t$ |
| $(0,0,0,-1,-1,0,1,1)$ | $(4)$ | $-\frac{9}{2}s+\frac{27}{2}t$ |
| $(0,0,0,-1,-1,0,1,1)$ | $(6)$ | $-\frac{9}{2}s+\frac{27}{2}t$ |
| $(0,0,0,-1,-1,0,1,1)$ | $(7)$ | $-\frac{9}{2}s+\frac{27}{2}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,0)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,1)$ | $6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,3)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,4)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,1)$ | $6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,3)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,4)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,6)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,7)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,3)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,4)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,6)$ | $12$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,7)$ | $12$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4,4)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4,6)$ | $12$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4,7)$ | $12$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6,7)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,1)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,3)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,4)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3,3)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3,4)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3,6)$ | $12$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,0,0,0,0)$ | $(1,3,7)$ | $12$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4,4)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4,6)$ | $12$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4,7)$ | $12$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6,7)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,7,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,3)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,4)$ | $6$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4,4)$ | $6$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4,6)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4,7)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6,7)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4,4)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6,6)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6,7)$ | $-6$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7,7)$ | $-3$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6,6)$ | $2$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6,7)$ | $6$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7,7)$ | $6$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7,7)$ | $2$ |
| $(0,0,0,1,1,0,-1,-1)$ | $(0)$ | $\frac{9}{10}s+\frac{27}{10}t$ |
| $(0,0,0,1,1,0,-1,-1)$ | $(1)$ | $\frac{9}{10}s+\frac{27}{10}t$ |
| $(0,0,0,1,1,0,-1,-1)$ | $(3)$ | $-\frac{9}{20}s-\frac{27}{20}t$ |
| $(0,0,0,1,1,0,-1,-1)$ | $(4)$ | $-\frac{9}{20}s-\frac{27}{20}t$ |
| $(0,0,0,1,1,0,-1,-1)$ | $(6)$ | $-\frac{9}{20}s-\frac{27}{20}t$ |
| $(0,0,0,1,1,0,-1,-1)$ | $(7)$ | $-\frac{9}{20}s-\frac{27}{20}t$ |
| $(1,1,0,-1,-1,0,0,0)$ | $(0)$ | $\frac{63}{20}+\frac{27}{20}st$ |
| $(1,1,0,-1,-1,0,0,0)$ | $(1)$ | $\frac{63}{20}+\frac{27}{20}st$ |
| $(1,1,0,-1,-1,0,0,0)$ | $(3)$ | $\frac{63}{20}+\frac{27}{20}st$ |
| $(1,1,0,-1,-1,0,0,0)$ | $(4)$ | $\frac{63}{20}+\frac{27}{20}st$ |
| $(1,1,0,-1,-1,0,0,0)$ | $(6)$ | $-\frac{63}{10}-\frac{27}{10}st$ |
| $(1,1,0,-1,-1,0,0,0)$ | $(7)$ | $-\frac{63}{10}-\frac{27}{10}st$ |
| $(1,1,0,0,0,0,-1,-1)$ | $(0)$ | $-\frac{9}{20}s+\frac{27}{20}t$ |
| $(1,1,0,0,0,0,-1,-1)$ | $(1)$ | $-\frac{9}{20}s+\frac{27}{20}t$ |
| $(1,1,0,0,0,0,-1,-1)$ | $(3)$ | $\frac{9}{10}s-\frac{27}{10}t$ |
| $(1,1,0,0,0,0,-1,-1)$ | $(4)$ | $\frac{9}{10}s-\frac{27}{10}t$ |
| $(1,1,0,0,0,0,-1,-1)$ | $(6)$ | $-\frac{9}{20}s+\frac{27}{20}t$ |
| $(1,1,0,0,0,0,-1,-1)$ | $(7)$ | $-\frac{9}{20}s+\frac{27}{20}t$ |

## S24.10 Field w

$w_{\mathrm{PF13}}$: 342 nonzero monomials. The export name `w` denotes the *unnormalized* parafermion weight-three field $w$. The normalized reconstruction generator is $a=\sigma w$, where

$$\sigma^2=-\frac{7+3st}{9223500},\qquad w_5w=(-86086+36894st)\mathbf{1},\qquad a_5a=\frac{8}{15}\mathbf{1}.$$

The coefficients below are those of $w$, without the factor $\sigma$. It is also distinct from the Potts generator `w_P`. The complex number in $w_5w$ is a bilinear self-OPE coefficient; it is not the positive Hermitian squared norm of the raw vector.

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(-1,-1,0,0,0,0,1,1)$ | (0) | $-\frac{63}{2}+\frac{27}{2}st$ |
| $(-1,-1,0,0,0,0,1,1)$ | (1) | $\frac{63}{2}-\frac{27}{2}st$ |
| $(-1,-1,0,0,0,0,1,1)$ | (2) | $7-3st$ |
| $(-1,-1,0,0,0,0,1,1)$ | (3) | $-14+6st$ |
| $(-1,-1,0,0,0,0,1,1)$ | (4) | $14-6st$ |
| $(-1,-1,0,0,0,0,1,1)$ | (6) | $-\frac{7}{2}+\frac{3}{2}st$ |
| $(-1,-1,0,0,0,0,1,1)$ | (7) | $-\frac{7}{2}+\frac{3}{2}st$ |
| $(-1,0,0,-1,0,0,1,1)$ | $\varnothing$ | $-247+13st$ |
| $(-1,0,0,0,0,0,0,1)$ | (0) | $-27+\frac{9}{2}s-9t-\frac{9}{2}st$ |
| $(-1,0,0,0,0,0,0,1)$ | (1) | $-12+2s-4t-2st$ |
| $(-1,0,0,0,0,0,0,1)$ | (2) | $6-s+2t+st$ |
| $(-1,0,0,0,0,0,0,1)$ | (3) | $-12+2s-4t-2st$ |
| $(-1,0,0,0,0,0,0,1)$ | (4) | $12-2s+4t+2st$ |
| $(-1,0,0,0,0,0,0,1)$ | (6) | $36-6s+12t+6st$ |
| $(-1,0,0,0,0,0,0,1)$ | (7) | $-3+\frac{1}{2}s-t-\frac{1}{2}st$ |
| $(-1,0,0,0,0,0,1,0)$ | (0) | $27+\frac{9}{2}s-9t+\frac{9}{2}st$ |
| $(-1,0,0,0,0,0,1,0)$ | (1) | $12+2s-4t+2st$ |
| $(-1,0,0,0,0,0,1,0)$ | (2) | $-6-s+2t-st$ |
| $(-1,0,0,0,0,0,1,0)$ | (3) | $12+2s-4t+2st$ |
| $(-1,0,0,0,0,0,1,0)$ | (4) | $-12-2s+4t-2st$ |
| $(-1,0,0,0,0,0,1,0)$ | (6) | $3+\frac{1}{2}s-t+\frac{1}{2}st$ |
| $(-1,0,0,0,0,0,1,0)$ | (7) | $-36-6s+12t-6st$ |
| $(-1,0,0,0,1,0,0,0)$ | (0) | $\frac{63}{10}s+\frac{99}{10}t$ |
| $(-1,0,0,0,1,0,0,0)$ | (1) | $\frac{14}{5}s+\frac{22}{5}t$ |
| $(-1,0,0,0,1,0,0,0)$ | (2) | $-\frac{7}{5}s-\frac{11}{5}t$ |
| $(-1,0,0,0,1,0,0,0)$ | (3) | $\frac{14}{5}s+\frac{22}{5}t$ |
| $(-1,0,0,0,1,0,0,0)$ | (4) | $\frac{63}{10}s+\frac{99}{10}t$ |
| $(-1,0,0,0,1,0,0,0)$ | (6) | $-\frac{42}{5}s-\frac{66}{5}t$ |
| $(-1,0,0,0,1,0,0,0)$ | (7) | $-\frac{42}{5}s-\frac{66}{5}t$ |
| $(-1,0,1,0,0,0,0,0)$ | (0) | $-3s+6t$ |
| $(-1,0,1,0,0,0,0,0)$ | (1) | $-\frac{4}{3}s+\frac{8}{3}t$ |
| $(-1,0,1,0,0,0,0,0)$ | (2) | $-\frac{11}{3}s+\frac{22}{3}t$ |
| $(-1,0,1,0,0,0,0,0)$ | (3) | $-\frac{4}{3}s+\frac{8}{3}t$ |
| $(-1,0,1,0,0,0,0,0)$ | (4) | $\frac{4}{3}s-\frac{8}{3}t$ |
| $(-1,0,1,0,0,0,0,0)$ | (6) | $4s-8t$ |
| $(-1,0,1,0,0,0,0,0)$ | (7) | $4s-8t$ |
| $(0,-1,-1,0,0,0,1,1)$ | (0) | $-33-3st$ |
| $(0,-1,-1,0,0,0,1,1)$ | (1) | $-\frac{297}{4}-\frac{27}{4}st$ |
| $(0,-1,-1,0,0,0,1,1)$ | (2) | $\frac{363}{4}+\frac{33}{4}st$ |
| $(0,-1,-1,0,0,0,1,1)$ | (3) | $33+3st$ |
| $(0,-1,-1,0,0,0,1,1)$ | (4) | $-33-3st$ |
| $(0,-1,-1,0,0,0,1,1)$ | (6) | $\frac{33}{4}+\frac{3}{4}st$ |
| $(0,-1,-1,0,0,0,1,1)$ | (7) | $\frac{33}{4}+\frac{3}{4}st$ |
| $(0,-1,0,0,-1,0,1,1)$ | $\varnothing$ | $-\frac{325}{2}+\frac{65}{2}st$ |
| $(0,-1,0,0,0,0,0,1)$ | (0) | $11+3s-11t+st$ |
| $(0,-1,0,0,0,0,0,1)$ | (1) | $\frac{99}{4}+\frac{27}{4}s-\frac{99}{4}t+\frac{9}{4}st$ |
| $(0,-1,0,0,0,0,0,1)$ | (2) | $\frac{11}{2}+\frac{3}{2}s-\frac{11}{2}t+\frac{1}{2}st$ |
| $(0,-1,0,0,0,0,0,1)$ | (3) | $-11-3s+11t-st$ |
| $(0,-1,0,0,0,0,0,1)$ | (4) | $11+3s-11t+st$ |
| $(0,-1,0,0,0,0,0,1)$ | (6) | $-\frac{77}{2}-\frac{21}{2}s+\frac{77}{2}t-\frac{7}{2}st$ |
| $(0,-1,0,0,0,0,0,1)$ | (7) | $-\frac{11}{4}-\frac{3}{4}s+\frac{11}{4}t-\frac{1}{4}st$ |
| $(0,-1,0,0,0,0,1,0)$ | (0) | $-11+3s-11t-st$ |
| $(0,-1,0,0,0,0,1,0)$ | (1) | $-\frac{99}{4}+\frac{27}{4}s-\frac{99}{4}t-\frac{9}{4}st$ |
| $(0,-1,0,0,0,0,1,0)$ | (2) | $-\frac{11}{2}+\frac{3}{2}s-\frac{11}{2}t-\frac{1}{2}st$ |
| $(0,-1,0,0,0,0,1,0)$ | (3) | $11-3s+11t+st$ |
| $(0,-1,0,0,0,0,1,0)$ | (4) | $-11+3s-11t-st$ |
| $(0,-1,0,0,0,0,1,0)$ | (6) | $\frac{11}{4}-\frac{3}{4}s+\frac{11}{4}t+\frac{1}{4}st$ |
| $(0,-1,0,0,0,0,1,0)$ | (7) | $\frac{77}{2}-\frac{21}{2}s+\frac{77}{2}t+\frac{7}{2}st$ |
| $(0,-1,0,1,0,0,0,0)$ | (0) | $-\frac{7}{2}s-\frac{1}{2}t$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,-1,0,1,0,0,0,0)$ | (1) | $-\frac{63}{8}s-\frac{9}{8}t$ |
| $(0,-1,0,1,0,0,0,0)$ | (2) | $-\frac{7}{4}s-\frac{1}{4}t$ |
| $(0,-1,0,1,0,0,0,0)$ | (3) | $-\frac{63}{8}s-\frac{9}{8}t$ |
| $(0,-1,0,1,0,0,0,0)$ | (4) | $-\frac{7}{2}s-\frac{1}{2}t$ |
| $(0,-1,0,1,0,0,0,0)$ | (6) | $\frac{49}{4}s+\frac{7}{4}t$ |
| $(0,-1,0,1,0,0,0,0)$ | (7) | $\frac{49}{4}s+\frac{7}{4}t$ |
| $(0,0,-1,-1,0,0,1,1)$ | (0) | $-6-6st$ |
| $(0,0,-1,-1,0,0,1,1)$ | (1) | $6+6st$ |
| $(0,0,-1,-1,0,0,1,1)$ | (2) | $\frac{33}{2}+\frac{33}{2}st$ |
| $(0,0,-1,-1,0,0,1,1)$ | (3) | $-\frac{27}{2}-\frac{27}{2}st$ |
| $(0,0,-1,-1,0,0,1,1)$ | (4) | $-6-6st$ |
| $(0,0,-1,-1,0,0,1,1)$ | (6) | $\frac{3}{2}+\frac{3}{2}st$ |
| $(0,0,-1,-1,0,0,1,1)$ | (7) | $\frac{3}{2}+\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,0,1)$ | (0) | $-\frac{27}{2}-\frac{3}{2}s-\frac{9}{2}t+\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,0,1)$ | (1) | $\frac{27}{2}+\frac{3}{2}s+\frac{9}{2}t-\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,0,1)$ | (2) | $\frac{297}{8}+\frac{33}{8}s+\frac{99}{8}t-\frac{33}{8}st$ |
| $(0,0,-1,0,0,0,0,1)$ | (3) | $\frac{27}{2}+\frac{3}{2}s+\frac{9}{2}t-\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,0,1)$ | (4) | $-\frac{27}{2}-\frac{3}{2}s-\frac{9}{2}t+\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,0,1)$ | (6) | $-\frac{81}{2}-\frac{9}{2}s-\frac{27}{2}t+\frac{9}{2}st$ |
| $(0,0,-1,0,0,0,0,1)$ | (7) | $\frac{27}{8}+\frac{3}{8}s+\frac{9}{8}t-\frac{3}{8}st$ |
| $(0,0,-1,0,0,0,1,0)$ | (0) | $\frac{27}{2}-\frac{3}{2}s-\frac{9}{2}t-\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,1,0)$ | (1) | $-\frac{27}{2}+\frac{3}{2}s+\frac{9}{2}t+\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,1,0)$ | (2) | $-\frac{297}{8}+\frac{33}{8}s+\frac{99}{8}t+\frac{33}{8}st$ |
| $(0,0,-1,0,0,0,1,0)$ | (3) | $-\frac{27}{2}+\frac{3}{2}s+\frac{9}{2}t+\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,1,0)$ | (4) | $\frac{27}{2}-\frac{3}{2}s-\frac{9}{2}t-\frac{3}{2}st$ |
| $(0,0,-1,0,0,0,1,0)$ | (6) | $-\frac{27}{8}+\frac{3}{8}s+\frac{9}{8}t+\frac{3}{8}st$ |
| $(0,0,-1,0,0,0,1,0)$ | (7) | $\frac{81}{2}-\frac{9}{2}s-\frac{27}{2}t-\frac{9}{2}st$ |
| $(0,0,-1,0,1,0,0,0)$ | (0) | $\frac{12}{5}s-\frac{24}{5}t$ |
| $(0,0,-1,0,1,0,0,0)$ | (1) | $-\frac{12}{5}s+\frac{24}{5}t$ |
| $(0,0,-1,0,1,0,0,0)$ | (2) | $-\frac{33}{5}s+\frac{66}{5}t$ |
| $(0,0,-1,0,1,0,0,0)$ | (3) | $-\frac{12}{5}s+\frac{24}{5}t$ |
| $(0,0,-1,0,1,0,0,0)$ | (4) | $-\frac{27}{5}s+\frac{54}{5}t$ |
| $(0,0,-1,0,1,0,0,0)$ | (6) | $\frac{36}{5}s-\frac{72}{5}t$ |
| $(0,0,-1,0,1,0,0,0)$ | (7) | $\frac{36}{5}s-\frac{72}{5}t$ |
| $(0,0,0,-1,-1,0,1,1)$ | (0) | $-40$ |
| $(0,0,0,-1,-1,0,1,1)$ | (1) | $40$ |
| $(0,0,0,-1,-1,0,1,1)$ | (2) | $-20$ |
| $(0,0,0,-1,-1,0,1,1)$ | (3) | $-90$ |
| $(0,0,0,-1,-1,0,1,1)$ | (4) | $90$ |
| $(0,0,0,-1,-1,0,1,1)$ | (6) | $10$ |
| $(0,0,0,-1,-1,0,1,1)$ | (7) | $10$ |
| $(0,0,0,-1,0,0,0,1)$ | (0) | $-2-6s+2t-2st$ |
| $(0,0,0,-1,0,0,0,1)$ | (1) | $2+6s-2t+2st$ |
| $(0,0,0,-1,0,0,0,1)$ | (2) | $-1-3s+t-st$ |
| $(0,0,0,-1,0,0,0,1)$ | (3) | $-\frac{9}{2}-\frac{27}{2}s+\frac{9}{2}t-\frac{9}{2}st$ |
| $(0,0,0,-1,0,0,0,1)$ | (4) | $-2-6s+2t-2st$ |
| $(0,0,0,-1,0,0,0,1)$ | (6) | $7+21s-7t+7st$ |
| $(0,0,0,-1,0,0,0,1)$ | (7) | $\frac{1}{2}+\frac{3}{2}s-\frac{1}{2}t+\frac{1}{2}st$ |
| $(0,0,0,-1,0,0,1,0)$ | (0) | $2-6s+2t+2st$ |
| $(0,0,0,-1,0,0,1,0)$ | (1) | $-2+6s-2t-2st$ |
| $(0,0,0,-1,0,0,1,0)$ | (2) | $1-3s+t+st$ |
| $(0,0,0,-1,0,0,1,0)$ | (3) | $\frac{9}{2}-\frac{27}{2}s+\frac{9}{2}t+\frac{9}{2}st$ |
| $(0,0,0,-1,0,0,1,0)$ | (4) | $2-6s+2t+2st$ |
| $(0,0,0,-1,0,0,1,0)$ | (6) | $-\frac{1}{2}+\frac{3}{2}s-\frac{1}{2}t-\frac{1}{2}st$ |
| $(0,0,0,-1,0,0,1,0)$ | (7) | $-7+21s-7t-7st$ |
| $(0,0,0,0,-1,0,0,1)$ | (0) | $-\frac{15}{2}+\frac{5}{2}s-\frac{5}{2}t-\frac{5}{2}st$ |
| $(0,0,0,0,-1,0,0,1)$ | (1) | $\frac{15}{2}-\frac{5}{2}s+\frac{5}{2}t+\frac{5}{2}st$ |
| $(0,0,0,0,-1,0,0,1)$ | (2) | $-\frac{15}{4}+\frac{5}{4}s-\frac{5}{4}t-\frac{5}{4}st$ |
| $(0,0,0,0,-1,0,0,1)$ | (3) | $\frac{15}{2}-\frac{5}{2}s+\frac{5}{2}t+\frac{5}{2}st$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,-1,0,0,1)$ | $(4)$ | $\frac{135}{8} - \frac{45}{8}s + \frac{45}{8}t + \frac{45}{8}st$ |
| $(0,0,0,0,-1,0,0,1)$ | $(6)$ | $-\frac{45}{2} + \frac{15}{2}s - \frac{15}{2}t - \frac{15}{2}st$ |
| $(0,0,0,0,-1,0,0,1)$ | $(7)$ | $\frac{15}{8} - \frac{5}{8}s + \frac{5}{8}t + \frac{5}{8}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $(0)$ | $\frac{15}{2} + \frac{5}{2}s - \frac{5}{2}t + \frac{5}{2}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $(1)$ | $-\frac{15}{2} - \frac{5}{2}s + \frac{5}{2}t - \frac{5}{2}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $(2)$ | $\frac{15}{4} + \frac{5}{4}s - \frac{5}{4}t + \frac{5}{4}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $(3)$ | $-\frac{15}{2} - \frac{5}{2}s + \frac{5}{2}t - \frac{5}{2}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $(4)$ | $-\frac{135}{8} - \frac{45}{8}s + \frac{45}{8}t - \frac{45}{8}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $(6)$ | $-\frac{15}{8} - \frac{5}{8}s + \frac{5}{8}t - \frac{5}{8}st$ |
| $(0,0,0,0,-1,0,1,0)$ | $(7)$ | $\frac{45}{2} + \frac{15}{2}s - \frac{15}{2}t + \frac{15}{2}st$ |
| $(0,0,0,0,0,0,-1,1)$ | $\varnothing$ | $\frac{117}{2} - \frac{13}{2}s - \frac{39}{2}t - \frac{13}{2}st$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,0)$ | $\frac{33}{26}s + \frac{99}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,1)$ | $\frac{9}{13}s + \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,2)$ | $-\frac{9}{26}s - \frac{27}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,3)$ | $\frac{9}{13}s + \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,4)$ | $-\frac{9}{13}s - \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,6)$ | $-\frac{27}{13}s - \frac{81}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,0,7)$ | $-\frac{27}{13}s - \frac{81}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,1)$ | $-\frac{9}{13}s - \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,2)$ | $-\frac{4}{13}s - \frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,3)$ | $\frac{8}{13}s + \frac{24}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,4)$ | $-\frac{8}{13}s - \frac{24}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,6)$ | $\frac{2}{13}s + \frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,1,7)$ | $\frac{2}{13}s + \frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2,2)$ | $-\frac{11}{26}s - \frac{33}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2,3)$ | $-\frac{4}{13}s - \frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2,4)$ | $\frac{4}{13}s + \frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2,6)$ | $\frac{12}{13}s + \frac{36}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,2,7)$ | $\frac{12}{13}s + \frac{36}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,3)$ | $-\frac{9}{13}s - \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,4)$ | $-\frac{8}{13}s - \frac{24}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,6)$ | $\frac{2}{13}s + \frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,3,7)$ | $\frac{2}{13}s + \frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4,4)$ | $-\frac{9}{13}s - \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4,6)$ | $\frac{24}{13}s + \frac{72}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,4,7)$ | $\frac{24}{13}s + \frac{72}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6,6)$ | $-\frac{3}{13}s - \frac{9}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,6,7)$ | $\frac{20}{13}s + \frac{60}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(0,7,7)$ | $-\frac{3}{13}s - \frac{9}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,1)$ | $-\frac{33}{26}s - \frac{99}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,2)$ | $-\frac{9}{26}s - \frac{27}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,3)$ | $\frac{9}{13}s + \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,4)$ | $-\frac{9}{13}s - \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,6)$ | $\frac{63}{26}s + \frac{189}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,1,7)$ | $\frac{63}{26}s + \frac{189}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2,2)$ | $\frac{11}{26}s + \frac{33}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2,3)$ | $\frac{4}{13}s + \frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2,4)$ | $-\frac{4}{13}s - \frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2,6)$ | $\frac{1}{13}s + \frac{3}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,2,7)$ | $\frac{1}{13}s + \frac{3}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3,3)$ | $\frac{9}{13}s + \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3,4)$ | $\frac{8}{13}s + \frac{24}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3,6)$ | $-\frac{28}{13}s - \frac{84}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,3,7)$ | $-\frac{28}{13}s - \frac{84}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4,4)$ | $\frac{9}{13}s + \frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4,6)$ | $\frac{2}{13}s + \frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,4,7)$ | $\frac{2}{13}s + \frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6,6)$ | $-\frac{7}{26}s - \frac{21}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(1,6,7)$ | $-\frac{33}{13}s - \frac{99}{13}t$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,0,0,0,0,0)$ | $(1,7,7)$ | $-\frac{7}{26}s-\frac{21}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2,2)$ | $\frac{11}{13}s+\frac{33}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2,3)$ | $\frac{11}{26}s+\frac{33}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2,4)$ | $-\frac{11}{26}s-\frac{33}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2,6)$ | $-\frac{33}{26}s-\frac{99}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,2,7)$ | $-\frac{33}{26}s-\frac{99}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3,3)$ | $-\frac{9}{26}s-\frac{27}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3,4)$ | $-\frac{4}{13}s-\frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3,6)$ | $\frac{1}{13}s+\frac{3}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,3,7)$ | $\frac{1}{13}s+\frac{3}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4,4)$ | $-\frac{9}{26}s-\frac{27}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4,6)$ | $\frac{12}{13}s+\frac{36}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,4,7)$ | $\frac{12}{13}s+\frac{36}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,6,6)$ | $-\frac{3}{26}s-\frac{9}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,6,7)$ | $\frac{10}{13}s+\frac{30}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(2,7,7)$ | $-\frac{3}{26}s-\frac{9}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,3)$ | $-\frac{33}{26}s-\frac{99}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,4)$ | $-\frac{9}{13}s-\frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,6)$ | $\frac{63}{26}s+\frac{189}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,3,7)$ | $\frac{63}{26}s+\frac{189}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4,4)$ | $\frac{9}{13}s+\frac{27}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4,6)$ | $\frac{2}{13}s+\frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,4,7)$ | $\frac{2}{13}s+\frac{6}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6,6)$ | $-\frac{7}{26}s-\frac{21}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,6,7)$ | $-\frac{33}{13}s-\frac{99}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(3,7,7)$ | $-\frac{7}{26}s-\frac{21}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4,4)$ | $\frac{33}{26}s+\frac{99}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4,6)$ | $-\frac{27}{13}s-\frac{81}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,4,7)$ | $-\frac{27}{13}s-\frac{81}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6,6)$ | $-\frac{3}{13}s-\frac{9}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,6,7)$ | $\frac{20}{13}s+\frac{60}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(4,7,7)$ | $-\frac{3}{13}s-\frac{9}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6,6)$ | $\frac{7}{26}s+\frac{21}{26}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,6,7)$ | $\frac{4}{13}s+\frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(6,7,7)$ | $\frac{4}{13}s+\frac{12}{13}t$ |
| $(0,0,0,0,0,0,0,0)$ | $(7,7,7)$ | $\frac{7}{26}s+\frac{21}{26}t$ |
| $(0,0,0,0,0,0,1,-1)$ | $\varnothing$ | $\frac{117}{2}+\frac{13}{2}s+\frac{39}{2}t-\frac{13}{2}st$ |
| $(0,0,0,0,1,0,-1,0)$ | $(0)$ | $\frac{24}{5}-\frac{4}{5}s+\frac{8}{5}t+\frac{4}{5}st$ |
| $(0,0,0,0,1,0,-1,0)$ | $(1)$ | $-\frac{24}{5}+\frac{4}{5}s-\frac{8}{5}t-\frac{4}{5}st$ |
| $(0,0,0,0,1,0,-1,0)$ | $(2)$ | $\frac{12}{5}-\frac{2}{5}s+\frac{4}{5}t+\frac{2}{5}st$ |
| $(0,0,0,0,1,0,-1,0)$ | $(3)$ | $-\frac{24}{5}+\frac{4}{5}s-\frac{8}{5}t-\frac{4}{5}st$ |
| $(0,0,0,0,1,0,-1,0)$ | $(4)$ | $-\frac{54}{5}+\frac{9}{5}s-\frac{18}{5}t-\frac{9}{5}st$ |
| $(0,0,0,0,1,0,-1,0)$ | $(6)$ | $-\frac{6}{5}+\frac{1}{5}s-\frac{2}{5}t-\frac{1}{5}st$ |
| $(0,0,0,0,1,0,-1,0)$ | $(7)$ | $\frac{72}{5}-\frac{12}{5}s+\frac{24}{5}t+\frac{12}{5}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $(0)$ | $-\frac{24}{5}-\frac{4}{5}s+\frac{8}{5}t-\frac{4}{5}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $(1)$ | $\frac{24}{5}+\frac{4}{5}s-\frac{8}{5}t+\frac{4}{5}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $(2)$ | $-\frac{12}{5}-\frac{2}{5}s+\frac{4}{5}t-\frac{2}{5}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $(3)$ | $\frac{24}{5}+\frac{4}{5}s-\frac{8}{5}t+\frac{4}{5}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $(4)$ | $\frac{54}{5}+\frac{9}{5}s-\frac{18}{5}t+\frac{9}{5}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $(6)$ | $-\frac{72}{5}-\frac{12}{5}s+\frac{24}{5}t-\frac{12}{5}st$ |
| $(0,0,0,0,1,0,0,-1)$ | $(7)$ | $\frac{6}{5}+\frac{1}{5}s-\frac{2}{5}t+\frac{1}{5}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $(0)$ | $\frac{11}{4}+\frac{3}{4}s-\frac{11}{4}t+\frac{1}{4}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $(1)$ | $-\frac{11}{4}-\frac{3}{4}s+\frac{11}{4}t-\frac{1}{4}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $(2)$ | $\frac{11}{8}+\frac{3}{8}s-\frac{11}{8}t+\frac{1}{8}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $(3)$ | $\frac{99}{16}+\frac{27}{16}s-\frac{99}{16}t+\frac{9}{16}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $(4)$ | $\frac{11}{4}+\frac{3}{4}s-\frac{11}{4}t+\frac{1}{4}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $(6)$ | $-\frac{11}{16}-\frac{3}{16}s+\frac{11}{16}t-\frac{1}{16}st$ |
| $(0,0,0,1,0,0,-1,0)$ | $(7)$ | $-\frac{77}{8}-\frac{21}{8}s+\frac{77}{8}t-\frac{7}{8}st$ |
| $(0,0,0,1,0,0,0,-1)$ | $(0)$ | $-\frac{11}{4}+\frac{3}{4}s-\frac{11}{4}t-\frac{1}{4}st$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,0,0,1,0,0,0,-1)$ | (1) | $\frac{11}{4} - \frac{3}{4}s + \frac{11}{4}t + \frac{1}{4}st$ |
| $(0,0,0,1,0,0,0,-1)$ | (2) | $-\frac{11}{8} + \frac{3}{8}s - \frac{11}{8}t - \frac{1}{8}st$ |
| $(0,0,0,1,0,0,0,-1)$ | (3) | $-\frac{99}{16} + \frac{27}{16}s - \frac{99}{16}t - \frac{9}{16}st$ |
| $(0,0,0,1,0,0,0,-1)$ | (4) | $-\frac{11}{4} + \frac{3}{4}s - \frac{11}{4}t - \frac{1}{4}st$ |
| $(0,0,0,1,0,0,0,-1)$ | (6) | $\frac{77}{8} - \frac{21}{8}s + \frac{77}{8}t + \frac{7}{8}st$ |
| $(0,0,0,1,0,0,0,-1)$ | (7) | $\frac{11}{16} - \frac{3}{16}s + \frac{11}{16}t + \frac{1}{16}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | (0) | $\frac{7}{5} - \frac{3}{5}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | (1) | $-\frac{7}{5} + \frac{3}{5}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | (2) | $\frac{7}{10} - \frac{3}{10}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | (3) | $\frac{63}{20} - \frac{27}{20}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | (4) | $-\frac{63}{20} + \frac{27}{20}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | (6) | $-\frac{7}{20} + \frac{3}{20}st$ |
| $(0,0,0,1,1,0,-1,-1)$ | (7) | $-\frac{7}{20} + \frac{3}{20}st$ |
| $(0,0,1,0,-1,0,0,0)$ | (0) | $-\frac{5}{3}s + \frac{5}{3}t$ |
| $(0,0,1,0,-1,0,0,0)$ | (1) | $\frac{5}{3}s - \frac{5}{3}t$ |
| $(0,0,1,0,-1,0,0,0)$ | (2) | $\frac{55}{12}s - \frac{55}{12}t$ |
| $(0,0,1,0,-1,0,0,0)$ | (3) | $\frac{5}{3}s - \frac{5}{3}t$ |
| $(0,0,1,0,-1,0,0,0)$ | (4) | $\frac{15}{4}s - \frac{15}{4}t$ |
| $(0,0,1,0,-1,0,0,0)$ | (6) | $-5s + 5t$ |
| $(0,0,1,0,-1,0,0,0)$ | (7) | $-5s + 5t$ |
| $(0,0,1,0,0,0,-1,0)$ | (0) | $-3 - \frac{1}{3}s - t + \frac{1}{3}st$ |
| $(0,0,1,0,0,0,-1,0)$ | (1) | $3 + \frac{1}{3}s + t - \frac{1}{3}st$ |
| $(0,0,1,0,0,0,-1,0)$ | (2) | $\frac{33}{4} + \frac{11}{12}s + \frac{11}{4}t - \frac{11}{12}st$ |
| $(0,0,1,0,0,0,-1,0)$ | (3) | $3 + \frac{1}{3}s + t - \frac{1}{3}st$ |
| $(0,0,1,0,0,0,-1,0)$ | (4) | $-3 - \frac{1}{3}s - t + \frac{1}{3}st$ |
| $(0,0,1,0,0,0,-1,0)$ | (6) | $\frac{3}{4} + \frac{1}{12}s + \frac{1}{4}t - \frac{1}{12}st$ |
| $(0,0,1,0,0,0,-1,0)$ | (7) | $-9 - s - 3t + st$ |
| $(0,0,1,0,0,0,0,-1)$ | (0) | $3 - \frac{1}{3}s - t - \frac{1}{3}st$ |
| $(0,0,1,0,0,0,0,-1)$ | (1) | $-3 + \frac{1}{3}s + t + \frac{1}{3}st$ |
| $(0,0,1,0,0,0,0,-1)$ | (2) | $-\frac{33}{4} + \frac{11}{12}s + \frac{11}{4}t + \frac{11}{12}st$ |
| $(0,0,1,0,0,0,0,-1)$ | (3) | $-3 + \frac{1}{3}s + t + \frac{1}{3}st$ |
| $(0,0,1,0,0,0,0,-1)$ | (4) | $3 - \frac{1}{3}s - t - \frac{1}{3}st$ |
| $(0,0,1,0,0,0,0,-1)$ | (6) | $9 - s - 3t - st$ |
| $(0,0,1,0,0,0,0,-1)$ | (7) | $-\frac{3}{4} + \frac{1}{12}s + \frac{1}{4}t + \frac{1}{12}st$ |
| $(0,0,1,1,0,0,-1,-1)$ | (0) | $-\frac{11}{6} - \frac{1}{6}st$ |
| $(0,0,1,1,0,0,-1,-1)$ | (1) | $\frac{11}{6} + \frac{1}{6}st$ |
| $(0,0,1,1,0,0,-1,-1)$ | (2) | $\frac{121}{24} + \frac{11}{24}st$ |
| $(0,0,1,1,0,0,-1,-1)$ | (3) | $-\frac{33}{8} - \frac{3}{8}st$ |
| $(0,0,1,1,0,0,-1,-1)$ | (4) | $-\frac{11}{6} - \frac{1}{6}st$ |
| $(0,0,1,1,0,0,-1,-1)$ | (6) | $\frac{11}{24} + \frac{1}{24}st$ |
| $(0,0,1,1,0,0,-1,-1)$ | (7) | $\frac{11}{24} + \frac{1}{24}st$ |
| $(0,1,0,-1,0,0,0,0)$ | (0) | $s - 7t$ |
| $(0,1,0,-1,0,0,0,0)$ | (1) | $\frac{9}{4}s - \frac{63}{4}t$ |
| $(0,1,0,-1,0,0,0,0)$ | (2) | $\frac{1}{2}s - \frac{7}{2}t$ |
| $(0,1,0,-1,0,0,0,0)$ | (3) | $\frac{9}{4}s - \frac{63}{4}t$ |
| $(0,1,0,-1,0,0,0,0)$ | (4) | $s - 7t$ |
| $(0,1,0,-1,0,0,0,0)$ | (6) | $-\frac{7}{2}s + \frac{49}{2}t$ |
| $(0,1,0,-1,0,0,0,0)$ | (7) | $-\frac{7}{2}s + \frac{49}{2}t$ |
| $(0,1,0,0,0,0,-1,0)$ | (0) | $-\frac{1}{2} - \frac{3}{2}s + \frac{1}{2}t - \frac{1}{2}st$ |
| $(0,1,0,0,0,0,-1,0)$ | (1) | $-\frac{9}{8} - \frac{27}{8}s + \frac{9}{8}t - \frac{9}{8}st$ |
| $(0,1,0,0,0,0,-1,0)$ | (2) | $-\frac{1}{4} - \frac{3}{4}s + \frac{1}{4}t - \frac{1}{4}st$ |
| $(0,1,0,0,0,0,-1,0)$ | (3) | $\frac{1}{2} + \frac{3}{2}s - \frac{1}{2}t + \frac{1}{2}st$ |
| $(0,1,0,0,0,0,-1,0)$ | (4) | $-\frac{1}{2} - \frac{3}{2}s + \frac{1}{2}t - \frac{1}{2}st$ |
| $(0,1,0,0,0,0,-1,0)$ | (6) | $\frac{1}{8} + \frac{3}{8}s - \frac{1}{8}t + \frac{1}{8}st$ |
| $(0,1,0,0,0,0,-1,0)$ | (7) | $\frac{7}{4} + \frac{21}{4}s - \frac{7}{4}t + \frac{7}{4}st$ |
| $(0,1,0,0,0,0,0,-1)$ | (0) | $\frac{1}{2} - \frac{3}{2}s + \frac{1}{2}t + \frac{1}{2}st$ |
| $(0,1,0,0,0,0,0,-1)$ | (1) | $\frac{9}{8} - \frac{27}{8}s + \frac{9}{8}t + \frac{9}{8}st$ |
| $(0,1,0,0,0,0,0,-1)$ | (2) | $\frac{1}{4} - \frac{3}{4}s + \frac{1}{4}t + \frac{1}{4}st$ |
| $(0,1,0,0,0,0,0,-1)$ | (3) | $-\frac{1}{2} + \frac{3}{2}s - \frac{1}{2}t - \frac{1}{2}st$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Exact coefficient |
|---|---|---|
| $(0,1,0,0,0,0,0,-1)$ | (4) | $\frac{1}{2}-\frac{3}{2}s+\frac{1}{2}t+\frac{1}{2}st$ |
| $(0,1,0,0,0,0,0,-1)$ | (6) | $-\frac{7}{4}+\frac{21}{4}s-\frac{7}{4}t-\frac{7}{4}st$ |
| $(0,1,0,0,0,0,0,-1)$ | (7) | $-\frac{1}{8}+\frac{3}{8}s-\frac{1}{8}t-\frac{1}{8}st$ |
| $(0,1,0,0,1,0,-1,-1)$ | $\varnothing$ | $-\frac{247}{10}+\frac{13}{10}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | (0) | $-\frac{1}{3}-\frac{1}{3}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | (1) | $-\frac{3}{4}-\frac{3}{4}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | (2) | $\frac{11}{12}+\frac{11}{12}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | (3) | $\frac{1}{3}+\frac{1}{3}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | (4) | $-\frac{1}{3}-\frac{1}{3}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | (6) | $\frac{1}{12}+\frac{1}{12}st$ |
| $(0,1,1,0,0,0,-1,-1)$ | (7) | $\frac{1}{12}+\frac{1}{12}st$ |
| $(1,0,-1,0,0,0,0,0)$ | (0) | $\frac{27}{4}s-\frac{27}{4}t$ |
| $(1,0,-1,0,0,0,0,0)$ | (1) | $3s-3t$ |
| $(1,0,-1,0,0,0,0,0)$ | (2) | $\frac{33}{4}s-\frac{33}{4}t$ |
| $(1,0,-1,0,0,0,0,0)$ | (3) | $3s-3t$ |
| $(1,0,-1,0,0,0,0,0)$ | (4) | $-3s+3t$ |
| $(1,0,-1,0,0,0,0,0)$ | (6) | $-9s+9t$ |
| $(1,0,-1,0,0,0,0,0)$ | (7) | $-9s+9t$ |
| $(1,0,0,0,-1,0,0,0)$ | (0) | $\frac{9}{4}s+\frac{63}{4}t$ |
| $(1,0,0,0,-1,0,0,0)$ | (1) | $s+7t$ |
| $(1,0,0,0,-1,0,0,0)$ | (2) | $-\frac{1}{2}s-\frac{7}{2}t$ |
| $(1,0,0,0,-1,0,0,0)$ | (3) | $s+7t$ |
| $(1,0,0,0,-1,0,0,0)$ | (4) | $\frac{9}{4}s+\frac{63}{4}t$ |
| $(1,0,0,0,-1,0,0,0)$ | (6) | $-3s-21t$ |
| $(1,0,0,0,-1,0,0,0)$ | (7) | $-3s-21t$ |
| $(1,0,0,0,0,0,-1,0)$ | (0) | $\frac{27}{4}-\frac{9}{4}s+\frac{9}{4}t+\frac{9}{4}st$ |
| $(1,0,0,0,0,0,-1,0)$ | (1) | $3-s+t+st$ |
| $(1,0,0,0,0,0,-1,0)$ | (2) | $-\frac{3}{2}+\frac{1}{2}s-\frac{1}{2}t-\frac{1}{2}st$ |
| $(1,0,0,0,0,0,-1,0)$ | (3) | $3-s+t+st$ |
| $(1,0,0,0,0,0,-1,0)$ | (4) | $-3+s-t-st$ |
| $(1,0,0,0,0,0,-1,0)$ | (6) | $\frac{3}{4}-\frac{1}{4}s+\frac{1}{4}t+\frac{1}{4}st$ |
| $(1,0,0,0,0,0,-1,0)$ | (7) | $-9+3s-3t-3st$ |
| $(1,0,0,0,0,0,0,-1)$ | (0) | $-\frac{27}{4}-\frac{9}{4}s+\frac{9}{4}t-\frac{9}{4}st$ |
| $(1,0,0,0,0,0,0,-1)$ | (1) | $-3-s+t-st$ |
| $(1,0,0,0,0,0,0,-1)$ | (2) | $\frac{3}{2}+\frac{1}{2}s-\frac{1}{2}t+\frac{1}{2}st$ |
| $(1,0,0,0,0,0,0,-1)$ | (3) | $-3-s+t-st$ |
| $(1,0,0,0,0,0,0,-1)$ | (4) | $3+s-t+st$ |
| $(1,0,0,0,0,0,0,-1)$ | (6) | $9+3s-3t+3st$ |
| $(1,0,0,0,0,0,0,-1)$ | (7) | $-\frac{3}{4}-\frac{1}{4}s+\frac{1}{4}t-\frac{1}{4}st$ |
| $(1,0,0,1,0,0,-1,-1)$ | $\varnothing$ | $-\frac{65}{4}+\frac{13}{4}st$ |
| $(1,1,0,0,0,0,-1,-1)$ | (0) | $9$ |
| $(1,1,0,0,0,0,-1,-1)$ | (1) | $-9$ |
| $(1,1,0,0,0,0,-1,-1)$ | (2) | $-2$ |
| $(1,1,0,0,0,0,-1,-1)$ | (3) | $4$ |
| $(1,1,0,0,0,0,-1,-1)$ | (4) | $-4$ |
| $(1,1,0,0,0,0,-1,-1)$ | (6) | $1$ |
| $(1,1,0,0,0,0,-1,-1)$ | (7) | $1$ |

## S25 Complete finite top witness atlas

This appendix fills in the finite coordinate and cyclic-generation records that the integrated guide illustrates with its first few rows. The tables are literal saved witnesses. Checking each row's mode action and elimination, together with distinct pivots, is the finite cyclicity argument. Merely counting the rows is not that verification. The ambient VOA interpretation, positive metric, invariant subspace, and all-weight module conclusion retain the separate hypotheses explained in Appendix S5.

## S25.1 All six joint spectral eigenvectors

The source is `c8_reviewability_20260927/core_spectral_witnesses.tsv`. Each of the six vectors has 21 nonzero terms, and has parafermion weight 12/13. The current charges distinguish the blocks. Coefficients use $s = \sqrt{13}$, $t = i\sqrt{3}$ as in the coordinate atlas; oscillator indices have the same meaning. Each subsection below specifies one entire vector as the sum of its displayed monomials.

**Cartan weight** $(-1, -5)$**.**

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,0,1,1,1,1,0,0)$ | $\varnothing$ | $\frac{35}{16} + \frac{175}{1872}st$ |
| $(0,0,0,0,0,1,1,1)$ | $\varnothing$ | $-\frac{203}{208}s - \frac{161}{48}t$ |
| $(0,0,0,1,1,1,0,0)$ | $\varnothing$ | $\frac{7}{8} - \frac{49}{156}st$ |
| $(0,0,1,0,0,1,0,1)$ | $\varnothing$ | $\frac{161}{48} + \frac{203}{624}s + \frac{161}{144}t - \frac{203}{624}st$ |
| $(0,0,1,0,0,1,1,0)$ | $\varnothing$ | $\frac{161}{48} - \frac{203}{624}s - \frac{161}{144}t - \frac{203}{624}st$ |
| $(0,0,1,1,0,1,0,0)$ | (3) | $\frac{91}{48} + \frac{371}{1872}st$ |
| $(0,0,1,1,0,1,0,0)$ | (4) | $\frac{91}{48} + \frac{371}{1872}st$ |
| $(0,0,1,1,0,1,0,0)$ | (6) | $-\frac{91}{48} - \frac{371}{1872}st$ |
| $(0,0,1,1,0,1,0,0)$ | (7) | $-\frac{91}{48} - \frac{371}{1872}st$ |
| $(0,0,1,1,1,1,-1,0)$ | $\varnothing$ | $\frac{7}{24} - \frac{49}{156}s - \frac{7}{24}t - \frac{49}{468}st$ |
| $(0,0,1,1,1,1,0,-1)$ | $\varnothing$ | $\frac{7}{24} + \frac{49}{156}s + \frac{7}{24}t - \frac{49}{468}st$ |
| $(0,1,1,0,0,1,0,0)$ | (0) | $\frac{7}{32} + \frac{1351}{3744}st$ |
| $(0,1,1,0,0,1,0,0)$ | (1) | $\frac{7}{32} + \frac{1351}{3744}st$ |
| $(0,1,1,0,0,1,0,0)$ | (6) | $-\frac{7}{32} - \frac{1351}{3744}st$ |
| $(0,1,1,0,0,1,0,0)$ | (7) | $-\frac{7}{32} - \frac{1351}{3744}st$ |
| $(0,1,1,1,1,1,-1,-1)$ | $\varnothing$ | $\frac{63}{416}s - \frac{77}{288}t$ |
| $(1,1,0,0,0,1,0,0)$ | $\varnothing$ | $-\frac{343}{160} - \frac{133}{6240}st$ |
| $(1,1,1,0,-1,1,0,0)$ | $\varnothing$ | $\frac{119}{240} - \frac{3311}{9360}st$ |
| $(1,1,1,0,0,1,-1,0)$ | $\varnothing$ | $\frac{343}{480} + \frac{133}{6240}s - \frac{343}{480}t + \frac{133}{18720}st$ |
| $(1,1,1,0,0,1,0,-1)$ | $\varnothing$ | $\frac{343}{480} - \frac{133}{6240}s + \frac{343}{480}t + \frac{133}{18720}st$ |
| $(1,1,1,1,0,1,-1,-1)$ | $\varnothing$ | $\frac{287}{1560}s - \frac{7}{45}t$ |

**Cartan weight** $(-1, 5)$**.**

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,1,2,1,1,1,1,1)$ | $\varnothing$ | $\frac{1013}{488} - \frac{289}{488}s + \frac{735}{488}t + \frac{185}{488}st$ |
| $(0,1,2,2,1,1,0,1)$ | $\varnothing$ | $-\frac{41}{122} - \frac{145}{244}s - \frac{23}{61}t + \frac{17}{244}st$ |
| $(0,1,2,2,1,1,1,0)$ | $\varnothing$ | $\frac{179}{244} + \frac{47}{244}s + \frac{5}{244}t - \frac{81}{244}st$ |
| $(0,1,2,2,2,1,0,0)$ | $\varnothing$ | $-\frac{2593}{2440} - \frac{163}{2440}s + \frac{1361}{2440}t - \frac{33}{2440}st$ |
| $(0,1,3,2,1,1,0,0)$ | $\varnothing$ | $\frac{97}{732} - \frac{49}{366}s - \frac{29}{244}t - \frac{16}{183}st$ |
| $(1,1,2,1,0,1,1,1)$ | $\varnothing$ | $\frac{376}{305} - \frac{481}{610}s + \frac{319}{305}t + \frac{283}{610}st$ |
| $(1,1,2,2,1,1,0,0)$ | (0) | $-\frac{2399}{2440} - \frac{359}{2440}s + \frac{1187}{2440}t - \frac{161}{2440}st$ |
| $(1,1,2,2,1,1,0,0)$ | (1) | $-\frac{2399}{2440} - \frac{359}{2440}s + \frac{1187}{2440}t - \frac{161}{2440}st$ |
| $(1,1,2,2,1,1,0,0)$ | (6) | $\frac{2399}{2440} + \frac{359}{2440}s - \frac{1187}{2440}t + \frac{161}{2440}st$ |
| $(1,1,2,2,1,1,0,0)$ | (7) | $\frac{2399}{2440} + \frac{359}{2440}s - \frac{1187}{2440}t + \frac{161}{2440}st$ |
| $(1,2,2,1,0,1,0,1)$ | $\varnothing$ | $-\frac{2563}{4880} - \frac{187}{4880}s - \frac{6299}{4880}t - \frac{127}{4880}st$ |
| $(1,2,2,1,0,1,1,0)$ | $\varnothing$ | $\frac{1073}{488} + \frac{71}{1220}s + \frac{467}{1220}t - \frac{3}{488}st$ |
| $(1,2,2,1,1,1,0,0)$ | (3) | $-\frac{1429}{4880} - \frac{1339}{4880}s + \frac{317}{4880}t - \frac{801}{4880}st$ |
| $(1,2,2,1,1,1,0,0)$ | (4) | $-\frac{1429}{4880} - \frac{1339}{4880}s + \frac{317}{4880}t - \frac{801}{4880}st$ |
| $(1,2,2,1,1,1,0,0)$ | (6) | $\frac{1429}{4880} + \frac{1339}{4880}s - \frac{317}{4880}t + \frac{801}{4880}st$ |
| $(1,2,2,1,1,1,0,0)$ | (7) | $\frac{1429}{4880} + \frac{1339}{4880}s - \frac{317}{4880}t + \frac{801}{4880}st$ |
| $(1,2,2,2,1,1,-1,0)$ | $\varnothing$ | $\frac{2713}{4880} - \frac{71}{4880}s - \frac{467}{4880}t - \frac{343}{4880}st$ |
| $(1,2,2,2,1,1,0,-1)$ | $\varnothing$ | $\frac{41}{305} - \frac{55}{488}s - \frac{159}{488}t - \frac{17}{610}st$ |
| $(1,2,3,1,0,1,0,0)$ | $\varnothing$ | $-\frac{8167}{14640} - \frac{97}{14640}s + \frac{1477}{4880}t + \frac{157}{14640}st$ |
| $(1,2,3,2,1,1,-1,-1)$ | $\varnothing$ | $-\frac{2057}{14640} - \frac{479}{14640}s - \frac{1123}{14640}t + \frac{69}{4880}st$ |

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(2,2,2,1,0,1,0,0)$ | $\varnothing$ | $-\frac{511}{12200}+\frac{3299}{12200}s+\frac{1423}{12200}t+\frac{2081}{12200}st$ |

**Cartan weight** $(0,-8)$**.**

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,-2,-1,0,0,0,0,0)$ | $\varnothing$ | $\frac{1181}{1660}-\frac{201}{1660}s-\frac{1343}{1660}t+\frac{391}{1660}st$ |
| $(-1,-1,-1,-1,-1,0,0,1)$ | $\varnothing$ | $\frac{103}{332}-\frac{39}{332}s+\frac{505}{332}t-\frac{117}{332}st$ |
| $(-1,-1,-1,-1,-1,0,1,0)$ | $\varnothing$ | $-\frac{809}{332}+\frac{195}{332}s-\frac{201}{332}t+\frac{39}{332}st$ |
| $(-1,-1,-1,-1,0,0,0,0)$ | (3) | $\frac{292}{415}-\frac{189}{830}s+\frac{583}{830}t-\frac{98}{415}st$ |
| $(-1,-1,-1,-1,0,0,0,0)$ | (4) | $\frac{292}{415}-\frac{189}{830}s+\frac{583}{830}t-\frac{98}{415}st$ |
| $(-1,-1,-1,-1,0,0,0,0)$ | (6) | $-\frac{292}{415}+\frac{189}{830}s-\frac{583}{830}t+\frac{98}{415}st$ |
| $(-1,-1,-1,-1,0,0,0,0)$ | (7) | $-\frac{292}{415}+\frac{189}{830}s-\frac{583}{830}t+\frac{98}{415}st$ |
| $(-1,-1,-1,0,0,0,-1,0)$ | $\varnothing$ | $-\frac{1263}{1660}+\frac{289}{1660}s-\frac{303}{1660}t+\frac{97}{1660}st$ |
| $(-1,-1,-1,0,0,0,0,-1)$ | $\varnothing$ | $-\frac{543}{830}+\frac{29}{166}s+\frac{24}{83}t-\frac{24}{415}st$ |
| $(-1,-1,0,-1,-1,0,0,0)$ | $\varnothing$ | $\frac{353}{498}-\frac{13}{83}s-\frac{76}{249}t+\frac{13}{166}st$ |
| $(-1,-1,0,0,0,0,-1,-1)$ | $\varnothing$ | $\frac{59}{1660}+\frac{1}{4980}s+\frac{261}{1660}t-\frac{193}{4980}st$ |
| $(0,-1,-1,-1,-1,0,0,0)$ | (0) | $\frac{937}{830}-\frac{267}{830}s+\frac{431}{830}t-\frac{157}{830}st$ |
| $(0,-1,-1,-1,-1,0,0,0)$ | (1) | $\frac{937}{830}-\frac{267}{830}s+\frac{431}{830}t-\frac{157}{830}st$ |
| $(0,-1,-1,-1,-1,0,0,0)$ | (6) | $-\frac{937}{830}+\frac{267}{830}s-\frac{431}{830}t+\frac{157}{830}st$ |
| $(0,-1,-1,-1,-1,0,0,0)$ | (7) | $-\frac{937}{830}+\frac{267}{830}s-\frac{431}{830}t+\frac{157}{830}st$ |
| $(0,-1,-1,0,0,0,-1,-1)$ | $\varnothing$ | $-\frac{147}{332}+\frac{47}{332}s+\frac{77}{1660}t-\frac{37}{1660}st$ |
| $(0,0,-1,-2,-1,0,0,0)$ | $\varnothing$ | $\frac{1351}{830}-\frac{657}{1660}s-\frac{329}{1660}t+\frac{19}{830}st$ |
| $(0,0,-1,-1,-1,0,-1,0)$ | $\varnothing$ | $-\frac{2019}{3320}+\frac{95}{664}s+\frac{183}{664}t-\frac{311}{3320}st$ |
| $(0,0,-1,-1,-1,0,0,-1)$ | $\varnothing$ | $\frac{363}{3320}-\frac{229}{3320}s+\frac{1467}{3320}t-\frac{393}{3320}st$ |
| $(0,0,-1,-1,0,0,-1,-1)$ | $\varnothing$ | $-\frac{279}{1660}+\frac{59}{830}s+\frac{43}{166}t-\frac{23}{332}st$ |
| $(0,0,0,-1,-1,0,-1,-1)$ | $\varnothing$ | $-\frac{397}{1660}+\frac{88}{1245}s-\frac{23}{415}t+\frac{41}{4980}st$ |

**Cartan weight** $(0,8)$**.**

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(0,0,0,1,1,0,1,1)$ | $\varnothing$ | $\frac{177}{520}-\frac{9}{40}s-\frac{277}{1560}t-\frac{7}{120}st$ |
| $(0,0,1,1,0,0,1,1)$ | $\varnothing$ | $-\frac{503}{1560}-\frac{197}{1560}s+\frac{2143}{4680}t-\frac{311}{4680}st$ |
| $(0,0,1,1,1,0,0,1)$ | $\varnothing$ | $\frac{5}{78}+\frac{2}{15}s+\frac{101}{585}t-\frac{1}{18}st$ |
| $(0,0,1,1,1,0,1,0)$ | $\varnothing$ | $\frac{227}{780}-\frac{1}{60}s+\frac{127}{2340}t-\frac{17}{180}st$ |
| $(0,0,1,2,1,0,0,0)$ | $\varnothing$ | $-\frac{899}{3120}+\frac{43}{3120}s-\frac{857}{9360}t+\frac{37}{9360}st$ |
| $(0,1,1,0,0,0,1,1)$ | $\varnothing$ | $\frac{191}{1560}-\frac{391}{1560}s+\frac{379}{4680}t-\frac{383}{4680}st$ |
| $(0,1,1,1,1,0,0,0)$ | (0) | $\frac{311}{1560}-\frac{67}{1560}s+\frac{163}{4680}t+\frac{157}{4680}st$ |
| $(0,1,1,1,1,0,0,0)$ | (1) | $\frac{311}{1560}-\frac{67}{1560}s+\frac{163}{4680}t+\frac{157}{4680}st$ |
| $(0,1,1,1,1,0,0,0)$ | (6) | $-\frac{311}{1560}+\frac{67}{1560}s-\frac{163}{4680}t-\frac{157}{4680}st$ |
| $(0,1,1,1,1,0,0,0)$ | (7) | $-\frac{311}{1560}+\frac{67}{1560}s-\frac{163}{4680}t-\frac{157}{4680}st$ |
| $(1,1,0,0,0,0,1,1)$ | $\varnothing$ | $-\frac{1197}{2600}-\frac{3}{2600}s+\frac{3907}{7800}t-\frac{239}{7800}st$ |
| $(1,1,0,1,1,0,0,0)$ | $\varnothing$ | $-\frac{147}{650}-\frac{3}{325}s-\frac{347}{3900}t+\frac{97}{3900}st$ |
| $(1,1,1,0,0,0,0,1)$ | $\varnothing$ | $\frac{271}{780}-\frac{121}{3900}s+\frac{3749}{11700}t-\frac{23}{2340}st$ |
| $(1,1,1,0,0,0,1,0)$ | $\varnothing$ | $\frac{638}{975}-\frac{59}{1950}s-\frac{79}{5850}t+\frac{31}{2925}st$ |
| $(1,1,1,1,0,0,0,0)$ | (3) | $\frac{967}{7800}-\frac{359}{7800}s+\frac{121}{23400}t+\frac{979}{23400}st$ |
| $(1,1,1,1,0,0,0,0)$ | (4) | $\frac{967}{7800}-\frac{359}{7800}s+\frac{121}{23400}t+\frac{979}{23400}st$ |
| $(1,1,1,1,0,0,0,0)$ | (6) | $-\frac{967}{7800}+\frac{359}{7800}s-\frac{121}{23400}t-\frac{979}{23400}st$ |
| $(1,1,1,1,0,0,0,0)$ | (7) | $-\frac{967}{7800}+\frac{359}{7800}s-\frac{121}{23400}t-\frac{979}{23400}st$ |
| $(1,1,1,1,1,0,-1,0)$ | $\varnothing$ | $\frac{641}{3900}-\frac{17}{780}s-\frac{107}{2340}t-\frac{133}{11700}st$ |
| $(1,1,1,1,1,0,0,-1)$ | $\varnothing$ | $-\frac{53}{3900}+\frac{109}{3900}s+\frac{1229}{11700}t-\frac{61}{11700}st$ |
| $(1,2,1,0,0,0,0,0)$ | $\varnothing$ | $-\frac{1973}{15600}-\frac{479}{15600}s-\frac{3349}{46800}t+\frac{1949}{46800}st$ |

**Cartan weight** $(1,-5)$**.**

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-2,-2,-2,-1,0,-1,0,0)$ | $\varnothing$ | $\frac{35}{16}+\frac{175}{1872}st$ |
| $(-1,-2,-3,-2,-1,-1,1,1)$ | $\varnothing$ | $-\frac{203}{208}s-\frac{161}{48}t$ |
| $(-1,-2,-3,-1,0,-1,0,0)$ | $\varnothing$ | $\frac{7}{8}-\frac{49}{156}st$ |
| $(-1,-2,-2,-2,-1,-1,0,1)$ | $\varnothing$ | $-\frac{161}{48}-\frac{203}{624}s-\frac{161}{144}t+\frac{203}{624}st$ |
| $(-1,-2,-2,-2,-1,-1,1,0)$ | $\varnothing$ | $-\frac{161}{48}+\frac{203}{624}s+\frac{161}{144}t+\frac{203}{624}st$ |
| $(-1,-2,-2,-1,-1,-1,0,0)$ | (3) | $\frac{91}{48}+\frac{371}{1872}st$ |
| $(-1,-2,-2,-1,-1,-1,0,0)$ | (4) | $\frac{91}{48}+\frac{371}{1872}st$ |
| $(-1,-2,-2,-1,-1,-1,0,0)$ | (6) | $-\frac{91}{48}-\frac{371}{1872}st$ |
| $(-1,-2,-2,-1,-1,-1,0,0)$ | (7) | $-\frac{91}{48}-\frac{371}{1872}st$ |
| $(-1,-2,-2,-1,0,-1,-1,0)$ | $\varnothing$ | $-\frac{7}{24}+\frac{49}{156}s+\frac{7}{24}t+\frac{49}{468}st$ |
| $(-1,-2,-2,-1,0,-1,0,-1)$ | $\varnothing$ | $-\frac{7}{24}-\frac{49}{156}s-\frac{7}{24}t+\frac{49}{468}st$ |
| $(-1,-1,-2,-2,-1,-1,0,0)$ | (0) | $\frac{7}{32}+\frac{1351}{3744}st$ |
| $(-1,-1,-2,-2,-1,-1,0,0)$ | (1) | $\frac{7}{32}+\frac{1351}{3744}st$ |
| $(-1,-1,-2,-2,-1,-1,0,0)$ | (6) | $-\frac{7}{32}-\frac{1351}{3744}st$ |
| $(-1,-1,-2,-2,-1,-1,0,0)$ | (7) | $-\frac{7}{32}-\frac{1351}{3744}st$ |
| $(-1,-1,-2,-1,0,-1,-1,-1)$ | $\varnothing$ | $\frac{63}{416}s-\frac{77}{288}t$ |
| $(0,-1,-3,-2,-1,-1,0,0)$ | $\varnothing$ | $-\frac{343}{160}-\frac{133}{6240}st$ |
| $(0,-1,-2,-2,-2,-1,0,0)$ | $\varnothing$ | $\frac{119}{240}-\frac{3311}{9360}st$ |
| $(0,-1,-2,-2,-1,-1,-1,0)$ | $\varnothing$ | $-\frac{343}{480}-\frac{133}{6240}s+\frac{343}{480}t-\frac{133}{18720}st$ |
| $(0,-1,-2,-2,-1,-1,0,-1)$ | $\varnothing$ | $-\frac{343}{480}+\frac{133}{6240}s-\frac{343}{480}t-\frac{133}{18720}st$ |
| $(0,-1,-2,-1,-1,-1,-1,-1)$ | $\varnothing$ | $\frac{287}{1560}s-\frac{7}{45}t$ |

**Cartan weight** $(1,5)$**.**

| Momentum $\lambda$ | Oscillators $\mathcal{O}$ | Coefficient |
|---|---|---|
| $(-1,-1,-1,-1,0,-1,1,1)$ | $\varnothing$ | $-\frac{1013}{488}+\frac{289}{488}s-\frac{735}{488}t-\frac{185}{488}st$ |
| $(-1,-1,-1,0,0,-1,0,1)$ | $\varnothing$ | $-\frac{41}{122}-\frac{145}{244}s-\frac{23}{61}t+\frac{17}{244}st$ |
| $(-1,-1,-1,0,0,-1,1,0)$ | $\varnothing$ | $\frac{179}{244}+\frac{47}{244}s+\frac{5}{244}t-\frac{81}{244}st$ |
| $(-1,-1,-1,0,1,-1,0,0)$ | $\varnothing$ | $\frac{2593}{2440}+\frac{163}{2440}s-\frac{1361}{2440}t+\frac{33}{2440}st$ |
| $(-1,-1,0,0,0,-1,0,0)$ | $\varnothing$ | $-\frac{97}{732}+\frac{49}{366}s+\frac{29}{244}t+\frac{16}{183}st$ |
| $(0,-1,-1,-1,-1,-1,1,1)$ | $\varnothing$ | $-\frac{376}{305}+\frac{481}{610}s-\frac{319}{305}t-\frac{283}{610}st$ |
| $(0,-1,-1,0,0,-1,0,0)$ | (0) | $\frac{2399}{2440}+\frac{359}{2440}s-\frac{1187}{2440}t+\frac{161}{2440}st$ |
| $(0,-1,-1,0,0,-1,0,0)$ | (1) | $\frac{2399}{2440}+\frac{359}{2440}s-\frac{1187}{2440}t+\frac{161}{2440}st$ |
| $(0,-1,-1,0,0,-1,0,0)$ | (6) | $-\frac{2399}{2440}-\frac{359}{2440}s+\frac{1187}{2440}t-\frac{161}{2440}st$ |
| $(0,-1,-1,0,0,-1,0,0)$ | (7) | $-\frac{2399}{2440}-\frac{359}{2440}s+\frac{1187}{2440}t-\frac{161}{2440}st$ |
| $(0,0,-1,-1,-1,-1,0,1)$ | $\varnothing$ | $-\frac{2563}{4880}-\frac{187}{4880}s-\frac{6299}{4880}t-\frac{127}{4880}st$ |
| $(0,0,-1,-1,-1,-1,1,0)$ | $\varnothing$ | $\frac{1073}{488}+\frac{71}{1220}s+\frac{467}{1220}t-\frac{3}{488}st$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (3) | $\frac{1429}{4880}+\frac{1339}{4880}s-\frac{317}{4880}t+\frac{801}{4880}st$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (4) | $\frac{1429}{4880}+\frac{1339}{4880}s-\frac{317}{4880}t+\frac{801}{4880}st$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (6) | $-\frac{1429}{4880}-\frac{1339}{4880}s+\frac{317}{4880}t-\frac{801}{4880}st$ |
| $(0,0,-1,-1,0,-1,0,0)$ | (7) | $-\frac{1429}{4880}-\frac{1339}{4880}s+\frac{317}{4880}t-\frac{801}{4880}st$ |
| $(0,0,-1,0,0,-1,-1,0)$ | $\varnothing$ | $\frac{2713}{4880}-\frac{71}{4880}s-\frac{467}{4880}t-\frac{343}{4880}st$ |
| $(0,0,-1,0,0,-1,0,-1)$ | $\varnothing$ | $\frac{41}{305}-\frac{55}{488}s-\frac{159}{488}t-\frac{17}{610}st$ |
| $(0,0,0,-1,-1,-1,0,0)$ | $\varnothing$ | $\frac{8167}{14640}+\frac{97}{14640}s-\frac{1477}{4880}t-\frac{157}{14640}st$ |
| $(0,0,0,0,0,-1,-1,-1)$ | $\varnothing$ | $\frac{2057}{14640}+\frac{479}{14640}s+\frac{1123}{14640}t-\frac{69}{4880}st$ |
| $(1,0,-1,-1,-1,-1,0,0)$ | $\varnothing$ | $\frac{511}{12200}-\frac{3299}{12200}s-\frac{1423}{12200}t-\frac{2081}{12200}st$ |

## S25.2 The 86-state ambient basis and the eighty-current complement

Source: `c8_reviewability_20260927/current80_witness.json`. The 86 ambient basis states are ordered as follows. Oscillator rows are Cartan states; the remaining rows are root exponentials. The positive Hermitian form is the lattice Gram form on the Cartan states and the transported torus norm on the mutually orthogonal root states, as specified in the integrated guide. These ambient basis indices are the $e_j$ used in the subsequent complement and cyclic-row tables.

| $j$ | Momentum $\lambda$ | Oscillators $\mathcal{O}$ |
|---|---|---|
| 0 | $(0,0,0,0,0,0,0,0)$ | $(0)$ |
| 1 | $(0,0,0,0,0,0,0,0)$ | $(1)$ |
| 2 | $(0,0,0,0,0,0,0,0)$ | $(2)$ |
| 3 | $(0,0,0,0,0,0,0,0)$ | $(3)$ |
| 4 | $(0,0,0,0,0,0,0,0)$ | $(4)$ |
| 5 | $(0,0,0,0,0,0,0,0)$ | $(5)$ |
| 6 | $(0,0,0,0,0,0,0,0)$ | $(6)$ |
| 7 | $(0,0,0,0,0,0,0,0)$ | $(7)$ |
| 8 | $(-1,-2,-3,-2,-1,-2,0,0)$ | $\varnothing$ |
| 9 | $(-1,-2,-3,-2,-1,-1,0,0)$ | $\varnothing$ |
| 10 | $(-1,-2,-2,-2,-1,-1,0,0)$ | $\varnothing$ |
| 11 | $(-1,-2,-2,-1,-1,-1,0,0)$ | $\varnothing$ |
| 12 | $(-1,-2,-2,-1,0,-1,0,0)$ | $\varnothing$ |
| 13 | $(-1,-1,-2,-2,-1,-1,0,0)$ | $\varnothing$ |
| 14 | $(-1,-1,-2,-1,-1,-1,0,0)$ | $\varnothing$ |
| 15 | $(-1,-1,-2,-1,0,-1,0,0)$ | $\varnothing$ |
| 16 | $(-1,-1,-1,-1,-1,-1,0,0)$ | $\varnothing$ |
| 17 | $(-1,-1,-1,-1,-1,0,0,0)$ | $\varnothing$ |
| 18 | $(-1,-1,-1,-1,0,-1,0,0)$ | $\varnothing$ |
| 19 | $(-1,-1,-1,-1,0,0,0,0)$ | $\varnothing$ |
| 20 | $(-1,-1,-1,0,0,-1,0,0)$ | $\varnothing$ |
| 21 | $(-1,-1,-1,0,0,0,0,0)$ | $\varnothing$ |
| 22 | $(-1,-1,0,0,0,0,0,0)$ | $\varnothing$ |
| 23 | $(-1,0,0,0,0,0,0,0)$ | $\varnothing$ |
| 24 | $(0,-1,-2,-2,-1,-1,0,0)$ | $\varnothing$ |
| 25 | $(0,-1,-2,-1,-1,-1,0,0)$ | $\varnothing$ |
| 26 | $(0,-1,-2,-1,0,-1,0,0)$ | $\varnothing$ |
| 27 | $(0,-1,-1,-1,-1,-1,0,0)$ | $\varnothing$ |
| 28 | $(0,-1,-1,-1,-1,0,0,0)$ | $\varnothing$ |
| 29 | $(0,-1,-1,-1,0,-1,0,0)$ | $\varnothing$ |
| 30 | $(0,-1,-1,-1,0,0,0,0)$ | $\varnothing$ |
| 31 | $(0,-1,-1,0,0,-1,0,0)$ | $\varnothing$ |
| 32 | $(0,-1,-1,0,0,0,0,0)$ | $\varnothing$ |
| 33 | $(0,-1,0,0,0,0,0,0)$ | $\varnothing$ |
| 34 | $(0,0,-1,-1,-1,-1,0,0)$ | $\varnothing$ |
| 35 | $(0,0,-1,-1,-1,0,0,0)$ | $\varnothing$ |
| 36 | $(0,0,-1,-1,0,-1,0,0)$ | $\varnothing$ |
| 37 | $(0,0,-1,-1,0,0,0,0)$ | $\varnothing$ |
| 38 | $(0,0,-1,0,0,-1,0,0)$ | $\varnothing$ |
| 39 | $(0,0,-1,0,0,0,0,0)$ | $\varnothing$ |
| 40 | $(0,0,0,-1,-1,0,0,0)$ | $\varnothing$ |
| 41 | $(0,0,0,-1,0,0,0,0)$ | $\varnothing$ |
| 42 | $(0,0,0,0,-1,0,0,0)$ | $\varnothing$ |
| 43 | $(0,0,0,0,0,-1,0,0)$ | $\varnothing$ |
| 44 | $(0,0,0,0,0,0,-1,-1)$ | $\varnothing$ |
| 45 | $(0,0,0,0,0,0,-1,0)$ | $\varnothing$ |
| 46 | $(0,0,0,0,0,0,0,-1)$ | $\varnothing$ |
| 47 | $(0,0,0,0,0,0,0,1)$ | $\varnothing$ |
| 48 | $(0,0,0,0,0,0,1,0)$ | $\varnothing$ |
| 49 | $(0,0,0,0,0,0,1,1)$ | $\varnothing$ |
| 50 | $(0,0,0,0,0,1,0,0)$ | $\varnothing$ |
| 51 | $(0,0,0,0,1,0,0,0)$ | $\varnothing$ |
| 52 | $(0,0,0,1,0,0,0,0)$ | $\varnothing$ |
| 53 | $(0,0,0,1,1,0,0,0)$ | $\varnothing$ |
| 54 | $(0,0,1,0,0,0,0,0)$ | $\varnothing$ |
| 55 | $(0,0,1,0,0,1,0,0)$ | $\varnothing$ |
| 56 | $(0,0,1,1,0,0,0,0)$ | $\varnothing$ |
| 57 | $(0,0,1,1,0,1,0,0)$ | $\varnothing$ |
| 58 | $(0,0,1,1,1,0,0,0)$ | $\varnothing$ |
| 59 | $(0,0,1,1,1,1,0,0)$ | $\varnothing$ |

| $j$ | Momentum $\lambda$ | Oscillators $\mathcal{O}$ |
|---|---|---|
| 60 | $(0,1,0,0,0,0,0,0)$ | $\varnothing$ |
| 61 | $(0,1,1,0,0,0,0,0)$ | $\varnothing$ |
| 62 | $(0,1,1,0,0,1,0,0)$ | $\varnothing$ |
| 63 | $(0,1,1,1,0,0,0,0)$ | $\varnothing$ |
| 64 | $(0,1,1,1,0,1,0,0)$ | $\varnothing$ |
| 65 | $(0,1,1,1,1,0,0,0)$ | $\varnothing$ |
| 66 | $(0,1,1,1,1,1,0,0)$ | $\varnothing$ |
| 67 | $(0,1,2,1,0,1,0,0)$ | $\varnothing$ |
| 68 | $(0,1,2,1,1,1,0,0)$ | $\varnothing$ |
| 69 | $(0,1,2,2,1,1,0,0)$ | $\varnothing$ |
| 70 | $(1,0,0,0,0,0,0,0)$ | $\varnothing$ |
| 71 | $(1,1,0,0,0,0,0,0)$ | $\varnothing$ |
| 72 | $(1,1,1,0,0,0,0,0)$ | $\varnothing$ |
| 73 | $(1,1,1,0,0,1,0,0)$ | $\varnothing$ |
| 74 | $(1,1,1,1,0,0,0,0)$ | $\varnothing$ |
| 75 | $(1,1,1,1,0,1,0,0)$ | $\varnothing$ |
| 76 | $(1,1,1,1,1,0,0,0)$ | $\varnothing$ |
| 77 | $(1,1,1,1,1,1,0,0)$ | $\varnothing$ |
| 78 | $(1,1,2,1,0,1,0,0)$ | $\varnothing$ |
| 79 | $(1,1,2,1,1,1,0,0)$ | $\varnothing$ |
| 80 | $(1,1,2,2,1,1,0,0)$ | $\varnothing$ |
| 81 | $(1,2,2,1,0,1,0,0)$ | $\varnothing$ |
| 82 | $(1,2,2,1,1,1,0,0)$ | $\varnothing$ |
| 83 | $(1,2,2,2,1,1,0,0)$ | $\varnothing$ |
| 84 | $(1,2,3,2,1,1,0,0)$ | $\varnothing$ |
| 85 | $(1,2,3,2,1,2,0,0)$ | $\varnothing$ |

Every supplied complement row is printed next. This is an 80-row basis of $K_{\mathrm{cur}}^{\perp} \cap B_1$, not a second choice of ambient coordinates. The symbol $b_i$ is used here only for these rows; it is unrelated to the module label $b$. The recorded independent matrix check verifies orthogonality to the current plane, positive metric, and independence.

| $i$ | Complete complement row |
|---|---|
| 0 | $b_0 = \big(1\big)e_0 + \big(-\frac{4}{17}\big)e_1 + \big(-\frac{9}{34}\big)e_2 + \big(-\frac{4}{17}\big)e_3 + \big(-\frac{5}{34}\big)e_4 + \big(-\frac{1}{17}\big)e_6 + \big(-\frac{1}{17}\big)e_7$ |
| 1 | $b_1 = \big(1\big)e_1 + \big(-\frac{9}{26}\big)e_2 + \big(-\frac{4}{13}\big)e_3 + \big(-\frac{5}{26}\big)e_4 + \big(-\frac{1}{13}\big)e_6 + \big(-\frac{1}{13}\big)e_7$ |
| 2 | $b_2 = \big(1\big)e_2 + \big(-\frac{8}{17}\big)e_3 + \big(-\frac{5}{17}\big)e_4 + \big(-\frac{2}{17}\big)e_6 + \big(-\frac{2}{17}\big)e_7$ |
| 3 | $b_3 = \big(1\big)e_3 + \big(-\frac{5}{9}\big)e_4 + \big(-\frac{2}{9}\big)e_6 + \big(-\frac{2}{9}\big)e_7$ |
| 4 | $b_4 = \big(1\big)e_4 + \big(-\frac{1}{2}\big)e_6 + \big(-\frac{1}{2}\big)e_7$ |
| 5 | $b_5 = \big(1\big)e_6 + \big(-1\big)e_7$ |
| 6 | $b_6 = \big(1\big)e_9$ |
| 7 | $b_7 = \big(1\big)e_{10}$ |
| 8 | $b_8 = \big(1\big)e_{11}$ |
| 9 | $b_9 = \big(1\big)e_{13}$ |
| 10 | $b_{10} = \big(1\big)e_{12}$ |
| 11 | $b_{11} = \big(1\big)e_{14}$ |
| 12 | $b_{12} = \big(1\big)e_{24}$ |
| 13 | $b_{13} = \big(1\big)e_{15}$ |
| 14 | $b_{14} = \big(1\big)e_{16}$ |
| 15 | $b_{15} = \big(1\big)e_{25}$ |
| 16 | $b_{16} = \big(1\big)e_{17}$ |
| 17 | $b_{17} = \big(1\big)e_{18}$ |
| 18 | $b_{18} = \big(1\big)e_{26}$ |
| 19 | $b_{19} = \big(1\big)e_{27}$ |
| 20 | $b_{20} = \big(1\big)e_{19}$ |

| $i$ | Complete complement row |
|---|---|
| 21 | $b_{21} = \left(1\right)e_{28}$ |
| 22 | $b_{22} = \left(1\right)e_{20}$ |
| 23 | $b_{23} = \left(1\right)e_{29}$ |
| 24 | $b_{24} = \left(1\right)e_{34}$ |
| 25 | $b_{25} = \left(1\right)e_{21}$ |
| 26 | $b_{26} = \left(1\right)e_{30}$ |
| 27 | $b_{27} = \left(1\right)e_{35}$ |
| 28 | $b_{28} = \left(1\right)e_{22}$ |
| 29 | $b_{29} = \left(1\right)e_{32}$ |
| 30 | $b_{30} = \left(1\right)e_{37}$ |
| 31 | $b_{31} = \left(1\right)e_{40}$ |
| 32 | $b_{32} = \left(1\right)e_{44}$ |
| 33 | $b_{33} = \left(1\right)e_{23} + \left(-\frac{4}{17}\right)e_{33} + \left(-\frac{9}{34}\right)e_{39} + \left(-\frac{4}{17}\right)e_{41} + \left(-\frac{5}{34}\right)e_{42} + \left(-\frac{1}{17}\right)e_{45} + \left(-\frac{1}{17}\right)e_{46}$ |
| 34 | $b_{34} = \left(1\right)e_{33} + \left(-\frac{9}{26}\right)e_{39} + \left(-\frac{4}{13}\right)e_{41} + \left(-\frac{5}{26}\right)e_{42} + \left(-\frac{1}{13}\right)e_{45} + \left(-\frac{1}{13}\right)e_{46}$ |
| 35 | $b_{35} = \left(1\right)e_{39} + \left(-\frac{8}{17}\right)e_{41} + \left(-\frac{5}{17}\right)e_{42} + \left(-\frac{2}{17}\right)e_{45} + \left(-\frac{2}{17}\right)e_{46}$ |
| 36 | $b_{36} = \left(1\right)e_{41} + \left(-\frac{5}{9}\right)e_{42} + \left(-\frac{2}{9}\right)e_{45} + \left(-\frac{2}{9}\right)e_{46}$ |
| 37 | $b_{37} = \left(1\right)e_{42} + \left(-\frac{1}{2}\right)e_{45} + \left(-\frac{1}{2}\right)e_{46}$ |
| 38 | $b_{38} = \left(1\right)e_{45} + \left(-1\right)e_{46}$ |
| 39 | $b_{39} = \left(1\right)e_{31}$ |
| 40 | $b_{40} = \left(1\right)e_{36}$ |
| 41 | $b_{41} = \left(1\right)e_{38}$ |
| 42 | $b_{42} = \left(1\right)e_{43}$ |
| 43 | $b_{43} = \left(1\right)e_{47} + \left(-\frac{2}{37}\right)e_{48} + \left(-\frac{2}{37}\right)e_{51} + \left(-\frac{2}{37}\right)e_{52} + \left(-\frac{2}{37}\right)e_{54} + \left(-\frac{2}{37}\right)e_{60} + \left(-\frac{2}{37}\right)e_{70}$ |
| 44 | $b_{44} = \left(1\right)e_{48} + \left(-\frac{2}{35}\right)e_{51} + \left(-\frac{2}{35}\right)e_{52} + \left(-\frac{2}{35}\right)e_{54} + \left(-\frac{2}{35}\right)e_{60} + \left(-\frac{2}{35}\right)e_{70}$ |
| 45 | $b_{45} = \left(1\right)e_{51} + \left(-\frac{1}{6}\right)e_{52} + \left(-\frac{1}{6}\right)e_{54} + \left(-\frac{1}{6}\right)e_{60} + \left(-\frac{1}{6}\right)e_{70}$ |
| 46 | $b_{46} = \left(1\right)e_{52} + \left(-\frac{4}{11}\right)e_{54} + \left(-\frac{4}{11}\right)e_{60} + \left(-\frac{4}{11}\right)e_{70}$ |
| 47 | $b_{47} = \left(1\right)e_{54} + \left(-\frac{9}{13}\right)e_{60} + \left(-\frac{9}{13}\right)e_{70}$ |
| 48 | $b_{48} = \left(1\right)e_{60} + \left(-\frac{8}{5}\right)e_{70}$ |
| 49 | $b_{49} = \left(1\right)e_{49}$ |
| 50 | $b_{50} = \left(1\right)e_{53}$ |
| 51 | $b_{51} = \left(1\right)e_{56}$ |
| 52 | $b_{52} = \left(1\right)e_{61}$ |
| 53 | $b_{53} = \left(1\right)e_{71}$ |
| 54 | $b_{54} = \left(1\right)e_{50}$ |
| 55 | $b_{55} = \left(1\right)e_{55}$ |
| 56 | $b_{56} = \left(1\right)e_{57}$ |
| 57 | $b_{57} = \left(1\right)e_{62}$ |
| 58 | $b_{58} = \left(1\right)e_{58}$ |
| 59 | $b_{59} = \left(1\right)e_{63}$ |
| 60 | $b_{60} = \left(1\right)e_{72}$ |
| 61 | $b_{61} = \left(1\right)e_{59}$ |
| 62 | $b_{62} = \left(1\right)e_{64}$ |
| 63 | $b_{63} = \left(1\right)e_{73}$ |
| 64 | $b_{64} = \left(1\right)e_{65}$ |
| 65 | $b_{65} = \left(1\right)e_{74}$ |
| 66 | $b_{66} = \left(1\right)e_{66}$ |
| 67 | $b_{67} = \left(1\right)e_{67}$ |
| 68 | $b_{68} = \left(1\right)e_{75}$ |
| 69 | $b_{69} = \left(1\right)e_{68}$ |
| 70 | $b_{70} = \left(1\right)e_{77}$ |
| 71 | $b_{71} = \left(1\right)e_{78}$ |

| $i$ | Complete complement row |
|---|---|
| 72 | $b_{72} = \big(1\big)e_{69}$ |
| 73 | $b_{73} = \big(1\big)e_{79}$ |
| 74 | $b_{74} = \big(1\big)e_{81}$ |
| 75 | $b_{75} = \big(1\big)e_{76}$ |
| 76 | $b_{76} = \big(1\big)e_{80}$ |
| 77 | $b_{77} = \big(1\big)e_{82}$ |
| 78 | $b_{78} = \big(1\big)e_{83}$ |
| 79 | $b_{79} = \big(1\big)e_{84}$ |

## S25.3 Operator indices and all eighty cyclic recipes

The original current names $K0, \dots, K5$ denote $(H_L, E_L, F_L, H_R, E_R, F_R)$ in that order. The weight-two field names field0, ..., field7 denote the charged highest fields $a_0, \dots, a_7$ in the finite supplement, and field8 is the retained neutral stress $e$. Each even operator index is followed by its adjoint. The current operators use mode zero; the weight-two operators use mode one. The adjoints are with respect to the transported positive form, so replacing an adjoint by an unscaled Chevalley lowering operator can change a row.

| $j$ | Saved operator name |
|---|---|
| 0 | `K0` |
| 1 | `adjoint:K0` |
| 2 | `K1` |
| 3 | `adjoint:K1` |
| 4 | `K2` |
| 5 | `adjoint:K2` |
| 6 | `K3` |
| 7 | `adjoint:K3` |
| 8 | `K4` |
| 9 | `adjoint:K4` |
| 10 | `K5` |
| 11 | `adjoint:K5` |
| 12 | `field0` |
| 13 | `adjoint:field0` |
| 14 | `field1` |
| 15 | `adjoint:field1` |
| 16 | `field2` |
| 17 | `adjoint:field2` |
| 18 | `field3` |
| 19 | `adjoint:field3` |
| 20 | `field4` |
| 21 | `adjoint:field4` |
| 22 | `field5` |
| 23 | `adjoint:field5` |
| 24 | `field6` |
| 25 | `adjoint:field6` |
| 26 | `field7` |
| 27 | `adjoint:field7` |
| 28 | `field8` |

| $j$ | Saved operator name |
|---|---|
| 29 | `adjoint:field8` |

The symbols $b_i$ and $r_i$ in this appendix are local names for the complement and cyclic rows. They are not the $b_i$ membership basis or the $r_i$ short-generation basis in the finite supplement. In the next table, $r_i$ is normalized after subtracting the earlier rows, exactly in the order written. The pivot is an ambient basis index. The unique seed has Cartan weight $(0, 10)$ and ambient index 76. All referenced parents and reduction rows precede the new row. A list of recipes still requires the claimed identification of the matrices with field modes; those finite-mode obligations are separate from reading the scalar coefficients.

| $i$ | Pivot | Exact cyclic-row recipe |
|---|---|---|
| 0 | 76 | $r_0 = e_{76}$ |
| 1 | 65 | $r_1 = \left(\frac{1}{5}\right)\left[M_9 r_0\right]$ |
| 2 | 58 | $r_2 = \left(-\frac{6461}{118600} + \frac{931}{118600}st\right)\left[M_{13} r_0\right]$ |
| 3 | 47 | $r_3 = \left(-\frac{1}{2600} + \frac{1}{5200}s - \frac{1}{2600}t - \frac{1}{15600}st\right)\left[M_{15} r_0\right]$ |
| 4 | 48 | $r_4 = \left(\frac{1}{3200} + \frac{1}{28800}s + \frac{1}{9600}t - \frac{1}{28800}st\right)\left[M_{17} r_0 - \left(-945 + 45s - 135t - 105st\right) r_3\right]$ |
| 5 | 33 | $r_5 = \left(\frac{1}{288000} - \frac{1}{192000}s + \frac{1}{576000}t + \frac{1}{288000}st\right)\left[M_{19} r_0\right]$ |
| 6 | 84 | $r_6 = \left(-3456s + 18432t\right)\left[M_{21} r_0\right]$ |
| 7 | 80 | $r_7 = \left(-\frac{6804}{47}s - \frac{3780}{47}t\right)\left[M_{23} r_0\right]$ |
| 8 | 69 | $r_8 = \left(\frac{1062}{215}s + \frac{3546}{215}t\right)\left[M_{25} r_0\right]$ |
| 9 | 59 | $r_9 = \left(\frac{1}{80}s - \frac{3}{80}t\right)\left[M_{27} r_0\right]$ |
| 10 | 63 | $r_{10} = \left(-\frac{2314}{33535} - \frac{539}{201210}st\right)\left[M_9 r_1 - \left(8\right) r_2\right]$ |
| 11 | 53 | $r_{11} = \left(-\frac{6461}{213480} + \frac{931}{213480}st\right)\left[M_{13} r_1\right]$ |
| 12 | 0 | $r_{12} = \left(-\frac{67}{144800}s + \frac{77}{434400}t\right)\left[M_{15} r_1\right]$ |
| 13 | 1 | $r_{13} = \left(\frac{1}{12800}s - \frac{83}{115200}t\right)\left[M_{17} r_1 - \left(150s + 150t\right) r_{12}\right]$ |
| 14 | 22 | $r_{14} = \left(-\frac{1}{288000} + \frac{1}{192000}s - \frac{1}{576000}t - \frac{1}{288000}st\right)\left[M_{19} r_1\right]$ |
| 15 | 43 | $r_{15} = \left(-2s - \frac{32}{3}t\right)\left[M_{20} r_1\right]$ |
| 16 | 83 | $r_{16} = \left(-1920s + 10240t\right)\left[M_{21} r_1\right]$ |
| 17 | 79 | $r_{17} = \left(\frac{945}{944}s + \frac{310905}{944}t\right)\left[M_{23} r_1 - \left(-\frac{1}{2016}s + \frac{5}{18144}t\right) r_8\right]$ |
| 18 | 68 | $r_{18} = \left(-\frac{387}{110}s - \frac{1101}{110}t\right)\left[M_{25} r_1\right]$ |
| 19 | 57 | $r_{19} = \left(-\frac{1}{80}s + \frac{3}{80}t\right)\left[M_{27} r_1\right]$ |
| 20 | 74 | $r_{20} = \left(\frac{1}{2} - \frac{71}{798}st\right)\left[M_8 r_2 - \left(\frac{6393}{11860} + \frac{10241}{154180}st\right) r_1\right]$ |
| 21 | 51 | $r_{21} = \left(-\frac{7168048979}{546824905896} + \frac{417551771}{546824905896}st\right)\left[M_{13} r_2 - \left(\frac{6228333}{269815} - \frac{7654644}{501085}s - \frac{9534732}{269815}t - \frac{1666737}{501085}st\right) r_3 - \left(\frac{34832529}{539630} + \frac{27964143}{1002170}s + \frac{34832529}{539630}t - \frac{9321381}{1002170}st\right) r_4\right]$ |
| 22 | 23 | $r_{22} = \left(-\frac{132071}{474974400}s + \frac{919451}{1424923200}t\right)\left[M_{15} r_2 - \left(-\frac{1065708}{14825}s - \frac{7184316}{14825}t\right) r_5\right]$ |
| 23 | 39 | $r_{23} = \left(-\frac{24781637}{250186821120}s + \frac{1320293767}{750560463360}t\right)\left[M_{17} r_2 - \left(-\frac{425484}{7709}s - \frac{51636}{593}t\right) r_{22} - \left(\frac{67626}{593}s - \frac{1398}{593}t\right) r_5\right]$ |
| 24 | 21 | $r_{24} = \left(-\frac{271}{28339200} - \frac{77}{245606400}s - \frac{271}{56678400}t + \frac{77}{368409600}st\right)\left[M_{19} r_2\right]$ |
| 25 | 38 | $r_{25} = \left(\frac{482}{661}s - \frac{7982}{1983}t\right)\left[M_{20} r_2\right]$ |
| 26 | 82 | $r_{26} = \left(\frac{1141152}{631}s - \frac{2111904}{631}t\right)\left[M_{21} r_2 - \left(-\frac{68409}{2910918400}s + \frac{439553}{6045753600}t\right) r_7\right]$ |
| 27 | 77 | $r_{27} = \left(-\frac{59444280}{1397023}s + \frac{150608640}{1397023}t\right)\left[M_{23} r_2 - \left(\frac{191939}{233120160}s - \frac{20449}{17932320}t\right) r_{18}\right]$ |
| 28 | 66 | $r_{28} = \left(\frac{1635598}{302531}s + \frac{4199754}{302531}t\right)\left[M_{25} r_2\right]$ |
| 29 | 55 | $r_{29} = \left(\frac{133}{4160}s - \frac{71}{960}t\right)\left[M_{27} r_2\right]$ |
| 30 | 72 | $r_{30} = \left(-\frac{1564601}{375516} + \frac{626603}{2628612}st\right)\left[M_{28} r_2 - \left(\frac{317243}{1507406} + \frac{305501}{19596278}st\right) r_2 - \left(-\frac{71563899}{937144585} - \frac{58157176}{29051482135}st\right) r_{10}\right]$ |
| 31 | 49 | $r_{31} = \left(\frac{1}{2} - \frac{1}{2}t\right)\left[M_8 r_3 - \left(\frac{27}{40} + \frac{1}{520}s - \frac{9}{40}t + \frac{1}{520}st\right) r_{11}\right]$ |
| 32 | 41 | $r_{32} = \left(-\frac{5456231417}{2248411915712} + \frac{54360841717}{2248411915712}s - \frac{5456231417}{6745235747136}t - \frac{54360841717}{2248411915712}st\right)\left[M_{13} r_3 - \left(\frac{2733}{1183} - \frac{657}{1183}s - \frac{911}{1183}t - \frac{657}{1183}st\right) r_{22} - \left(-\frac{57}{13} + \frac{3147}{1183}s + \frac{19}{13}t + \frac{3147}{1183}st\right) r_5 - \left(-\frac{883386}{16723057} + \frac{27719784}{117061399}s + \frac{294462}{16723057}t + \frac{27719784}{117061399}st\right) r_{23}\right]$ |

| $i$ | Pivot | Exact cyclic-row recipe |
|---|---|---|
| 33 | 30 | $r_{33} = \left(-\frac{605371}{1359976320} - \frac{155441}{453325440}s + \frac{605371}{1359976320}t - \frac{155441}{1359976320}st\right)\left[M_{15}r_3 - \left(-\frac{2664}{13} + \frac{288}{13}s - \frac{2664}{13}t - \frac{96}{13}st\right)r_{24}\right]$ |
| 34 | 35 | $r_{34} = \left(-\frac{91}{625200} + \frac{1213}{937800}s + \frac{91}{625200}t + \frac{1213}{2813400}st\right)\left[M_{17}r_3 - \left(-\frac{630}{13} + \frac{810}{13}s - \frac{630}{13}t - \frac{270}{13}st\right)r_{24} - \left(-\frac{48429}{1066} + \frac{57483}{1066}s - \frac{48429}{1066}t - \frac{19161}{1066}st\right)r_{33}\right]$ |
| 35 | 17 | $r_{35} = \left(-\frac{13}{4032000} - \frac{13}{2419200}t\right)\left[M_{19}r_3\right]$ |
| 36 | 20 | $r_{36} = \left(-\frac{486837}{430720} - \frac{69069}{430720}s + \frac{486837}{430720}t - \frac{23023}{430720}st\right)\left[M_{20}r_3\right]$ |
| 37 | 78 | $r_{37} = \left(-\frac{486268848}{10609} - \frac{42835824}{10609}s + \frac{486268848}{10609}t - \frac{14278608}{10609}st\right)\left[M_{21}r_3 - \left(\frac{89}{17671680} + \frac{173}{10603008}s + \frac{89}{17671680}t - \frac{173}{31809024}st\right)r_{18} - \left(-\frac{99923}{1619904000} - \frac{5327}{539968000}s - \frac{99923}{1619904000}t + \frac{5327}{1619904000}st\right)r_{27}\right]$ |
| 38 | 64 | $r_{38} = \left(312 - 312t\right)\left[M_{23}r_3 - \left(\frac{137}{131040} - \frac{101}{131040}s + \frac{137}{131040}t + \frac{101}{393120}st\right)r_9\right]$ |
| 39 | 62 | $r_{39} = \left(\frac{117}{46} - \frac{579}{46}s - \frac{117}{46}t - \frac{193}{46}st\right)\left[M_{25}r_3 - \left(\frac{1069}{146640} + \frac{209}{48880}s + \frac{1069}{146640}t - \frac{209}{146640}st\right)r_{19}\right]$ |
| 40 | 52 | $r_{40} = \left(\frac{1606361015313301}{132356594576872} + \frac{399773767573843}{297802337797962}s + \frac{1606361015313301}{397069783730616}t - \frac{399773767573843}{297802337797962}st\right)\left[M_{28}r_3 - \left(\frac{1003}{2542} - \frac{349}{2542}t\right)r_3 - \left(\frac{1047}{2542} + \frac{349}{2542}t\right)r_4 - \left(-\frac{224}{1271} - \frac{1077}{33046}s + \frac{224}{3813}t - \frac{1077}{33046}st\right)r_{21}\right]$ |
| 41 | 56 | $r_{41} = \left(\frac{87133293}{361785616} + \frac{91727181}{361785616}s + \frac{87133293}{361785616}t - \frac{30575727}{361785616}st\right)\left[M_8r_4 - \left(-1\right)r_{31} - \left(\frac{23}{60} + \frac{2}{65}s + \frac{1}{15}t + \frac{3}{260}st\right)r_{11}\right]$ |
| 42 | 42 | $r_{42} = \left(-\frac{79937}{4536960} + \frac{371593}{28734080}s - \frac{79937}{4536960}t - \frac{371593}{86202240}st\right)\left[M_{13}r_4 - \left(-\frac{211}{2366} - \frac{3695}{2366}s + \frac{3855}{2366}t - \frac{6323}{7098}st\right)r_{22} - \left(-\frac{449}{182} - \frac{4817}{2366}s - \frac{83}{182}t + \frac{7771}{7098}st\right)r_5 - \left(-\frac{101425059}{234122798} - \frac{195278337}{234122798}s + \frac{93180123}{234122798}t - \frac{28133067}{234122798}st\right)r_{23} - \left(\frac{8295648957}{3293836910} + \frac{4643613357}{6117125690}s - \frac{8470912519}{3293836910}t + \frac{4790747077}{6117125690}st\right)r_{32}\right]$ |
| 43 | 29 | $r_{43} = \left(-\frac{3717677}{361180} - \frac{341033}{361180}s + \frac{3717677}{1083540}t - \frac{341033}{361180}st\right)\left[M_{20}r_4 - \left(\frac{1317}{61360} - \frac{421}{61360}s - \frac{7883}{61360}t - \frac{2279}{184080}st\right)r_{36}\right]$ |
| 44 | 73 | $r_{44} = \left(-\frac{71442}{863} - \frac{214326}{11219}s + \frac{23814}{863}t - \frac{214326}{11219}st\right)\left[M_{23}r_4 - \left(\frac{3511}{2358720} + \frac{23}{262080}s + \frac{8443}{7076160}t - \frac{473}{786240}st\right)r_9 - \left(\frac{17}{117936} + \frac{103}{176904}t\right)r_{38}\right]$ |
| 45 | 54 | $r_{45} = \left(-\frac{81116728393209849}{3599792168321216} - \frac{147333876920252613}{46797298188175808}s - \frac{81116728393209849}{3599792168321216}t + \frac{49111292306750871}{46797298188175808}st\right)\left[M_{28}r_4 - \left(-\frac{163}{2542} - \frac{7}{62}t\right)r_3 - \left(\frac{2027}{2542} + \frac{349}{2542}t\right)r_4 - \left(-\frac{1631}{19065} + \frac{522}{82615}s - \frac{203}{6355}t - \frac{1621}{82615}st\right)r_{21} - \left(\frac{6049024880463}{376466163338317} - \frac{1194379412559}{376466163338317}s + \frac{4450702145733}{376466163338317}t + \frac{763232897515}{376466163338317}st\right)r_{40}\right]$ |
| 46 | 2 | $r_{46} = \left(\frac{1180}{2867} + \frac{620}{8601}st\right)\left[M_8r_5 - \left(-1\right)r_{13}\right]$ |
| 47 | 60 | $r_{47} = \left(\frac{4465537531}{1021051284} + \frac{40498003}{92822844}st\right)\left[M_{12}r_5 - \left(-\frac{1251}{3640} + \frac{276}{455}s - \frac{92}{455}t - \frac{1251}{3640}st\right)r_3 - \left(\frac{957}{7280} - \frac{2871}{7280}s + \frac{957}{7280}t + \frac{957}{7280}st\right)r_4 - \left(-\frac{19979}{59150} + \frac{7191}{59150}st\right)r_{21} - \left(-\frac{4643856259779}{269539109864570} - \frac{7335869436352}{134769554932285}st\right)r_{40} - \left(\frac{3076461412688453}{18066675159743028} + \frac{882783969937693}{54200025479229084}st\right)r_{45}\right]$ |
| 48 | 15 | $r_{48} = \left(\frac{161}{110}s - \frac{1259}{330}t\right)\left[M_{20}r_5\right]$ |
| 49 | 34 | $r_{49} = \left(\frac{14343}{3068}s + \frac{7}{4}t\right)\left[M_{22}r_5 - \left(-\frac{2}{21}s - \frac{4}{9}t\right)r_{36} - \left(-\frac{483029}{4522560}s + \frac{4939639}{13567680}t\right)r_{43}\right]$ |
| 50 | 31 | $r_{50} = \left(\frac{107}{176}s + \frac{617}{528}t\right)\left[M_{24}r_5\right]$ |
| 51 | 50 | $r_{51} = \left(-\frac{10}{7}s - \frac{170}{49}t\right)\left[M_{25}r_5\right]$ |
| 52 | 36 | $r_{52} = \left(-\frac{1}{52} + \frac{105}{676}st\right)\left[M_2r_7 - \left(-1\right)r_{50}\right]$ |
| 53 | 67 | $r_{53} = \left(\frac{10258302691}{862871513040} - \frac{4186896077}{2588614539120}st\right)\left[M_{13}r_8 - \left(-\frac{64989}{3913} + \frac{15273}{3913}st\right)r_{28}\right]$ |
| 54 | 61 | $r_{54} = \left(\frac{27167872978}{959124501257}s - \frac{227860380134}{221336423367}t\right)\left[M_{20}r_8 - \left(-\frac{2130}{2537} + \frac{420}{2537}st\right)r_{31} - \left(-\frac{35781}{253700}s - \frac{11669}{126850}t\right)r_{11} - \left(-\frac{3266644}{58673199}s - \frac{137243541}{195577330}t\right)r_{41}\right]$ |
| 55 | 81 | $r_{55} = \left(-\frac{115661}{24138} - \frac{341899}{313794}st\right)\left[M_{28}r_8 - \left(\frac{15921}{54653} + \frac{532}{54653}st\right)r_8 - \left(-\frac{131928}{11750395} + \frac{652}{379045}st\right)r_{17}\right]$ |
| 56 | 12 | $r_{56} = \left(-1\right)\left[M_2r_9\right]$ |
| 57 | 75 | $r_{57} = \left(-\frac{2639}{16082} + \frac{47449}{627198}st\right)\left[M_8r_9 - \left(1\right)r_{28} - \left(-\frac{2610697}{2420248} - \frac{687133}{2420248}st\right)r_{53}\right]$ |
| 58 | 45 | $r_{58} = \left(-\frac{347361}{83191} + \frac{82705}{83191}st\right)\left[M_{20}r_9 - \left(\frac{29}{708}s + \frac{7}{118}t\right)r_{22} - \left(\frac{1}{708}s + \frac{31}{236}t\right)r_5 - \left(\frac{21348183}{934116320}s + \frac{85494879}{934116320}t\right)r_{23} - \left(-\frac{3151589789}{51253550160}s + \frac{790152389}{17084516720}t\right)r_{32} - \left(-\frac{16670531767}{6981910685632}s - \frac{2768777465519}{6981910685632}t\right)r_{42}\right]$ |
| 59 | 71 | $r_{59} = \left(\frac{3217201}{4540941}s - \frac{1176037}{4540941}t\right)\left[M_{24}r_9 - \left(-\frac{25}{188} - \frac{15}{188}st\right)r_{31} - \left(\frac{4}{47}s + \frac{3}{47}t\right)r_{11} - \left(\frac{254206}{1811615}s + \frac{551497}{1811615}t\right)r_{41} - \left(-\frac{143921835}{4250980988}s + \frac{39366717}{4250980988}t\right)r_{54}\right]$ |
| 60 | 3 | $r_{60} = \left(\frac{183129037}{2154706560} + \frac{14846741}{2154706560}st\right)\left[M_{13}r_{11} - \left(\frac{2478941}{269815} - \frac{349619}{115635}st\right)r_{12} - \left(\frac{485801488}{48836515} + \frac{209093776}{634874695}st\right)r_{13} - \left(\frac{333663}{1349075} - \frac{8445949}{17537975}st\right)r_{46}\right]$ |
| 61 | 32 | $r_{61} = \left(-\frac{224183}{1107164928}s + \frac{512473}{3321494784}t\right)\left[M_{15}r_{11} - \left(-\frac{2686916}{38545}s + \frac{43836}{2965}t\right)r_{14}\right]$ |
| 62 | 37 | $r_{62} = \left(-\frac{2649855649}{4540554912000}s + \frac{36653790619}{13621664736000}t\right)\left[M_{17}r_{11} - \left(-\frac{866618}{7709}s - \frac{45122}{593}t\right)r_{14} - \left(-\frac{4728816}{38545}s + \frac{361296}{2965}t\right)r_{61}\right]$ |
| 63 | 19 | $r_{63} = \left(-\frac{191}{113318400} - \frac{49}{17433600}s - \frac{191}{226636800}t + \frac{49}{26150400}st\right)\left[M_{19}r_{11}\right]$ |

| $i$ | Pivot | Exact cyclic-row recipe |
|---|---|---|
| 64 | 40 | $r_{64} = \left(-\frac{1581307}{115997888} - \frac{5325887}{1043980992}st\right)\left[M_{13}r_{12} - \left(-\frac{2753619}{82355} + \frac{326751}{82355}st\right)r_{14} - \left(-\frac{5452776}{294125} + \frac{7660728}{2058875}st\right)r_{61} - \left(-\frac{73371682557}{9132669035} + \frac{196548839649}{118724697455}st\right)r_{62}\right]$ |
| 65 | 28 | $r_{65} = \left(\frac{37}{288000}s - \frac{1973}{7776000}t\right)\left[M_{15}r_{12} - \left(-\frac{59940}{181}s - \frac{135756}{181}t\right)r_{63}\right]$ |
| 66 | 18 | $r_{66} = \left(\frac{28843}{878690}s - \frac{556039}{878690}t\right)\left[M_{20}r_{12}\right]$ |
| 67 | 4 | $r_{67} = \left(-\frac{87362185}{9311112} + \frac{199616905}{121044456}st\right)\left[M_{28}r_{12} - \left(\frac{39025}{460102} - \frac{3395}{460102}st\right)r_{12} - \left(\frac{2391528}{208196155} - \frac{142072}{5077955}st\right)r_{13} - \left(-\frac{1589829}{23005100} + \frac{248181}{23005100}st\right)r_{46} - \left(-\frac{249107742}{3297781085} - \frac{50629962}{3297781085}st\right)r_{60}\right]$ |
| 68 | 44 | $r_{68} = \left(-\frac{477441965}{16656094944}s - \frac{303826705}{5552031648}t\right)\left[M_{13}r_{13} - \left(-\frac{297}{182} - \frac{309}{182}st\right)r_{14} - \left(\frac{15768}{2275} + \frac{15348}{2275}st\right)r_{61} - \left(-\frac{2002212081}{807307760} + \frac{57567403407}{10495000880}st\right)r_{62} - \left(-\frac{107611240272}{1655410645} + \frac{35991831822}{4304067677}st\right)r_{64}\right]$ |
| 69 | 26 | $r_{69} = \left(\frac{8510832320}{31266341277}s + \frac{63961234720}{93799023831}t\right)\left[M_{20}r_{13} - \left(-\frac{334}{1475}s - \frac{10643}{35400}t\right)r_{66}\right]$ |
| 70 | 6 | $r_{70} = \left(-\frac{1004464945}{1939325948}s - \frac{273944985}{277046564}t\right)\left[M_{28}r_{13} - \left(\frac{6785}{20336} - \frac{23}{496}st\right)r_{12} - \left(\frac{855674}{1150255} + \frac{26497}{1150255}st\right)r_{13} - \left(-\frac{3417}{50840} - \frac{3163}{127100}st\right)r_{46} - \left(\frac{1205607}{7287914} + \frac{577989}{36439570}st\right)r_{60} - \left(-\frac{81846609}{2291358800} - \frac{63228141}{2291358800}st\right)r_{67}\right]$ |
| 71 | 14 | $r_{71} = \left(\frac{17}{48}s - \frac{23}{144}t\right)\left[M_{20}r_{14} - \left(\frac{5409}{29500}s + \frac{357}{29500}t\right)r_{56}\right]$ |
| 72 | 27 | $r_{72} = \left(-\frac{1284675}{1273064}s - \frac{33005}{97928}t\right)\left[M_{22}r_{14} - \left(\frac{4}{105}s + \frac{23}{45}t\right)r_{66} - \left(-\frac{18575777}{123016600}s + \frac{46179349}{123016600}t\right)r_{69}\right]$ |
| 73 | 24 | $r_{73} = \left(-\frac{1}{291600} + \frac{1}{842400}s - \frac{1}{583200}t - \frac{1}{1263600}st\right)\left[M_{19}r_{15} - \left(-\frac{224640}{7} - \frac{77760}{7}s + \frac{112320}{7}t - \frac{51840}{7}st\right)r_{56} - \left(-\frac{172800}{7} + \frac{86400}{7}t\right)r_{71}\right]$ |
| 74 | 11 | $r_{74} = \left(1\right)\left[M_{2}r_{19}\right]$ |
| 75 | 16 | $r_{75} = \left(-\frac{2143578782}{14604298825}s - \frac{311561186}{14604298825}t\right)\left[M_{20}r_{22} - \left(\frac{157099627}{1868232640}s + \frac{757800451}{1868232640}t\right)r_{48}\right]$ |
| 76 | 25 | $r_{76} = \left(-\frac{1625399704085}{2781170445222}s + \frac{953771098765}{641808564282}t\right)\left[M_{20}r_{23} - \left(-\frac{2984911921}{12813387540}s + \frac{3224844077}{76880325240}t\right)r_{48} - \left(\frac{106032828437}{451031241408}s - \frac{1432471320743}{1353093724224}t\right)r_{75}\right]$ |
| 77 | 13 | $r_{77} = \left(-\frac{126}{403}s + \frac{38}{527}t\right)\left[M_{20}r_{24} - \left(-\frac{10575}{77408}s + \frac{32851}{232224}t\right)r_{74}\right]$ |
| 78 | 10 | $r_{78} = \left(-1\right)\left[M_{2}r_{29}\right]$ |
| 79 | 9 | $r_{79} = \left(-\frac{2}{3}s - \frac{32}{9}t\right)\left[M_{20}r_{35}\right]$ |

## S25.4 All four mixed 81-dimensional tops: basis and recipes

Source: `hi_cft_20260923_next_81top_all.json`; its readable recipe export is `c8_reviewability_20260927/top81_pivots.tsv`. Each root basis is printed as the integer tuple $3\beta_j$. Thus the actual momentum is one third of that tuple, not the tuple itself. The listed norm is the exact diagonal Hermitian entry $\langle e^{\beta_j}, e^{\beta_j}\rangle$. Every table is specific to its glue and sector signs; no unproved identification of the four bases is used to discard repeated-looking rows. All four seeds have index 68 and Cartan weight $(0, 10)$.

The operator index order is the same interleaved field/adjoint order listed above, now acting in the indicated mixed lattice sector. Each sector uses its actual overlattice cocycle and quadratic gauge. The mode-on-root rules in the integrated guide specify the elementary calculation, while the archived matrix metadata binds the sector-specific action. The full sparse elimination rows and matrix hashes remain in the source; this appendix prints the complete basis, metric, pivot schedule, and all recipes needed to reconstruct those rows.

### S25.4.1 Glue and sector signs (1,1)

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 0 | $(-2,-4,-6,-5,-4,-3,-1,-2)$ | 1/4299816960000 |
| 1 | $(-2,-4,-6,-5,-4,-3,-1,1)$ | 1/2149908480000 |
| 2 | $(-2,-4,-6,-5,-4,-3,2,1)$ | 1/1074954240000 |
| 3 | $(-2,-4,-6,-5,-1,-3,-1,-2)$ | 1/859963392000 |
| 4 | $(-2,-4,-6,-5,-1,-3,-1,1)$ | 1/429981696000 |
| 5 | $(-2,-4,-6,-5,-1,-3,2,1)$ | 1/214990848000 |

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 6 | $(-2,-4,-6,-2,-1,-3,-1,-2)$ | 1/107495424000 |
| 7 | $(-2,-4,-6,-2,-1,-3,-1,1)$ | 1/53747712000 |
| 8 | $(-2,-4,-6,-2,-1,-3,2,1)$ | 1/26873856000 |
| 9 | $(-2,-4,-3,-2,-1,-3,-1,-2)$ | 1/11943936000 |
| 10 | $(-2,-4,-3,-2,-1,-3,-1,1)$ | 1/5971968000 |
| 11 | $(-2,-4,-3,-2,-1,-3,2,1)$ | 1/2985984000 |
| 12 | $(-2,-4,-3,-2,-1,0,-1,-2)$ | 1/11943936000 |
| 13 | $(-2,-4,-3,-2,-1,0,-1,1)$ | 1/5971968000 |
| 14 | $(-2,-4,-3,-2,-1,0,2,1)$ | 1/2985984000 |
| 15 | $(-2,-1,-3,-2,-1,-3,-1,-2)$ | 1/1492992000 |
| 16 | $(-2,-1,-3,-2,-1,-3,-1,1)$ | 1/746496000 |
| 17 | $(-2,-1,-3,-2,-1,-3,2,1)$ | 1/373248000 |
| 18 | $(-2,-1,-3,-2,-1,0,-1,-2)$ | 1/1492992000 |
| 19 | $(-2,-1,-3,-2,-1,0,-1,1)$ | 1/746496000 |
| 20 | $(-2,-1,-3,-2,-1,0,2,1)$ | 1/373248000 |
| 21 | $(-2,-1,0,-2,-1,0,-1,-2)$ | 1/165888000 |
| 22 | $(-2,-1,0,-2,-1,0,-1,1)$ | 1/82944000 |
| 23 | $(-2,-1,0,-2,-1,0,2,1)$ | 1/41472000 |
| 24 | $(-2,-1,0,1,-1,0,-1,-2)$ | 1/20736000 |
| 25 | $(-2,-1,0,1,-1,0,-1,1)$ | 1/10368000 |
| 26 | $(-2,-1,0,1,-1,0,2,1)$ | 1/5184000 |
| 27 | $(-2,-1,0,1,2,0,-1,-2)$ | 1/4147200 |
| 28 | $(-2,-1,0,1,2,0,-1,1)$ | 1/2073600 |
| 29 | $(-2,-1,0,1,2,0,2,1)$ | 1/1036800 |
| 30 | $(1,-1,-3,-2,-1,-3,-1,-2)$ | 1/298598400 |
| 31 | $(1,-1,-3,-2,-1,-3,-1,1)$ | 1/149299200 |
| 32 | $(1,-1,-3,-2,-1,-3,2,1)$ | 1/74649600 |
| 33 | $(1,-1,-3,-2,-1,0,-1,-2)$ | 1/298598400 |
| 34 | $(1,-1,-3,-2,-1,0,-1,1)$ | 1/149299200 |
| 35 | $(1,-1,-3,-2,-1,0,2,1)$ | 1/74649600 |
| 36 | $(1,-1,0,-2,-1,0,-1,-2)$ | 1/33177600 |
| 37 | $(1,-1,0,-2,-1,0,-1,1)$ | 1/16588800 |
| 38 | $(1,-1,0,-2,-1,0,2,1)$ | 1/8294400 |
| 39 | $(1,-1,0,1,-1,0,-1,-2)$ | 1/4147200 |
| 40 | $(1,-1,0,1,-1,0,-1,1)$ | 1/2073600 |
| 41 | $(1,-1,0,1,-1,0,2,1)$ | 1/1036800 |
| 42 | $(1,-1,0,1,2,0,-1,-2)$ | 1/829440 |
| 43 | $(1,-1,0,1,2,0,-1,1)$ | 1/414720 |
| 44 | $(1,-1,0,1,2,0,2,1)$ | 1/207360 |
| 45 | $(1,2,0,-2,-1,0,-1,-2)$ | 1/4147200 |
| 46 | $(1,2,0,-2,-1,0,-1,1)$ | 1/2073600 |
| 47 | $(1,2,0,-2,-1,0,2,1)$ | 1/1036800 |
| 48 | $(1,2,0,1,-1,0,-1,-2)$ | 1/518400 |
| 49 | $(1,2,0,1,-1,0,-1,1)$ | 1/259200 |
| 50 | $(1,2,0,1,-1,0,2,1)$ | 1/129600 |
| 51 | $(1,2,0,1,2,0,-1,-2)$ | 1/103680 |
| 52 | $(1,2,0,1,2,0,-1,1)$ | 1/51840 |
| 53 | $(1,2,0,1,2,0,2,1)$ | 1/25920 |
| 54 | $(1,2,3,1,-1,0,-1,-2)$ | 1/57600 |
| 55 | $(1,2,3,1,-1,0,-1,1)$ | 1/28800 |
| 56 | $(1,2,3,1,-1,0,2,1)$ | 1/14400 |
| 57 | $(1,2,3,1,-1,3,-1,-2)$ | 1/57600 |
| 58 | $(1,2,3,1,-1,3,-1,1)$ | 1/28800 |
| 59 | $(1,2,3,1,-1,3,2,1)$ | 1/14400 |
| 60 | $(1,2,3,1,2,0,-1,-2)$ | 1/11520 |
| 61 | $(1,2,3,1,2,0,-1,1)$ | 1/5760 |
| 62 | $(1,2,3,1,2,0,2,1)$ | 1/2880 |
| 63 | $(1,2,3,1,2,3,-1,-2)$ | 1/11520 |
| 64 | $(1,2,3,1,2,3,-1,1)$ | 1/5760 |
| 65 | $(1,2,3,1,2,3,2,1)$ | 1/2880 |
| 66 | $(1,2,3,4,2,0,-1,-2)$ | 1/1440 |

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 67 | $(1,2,3,4,2,0,-1,1)$ | $1/720$ |
| 68 | $(1,2,3,4,2,0,2,1)$ | $1/360$ |
| 69 | $(1,2,3,4,2,3,-1,-2)$ | $1/1440$ |
| 70 | $(1,2,3,4,2,3,-1,1)$ | $1/720$ |
| 71 | $(1,2,3,4,2,3,2,1)$ | $1/360$ |
| 72 | $(1,2,6,4,2,3,-1,-2)$ | $1/160$ |
| 73 | $(1,2,6,4,2,3,-1,1)$ | $1/80$ |
| 74 | $(1,2,6,4,2,3,2,1)$ | $1/40$ |
| 75 | $(1,5,6,4,2,3,-1,-2)$ | $1/20$ |
| 76 | $(1,5,6,4,2,3,-1,1)$ | $1/10$ |
| 77 | $(1,5,6,4,2,3,2,1)$ | $1/5$ |
| 78 | $(4,5,6,4,2,3,-1,-2)$ | $1/4$ |
| 79 | $(4,5,6,4,2,3,-1,1)$ | $1/2$ |
| 80 | $(4,5,6,4,2,3,2,1)$ | $1$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 0 | 68 | $(0,10)$ | $r_0 = e_{68}$ |
| 1 | 62 | $(0,8)$ | $r_1 = \left(\frac{1}{8}\right)\left[M_9 r_0\right]$ |
| 2 | 53 | $(0,6)$ | $r_2 = \left(-\frac{69393779}{959499936} - \frac{16057223}{959499936}s + \frac{2387749}{319833312}t - \frac{3103555}{319833312}st\right)\left[M_{13} r_0\right]$ |
| 3 | 29 | $(0,2)$ | $r_3 = \left(-\frac{143}{15140} - \frac{261}{121120}s - \frac{89}{363360}t + \frac{7}{90840}st\right)\left[M_{15} r_0\right]$ |
| 4 | 41 | $(0,2)$ | $r_4 = \left(\frac{241}{634240} - \frac{27931}{123676800}s - \frac{667}{3171200}t - \frac{1591}{24735360}st\right)\left[M_{17} r_0 - \left(-150 + 240t - 30st\right) r_3\right]$ |
| 5 | 23 | $(0,-2)$ | $r_5 = \left(\frac{1}{288000} + \frac{1}{192000}s + \frac{1}{576000}t - \frac{1}{288000}st\right)\left[M_{19} r_0\right]$ |
| 6 | 77 | $(-1,5)$ | $r_6 = \left(\frac{2454543}{14491} - \frac{1264977}{14491}s + \frac{595161}{14491}t + \frac{166509}{14491}st\right)\left[M_{23} r_0\right]$ |
| 7 | 74 | $(-1,3)$ | $r_7 = \left(-\frac{4977}{1348} + \frac{31149}{1348}s - \frac{11853}{1348}t + \frac{1413}{1348}st\right)\left[M_{25} r_0\right]$ |
| 8 | 59 | $(-1,-3)$ | $r_8 = \left(-\frac{1}{80}s - \frac{3}{80}t\right)\left[M_{27} r_0\right]$ |
| 9 | 56 | $(0,6)$ | $r_9 = \left(\frac{22949}{422560} - \frac{135}{84512}s - \frac{1013}{253536}t - \frac{2053}{1267680}st\right)\left[M_9 r_1 - \left(-9\right) r_2\right]$ |
| 10 | 44 | $(0,4)$ | $r_{10} = \left(\frac{69393779}{959499936} + \frac{16057223}{959499936}s - \frac{2387749}{319833312}t + \frac{3103555}{319833312}st\right)\left[M_{13} r_1\right]$ |
| 11 | 26 | $(0,0)$ | $r_{11} = \left(\frac{8177}{8049900} + \frac{3019}{4024950}s + \frac{11549}{8049900}t + \frac{91}{670825}st\right)\left[M_{15} r_1\right]$ |
| 12 | 28 | $(0,0)$ | $r_{12} = \left(\frac{2965217}{4392794400} - \frac{97549}{4392794400}s + \frac{2667893}{4392794400}t - \frac{592871}{1464264800}st\right)\left[M_{17} r_1 - \left(-\frac{945}{4} - \frac{285}{4}s - \frac{345}{4}t - \frac{105}{4}st\right) r_{11}\right]$ |
| 13 | 20 | $(0,-4)$ | $r_{13} = \left(-\frac{1}{43200} - \frac{1}{86400}t\right)\left[M_{19} r_1\right]$ |
| 14 | 76 | $(-1,3)$ | $r_{14} = \left(\frac{15101856}{248167} - \frac{12445272}{248167}s - \frac{18897480}{248167}t + \frac{2862720}{248167}st\right)\left[M_{23} r_1 - \left(-\frac{13}{18144} - \frac{1}{1296}s - \frac{1}{1008}t - \frac{5}{18144}st\right) r_7\right]$ |
| 15 | 71 | $(-1,1)$ | $r_{15} = \left(-\frac{1106}{337} + \frac{6922}{337}s - \frac{2634}{337}t + \frac{314}{337}st\right)\left[M_{25} r_1\right]$ |
| 16 | 58 | $(-1,-5)$ | $r_{16} = \left(\frac{1}{20}s + \frac{3}{20}t\right)\left[M_{27} r_1\right]$ |
| 17 | 67 | $(0,8)$ | $r_{17} = \left(\frac{39}{38} - \frac{345}{494}s - \frac{87}{38}t + \frac{27}{494}st\right)\left[M_8 r_2 - \left(\frac{2257138}{9994791} + \frac{139840}{3331597}s + \frac{4891360}{29984373}t + \frac{334438}{29984373}st\right) r_1\right]$ |
| 18 | 43 | $(0,2)$ | $r_{18} = \left(\frac{4209107406326711478135}{3892132325083765459557744} - \frac{816417127743365603819}{3892132325083765459557744}s - \frac{611226701487383176950 3}{3892132325083765459557744}t + \frac{150529642793009505052 3}{3892132325083765459557744}st\right)\left[M_{13} r_2 - \left(\frac{68427746}{303175327} + \frac{288173475}{43310761}s + \frac{9077438765}{909525981}t - \frac{7627980502}{909525981}st\right) r_3 - \left(\frac{5065154554027}{459007445078} - \frac{6324506293347}{459007445078}s - \frac{31532161180029}{459007445078}t - \frac{9555889988887}{459007445078}st\right) r_4\right]$ |
| 19 | 25 | $(0,-2)$ | $r_{19} = \left(-\frac{1510291547123907}{2953532232249927200} + \frac{115775485311147}{590706446449985440}s + \frac{80170230663631}{590706446449985440}t - \frac{383349910276747}{2953532232249927200}st\right)\left[M_{15} r_2 - \left(\frac{6278241320}{9994791} - \frac{257965832}{9994791}s - \frac{1536636088}{3331597}t + \frac{153214360}{3331597}st\right) r_5\right]$ |
| 20 | 27 | $(0,-2)$ | $r_{20} = \left(\frac{6978658750580763541 7071}{9900887554027889552 00602240} + \frac{1020512850742090894 8471}{9900887554027889552 00602240}s - \frac{1104547357326832702300729}{9900887554027889552 00602240}t - \frac{6104045470551337647795 43}{2970266266208366865601806720}st\right)\left[M_{17} r_2 - \left(\frac{558554860}{3331597} - \frac{847105100}{3331597}s - \frac{2303233900}{3331597}t + \frac{231252540}{3331597}st\right) r_5 - \left(\frac{1152444798}{3331597} + \frac{626444072}{3331597}s - \frac{5987552218}{9994791}t + \frac{981133076}{9994791}st\right) r_{19}\right]$ |
| 21 | 14 | $(0,-6)$ | $r_{21} = \left(\frac{260736019}{289844493096960} + \frac{747453749}{724611232742400}s + \frac{211704113}{241537077580800}t - \frac{462962227}{483074155161600}st\right)\left[M_{19} r_2\right]$ |
| 22 | 32 | $(1,7)$ | $r_{22} = \left(-\frac{3775}{59622} + \frac{10879}{59622}s + \frac{8105}{29811}t - \frac{200}{1569}st\right)\left[M_{20} r_2\right]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 23 | 79 | $(-1,5)$ | $r_{23} = \big(\frac{965768400}{43909} - \frac{2218896288}{570817}s + \frac{620367552}{43909}t - \frac{2125126800}{570817}st\big)\big[M_{21}r_2 - \big(\frac{504476999}{20379778640640} + \frac{1036279}{20379778640640}s - \frac{2484611551}{61139335921920}t - \frac{130550767}{61139335921920}st\big)r_6\big]$ |
| 24 | 73 | $(-1,1)$ | $r_{24} = \big(-\frac{4625839906337719566}{46880232503780887} - \frac{1048124782729379313 0}{609443022549151531}s - \frac{312002748743826420}{46880232503780887}t - \frac{790502293380431520}{609443022549151531}st\big)\big[M_{23}r_2 - \big(-\frac{85210123}{15112123992} + \frac{72371797}{60448495968}s + \frac{108744235}{60448495968}t - \frac{6885227}{7556061996}st\big)r_{15}\big]$ |
| 25 | 65 | $(-1,-1)$ | $r_{25} = \big(-\frac{4990106743191}{12710171598004} + \frac{26882500263453}{12710171598004}s + \frac{22001305098519}{12710171598004}t - \frac{2531878407441}{12710171598004}st\big)\big[M_{25}r_2\big]$ |
| 26 | 57 | $(-1,-7)$ | $r_{26} = \big(-\frac{19443}{183080} + \frac{543}{183080}s + \frac{1863}{7960}t - \frac{501}{7960}st\big)\big[M_{27}r_2\big]$ |
| 27 | 61 | $(0,6)$ | $r_{27} = \big(\frac{17340313665761643}{11885830292420716} - \frac{2386698723398202}{2971457573105179}s - \frac{831178029744497}{312785007695282}t - \frac{4275448153377961}{11885830292420716}st\big)\big[M_{28}r_2 - \big(\frac{12131611235}{76220276166} - \frac{2597278757}{76220276166}s - \frac{1018825657}{8468919574}t + \frac{44035825}{8468919574}st\big)r_2 - \big(\frac{2973573145379029}{84645081137939034} - \frac{811739789499649}{28215027045979678}s + \frac{7079533160948411}{84645081137939034}t + \frac{2821183164629173}{253935243413817102}st\big)r_9\big]$ |
| 28 | 50 | $(0,4)$ | $r_{28} = \big(-\frac{1429683987129}{51521839148435} + \frac{1963201104232}{51521839148435}s + \frac{13994269295236}{51521839148435}t - \frac{3095148706709}{51521839148435}st\big)\big[M_8r_3 - \big(\frac{13289}{15140} + \frac{3857}{15140}s + \frac{28139}{15140}t + \frac{3591}{15140}st\big)r_{10}\big]$ |
| 29 | 66 | $(0,6)$ | $r_{29} = \big(-\frac{5645877097263}{1428966551855} + \frac{1663979336859}{1428966551855}s - \frac{35393909658029}{4286899655565}t + \frac{10049794261381}{4286899655565}st\big)\big[M_{12}r_3 - \big(\frac{440053}{551096} + \frac{330427}{688870}s + \frac{158191}{105980}t + \frac{45665}{78728}st\big)r_2 - \big(\frac{105015070980059}{27540446704680} + \frac{26136842435713}{27540446704680}s - \frac{4744758963157}{3934349529240}t - \frac{8236088242789}{27540446704680}st\big)r_9 - \big(\frac{150486841}{95752930} + \frac{66900845}{153204688}s - \frac{486646705}{459614064}t - \frac{77716423}{287258790}st\big)r_{27}\big]$ |
| 30 | 35 | $(0,-2)$ | $r_{30} = \big(\frac{539966786230539423262219920979 7629}{354227572701164123537546372141280672} - \frac{661276536565530756894279521068 6129}{177113786350582061768773186070640336 0}s + \frac{167700712925534965245158129676 2083}{177113786350582061768773186070640336 0}t - \frac{467290760294765391615286545326 71}{354227572701164123537546372141280672}st\big)\big[M_{13}r_3 - \big(-\frac{6863718}{68887} - \frac{317115}{9841}s + \frac{1791585}{68887}t - \frac{9856}{9841}st\big)r_5 - \big(\frac{7964113}{98410} + \frac{13013549}{688870}s - \frac{2492961}{98410}t - \frac{327097}{52990}st\big)r_{19} - \big(\frac{9879083785262813821845}{508649937207501837566} + \frac{1997983369282853952 87}{391269182467309105 82}s - \frac{270514067092161356 4593}{726642767439288339 38}t - \frac{500788560639661102 7981}{508649937207501837566}st\big)r_{20}\big]$ |
| 31 | 19 | $(0,-6)$ | $r_{31} = \big(\frac{34485240132913470703}{456261507178373316564 00} - \frac{6959510861529665681}{456261507178373316564 00}s - \frac{23402546085374945821}{152087169059457772188 00}t + \frac{6656086015160983271}{152087169059457772188 00}st\big)\big[M_{15}r_3 - \big(-\frac{14488916}{3785} - \frac{4130008}{3785}s - \frac{2225416}{3785}t - \frac{1893164}{3785}st\big)r_{21}\big]$ |
| 32 | 21 | $(0,-6)$ | $r_{32} = \big(-\frac{3818770679663453907281}{29838341436225853749417600} - \frac{5757511868823413817 1439}{3878984386709360987424288 00}s - \frac{16862950528402270211 17}{59676682872451707498835 20}t - \frac{4242867179573571859403}{3878984386709360987424288 00}st\big)\big[M_{17}r_3 - \big(\frac{549216}{757} + \frac{80136}{757}s + \frac{1148472}{757}t + \frac{161064}{757}st\big)r_{21} - \big(\frac{3190391053282671}{4761551242934} + \frac{210361114536873}{2380775621467}s + \frac{467118279637371}{2380775621467}t + \frac{330697400617359}{4761551242934}st\big)r_{31}\big]$ |
| 33 | 12 | $(0,-10)$ | $r_{33} = \big(-\frac{7}{384000} - \frac{1}{1152000}s - \frac{497}{79488000}t + \frac{151}{79488000}st\big)\big[M_{19}r_3\big]$ |
| 34 | 11 | $(1,3)$ | $r_{34} = \big(-\frac{3465135273}{1521576578} + \frac{865181079}{1521576578}s - \frac{6410461}{1521576578}t - \frac{123361657}{1521576578}st\big)\big[M_{20}r_3\big]$ |
| 35 | 75 | $(-1,1)$ | $r_{35} = \big(\frac{4308310935946512}{367217837005} - \frac{2300229222013872}{954766376213}s - \frac{890830893214512}{73443567401}t + \frac{16178383876295376}{4773831881065}st\big)\big[M_{21}r_3 - \big(\frac{8045719}{30871065600} + \frac{716743}{6174213120}s - \frac{364699}{6174213120}t + \frac{108401}{30871065600}st\big)r_{15} - \big(\frac{7805856289}{11703992745600} + \frac{2476919909}{11703992745600}s - \frac{5012575907}{11703992745600}t - \frac{690328519}{11703992745600}st\big)r_{24}\big]$ |
| 36 | 64 | $(-1,-3)$ | $r_{36} = \big(-\frac{56566377}{713659} + \frac{321865299}{9277567}s + \frac{49146489}{713659}t - \frac{87825591}{9277567}st\big)\big[M_{23}r_3 - \big(\frac{176501}{17168760} + \frac{261031}{68675040}s + \frac{191237}{68675040}t + \frac{9269}{4905360}st\big)r_8\big]$ |
| 37 | 63 | $(-1,-5)$ | $r_{37} = \big(-\frac{180921}{18038} + \frac{337017}{117247}s - \frac{15193}{9019}t + \frac{229417}{234494}st\big)\big[M_{25}r_3 - \big(-\frac{112181}{2277056} - \frac{3740471}{170779200}s - \frac{5389367}{170779200}t - \frac{110267}{6831168}st\big)r_{16}\big]$ |
| 38 | 47 | $(0,2)$ | $r_{38} = \big(-\frac{2103695424755118019585207 3029493181}{1220736580340447141234971067094 0720} + \frac{9744584819406082147581618364 60937}{2441473160680894282469942134188144}s + \frac{1957273046515852232028842831 6098511}{2441473160680894282469942134188144}t - \frac{2638767635476315167873784009 7797751}{1220736580340447141234971067094 0720}st\big)\big[M_{28}r_3 - \big(\frac{3846031}{34637292} - \frac{1511605}{34637292}s - \frac{1281735}{3848588}t - \frac{271959}{3848588}st\big)r_3 - \big(-\frac{29343533398}{10925179185} - \frac{15542193319}{21850358370}s - \frac{32458435763}{21850358370}t - \frac{4225074917}{10925179185}st\big)r_4 - \big(\frac{27865549721}{9534876770} + \frac{19385963327}{24790679602}s + \frac{550741659}{1906975354}t + \frac{10485800737}{123953398010}st\big)r_{18}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 39 | 52 | $(0,4)$ | $r_{39} = \big(\frac{6112292834948821}{10471790908042492} - \frac{10943051262279513}{20943581816084984}s + \frac{29234168743122667}{62830745448254952}t - \frac{531506011565383}{31415372724127476}st\big)\big[M_8 r_4 - \big(\frac{1909}{3964} - \frac{54137}{257660}s + \frac{2513}{19820}t - \frac{3521}{51532}st\big)r_{10} - \big(-\frac{57738058093}{52825802032} + \frac{59719999775}{686735426416}s - \frac{8645300695}{52825802032}t - \frac{11091536431}{686735426416}st\big)r_{28}\big]$ |
| 40 | 37 | $(0,-2)$ | $r_{40} = \big(\frac{4705098139995842887554215256987235}{40449338005564379848669846808944128} - \frac{41722406511890392027925572151231265}{76853742210572321712472708936993843 2} s - \frac{88601618918110358507602519418438111}{76853742210572321712472708936993843 2}t + \frac{41777036974170827379417849935 4257}{25617914070190773904157569645664614 4}st\big)\big[M_{13} r_4 - \big(\frac{2352543}{90181} + \frac{35928}{6937}s - \frac{779638}{90181}t - \frac{141847}{90181}st\big)r_5 - \big(-\frac{5163231}{180362} - \frac{3662595}{2344706}s + \frac{95217}{25766}t + \frac{242545}{334958}st\big)r_{19} - \big(\frac{1082847801560227747815}{33294061279566335602 9} - \frac{2723962106643094785516}{61831828090623194691 1}s + \frac{4151171475773978738 54}{47562944685094765147}t + \frac{5091314168278271010255}{43282279663436236283 77}st\big)r_{20} - \big(-\frac{76209054188610265196649658573 35}{13951124070465454807647736032 1} - \frac{23948827126305354758697706591197}{18136461291605091249942056841 73}s - \frac{50502921863413882086736004721}{10731633900358042159729027717}t + \frac{13802767602961360415106213234 03}{25909230416578701785631509773 9}st\big)r_{30}\big]$ |
| 41 | 33 | $(0,-6)$ | $r_{41} = \big(\frac{1997548163517683}{6047532507495513600} + \frac{24504122512673}{211663637762342976}s + \frac{5424870578449}{10079220845825856}t - \frac{2198155165461083}{1411090918415619840 0}st\big)\big[M_{17} r_4 - \big(\frac{26895}{991} + \frac{1579485}{12883}s - \frac{588195}{991}t + \frac{555195}{12883}st\big)r_{21} - \big(-\frac{8245472183815035}{24933671237768} + \frac{63640106549856 75}{324137726090984}s - \frac{946307958088785}{24933671237768}t + \frac{19360838600163405}{324137726090984}st\big)r_{31} - \big(\frac{11072394836770932864602935}{25119730756320442039718} - \frac{51970039696702847590344 55}{32655649983216574651633 4}s - \frac{16775032911400451520205115}{25119730756320442039718}t + \frac{38440589082905211199509535}{32655649983216574651633 4}st\big)r_{32}\big]$ |
| 42 | 16 | $(1,3)$ | $r_{42} = \big(-\frac{6313899170717159}{1780895370106340} + \frac{13007936146487227}{23151639811382420}s - \frac{12348617717642501}{5342686110319020}t - \frac{25105106263421579}{69454919434147260}st\big)\big[M_{20} r_4 - \big(\frac{1002653}{14032560} + \frac{1820639}{36484656}s - \frac{124277}{935504}t - \frac{6722281}{182423280}st\big)r_{34}\big]$ |
| 43 | 69 | $(-1,-3)$ | $r_{43} = \big(\frac{382514616540}{3456258367} - \frac{308111158152}{3456258367}s - \frac{311960329884}{3456258367}t + \frac{22798812960}{3456258367}st\big)\big[M_{23} r_4 - \big(\frac{14177}{8990352} - \frac{10915}{17980704}s - \frac{132071}{17980704}t - \frac{65}{142704}st\big)r_8 - \big(-\frac{23467}{44951760} + \frac{1909}{6421680}s + \frac{373}{14983920}t + \frac{26113}{44951760}st\big)r_{36}\big]$ |
| 44 | 49 | $(0,2)$ | $r_{44} = \big(-\frac{4898038736083462257124222532165345827986839}{1381918879893140287334089680842298029117536 8} + \frac{3410217091780110223730769881074112187500113}{1381918879893140287334089680842298029117536 8}s + \frac{1873993343116073216960120512979350009737 20297}{4145756639679420862002269042526894087352610 4}t + \frac{2806222541740504897680790267500231355257 6917}{4145756639679420862002269042526894087352610 4}st\big)\big[M_{28} r_4 - \big(\frac{473429}{5038244} + \frac{218785}{5038244}s + \frac{516515}{5038244}t - \frac{53921}{5038244}st\big)r_3 - \big(\frac{40168134913}{34325556372} + \frac{5724357293}{34325556372}s - \frac{149882653}{3813950708}t + \frac{4070501}{123030668}st\big)r_4 - \big(-\frac{1632205644}{6241124755} - \frac{9928684179}{81134621815}s + \frac{400712911}{6241124755}t + \frac{3658982778}{81134621815}st\big)r_{18} - \big(\frac{3317984986337225291319258 3923}{1838391781318062264752110756 44} + \frac{7857687203596013882689270729}{1838391781318062264752110756 44}s + \frac{2245965836952303989313220 9337}{1838391781318062264752110756 44}t + \frac{1080793055988001403397891 8575}{1838391781318062264752110756 44}st\big)r_{38}\big]$ |
| 45 | 38 | $(0,0)$ | $r_{45} = \big(-\frac{16575187865}{33503395304} + \frac{1715651335}{16751697652}s + \frac{5528651965}{50255092956}t - \frac{314401795}{100510185912}st\big)\big[M_8 r_5 - \big(\frac{9}{10} - \frac{3}{20}s - \frac{1}{20}t + \frac{1}{10}st\big)r_{11} - \big(\frac{560827}{1073320} + \frac{48091}{1073320}s - \frac{346307}{1073320}t - \frac{32723}{214664}st\big)r_{12}\big]$ |
| 46 | 51 | $(0,2)$ | $r_{46} = \big(-\frac{6061116406181689313055383209999489079045839497838}{2310347752555853546255281156952026229434641899 7} - \frac{2173186614976594738042962916899886581477918456 1}{2310347752555853546255281156952026229434641899 7}s + \frac{6573634993758283481374531587907340395035281899}{7701159175186178487517603856506754098115472999}t - \frac{1943022042171917043280254626378829441656196 16}{7701159175186178487517603856506754098115472999}st\big)\big[M_{12} r_5 - \big(\frac{1249}{3640} - \frac{439}{3640}s - \frac{47}{104}t + \frac{583}{3640}st\big)r_3 - \big(-\frac{801186}{1722175} + \frac{369981}{6888700}s + \frac{40599}{529900}t + \frac{243631}{1722175}st\big)r_4 - \big(\frac{759019}{1803620} - \frac{2500859}{117235300}s + \frac{690939}{9018100}t - \frac{17663}{23447060}st\big)r_{18} - \big(-\frac{6229851924052211528114 07}{8514039461120736942781900} + \frac{1393891189946039152278716 7}{1106825129945695802561647 00}s - \frac{1750454385400721827915807}{8514039461120736942781900}t + \frac{6744832254233988935266811}{1106825129945695802561647 00}st\big)r_{38} - \big(-\frac{4454476172050657974019788265382151 3}{1424192677063854998107466244943084 00} + \frac{1219890708960306654964739213545379 07}{1851450480183011497539706118426009200}s + \frac{3730506347467329219008370676580 81}{1095532828510657690851897111494680 0}t - \frac{1522509084633655948663747452763365 1}{1851450480183011497539706118426009200}st\big)r_{44}\big]$ |
| 47 | 5 | $(1,-1)$ | $r_{47} = \big(-\frac{2401745}{7552276} + \frac{34665}{7552276}s - \frac{14000365}{22656828}t - \frac{382255}{22656828}st\big)\big[M_{20} r_5\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 48 | 30 | $(1,3)$ | $r_{48} = \big(-\frac{53698568}{3628507} - \frac{485398375}{94341182}s - \frac{257322275}{43542084}t - \frac{987663593}{566047092}st\big)\big[M_{22}r_5 - \big(-\frac{3}{140} - \frac{1}{28}s + \frac{1}{12}t - \frac{13}{1260}st\big)r_{34} - \big(\frac{327126980077}{958593244140} - \frac{11411961313}{136941892020}s + \frac{71221864103}{958593244140}t - \frac{11444282077}{958593244140}st\big)r_{42}\big]$ |
| 49 | 17 | $(1,5)$ | $r_{49} = \big(\frac{10405}{6124} - \frac{4070}{19903}s + \frac{1915}{9186}t - \frac{52375}{238836}st\big)\big[M_{24}r_5\big]$ |
| 50 | 31 | $(1,5)$ | $r_{50} = \big(-\frac{6749}{102764} - \frac{123}{333983}s + \frac{21049}{51382}t + \frac{47953}{1335932}st\big)\big[M_2r_6 - \big(-1\big)r_{49}\big]$ |
| 51 | 80 | $(-1,7)$ | $r_{51} = \big(-\frac{73}{30} - \frac{11}{30}s - \frac{53}{30}t - \frac{13}{30}st\big)\big[M_8r_6\big]$ |
| 52 | 70 | $(-1,-1)$ | $r_{52} = \big(\frac{388030549667728331}{22271875515417394080} + \frac{129113681413938221}{2227187551541739408 0}s - \frac{2315083558958893541}{66815626546252182240}t - \frac{46984302872305253}{3516611923486956960}st\big)\big[M_{13}r_7 - \big(\frac{681021}{30667} + \frac{26163}{2359}s + \frac{390279}{30667}t - \frac{444999}{30667}st\big)r_{25}\big]$ |
| 53 | 55 | $(0,4)$ | $r_{53} = \big(-\frac{2855938717683614553763}{3713050676601726603361} - \frac{869885210183391401749}{48269658795822445843693}s - \frac{1615029940088566767943}{37130506766017266033 61}t + \frac{3055855515810286849519}{48269658795822445843693}st\big)\big[M_{20}r_7 - \big(\frac{96129}{795320} + \frac{111081}{795320}s - \frac{444047}{795320}t + \frac{15909}{795320}st\big)r_{10} - \big(-\frac{1697344569535}{4239497161664} - \frac{621328814801}{4239497161664}s + \frac{606586736543}{4239497161664}t + \frac{64573387693}{4239497161664}st\big)r_{28} - \big(-\frac{482371961144242779}{13112431715690663744} - \frac{662715374082729 7623}{65562158578453318720}s + \frac{26060819600106260541}{65562158578453318720}t + \frac{5943933591958123 17}{13112431715690663744}st\big)r_{39}\big]$ |
| 54 | 78 | $(-1,3)$ | $r_{54} = \big(\frac{311181582577}{80156221704} + \frac{2596346689}{6165863208}s + \frac{139619791037}{26718740568}t + \frac{32381011529}{26718740568}st\big)\big[M_{28}r_7 - \big(\frac{866938}{3854943} + \frac{52519}{7709886}s + \frac{354913}{2569962}t - \frac{37808}{1284981}st\big)r_7 - \big(-\frac{346210631}{2309539184} + \frac{200219243}{2309539184}s + \frac{400860707}{2309539184}t - \frac{170502187}{2309539184}st\big)r_{14}\big]$ |
| 55 | 2 | $(1,-3)$ | $r_{55} = \big(-1\big)\big[M_2r_8\big]$ |
| 56 | 72 | $(-1,-1)$ | $r_{56} = \big(\frac{12248741}{10319492} + \frac{20432593}{134153396}s + \frac{94538741}{30958476}t + \frac{218709341}{402460188}st\big)\big[M_8r_8 - \big(1\big)r_{25} - \big(-\frac{8513351902931}{25420343196008} - \frac{818982162117}{127101715980040}s - \frac{10872775852471}{127101715980040}t + \frac{935605023501}{25420343196008}st\big)r_{52}\big]$ |
| 57 | 39 | $(0,-2)$ | $r_{57} = \big(\frac{3638405746974236563991080 93}{4689461444649093863981832} - \frac{7120493971864834365031033 31}{4689461444649093863989818 32}s + \frac{1354002510106949425057938487}{4689461444649093863989818 32}t + \frac{1818425222752730362531416619}{4689461444649093863989818 32}st\big)\big[M_{20}r_8 - \big(-\frac{103}{1770} - \frac{41}{708}s + \frac{11}{354}t + \frac{2}{295}st\big)r_5 - \big(-\frac{61}{2360} + \frac{13}{3540}s + \frac{29}{3540}t + \frac{7}{7080}st\big)r_{19} - \big(\frac{4965227156456257}{1742584017027457048} - \frac{326340565679533583}{26138760255411855720}s - \frac{90707766371366009}{26138760255411855720}t - \frac{6656689963698799}{52277520510823711 44}st\big)r_{20} - \big(\frac{9677367959286237676278413}{45636903569347303403777595} - \frac{5804513799083141272421912 3}{18254761427738921361511103 80}s + \frac{3507992445981953287587953 23}{54764284283216764084533311 40}t - \frac{1089517557746068814205114 1}{13691071070804191021133327 85}st\big)r_{30} - \big(\frac{1547985565653772471843100221358 68}{5729009536559397831336413365223563 5} + \frac{1771122618647045268642327599631973}{2291603814623759132534565346089425 40}s - \frac{2162619866933287248473967698450503 3}{6874811443871277397603696038266276 20}t + \frac{4912911440139653640168431909837 3}{3437405721935638698801848019134138 10}st\big)r_{40}\big]$ |
| 58 | 54 | $(0,2)$ | $r_{58} = \big(-\frac{3309115692918275918437221}{2203128362767559489 19720} - \frac{1556949911909388897650 5629}{2864066871597879335956360}s + \frac{4707690501423545634661551}{2203128362767559489 19720}t + \frac{2026608408270196000 4568627}{2864066871597879335956360}st\big)\big[M_{22}r_8 - \big(\frac{17}{252}s + \frac{31}{252}t\big)r_3 - \big(\frac{388765}{381528} + \frac{55723}{381528}s + \frac{83221}{381528}t + \frac{143}{381528}st\big)r_4 - \big(-\frac{13721}{41622} - \frac{45856}{579735}s + \frac{77183}{312165}t + \frac{52249}{1623258}st\big)r_{18} - \big(\frac{1582260170715627937419 47}{1021684735334488433133828} - \frac{4571415460322285596492 3}{1021684735334488433133828}s + \frac{1817165538308173110882 5}{3065054206003465299401484}t + \frac{1941713634982413174651 7}{4378648865719236142002 12}st\big)r_{38} - \big(\frac{1359414567716285770315367338689 53}{1971959091319183843533414800690424} + \frac{2054878573928115064114052903211 73}{2563546818714938996593439240897551 2}s - \frac{2100786877704818677308732933361 67}{1971959091319183843533414800690424}t + \frac{5140717496212454808116913699676 73}{2563546818714938996593439240897551 2}st\big)r_{44} - \big(\frac{3496420405430083795150691651455332535639}{2555728443659704626509544984384699129147 36} - \frac{4995251010356577747591536078880774 3128851}{3322446976757616014462408479700108867891568}s + \frac{1424692998974338678125345640062574 85671}{2555728443659704626509544984384699129147 36}t + \frac{5354009991678334587355092285973156994761}{3322446976757616014462408479700108867891568}st\big)r_{46}\big]$ |
| 59 | 60 | $(0,4)$ | $r_{59} = \big(\frac{51756051121}{2791074024} + \frac{1110614328055}{253987736184}s + \frac{9355499125}{25119666216}t + \frac{3766099670551}{2285889625656}st\big)\big[M_{24}r_8 - \big(-\frac{1093}{9400}s + \frac{461}{9400}t\big)r_{10} - \big(\frac{87662153}{1066111040} + \frac{7432863889}{50107218880}s + \frac{8170000021}{150321656640}t + \frac{72896929}{3198333120}st\big)r_{28} - \big(\frac{55046373384677013}{387444230396231200} - \frac{442631930830137}{387444230396231200}s - \frac{26862795824643951}{387444230396231200}t - \frac{4217749441458177}{387444230396231200}st\big)r_{39} - \big(-\frac{105445998557751}{13003280651994640} + \frac{18662139247761}{13003280651994640}s + \frac{373663183696503}{13003280651994640}t - \frac{44529984580101}{13003280651994640}st\big)r_{53}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 60 | 45 | $(0,-2)$ | $r_{60}=\big(\frac{16964687443799914703}{3453231075537716386200}+\frac{14420631419180968733}{6413143425998616145800}s-\frac{4933546017501358399}{3453231075537716386200}t+\frac{222696809221465367}{6413143425998616145800}st\big)\big[M_{15}r_9-\big(\frac{79729}{278}-\frac{4033}{278}s-\frac{92521}{278}t+\frac{10461}{278}st\big)r_5-\big(\frac{194399}{2780}+\frac{170747}{1390}s+\frac{195019}{1390}t+\frac{207771}{2780}st\big)r_{19}-\big(\frac{345356873331037206883}{4105409802827398808}-\frac{4252370563599715825}{4105409802827398808}s+\frac{14865434430243934752 7}{4105409802827398808}t-\frac{26750091963561381369}{4105409802827398808}st\big)r_{20}-\big(\frac{246453460238313538972934396271}{860139606256172904830523 1960}-\frac{380263806593015640050355865287}{860139606256172904830523 1960}s+\frac{18032109904978672900898944440 01}{860139606256172904830523 1960}t-\frac{17441253229853403603 1699104541}{860139606256172904830523 1960}st\big)r_{30}-\big(\frac{12081340200446683250675663240949701 25}{35992421495220623663537241028975156}+\frac{75496767033889853581384295525817 4263}{1799621074761031183176862051448757 80}s-\frac{23961606889986775233645147291812240147}{5398863224283093549530586154346273 40}t+\frac{23175941714887321426657663465449249 55}{1079772644856618709906117230869254 68}st\big)r_{40}-\big(-\frac{11668189517852286996530525235557 71}{48354747897861700508492857517710}+\frac{74098110162590632790115505857004 9}{48354747897861700508492857517710}s+\frac{35141943809116817537403249016096 73}{48354747897861700508492857517710}t-\frac{60794696889126157642325903920895 9}{48354747897861700508492857517710}st\big)r_{57}\big]$ |
| 61 | 40 | $(0,0)$ | $r_{61}=\big(-\frac{16163616168084543064356622 9}{48825657788799506224478782 80}-\frac{59828791659059171483925 19}{48825657788799506224478782 80}s-\frac{24617993923257912819059327}{4882565778879950622447878280}t-\frac{53477092299163855200617 83}{16275219262933168741492927 60}st\big)\big[M_{13}r_{10}-\big(\frac{16239460205}{1212701308}-\frac{692791205}{303175327}s+\frac{386114390}{303175327}t+\frac{1494686015}{1212701308}st\big)r_{11}-\big(\frac{218246838821645}{65080828395128}+\frac{278119865027245}{65080828395128}s-\frac{1086698500224215}{195242485185384}t-\frac{77085334456885}{27891783597912}st\big)r_{12}-\big(-\frac{7023092710897167}{554911199443237}-\frac{115524109015641}{79273028491891}s+\frac{7131148303475 11}{554911199443237}t-\frac{2482776247695353}{554911199443237}st\big)r_{45}\big]$ |
| 62 | 22 | $(0,-4)$ | $r_{62}=\big(\frac{228500447862849}{155975642793910900}+\frac{12313155391809}{31195128558782180}s+\frac{28873306193791}{935853856763465 40}t-\frac{42963662681713}{467926928381732700}st\big)\big[M_{15}r_{10}-\big(\frac{1748921896}{3331597}+\frac{622310771}{3331597}s+\frac{482944467}{3331597}t+\frac{191611134}{3331597}st\big)r_{13}\big]$ |
| 63 | 24 | $(0,-4)$ | $r_{63}=\big(\frac{2342731563098853080539 81}{6190973476687433980900640}+\frac{1958033845616870117229 33}{3095486738343716949045032 00}s+\frac{13601841460820149897447 67}{9286460215031150847135096 00}t+\frac{1794910601670945274996 3}{1857292043006230169427019 20}st\big)\big[M_{17}r_{10}-\big(\frac{1767480975}{6663194}-\frac{1724649195}{6663194}s-\frac{3813125715}{6663194}t+\frac{548322375}{6663194}st\big)r_{13}-\big(\frac{20512969815}{26652776}-\frac{7988284785}{26652776}s-\frac{4550765745}{26652776}t-\frac{10507365}{26652776}st\big)r_{62}\big]$ |
| 64 | 13 | $(0,-8)$ | $r_{64}=\big(\frac{3266417257}{137912496854400}-\frac{429307223}{275824993708800}s+\frac{1429035347}{275824993708800}t-\frac{9325979}{3447812421360}st\big)\big[M_{19}r_{10}\big]$ |
| 65 | 34 | $(0,-4)$ | $r_{65}=\big(\frac{19449858099493505152312370 29}{44275772390037324744321689182524}-\frac{25880261760919307601596518 3025}{18642430480015715681819658603168}s-\frac{51642434083163746286047242035195}{10626185373608957938637205403805 76}t+\frac{91207902419631527949693427 6871}{66413658585055987116482533773786}st\big)\big[M_{13}r_{11}-\big(-\frac{60458091}{4883606}-\frac{33632985}{4883606}s+\frac{4312807}{4883606}t+\frac{43712729}{4883606}st\big)r_{13}-\big(-\frac{242824509}{19534424}-\frac{106190631}{19534424}s+\frac{11506835}{58603272}t-\frac{64283771}{58603272}st\big)r_{62}-\big(\frac{41888782682986753954347}{38086179250110001735 27}+\frac{10365359030308695000477}{38086179250110001735 27}s+\frac{15233907166177413751439}{11425853775033000520581}t+\frac{12467444382546480726313}{11425853775033000520581}st\big)r_{63}\big]$ |
| 66 | 18 | $(0,-8)$ | $r_{66}=\big(\frac{1270343747}{83611103472}+\frac{870777929}{209027758680}s+\frac{24609607}{34837959780}t+\frac{718109}{27870367824}st\big)\big[M_{15}r_{11}-\big(\frac{5469128}{26833}+\frac{18000992}{134165}s-\frac{74421448}{402495}t+\frac{2908496}{80499}st\big)r_{64}\big]$ |
| 67 | 8 | $(1,1)$ | $r_{67}=\big(-\frac{3094675479}{11112498314}-\frac{2578427565}{11112498314}s+\frac{5696922655}{33337494942}t-\frac{7056493643}{33337494942}st\big)\big[M_{20}r_{11}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 68 | 42 | $(0,0)$ | $r_{68}=\big(\frac{3444723145429207088475658783314698702267 55}{25173158742046662741667541465956039331552}-\frac{9744936732181172669659917053331142191231 805}{3272510636466066156416780390574285113101 76}s+\frac{4813043524926201069034575344787552156864 35}{7551947622613998822500262439786811799465 6}t-\frac{2138365789416156873586048112012492820536 425}{98175319093981984692503411717228553393052 8}st\big)\big[M_{28}r_{11}-\big(\frac{109979029}{1227770748}-\frac{11744167}{1227770748}s+\frac{73787701}{1227770748}t+\frac{36763661}{1227770748}st\big)r_{11}-\big(\frac{2687629855429}{16472386240542}+\frac{690192785453}{16472386240542}s-\frac{890096487331}{16472386240542}t-\frac{240076340363}{16472386240542}st\big)r_{12}-\big(-\frac{60577731821656}{936344525398995}+\frac{5642424826204}{936344525398995}s-\frac{506838214853756}{2809033576196985}t-\frac{225958191512872}{2809033576196985}st\big)r_{45}-\big(-\frac{215043150040069309}{1428280858087908590}+\frac{21218215258080601}{1428280858087908590}s-\frac{269502176452836767}{12854527722791177310}t+\frac{3166455942641551}{313525066409540910}st\big)r_{61}\big]$ |
| 69 | 36 | $(0,-4)$ | $r_{69}=\big(\frac{26751960987616095673}{428437544321145367080}+\frac{384825154080450684031}{27420002836553303493120}s+\frac{4955291831188051912 13}{27420002836553303493120}t+\frac{7412361006694585607}{1713750177284581468320}st\big)\big[M_{13}r_{12}-\big(\frac{4837677567}{333120242}+\frac{1138610181}{333120242}s+\frac{12122620149}{333120242}t+\frac{854728827}{333120242}st\big)r_{13}-\big(-\frac{29636892527}{1332480968}+\frac{125092979}{1332480968}s+\frac{21397608507}{1332480968}t-\frac{3194177683}{1332480968}st\big)r_{62}-\big(\frac{9486029027003167251750 04}{2597932193680657757621 89}+\frac{9362971743330797060261 90}{2597932193680657757621 89}s+\frac{4404514907986538607590 0}{1998403979754352120555 3}t-\frac{5653612367576689422221 94}{2597932193680657757621 89}st\big)r_{63}-\big(-\frac{1957883805353454346220060416499190}{64448080711553104327836420546109}-\frac{8613718816069752395982741825256 98}{64448080711553104327836420546109}s+\frac{7520908706636352167739746306488 62}{64448080711553104327836420546109}t+\frac{4333126111476612766369472065778 58}{64448080711553104327836420546109}st\big)r_{65}\big]$ |
| 70 | 10 | $(1,1)$ | $r_{70}=\big(\frac{95595143660968799}{23038529553523056}-\frac{6211399972314235}{23038529553523056}s-\frac{62227185748809055}{23038529553523056}t+\frac{1322087582547759 1}{23038529553523056}st\big)\big[M_{20}r_{12}-\big(-\frac{1827805469}{4319581160}+\frac{30596041}{863916232}s+\frac{199588023}{863916232}t+\frac{154633621}{4319581160}st\big)r_{67}\big]$ |
| 71 | 46 | $(0,0)$ | $r_{71}=\big(\frac{1072420498438948173601988272062741048012431786384 15}{3060137762127095403058390682480217228725624251132 8}+\frac{2443907317546571442327028490880568050406098248111 59}{1989089545382612011987953943612141198671655763236320}s-\frac{1923897167784775700923418158541098315343567897283 11}{1530068881063547701529195341240108614362812125566 40}t+\frac{2454013312309735739708903143891354180099197541935 65}{3978179090765224023975907887224282397343311526472 64}st\big)\big[M_{28}r_{12}-\big(\frac{4088162255}{27916208412}-\frac{1787130881}{27916208412}s+\frac{1651430561}{9305402804}t+\frac{247163205}{9305402804}st\big)r_{11}-\big(\frac{422057805762487}{561806715239397}-\frac{32068280425840}{561806715239397}s-\frac{63838261587421}{561806715239397}t-\frac{14188736368760}{561806715239397}st\big)r_{12}-\big(-\frac{32652235910863924}{63869877073680465}+\frac{1686005454103948}{63869877073680465}s-\frac{312638904298059 64}{63869877073680465}t+\frac{2247216725574004}{63869877073680465}st\big)r_{45}-\big(\frac{1095191168480896 54787}{2922774268175269053 90}-\frac{2977595448378421 3223}{2922774268175269053 90}s-\frac{1347831400697249437}{9742580893917563513 0}t+\frac{3469696069968467461}{9742580893917563513 0}st\big)r_{61}-\big(\frac{6625777761389606369524248346056981 71}{2839640080593996031344251993160168570}+\frac{1887641964813111620729497616103515 9}{5679280161187992062688503986320337 14}s-\frac{1008407635271536771507546802342210423}{8518920241781988094032755979480505710}t+\frac{2818628400191848933045579332938367 41}{8518920241781988094032755979480505710}st\big)r_{68}\big]$ |
| 72 | 48 | $(0,0)$ | $r_{72}=\big(\frac{3482295614997657280875141 8}{1101522295897991113738720 45}+\frac{1101510358176773583603241 4}{1101522295897991137387204 5}s-\frac{1027031832033233591708 78}{1101522295897991137387204 5}t-\frac{1103734608009293774314522}{1101522295897991137387204 5}st\big)\big[M_{12}r_{13}-\big(\frac{185}{1456}-\frac{19}{1092}s+\frac{569}{2184}t-\frac{5}{624}st\big)r_{11}-\big(\frac{126865679}{586032720}+\frac{7835129}{97672120}s-\frac{19066523}{62789220}t-\frac{79773613}{1758098160}st\big)r_{12}-\big(-\frac{82959248}{118971515}+\frac{87753303}{1332480968}s-\frac{258889733}{6662404840}t-\frac{99817759}{2498401815}st\big)r_{45}-\big(\frac{366111486841}{6097617945328}-\frac{1257688950391}{15244044863320}s-\frac{8221571840431}{45732134589960}t+\frac{1513632587171}{18292853835984}st\big)r_{61}-\big(\frac{1027814133733461199909579015}{1367118418086386174285405918 4}-\frac{1156380892992163376528988 8}{7934169390679919761477802205}s+\frac{4496382992219022405499644 41}{4272245056519956794641893495}t-\frac{3016780593765026666141385 19}{2538934205017574323672896705 6}st\big)r_{68}-\big(-\frac{2970544869297607616287325124860406237359 69}{8810605559716331959583639513084613766043 20}+\frac{3988792668130607175649858866416083706293 33}{1145378722763123154745873136700999789585616 0}s+\frac{2311167252211180527589668683549086971397 1}{2643181667914899587875091853925384129812960}t+\frac{1075961876066731894721926237364746344118661}{3436136168289369464237619410102999368756848 0}st\big)r_{71}\big]$ |
| 73 | 4 | $(1,-3)$ | $r_{73}=\big(-\frac{25988}{8553}-\frac{1678}{8553}s+\frac{23894}{8553}t-\frac{1204}{2851}st\big)\big[M_{20}r_{13}-\big(\frac{397}{2832}-\frac{47}{1416}s-\frac{5}{1416}t-\frac{119}{2832}st\big)r_{55}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 74 | 15 | $(1,1)$ | $r_{74} = \big(-\frac{1480552983}{348776440} - \frac{1040424651}{906818744}s + \frac{46558911}{69755288}t + \frac{1767470271}{4534093720}st\big)\big[M_{22}r_{13} - \big(\frac{1}{84} - \frac{19}{168}s - \frac{71}{504}t - \frac{1}{63}st\big)r_{67} - \big(\frac{395094795623}{1050131090673} - \frac{124214829701}{4200524362692}s - \frac{623229116701}{4200524362692}t + \frac{84806819327}{2100262181346}st\big)r_{70}\big]$ |
| 75 | 1 | $(1,-5)$ | $r_{75} = \big(-1\big)\big[M_2r_{16}\big]$ |
| 76 | 7 | $(1,-1)$ | $r_{76} = \big(-\frac{1487904812926004534245013}{7558203246651042708531 00} - \frac{1862372611306246170251}{7558203246651042708531 00}s + \frac{2102576960888691449474969}{2267460973995312812559300}t - \frac{6066921866231853888997 81}{2267460973995312812559300}st\big)\big[M_{20}r_{19} - \big(-\frac{1070404654084383327}{4356460042568642620} + \frac{18921501165106471}{871292008513728524}s - \frac{453224675099381465}{871292008513728524}t + \frac{314092902810866553}{4356460042568642620}st\big)r_{47}\big]$ |
| 77 | 9 | $(1,-1)$ | $r_{77} = \big(\frac{861706815200588117000 2048789654}{124251369644076365002047884 4215} - \frac{550812533905415557872018342 1077}{161526780537299274502662249747 95}s + \frac{396607606933487539724002693873 73}{7455082178644581900122873065290}t + \frac{102857543882284596295315155387127}{969160683223795647015973498487 70}st\big)\big[M_{20}r_{20} - \big(-\frac{2206882138254408199284908 47}{365095228554778427230222076} + \frac{118122474544406311950661273}{365095228554778427230222076}s + \frac{296153014327438040916398817}{365095228554778427230222076}t + \frac{984199239396792607053233 3}{365095228554778427230222076}st\big)r_{47} - \big(-\frac{381592145315502821284378153144501}{275729993232876780128855265924497 6} + \frac{270952494993286395990102477284 11}{275729993232876780128855265924497 6}s - \frac{159259936334483823504645784432346 3}{827189979698630340386565797773492 8}t - \frac{437933575337334118855976828685331}{827189979698630340386565797773492 8}st\big)r_{76}\big]$ |
| 78 | 3 | $(1,-5)$ | $r_{78} = \big(\frac{652601}{260789} - \frac{13489187}{3390257}s + \frac{9090887}{260789}t - \frac{25444113}{3390257}st\big)\big[M_{20}r_{21} - \big(\frac{379990669951}{17813359471584} - \frac{123450203585}{8906679735792}s + \frac{67201688859}{2968893245264}t + \frac{12437118671}{17813359471584}st\big)r_{75}\big]$ |
| 79 | 0 | $(1,-7)$ | $r_{79} = \big(1\big)\big[M_2r_{26}\big]$ |
| 80 | 6 | $(1,-3)$ | $r_{80} = \big(-\frac{9737}{31896} + \frac{12175}{414648}s + \frac{21835}{31896}t + \frac{4471}{414648}st\big)\big[M_{24}r_{33} - \big(-1\big)r_{55} - \big(\frac{373}{1880} + \frac{253}{1880}s + \frac{319}{1880}t - \frac{77}{1880}st\big)r_{73}\big]$ |

### S25.4.2 Glue and sector signs (1,-1)

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 0 | $(-4,-5,-6,-4,-2,-3,-2,-1)$ | 1 |
| 1 | $(-4,-5,-6,-4,-2,-3,1,-1)$ | 2 |
| 2 | $(-4,-5,-6,-4,-2,-3,1,2)$ | 4 |
| 3 | $(-1,-5,-6,-4,-2,-3,-2,-1)$ | 5 |
| 4 | $(-1,-5,-6,-4,-2,-3,1,-1)$ | 10 |
| 5 | $(-1,-5,-6,-4,-2,-3,1,2)$ | 20 |
| 6 | $(-1,-2,-6,-4,-2,-3,-2,-1)$ | 40 |
| 7 | $(-1,-2,-6,-4,-2,-3,1,-1)$ | 80 |
| 8 | $(-1,-2,-6,-4,-2,-3,1,2)$ | 160 |
| 9 | $(-1,-2,-3,-4,-2,-3,-2,-1)$ | 360 |
| 10 | $(-1,-2,-3,-4,-2,-3,1,-1)$ | 720 |
| 11 | $(-1,-2,-3,-4,-2,-3,1,2)$ | 1440 |
| 12 | $(-1,-2,-3,-4,-2,0,-2,-1)$ | 360 |
| 13 | $(-1,-2,-3,-4,-2,0,1,-1)$ | 720 |
| 14 | $(-1,-2,-3,-4,-2,0,1,2)$ | 1440 |
| 15 | $(-1,-2,-3,-1,-2,-3,-2,-1)$ | 2880 |
| 16 | $(-1,-2,-3,-1,-2,-3,1,-1)$ | 5760 |
| 17 | $(-1,-2,-3,-1,-2,-3,1,2)$ | 11520 |
| 18 | $(-1,-2,-3,-1,-2,0,-2,-1)$ | 2880 |
| 19 | $(-1,-2,-3,-1,-2,0,1,-1)$ | 5760 |
| 20 | $(-1,-2,-3,-1,-2,0,1,2)$ | 11520 |
| 21 | $(-1,-2,-3,-1,1,-3,-2,-1)$ | 14400 |
| 22 | $(-1,-2,-3,-1,1,-3,1,-1)$ | 28800 |
| 23 | $(-1,-2,-3,-1,1,-3,1,2)$ | 57600 |
| 24 | $(-1,-2,-3,-1,1,0,-2,-1)$ | 14400 |
| 25 | $(-1,-2,-3,-1,1,0,1,-1)$ | 28800 |
| 26 | $(-1,-2,-3,-1,1,0,1,2)$ | 57600 |
| 27 | $(-1,-2,0,-1,-2,0,-2,-1)$ | 25920 |
| 28 | $(-1,-2,0,-1,-2,0,1,-1)$ | 51840 |

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 29 | $(-1,-2,0,-1,-2,0,1,2)$ | 103680 |
| 30 | $(-1,-2,0,-1,1,0,-2,-1)$ | 129600 |
| 31 | $(-1,-2,0,-1,1,0,1,-1)$ | 259200 |
| 32 | $(-1,-2,0,-1,1,0,1,2)$ | 518400 |
| 33 | $(-1,-2,0,2,1,0,-2,-1)$ | 1036800 |
| 34 | $(-1,-2,0,2,1,0,1,-1)$ | 2073600 |
| 35 | $(-1,-2,0,2,1,0,1,2)$ | 4147200 |
| 36 | $(-1,1,0,-1,-2,0,-2,-1)$ | 207360 |
| 37 | $(-1,1,0,-1,-2,0,1,-1)$ | 414720 |
| 38 | $(-1,1,0,-1,-2,0,1,2)$ | 829440 |
| 39 | $(-1,1,0,-1,1,0,-2,-1)$ | 1036800 |
| 40 | $(-1,1,0,-1,1,0,1,-1)$ | 2073600 |
| 41 | $(-1,1,0,-1,1,0,1,2)$ | 4147200 |
| 42 | $(-1,1,0,2,1,0,-2,-1)$ | 8294400 |
| 43 | $(-1,1,0,2,1,0,1,-1)$ | 16588800 |
| 44 | $(-1,1,0,2,1,0,1,2)$ | 33177600 |
| 45 | $(-1,1,3,2,1,0,-2,-1)$ | 74649600 |
| 46 | $(-1,1,3,2,1,0,1,-1)$ | 149299200 |
| 47 | $(-1,1,3,2,1,0,1,2)$ | 298598400 |
| 48 | $(-1,1,3,2,1,3,-2,-1)$ | 74649600 |
| 49 | $(-1,1,3,2,1,3,1,-1)$ | 149299200 |
| 50 | $(-1,1,3,2,1,3,1,2)$ | 298598400 |
| 51 | $(2,1,0,-1,-2,0,-2,-1)$ | 1036800 |
| 52 | $(2,1,0,-1,-2,0,1,-1)$ | 2073600 |
| 53 | $(2,1,0,-1,-2,0,1,2)$ | 4147200 |
| 54 | $(2,1,0,-1,1,0,-2,-1)$ | 5184000 |
| 55 | $(2,1,0,-1,1,0,1,-1)$ | 10368000 |
| 56 | $(2,1,0,-1,1,0,1,2)$ | 20736000 |
| 57 | $(2,1,0,2,1,0,-2,-1)$ | 41472000 |
| 58 | $(2,1,0,2,1,0,1,-1)$ | 82944000 |
| 59 | $(2,1,0,2,1,0,1,2)$ | 165888000 |
| 60 | $(2,1,3,2,1,0,-2,-1)$ | 373248000 |
| 61 | $(2,1,3,2,1,0,1,-1)$ | 746496000 |
| 62 | $(2,1,3,2,1,0,1,2)$ | 1492992000 |
| 63 | $(2,1,3,2,1,3,-2,-1)$ | 373248000 |
| 64 | $(2,1,3,2,1,3,1,-1)$ | 746496000 |
| 65 | $(2,1,3,2,1,3,1,2)$ | 1492992000 |
| 66 | $(2,4,3,2,1,0,-2,-1)$ | 2985984000 |
| 67 | $(2,4,3,2,1,0,1,-1)$ | 5971968000 |
| 68 | $(2,4,3,2,1,0,1,2)$ | 11943936000 |
| 69 | $(2,4,3,2,1,3,-2,-1)$ | 2985984000 |
| 70 | $(2,4,3,2,1,3,1,-1)$ | 5971968000 |
| 71 | $(2,4,3,2,1,3,1,2)$ | 11943936000 |
| 72 | $(2,4,6,2,1,3,-2,-1)$ | 26873856000 |
| 73 | $(2,4,6,2,1,3,1,-1)$ | 53747712000 |
| 74 | $(2,4,6,2,1,3,1,2)$ | 107495424000 |
| 75 | $(2,4,6,5,1,3,-2,-1)$ | 214990848000 |
| 76 | $(2,4,6,5,1,3,1,-1)$ | 429981696000 |
| 77 | $(2,4,6,5,1,3,1,2)$ | 859963392000 |
| 78 | $(2,4,6,5,4,3,-2,-1)$ | 1074954240000 |
| 79 | $(2,4,6,5,4,3,1,-1)$ | 2149908480000 |
| 80 | $(2,4,6,5,4,3,1,2)$ | 4299816960000 |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 0 | 68 | $(0,10)$ | $r_0 = e_{68}$ |
| 1 | 62 | $(0,8)$ | $r_1 = \left(-\frac{1}{8}\right)\left[M_9 r_0\right]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 2 | 47 | $(0,6)$ | $r_2 = \big(-\frac{3473131389}{70339709920} - \frac{698759789}{70339709920}s - \frac{1318806853}{70339709920}t + \frac{265143879}{70339709920}st\big)\big[M_{13}r_0\big]$ |
| 3 | 35 | $(0,2)$ | $r_3 = \big(\frac{9}{16480}s + \frac{31}{49440}t\big)\big[M_{15}r_0\big]$ |
| 4 | 41 | $(0,2)$ | $r_4 = \big(\frac{46193}{39785280} - \frac{39191}{258604320}s - \frac{4311}{6630880}t + \frac{7655}{103441728}st\big)\big[M_{17}r_0 - \big(-120s + 240t\big)r_3\big]$ |
| 5 | 26 | $(0,-2)$ | $r_5 = \big(\frac{1}{172800} + \frac{1}{499200}s + \frac{1}{345600}t - \frac{1}{748800}st\big)\big[M_{19}r_0\big]$ |
| 6 | 77 | $(-1,5)$ | $r_6 = \big(-\frac{2454543}{14491} + \frac{1264977}{14491}s + \frac{595161}{14491}t + \frac{166509}{14491}st\big)\big[M_{23}r_0\big]$ |
| 7 | 74 | $(-1,3)$ | $r_7 = \big(-\frac{7893}{1348} + \frac{531}{1348}s - \frac{65907}{1348}t + \frac{1017}{1348}st\big)\big[M_{25}r_0\big]$ |
| 8 | 50 | $(-1,-3)$ | $r_8 = \big(-\frac{1}{80}s + \frac{3}{80}t\big)\big[M_{27}r_0\big]$ |
| 9 | 59 | $(0,6)$ | $r_9 = \big(-\frac{20521}{253536} - \frac{75}{84512}s + \frac{5065}{2281824}t + \frac{2053}{2281824}st\big)\big[M_9r_1 - \big(-5\big)r_2\big]$ |
| 10 | 44 | $(0,4)$ | $r_{10} = \big(-\frac{943224373}{14753895480} - \frac{155310883}{14753895480}s - \frac{504831691}{14753895480}t + \frac{62325263}{14753895480}st\big)\big[M_{13}r_1\big]$ |
| 11 | 32 | $(0,0)$ | $r_{11} = \big(\frac{13}{3840} + \frac{1}{3840}s - \frac{1}{3840}t - \frac{1}{3840}st\big)\big[M_{15}r_1\big]$ |
| 12 | 34 | $(0,0)$ | $r_{12} = \big(\frac{271585}{295936176} + \frac{6997}{493226960}s - \frac{59297}{1479680880}t - \frac{56123}{295936176}st\big)\big[M_{17}r_1 - \big(150 + 90s + 330t + 30st\big)r_{11}\big]$ |
| 13 | 20 | $(0,-4)$ | $r_{13} = \big(\frac{1}{43200} + \frac{1}{86400}t\big)\big[M_{19}r_1\big]$ |
| 14 | 76 | $(-1,3)$ | $r_{14} = \big(\frac{15101856}{248167} - \frac{12445272}{248167}s + \frac{18897480}{248167}t - \frac{2862720}{248167}st\big)\big[M_{23}r_1 - \big(-\frac{13}{18144} - \frac{1}{1296}s + \frac{1}{1008}t + \frac{5}{18144}st\big)r_7\big]$ |
| 15 | 71 | $(-1,1)$ | $r_{15} = \big(-\frac{1754}{337} + \frac{118}{337}s - \frac{14646}{337}t + \frac{226}{337}st\big)\big[M_{25}r_1\big]$ |
| 16 | 49 | $(-1,-5)$ | $r_{16} = \big(-\frac{1}{20}s + \frac{3}{20}t\big)\big[M_{27}r_1\big]$ |
| 17 | 67 | $(0,8)$ | $r_{17} = \big(\frac{43}{19} - \frac{129}{247}s + \frac{203}{57}t + \frac{32}{741}st\big)\big[M_8r_2 - \big(-\frac{378503722}{2198115935} - \frac{15512740}{439623187}s + \frac{33222412}{439623187}t - \frac{13268878}{2198115935}st\big)r_1\big]$ |
| 18 | 43 | $(0,2)$ | $r_{18} = \big(\frac{207127596150535522376551}{284040505395047568794 3928} + \frac{191107169955454495793 05}{284040505395047568794 3928}s + \frac{388585917003719685021 59}{284040505395047568794 3928}t + \frac{180372109956510114736 73}{284040505395047568794 3928}st\big)\big[M_{13}r_2 - \big(\frac{38057456100}{5715101431} - \frac{129291690504}{40005710017}s + \frac{68785628712}{40005710017}t + \frac{168336514260}{40005710017}st\big)r_3 - \big(\frac{2980377115491}{4120588131751} - \frac{6713712060708}{4120588131751}s + \frac{5758478197496}{4120588131751}t + \frac{1110908377399}{4120588131751}st\big)r_4\big]$ |
| 19 | 29 | $(0,-2)$ | $r_{19} = \big(\frac{10140104961311}{33684696982742400} - \frac{8138073397291}{33684696982742400}s - \frac{13074858081493}{33684696982742400}t - \frac{1558482375619}{33684696982742400}st\big)\big[M_{15}r_2 - \big(-\frac{111687692064}{439623187} + \frac{1317944901 84}{439623187}s + \frac{282729060168}{439623187}t - \frac{8653271136}{439623187}st\big)r_5\big]$ |
| 20 | 31 | $(0,-2)$ | $r_{20} = \big(\frac{196142923973326803 1143}{130747599597455172407040} + \frac{259500035381923618 9}{435825331991517241356 80}s - \frac{222042147587696687537}{392242798792365517221120}t - \frac{224198152169944618021}{392242798792365517221120}st\big)\big[M_{17}r_2 - \big(\frac{9878124888}{439623187} - \frac{45951277572}{439623187}s - \frac{14648738724}{439623187}t - \frac{10248657768}{439623187}st\big)r_5 - \big(-\frac{201423935904}{439623187} + \frac{1308737003748}{5715101431}s + \frac{85181228292}{439623187}t - \frac{19202506608}{5715101431}st\big)r_{19}\big]$ |
| 21 | 14 | $(0,-6)$ | $r_{21} = \big(\frac{5493490927}{1449222465484800} - \frac{3400977941}{2173833698227200}s + \frac{3840887987}{2173833698227200}t + \frac{1705730377}{1449222465484800}st\big)\big[M_{19}r_2\big]$ |
| 22 | 23 | $(1,7)$ | $r_{22} = \big(-\frac{63475}{89433} - \frac{3671}{89433}s + \frac{14409}{19874}t - \frac{5185}{59622}st\big)\big[M_{20}r_2\big]$ |
| 23 | 79 | $(-1,5)$ | $r_{23} = \big(\frac{251529360}{43909} + \frac{61824480}{30043}s - \frac{955351680}{43909}t + \frac{190395120}{30043}st\big)\big[M_{21}r_2 - \big(-\frac{37763765989}{4482046316102400} - \frac{5185443121}{896409263220480}s - \frac{7823230069}{298803087740160}t + \frac{768017069}{4482046316102400}st\big)r_6\big]$ |
| 24 | 73 | $(-1,1)$ | $r_{24} = \big(\frac{6369247059029449056}{46880232503780887} + \frac{149195928116697618 24}{609443022549151531}s - \frac{195953575701913734 6}{46880232503780887}t - \frac{698498555799723524 42}{609443022549151531}st\big)\big[M_{23}r_2 - \big(-\frac{63805713179}{13294205174880} + \frac{391335178}{415443911715}s - \frac{2469810001}{1661775646860}t + \frac{7736986829}{13294205174880}st\big)r_{15}\big]$ |
| 25 | 65 | $(-1,-1)$ | $r_{25} = \big(-\frac{180862138424 63}{25420343196008} - \frac{599393704225 49}{25420343196008}s - \frac{123545028360627}{25420343196008}t + \frac{1413016415427}{25420343196008}st\big)\big[M_{25}r_2\big]$ |
| 26 | 48 | $(-1,-7)$ | $r_{26} = \big(\frac{2039}{18308} + \frac{99}{9154}s + \frac{7853}{27462}t - \frac{1199}{13731}st\big)\big[M_{27}r_2\big]$ |
| 27 | 61 | $(0,6)$ | $r_{27} = \big(\frac{24214346263634355}{5942951461 0210358} - \frac{4311074948998119}{1188583029242071 6}s + \frac{2379931352578339 3}{5630130138515076}t + \frac{165188505507493 15}{267431181576466 11}st\big)\big[M_{28}r_2 - \big(\frac{1106418520303}{5028849636093} - \frac{229709077651}{5028849636093}s + \frac{104033866426}{1676283212031}t - \frac{17976721579}{1676283212031}st\big)r_2 - \big(\frac{29835473229607900947}{491288645325109975198} - \frac{99335892632804820 57}{245644322662554987 5990}s - \frac{89558515443140491889}{245644322662554987 5990}t - \frac{46611771435765668 9}{491288645325109975198}st\big)r_9\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 28 | 46 | $(0,4)$ | $r_{28} = \left(\frac{67131326339253}{38966568357038} + \frac{7136881191369}{38966568357038}s - \frac{24625752425565}{38966568357038}t - \frac{13521427045353}{38966568357038}st\right)\left[M_8 r_3 - \left(-\frac{39}{206} + \frac{19}{412}s - \frac{38}{309}t - \frac{55}{1236}st\right)r_{10}\right]$ |
| 29 | 66 | $(0,6)$ | $r_{29} = \left(-\frac{3078578527947}{285793310371} + \frac{1105339046943}{285793310371}s - \frac{1383288507263}{857379931113}t + \frac{457870976335}{857379931113}st\right)\left[M_{12} r_3 - \left(\frac{140797}{224952} - \frac{1285}{4326}s - \frac{349}{25956}t - \frac{124477}{674856}st\right)r_2 - \left(\frac{13715279983855}{32964705054008} + \frac{10519215959869}{164823525270040}s - \frac{67156539437051}{494470575810120}t + \frac{295934993509}{14127730737432}st\right)r_9 - \left(\frac{39879397}{104227760} + \frac{29513}{501095}s - \frac{291062}{6514235}t - \frac{4310771}{312683280}st\right)r_{27}\right]$ |
| 30 | 33 | $(0,-2)$ | $r_{30} = \left(-\frac{633009761703611316564421 3633}{4162539601364787607706335169 28} - \frac{2902138817150841636193049497}{4162539601364787607706335169 28}s - \frac{247992409826848180078975052 39}{4162539601364787607706335169 28}t - \frac{1290911393714531438218340699}{4162539601364787607706335169 28}st\right)\left[M_{13} r_3 - \left(\frac{416937}{9373} - \frac{10776}{721}s + \frac{144290}{9373}t - \frac{145115}{9373}st\right)r_5 - \left(\frac{68358}{9373} - \frac{24504}{17407}s - \frac{9472}{2163}t + \frac{247678}{365547}st\right)r_{19} - \left(-\frac{73771373613980057}{10962731417334879} - \frac{22028527244626979}{10962731417334879}s + \frac{6323471252486065}{3654243805778293}t - \frac{3591380249711699}{10962731417334879}st\right)r_{20}\right]$ |
| 31 | 19 | $(0,-6)$ | $r_{31} = \left(-\frac{4636746580984083569}{45626150717837331656400} + \frac{16802526283947630199}{45626150717837331656400}s - \frac{73757330438982632473}{45626150717837331656400}t + \frac{12287236864918049651}{45626150717837331656400}st\right)\left[M_{15} r_3 - \left(-\frac{74620}{103} - \frac{12056}{103}s + \frac{227464}{309}t - \frac{59140}{309}st\right)r_{21}\right]$ |
| 32 | 24 | $(0,-6)$ | $r_{32} = \left(-\frac{417472737327302107791 61}{26237640196632388582159008} - \frac{2122514564575791082389773}{68217864511244210313613420 80}s + \frac{151827633726269581591829 3}{47227752353938299447886214 40}t + \frac{5206456838081977189053 7}{76745097575149736028150984}st\right)\left[M_{17} r_3 - \left(\frac{39150}{103} - \frac{16488}{103}s - \frac{22776}{103}t + \frac{14550}{103}st\right)r_{21} - \left(\frac{65775603472647}{647872890386} - \frac{43193459361345}{647872890386}s + \frac{64925050820035}{647872890386}t - \frac{8959173154837}{647872890386}st\right)r_{31}\right]$ |
| 33 | 12 | $(0,-10)$ | $r_{33} = \left(\frac{131}{1766400} + \frac{53}{2944000}s - \frac{2467}{79488000}t - \frac{83}{15897600}st\right)\left[M_{19} r_3\right]$ |
| 34 | 11 | $(1,3)$ | $r_{34} = \left(-\frac{238755645}{1521576578} - \frac{1212751305}{1521576578}s + \frac{242977717}{1521576578}t + \frac{675632581}{1521576578}st\right)\left[M_{20} r_3\right]$ |
| 35 | 75 | $(-1,1)$ | $r_{35} = \left(-\frac{689454826961328}{73443567401} + \frac{1060743755903760}{954766376213}s + \frac{1273707313409520}{73443567401}t - \frac{1119104654503824}{954766376213}st\right)\left[M_{21} r_3 - \left(-\frac{85213}{4200422400} + \frac{6463}{4200422400}s + \frac{537353}{12601267200}t - \frac{326831}{12601267200}st\right)r_{15} - \left(-\frac{256645349}{1592485142400} + \frac{32769563}{1592485142400}s - \frac{234226097}{4777455427200}t + \frac{233189087}{4777455427200}st\right)r_{24}\right]$ |
| 36 | 64 | $(-1,-3)$ | $r_{36} = \left(-\frac{254809611}{713659} - \frac{544104351}{9277567}s + \frac{14064309}{713659}t - \frac{213342003}{9277567}st\right)\left[M_{23} r_3 - \left(\frac{2353}{622944} - \frac{187}{155736}s + \frac{61}{400464}t - \frac{3623}{5606496}st\right)r_8\right]$ |
| 37 | 63 | $(-1,-5)$ | $r_{37} = \left(-\frac{1839}{622} + \frac{34035}{8086}s + \frac{138565}{18038}t + \frac{300787}{234494}st\right)\left[M_{25} r_3 - \left(\frac{41069}{1549120} - \frac{1349}{4647360}s - \frac{262879}{13942080}t + \frac{58009}{13942080}st\right)r_{16}\right]$ |
| 38 | 45 | $(0,2)$ | $r_{38} = \left(-\frac{125767343062479613071858806208 0692455}{977027675125732673190458325675885 07} + \frac{123203732499687673472146575864208838 29}{254027195532690495029519164675730118 2}s + \frac{372953962648222233750125218523393 72}{977027675125732673190458325675885 07}t - \frac{549230558200138466543663961497924906 3}{254027195532690495029519164675730118 2}st\right)\left[M_{28} r_3 - \left(\frac{350315}{2356434} - \frac{30497}{2356434}s - \frac{52403}{2356434}t - \frac{61747}{2356434}st\right)r_3 - \left(-\frac{74909}{11839644} - \frac{9810005}{485425404}s + \frac{6620029}{161808468}t - \frac{2899927}{161808468}st\right)r_4 - \left(\frac{1546740679}{65105129508} - \frac{22807227835}{846366683604}s + \frac{2355128599}{195315388524}t - \frac{28110000619}{2539100050812}st\right)r_{18}\right]$ |
| 39 | 56 | $(0,4)$ | $r_{39} = \left(-\frac{1171580059717531}{18216598725207184} + \frac{904289519171813}{4554149681301796}s + \frac{5727025120224041}{6831224521952694}t + \frac{11678716085757947}{54649796175621552}st\right)\left[M_8 r_4 - \left(-\frac{46477}{331544} - \frac{576233}{4310072}s - \frac{112691}{331544}t + \frac{349437}{4310072}st\right)r_{10} - \left(-\frac{22121093292803}{40763046025176} + \frac{39417569166485}{529919598327288}s - \frac{70849381265}{40763046025176}t - \frac{8847516124679}{176639866109096}st\right)r_{28}\right]$ |
| 40 | 37 | $(0,-2)$ | $r_{40} = \left(-\frac{251355748382374710667511570308 93}{312713158539213970093912030532160} - \frac{105466542063595040542266946327 07}{312713158539213970093912030532160}s + \frac{263974932794577968482961702257 27}{938139475617641910281736091596480}t + \frac{356835532684873907779307335079}{493757618746127321200913732419 20}st\right)\left[M_{13} r_4 - \left(-\frac{192813888}{3771313} + \frac{57220077}{3771313}s - \frac{139274085}{3771313}t + \frac{32642760}{3771313}st\right)r_5 - \left(-\frac{41617270}{3771313} + \frac{106335064}{49027069}s - \frac{32194424}{3771313}t + \frac{128004790}{49027069}st\right)r_{19} - \left(\frac{84100288945801465019}{4410956098336013499} - \frac{126430993448478029611}{57342429278368175487}s + \frac{4229827643177619943}{4410956098336013499}t + \frac{7783783480445102960 5}{57342429278368175487}st\right)r_{20} - \left(\frac{48242801334456213884769956}{18344126565501393549699079} - \frac{12303316070287687389645578 0}{16693155174600626813022616189}s - \frac{44596398500528888611941429 2}{55032379696504180649097237}t + \frac{10011864179374784371704344996}{50079465523818804390678485 67}st\right)r_{30}\right]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 41 | 27 | $(0,-6)$ | $r_{41} = \big(-\frac{41134269857819}{76799135850361344} - \frac{604145547358231}{1919978396259033600}s - \frac{1548809812077589}{5759935188777100800}t + \frac{60382305424151}{11519870377554201 60}st\big)\big[M_{17}r_4 - \big(\frac{44902140}{41443} - \frac{44575068}{538759}s - \frac{14738772}{41443}t + \frac{17844180}{538759}st\big)r_{21} - \big(\frac{11933858680986261}{260677632973466} - \frac{255478913489170803}{3388809228655058}s - \frac{80831877946264137}{260677632973466}t + \frac{89918173884619347}{3388809228655058}st\big)r_{31} - \big(\frac{16287249482253907 4832792810}{52524571227759237 1065607} + \frac{17842118922096130322694 1794}{68281942596087008238528 91}s + \frac{65150248833309854171143386}{52524571227759237 1065607}t - \frac{86814062238268914087682 2870}{68281942596087008238528 91}st\big)r_{32}\big]$ |
| 42 | 16 | $(1,3)$ | $r_{42} = \big(-\frac{44532263102155}{24231469368686} - \frac{174108581307221}{157504550896459}s + \frac{2534027316136}{36347204053029}t + \frac{333362692150733}{945027305378754}st\big)\big[M_{20}r_4 - \big(-\frac{19846373}{58683288} + \frac{32010593}{762882744}s + \frac{351981}{19561096}t + \frac{6809785}{762882744}st\big)r_{34}\big]$ |
| 43 | 69 | $(-1,-3)$ | $r_{43} = \big(\frac{60680871657}{470270293} - \frac{5705477379}{470270293}s + \frac{6260376843}{470270293}t + \frac{14649870627}{470270293}st\big)\big[M_{23}r_4 - \big(-\frac{7050929}{375970896} + \frac{1236775}{751941792}s + \frac{1832531}{250647264}t - \frac{354943}{375970896}st\big)r_8 - \big(\frac{261661}{46996362} - \frac{731153}{375970896}s - \frac{174383}{375970896}t - \frac{2017}{15665454}st\big)r_{36}\big]$ |
| 44 | 53 | $(0,2)$ | $r_{44} = \big(-\frac{983716668861281276132196574849769039231614723 7}{574709527229587696344383092668548734795072 8160} - \frac{171316632507296360341420029300028888979700908321}{574709527229587696344383092668548734795072 8160}s - \frac{166492707691494946475678434902807668659468610797}{517238574506628926709944783401693861311456553440}t - \frac{746991459133426459346670427564119329878367392 93}{517238574506628926709944783401693861311456553440}st\big)\big[M_{28}r_4 - \big(\frac{128331235}{316044318} - \frac{61011391}{316044318}s - \frac{22367113}{316044318}t + \frac{22514569}{316044318}st\big)r_3 - \big(\frac{52628519159}{97657694262} - \frac{2145564239}{97657694262}s - \frac{865124935}{3150248202}t + \frac{9037973029}{97657694262}st\big)r_4 - \big(-\frac{16461175816493}{26195649341748} + \frac{51729012551063}{340543441442724}s - \frac{3564454692883}{26195649341748}t + \frac{18874875153517}{340543441442724}st\big)r_{18} - \big(\frac{4066587703929468984644806871 05}{7480782317662760788098496225 0092} + \frac{259197523559844245920942352293 79}{9725017012961589024528045092511 96}s + \frac{838062961224535821943382575361}{831198035295862309788721802778 8}t - \frac{4536600138409547893961608471521}{1080557445884621002725338343612 44}st\big)r_{38}\big]$ |
| 45 | 38 | $(0,0)$ | $r_{45} = \big(\frac{1788144657}{1616775620} - \frac{21209567}{1616775620}s - \frac{181321361}{1616775620}t - \frac{107995933}{1616775620}st\big)\big[M_8r_5 - \big(-\frac{2}{3} + \frac{2}{39}st\big)r_{11} - \big(\frac{31}{96} - \frac{853}{6240}s + \frac{113}{480}t - \frac{29}{416}st\big)r_{12}\big]$ |
| 46 | 55 | $(0,2)$ | $r_{46} = \big(-\frac{3234993404398315244751586841241731222915774287835}{1437337456751513002682208695382251884407493702104} + \frac{4789994448591307047852316207082902475022235375 25}{1437337456751513002682208695382251884407493702104}s - \frac{866644524620042734906276049832207418807094610729 7}{1437337456751513002682208695382251884407493702104}t - \frac{459532060054603294609239691593003952262298689869}{1437337456751513002682208695382251884407493702104}st\big)\big[M_{12}r_5 - \big(\frac{127}{1092} - \frac{19}{507}s + \frac{19}{273}t + \frac{363}{4732}st\big)r_3 - \big(\frac{27959}{74984} + \frac{543}{10712}s + \frac{110965}{674856}t + \frac{294815}{8773128}st\big)r_4 - \big(\frac{60371525}{294162414} - \frac{935381}{22627878}s - \frac{72551}{49027069}t + \frac{222302}{147081207}st\big)r_{18} - \big(-\frac{14751537221028529151655499}{25847685990949328760289 7448} + \frac{40421831463101290403268415}{3360199178823412738837666824}s - \frac{53396341952889535457021 51}{25847685990949328760289 7448}t + \frac{22137701587936251809028727}{3360199178823412738837666824}st\big)r_{38} - \big(-\frac{25128320163288650308808065880129226513}{17781903687288334652066341527301108274 0} + \frac{18427983483120705261225724849716406608 7}{2311647479347483504768624398549144075620}s + \frac{32791849489040499793629610116947893 39}{25402719553269049502951916467573011820}t - \frac{42184552533550661679331904689093418615 3}{69349424380424505143058731956474322268 60}st\big)r_{44}\big]$ |
| 47 | 5 | $(1,-1)$ | $r_{47} = \big(-\frac{1400685}{7552276} - \frac{4845387}{7552276}s - \frac{461149}{7552276}t - \frac{847279}{7552276}st\big)\big[M_{20}r_5\big]$ |
| 48 | 21 | $(1,3)$ | $r_{48} = \big(\frac{49019901}{7257014} + \frac{13684566}{3628507}s - \frac{366798803}{14514028}t - \frac{91057169}{14514028}st\big)\big[M_{22}r_5 - \big(\frac{1}{36} - \frac{1}{1092}s + \frac{13}{252}t - \frac{23}{9828}st\big)r_{34} - \big(\frac{51692257523}{575155946484} - \frac{9402419921}{575155946484}s - \frac{46197864497}{575155946484}t + \frac{7647901211}{356048919252}st\big)r_{42}\big]$ |
| 49 | 17 | $(1,5)$ | $r_{49} = \big(-\frac{8901}{24496} - \frac{1755}{24496}s - \frac{24851}{24496}t + \frac{18463}{24496}st\big)\big[M_{24}r_5\big]$ |
| 50 | 22 | $(1,5)$ | $r_{50} = \big(\frac{6749}{102764} + \frac{123}{333983}s + \frac{21049}{51382}t + \frac{47953}{1335932}st\big)\big[M_2r_6 - \big(1\big)r_{49}\big]$ |
| 51 | 80 | $(-1,7)$ | $r_{51} = \big(\frac{73}{30} + \frac{11}{30}s - \frac{53}{30}t - \frac{13}{30}st\big)\big[M_8r_6\big]$ |
| 52 | 70 | $(-1,-1)$ | $r_{52} = \big(\frac{388030549667728331}{22271875515417394080} + \frac{129113681413938221}{22271875515417394080}s + \frac{2315083558958893541}{66815626546252182240}t + \frac{469843028723052 53}{35166119234869569 60}st\big)\big[M_{13}r_7 - \big(\frac{681021}{30667} + \frac{26163}{2359}s - \frac{390279}{30667}t + \frac{444999}{30667}st\big)r_{25}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 53 | 58 | $(0,4)$ | $r_{53} = \big(\frac{12283067792669915207443}{3341745608941553943024 9} - \frac{20274717002064750836279}{4344269291624020125932 37}s + \frac{16376225028903185593 03}{11139152029805179810083}t - \frac{55558579502104188687 23}{14480897638746733753107 9}st\big)\big[M_{20}r_7 - \big(-\frac{1664917}{6362560} - \frac{17783}{6362560}s - \frac{1878609}{6362560}t + \frac{220793}{6362560}st\big)r_{10} - \big(-\frac{287232536274191}{782271210210240} + \frac{16978295866021}{260757070070080}s - \frac{73062801261617}{260757070070080}t - \frac{7656830793587}{260757070070080}st\big)r_{28} - \big(\frac{26621202788470159007}{30990891145719462160} + \frac{3814982052659671123}{30990891145719462160}s + \frac{34130712708936975279}{30990891145719462160}t + \frac{200806202307537467}{30990891145719462160}st\big)r_{39}\big]$ |
| 54 | 78 | $(-1,3)$ | $r_{54} = \big(\frac{311181582577}{80156221704} + \frac{2596346689}{6165863208}s - \frac{139619791037}{26718740568}t - \frac{32381011529}{26718740568}st\big)\big[M_{28}r_7 - \big(\frac{866938}{3854943} + \frac{52519}{7709886}s - \frac{354913}{2569962}t + \frac{37808}{1284981}st\big)r_7 - \big(-\frac{346210631}{2309539184} + \frac{200219243}{2309539184}s - \frac{400860707}{2309539184}t + \frac{170502187}{2309539184}st\big)r_{14}\big]$ |
| 55 | 2 | $(1,-3)$ | $r_{55} = \big(1\big)\big[M_2r_8\big]$ |
| 56 | 72 | $(-1,-1)$ | $r_{56} = \big(-\frac{12248741}{10319492} - \frac{20432593}{134153396}s + \frac{94538741}{30958476}t + \frac{218709341}{402460188}st\big)\big[M_8r_8 - \big(-1\big)r_{25} - \big(\frac{8513351902931}{25420343196008} + \frac{818982162117}{127101715980040}s - \frac{10872775852471}{127101715980040}t + \frac{935605023501}{25420343196008}st\big)r_{52}\big]$ |
| 57 | 39 | $(0,-2)$ | $r_{57} = \big(-\frac{24804874418342525506555394910 61}{63505020262695174050778830634 5} + \frac{38568010736790032040089827871 3}{63505020262695174050778830634 5}s + \frac{47602869671183285760084471505 1}{63505020262695174050778830634 5}t - \frac{68039266309883015169772126201 9}{63505020262695174050778830634 5}st\big)\big[M_{20}r_8 - \big(\frac{32}{295} + \frac{79}{590}s + \frac{239}{1180}t + \frac{2}{295}st\big)r_5 - \big(\frac{1}{118} - \frac{175}{9204}s + \frac{37}{708}t - \frac{8}{2301}st\big)r_{19} - \big(\frac{163672936020607}{8280821341590840} - \frac{5446339771241}{8280821341590840}s + \frac{1466647737152911}{24842464024772520}t + \frac{112895764217191}{24842464024772520}st\big)r_{20} - \big(-\frac{154711128859044184179 49}{48213177351561594825096 0} - \frac{46352308240197290263 9}{96426354703123189650192}s - \frac{33550167028113468840 01}{96426354703123189650192}t + \frac{64967543547461907426 7}{16071059117187198275032 0}st\big)r_{30} - \big(\frac{10651301509369440969463845296 9}{64628904336979597067019414470 4} - \frac{40146991289704439289875899 93}{16157226084244899266754853617 6}s + \frac{29460460973641958622581344 91}{80786130421224496333774268088}t - \frac{11106236618900585625253606649}{64628904336979597067019414470 4}st\big)r_{40}\big]$ |
| 58 | 57 | $(0,2)$ | $r_{58} = \big(-\frac{71213007695730115781844 3}{44062567255351989783944} - \frac{26713336822314520124354 19}{57281337431957586719127 2}s - \frac{79100745040154731711672 9}{44062567255351989783944}t - \frac{31269735077584319311371 97}{57281337431957586719127 2}st\big)\big[M_{22}r_8 - \big(-\frac{13}{252} + \frac{3}{28}s - \frac{31}{252}t + \frac{5}{252}st\big)r_3 - \big(-\frac{832}{2163} + \frac{817}{25956}s - \frac{11995}{77868}t - \frac{739}{38934}st\big)r_4 - \big(-\frac{52063}{3481212} + \frac{5067977}{135767268}s + \frac{1115165}{10443636}t - \frac{288089}{15085252}st\big)r_{18} - \big(\frac{13986896529478187864036}{74560632666199868052811} - \frac{51806276658612030476 0857}{14912126533239997361705622}s + \frac{53490239653748715637 97}{10651518952314283829789 73}t - \frac{69257265853263007941811}{14912126533239997361705622}st\big)r_{38} - \big(\frac{44181766455973036363500017297857597723}{12310548706584231682199774903516151882 0} + \frac{12441227537477952726073693104235366829}{12310548706584231682199774903516151882 0}s - \frac{18506783340654580092823121629076294167}{12310548706584231682199774903516151882 0}t + \frac{52244435737457100153121185421435452 67}{12310548706584231682199774903516151882 0}st\big)r_{44} - \big(\frac{16043319515435983775132660919756748941819680}{21970085082199044217048625508567093623666237 1} - \frac{82009076516690403558376320617042812250111 83}{43940170164398088434097251017134187247332 4742}s - \frac{17014337094358305314076518330951388413413 79}{14646723388132696144699083672378062415777 4914}t - \frac{27870233057373320767643528456252010177582 5}{73233616940663480723495418361890312078887457}st\big)r_{46}\big]$ |
| 59 | 60 | $(0,4)$ | $r_{59} = \big(-\frac{69579817427}{9768759084} + \frac{68264602951}{126993868092}s - \frac{794887517689}{87918831756}t - \frac{3655715356831}{1142944812828}st\big)\big[M_{24}r_8 - \big(\frac{27}{3760} - \frac{1}{3760}s + \frac{691}{11280}t + \frac{11}{11280}st\big)r_{10} - \big(\frac{1624280749}{154096241680} - \frac{17629068581}{462288725040}s + \frac{41446963117}{462288725040}t + \frac{1498316831}{1386866175120}st\big)r_{28} - \big(\frac{192293561198763}{18314287127807860} - \frac{413007951762243}{18314287127807860}s - \frac{2294088383957 49}{18314287127807860}t - \frac{14170206989628 3}{18314287127807860}st\big)r_{39} - \big(-\frac{1070252476547499}{20781071361280040} + \frac{96439436843193}{20781071361280040}s - \frac{435831013795077}{20781071361280040}t - \frac{387785024664309}{20781071361280040}st\big)r_{53}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 60 | 51 | $(0,-2)$ | $r_{60}=\big(\frac{3002647107947248015 37}{2025215535939775851112 00}+\frac{6048966048269227734 31}{3290975245902135758057 00}s+\frac{37652895443811766077}{5063038839849439627780 0}t-\frac{2640570252358077525099}{2632780196721708606445600}st\big)\big[M_{15}r_9-\big(260-140s-260t+20st\big)r_5-\big(\frac{34367}{278}+\frac{17203}{834}s-\frac{152257}{2502}t-\frac{12763}{2502}st\big)r_{19}-\big(\frac{682163861522268745}{2926358050375746}-\frac{65874866879869745}{2926358050375746}s+\frac{16280597801229955}{975452683458582}t-\frac{3276599620331735}{975452683458582}st\big)r_{20}-\big(-\frac{3431223007645618882438 58}{2028338877683735375511}+\frac{1537874674814880499821 25}{4056677755367470751022}s+\frac{2753893578195369750452 5}{1352225918455823583674}t+\frac{1467884502021662112748}{2028338877683735375511}st\big)r_{30}-\big(\frac{2792779656329405428015398190888 37}{7613065849864545756199744585 92}-\frac{1068966098883877711655176133515 03}{7613065849864545756199744585 92}s-\frac{1729561480013903698847991860206 81}{7613065849864545756199744585 92}t+\frac{1358260405729372092843558292554 1}{2537688616621515252066581528 64}st\big)r_{40}-\big(-\frac{6517726648645289382361998046271 5}{5552775809523600133246521748 08}+\frac{4217536116604166666514680349974 5}{5552775809523600133246521748 08}s+\frac{2253026996224336644642018000521 5}{5552775809523600133246521748 08}t-\frac{1229376310450613408245609208802 5}{5552775809523600133246521748 08}st\big)r_{57}\big]$ |
| 61 | 40 | $(0,0)$ | $r_{61}=\big(-\frac{6560618777567060342653 79}{3115901647611164149795 2938}+\frac{1394471440527361234737881}{1246360659044656599181 1752}s+\frac{6049128389844850399171 81}{9347704942833492449385881 4}t+\frac{2813865820201301957994175}{3739081977133396979754352 56}st\big)\big[M_{13}r_{10}-\big(-\frac{161800822}{286881301}-\frac{3498048314}{532779559}s+\frac{47725771946}{3729456913}t-\frac{8542392134}{3729456913}st\big)r_{11}-\big(-\frac{660911215093}{29835655304}+\frac{45720754721}{21311182360}s-\frac{826283989751}{149178276520}t+\frac{72694023497}{29835655304}st\big)r_{12}-\big(\frac{60322369457095716}{22993358695624681}+\frac{2093946568705554}{3284765527946383}s-\frac{102490514924200924}{22993358695624681}t+\frac{22466706048519634}{22993358695624681}st\big)r_{45}\big]$ |
| 62 | 25 | $(0,-4)$ | $r_{62}=\big(\frac{330834174269}{3238563072753656}+\frac{8841174696529}{48578446091304840}s-\frac{7776431778029}{24289223045652420}t+\frac{971568233}{14575333827391452}st\big)\big[M_{15}r_{10}-\big(-\frac{298470578}{204915215}-\frac{1674371186}{40983043}s+\frac{13914412678}{40983043}t+\frac{13861405438}{204915215}st\big)r_{13}\big]$ |
| 63 | 28 | $(0,-4)$ | $r_{63}=\big(\frac{2581572337899281635722 3}{9523954391937754839657 7024}+\frac{2054911418540585543078 73}{2380988597984438709914425600}s+\frac{1163866523136516395230 4137}{7142965793953316129743276800}t+\frac{9006011864757387207793 9}{2857186317581326451897310 72}st\big)\big[M_{17}r_{10}-\big(-\frac{38663265201}{40983043}+\frac{12958189299}{40983043}s+\frac{18840390723}{40983043}t-\frac{1342361829}{40983043}st\big)r_{13}-\big(\frac{22303445787}{81966086}+\frac{2235671235}{81966086}s-\frac{51680816745}{81966086}t+\frac{5339460603}{81966086}st\big)r_{62}\big]$ |
| 64 | 13 | $(0,-8)$ | $r_{64}=\big(\frac{8716946933}{551649987417600}-\frac{1830746197}{1103299974835200}s+\frac{15356044139}{1103299974835200}t-\frac{792547}{91941664569600}st\big)\big[M_{19}r_{10}\big]$ |
| 65 | 30 | $(0,-4)$ | $r_{65}=\big(\frac{1505127827675098954558953778322 63}{3062176336951908142104277605298232}-\frac{7137319491140448448511953466529}{3062176336951908142104277605298232}s+\frac{6600373568170394968805443183149}{9186529010855724426312832815894696}t-\frac{3082623664893216027827462355 11}{9186529010855724426312832815894696}st\big)\big[M_{13}r_{11}-\big(-\frac{85767}{29120}-\frac{478047}{29120}s-\frac{82569}{29120}t+\frac{122907}{29120}st\big)r_{13}-\big(\frac{1099989}{116480}+\frac{4203}{16640}s-\frac{1214613}{116480}t-\frac{127089}{116480}st\big)r_{62}-\big(\frac{201698571046149637}{64771261455073120}+\frac{1031520336978315719}{842026398915950560}s-\frac{1030692453299189324 27}{1768255437723496176 0}t-\frac{7639229006216707303}{2526079196747851680}st\big)r_{63}\big]$ |
| 66 | 18 | $(0,-8)$ | $r_{66}=\big(-\frac{1969657444}{130642349175}-\frac{1119147569}{261284698350}s+\frac{173016821}{87094899450}t+\frac{29976592}{43547449725}st\big)\big[M_{15}r_{11}-\big(\frac{15823}{40}+\frac{12931}{200}s-\frac{26963}{200}t+\frac{2917}{40}st\big)r_{64}\big]$ |
| 67 | 8 | $(1,1)$ | $r_{67}=\big(\frac{3458843460}{5556249157}+\frac{1440488132}{5556249157}s-\frac{8185326452}{16668747471}t+\frac{1600289980}{16668747471}st\big)\big[M_{20}r_{11}\big]$ |
| 68 | 42 | $(0,0)$ | $r_{68}=\big(-\frac{4090479034231849874506571891802 9754341}{1363039932039938103881604039073113 4682}-\frac{8418727960130392233671435845609860520041}{1771951911651919535046085250795047508 66}s+\frac{4270740365593020605946917840233448813 77}{4089119796119814311644812117219340404 6}t-\frac{4300449548471005485741688637525732893 19}{5315855734955758605138255752385142525 98}st\big)\big[M_{28}r_{11}-\big(\frac{11945}{91512}-\frac{12557}{228780}s-\frac{4043}{76260}t-\frac{115}{30504}st\big)r_{11}-\big(-\frac{6526231}{48806400}+\frac{112453}{1952256}s-\frac{6735}{650752}t-\frac{23783}{16268800}st\big)r_{12}-\big(-\frac{66588671833}{2507565864640}+\frac{4503728791}{156722866540}s+\frac{7755445041}{78361433270}t+\frac{1300065773}{61160143040}st\big)r_{45}-\big(-\frac{2011244396393}{197272494049920}-\frac{4851377212261}{197272494049920}s+\frac{12945240936013}{197272494049920}t+\frac{561241729973}{197272494049920}st\big)r_{61}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 69 | 36 | $(0,-4)$ | $r_{69}=\big(\frac{843070131550419383}{20437523077220812800}-\frac{1002909477197013319}{20437523077220812800}s-\frac{438984124277399233}{20437523077220812800}t+\frac{339242331833236277}{20437523077220812800}st\big)\big[M_{13}r_{12}-\big(\frac{39293102496}{561045667}+\frac{3690507777}{561045667}s-\frac{13426678521}{561045667}t-\frac{740124396}{561045667}st\big)r_{13}-\big(-\frac{25296704757}{1122091334}-\frac{7273067463}{2244182668}s+\frac{19145721375}{2244182668}t+\frac{235283337}{86314718}st\big)r_{62}-\big(-\frac{16930598112532581703329 23}{22712272240933055715 1069}-\frac{64546420826058340809 2839}{22712272240933055715 1069}s+\frac{43031668668914557955906627}{22712272240933055715 1069}t+\frac{28367439172213300415 5203}{22712272240933055715 1069}st\big)r_{63}-\big(-\frac{24761278719565131638232804298836 70}{20872552126180847994133725369904 3}-\frac{13424248717911415662075788091286 70}{20872552126180847994133725369904 3}s-\frac{32844008646469124266499416959278 2}{20872552126180847994133725369904 3}t+\frac{66673121820424432024082415110450}{20872552126180847994133725369904 3}st\big)r_{65}\big]$ |
| 70 | 10 | $(1,1)$ | $r_{70}=\big(-\frac{9557623030385}{858589406832}+\frac{38715427917449}{17171788136640}s-\frac{88126372383617}{17171788136640}t+\frac{1009821911969}{858589406832}st\big)\big[M_{20}r_{12}-\big(-\frac{1534200217}{5820078128}-\frac{11887183}{98645392}s+\frac{8796301}{98645392}t-\frac{360176793}{5820078128}st\big)r_{67}\big]$ |
| 71 | 52 | $(0,0)$ | $r_{71}=\big(\frac{661503046313664981543160638792456928244685184 7}{887568820484961767657573878359427242921062188 80}-\frac{319148420732146676653714958410311086042537582623}{115383946663045029795484604186725541579738084544 0}s-\frac{841001891084863078772944521767718633849777283333}{887568820484961767657573878359427242921062188 80}t+\frac{269519627035922185796324648112076419485391941993}{115383946663045029795484604186725541579738084544 0}st\big)\big[M_{28}r_{12}-\big(\frac{6711842219}{23508429981}+\frac{2033944567}{23508429981}s+\frac{844658977}{7836143327}t+\frac{128214345}{7836143327}st\big)r_{11}-\big(\frac{180294860759}{564202319544}+\frac{1325442509}{34402580460}s-\frac{29065807229}{470168599620}t+\frac{1892834867}{188067439848}st\big)r_{12}-\big(-\frac{27905255727163577}{193249857565024796}+\frac{2602609948864457}{193249857565024796}s-\frac{51785202677831}{6233866373065316}t+\frac{1200433062709477}{193249857565024796}st\big)r_{45}-\big(\frac{148010913217326557}{1900392822887893161}+\frac{326292187892677 43}{3800785645775786322}s+\frac{51127903299465811}{3800785645775786322}t-\frac{3387058273854866}{1900392822887893161}st\big)r_{61}-\big(\frac{3861151318879584957035211 3954066791}{1464999114210991766587201195612468356}+\frac{1461552643484705624243456821510885}{1464999114210991766587201195612468356}s+\frac{207368519601242535022692144607921961}{4394997342632975299761603586837405068}t+\frac{855226818547231991746185718181 01727}{4394997342632975299761603586837405068}st\big)r_{68}\big]$ |
| 72 | 54 | $(0,0)$ | $r_{72}=\big(-\frac{6943853750402266137827882}{2203044591795982274774409}+\frac{6627048902158778670981 05}{2203044591795982274774409}s-\frac{6977178288757115687953915}{7343481972653274249248 03}t+\frac{1376841426885132532520 14}{7343481972653274249248 03}st\big)\big[M_{12}r_{13}-\big(\frac{529}{2730}-\frac{2381}{10920}s+\frac{4037}{10920}t+\frac{179}{2730}st\big)r_{11}-\big(-\frac{6535}{19968}-\frac{6663}{232960}s+\frac{4467}{46592}t+\frac{4829}{139776}st\big)r_{12}-\big(\frac{10183574325}{35906922688}+\frac{4903298691}{35906922688}s-\frac{14901979465}{107720768064}t+\frac{1512247655}{35906922688}st\big)r_{45}-\big(\frac{156496319719}{441379744260}+\frac{2593210825}{117701265136}s+\frac{1595719487}{117701265136}t-\frac{1718051448}{36781645355}st\big)r_{61}-\big(\frac{1369065869189465091440284985 89}{1361025839676556500630584331840}+\frac{3047065107811614550774771631 3}{1361025839676556500630584331840}s-\frac{2222870784555648616212363936 7}{4536752798921855002101947772 80}t+\frac{1038694828861464696301836239}{6481075427031221431574211104 0}st\big)r_{68}-\big(\frac{5220662296261949042801419993821230 3221}{2180863891263900966210566462516981549 12}+\frac{1541205793072346477098231482924761814265}{1984586141050149879251615480890453209699 2}s+\frac{1869712904512743975160014737343158234041}{4579814171654192029042189571285661253152}t+\frac{7551092189752805368388747307614673945 77}{8505369175929213768221209203816228041568}st\big)r_{71}\big]$ |
| 73 | 4 | $(1,-3)$ | $r_{73}=\big(\frac{3836}{8553}+\frac{9010}{8553}s+\frac{5070}{2851}t-\frac{5476}{8553}st\big)\big[M_{20}r_{13}-\big(\frac{1057}{14160}+\frac{181}{7080}s+\frac{313}{7080}t-\frac{653}{14160}st\big)r_{55}\big]$ |
| 74 | 15 | $(1,1)$ | $r_{74}=\big(\frac{1480552983}{348776440}+\frac{1040424651}{90681874 4}s+\frac{46558911}{69755288}t+\frac{1767470271}{4534093720}st\big)\big[M_{22}r_{13}-\big(-\frac{1}{84}+\frac{19}{168}s-\frac{71}{504}t-\frac{1}{63}st\big)r_{67}-\big(-\frac{395094795623}{1050131090673}+\frac{124214829701}{4200524362692}s-\frac{623229116701}{4200524362692}t+\frac{84806819327}{2100262181346}st\big)r_{70}\big]$ |
| 75 | 1 | $(1,-5)$ | $r_{75}=\big(-1\big)\big[M_{2}r_{16}\big]$ |
| 76 | 7 | $(1,-1)$ | $r_{76}=\big(\frac{42736967548896837447 98}{25194010822170142361 77}-\frac{20662433224514922318 28}{32752214068821185070301}s+\frac{51774787619512017662 84}{25194010822170142361 77}t-\frac{13482680295338391419130}{32752214068821185070301}st\big)\big[M_{20}r_{19}-\big(-\frac{227614992804767}{1380136890265140}+\frac{534459391825553}{2760273780530280}s+\frac{184903447154011}{4140410670795420}t-\frac{378179077612627}{8280821341590840}st\big)r_{47}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 77 | 9 | $(1,-1)$ | $r_{77} = \big(-\frac{1229653382846923560055976141}{3149050325009296465339596 84} - \frac{3566975266764387750807 3699613}{2046882711256042702470737 9460}s + \frac{1919708444791532003538494 4619}{4723575487513944698009395260}t + \frac{1651876455542143301125466 3279}{1228129626753625621482442767 6}st\big)\big[M_{20}r_{20} - \big(-\frac{7296339397667860679 2819}{4821317735156159482509 6} + \frac{2228150525748921411270 7}{4821317735156159482509 6}s - \frac{9438863811670800643229}{2410658867578079741254 8}t + \frac{2865064642006194881957}{2410658867578079741254 8}st\big)r_{47} - \big(-\frac{6605444414201928553341 9457529}{3641192221959421951192 96718496} - \frac{2936751889191185428540 3105549}{3641192221959421951192 96718496}s + \frac{1861383139301258516563 3700653}{3641192221959421951192 96718496}t + \frac{2843900645746892307384 535301}{3641192221959421951192 96718496}st\big)r_{76}\big]$ |
| 78 | 3 | $(1,-5)$ | $r_{78} = \big(-\frac{8760167}{260789} + \frac{48381629}{3390257}s + \frac{5223961}{260789}t - \frac{17796239}{3390257}st\big)\big[M_{20}r_{21} - \big(\frac{332133896545}{17813359471584} - \frac{126010301467}{44533398678960}s - \frac{1611925154891}{44533398678960}t + \frac{10690222245}{5937786490528}st\big)r_{75}\big]$ |
| 79 | 0 | $(1,-7)$ | $r_{79} = \big(-1\big)\big[M_2r_{26}\big]$ |
| 80 | 6 | $(1,-3)$ | $r_{80} = \big(\frac{3709}{31896} + \frac{132541}{414648}s + \frac{5489}{31896}t - \frac{19267}{414648}st\big)\big[M_{24}r_{33} - \big(\frac{307}{1880} + \frac{297}{1880}st\big)r_{55} - \big(\frac{827}{3760} - \frac{77}{752}s - \frac{187}{752}t - \frac{143}{3760}st\big)r_{73}\big]$ |

### S25.4.3 Glue and sector signs (-1,1)

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 0 | $(-2,-4,-6,-5,-4,-3,-2,-1)$ | 1/1074954240000 |
| 1 | $(-2,-4,-6,-5,-4,-3,1,-1)$ | 1/537477120000 |
| 2 | $(-2,-4,-6,-5,-4,-3,1,2)$ | 1/268738560000 |
| 3 | $(-2,-4,-6,-5,-1,-3,-2,-1)$ | 1/214990848000 |
| 4 | $(-2,-4,-6,-5,-1,-3,1,-1)$ | 1/107495424000 |
| 5 | $(-2,-4,-6,-5,-1,-3,1,2)$ | 1/53747712000 |
| 6 | $(-2,-4,-6,-2,-1,-3,-2,-1)$ | 1/26873856000 |
| 7 | $(-2,-4,-6,-2,-1,-3,1,-1)$ | 1/13436928000 |
| 8 | $(-2,-4,-6,-2,-1,-3,1,2)$ | 1/6718464000 |
| 9 | $(-2,-4,-3,-2,-1,-3,-2,-1)$ | 1/2985984000 |
| 10 | $(-2,-4,-3,-2,-1,-3,1,-1)$ | 1/1492992000 |
| 11 | $(-2,-4,-3,-2,-1,-3,1,2)$ | 1/746496000 |
| 12 | $(-2,-4,-3,-2,-1,0,-2,-1)$ | 1/2985984000 |
| 13 | $(-2,-4,-3,-2,-1,0,1,-1)$ | 1/1492992000 |
| 14 | $(-2,-4,-3,-2,-1,0,1,2)$ | 1/746496000 |
| 15 | $(-2,-1,-3,-2,-1,-3,-2,-1)$ | 1/373248000 |
| 16 | $(-2,-1,-3,-2,-1,-3,1,-1)$ | 1/186624000 |
| 17 | $(-2,-1,-3,-2,-1,-3,1,2)$ | 1/93312000 |
| 18 | $(-2,-1,-3,-2,-1,0,-2,-1)$ | 1/373248000 |
| 19 | $(-2,-1,-3,-2,-1,0,1,-1)$ | 1/186624000 |
| 20 | $(-2,-1,-3,-2,-1,0,1,2)$ | 1/93312000 |
| 21 | $(-2,-1,0,-2,-1,0,-2,-1)$ | 1/41472000 |
| 22 | $(-2,-1,0,-2,-1,0,1,-1)$ | 1/20736000 |
| 23 | $(-2,-1,0,-2,-1,0,1,2)$ | 1/10368000 |
| 24 | $(-2,-1,0,1,-1,0,-2,-1)$ | 1/5184000 |
| 25 | $(-2,-1,0,1,-1,0,1,-1)$ | 1/2592000 |
| 26 | $(-2,-1,0,1,-1,0,1,2)$ | 1/1296000 |
| 27 | $(-2,-1,0,1,2,0,-2,-1)$ | 1/1036800 |
| 28 | $(-2,-1,0,1,2,0,1,-1)$ | 1/518400 |
| 29 | $(-2,-1,0,1,2,0,1,2)$ | 1/259200 |
| 30 | $(1,-1,-3,-2,-1,-3,-2,-1)$ | 1/74649600 |
| 31 | $(1,-1,-3,-2,-1,-3,1,-1)$ | 1/37324800 |
| 32 | $(1,-1,-3,-2,-1,-3,1,2)$ | 1/18662400 |
| 33 | $(1,-1,-3,-2,-1,0,-2,-1)$ | 1/74649600 |
| 34 | $(1,-1,-3,-2,-1,0,1,-1)$ | 1/37324800 |
| 35 | $(1,-1,-3,-2,-1,0,1,2)$ | 1/18662400 |
| 36 | $(1,-1,0,-2,-1,0,-2,-1)$ | 1/8294400 |
| 37 | $(1,-1,0,-2,-1,0,1,-1)$ | 1/4147200 |
| 38 | $(1,-1,0,-2,-1,0,1,2)$ | 1/2073600 |

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 39 | $(1,-1,0,1,-1,0,-2,-1)$ | 1/1036800 |
| 40 | $(1,-1,0,1,-1,0,1,-1)$ | 1/518400 |
| 41 | $(1,-1,0,1,-1,0,1,2)$ | 1/259200 |
| 42 | $(1,-1,0,1,2,0,-2,-1)$ | 1/207360 |
| 43 | $(1,-1,0,1,2,0,1,-1)$ | 1/103680 |
| 44 | $(1,-1,0,1,2,0,1,2)$ | 1/51840 |
| 45 | $(1,2,0,-2,-1,0,-2,-1)$ | 1/1036800 |
| 46 | $(1,2,0,-2,-1,0,1,-1)$ | 1/518400 |
| 47 | $(1,2,0,-2,-1,0,1,2)$ | 1/259200 |
| 48 | $(1,2,0,1,-1,0,-2,-1)$ | 1/129600 |
| 49 | $(1,2,0,1,-1,0,1,-1)$ | 1/64800 |
| 50 | $(1,2,0,1,-1,0,1,2)$ | 1/32400 |
| 51 | $(1,2,0,1,2,0,-2,-1)$ | 1/25920 |
| 52 | $(1,2,0,1,2,0,1,-1)$ | 1/12960 |
| 53 | $(1,2,0,1,2,0,1,2)$ | 1/6480 |
| 54 | $(1,2,3,1,-1,0,-2,-1)$ | 1/14400 |
| 55 | $(1,2,3,1,-1,0,1,-1)$ | 1/7200 |
| 56 | $(1,2,3,1,-1,0,1,2)$ | 1/3600 |
| 57 | $(1,2,3,1,-1,3,-2,-1)$ | 1/14400 |
| 58 | $(1,2,3,1,-1,3,1,-1)$ | 1/7200 |
| 59 | $(1,2,3,1,-1,3,1,2)$ | 1/3600 |
| 60 | $(1,2,3,1,2,0,-2,-1)$ | 1/2880 |
| 61 | $(1,2,3,1,2,0,1,-1)$ | 1/1440 |
| 62 | $(1,2,3,1,2,0,1,2)$ | 1/720 |
| 63 | $(1,2,3,1,2,3,-2,-1)$ | 1/2880 |
| 64 | $(1,2,3,1,2,3,1,-1)$ | 1/1440 |
| 65 | $(1,2,3,1,2,3,1,2)$ | 1/720 |
| 66 | $(1,2,3,4,2,0,-2,-1)$ | 1/360 |
| 67 | $(1,2,3,4,2,0,1,-1)$ | 1/180 |
| 68 | $(1,2,3,4,2,0,1,2)$ | 1/90 |
| 69 | $(1,2,3,4,2,3,-2,-1)$ | 1/360 |
| 70 | $(1,2,3,4,2,3,1,-1)$ | 1/180 |
| 71 | $(1,2,3,4,2,3,1,2)$ | 1/90 |
| 72 | $(1,2,6,4,2,3,-2,-1)$ | 1/40 |
| 73 | $(1,2,6,4,2,3,1,-1)$ | 1/20 |
| 74 | $(1,2,6,4,2,3,1,2)$ | 1/10 |
| 75 | $(1,5,6,4,2,3,-2,-1)$ | 1/5 |
| 76 | $(1,5,6,4,2,3,1,-1)$ | 2/5 |
| 77 | $(1,5,6,4,2,3,1,2)$ | 4/5 |
| 78 | $(4,5,6,4,2,3,-2,-1)$ | 1 |
| 79 | $(4,5,6,4,2,3,1,-1)$ | 2 |
| 80 | $(4,5,6,4,2,3,1,2)$ | 4 |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 0 | 68 | $(0,10)$ | $r_0 = e_{68}$ |
| 1 | 62 | $(0,8)$ | $r_1 = \left(\frac{1}{8}\right)\left[M_9 r_0\right]$ |
| 2 | 53 | $(0,6)$ | $r_2 = \left(\frac{69393779}{959499936} - \frac{16057223}{959499936}s + \frac{2387749}{319833312}t + \frac{3103555}{319833312}st\right)\left[M_{13} r_0\right]$ |
| 3 | 29 | $(0,2)$ | $r_3 = \left(\frac{143}{15140} - \frac{261}{121120}s - \frac{89}{363360}t - \frac{7}{90840}st\right)\left[M_{15} r_0\right]$ |
| 4 | 41 | $(0,2)$ | $r_4 = \left(-\frac{241}{634240} - \frac{27931}{123676800}s - \frac{667}{3171200}t + \frac{1591}{24735360}st\right)\left[M_{17} r_0 - \left(150 + 240t + 30st\right) r_3\right]$ |
| 5 | 23 | $(0,-2)$ | $r_5 = \left(\frac{1}{288000} + \frac{1}{192000}s + \frac{1}{576000}t - \frac{1}{288000}st\right)\left[M_{19} r_0\right]$ |
| 6 | 77 | $(-1,5)$ | $r_6 = \left(\frac{2454543}{14491} + \frac{1264977}{14491}s - \frac{595161}{14491}t + \frac{166509}{14491}st\right)\left[M_{23} r_0\right]$ |
| 7 | 74 | $(-1,3)$ | $r_7 = \left(\frac{4977}{1348} + \frac{31149}{1348}s - \frac{11853}{1348}t - \frac{1413}{1348}st\right)\left[M_{25} r_0\right]$ |
| 8 | 59 | $(-1,-3)$ | $r_8 = \left(\frac{1}{80}s + \frac{3}{80}t\right)\left[M_{27} r_0\right]$ |
| 9 | 56 | $(0,6)$ | $r_9 = \left(\frac{22949}{422560} + \frac{135}{84512}s + \frac{1013}{253536}t - \frac{2053}{1267680}st\right)\left[M_9 r_1 - \left(9\right) r_2\right]$ |
| 10 | 44 | $(0,4)$ | $r_{10} = \left(\frac{69393779}{959499936} - \frac{16057223}{959499936}s + \frac{2387749}{319833312}t + \frac{3103555}{319833312}st\right)\left[M_{13} r_1\right]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 11 | 26 | $(0,0)$ | $r_{11} = \left(-\frac{8177}{8049900} + \frac{3019}{4024950}s + \frac{11549}{8049900}t - \frac{91}{670825}st\right)\left[M_{15}r_1\right]$ |
| 12 | 28 | $(0,0)$ | $r_{12} = \left(-\frac{2965217}{4392794400} - \frac{97549}{4392794400}s + \frac{2667893}{4392794400}t + \frac{592871}{1464264800}st\right)\left[M_{17}r_1 - \left(\frac{945}{4} - \frac{285}{4}s - \frac{345}{4}t + \frac{105}{4}st\right)r_{11}\right]$ |
| 13 | 20 | $(0,-4)$ | $r_{13} = \left(\frac{1}{43200} + \frac{1}{86400}t\right)\left[M_{19}r_1\right]$ |
| 14 | 76 | $(-1,3)$ | $r_{14} = \left(-\frac{15101856}{248167} - \frac{12445272}{248167}s - \frac{18897480}{248167}t - \frac{2862720}{248167}st\right)\left[M_{23}r_1 - \left(\frac{13}{18144} - \frac{1}{1296}s - \frac{1}{1008}t + \frac{5}{18144}st\right)r_7\right]$ |
| 15 | 71 | $(-1,1)$ | $r_{15} = \left(-\frac{1106}{337} - \frac{6922}{337}s + \frac{2634}{337}t + \frac{314}{337}st\right)\left[M_{25}r_1\right]$ |
| 16 | 58 | $(-1,-5)$ | $r_{16} = \left(\frac{1}{20}s + \frac{3}{20}t\right)\left[M_{27}r_1\right]$ |
| 17 | 67 | $(0,8)$ | $r_{17} = \left(-\frac{39}{38} - \frac{345}{494}s - \frac{87}{38}t - \frac{27}{494}st\right)\left[M_8r_2 - \left(-\frac{2257138}{9994791} + \frac{139840}{3331597}s + \frac{4891360}{29984373}t - \frac{334438}{29984373}st\right)r_1\right]$ |
| 18 | 43 | $(0,2)$ | $r_{18} = \left(-\frac{4209107406326711478135}{389213232508376545955744} - \frac{8164171277433656038 19}{389213232508376545955744}s - \frac{611226701487383176950 3}{389213232508376545955744}t - \frac{150529642793009505052 3}{389213232508376545955744}st\right)\left[M_{13}r_2 - \left(-\frac{68427746}{303175327} + \frac{288173475}{43310761}s + \frac{9077438765}{909525981}t + \frac{7627980502}{909525981}st\right)r_3 - \left(-\frac{5065154554027}{459007445078} - \frac{6324506293347}{459007445078}s - \frac{31532161180029}{459007445078}t + \frac{9555889988887}{459007445078}st\right)r_4\right]$ |
| 19 | 25 | $(0,-2)$ | $r_{19} = \left(-\frac{1510291547123907}{2953532232249927200} - \frac{115775485311147}{590706446449985440}s - \frac{80170230663631}{590706446449985440}t - \frac{383349910276747}{2953532232249927200}st\right)\left[M_{15}r_2 - \left(\frac{6278241320}{9994791} + \frac{257965832}{9994791}s + \frac{1536636088}{3331597}t + \frac{153214360}{3331597}st\right)r_5\right]$ |
| 20 | 27 | $(0,-2)$ | $r_{20} = \left(\frac{697865875058076354170 71}{990088755402788955200602240} - \frac{102051285074209808948471}{990088755402788955200602240}s + \frac{110454735732683270230072 9}{990088755402788955200602240}t - \frac{610404547055133764779543}{297026626620836865601806720}st\right)\left[M_{17}r_2 - \left(\frac{558554860}{3331597} + \frac{847105100}{3331597}s + \frac{2303233900}{3331597}t + \frac{231252540}{3331597}st\right)r_5 - \left(\frac{1152444798}{3331597} - \frac{626444072}{3331597}s + \frac{5987552218}{9994791}t + \frac{981133076}{9994791}st\right)r_{19}\right]$ |
| 21 | 14 | $(0,-6)$ | $r_{21} = \left(-\frac{2363768033}{1449222465484800} - \frac{1083695191}{724611232742400}s - \frac{281749693}{724611232742400}t + \frac{1052645239}{1449222465484800}st\right)\left[M_{19}r_2\right]$ |
| 22 | 32 | $(1,7)$ | $r_{22} = \left(-\frac{3775}{59622} - \frac{10879}{59622}s - \frac{8105}{29811}t - \frac{200}{1569}st\right)\left[M_{20}r_2\right]$ |
| 23 | 79 | $(-1,5)$ | $r_{23} = \left(\frac{965768400}{43909} + \frac{2218896288}{570817}s - \frac{620367552}{43909}t - \frac{2125126800}{570817}st\right)\left[M_{21}r_2 - \left(-\frac{504476999}{20379778640640} + \frac{1036279}{20379778640640}s - \frac{2484611551}{61139335921920}t + \frac{130550767}{61139335921920}st\right)r_6\right]$ |
| 24 | 73 | $(-1,1)$ | $r_{24} = \left(-\frac{4625839906337719566}{46880232503780887} + \frac{10481247827293793130}{609443022549151531}s + \frac{312002748743826420}{46880232503780887}t - \frac{790502293380431520}{609443022549151531}st\right)\left[M_{23}r_2 - \left(\frac{85210123}{15112123992} + \frac{72371797}{60448495968}s + \frac{108744235}{60448495968}t + \frac{6885227}{7556061996}st\right)r_{15}\right]$ |
| 25 | 65 | $(-1,-1)$ | $r_{25} = \left(\frac{4990106743191}{12710171598004} + \frac{26882500263453}{12710171598004}s + \frac{22001305098519}{12710171598004}t + \frac{2531878407441}{12710171598004}st\right)\left[M_{25}r_2\right]$ |
| 26 | 57 | $(-1,-7)$ | $r_{26} = \left(\frac{19443}{183080} + \frac{543}{183080}s + \frac{1863}{7960}t + \frac{501}{7960}st\right)\left[M_{27}r_2\right]$ |
| 27 | 61 | $(0,6)$ | $r_{27} = \left(-\frac{17340313665761643}{11885830292420716} - \frac{2386698723398202}{2971457573105179}s - \frac{831178029744497}{312785007695282}t + \frac{4275448153377961}{11885830292420716}st\right)\left[M_{28}r_2 - \left(\frac{12131611235}{76220276166} + \frac{2597278757}{76220276166}s + \frac{1018825657}{8468919574}t + \frac{44035825}{8468919574}st\right)r_2 - \left(-\frac{2973573145379029}{84645081137939034} - \frac{811739789499649}{28215027045979678}s + \frac{7079533160948411}{84645081137939034}t - \frac{2821183164629173}{253935243413817102}st\right)r_9\right]$ |
| 28 | 50 | $(0,4)$ | $r_{28} = \left(\frac{1429683987129}{51521839148435} + \frac{1963201104232}{51521839148435}s + \frac{13994269295236}{51521839148435}t + \frac{3095148706709}{51521839148435}st\right)\left[M_8r_3 - \left(\frac{13289}{15140} - \frac{3857}{15140}s - \frac{28139}{15140}t + \frac{3591}{15140}st\right)r_{10}\right]$ |
| 29 | 66 | $(0,6)$ | $r_{29} = \left(-\frac{5645877097263}{1428966551855} - \frac{1663979336859}{1428966551855}s + \frac{35393909658029}{4286899655565}t + \frac{10049794261381}{4286899655565}st\right)\left[M_{12}r_3 - \left(-\frac{440053}{551096} + \frac{330427}{688870}s + \frac{158191}{105980}t - \frac{45665}{78728}st\right)r_2 - \left(\frac{105015070980059}{27540446704680} - \frac{26136842435713}{27540446704680}s + \frac{4744758963157}{3934349529240}t - \frac{8236088242789}{27540446704680}st\right)r_9 - \left(\frac{150486841}{95752930} - \frac{66900845}{153204688}s + \frac{486646705}{459614064}t - \frac{77716423}{287258790}st\right)r_{27}\right]$ |
| 30 | 35 | $(0,-2)$ | $r_{30} = \left(-\frac{5399667862305394232622199209797629}{3542275727011641235375463721412806 72} - \frac{661276536565530756894279521068 6129}{1771137863505820617687731860706403360}s + \frac{167700712925534965245158129676 2083}{1771137863505820617687731860706403360}t + \frac{467290760294765391615286546 32671}{3542275727011641235375463721412806 72}st\right)\left[M_{13}r_3 - \left(-\frac{6863718}{68887} + \frac{317115}{9841}s - \frac{1791585}{68887}t - \frac{9856}{9841}st\right)r_5 - \left(\frac{7964113}{98410} - \frac{13013549}{688870}s + \frac{2492961}{98410}t - \frac{327097}{52990}st\right)r_{19} - \left(\frac{9879083785262813821845}{508649937207501837566} - \frac{199798336928285395287}{39126918246730910582}s + \frac{270514067092161356 4593}{726642767439288339 38}t - \frac{500788560639661102 7981}{508649937207501837566}st\right)r_{20}\right]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 31 | 19 | $(0,-6)$ | $r_{31} = \big(-\frac{34485240132913470703}{45626150717837331656400} - \frac{6959510861529665681}{45626150717837331656400}s - \frac{23402546085374945821}{15208716905945777218800}t - \frac{6656086015160983271}{15208716905945777218800}st\big)\big[M_{15}r_3 - \big(\frac{14488916}{3785} - \frac{4130008}{3785}s - \frac{2225416}{3785}t + \frac{1893164}{3785}st\big)r_{21}\big]$ |
| 32 | 21 | $(0,-6)$ | $r_{32} = \big(\frac{3818770679663453907281}{29838341436225853749417600} - \frac{57575118688234138171439}{3878984386709360987424288800}s - \frac{1686295052840227021117}{59676682872451707498835 20}t + \frac{42428671795735718594 03}{3878984386709360987424288800}st\big)\big[M_{17}r_3 - \big(-\frac{549216}{757} + \frac{80136}{757}s + \frac{1148472}{757}t - \frac{161064}{757}st\big)r_{21} - \big(-\frac{3190391053282671}{4761551242934} + \frac{210361114536873}{2380775621467}s + \frac{467118279637371}{2380775621467}t - \frac{330697400617359}{4761551242934}st\big)r_{31}\big]$ |
| 33 | 12 | $(0,-10)$ | $r_{33} = \big(-\frac{353}{26496000} - \frac{83}{26496000}s - \frac{757}{79488000}t + \frac{61}{79488000}st\big)\big[M_{19}r_3\big]$ |
| 34 | 11 | $(1,3)$ | $r_{34} = \big(-\frac{3465135273}{1521576578} - \frac{865181079}{1521576578}s + \frac{6410461}{1521576578}t - \frac{123361657}{1521576578}st\big)\big[M_{20}r_3\big]$ |
| 35 | 75 | $(-1,1)$ | $r_{35} = \big(\frac{4308310935946512}{367217837005} + \frac{2300229222013872}{954766376213}s + \frac{890830893214512}{73443567401}t + \frac{16178383876295376}{4773831881065}st\big)\big[M_{21}r_3 - \big(\frac{8045719}{30871065600} - \frac{716743}{6174213120}s + \frac{364699}{6174213120}t + \frac{108401}{30871065600}st\big)r_{15} - \big(-\frac{7805856289}{11703992745600} + \frac{2476919909}{11703992745600}s - \frac{5012575907}{11703992745600}t + \frac{690328519}{11703992745600}st\big)r_{24}\big]$ |
| 36 | 64 | $(-1,-3)$ | $r_{36} = \big(\frac{56566377}{713659} + \frac{321865299}{9277567}s + \frac{49146489}{713659}t + \frac{87825591}{9277567}st\big)\big[M_{23}r_3 - \big(\frac{176501}{17168760} - \frac{261031}{68675040}s - \frac{191237}{68675040}t + \frac{9269}{4905360}st\big)r_8\big]$ |
| 37 | 63 | $(-1,-5)$ | $r_{37} = \big(-\frac{180921}{18038} - \frac{337017}{117247}s + \frac{15193}{9019}t + \frac{229417}{234494}st\big)\big[M_{25}r_3 - \big(\frac{112181}{2277056} - \frac{3740471}{170779200}s - \frac{5389367}{170779200}t + \frac{110267}{6831168}st\big)r_{16}\big]$ |
| 38 | 47 | $(0,2)$ | $r_{38} = \big(\frac{21036954247551180195852073029493181}{12207365803404471412349710670940720} + \frac{9744584819406082147581618364600 37}{24414731606808942824699421341881 44}s + \frac{1957273046515852232028842831609 8511}{24414731606808942824699421341881 44}t + \frac{2638767635476315167873784009779 7751}{12207365803404471412349710670940720}st\big)\big[M_{28}r_3 - \big(\frac{3846031}{34637292} + \frac{1511605}{34637292}s + \frac{1281735}{3848588}t - \frac{271959}{3848588}st\big)r_3 - \big(-\frac{29343533398}{10925179185} + \frac{15542193319}{21850358370}s + \frac{32458435763}{21850358370}t - \frac{4225074917}{10925179185}st\big)r_4 - \big(\frac{27865549721}{9534876770} - \frac{19385963327}{24790679602}s - \frac{550741659}{1906975354}t + \frac{10485800737}{123953398010}st\big)r_{18}\big]$ |
| 39 | 52 | $(0,4)$ | $r_{39} = \big(\frac{6112292834948821}{10471790908042492} + \frac{10943051262279513}{20943581816084984}s - \frac{29234168743122667}{62830745448254952}t - \frac{531506011565383}{31415372724127476}st\big)\big[M_8r_4 - \big(\frac{1909}{3964} + \frac{54137}{257660}s - \frac{2513}{19820}t - \frac{3521}{51532}st\big)r_{10} - \big(\frac{57738058093}{52825802032} + \frac{59719999775}{686735426416}s - \frac{8645300695}{52825802032}t + \frac{11091536431}{686735426416}st\big)r_{28}\big]$ |
| 40 | 37 | $(0,-2)$ | $r_{40} = \big(\frac{4705098139995842887554215256987235}{40449338005564379848669846808944128} + \frac{41722406511890392027925572151231 23265}{76853742210572321712472708936993 8432}s + \frac{8860161891811035850760251941843 8111}{76853742210572321712472708936993 8432}t + \frac{4177703697417082737941784993542 57}{25617914070190773904157569645664 6144}st\big)\big[M_{13}r_4 - \big(\frac{2352543}{90181} - \frac{35928}{6937}s + \frac{779638}{90181}t - \frac{141847}{90181}st\big)r_5 - \big(-\frac{5163231}{180362} + \frac{3662595}{2344706}s - \frac{95217}{25766}t + \frac{242545}{334958}st\big)r_{19} - \big(\frac{1082847801560227747815}{33294061279566335602 9} + \frac{2723962106643094785516}{6183182809062319469 11}s - \frac{4151171475773978738 54}{4756294468509476514 7}t + \frac{5091314168278271010255}{4328227966343623628377}st\big)r_{20} - \big(\frac{7620905418861026519664965857335}{13951124070465454807647736 0321} - \frac{2394882712630535475869770659 1197}{18136461291605091249942056841 73}s - \frac{5050292186341388208673600472 1}{10731633900358042159729027717}t - \frac{1380276760296136041510621323403}{2590923041657870178563150977 39}st\big)r_{30}\big]$ |
| 41 | 33 | $(0,-6)$ | $r_{41} = \big(\frac{1997548163517683}{6047532507495513600} - \frac{24504122512673}{211663637762342976}s - \frac{5424870578449}{10079220845825856}t - \frac{2198155165461083}{14110909184156198400}st\big)\big[M_{17}r_4 - \big(-\frac{26895}{991} + \frac{1579485}{12883}s - \frac{588195}{991}t - \frac{555195}{12883}st\big)r_{21} - \big(\frac{8245472183815035}{24933671237768} + \frac{6364010654985675}{324137726090984}s - \frac{946307958088785}{24933671237768}t - \frac{19360838600163405}{324137726090984}st\big)r_{31} - \big(-\frac{1107239483677093286460 2935}{2511973075632044203971 8} - \frac{5197003969670284759034455}{3265564998321657465163 34}s - \frac{1677503291140045152020 5115}{2511973075632044203971 8}t - \frac{3844058908290521119950 9535}{3265564998321657465163 34}st\big)r_{32}\big]$ |
| 42 | 16 | $(1,3)$ | $r_{42} = \big(\frac{6313899170717159}{1780895370106340} + \frac{13007936146487227}{23151639811382420}s - \frac{12348617717642501}{5342686110319020}t + \frac{25105106263421579}{69454919434147260}st\big)\big[M_{20}r_4 - \big(\frac{1002653}{14032560} - \frac{1820639}{36484656}s + \frac{124277}{935504}t - \frac{6722281}{182423280}st\big)r_{34}\big]$ |
| 43 | 69 | $(-1,-3)$ | $r_{43} = \big(\frac{382514616540}{3456258367} + \frac{308111158152}{3456258367}s + \frac{311960329884}{3456258367}t + \frac{22798812960}{3456258367}st\big)\big[M_{23}r_4 - \big(\frac{14177}{8990352} + \frac{10915}{17980704}s + \frac{132071}{17980704}t - \frac{65}{142704}st\big)r_8 - \big(\frac{23467}{44951760} + \frac{1909}{6421680}s + \frac{373}{14983920}t - \frac{26113}{44951760}st\big)r_{36}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 44 | 49 | $(0,2)$ | $r_{44} = \big(-\frac{4898038736083462257124222532165345827986839}{1381918879893140287334089680842298029117536 8} - \frac{3410217091780110223730769881074112187500113}{1381918879893140287334089680842298029117536 8}s - \frac{1873993334311607321696012051297935009737202 97}{4145756639679420862002269042526894087352610 4}t + \frac{2806222541740504897680790267500231355257691 7}{4145756639679420862002269042526894087352610 4}st\big)\big[M_{28}r_4 - \big(\frac{473429}{5038244} - \frac{218785}{5038244}s - \frac{516515}{5038244}t - \frac{53921}{5038244}st\big)r_3 - \big(\frac{40168134913}{34325556372} - \frac{5724357293}{34325556372}s + \frac{149882653}{3813950708}t + \frac{4070501}{123030668}st\big)r_4 - \big(-\frac{1632205644}{6241124755} + \frac{9928684179}{81134621815}s - \frac{400712911}{6241124755}t + \frac{3658982778}{81134621815}st\big)r_{18} - \big(-\frac{331798498633722252913192583923}{183839178131806226475211075644} + \frac{785768720359601388268927072 9}{183839178131806226475211075644}s + \frac{2245965836952303989313220933 7}{183839178131806226475211075644}t - \frac{1080793055988001403397891857 5}{183839178131806226475211075644}st\big)r_{38}\big]$ |
| 45 | 38 | $(0,0)$ | $r_{45} = \big(-\frac{16575187865}{33503395304} - \frac{1715651335}{16751697652}s - \frac{5528651965}{50255092956}t - \frac{314401795}{100510185912}st\big)\big[M_8r_5 - \big(\frac{9}{10} + \frac{3}{20}s + \frac{1}{20}t + \frac{1}{10}st\big)r_{11} - \big(\frac{560827}{1073320} - \frac{48091}{1073320}s + \frac{346307}{1073320}t - \frac{32723}{214664}st\big)r_{12}\big]$ |
| 46 | 51 | $(0,2)$ | $r_{46} = \big(\frac{6061116406181689313055383209999489074583949783 8}{2310347752555585354625528115695202622943464189 97} - \frac{2173186614976594738042962916899886581477918456 1}{2310347752555585354625528115695202622943464189 97}s + \frac{6573634993758283481374531587907340395035281899}{7701159175186178487517603856506754098115472999}t + \frac{1943022042171917043280254626378829441656196 16}{7701159175186178487517603856506754098115472999}st\big)\big[M_{12}r_5 - \big(\frac{1249}{3640} + \frac{439}{3640}s + \frac{47}{104}t + \frac{583}{3640}st\big)r_3 - \big(-\frac{801186}{1722175} - \frac{369981}{6888700}s - \frac{40599}{529900}t + \frac{243631}{1722175}st\big)r_4 - \big(\frac{759019}{1803620} + \frac{2500859}{117235300}s - \frac{690939}{9018100}t - \frac{17663}{23447060}st\big)r_{18} - \big(\frac{6229851924052211528114 07}{8514039461120736942781900} + \frac{1393891189946039152278716 7}{8514039461120736942781900}s - \frac{1750454385400721827915807}{8514039461120736942781900}t - \frac{6744832254233988935266811}{1106825129945695802561647 00}st\big)r_{38} - \big(-\frac{4454476172050657974019788265382151 3}{1424192677063854998107466244943084 00} - \frac{1219890708960306654964739213545379 07}{1851450480183011497539706118426009200}s - \frac{3730506347467329219008370676580 81}{1095532828510657690851897111494680 0}t - \frac{1522509084633655948663747452763365 1}{1851450480183011497539706118426009200}st\big)r_{44}\big]$ |
| 47 | 5 | $(1,-1)$ | $r_{47} = \big(\frac{2401745}{7552276} + \frac{34665}{7552276}s - \frac{14000365}{22656828}t + \frac{382255}{22656828}st\big)\big[M_{20}r_5\big]$ |
| 48 | 30 | $(1,3)$ | $r_{48} = \big(\frac{53698568}{3628507} - \frac{485398375}{94341182}s - \frac{257322275}{43542084}t + \frac{987663593}{566047092}st\big)\big[M_{22}r_5 - \big(-\frac{3}{140} + \frac{1}{28}s - \frac{1}{12}t - \frac{13}{1260}st\big)r_{34} - \big(-\frac{327126980077}{958593244140} - \frac{11411961313}{136941892020}s + \frac{71221864103}{958593244140}t + \frac{11444282077}{958593244140}st\big)r_{42}\big]$ |
| 49 | 17 | $(1,5)$ | $r_{49} = \big(-\frac{10405}{6124} - \frac{4070}{19903}s + \frac{1915}{9186}t + \frac{52375}{238836}st\big)\big[M_{24}r_5\big]$ |
| 50 | 31 | $(1,5)$ | $r_{50} = \big(\frac{6749}{102764} - \frac{123}{333983}s + \frac{21049}{51382}t - \frac{47953}{1335932}st\big)\big[M_2r_6 - \big(1\big)r_{49}\big]$ |
| 51 | 80 | $(-1,7)$ | $r_{51} = \big(-\frac{73}{30} + \frac{11}{30}s + \frac{53}{30}t - \frac{13}{30}st\big)\big[M_8r_6\big]$ |
| 52 | 70 | $(-1,-1)$ | $r_{52} = \big(\frac{388030549667728331}{22271875515417394080} - \frac{129113681413938221}{22271875515417394080}s + \frac{2315083558958893541}{66815626546252182240}t - \frac{469843028723052 53}{3516611923486956960}st\big)\big[M_{13}r_7 - \big(-\frac{681021}{30667} + \frac{26163}{2359}s + \frac{390279}{30667}t + \frac{444999}{30667}st\big)r_{25}\big]$ |
| 53 | 55 | $(0,4)$ | $r_{53} = \big(\frac{28559387176836145537 63}{37130506766017266033 61} - \frac{869885210183391401749}{48269658795822445843693}s - \frac{1615029940088566767943}{37130506766017266033 61}t - \frac{30558551510286849519}{48269658795822445843693}st\big)\big[M_{20}r_7 - \big(-\frac{96129}{795320} + \frac{111081}{795320}s - \frac{444047}{795320}t - \frac{15909}{795320}st\big)r_{10} - \big(-\frac{1697344569535}{4239497161664} + \frac{621328814801}{4239497161664}s - \frac{606586736543}{4239497161664}t + \frac{64573387693}{4239497161664}st\big)r_{28} - \big(\frac{482371961144242779}{13112431715690663744} - \frac{6627153740827297623}{65562158578453318720}s + \frac{260608196001062605 41}{65562158578453318720}t - \frac{594393359195812317}{13112431715690663744}st\big)r_{39}\big]$ |
| 54 | 78 | $(-1,3)$ | $r_{54} = \big(-\frac{311181582577}{80156221704} + \frac{2596346689}{6165863208}s + \frac{139619791037}{26718740568}t - \frac{32381011529}{26718740568}st\big)\big[M_{28}r_7 - \big(\frac{866938}{3854943} - \frac{52519}{7709886}s - \frac{354913}{2569962}t - \frac{37808}{1284981}st\big)r_7 - \big(-\frac{346210631}{2309539184} - \frac{200219243}{2309539184}s - \frac{400860707}{2309539184}t - \frac{170502187}{2309539184}st\big)r_{14}\big]$ |
| 55 | 2 | $(1,-3)$ | $r_{55} = \big(1\big)\big[M_2r_8\big]$ |
| 56 | 72 | $(-1,-1)$ | $r_{56} = \big(-\frac{12248741}{10319492} + \frac{20432593}{134153396}s + \frac{94538741}{30958476}t - \frac{218709341}{402460188}st\big)\big[M_8r_8 - \big(1\big)r_{25} - \big(\frac{8513351902931}{25420343196008} - \frac{818982162117}{127101715980040}s - \frac{10872775852471}{127101715980040}t - \frac{935605023501}{25420343196008}st\big)r_{52}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 57 | 39 | $(0,-2)$ | $r_{57}=\big(\frac{3638405746974236563991080 93}{4689461444649093863989818 32}+\frac{7120493971864834365031033 31}{4689461444649093863989818 32}s-\frac{1354002510106949425057938487}{4689461444649093863989818 32}t+\frac{1818425222752730362531416619}{4689461444649093863989818 32}st\big)\big[M_{20}r_8-\big(-\frac{103}{1770}+\frac{41}{708}s-\frac{11}{354}t+\frac{2}{295}st\big)r_5-\big(-\frac{61}{2360}-\frac{13}{3540}s-\frac{29}{3540}t+\frac{7}{7080}st\big)r_{19}-\big(\frac{4965227156456257}{1742584017027457048}+\frac{326340565679533583}{26138760255411855720}s+\frac{90707766371366009}{26138760255411855720}t-\frac{6656689963698799}{5227752051082371144}st\big)r_{20}-\big(-\frac{9677367959286237676278413}{4563690356934730340377775 95}-\frac{5804513799083141272421912 3}{1825476142773892136151110380}s+\frac{3507992445981953285787953 23}{5476428428321676408453331140}t+\frac{1089517557746068814205114 1}{1369107107080419102113332785}st\big)r_{30}-\big(\frac{1547985566563772471843100221358 68}{5729009536559397831336413652235635}-\frac{1771122618647045268642327599631973}{2291603814623759132534565346089425 40}s+\frac{2162619866933287248473967698450503 3}{6874811443871277397603696038266276 20}t+\frac{4912911440139653640168431909837 3}{3437405721935638698801848019134138 10}st\big)r_{40}\big]$ |
| 58 | 54 | $(0,2)$ | $r_{58}=\big(-\frac{3309115692918275918437221}{2203128362767599489197 20}+\frac{1556949911909388897650562 9}{2864066871597879335956360}s-\frac{4707690501423545634661551}{2203128362767599489197 20}t+\frac{2026608408270196000456862 7}{2864066871597879335956360}st\big)\big[M_{22}r_8-\big(-\frac{17}{252}s-\frac{31}{252}t\big)r_3-\big(\frac{388765}{381528}-\frac{55723}{381528}s-\frac{83221}{381528}t+\frac{143}{381528}st\big)r_4-\big(-\frac{13721}{41622}+\frac{45856}{579735}s-\frac{77183}{312165}t+\frac{52249}{1623258}st\big)r_{18}-\big(-\frac{158226017071562793741947}{102168473533448843313382 8}-\frac{457141546032228559649 23}{102168473533448843313382 8}s+\frac{181716553830817311088 25}{306505420600346529940148 4}t-\frac{194171363498241317465 17}{437864886571923614200212}st\big)r_{38}-\big(\frac{135941456771628577031536733868953}{197195909131918384353341480069042 4}-\frac{205487857392811506411405290321173}{256354681871493899659343924089755 12}s+\frac{210078687770481867730873293336167}{197195909131918384353341480069042 4}t+\frac{514071749621245480811691369967673}{256354681871493899659343924089755 12}st\big)r_{44}-\big(-\frac{349642040543008379515069165145533253 5639}{255572844365970462650954498438469912914736}-\frac{499525101035657774759153607888077431288 51}{332244697675761601446240847970010886789 1568}s+\frac{142469299897433867812534564006257485671}{255572844365970462650954498438469912914736}t-\frac{535400999167833458735509228597315699476 1}{332244697675761601446240847970010886789 1568}st\big)r_{46}\big]$ |
| 59 | 60 | $(0,4)$ | $r_{59}=\big(\frac{51756051121}{2791074024}-\frac{1110614328055}{253987736184}s-\frac{9355499125}{25119666216}t+\frac{3766099670551}{2285889625656}st\big)\big[M_{24}r_8-\big(\frac{1093}{9400}s-\frac{461}{9400}t\big)r_{10}-\big(-\frac{87662153}{1066111040}+\frac{7432863889}{50107218880}s+\frac{8170000021}{150321656640}t-\frac{72896929}{3198333120}st\big)r_{28}-\big(\frac{55046373384677013}{387444230396231200}+\frac{442631930830137}{387444230396231200}s+\frac{26862795824643951}{387444230396231200}t-\frac{4217749441458177}{387444230396231200}st\big)r_{39}-\big(-\frac{105445998557751}{13003280651994640}-\frac{18662139247761}{13003280651994640}s-\frac{373663183696503}{13003280651994640}t-\frac{44529984580101}{13003280651994640}st\big)r_{53}\big]$ |
| 60 | 45 | $(0,-2)$ | $r_{60}=\big(\frac{16964687443799914703}{3453231075537716386200}-\frac{14420631419180968733}{6413143425998616145800}s+\frac{4933546017501358399}{3453231075537716386200}t+\frac{2226968092214665367}{6413143425998616145800}st\big)\big[M_{15}r_9-\big(-\frac{79729}{278}-\frac{4033}{278}s-\frac{92521}{278}t-\frac{10461}{278}st\big)r_5-\big(-\frac{194399}{2780}+\frac{170747}{1390}s+\frac{195019}{1390}t-\frac{207771}{2780}st\big)r_{19}-\big(-\frac{345356873331037206883}{4105409802827398808}-\frac{4252370563599715825}{4105409802827398808}s+\frac{14865434430243934752 7}{4105409802827398808}t+\frac{2675009196356138136 9}{4105409802827398808}st\big)r_{20}-\big(\frac{2464534602383135389729343962 71}{8601396062561729048305231960}+\frac{3802638065930156400503558652 87}{8601396062561729048305231960}s-\frac{1803210990497867290089894444001}{8601396062561729048305231960}t-\frac{1744125322985340360316991045 41}{8601396062561729048305231960}st\big)r_{30}-\big(-\frac{1208134020044668325067566324094970125}{3599242149522062366353724102897515 6}+\frac{7549676703388985358138429552581742 63}{1799621074761031183176862051448757 80}s-\frac{2396160688998677523364514729181224 0147}{5398863224283093549530586154346273 40}t-\frac{2317594171488732142665766346544924955}{1079772644856618709906117230869254 68}st\big)r_{40}-\big(\frac{1166818951785228699653052523555771}{4835474789786170050849285751771 0}+\frac{7409811016259063279011550585700 49}{4835474789786170050849285751771 0}s+\frac{3514194380911681753740324901609673}{4835474789786170050849285751771 0}t+\frac{6079469688912615764232590392089 59}{4835474789786170050849285751771 0}st\big)r_{57}\big]$ |
| 61 | 40 | $(0,0)$ | $r_{61}=\big(-\frac{1616361616808454306435662 29}{4882565778879950622447878280}+\frac{5982879165905917148392519}{4882565778879950622447878280}s+\frac{2461799392325791281905932 7}{4882565778879950622447878280}t-\frac{5347709229916385520061783}{1627521926293316874149292760}st\big)\big[M_{13}r_{10}-\big(\frac{16239460205}{1212701308}+\frac{692791205}{303175327}s-\frac{386114390}{303175327}t+\frac{1494686015}{1212701308}st\big)r_{11}-\big(\frac{2182468388216 45}{6508082839512 8}-\frac{2781198650272 45}{6508082839512 8}s+\frac{1086698500224215}{195242485185384}t-\frac{77085334456885}{27891783597912}st\big)r_{12}-\big(-\frac{7023092710897167}{554911199443237}+\frac{115524109015641}{79273028491891}s-\frac{713114830347511}{554911199443237}t-\frac{2482776247695353}{554911199443237}st\big)r_{45}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 62 | 22 | $(0,-4)$ | $r_{62} = \big(-\frac{228500447862849}{155975642793910900} + \frac{12313155391809}{31195128558782180}s + \frac{28873306193791}{9358538567634654 0}t + \frac{42963662681713}{467926928381732700}st\big)\big[M_{15}r_{10} - \big(\frac{1748921896}{3331597} - \frac{622310771}{3331597}s - \frac{482944467}{3331597}t + \frac{191611134}{3331597}st\big)r_{13}\big]$ |
| 63 | 24 | $(0,-4)$ | $r_{63} = \big(-\frac{2342731563098853080 53981}{619097347668743389809006 40} + \frac{1958033845616870117229 33}{309548673834371694904503200}s + \frac{1360184146082014989744767}{928646021503115084713509600}t - \frac{1794910601670945274996 3}{185729204300623016942701920}st\big)\big[M_{17}r_{10} - \big(\frac{1767480975}{6663194} + \frac{1724649195}{6663194}s + \frac{3813125715}{6663194}t + \frac{548322375}{6663194}st\big)r_{13} - \big(-\frac{20512969815}{26652776} - \frac{7988284785}{26652776}s - \frac{4550765745}{26652776}t + \frac{10507365}{26652776}st\big)r_{62}\big]$ |
| 64 | 13 | $(0,-8)$ | $r_{64} = \big(\frac{1691518477}{137912496854400} + \frac{1340321009}{275824993708800}s + \frac{392100043}{30647221523200}t + \frac{34683949}{68956248427200}st\big)\big[M_{19}r_{10}\big]$ |
| 65 | 34 | $(0,-4)$ | $r_{65} = \big(\frac{1944985809949350515231237018029}{4427577239003732474432168918 2524} + \frac{2588026176091930760159651830 25}{1864243048001571568181965860316 8}s + \frac{5164243408316374628604724203 5195}{1062618537360895793863720540380576}t + \frac{9120790241963152794969342768 71}{6641365858505598711648253377378 6}st\big)\big[M_{13}r_{11} - \big(\frac{60458091}{4883606} - \frac{33632985}{4883606}s + \frac{4312807}{4883606}t - \frac{43712729}{4883606}st\big)r_{13} - \big(-\frac{242824509}{19534424} + \frac{106190631}{19534424}s - \frac{11506835}{58603272}t - \frac{64283771}{58603272}st\big)r_{62} - \big(\frac{41888782682986753954347}{380861792501100017352 7} - \frac{10365359030308695000477}{380861792501100017352 7}s - \frac{1523390716617741375143 9}{1142585377503300052058 1}t + \frac{1246744438254648072631 3}{1142585377503300052058 1}st\big)r_{63}\big]$ |
| 66 | 18 | $(0,-8)$ | $r_{66} = \big(\frac{1270343747}{83611103472} - \frac{870777929}{209027758680}s - \frac{24609607}{34837959780}t + \frac{718109}{27870367824}st\big)\big[M_{15}r_{11} - \big(-\frac{5469128}{26833} + \frac{18000992}{134165}s - \frac{74421448}{402495}t - \frac{2908496}{80499}st\big)r_{64}\big]$ |
| 67 | 8 | $(1,1)$ | $r_{67} = \big(\frac{3094675479}{11112498314} - \frac{2578427565}{11112498314}s + \frac{5696922655}{33337494942}t + \frac{7056493643}{33337494942}st\big)\big[M_{20}r_{11}\big]$ |
| 68 | 42 | $(0,0)$ | $r_{68} = \big(\frac{3444723145429207088475658783314 69870226755}{2517315874204666274166754146595603933 1552} + \frac{9744936732181172669659917053331 14219123805}{3272510636466066156416780390574285113 10176}s - \frac{4813043524926201069034575344787552156 86435}{7551947622613998822500262439786811799 4656}t - \frac{2138365789416156873586048112012492820536425}{9817531909398198469250341171722855339305 28}st\big)\big[M_{28}r_{11} - \big(\frac{109979029}{1227770748} + \frac{11744167}{1227770748}s - \frac{73787701}{1227770748}t + \frac{36763661}{1227770748}st\big)r_{11} - \big(\frac{2687629855429}{16472386240542} - \frac{690192785453}{16472386240542}s + \frac{890096487331}{16472386240542}t - \frac{240076340363}{16472386240542}st\big)r_{12} - \big(-\frac{60577731821656}{936344525398995} - \frac{5642424826204}{936344525398995}s + \frac{506838214853756}{2809033576196985}t - \frac{225958191512872}{2809033576196985}st\big)r_{45} - \big(-\frac{215043150040069309}{14282808580879085 90} - \frac{212182152580806 01}{14282808580879085 90}s + \frac{269502176452836767}{12854527722791177310}t + \frac{3166455942641551}{313525066409540910}st\big)r_{61}\big]$ |
| 69 | 36 | $(0,-4)$ | $r_{69} = \big(\frac{26751960987616095673}{428437544321145367080} - \frac{384825154080450684031}{2742000283655330349312 0}s - \frac{495529183118805191213}{2742000283655330349312 0}t + \frac{7412361006694585607}{1713750177284581468320}st\big)\big[M_{13}r_{12} - \big(-\frac{4837677567}{333120242} + \frac{1138610181}{333120242}s + \frac{12122620149}{333120242}t - \frac{854728827}{333120242}st\big)r_{13} - \big(-\frac{29636892527}{1332480968} - \frac{125092979}{1332480968}s - \frac{21397608507}{1332480968}t - \frac{3194177683}{1332480968}st\big)r_{62} - \big(\frac{94860290270031672517 5004}{25979321936806577567 2189} - \frac{93629717433307970602 6190}{25979321936806577567 2189}s - \frac{44045149079865386075900}{19984093797543521205553}t - \frac{56536123675766894222219 4}{25979321936806577567 2189}st\big)r_{63} - \big(-\frac{19578838053534543462200 60416499190}{64448080711553104327836420546109} + \frac{86137188160697523959827 4182525698}{64448080711553104327836420546109}s - \frac{75209087066363521677397 4630648862}{64448080711553104327836420546109}t + \frac{43331261114766127663694720 6577858}{64448080711553104327836420546109}st\big)r_{65}\big]$ |
| 70 | 10 | $(1,1)$ | $r_{70} = \big(-\frac{95595143660968799}{23038529553523056} - \frac{6211399972314235}{23038529553523056}s - \frac{62227185748809055}{23038529553523056}t - \frac{13220875825477591}{23038529553523056}st\big)\big[M_{20}r_{12} - \big(\frac{1827805469}{4319581160} + \frac{30596041}{863916232}s + \frac{199588023}{863916232}t - \frac{154633621}{4319581160}st\big)r_{67}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 71 | 46 | $(0,0)$ | $r_{71} = \big(\frac{10724204984389481736019882720627410480124317863841 5}{3060137762127095403058390682480217228725624251132 8} - \frac{24439073175465714423270284908805680504060924811159}{19890895453826120119879539436121411986716557632363 20}s + \frac{19238971677847757009234181585410983153435678972831 1}{15300688810635477015291953412401086143628121255664 0}t + \frac{24540133123097357397089031438913541800991975419356 5}{39781790907652240239759078872242823973433115264726 4}st\big)\big[M_{28}r_{12} - \big(\frac{4088162255}{27916208412} + \frac{1787130881}{27916208412}s - \frac{1651430561}{9305402804}t + \frac{247163205}{9305402804}st\big)r_{11} - \big(\frac{422057805762487}{561806715239397} + \frac{32068280425840}{561806715239397}s + \frac{63838261587421}{561806715239397}t - \frac{14188736368760}{561806715239397}st\big)r_{12} - \big(-\frac{32652235910863924}{63869877073680465} - \frac{1686005454103948}{63869877073680465}s + \frac{31263890429805964}{63869877073680465}t + \frac{2247216725574004}{63869877073680465}st\big)r_{45} - \big(\frac{109519116848089654787}{292277426817526905390} + \frac{29775954483784213223}{292277426817526905390}s + \frac{134783140069724943 7}{97425808939175635130}t + \frac{34696960699684674 61}{97425808939175635130}st\big)r_{61} - \big(\frac{66257777613896063695242483460569817 1}{28396400805939960313442519931601685 70} - \frac{18876419648131116207294976161035159}{56792801611879920626885039863203371 4}s + \frac{10084076352715367715075468023422104 23}{85189202417819880940327559794805057 10}t + \frac{28186284001918489330455793329338367 41}{85189202417819880940327559794805057 10}st\big)r_{68}\big]$ |
| 72 | 48 | $(0,0)$ | $r_{72} = \big(\frac{34822956149976572808751418}{11015222958979911373872045} - \frac{11015103581767735836032414}{11015222958979911373872045}s + \frac{10270318320332323591708 78}{11015222958979911373872045}t - \frac{11037346080092937743145 22}{11015222958979911373872045}st\big)\big[M_{12}r_{13} - \big(-\frac{185}{1456} - \frac{19}{1092}s + \frac{569}{2184}t + \frac{5}{624}st\big)r_{11} - \big(-\frac{126865679}{586032720} + \frac{7835129}{97672120}s - \frac{19066523}{62789220}t + \frac{79773613}{1758098160}st\big)r_{12} - \big(\frac{82959248}{118971515} + \frac{87753303}{1332480968}s - \frac{258889733}{6662404840}t + \frac{99817759}{2498401815}st\big)r_{45} - \big(-\frac{366111486841}{6097617945328} - \frac{1257688950391}{15244044863320}s - \frac{8221571840431}{45732134589960}t - \frac{1513632587171}{18292853835984}st\big)r_{61} - \big(-\frac{10278141337334611999095790 15}{13671184180863861742854059184} - \frac{11563808929921633765289888}{79341693906799197614778022 05}s + \frac{44963829922190224054996444 1}{42722450565199567946418934 95}t + \frac{30167805937650266661413851 9}{25389342050175743236728967056}st\big)r_{68} - \big(\frac{29705448692976076162873251248604062337596 9}{88106055597163319595836395130846137660432 0} + \frac{39887926681306071756498588664160837062933 3}{11453787227631231547458731367009997895856160}s + \frac{23111672522111805275896686835489086971397 1}{26431816679148995877509185392538412981296 0}t - \frac{10759618760667318947219262373647463441186 61}{34361361682893694642376194101029993687568480}st\big)r_{71}\big]$ |
| 73 | 4 | $(1,-3)$ | $r_{73} = \big(-\frac{25988}{8553} + \frac{1678}{8553}s - \frac{23894}{8553}t - \frac{1204}{2851}st\big)\big[M_{20}r_{13} - \big(\frac{397}{2832} + \frac{47}{1416}s + \frac{5}{1416}t - \frac{119}{2832}st\big)r_{55}\big]$ |
| 74 | 15 | $(1,1)$ | $r_{74} = \big(-\frac{1480552983}{348776440} + \frac{1040424651}{906818744}s - \frac{46558911}{69755288}t + \frac{1767470271}{4534093720}st\big)\big[M_{22}r_{13} - \big(\frac{1}{84} + \frac{19}{168}s + \frac{71}{504}t - \frac{1}{63}st\big)r_{67} - \big(\frac{395094795623}{1050131090673} + \frac{124214829701}{4200524362692}s + \frac{623229116701}{4200524362692}t + \frac{84806819327}{2100262181346}st\big)r_{70}\big]$ |
| 75 | 1 | $(1,-5)$ | $r_{75} = \big(-1\big)\big[M_2r_{16}\big]$ |
| 76 | 7 | $(1,-1)$ | $r_{76} = \big(\frac{14879048129260045342450 13}{75582032466510427085310 0} - \frac{18623726113062461702 51}{75582032466510427085310 0}s + \frac{21025769608886914494749 69}{22674609739953128125593 00}t + \frac{60669218662318538889978 1}{22674609739953128125593 00}st\big)\big[M_{20}r_{19} - \big(\frac{10704046540840383 27}{43564600425686426 20} + \frac{18921501165106471}{87129200851372852 4}s - \frac{453224675099381465}{87129200851372852 4}t - \frac{314092902810866553}{43564600425686426 20}st\big)r_{47}\big]$ |
| 77 | 9 | $(1,-1)$ | $r_{77} = \big(\frac{86170681520058811700020 48789654}{12425136964407636500204 78844215} + \frac{55081253390541555787201834210 77}{16152678053729927450266224974795}s - \frac{39660760693348753972400269387373}{74550821786445819001228730652 90}t + \frac{10285754388228459629531515538712 7}{96916068322379564701597349848770}st\big)\big[M_{20}r_{20} - \big(\frac{22068821382544081992849 0847}{36509522855477842723022 2076} + \frac{11812247454440631195066 1273}{36509522855477842723022 2076}s + \frac{29615301432743804916398817}{36509522855477842723022 2076}t - \frac{98419923939679260705323 33}{36509522855477842723022 2076}st\big)r_{47} - \big(\frac{38159214531550282128437815 3144501}{27572999323287678012885526 59244976} + \frac{27095249499328639599010247728411}{27572999323287678012885526 59244976}s - \frac{15925993633448382350464578443234 63}{82718997969863034038656579777349 28}t + \frac{43793357533733411885597682868533 1}{82718997969863034038656579777349 28}st\big)r_{76}\big]$ |
| 78 | 3 | $(1,-5)$ | $r_{78} = \big(-\frac{652601}{260789} - \frac{13489187}{3390257}s + \frac{9090887}{260789}t + \frac{25444113}{3390257}st\big)\big[M_{20}r_{21} - \big(-\frac{379990669951}{17813359471584} - \frac{123450203585}{8906679735792}s + \frac{67201688859}{2968893245264}t - \frac{12437118671}{17813359471584}st\big)r_{75}\big]$ |
| 79 | 0 | $(1,-7)$ | $r_{79} = \big(-1\big)\big[M_2r_{26}\big]$ |
| 80 | 6 | $(1,-3)$ | $r_{80} = \big(\frac{9737}{31896} + \frac{12175}{414648}s + \frac{21835}{31896}t - \frac{4471}{414648}st\big)\big[M_{24}r_{33} - \big(1\big)r_{55} - \big(-\frac{373}{1880} + \frac{253}{1880}s + \frac{319}{1880}t + \frac{77}{1880}st\big)r_{73}\big]$ |

### S25.4.4 Glue and sector signs (-1,-1)

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 0 | $(-4,-5,-6,-4,-2,-3,-1,-2)$ | 1/4 |
| 1 | $(-4,-5,-6,-4,-2,-3,-1,1)$ | 1/2 |
| 2 | $(-4,-5,-6,-4,-2,-3,2,1)$ | 1 |
| 3 | $(-1,-5,-6,-4,-2,-3,-1,-2)$ | 5/4 |
| 4 | $(-1,-5,-6,-4,-2,-3,-1,1)$ | 5/2 |
| 5 | $(-1,-5,-6,-4,-2,-3,2,1)$ | 5 |
| 6 | $(-1,-2,-6,-4,-2,-3,-1,-2)$ | 10 |
| 7 | $(-1,-2,-6,-4,-2,-3,-1,1)$ | 20 |
| 8 | $(-1,-2,-6,-4,-2,-3,2,1)$ | 40 |
| 9 | $(-1,-2,-3,-4,-2,-3,-1,-2)$ | 90 |
| 10 | $(-1,-2,-3,-4,-2,-3,-1,1)$ | 180 |
| 11 | $(-1,-2,-3,-4,-2,-3,2,1)$ | 360 |
| 12 | $(-1,-2,-3,-4,-2,0,-1,-2)$ | 90 |
| 13 | $(-1,-2,-3,-4,-2,0,-1,1)$ | 180 |
| 14 | $(-1,-2,-3,-4,-2,0,2,1)$ | 360 |
| 15 | $(-1,-2,-3,-1,-2,-3,-1,-2)$ | 720 |
| 16 | $(-1,-2,-3,-1,-2,-3,-1,1)$ | 1440 |
| 17 | $(-1,-2,-3,-1,-2,-3,2,1)$ | 2880 |
| 18 | $(-1,-2,-3,-1,-2,0,-1,-2)$ | 720 |
| 19 | $(-1,-2,-3,-1,-2,0,-1,1)$ | 1440 |
| 20 | $(-1,-2,-3,-1,-2,0,2,1)$ | 2880 |
| 21 | $(-1,-2,-3,-1,1,-3,-1,-2)$ | 3600 |
| 22 | $(-1,-2,-3,-1,1,-3,-1,1)$ | 7200 |
| 23 | $(-1,-2,-3,-1,1,-3,2,1)$ | 14400 |
| 24 | $(-1,-2,-3,-1,1,0,-1,-2)$ | 3600 |
| 25 | $(-1,-2,-3,-1,1,0,-1,1)$ | 7200 |
| 26 | $(-1,-2,-3,-1,1,0,2,1)$ | 14400 |
| 27 | $(-1,-2,0,-1,-2,0,-1,-2)$ | 6480 |
| 28 | $(-1,-2,0,-1,-2,0,-1,1)$ | 12960 |
| 29 | $(-1,-2,0,-1,-2,0,2,1)$ | 25920 |
| 30 | $(-1,-2,0,-1,1,0,-1,-2)$ | 32400 |
| 31 | $(-1,-2,0,-1,1,0,-1,1)$ | 64800 |
| 32 | $(-1,-2,0,-1,1,0,2,1)$ | 129600 |
| 33 | $(-1,-2,0,2,1,0,-1,-2)$ | 259200 |
| 34 | $(-1,-2,0,2,1,0,-1,1)$ | 518400 |
| 35 | $(-1,-2,0,2,1,0,2,1)$ | 1036800 |
| 36 | $(-1,1,0,-1,-2,0,-1,-2)$ | 51840 |
| 37 | $(-1,1,0,-1,-2,0,-1,1)$ | 103680 |
| 38 | $(-1,1,0,-1,-2,0,2,1)$ | 207360 |
| 39 | $(-1,1,0,-1,1,0,-1,-2)$ | 259200 |
| 40 | $(-1,1,0,-1,1,0,-1,1)$ | 518400 |
| 41 | $(-1,1,0,-1,1,0,2,1)$ | 1036800 |
| 42 | $(-1,1,0,2,1,0,-1,-2)$ | 2073600 |
| 43 | $(-1,1,0,2,1,0,-1,1)$ | 4147200 |
| 44 | $(-1,1,0,2,1,0,2,1)$ | 8294400 |
| 45 | $(-1,1,3,2,1,0,-1,-2)$ | 18662400 |
| 46 | $(-1,1,3,2,1,0,-1,1)$ | 37324800 |
| 47 | $(-1,1,3,2,1,0,2,1)$ | 74649600 |
| 48 | $(-1,1,3,2,1,3,-1,-2)$ | 18662400 |
| 49 | $(-1,1,3,2,1,3,-1,1)$ | 37324800 |
| 50 | $(-1,1,3,2,1,3,2,1)$ | 74649600 |
| 51 | $(2,1,0,-1,-2,0,-1,-2)$ | 259200 |
| 52 | $(2,1,0,-1,-2,0,-1,1)$ | 518400 |
| 53 | $(2,1,0,-1,-2,0,2,1)$ | 1036800 |
| 54 | $(2,1,0,-1,1,0,-1,-2)$ | 1296000 |
| 55 | $(2,1,0,-1,1,0,-1,1)$ | 2592000 |
| 56 | $(2,1,0,-1,1,0,2,1)$ | 5184000 |
| 57 | $(2,1,0,2,1,0,-1,-2)$ | 10368000 |

| $j$ | $3\beta_j$ | Squared norm |
|---|---|---|
| 58 | $(2,1,0,2,1,0,-1,1)$ | 20736000 |
| 59 | $(2,1,0,2,1,0,2,1)$ | 41472000 |
| 60 | $(2,1,3,2,1,0,-1,-2)$ | 93312000 |
| 61 | $(2,1,3,2,1,0,-1,1)$ | 186624000 |
| 62 | $(2,1,3,2,1,0,2,1)$ | 373248000 |
| 63 | $(2,1,3,2,1,3,-1,-2)$ | 93312000 |
| 64 | $(2,1,3,2,1,3,-1,1)$ | 186624000 |
| 65 | $(2,1,3,2,1,3,2,1)$ | 373248000 |
| 66 | $(2,4,3,2,1,0,-1,-2)$ | 746496000 |
| 67 | $(2,4,3,2,1,0,-1,1)$ | 1492992000 |
| 68 | $(2,4,3,2,1,0,2,1)$ | 2985984000 |
| 69 | $(2,4,3,2,1,3,-1,-2)$ | 746496000 |
| 70 | $(2,4,3,2,1,3,-1,1)$ | 1492992000 |
| 71 | $(2,4,3,2,1,3,2,1)$ | 2985984000 |
| 72 | $(2,4,6,2,1,3,-1,-2)$ | 6718464000 |
| 73 | $(2,4,6,2,1,3,-1,1)$ | 13436928000 |
| 74 | $(2,4,6,2,1,3,2,1)$ | 26873856000 |
| 75 | $(2,4,6,5,1,3,-1,-2)$ | 53747712000 |
| 76 | $(2,4,6,5,1,3,-1,1)$ | 107495424000 |
| 77 | $(2,4,6,5,1,3,2,1)$ | 214990848000 |
| 78 | $(2,4,6,5,4,3,-1,-2)$ | 268738560000 |
| 79 | $(2,4,6,5,4,3,-1,1)$ | 537477120000 |
| 80 | $(2,4,6,5,4,3,2,1)$ | 1074954240000 |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 0 | 68 | $(0,10)$ | $r_0 = e_{68}$ |
| 1 | 62 | $(0,8)$ | $r_1 = \left(\frac{1}{8}\right)\left[M_9 r_0\right]$ |
| 2 | 47 | $(0,6)$ | $r_2 = \left(\frac{3473131389}{70339709920} - \frac{698759789}{70339709920}s - \frac{1318806853}{70339709920}t - \frac{265143879}{70339709920}st\right)\left[M_{13} r_0\right]$ |
| 3 | 35 | $(0,2)$ | $r_3 = \left(-\frac{9}{16480}s - \frac{31}{49440}t\right)\left[M_{15} r_0\right]$ |
| 4 | 41 | $(0,2)$ | $r_4 = \left(-\frac{46193}{39785280} - \frac{39191}{258604320}s - \frac{4311}{6630880}t - \frac{7655}{103441728}st\right)\left[M_{17} r_0 - \left(120s - 240t\right) r_3\right]$ |
| 5 | 26 | $(0,-2)$ | $r_5 = \left(\frac{1}{172800} + \frac{1}{499200}s + \frac{1}{345600}t - \frac{1}{748800}st\right)\left[M_{19} r_0\right]$ |
| 6 | 77 | $(-1,5)$ | $r_6 = \left(\frac{2454543}{14491} + \frac{1264977}{14491}s + \frac{595161}{14491}t - \frac{166509}{14491}st\right)\left[M_{23} r_0\right]$ |
| 7 | 74 | $(-1,3)$ | $r_7 = \left(\frac{7893}{1348} + \frac{531}{1348}s - \frac{65907}{1348}t - \frac{1017}{1348}st\right)\left[M_{25} r_0\right]$ |
| 8 | 50 | $(-1,-3)$ | $r_8 = \left(-\frac{1}{80}s + \frac{3}{80}t\right)\left[M_{27} r_0\right]$ |
| 9 | 59 | $(0,6)$ | $r_9 = \left(\frac{20521}{253536} - \frac{75}{84512}s + \frac{5065}{2281824}t - \frac{2053}{2281824}st\right)\left[M_9 r_1 - \left(-5\right) r_2\right]$ |
| 10 | 44 | $(0,4)$ | $r_{10} = \left(\frac{943224373}{14753895480} - \frac{155310883}{14753895480}s - \frac{504831691}{14753895480}t - \frac{62325263}{14753895480}st\right)\left[M_{13} r_1\right]$ |
| 11 | 32 | $(0,0)$ | $r_{11} = \left(-\frac{13}{3840} + \frac{1}{3840}s - \frac{1}{3840}t + \frac{1}{3840}st\right)\left[M_{15} r_1\right]$ |
| 12 | 34 | $(0,0)$ | $r_{12} = \left(\frac{271585}{295936176} - \frac{6997}{493226960}s + \frac{59297}{1479680880}t - \frac{56123}{295936176}st\right)\left[M_{17} r_1 - \left(-150 + 90s + 330t - 30st\right) r_{11}\right]$ |
| 13 | 20 | $(0,-4)$ | $r_{13} = \left(-\frac{1}{43200} - \frac{1}{86400}t\right)\left[M_{19} r_1\right]$ |
| 14 | 76 | $(-1,3)$ | $r_{14} = \left(\frac{15101856}{248167} + \frac{12445272}{248167}s - \frac{18897480}{248167}t - \frac{2862720}{248167}st\right)\left[M_{23} r_1 - \left(-\frac{13}{18144} + \frac{1}{1296}s - \frac{1}{1008}t + \frac{5}{18144}st\right) r_7\right]$ |
| 15 | 71 | $(-1,1)$ | $r_{15} = \left(\frac{1754}{337} + \frac{118}{337}s - \frac{14646}{337}t - \frac{226}{337}st\right)\left[M_{25} r_1\right]$ |
| 16 | 49 | $(-1,-5)$ | $r_{16} = \left(\frac{1}{20}s - \frac{3}{20}t\right)\left[M_{27} r_1\right]$ |
| 17 | 67 | $(0,8)$ | $r_{17} = \left(-\frac{43}{19} - \frac{129}{247}s + \frac{203}{57}t - \frac{32}{741}st\right)\left[M_8 r_2 - \left(-\frac{378503722}{2198115935} + \frac{15512740}{439623187}s - \frac{33222412}{439623187}t - \frac{13268878}{2198115935}st\right) r_1\right]$ |
| 18 | 43 | $(0,2)$ | $r_{18} = \left(-\frac{20712759615053552237 6551}{28404050539504756879439 28} + \frac{19110716995545449579305}{28404050539504756879439 28}s + \frac{38858591700371968502159}{28404050539504756879439 28}t - \frac{18037210995651011473673}{28404050539504756879439 28}st\right)\left[M_{13} r_2 - \left(\frac{38057456100}{5715101431} + \frac{129291690504}{40005710017}s - \frac{68785628712}{40005710017}t + \frac{168336514260}{40005710017}st\right) r_3 - \left(-\frac{2980377115491}{4120588131751} - \frac{6713712060708}{4120588131751}s + \frac{5758478197496}{4120588131751}t - \frac{1110908377399}{4120588131751}st\right) r_4\right]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 19 | 29 | $(0,-2)$ | $r_{19}=\big(-\frac{10140104961311}{33684696982742400}-\frac{8138073397291}{33684696982742400}s-\frac{13074858081493}{33684696982742400}t+\frac{1558482375619}{33684696982742400}st\big)\big[M_{15}r_2-\big(-\frac{111687692064}{439623187}-\frac{131794490184}{439623187}s-\frac{282729060168}{439623187}t-\frac{8653271136}{439623187}st\big)r_5\big]$ |
| 20 | 31 | $(0,-2)$ | $r_{20}=\big(\frac{196149239733268031143}{13074759959745517240704 0}-\frac{259500035381923618 9}{435825331991517241356 80}s+\frac{222042147587696687537}{39224279879236551722112 0}t-\frac{224198152169944618021}{39224279879236551722112 0}st\big)\big[M_{17}r_2-\big(\frac{9878124888}{439623187}+\frac{45951277572}{439623187}s+\frac{14648738724}{439623187}t-\frac{10248657768}{439623187}st\big)r_5-\big(\frac{201423935904}{439623187}+\frac{1308737003748}{5715101431}s+\frac{85181228292}{439623187}t+\frac{19202506608}{5715101431}st\big)r_{19}\big]$ |
| 21 | 14 | $(0,-6)$ | $r_{21}=\big(-\frac{5174370689}{1449222465484800}+\frac{780061967}{434766739645440}s-\frac{166400329}{86953347929088}t-\frac{4617859237}{4347667396454400}st\big)\big[M_{19}r_2\big]$ |
| 22 | 23 | $(1,7)$ | $r_{22}=\big(\frac{63475}{89433}-\frac{3671}{89433}s+\frac{14409}{19874}t+\frac{5185}{59622}st\big)\big[M_{20}r_2\big]$ |
| 23 | 79 | $(-1,5)$ | $r_{23}=\big(\frac{251529360}{43909}-\frac{61824480}{30043}s+\frac{955351680}{43909}t+\frac{190395120}{30043}st\big)\big[M_{21}r_2-\big(-\frac{37763765989}{4482046316102400}+\frac{5185443121}{896409263220480}s+\frac{7823230069}{298803087740160}t+\frac{768017069}{4482046316102400}st\big)r_6\big]$ |
| 24 | 73 | $(-1,1)$ | $r_{24}=\big(\frac{6369247059029449056}{46880232503780887}-\frac{14919592811669761824}{609443022549151531}s+\frac{19595357570191373 46}{46880232503780887}t-\frac{69849855799723524 42}{60944302254915153 1}st\big)\big[M_{23}r_2-\big(\frac{63805713179}{13294205174880}+\frac{391335178}{415443911715}s-\frac{2469810001}{1661775646860}t-\frac{7736986829}{13294205174880}st\big)r_{15}\big]$ |
| 25 | 65 | $(-1,-1)$ | $r_{25}=\big(-\frac{18086213842463}{25420343196008}+\frac{59939370422549}{25420343196008}s+\frac{123545028360627}{25420343196008}t+\frac{1413016415427}{25420343196008}st\big)\big[M_{25}r_2\big]$ |
| 26 | 48 | $(-1,-7)$ | $r_{26}=\big(\frac{2039}{18308}-\frac{99}{9154}s-\frac{7853}{27462}t-\frac{1199}{13731}st\big)\big[M_{27}r_2\big]$ |
| 27 | 61 | $(0,6)$ | $r_{27}=\big(-\frac{24214346263634355}{5942915146210358}-\frac{4311074948998119}{11885830292420716}s+\frac{23799313525783393}{5630130138515076}t-\frac{16518850550749315}{26743118157946611}st\big)\big[M_{28}r_2-\big(\frac{1106418520303}{5028849636093}+\frac{229709077651}{5028849636093}s-\frac{104033866426}{1676283212031}t-\frac{17976721579}{1676283212031}st\big)r_2-\big(-\frac{29835473229607900947}{491288645325109975198}-\frac{99335892632804820 57}{24564432266255498750 90}s-\frac{89558515443140491889}{24564432266255498759 90}t+\frac{46611771435765668 89}{49128864532510997519 8}st\big)r_9\big]$ |
| 28 | 46 | $(0,4)$ | $r_{28}=\big(-\frac{67131326339253}{38966568357038}+\frac{7136881191369}{38966568357038}s-\frac{24625752425565}{38966568357038}t+\frac{13521427045353}{38966568357038}st\big)\big[M_8r_3-\big(\frac{39}{206}+\frac{19}{412}s-\frac{38}{309}t+\frac{55}{1236}st\big)r_{10}\big]$ |
| 29 | 66 | $(0,6)$ | $r_{29}=\big(\frac{3078578527947}{285793310371}+\frac{1105339046943}{285793310371}s-\frac{1383288507263}{857379931113}t-\frac{457870976335}{857379931113}st\big)\big[M_{12}r_3-\big(\frac{140797}{224952}+\frac{1285}{4326}s+\frac{349}{25956}t-\frac{124477}{674856}st\big)r_2-\big(-\frac{13715279983855}{32964705054008}+\frac{10519215959869}{164823525270040}s-\frac{67156539437051}{494470575810120}t-\frac{295934993509}{14127730737432}st\big)r_9-\big(-\frac{39879397}{104227760}+\frac{29513}{501095}s-\frac{291062}{6514235}t+\frac{4310771}{312683280}st\big)r_{27}\big]$ |
| 30 | 33 | $(0,-2)$ | $r_{30}=\big(-\frac{6330097617036113165644213633}{41625396013647876077063351 6928}+\frac{2902138817150841636193049497}{41625396013647876077063351 6928}s+\frac{2479924098268481800789750 5239}{41625396013647876077063351 6928}t-\frac{1290911393714531438218340699}{41625396013647876077063351 6928}st\big)\big[M_{13}r_3-\big(-\frac{416937}{9373}-\frac{10776}{721}s+\frac{144290}{9373}t+\frac{145115}{9373}st\big)r_5-\big(\frac{68358}{9373}+\frac{24504}{17407}s+\frac{9472}{2163}t+\frac{247678}{365547}st\big)r_{19}-\big(\frac{73771373613980057}{10962731417334879}-\frac{22028527244626979}{10962731417334879}s+\frac{6323471252486065}{3654243805778293}t+\frac{3591380249711699}{10962731417334879}st\big)r_{20}\big]$ |
| 31 | 19 | $(0,-6)$ | $r_{31}=\big(-\frac{4636746580984083569}{45626150717837331656400}-\frac{16802526283947630199}{45626150717837331656400}s+\frac{73757330438982632473}{45626150717837331656400}t+\frac{12287236864918049651}{45626150717837331656400}st\big)\big[M_{15}r_3-\big(-\frac{74620}{103}+\frac{12056}{103}s-\frac{227464}{309}t-\frac{59140}{309}st\big)r_{21}\big]$ |
| 32 | 24 | $(0,-6)$ | $r_{32}=\big(-\frac{41747273732730210779161}{26237640196632388582159008}+\frac{21225145645757910823897 73}{68217864511244210313613420 80}s-\frac{15182763372626958159182 93}{47227752353938299447886214 40}t+\frac{52064568380819771890537}{76745097575149736602815098 4}st\big)\big[M_{17}r_3-\big(\frac{39150}{103}+\frac{16488}{103}s+\frac{22776}{103}t+\frac{14550}{103}st\big)r_{21}-\big(\frac{65775603472647}{647872890386}+\frac{43193459361345}{647872890386}s-\frac{64925050820035}{647872890386}t-\frac{8959173154837}{647872890386}st\big)r_{31}\big]$ |
| 33 | 12 | $(0,-10)$ | $r_{33}=\big(\frac{1129}{26496000}-\frac{169}{26496000}s-\frac{3721}{79488000}t+\frac{877}{79488000}st\big)\big[M_{19}r_3\big]$ |
| 34 | 11 | $(1,3)$ | $r_{34}=\big(\frac{238755645}{1521576578}-\frac{1212751305}{1521576578}s+\frac{242977717}{1521576578}t-\frac{675632581}{1521576578}st\big)\big[M_{20}r_3\big]$ |
| 35 | 75 | $(-1,1)$ | $r_{35}=\big(-\frac{689454826961328}{73443567401}-\frac{1060743755903760}{954766376213}s-\frac{1273707313409520}{73443567401}t-\frac{1119104654503824}{954766376213}st\big)\big[M_{21}r_3-\big(\frac{85213}{4200422400}+\frac{6463}{4200422400}s+\frac{537353}{12601267200}t+\frac{326831}{12601267200}st\big)r_{15}-\big(-\frac{256645349}{1592485142400}-\frac{32769563}{1592485142400}s+\frac{234226097}{4777455427200}t+\frac{233189087}{4777455427200}st\big)r_{24}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 36 | 64 | $(-1,-3)$ | $r_{36} = \big(-\frac{254809611}{713659} + \frac{544104351}{9277567}s - \frac{14064309}{713659}t - \frac{213342003}{9277567}st\big)\big[M_{23}r_3 - \big(\frac{2353}{622944} + \frac{187}{155736}s - \frac{61}{400464}t - \frac{3623}{5606496}st\big)r_8\big]$ |
| 37 | 63 | $(-1,-5)$ | $r_{37} = \big(-\frac{1839}{622} - \frac{34035}{8086}s - \frac{138565}{18038}t + \frac{300787}{234494}st\big)\big[M_{25}r_3 - \big(\frac{41069}{1549120} + \frac{1349}{4647360}s + \frac{262879}{13942080}t + \frac{58009}{13942080}st\big)r_{16}\big]$ |
| 38 | 45 | $(0,2)$ | $r_{38} = \big(-\frac{125767343062479613071858806208069245 5}{977027675125732673190458325675885 07} - \frac{123203732499687673472146575864208838 29}{254027195532690495029519164675730118 2}s - \frac{372953962648222233750125218523393 72}{977027675125732673190458325675885 07}t - \frac{549230558200138466543663961497924906 3}{254027195532690495029519164675730118 2}st\big)\big[M_{28}r_3 - \big(\frac{350315}{2356434} + \frac{30497}{2356434}s + \frac{52403}{2356434}t - \frac{61747}{2356434}st\big)r_3 - \big(\frac{74909}{11839644} - \frac{9810005}{485425404}s + \frac{6620029}{161808468}t + \frac{2899927}{161808468}st\big)r_4 - \big(-\frac{1546740679}{65105129508} - \frac{22807227835}{846366683604}s + \frac{2355128599}{195315388524}t + \frac{28110000619}{2539100050812}st\big)r_{18}\big]$ |
| 39 | 56 | $(0,4)$ | $r_{39} = \big(-\frac{1171580059717531}{18216598725207184} - \frac{904289519171813}{4554149681301796}s - \frac{5727025120224041}{6831224521952694}t + \frac{11678716085757947}{54649796175621552}st\big)\big[M_8r_4 - \big(-\frac{46477}{331544} + \frac{576233}{4310072}s + \frac{112691}{331544}t + \frac{349437}{4310072}st\big)r_{10} - \big(-\frac{22121093292803}{40763046025176} - \frac{39417569166485}{529919598327288}s + \frac{70849381265}{40763046025176}t - \frac{8847516124679}{176639866109096}st\big)r_{28}\big]$ |
| 40 | 37 | $(0,-2)$ | $r_{40} = \big(-\frac{25135574838237471066751157030893}{31271315853921397009391203053216 0} + \frac{10546654206359504054226694632707}{31271315853921397009391203053216 0}s - \frac{26397493279457796848296170225727}{93813947561764191028173609159648 0}t + \frac{35683553268487397077930733507 9}{49375761874612732120091373241920}st\big)\big[M_{13}r_4 - \big(-\frac{192813888}{3771313} - \frac{57220077}{3771313}s + \frac{139274085}{3771313}t + \frac{32642760}{3771313}st\big)r_5 - \big(\frac{41617270}{3771313} + \frac{106335064}{49027069}s - \frac{32194424}{3771313}t - \frac{128004790}{49027069}st\big)r_{19} - \big(\frac{84100288945801465019}{4410956098336013499} + \frac{126430993448478029611}{57342429278368175487}s - \frac{4229827643177619943}{4410956098336013499}t + \frac{77837834804451029605}{57342429278368175487}st\big)r_{20} - \big(-\frac{48242801334456213884769956}{18344126565501393549699079} - \frac{12303316070287687389645578 0}{16693155174606268130226161 89}s - \frac{44596398500528888611941429 2}{55032379696504180649097237}t - \frac{10011864179374784371704344996}{5007946552381880439067848567}st\big)r_{30}\big]$ |
| 41 | 27 | $(0,-6)$ | $r_{41} = \big(-\frac{41134269857819}{76799135850361344} + \frac{604145547358231}{1919978396259033600}s + \frac{1548809812077589}{57599351887771008 00}t + \frac{60382305424151}{11519870377554201 60}st\big)\big[M_{17}r_4 - \big(-\frac{44902140}{41443} - \frac{44575068}{538759}s - \frac{14738772}{41443}t - \frac{17844180}{538759}st\big)r_{21} - \big(-\frac{11933858680986261}{260677632973466} - \frac{255478913489170803}{3388809228655058}s - \frac{80831877946264137}{260677632973466}t - \frac{899181738841619347}{3388809228655058}st\big)r_{31} - \big(-\frac{16287249482253907483279281 0}{52524571227759237106560 7} + \frac{17842118922096130322694179 4}{68281942596087008238528 91}s + \frac{65150248833309854171143386}{52524571227759237106560 7}t + \frac{86814062238268914087682287 0}{68281942596087008238528 91}st\big)r_{32}\big]$ |
| 42 | 16 | $(1,3)$ | $r_{42} = \big(\frac{44532263102155}{24231469368686} - \frac{174108581307221}{157504550896459}s + \frac{2534027316136}{36347204053029}t - \frac{333362692150733}{945027305378754}st\big)\big[M_{20}r_4 - \big(-\frac{19846373}{58683288} - \frac{32010593}{762882744}s - \frac{351981}{19561096}t + \frac{6809785}{762882744}st\big)r_{34}\big]$ |
| 43 | 69 | $(-1,-3)$ | $r_{43} = \big(\frac{60680871657}{470270293} + \frac{5705477379}{470270293}s - \frac{6260376843}{470270293}t + \frac{14649870627}{470270293}st\big)\big[M_{23}r_4 - \big(\frac{7050929}{375970896} + \frac{1236775}{751941792}s + \frac{1832531}{250647264}t + \frac{354943}{375970896}st\big)r_8 - \big(-\frac{261661}{46996362} - \frac{731153}{375970896}s - \frac{174383}{375970896}t + \frac{2017}{15665454}st\big)r_{36}\big]$ |
| 44 | 53 | $(0,2)$ | $r_{44} = \big(-\frac{98371666886128127613219657484976903923161 47237}{57470952722958769634438309266854873479050728160} + \frac{17131663250729636034142002930002888979700908321}{57470952722958769634438309266854873479050728160}s + \frac{16649270769149494647567843490280766865946861 0797}{51723857450662892670994478340169386131145655344 0}t - \frac{74699145913342645934667042756411932987836739293}{51723857450662892670994478340169386131145655344 0}st\big)\big[M_{28}r_4 - \big(-\frac{128331235}{316044318} - \frac{61011391}{316044318}s - \frac{22367113}{316044318}t - \frac{22514569}{316044318}st\big)r_3 - \big(\frac{52628519159}{97657694262} + \frac{2145564239}{97657694262}s + \frac{865124935}{3150248202}t + \frac{9037973029}{97657694262}st\big)r_4 - \big(-\frac{16461175816493}{26195649341748} - \frac{51729012551063}{340543441442724}s + \frac{3564454692883}{26195649341748}t + \frac{18874875153517}{340543441442724}st\big)r_{18} - \big(-\frac{406658770392946898464480687105}{74807823176627607880984962250092} + \frac{25919752355984424592094235229379}{97250170129615890245280450925119 6}s + \frac{83806296122453582194338257536 1}{83119803529586230978872180277 88}t + \frac{45366001384095478939616084715 21}{10805574458846210027253383436124 4}st\big)r_{38}\big]$ |
| 45 | 38 | $(0,0)$ | $r_{45} = \big(\frac{1788144657}{1616775620} + \frac{21209567}{1616775620}s + \frac{181321361}{1616775620}t - \frac{107995933}{1616775620}st\big)\big[M_8r_5 - \big(\frac{2}{3} - \frac{2}{39}st\big)r_{11} - \big(\frac{31}{96} + \frac{853}{6240}s - \frac{113}{480}t - \frac{29}{416}st\big)r_{12}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 46 | 55 | $(0,2)$ | $r_{46} = \big(-\frac{32349934043983152447515868412417312229157742878 35}{14373374567515130026822086953822518844074937021 04} - \frac{47899944485913070478523162070829024750222353772 5}{14373374567515130026822086953822518844074937021 04}s + \frac{86664452462004273490627604983220741880709461072 97}{14373374567515130026822086953822518844074937021 04}t - \frac{45953206005460329460923969159300395226229868986 9}{14373374567515130026822086953822518844074937021 04}st\big)\big[M_{12}r_5 - \big(-\frac{127}{1092} - \frac{19}{507}s + \frac{19}{273}t - \frac{363}{4732}st\big)r_3 - \big(\frac{27959}{74984} - \frac{543}{10712}s - \frac{110965}{674856}t + \frac{294815}{8773128}st\big)r_4 - \big(\frac{60371525}{294162414} + \frac{935381}{22627878}s + \frac{72551}{49027069}t + \frac{222302}{147081207}st\big)r_{18} - \big(\frac{14751537221028529151655499}{25847685990949328760289 7448} + \frac{40421831463101290403268415}{33601991788234127388376668 24}s - \frac{53396341952889535457021 51}{25847685990949328760289 7448}t - \frac{22137701587936251809028727}{33601991788234127388376668 24}st\big)r_{38} - \big(-\frac{25128320163288650308808065880129226513}{17781903687288334652066341527301108274 0} - \frac{18427983483120705261225724849716406608 7}{23116474793474835047686243985491440756 20}s - \frac{32791849489040499793629610116947893 39}{25402719553269049502951916467573011820}t - \frac{42184552533550661679331904689093418615 3}{69349424380424505143058731956474322268 60}st\big)r_{44}\big]$ |
| 47 | 5 | $(1,-1)$ | $r_{47} = \big(-\frac{1400685}{7552276} + \frac{4845387}{7552276}s + \frac{461149}{7552276}t - \frac{847279}{7552276}st\big)\big[M_{20}r_5\big]$ |
| 48 | 21 | $(1,3)$ | $r_{48} = \big(\frac{49019901}{7257014} - \frac{13684566}{3628507}s + \frac{366798803}{14514028}t - \frac{91057169}{14514028}st\big)\big[M_{22}r_5 - \big(\frac{1}{36} + \frac{1}{1092}s - \frac{13}{252}t - \frac{23}{9828}st\big)r_{34} - \big(-\frac{51692257523}{575155946484} - \frac{9402419921}{575155946484}s - \frac{46197864497}{575155946484}t - \frac{7647901211}{356048919252}st\big)r_{42}\big]$ |
| 49 | 17 | $(1,5)$ | $r_{49} = \big(-\frac{8901}{24496} + \frac{1755}{24496}s + \frac{24851}{24496}t + \frac{18463}{24496}st\big)\big[M_{24}r_5\big]$ |
| 50 | 22 | $(1,5)$ | $r_{50} = \big(\frac{6749}{102764} - \frac{123}{333983}s - \frac{21049}{51382}t + \frac{47953}{1335932}st\big)\big[M_2r_6 - \big(-1\big)r_{49}\big]$ |
| 51 | 80 | $(-1,7)$ | $r_{51} = \big(\frac{73}{30} - \frac{11}{30}s + \frac{53}{30}t - \frac{13}{30}st\big)\big[M_8r_6\big]$ |
| 52 | 70 | $(-1,-1)$ | $r_{52} = \big(\frac{388030549667728331}{22271875515417394080} - \frac{129113681413938221}{22271875515417394080}s - \frac{2315083558958893541}{66815626546252182240}t + \frac{46984302872305253}{3516611923486956960}st\big)\big[M_{13}r_7 - \big(\frac{681021}{30667} - \frac{26163}{2359}s + \frac{390279}{30667}t + \frac{444999}{30667}st\big)r_{25}\big]$ |
| 53 | 58 | $(0,4)$ | $r_{53} = \big(-\frac{12283067792669915207443}{33417456089415539430249} - \frac{20274717002064750836279}{43442692916240201259323 7}s + \frac{16376225028903185593 03}{11139152029805179810083}t + \frac{55558579502104188687 23}{14480897638746733753107 9}st\big)\big[M_{20}r_7 - \big(\frac{1664917}{6362560} - \frac{17783}{6362560}s - \frac{1878609}{6362560}t - \frac{220793}{6362560}st\big)r_{10} - \big(\frac{287232536274191}{782271210210240} + \frac{16978295866021}{260757070070080}s - \frac{73062801261617}{260757070070080}t + \frac{7656830793587}{260757070070080}st\big)r_{28} - \big(-\frac{26621202788470159007}{309908911457194621 60} + \frac{38149820526596711 23}{309908911457194621 60}s + \frac{34130712708936975279}{309908911457194621 60}t - \frac{200806202307537467}{309908911457194621 60}st\big)r_{39}\big]$ |
| 54 | 78 | $(-1,3)$ | $r_{54} = \big(\frac{311181582577}{80156221704} - \frac{2596346689}{6165863208}s + \frac{139619791037}{26718740568}t - \frac{32381011529}{26718740568}st\big)\big[M_{28}r_7 - \big(\frac{866938}{3854943} - \frac{52519}{7709886}s + \frac{354913}{2569962}t + \frac{37808}{1284981}st\big)r_7 - \big(-\frac{346210631}{2309539184} - \frac{200219243}{2309539184}s + \frac{400860707}{2309539184}t + \frac{170502187}{2309539184}st\big)r_{14}\big]$ |
| 55 | 2 | $(1,-3)$ | $r_{55} = \big(-1\big)\big[M_2r_8\big]$ |
| 56 | 72 | $(-1,-1)$ | $r_{56} = \big(-\frac{12248741}{10319492} + \frac{20432593}{134153396}s - \frac{94538741}{30958476}t + \frac{218709341}{402460188}st\big)\big[M_8r_8 - \big(-1\big)r_{25} - \big(\frac{8513351902931}{25420343196008} - \frac{818982162117}{127101715980040}s + \frac{10872775852471}{127101715980040}t + \frac{935605023501}{25420343196008}st\big)r_{52}\big]$ |
| 57 | 39 | $(0,-2)$ | $r_{57} = \big(\frac{24804874418342525506555394910 61}{63505020262695174050778830 6345} + \frac{38568010736790032040089827871 3}{63505020262695174050778830 6345}s + \frac{47602869671183285760084471505 1}{63505020262695174050778830 6345}t + \frac{68039266309883015169772126 2019}{63505020262695174050778830 6345}st\big)\big[M_{20}r_8 - \big(-\frac{32}{295} + \frac{79}{590}s + \frac{239}{1180}t - \frac{2}{295}st\big)r_5 - \big(\frac{1}{118} + \frac{175}{9204}s - \frac{37}{708}t - \frac{8}{2301}st\big)r_{19} - \big(-\frac{163672936020607}{8280821341590840} - \frac{5446339771241}{8280821341590840}s + \frac{1466647737152911}{24842464024772520}t - \frac{112895764217191}{24842464024772520}st\big)r_{20} - \big(-\frac{154711128859044184179 49}{48213177351561594825096 0} + \frac{46352308240197290263 9}{96426354703123189650192}s + \frac{33550167028113468840 01}{96426354703123189650192}t + \frac{64967543547461907426 7}{16071059117187198275032 0}st\big)r_{30} - \big(-\frac{10651301509369440969463845296 9}{64628904336979597067019414470 4} - \frac{40146991289704439289875899 93}{16157226084244899266754853617 6}s + \frac{29460460973614958622585134491}{80786130421224496333774268088}t + \frac{11106236618900585625253606649}{64628904336979597067019414470 4}st\big)r_{40}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 58 | 57 | $(0,2)$ | $r_{58} = \big(\frac{712130076957301157818443}{4406256725535198978394 4} - \frac{267133368223145201243541 9}{5728133743195758671912 72}s - \frac{791007450401547317116729}{4406256725535198978394 4}t + \frac{312697350775843193113719 7}{5728133743195758671912 72}st\big)\big[M_{22}r_8 - \big(-\frac{13}{252} - \frac{3}{28}s + \frac{31}{252}t + \frac{5}{252}st\big)r_3 - \big(\frac{832}{2163} + \frac{817}{25956}s - \frac{11995}{77868}t + \frac{739}{38934}st\big)r_4 - \big(\frac{52063}{3481212} + \frac{5067977}{135767268}s + \frac{1115165}{10443636}t + \frac{288089}{15085252}st\big)r_{18} - \big(\frac{139868965294781878640 36}{7456063266619998680852811} + \frac{518062766586120304760857}{1491212653323999736170562 2}s - \frac{534902396537487156379 7}{1065151895231428382978973}t - \frac{692572658532630079418 11}{1491212653323999736170562 2}st\big)r_{38} - \big(-\frac{441817664559730363635000172978575977 23}{1231054870658423168219977490351615188 20} + \frac{124412275374779527260736931042353668 29}{1231054870658423168219977490351615188 20}s - \frac{185067833406545800928231216290762941 67}{1231054870658423168219977490351615188 20}t - \frac{522444357374571001531211854214354526 7}{1231054870658423168219977490351615188 20}st\big)r_{44} - \big(-\frac{160433195154359837751326609197567489418196 80}{2197008508219904421704862550856709362366623 71} - \frac{820090765166904035583763206170428122501118 3}{4394017016439808843409725101713418724733247 42}s - \frac{170143370943583053140765183309513884413413 79}{1464672338813269614469908367237806241577749 14}t + \frac{278702330573733207676435284562520101775825}{7323361694066348072349541836189031207888745 7}st\big)r_{46}\big]$ |
| 59 | 60 | $(0,4)$ | $r_{59} = \big(-\frac{69579817427}{9768759084} - \frac{68264602951}{126993868092}s + \frac{794887517689}{87918831756}t - \frac{3655715356831}{1142944812828}st\big)\big[M_{24}r_8 - \big(-\frac{27}{3760} - \frac{1}{3760}s + \frac{691}{11280}t - \frac{11}{11280}st\big)r_{10} - \big(-\frac{1624280749}{154096241680} - \frac{17629068581}{462288725040}s + \frac{41446963117}{462288725040}t - \frac{1498316831}{1386866175120}st\big)r_{28} - \big(-\frac{1922935611987 63}{1831428712780 7860} - \frac{4130079517622 43}{1831428712780 7860}s - \frac{2294088383957 49}{1831428712780 7860}t + \frac{1417020698962 83}{1831428712780 7860}st\big)r_{39} - \big(\frac{1070252476547499}{20781071361280040} + \frac{96439436843193}{20781071361280040}s - \frac{435831013795077}{20781071361280040}t + \frac{387785024664309}{20781071361280040}st\big)r_{53}\big]$ |
| 60 | 51 | $(0,-2)$ | $r_{60} = \big(-\frac{300264710794724801537}{2025215535939775851112 00} + \frac{604896604826922773431}{3290975245902135758057 00}s + \frac{376528954438117660 77}{5063038839849439627780 0}t + \frac{264057025235807752509 9}{2632780196721708606445600}st\big)\big[M_{15}r_9 - \big(-260 - 140s - 260t - 20st\big)r_5 - \big(\frac{34367}{278} - \frac{17203}{834}s + \frac{152257}{2502}t - \frac{12763}{2502}st\big)r_{19} - \big(-\frac{682163861522268745}{2926358050375746} - \frac{65874866879869745}{2926358050375746}s + \frac{162805978012299 55}{9754526834585 82}t + \frac{327659962033173 5}{9754526834585 82}st\big)r_{20} - \big(-\frac{343122300764561888243858}{2028338877683735375511} - \frac{153787467481488049982125}{4056677755367470751022}s - \frac{275389357819536975045 25}{1352225918455823583674}t + \frac{146788450206216621127 48}{2028338877683735375511}st\big)r_{30} - \big(-\frac{279277965632940542801539819088837}{7613065849864545756199744585 92} - \frac{106896609888387771165517613351503}{7613065849864545756199744585 92}s - \frac{172956148001390369884799186020681}{7613065849864545756199744585 92}t - \frac{135826040572937209284355825 41}{2537688616621515252066581528 64}st\big)r_{40} - \big(\frac{651772664864528938236199804627 15}{5552775809523600133246521748 08} + \frac{421753611660416666651468034997 45}{5552775809523600133246521748 08}s + \frac{225302699622433664464201800052 15}{5552775809523600133246521748 08}t + \frac{122937631045061340824560920880 25}{5552775809523600133246521748 08}st\big)r_{57}\big]$ |
| 61 | 40 | $(0,0)$ | $r_{61} = \big(-\frac{656061877756706034265379}{3115901647611164149795 2938} - \frac{139447144052736123437 3881}{1246360659044656599181175 2}s - \frac{604912838984485039917181}{9347704942833492449385881 4}t + \frac{281386582020130195799 4175}{3739081977133396979754352 56}st\big)\big[M_{13}r_{10} - \big(\frac{161800822}{286881301} - \frac{3498048314}{532779559}s + \frac{47725771946}{3729456913}t + \frac{8542392134}{3729456913}st\big)r_{11} - \big(-\frac{660911215093}{29835655304} - \frac{45720754721}{21311182360}s + \frac{826283989751}{149178276520}t + \frac{72694023497}{29835655304}st\big)r_{12} - \big(\frac{60322369457095716}{22993358695624681} - \frac{2093946568705554}{3284765527946383}s + \frac{102490514924200924}{22993358695624681}t + \frac{22466706048519634}{22993358695624681}st\big)r_{45}\big]$ |
| 62 | 25 | $(0,-4)$ | $r_{62} = \big(-\frac{330834174269}{3238563072753656} + \frac{8841174696529}{4857844609130484 0}s - \frac{7776431778029}{2428922304565242 0}t - \frac{971568233}{1457353382739145 2}st\big)\big[M_{15}r_{10} - \big(\frac{298470578}{204915215} - \frac{1674371186}{40983043}s + \frac{13914412678}{40983043}t - \frac{13861405438}{204915215}st\big)r_{13}\big]$ |
| 63 | 28 | $(0,-4)$ | $r_{63} = \big(-\frac{258157233789928163572 23}{9523954391937754839657 7024} + \frac{205491141854058554307873}{2380988597984438709914425600}s + \frac{116386652313651639523041 37}{7142965793953316129743276800}t - \frac{900601186475738720779 39}{2857186317581326451897310 72}st\big)\big[M_{17}r_{10} - \big(\frac{38663265201}{40983043} + \frac{12958189299}{40983043}s + \frac{18840390723}{40983043}t + \frac{1342361829}{40983043}st\big)r_{13} - \big(-\frac{22303445787}{81966086} + \frac{2235671235}{81966086}s - \frac{51680816745}{81966086}t - \frac{5339460603}{81966086}st\big)r_{62}\big]$ |
| 64 | 13 | $(0,-8)$ | $r_{64} = \big(\frac{14407601681}{551649987417600} + \frac{55567799}{220659994967040}s + \frac{310740179}{44131998993408}t + \frac{65298877}{68956248427200}st\big)\big[M_{19}r_{10}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 65 | 30 | $(0,-4)$ | $r_{65} = \big(\frac{150512782767509895455895377832263}{3062176336951908142104277605298232} + \frac{7137319491140448448511953466529}{3062176336951908142104277605298232}s - \frac{66003735681703949688058443183149}{91865290108557244263128328158946 96}t - \frac{308262366489321602782274623551 1}{9186529010855724426312832815894696}st\big)\big[M_{13}r_{11} - \big(\frac{85767}{29120} - \frac{478047}{29120}s - \frac{82569}{29120}t - \frac{122907}{29120}st\big)r_{13} - \big(-\frac{1099989}{116480} + \frac{4203}{16640}s - \frac{1214613}{116480}t + \frac{127089}{116480}st\big)r_{62} - \big(-\frac{201698571046149637}{64771261455073120} + \frac{1031520336978315719}{842026398915950560}s - \frac{1030692453299189324 27}{17682554377234961760}t + \frac{7639229006216707303}{2526079196747851680}st\big)r_{63}\big]$ |
| 66 | 18 | $(0,-8)$ | $r_{66} = \big(-\frac{1969657444}{130642349175} + \frac{1119147569}{261284698350}s - \frac{173016821}{87094899450}t + \frac{29976592}{43547449725}st\big)\big[M_{15}r_{11} - \big(\frac{15823}{40} - \frac{12931}{200}s + \frac{26963}{200}t + \frac{2917}{40}st\big)r_{64}\big]$ |
| 67 | 8 | $(1,1)$ | $r_{67} = \big(\frac{3458843460}{5556249157} - \frac{1440488132}{5556249157}s + \frac{8185326452}{16668747471}t + \frac{1600289980}{16668747471}st\big)\big[M_{20}r_{11}\big]$ |
| 68 | 42 | $(0,0)$ | $r_{68} = \big(\frac{4090479034231849874506571891802975434 1}{1363039932039938103881604039073113468 2} - \frac{8418727960130392233671435845600986520041}{1771951911651919535046085250795047508 66}s + \frac{4270740365593020605946917840233448813 77}{4089119796119814311644812117219340404 6}t + \frac{4300449548471005485741688637525732893 19}{5315855734955758605138255752385142525 98}st\big)\big[M_{28}r_{11} - \big(\frac{11945}{91512} + \frac{12557}{228780}s + \frac{4043}{76260}t - \frac{115}{30504}st\big)r_{11} - \big(\frac{6526231}{48806400} + \frac{112453}{1952256}s - \frac{6735}{650752}t + \frac{23783}{16268800}st\big)r_{12} - \big(\frac{66588671833}{2507565864640} + \frac{4503728791}{156722866540}s + \frac{7755445041}{78361433270}t - \frac{1300065773}{61160143040}st\big)r_{45} - \big(\frac{2011244396393}{197272494049920} - \frac{4851377212261}{197272494049920}s + \frac{12945240936013}{197272494049920}t - \frac{561241729973}{197272494049920}st\big)r_{61}\big]$ |
| 69 | 36 | $(0,-4)$ | $r_{69} = \big(\frac{843070131550419383}{20437523077220812800} + \frac{1002909477197013319}{20437523077220812800}s + \frac{438984124277399233}{20437523077220812800}t + \frac{3392423318332362 77}{20437523077220812800}st\big)\big[M_{13}r_{12} - \big(\frac{39293102496}{561045667} - \frac{3690507777}{561045667}s + \frac{13426678521}{561045667}t - \frac{740124396}{561045667}st\big)r_{13} - \big(-\frac{25296704757}{1122091334} + \frac{7273067463}{2244182668}s - \frac{19145721375}{2244182668}t + \frac{235283337}{86314718}st\big)r_{62} - \big(-\frac{16930598112532581703329 23}{22712272240933055715106 9} + \frac{6454642082605834080928 39}{22712272240933055715106 9}s - \frac{4303166866891455795590627}{22712272240933055715106 9}t + \frac{2836743917722133004155203}{22712272240933055715106 9}st\big)r_{63} - \big(\frac{24761278719565131638232804298836 70}{20872552126180847994133725369904 3} - \frac{1342424871791141566207578809128670}{20872552126180847994133725369904 3}s - \frac{3284400864646912426649941695927 82}{20872552126180847994133725369904 3}t - \frac{6667312182042443202408241511045 0}{20872552126180847994133725369904 3}st\big)r_{65}\big]$ |
| 70 | 10 | $(1,1)$ | $r_{70} = \big(\frac{9557623030385}{858589406832} + \frac{38715427917449}{17171788136640}s - \frac{881263723836 17}{17171788136640}t - \frac{1009821911969}{858589406832}st\big)\big[M_{20}r_{12} - \big(\frac{1534200217}{5820078128} - \frac{11887183}{98645392}s + \frac{8796301}{98645392}t + \frac{360176793}{5820078128}st\big)r_{67}\big]$ |
| 71 | 52 | $(0,0)$ | $r_{71} = \big(\frac{661503046313664981543160638792456928244685184 7}{8875688204849617676575738783594272429210621888 0} + \frac{3191484207321466766537149584103110860425375826 23}{1153839466630450297954846041867255415797380845440}s + \frac{8410018910848630787729445217677186338497772833 3}{8875688204849617676575738783594272429210621888 0}t + \frac{2695196270359221857963246481120764194853919419 93}{1153839466630450297954846041867255415797380845440}st\big)\big[M_{28}r_{12} - \big(-\frac{6711842219}{23508429981} + \frac{2033944567}{23508429981}s + \frac{844658977}{7836143327}t - \frac{128214345}{7836143327}st\big)r_{11} - \big(\frac{180294860759}{564202319544} - \frac{1325442509}{34402580460}s + \frac{29065807229}{470168599620}t + \frac{1892834867}{188067439848}st\big)r_{12} - \big(-\frac{27905255727163577}{193249857565024796} - \frac{2602609948864457}{193249857565024796}s + \frac{51785202677831}{6233866373065316}t + \frac{1200433062709477}{193249857565024796}st\big)r_{45} - \big(\frac{148010913217326557}{1900392822887893161} - \frac{32629218789267743}{3800785645775786322}s - \frac{5112790329946581 1}{3800785645775786322}t - \frac{3387058273854866}{1900392822887893161}st\big)r_{61} - \big(\frac{3861151318879584957035211395406679 1}{1464999114210991766587201195612468356} - \frac{1461552643484705624243456821510885}{1464999114210991766587201195612468356}s - \frac{2073685196012425350226921460792196 1}{4394997342632975299761603586837405068}t + \frac{8552268185472319917461857181810172 7}{4394997342632975299761603586837405068}st\big)r_{68}\big]$ |

| $i$ | Pivot | Cartan weight | Exact cyclic-row recipe |
|---|---|---|---|
| 72 | 54 | $(0,0)$ | $r_{72} = \big(-\frac{69438537504022661378278 82}{22030445917959822747744 09} - \frac{66230489021587786709810 5}{22030445917959822747744 09}s + \frac{69771782887571168795391 5}{73434819726532742492480 3}t + \frac{13768414268851325325201 4}{73434819726532742492480 3}st\big)\big[M_{12}r_{13} - \big(-\frac{529}{2730} - \frac{2381}{10920}s + \frac{4037}{10920}t - \frac{179}{2730}st\big)r_{11} - \big(-\frac{6535}{19968} + \frac{6663}{232960}s - \frac{4467}{46592}t + \frac{4829}{139776}st\big)r_{12} - \big(\frac{10183574325}{35906922688} - \frac{4903298691}{35906922688}s + \frac{14901979465}{107720768064}t + \frac{1512247655}{35906922688}st\big)r_{45} - \big(\frac{156496319719}{441379744260} - \frac{2593210825}{117701265136}s - \frac{1595719487}{117701265136}t - \frac{1718051448}{36781645355}st\big)r_{61} - \big(\frac{13690658691894650914402849 8589}{13610258396765650063058433 1840} - \frac{30470651078116145507747716313}{13610258396765650063058433 1840}s + \frac{22228707845556486162123639367}{45367527989218550021019477 7280}t + \frac{10386948288614646963018362 39}{64810754270312214315742111 040}st\big)r_{68} - \big(\frac{52206622962619490428014199938212303221}{21808638912639009662105664625169815491 2} - \frac{15412057930723464770982314829247618142 65}{19845861410501498792516154808904532096992}s - \frac{18697129045127439751600147373431582340 41}{45798141716541920290421895712856612531 52}t + \frac{75510921897528053683887473076146739457 7}{85053691759292137682212092038162280415 68}st\big)r_{71}\big]$ |
| 73 | 4 | $(1,-3)$ | $r_{73} = \big(-\frac{3836}{8553} + \frac{9010}{8553}s + \frac{5070}{2851}t + \frac{5476}{8553}st\big)\big[M_{20}r_{13} - \big(\frac{1057}{14160} - \frac{181}{7080}s - \frac{313}{7080}t - \frac{653}{14160}st\big)r_{55}\big]$ |
| 74 | 15 | $(1,1)$ | $r_{74} = \big(\frac{1480552983}{348776440} - \frac{1040424651}{906818744}s - \frac{46558911}{69755288}t + \frac{1767470271}{4534093720}st\big)\big[M_{22}r_{13} - \big(\frac{1}{84} + \frac{19}{168}s - \frac{71}{504}t + \frac{1}{63}st\big)r_{67} - \big(\frac{395094795623}{1050131090673} + \frac{124214829701}{4200524362692}s - \frac{623229116701}{4200524362692}t - \frac{84806819327}{2100262181346}st\big)r_{70}\big]$ |
| 75 | 1 | $(1,-5)$ | $r_{75} = \big(-1\big)\big[M_2r_{16}\big]$ |
| 76 | 7 | $(1,-1)$ | $r_{76} = \big(\frac{42736967548896837447 98}{25194010822170142361 77} + \frac{20662433224514922318 28}{32752210688211850703 01}s - \frac{51774787619512017662 84}{25194010822170142361 77}t - \frac{13482680295338391419130}{32752210688211850703 01}st\big)\big[M_{20}r_{19} - \big(\frac{227614992804767}{138013689026514 0} + \frac{534459391825553}{2760273780530280}s + \frac{184903447154011}{4140410670795420}t + \frac{378179077612627}{8280821341590840}st\big)r_{47}\big]$ |
| 77 | 9 | $(1,-1)$ | $r_{77} = \big(-\frac{12296533828469235600559761 41}{31490503250092964653395968 4} + \frac{35669752667643877508073699613}{20468827112560427024707379460}s - \frac{19197084447915320035384944619}{47235754875139446980093952 60}t + \frac{16518764555421433011254663279}{12281296267536256214824427676}st\big)\big[M_{20}r_{20} - \big(-\frac{72963393976678606792819}{48213177351561594825096} - \frac{22281505257489214112707}{48213177351561594825096}s + \frac{94388638116708006432 29}{24106588675780797412548}t + \frac{28650646420061948819 57}{24106588675780797412548}st\big)r_{47} - \big(\frac{66054444142019285533419457529}{36411922219594219511929671 8496} - \frac{29367518891911854285403105549}{36411922219594219511929671 8496}s + \frac{18613831393012585165633700653}{36411922219594219511929671 8496}t - \frac{28439006457468923073845353 01}{36411922219594219511929671 8496}st\big)r_{76}\big]$ |
| 78 | 3 | $(1,-5)$ | $r_{78} = \big(-\frac{8760167}{260789} - \frac{48381629}{3390257}s - \frac{5223961}{260789}t - \frac{17796239}{3390257}st\big)\big[M_{20}r_{21} - \big(-\frac{332133896545}{17813359471584} - \frac{126010301467}{44533398678960}s - \frac{1611925154891}{44533398678960}t - \frac{10690222245}{5937786490528}st\big)r_{75}\big]$ |
| 79 | 0 | $(1,-7)$ | $r_{79} = \big(1\big)\big[M_2r_{26}\big]$ |
| 80 | 6 | $(1,-3)$ | $r_{80} = \big(-\frac{3709}{31896} + \frac{132541}{414648}s + \frac{5489}{31896}t + \frac{19267}{414648}st\big)\big[M_{24}r_{33} - \big(\frac{307}{1880} + \frac{297}{1880}st\big)r_{55} - \big(-\frac{827}{3760} - \frac{77}{752}s - \frac{187}{752}t + \frac{143}{3760}st\big)r_{73}\big]$ |

# Part VI
# Formalization scope and source concordance

## S26 Lean formalization: present coverage and remaining work

The present Lean development does not formalize the central-charge-eight construction or its main theorem. It checks several useful implications inside the finite-closure and top-bound arguments. A separate, much larger development proves existence of coefficients satisfying the Evans–Gannon equations; its categorical reconstruction and its connection to this VOA are not formalized. These are substantive results with different scopes. Neither should be described as a formal verification of the entire CFT manuscript.

This assessment reads the revised manuscript, the local Lean sources, their saved check reports, and the installed Mathlib sources. The pinned toolchain is Lean `v4.33.1`; the Mathlib revision in `lean/lakefile.toml` is `0df444a360eaa60ab8c11dca51a86af692955474`. Paths below are relative to the repository; paths beginning `Mathlib/` are relative to `lean/.lake/packages/mathlib/`. Declaration names were checked against that checkout. This is a source and scope audit, not a new compiler run or an audit of every transitive dependency.

### S26.1 What is already checked

All five declarations below are in the namespace `HaagerupIzumi.CFT`. The two source files are independent of the Evans–Gannon umbrella import `HaagerupIzumi.lean`.

| Declaration and source | Exact scope of the conclusion |
|---|---|
| `nilpotent_raising_saturation`, `CFT/FiniteClosure.lean` | A submodule stable under two linear operators is the whole module, assuming a spanning equation modulo their two images and vanishing of every operator word of one fixed length. Joint nilpotence is an input; the theorem does not derive it from the actual current weights. |
| `projected_residual_recovery`, same file | From $v - Pv \in C$ and $Pv \in C$, deduce $v \in C$. No idempotence of $P$ is needed. The actual projector coefficients and the two membership assertions are not checked by this theorem. |
| `jacobi_bottom_witness`, same file | From $f(v) = g(u)$, $u = 0$, $v = cq$, and $c \neq 0$, deduce $f(q) = 0$. The Jacobi identity, the relevant quotient, and the particular bottom coefficient are explicit inputs, not reconstructed VOA facts. |
| `core_top_deficit_identity`, `CFT/TopBound.lean` | The printed fourteen-variable polynomial identity, with the exact dimension and integer-scaled trace coefficient definitions. |
| `core_top_bound`, same file | Nonnegative multiplicities and the two stated current-character inequalities imply the cleared-denominator 17/10 bound. It does not derive the fourteen tops, their multiplicities, or their stress traces. |

The full file paths have prefix `lean/HaagerupIzumi/`. The reports `c8_revision_20260927/lean_finite_closure_final.log` and `c8_revision_20260927/lean_top_bound.log` record successful checks. Their axiom reports contain only `propext`, `Quot.sound`, and, where used, `Classical.choice`. The projected-residual lemma has an empty axiom report. These are the usual logical foundations of the stated Lean proofs, rather than new CFT-specific assumptions. The rational reconstruction of the top table in `c8_revision_20260927/check_top_bound.py` is separate from the Lean proof of the inequality.

### S26.2 Assessment of every manuscript section

The difficulty descriptions distinguish a *local conditional formalization*, in which the surrounding mathematical structures and theorems are supplied explicitly, from an *end-to-end formalization*, which constructs those structures and proves the inputs. "Focused" means a bounded algebraic or analytic lemma with nearby library support. "Substantial" means several interacting new definitions and proofs. "Foundational" means that major mathematical theories or deep external theorems still have to be formalized. These are assessments of dependency scope, not estimates of working hours.

| Current anchor | Existing CFT coverage | Missing work and conversion difficulty |
|---|---|---|
| Section 1 p. 11 | No formal main theorem or definition of the constructed $U$. | A theorem assembling explicit interfaces would be focused once its inputs exist. A proof of the stated existence theorem inherits all the foundational dependencies below; packaging them as assumptions does not discharge them. |
| Appendix A p. 35 | No lattice VOA, cocycle-to-field construction, or embedding is formalized here. | Finite lattice arithmetic, the four-component coefficient algebra, and low-weight identities are substantial but concrete targets. Construction of the Fock space, lattice fields, PCT, and their VOA identities is foundational. |
| Appendix B p. 38 | No filtered Wick theorem or all-weight recursion is formalized. | A graded mode-algebra interface could support the induction, filtration, skew-symmetry, and Wick reductions. This is a substantial local project; deriving that interface for the actual lattice VOA adds foundational work. |
| Appendix C p. 41 | Raising saturation and projected-residual recovery are checked as abstract implications. | Prove bounded-current-weight joint nilpotence, the singlet projection identities, and the actual rank and residual certificates. These are focused to substantial once the finite spaces and current actions are represented faithfully. |
| Appendix D p. 43 | Only the abstract bottom-witness scalar cancellation is checked. | Derive the quotient Jacobi relation from mode identities, justify the charged spanning and neutral-family reductions, and verify all nonzero coefficients. The local proof is substantial; most scalar identities are focused. |
| Appendix E p. 45 | No theorem applies the gates to the actual subVOA. | Given a formal recursive criterion and genuine gate inputs, the induction, dimension comparison, and properness deduction are focused. Without the bridge from finite data to actual modes, they remain conditional. |

| Current anchor | Existing CFT coverage | Missing work and conversion difficulty |
|---|---|---|
| Appendix F p. 46 | No embedded Potts/parafermion VOA or regularity theorem is formalized. | Finite positive-operator inequalities and the displayed stress identity are accessible local targets. Their application requires conformal vectors, $C_2$ spaces, regular modules, cosets, and the relevant rationality theorems: foundational work. |
| Appendix G p. 51 | The final fourteen-variable top-bound identity and inequality are checked. | Derive the core-top table and current inequalities; formalize branching, the rank-four exclusion, rank-six based-ring argument, formal codegrees, and center data. Conditional finite arithmetic is focused to substantial. VOA tensor theory and fusion-category classification remain foundational. The optional classification route must retain its extra hypothesis. |
| Appendix H p. 65 | No theorem identifies the full character vector. | Finite twist arithmetic and the equation $6 = 6 + 13\gamma$ are focused. The proof needs trace characters, Zhu modularity, the specified modular representation, finite image, the principal-part uniqueness argument, and the Evans–Gannon fundamental matrix. Their complete formalization is foundational, despite existing scalar modular-form infrastructure. |
| Appendix I p. 66 | None of the analytic comparison propositions is formalized. | Common-core reduction, dense-domain transport, analytic continuation, and the Haag-duality squeeze are bounded local targets. Conformal nets, DHR sectors, reconstruction, annulus integration, sewing, and fusion positivity constitute the principal foundational burden. Detailed distinctions appear below. |
| Appendix J p. 74 | No unitary ribbon identification is checked. | The ring-level scalar arguments and cyclic-group obstruction calculations can be isolated. A proof of the categorical conclusion needs unitary fusion categories, the relevant Izumi classification, Morita equivalence, and the center/ribbon comparison. General rigid and braided categories in Mathlib supply only part of this framework. |
| Appendix K p. 76 | No Q-system extension, diagonal full theory, or $E_8$ identification is formalized. | Once extension correspondences and categorical duality are inputs, transport of the chosen algebra and finite dimension checks are substantial local targets. Proving the correspondences, the full-net construction, and the holomorphic-VOA classification is foundational. |

| Current anchor | Existing CFT coverage | Missing work and conversion difficulty |
|---|---|---|
| Appendix L p. 78 | No formal identification with another physical construction. | Most assertions are scope statements, not independent formal theorems. One can formalize the conditional character and extension comparisons after the earlier interfaces exist. Neither those interfaces nor current Lean files imply a spin-chain continuum limit or uniqueness of all $c = 8$ theories. |
| Appendix M p. 78 | No Lean authentication or mathematical replay of the frozen data. | A small sound certificate checker with explicit data is a concrete, substantial target. Its soundness must connect encoded vectors and matrices to the actual fields. Hash agreement binds files but proves neither their residual identities nor their VOA interpretation. |
| Appendix N p. 81 | The five construction declarations listed above, plus elementary support lemmas in their two files. The current appendix also records the two thermal lemmas assessed in the physics supplement below. | Keep theorem names, assumptions, source hashes, toolchain, and axiom reports synchronized. This is focused maintenance; the appendix must not enlarge the scope of the checked declarations. |
| Appendix O p. 82 | Successful TeX compilation is not a Lean result. | A reproducible build manifest can record the checked artifacts and interfaces. Mathematical review, compiler checking, finite replay, and PDF compilation are separate checks with separate conclusions. |

### S26.3 Finite certificates and the algebraic sections

The new companion now supplies concrete data for this next stage: a 120-row short-generation graph with normalization and target expansions, 33 projected Gate 4 vectors and all twelve relations, the explicit Potts and parafermion membership expressions, and the eighty-current space's matrices and cyclic-basis recipes. These are in `c8_reviewability_20260927/`, with separate successful Python coordinate, residual, and matrix checks. They remove the need to extract those witnesses again. They introduce no Lean proofs: formal parsers, checker soundness, and the interpretation of the supplied mode recipes and matrices remain to be established.

For Appendices A–E and the ledger in Appendix M, the most valuable next boundary is between the sparse finite data and the mode calculations that they purport to represent. A certificate checker should have a proved interpretation map from its encoded vectors into a mathematical vector space. It should prove that its addition, scalar multiplication, mode actions, and selected column combinations have the stated meanings. Only then does a zero residual certify the required identity in that space. The further identification of this space and action with the actual lattice fields is a distinct theorem.

The coefficient algebra uses $1, s, t, st$ with $s^2 = 13$ and $t^2 = -3$. One can first verify

multiplication and residual identities in this four-dimensional rational algebra. Rank arguments additionally need a field, or an appropriately injective embedding into the intended complex coefficient field; merely writing a quotient by the two polynomials does not establish those properties. Sparse vectors fit `Finsupp.linearCombination` in `Mathlib/LinearAlgebra/Finsupp/LinearCombination.lean`. The declarations `Matrix.rank_of_det_ne_zero`, `Matrix.rank_submatrix_le`, and `Matrix.rank_eq_finrank_span_cols` in `Mathlib/LinearAlgebra/Matrix/Rank.lean` provide useful rank interfaces. A nonzero minor together with an independently proved upper bound gives an economical exact rank certificate. A modular nonzero minor may also prove a characteristic-zero lower bound once the reduction map and cleared denominators are justified; a modular zero residual is insufficient for characteristic-zero containment.

The current-algebra part has more reusable support than a new representation-theory library would suggest. In `Mathlib/Algebra/Lie/Sl2.lean`, `IsSl2Triple.HasPrimitiveVectorWith` packages a highest-weight vector. Its lemmas `lie_h_pow_toEnd_f`, `lie_e_pow_succ_toEnd_f`, and `pow_toEnd_f_eq_zero_of_eq_nat` give the standard weight, raising-coefficient, and finite-dimensional nilpotence calculations. These can support the singlet polynomials and the bounded-weight argument supplying `nilpotent_raising_saturation`. They do not identify the present current actions with integrable affine-VOA modules. Nor does the generic saturation theorem verify the saved spanning matrix.

Mathlib already has formal vertex *operators*: `VertexOperator`, `VertexOperator.ncoeff`, and `VertexOperator.of_coeff` are in `Mathlib/Algebra/Vertex/VertexOperator.lean`. They use Hahn-series linear maps from `Mathlib/Algebra/Vertex/HVertexOperator.lean`, which provides coefficient and composition operations. This is relevant groundwork for mode indexing and lower truncation. The inspected file does not provide a complete conformal vertex algebra with vacuum, translation, Jacobi/locality, modules, and invariant forms; its own TODO list includes locality and residue products. Directory and declaration searches, followed by inspection of these files, located no existing implementation of the manuscript's complete VOA interface. That is a statement about the audited checkout, not a claim that Mathlib contains no relevant vertex-algebra ingredients.

The filtered Wick criterion is therefore a worthwhile intermediate deliverable. Its hypotheses should explicitly state the mode identities and filtration controls used in the proof. Formalizing only an abstract implication is useful, provided the final theorem continues to display those hypotheses until the actual lattice construction supplies them.

For Appendix F, the positive stress identity separates similarly. Finite-grade self-adjoint operators and their nonnegative quadratic forms can be treated with the inner-product-space library. For example, `LinearMap.IsSymmetric.eigenvectorBasis` in `Mathlib/Analysis/InnerProductSpace/Spectrum.lean` supports finite-dimensional diagonalization. Turning the inequality into regularity requires the specified conformal embeddings and the module-theoretic finiteness results. A formal proof of a positive matrix identity alone cannot establish those embeddings or $C_2$-cofiniteness.

Appendix G has three distinct layers. The first is the core-top enumeration and exact trace inequality; only the last linear implication is currently checked. The second is finite algebra: branching multiplicities, a based ring on four or six symbols, dimension equations, and root-of-unity sums. These are credible conditional targets using finite types, matrices, polynomial identities, and number theory. The third is the interpretation through semisimple tensor categories, formal codegrees, indicators, induction, and a modular center. It needs structures and theorems beyond a table of fusion coefficients. In particular, verifying a fusion table does not verify a pentagon, construct its associator, or classify its categorifications.

The optional algebraic classification Proposition G.13 (p. 64) must have an explicit parame-

ter expressing exhaustion by the four categories in the cited list. Its pivotal-scalar exclusion can then be formalized separately. The repository's Evans–Gannon coefficient construction does not discharge that exhaustion hypothesis. The manuscript's unitary route avoids this optional assumption, but a Lean version of that route still needs the separate unitary classification and Morita-equivalence theorems. Avoiding one external premise does not automatically formalize the replacement theorem.

For Appendix H, the installed library has `ModularForm` in `Mathlib/NumberTheory/ModularForms/Basic.lean` and `UpperHalfPlane.qExpansion`, `UpperHalfPlane.qExpansion_eq_zero_iff`, and `UpperHalfPlane.qExpansion_coeff_unique` in `Mathlib/NumberTheory/ModularForms/QExpansion.lean`. There are also scalar theta, eta, congruence-subgroup, and level-one developments. They do not directly supply Zhu's vector-valued characters or the exact finite-image representation used here. An appropriate first theorem would prove principal-part uniqueness under explicit transformation, holomorphy, cusp, and invariant-vector hypotheses, and then distinguish its application to the actual trace functions from finite coefficient arithmetic.

## S26.4 The analytic comparison in detail: A2, A3, and A1

The target is Appendix I, *From vacuum unitarity to complete unitarity*, beginning on p. 66.

The manuscript now gives mathematical proofs of all three comparison steps. Their lack of a Lean formalization is not an assertion that they remain mathematical conjectures. Conversely, replacing each of them by a Lean proposition whose proof is an assumed field would only record the dependency graph.

**A2: the actual subnet and sector count.** Proposition I.1 (p. 67) is the precise target. A local formalization can separate the branching multiplicities $1, 1, 2$, the endomorphism algebra $\mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C})$, and the rank-six based-ring argument. Those are finite algebra once the representation objects and their decompositions are supplied. The passage from an algebraic VOA summand to a reducing Hilbert subspace requires the A3 common-core lemma, energy bounds, and an actual represented subnet. The proof of local irreducibility uses locality, strong additivity, the shared rotation action, and the separating vacuum; all four inputs must be visible in a formal statement.

End-to-end A2 then needs finite-index local extensions, DHR sectors, statistical dimension, the canonical algebra, center induction, and complete rationality. The index-two exclusion also needs the precise net-automorphism/VOA-automorphism correspondence and the finite-order Lie-algebra fixed-rank argument. Its untwisted $\mathbb{Z}/3$ subcategory comes from the transverse diagonal and antidiagonal isotropic lines, followed by free-module induction. A formal proof should preserve that construction: an abstract group of three invertible symbols does not determine a trivial associator.

Mathlib does provide `VonNeumannAlgebra` and `VonNeumannAlgebra.commutant` in `Mathlib/Analysis/VonNeumannAlgebra/Basic.lean`. Here a concrete von Neumann algebra is packaged by its bicommutant property. This is useful support for locality and commutant arguments. The file is not a conformal-net, DHR, subfactor-index, or Connes-fusion library. Those theories were not located in the audited checkout. The distinction matters: the basic operator-algebra vocabulary exists, while the net-theoretic theorems used by A2 still need formalization.

**A3: reconstruction, common cores, and exhaustion.** The reconstruction statement is Proposition I.3 (p. 69), with its strong-unitarity consequence in Proposition I.4. The most

accessible analytic target is the graph-reduction Lemma I.2 (p. 69). Represent a densely defined operator by `LinearPMap`, require closability, and assume explicitly that the bounded orthogonal projection preserves its domain and commutes there. The map $(\xi, \eta) \mapsto (P\xi, P\eta)$ preserves the graph and therefore its closure. This proves commutation on the closed domain. The adjoint claim follows from the defining inner-product identity, using $P^* = P$ and domain invariance. Both arguments can be formalized without claiming that the original core is a core for the adjoint.

The existing declarations `LinearPMap.IsClosable.graph_closure_eq_closure_graph`, `LinearPMap.IsClosable.closure_isClosed`, and `LinearPMap.HasCore` are in `Mathlib/Topology/Algebra/Module/LinearPMap.lean`. The complementary file `Mathlib/Analysis/InnerProductSpace/LinearPMap.lean` provides `LinearPMap.IsFormalAdjoint` and `LinearPMap.adjoint`. Projection APIs include `Submodule.orthogonalProjectionOnto` and `Submodule.starProjection` in `Mathlib/Analysis/InnerProductSpace/Projection/Basic.lean`. These are actual reusable declarations, not proposed abstractions. The further conclusion about the von Neumann algebra generated by an unbounded field requires an affiliation or polar/spectral-generation interface in addition to the graph lemma.

The next target is the energy-domain argument. Starting with a positive energy representation whose bounded-energy vectors form a weak module, regularity gives an ordinary simple summand $S$. One must prove that its Hilbert closure reduces $L_0$ and that its projection sends every bounded-energy vector into a finite sum of finite-dimensional grades of $S$. This establishes $PM \subset S \subset M$ before any field commutation is used. Assuming finite multiplicities for the original net representation would remove precisely the issue the proof resolves. Likewise, a smeared field need not map the bounded-energy core to itself; its value is a Hilbert-space vector. A Lean signature that accidentally makes every smeared field an endomorphism of that core would encode a stronger and inappropriate premise.

The grade-by-grade comparison of two positive invariant forms is another bounded target. It first constructs an algebraic module endomorphism, uses an eigenvector on a finite-dimensional lowest grade and simplicity to prove that endomorphism scalar, and only then obtains boundedness of the module isomorphism. This order avoids importing a Hilbert-space Schur lemma before boundedness is known. Finite direct sums and the counting argument that twelve reconstructed simple modules exhaust twelve algebraic simples are comparatively small after the interfaces are fixed.

Full A3 still requires the Henriques–Tener reconstruction theorem, including its bounded-energy extension, equality of closed smeared fields, and reconstruction of the original vacuum VOA. Formalizing the consequences of an explicitly supplied reconstruction interface would be useful. It would not be a formal proof of that theorem.

**A1: pairwise annulus compatibility and positivity.** The compatibility assertion is Proposition I.5 (p. 71); its domain and continuation steps are Lemmas I.6 and I.7. The final application is Theorem I.8. The revised proof deliberately constructs a lift for each compatible annulus pair. It does not require a global canonical choice of lifts for every annulus. A suitable formal interface would record the following data and laws:

1. the centrally extended annulus semigroup and its localized subsemigroups, together with their positive-energy representations;
2. the common localizer $\delta$, localized lifts $D_A, D_B$, and the pair $\widetilde{A} = \widetilde{\delta}^{-1} D_A$, $\widetilde{B} = D_B \widetilde{\delta}$, so that $\widetilde{B}\widetilde{A} = D_B D_A$ is canonically localized;
3. the equality of local annulus actions with the represented vacuum operators, incoming

insertions, sewing, and the outgoing unitary adjoint formula;

4. the round-annulus factorization giving both finite-energy range inclusion and density of the transported form domain;

5. holomorphy on the sewn interior, coverage by all allowed geometric representatives, and the no-insertion generators.

The deck ambiguity must be retained: changing the common lift conjugates the inserted field by $e^{2\pi i m L_0}$. Integer VOA weights make this conjugation trivial even when module lowest weights are fractional. Normalizing a vacuum operator alone does not eliminate the deck group on those modules. This small algebraic cancellation is a realistic formal target once the grading and representation laws are explicit.

The analytic continuation step can use `AnalyticOnNhd.eqOn_of_preconnected_of_eventuallyEq` from `Mathlib/Analysis/Analytic/Uniqueness.lean`, applied to scalar matrix coefficients after all holomorphy and component hypotheses are supplied. The finite-energy range and dense-domain lemmas can also be isolated from annulus geometry. The final equality of concrete nets is an elementary conditional commutant argument:

$$\mathcal{A}^{\mathrm{bl}}(I) \subset \mathcal{N}(I) \subset \mathcal{A}^{\mathrm{bl}}(I')' = \mathcal{A}^{\mathrm{bl}}(I).$$

It explicitly uses locality and Haag duality on the same vacuum Hilbert space; an abstract isomorphism of the two nets would not suffice.

These smaller arguments do not build the central extension, integrate Virasoro representations, prove geometric sewing, or prove positivity of Connes fusion and the resulting canonical VOA fusion pairings. Those are the large missing foundations. An honest intermediate formalization would name the exact integration, reconstruction, and sewing hypotheses, prove their pairwise consequence, and keep the positive-fusion theorem separate until it too is formalized.

## S26.5 Categories, extensions, and the independent Evans–Gannon proof

Mathlib has substantial general categorical infrastructure. Verified examples include `CategoryTheory.BraidedCategory` in `Mathlib/CategoryTheory/Monoidal/Braided/Basic.lean`, `CategoryTheory.RigidCategory` and duality interfaces in `Mathlib/CategoryTheory/Monoidal/Rigid/Basic.lean`, and `CategoryTheory.HalfBraiding`, `CategoryTheory.Center`, `CategoryTheory.Center.forget`, and `CategoryTheory.Center.ofBraided` in `Mathlib/CategoryTheory/Monoidal/Center.lean`. These permit genuine reuse in Appendices G, J, and K. They do not by themselves provide a unitary fusion category with finite simple-object data, positive spherical dimensions, modular $S, T$, a Q-system correspondence, or the Haagerup classification. No such complete manuscript-level framework was located in the local source audit.

A finite based ring can be formalized before its categorical realization. One can then prove the conditional dimension equations and obstruction calculations, keeping separate the theorem that a given unitary category realizes that ring. Similarly, transporting an already constructed algebra along a braided equivalence is distinct from proving that it yields a local net or a unitary VOA extension. The formal target for Appendix K must preserve the manuscript's scope: the original embedded extension and the chosen $\mathcal{H}_3$ extension need not be the same algebra; the 36 bimodule defects preserve $U$; and the holomorphic-net construction is not identified with the $E_8$ net by the VOA classification theorem.

The existing Evans–Gannon development should be credited precisely. In `lean/HaagerupIzumi/Analytic/GammaConstruction.lean`, `HaagerupIzumi.GammaContinuation.hypGamma`

constructs the hyperbolic-gamma interface from the proved integral development. `HaagerupIzumi.ruijsenaarsHypGamma` specializes it to the required periods, and `HaagerupIzumi.EGEquations_ruijsenaars` proves the equations for the resulting concrete coefficient data. The theorem `HaagerupIzumi.exists_EGEquations_odd` assumes only `Oddn` and produces complex coefficient data satisfying all the packaged Evans–Gannon equations. The source genuinely supplies the earlier analytic interface; this final theorem does not assume an unconstructed gamma function.

The packaged proposition `HaagerupIzumi.CoeffData.EGEquations` is defined in `lean/HaagerupIzumi/Completion.lean`; it records the linear symmetries and axis sums together with the quadratic, cubic, and quartic equations. The project README explicitly stops before the reconstruction theorem and categorical conclusions of the separate infinitude manuscript. Thus this development does not prove exhaustion of rank-six fusion categories, construct the present lattice subVOA, or establish its complete unitarity. Its finite Fourier infrastructure can be reused: the local interpolation proof already uses `ZMod.dft` and related declarations from `Mathlib/Analysis/Fourier/ZMod.lean`. Its contour machinery may be useful for appropriate scalar analytic sublemmas, but it is not an annulus or conformal-net construction.

### S26.6 A staged program with explicit completion criteria

The following stages give reviewable deliverables without suggesting that a short final Lean file would amount to a short foundational proof. Some stages can proceed independently; their applications must retain unproved inputs visibly.

1. **Certify finite data.** Define the exact coefficient algebra, sparse vectors, and matrix witnesses; prove a small checker's soundness; import and verify selected residual and rank certificates. The deliverable is a theorem about named finite data, with the interpretation map stated. It is not yet a theorem about the actual VOA.

2. **Complete the current-algebra layer.** Formalize the bounded weight argument for joint nilpotence, the singlet projection, the charged-module spanning deductions, and the top-table arithmetic. Connect these to the already checked saturation and inequality lemmas. The deliverable removes the corresponding finite algebraic assumptions.

3. **Formalize the recursive closure criterion.** Give a precise graded mode-algebra interface and prove the filtered Wick and Jacobi reductions. Instantiate the finite gates through the certified data. Construction of the genuine lattice VOA and validation of that instantiation remain explicit subsequent obligations.

4. **Separate representation theory from finite consequences.** Formalize the core-top and branching deductions, the based-ring classification, and character uniqueness under stated VOA and categorical interfaces. Keep optional categorical exhaustion as a separate parameter. The deliverable should make clear which deep representation-theory theorems remain inputs.

5. **Formalize the analytic lemmas.** Prove common-core reduction, adjoint commutation, dense-domain transport, grade-wise form comparison, deck cancellation, and the conditional Haag-duality squeeze. State reconstruction and annulus comparison with their full domains and normalizations. These supply reusable analysis without pretending to formalize the entire net/VOA correspondence.

6. **Discharge the foundational interfaces.** Build the required VOA, module tensor, conformal-net, annulus, and unitary fusion theories; formalize the cited reconstruction,

classification, positivity, and extension theorems; construct the actual objects and connect the finite certificates. Only completion of these dependencies supports an end-to-end Lean theorem matching the manuscript's full conclusion.

### S26.7 Assumptions, computation, and the trust boundary

A theorem with explicit parameters such as a spanning identity or a reconstruction theorem can be fully kernel-checked as an implication. Its parameters are not new global logical axioms. Nevertheless, the application to the CFT is unproved until those parameters are supplied. An interface whose fields simply repeat the desired conclusion adds little verification; useful interfaces expose the smaller mathematical laws from which the conclusion is actually derived.

An admitted `sorry` introduces `sorryAx`; a declared `axiom` installs an unproved proposition. Neither is equivalent to a theorem with all premises discharged. A source search of the local `HaagerupIzumi` Lean files found no uses of `sorry`, new `axiom` declarations, or `native_decide`; the saved axiom reports give the more relevant dependency information for the specific checked conclusions. A text search alone is not a substitute for an axiom audit of the final theorem.

There is a version-specific point about computation. In this Lean 4.33.1 installation, `native_decide` calls `Lean.Meta.nativeEqTrue`; the implementation in `Lean/Meta/Native.lean` compiles and evaluates a closed Boolean and, on success, introduces an auxiliary axiom asserting that its value is true. The kernel then checks a proof relative to that axiom; it does not independently replay the native computation by ordinary reduction. The older `Lean.ofReduceBool` mechanism is deprecated in this toolchain. Consequently a future certificate replay using `native_decide` would need an explicit expanded trust statement, even if it is useful operationally.

By contrast, ordinary proof-producing tactics and a `decide` proof checked by kernel reduction can verify an exact certificate without that native-evaluation axiom, subject to the final dependency report. For large data, the practical target is a small sound checker and manageable proof certificates, not necessarily replaying the original search algorithm inside the kernel. A Boolean printed by `#eval`, a Python exact calculation, a successful hash check, and a theorem with a kernel-checked proof are different artifacts. They should remain separately labeled in the ledger.

Future compiler and certificate runs must retain the repository's shared-desktop resource policy and run serially through its guarded launcher. The preserved baseline assessment introduced no new Lean declarations and does not claim any additional kernel-checked results beyond the recorded files above.

## S27 Additional assessment of the physics revision

The new physics sections provide much more explicit mathematical data: character coefficients, Virasoro-primary multiplicities, spin-resolved spectra, thermal intervals, low-weight couplings, rational correlators, fusion coefficients, defects, and boundary partition functions. None of the Python producers in `c8_physics_20260927/` is a Lean proof or has a Lean-verified implementation. Their exact arithmetic and internal checks improve reviewability and supply concrete inputs for formalization; they do not enlarge the scopes of the five existing CFT declarations listed in the preceding assessment. A separate new file adds the two small thermal-comparison lemmas stated below, bringing the selected CFT coverage to seven principal declarations in three files. The five baseline declarations retain their original scope.

## S27.1 Current manuscript anchors and historical concordance

The linked numbers, titles, and pages in the following table refer to the current manuscript. The first column is only a concordance for readers of the archived 55-page PDF; its former section numbers are not operative references in this guide. The current document places the mathematical development in appendices after the physics sections.

| Archived label | Current manuscript title | Current anchor |
|---|---|---|
| 1 | Introduction and statement of results | Section 1, p. 11 |
| 2 | The explicit lattice construction | Appendix A, p. 35 |
| 3 | Finite closure and the all-weight criterion | Appendix B, p. 38 |
| 4 | Gate 4: exact seeds, raising, and singlet recovery | Appendix C, p. 41 |
| 5 | Gate 5 from Jacobi identities | Appendix D, p. 43 |
| 6 | From the finite gates to properness | Appendix E, p. 45 |
| 7 | Regularity from a positive conformal-vector identity | Appendix F, p. 46 |
| 8 | Branching and the boundary fusion ring | Appendix G, p. 51 |
| 9 | The complete character vector | Appendix H, p. 65 |
| 10 | From vacuum unitarity to complete unitarity | Appendix I, p. 66 |
| 11 | Unitary categorical identifications | Appendix J, p. 74 |
| 12 | Full net theory and a chosen $E_8$ VOA extension | Appendix K, p. 76 |
| 13 | Relation to the proposed Haagerup theories | Appendix L, p. 78 |
| A | Certificate ledger and reproducibility | Appendix M, p. 78 |
| B | Limited formalization | Appendix N, p. 81 |
| C | Compilation and review scope of the base revision | Appendix O, p. 82 |

The added physics material has the following current anchors. “Focused”, “substantial”, and “foundational” retain the dependency-based meanings stated earlier.

| Current section title | Current anchor |
|---|---|
| Characters and a spin-resolved physical spectrum | Section 2, p. 14 |
| Exact correlators and channel data | Section 3, p. 20 |
| A complete weight-two chiral three-point dataset | Section 4, p. 24 |
| Modular data, topological defects, and conformal boundaries | Section 5, p. 27 |
| The Haagerup symmetry of the holomorphic extension | Section 6, p. 30 |
| Perturbations, bootstrap coordinates, and use of the data | Section 7, p. 33 |

## S27.2 Characters, Virasoro primaries, and the full spectrum

Section 2, *Characters and a spin-resolved physical spectrum*, is in `c8_physics_20260927/spectrum.tex`. Its producer `spectrum.py` checks the published initial coefficients, the rational matrix recurrence, nonnegative integrality through the requested grade, descendant reconstruction, parity of the resulting spin spectrum, and an independent $E_4/\eta^8$ identity. The default data include 25 coefficients for each of twelve characters, complete full-CFT spectra through $\Delta = 24$, and an independent $E_8$ spectrum. These are finite arithmetic conclusions conditional on the input matrices and their character interpretation.

| Physics argument | Missing Lean theorem and difficulty |
|---|---|
| Matrix recurrence | Define the twelve-dimensional exponent/characteristic data and a finite rational recurrence, prove nonzero denominators, and prove that its finite output agrees with the formal differential equation. This is focused to substantial. Identifying the resulting vector with the actual VOA trace functions still requires the foundational character theorem assessed under Appendix H. |
| Virasoro-primary subtraction | The finite Euler-product convolution and its inverse are focused. Their interpretation needs the irreducible $c = 8$ Virasoro characters, the Kac determinant or an equivalent irreducibility theorem, and the unitary highest-weight decomposition of each module. These are substantial to foundational representation-theoretic inputs. |
| Complete spin-resolved cutoff | Given nonnegative lowest weights and coefficient functions, prove the product formula and that no grade above $N$ contributes to $\Delta \leq N$. The finite sums, parity identity, and relevant/marginal counts are focused. The tensor-product Hilbert-space interpretation and the diagonal modular invariant are additional premises. |
| $E_8$ comparison and invariant counts | A finite identity between the oscillator series and the selected character combination is focused. It does not identify the extension algebra. Counts $1488 = 6 \cdot 248$ for one chiral Haagerup action and $36 = 6^2$ for both independent actions additionally require the precise invariant-space theorem. That categorical interpretation belongs to the separate defect formalization below. |

There is useful existing power-series infrastructure. `PowerSeries.coeff_mk` and `PowerSeries.coeff_mul` occur in `Mathlib/RingTheory/PowerSeries/Basic.lean`. The file `Mathlib/Combinatorics/Enumerative/Pentagonal/PowerSeries.lean` contains the Euler-product development, including `PowerSeries.coeff_prod_one_sub_X_pow_eventually_eq` and `PowerSeries.WithPiTopology.hasProd_one_sub_X_pow`. These declarations were read in the pinned checkout. They support formal coefficient manipulations; they do not state that these coefficients count this CFT's Virasoro primaries.

An economical formalization would first take an abstract sequence of graded dimensions and prove the descendant-removal and tensor-product counting lemmas. A second layer would connect that sequence to the Evans–Gannon recurrence. The final interpretation theorem must connect both layers to the actual modules. Merely importing the printed integers as definitions and proving that their sum is 2941 would check arithmetic but leave the substantial spectral assertion open.

### S27.3 Thermal bounds and their numerical intervals

Proposition 2.1 (p. 18) proves the modular heavy-tail estimate in Section 2. For $E = \Delta - 2/3$, an exact cutoff $D > 2/3$, and $b' = 4\pi^2/b$, it uses

$$Z = L_D + H_D, \qquad Z(b) = Z(b'), \qquad H_D(b) \leq e^{-(b-b')(D-2/3)} H_D(b')$$

to bound the tail at $b > 2\pi$. Two real-number steps are now checked in `lean/HaagerupIzumi/CFT/PhysicsBounds.lean`, in namespace `HaagerupIzumi.CFT`:

- `modular_heavy_tail_comparison` assumes $L + H = L_d + H_d$, $H \le rH_d$, and $0 \le r < 1$, and proves $H \le r(L - L_d)/(1 - r)$;
- `selfdual_tail_first_moment_comparison` assumes $E_{\mathrm{cut}} > 0$ and $E_{\mathrm{cut}}H \le L'$, and proves $H \le L'/E_{\mathrm{cut}}$.

The successful guarded compiler record is `c8_physics_20260927/physics_lean.log`, with sampled peak group RSS 1.549 GiB. Both axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`; the source contains neither `sorry` nor new axioms. The nonnegative contraction range is retained explicitly, although the first rearrangement only needs $r < 1$ once its comparison is supplied.

These lemmas do not derive the comparison from a spectral sum or identify $r$ with the exponential in the application. The actual infinite-sum theorem remains more substantial: it needs convergence, termwise positivity, completeness of the finite cutoff, and the modular functional equation. Its self-dual form also needs differentiability and the identity $Z'(2\pi) = 0$. The first and second moment bounds require justified moment convergence and the elementary optimization of $E^k e^{-aE}$ on a positive half-line.

The Python output is a conditional interval certificate. Its positive spectrum and modular invariance are mathematical premises, not facts established by evaluating finitely many coefficients. The implementation uses directed decimal arithmetic, a fixed rational enclosure of $\pi$, and correctly rounded exponential/logarithm functions; the displayed six-decimal table is accepted only when every rigorous interval lies within $10^{-6}$ of its printed value. None of that implementation is currently kernel checked.

For a Lean version, a small rational interval checker with proved exponential and logarithm enclosures would be preferable to treating Python's decimal library as an axiom. One can separately formalize the analytic tail theorem over the real numbers and feed it certified finite moment bounds. Existing real analysis and infinite-sum infrastructure is relevant, but the complete pipeline from characters to the saved numerical intervals is new work. The two checked implications are the completed local algebraic part; sound interval evaluation and its analytic interpretation are substantial. The CFT assumptions still inherit the foundational requirements of the earlier sections.

### S27.4 Currents, lattice correlators, and low-weight couplings

The correlator and coupling material in Sections 3 and 4 is supplied by `c8_physics_20260927/correlators.tex` and `c8_physics_20260927/couplings.tex`. Its data distinguish direct finite lattice-mode products, rational identities reconstructed from saved singular OPEs, and coefficients completed using current Ward identities. Those three forms of evidence have different interpretation obligations; the final reports specify the supplied bases and direct probes.

In particular the completed weight-two dataset in `c8_physics_20260927/full_weight_two/` uses all 113 chiral Virasoro primaries. Its saved report records 1,442,897 ordered tensor slots, 18,365 nonzero entries, 51,076 current Ward checks, and 33 additional direct lattice-mode probes beyond the seed calculation. This supplies substantially more explicit input than a list of summed OPE coefficients. Those successful Python checks remain separate from the Lean obligations below.

The following finite targets are realistic once the coefficient algebra and basis conventions are represented in Lean:

- singular current brackets, levels, gauge factors, pairings, and the orthogonality and positivity of the stated finite Gram matrices;
- polynomial identities obtained by clearing denominators in the displayed rational four-point functions, with collision loci excluded when interpreting them as functions;
- exact extraction of three-point coefficients and their squared unit-norm values, retaining the specified invariant/Hermitian pairing and adjoint conventions;
- the finite Ward recursion, its compatibility with both current factors, and uniqueness of its completion from the supplied seed slice;
- the completeness of the chosen low-weight primary subspace, provided its exact rank and descendant-complement certificates.

These are substantial finite algebra rather than a need to formalize all CFT at once. The bridge from encoded lattice words to actual mode operators is the same missing bridge described for the finite gates. A successful check of a completed tensor against Ward identities establishes consistency of the completion. Its identification with all actual mode products also needs a proved determination theorem and authenticated seed products; a finite selection of direct probes does not by itself check every omitted product.

Beyond that finite layer, one needs the state-field correspondence, locality, conformal covariance, OPE convergence or formal expansion identities, and an explanation of how the chiral functions assemble into local bulk correlators. The known Potts and parafermion inputs also need their formal embeddings. These are foundational. A chiral vacuum correlator or a chosen lattice intertwiner block is not by itself the four-point function of the lightest diagonal bulk scalar. The formal statement must preserve the external fields and their normalization rather than silently promote one kind of block to the other.

No claim that a dimension-two scalar is exactly marginal follows from its existence or its three-point coefficient alone. A formal statement about a conformal manifold would additionally require a precise deformation theory, beta functions, and the appropriate all-orders argument. That subject is outside both the present Lean files and the finite arithmetic producer.

## S27.5 Modular data, defects, and conformal boundaries

Sections 5 and 6, *Modular data, topological defects, and conformal boundaries* and *The Haagerup symmetry of the holomorphic extension*, are in `c8_physics_20260927/defects.tex`. Their producer `defects.py` computes in $\mathbb{Q}(\zeta_{13})$, with exact Verlinde coefficients, modular relations, the forgetful multiplicities, and boundary/defect spectral tables. These checks establish finite algebraic properties of the input matrices and based rings, not the existence of the required associator or a conformal defect.

| Physics argument | Missing Lean theorem and difficulty |
|---|---|
| Cyclotomic modular arithmetic | Prove that the twelve-coordinate quotient representation is a field embedded in $\mathbb{C}$, fix the primitive root and conjugation, and verify the Gauss-sum square root and sign. Then $S^2 = 1$, the twist relation, Verlinde integrality, and fusion associativity are substantial but bounded exact computations. |
| Diagonal Verlinde lines | Given a unitary modular tensor category and the diagonal full CFT, derive line eigenvalues, twisted character matrices, and their sector multiplicities. The scalar sums are focused; the theorem that these matrices describe actual topological defects requires the categorical/full-CFT construction and is foundational. |
| Cardy boundaries and annuli | Formalize the boundary-state normalization, open/closed-channel identity, and the positive square roots entering $g$ factors. Finite character multiplicities are focused. Their interpretation as boundary Hilbert spaces requires the Cardy/extension theorem. Formal verification of annulus integers is not that theorem. |
| Selected chiral $\mathcal{H}_3$ action | Prove the center-forgetful correspondence for the chosen extension, the bimodule/full-defect equivalence, and the identification of tube multiplicity spaces with $M_\mu \otimes \overline{H'}$. These are foundational categorical inputs. Once supplied, the $6 \times 12$ forgetful ring-map check and charge matrices are substantial finite algebra. |
| Invariant and charged gaps | Under the correct action, derive the fixed spaces and count their lowest scalar grades. This is focused after the sector assignment. It must distinguish one chiral action from both independent factors, and tube multiplicity spaces from their multiple copies in actual defect Hilbert spaces. Modular covariance alone is insufficient. |
| Six extension boundaries | Verify the finite annulus coefficients from the forgetful functor and the six-object fusion ring. The data-to-category interface must retain the actual module category, duals, and boundary normalization. The finite arithmetic is substantial; constructing these boundaries as part of the full CFT remains foundational. |

The pinned checkout defines `CyclotomicField` and `IsCyclotomicExtension` in `Mathlib/NumberTheory/Cyclotomic/Basic.lean`; it also has primitive-root, complex-root-of-unity, and cyclotomic-polynomial developments. Thus the field-theory layer need not start from scratch. It still takes work to prove that the producer's concrete coordinate arithmetic and chosen complex embedding agree with those structures. An arbitrary root of the polynomial without the intended complex embedding does not determine the positive quantum dimensions used by the physics interpretation.

### S27.6 Bootstrap placement and observable scope

The physics reading guide in Section 7 distinguishes the diagonal full theory from the selected holomorphic extension and states which published bootstrap plots use which symmetry sectors. Formalizing the label correspondence, the central-charge convention $c_L = c_R = 8$, and a rational gap extracted from a proved spectrum would be focused. It would not formally verify a published numerical exclusion curve. That requires the original functional, its positivity on every required conformal block, the truncation/error bounds, and a sound numerical or exact certificate. A plotted point read from a figure is evidence for comparison, not a kernel-checked exclusion or inclusion theorem.

The observable inventory is also an interface specification. Characters determine unflavoured multiplicities and, with analytic control, partition functions. They do not determine every normalized OPE coefficient or defect eigenvector. The new coupling and defect datasets close specific parts of that gap; each needs its own sound checker and interpretation theorem. Keeping those claims separate is essential to an eventual formal treatment that a physicist can use without reading the implementation.

### S27.7 A concrete order of formalization

A useful first deliverable is a small exact coefficient library covering the rational recurrence, finite Euler product, primary reconstruction, and cutoff-completeness lemma. Its theorem should expose the input character interpretation explicitly. A second deliverable is a sound finite algebra checker for the cyclotomic modular data and the supplied low-weight tensors, with an explicit interpretation map. A third is the real-analytic tail theorem and a certified interval evaluator. Each can be independently reviewed before the foundational VOA and defect interfaces are complete.

Completion criteria must be semantic: the formal theorem should refer to the actual character coefficients, mode products, or represented sectors through proved maps. File hashes, successful Python assertions, or a Lean theorem about unrelated constants cannot replace that connection. The two explicitly stated thermal-comparison lemmas are the additional Lean results of this physics revision; the much larger analytic and physical interpretations remain to be formalized.

## S28 Source concordance, version decisions, and assembly record

The base is `hi_cft_c8_proof.tex`, together with its structural, analytic, physics, and bibliography inputs as they stood on 29 September 2026. The base PDF has 78 pages. The two companion PDFs have 20 and 17 pages. This edition folds their relevant source text into the same document and adds the supplements named below. Its page numbers should not be substituted into historical audit reports for those earlier PDFs.

### S28.1 Result-to-source concordance

The following table identifies the principal data families. The individual supplements give more precise file names, formulas, and scope boundaries. The assembly manifest supplies full SHA-256 hashes. A hash fixes the bytes used by the transcription; it is not a proof of the mathematics in those bytes.

| Printed material | Saved source family |
|---|---|
| Main construction and proof | `hi_cft_c8_proof.tex`, `hi_cft_c8_manuscript_structure.tex`, `hi_cft_c8_manuscript_analytic.tex`. |
| Physics exposition | `c8_physics_20260927/spectrum.tex`, `correlators.tex`, `couplings.tex`, `defects.tex`, `physics_guide.tex` in the same directory. |
| Small independent verification guide | `hi_cft_c8_verification_guide.tex`, `c8_physics_20260927/verification_finite_inputs.tex`, `c8_reviewability_20260927/structural_inputs.tex`. |
| Finite fields and generation | `hi_cft_20260923_c8_exact_fields.json`, `hi_cft_20260923_c8_minimal_generators.json`, `c8_reviewability_20260927/minimal_generation_witness.json`, the exact grade-three records, and the Gate 4/5 records cited in the finite supplement. |
| Small stress and structural coordinates | `c8_reviewability_20260927/grade2_fields.tsv`, `c8_reviewability_20260927/structural_fields.tsv`. |
| Stress-factor and commutant records | The `hi_cft_20260923_proof_` and `hi_cft_20260923_next_structure_` result families, with the individual artifacts named in the structural supplement. |
| Character and recurrence tables | `c8_physics_20260927/data/characters.json` and `c8_physics_20260927/data/fundamental_matrix.json`. |
| Full spectra and scalars | `c8_physics_20260927/data/diagonal_spectrum.tsv`, `e8_spectrum.tsv`, `scalars.tsv` in the same directory. |
| Thermal intervals | `c8_physics_20260927/data/thermal.json`; the separate `thermal_input_check.json` in the parent directory records the earlier check of the supplied enclosure of $\pi$. |
| Complete primary basis and coupling atlas | `c8_physics_20260927/full_weight_two/low_weight_basis.json`, `complete_primary_couplings.json` and `complete_primary_couplings.tsv` in that same subdirectory. The parent-directory export is the older 96-field record. |
| Four-point and block tables | `c8_physics_20260927/correlators.json`, `correlators.global_blocks.tsv`, `correlators.terms.tsv`, and `correlators.samples.tsv` in the same directory. |
| Fusion, defects and boundaries | `c8_physics_20260927/defect_data.json`, `Cardy_annuli.tsv`, `E8_U_preserving_annuli.tsv`, and the remaining named defect TSV exports. |
| Original scoped replay ledgers | `c8_revision_20260927/finite_input_ledger_final.json`, `c8_reviewability_20260927/reviewer_manifest.json`, `c8_physics_20260927/release_audit.json`. |
| Formalization assessment | `c8_physics_20260927/lean_baseline_assessment.tex` and `c8_physics_20260927/lean_assessment.tex`. |

## S28.2 Specific version and interpretation decisions

The older search records retain their original conclusions and caveats. The new exposition uses the current manuscript for the main theorem's status. In particular, the initial list of missing annulus comparison arguments is superseded by the printed A1–A3 arguments in the current analytic appendix. The reader should assess those arguments directly; neither a finite output file nor the current edition's document build certifies them.

The distinction between the original embedded extension and the selected extension persists throughout. The original boundary category is only identified up to the stated $\mathcal{H}_2/\mathcal{H}_3$ alternative. The separately selected extension is the one used for the specified Haagerup-action observables. A scalar character identity does not choose an extension algebra, and identical modular matrices do not resolve every equivalence of tensor categories.

The exact 956-dimensional grade-three calculation is a finite vector-space statement. The grade-four coefficient 6010 in the character table is a consequence of the full character identification. It is not relabeled as a direct finite rank computation. Likewise, the finite Gate 4 and Gate 5 records support their named reduction arguments; their counts do not mean that every required channel was separately recomputed as an ambient high-weight vector.

The character table uses $h_{d_3} = 15/13$, with grade measured above this actual ground weight. Replacing it with the congruent twist representative $2/13$ changes the physical spectrum and is incorrect. The $a$ and $c_0$ characters agree, but both labels remain in sums over simple sectors. For noninvertible symmetry, invariance means eigenvalue equal to the vacuum eigenvalue, rather than equal to one. The different first-factor and both-factor invariant multiplicities are retained.

The complete 113-field primary tensor remains a chiral vacuum-module tensor. It does not provide the nonvacuum bulk three-point constants of the diagonal theory. Selected fractional lattice blocks likewise do not constitute sewn bulk four-point functions. No extra scattering, RG-flow, or exact-marginality claim is inferred from printing additional entries.

## S28.3 What is deliberately represented compactly

Large families are reduced only when the text gives a complete recovery rule. Opposite spin rows are identical and are folded with their individual multiplicities specified. The center fusion tensor is symmetric in its two input labels, so the unordered products determine every ordered entry. The primary coupling atlas uses one nonzero component for every nonzero ordered multiplet map, its exact phase and magnitude, and the explicit current-equivariance reconstruction. Its accompanying support table distinguishes a zero map from an omitted or uncomputed one. A finite recurrence specifies the remaining matrix coefficients without listing every matrix at every grade.

Raw high-weight sparse vector stores, SQLite elimination databases, and all historical exploratory logs are not reproduced wholesale. Their mathematical identities, the recorded dimensions, the selected exact witnesses, and the remaining provenance obligations are stated. This keeps the document a verification draft rather than a line-for-line printout of the repository.

## S28.4 Rebuilding and checking this edition

The main source is `c8_expanded_20260929/haagerup_c8_expanded.tex`. Its copied figures are in `assets/`; the new input fragments and tables are beside it. Compile from that directory, with no shell escape. The README supplies guarded commands. No research scripts are imported or executed by document compilation.

`assemble_draft.py` transcribes the existing character, spectrum, thermal, and coordinate records and incorporates the base source and companions. `physics_tables/build_physics_tables.py` produces the new physics table fragments from the completed release. Their input manifests record source identities. The table and document audit report is separate from the historical research replay reports. Rerunning these transcription tools requires the original repository inputs; compiling the distributed TeX source package does not.

All original source and data files are opened read-only. A copy of the base TeX inputs is preserved under `base/`. The permanent shared desktop policy remains in force: the September 15 memory exception has expired. No new numerical research calculation is needed for this edition.

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